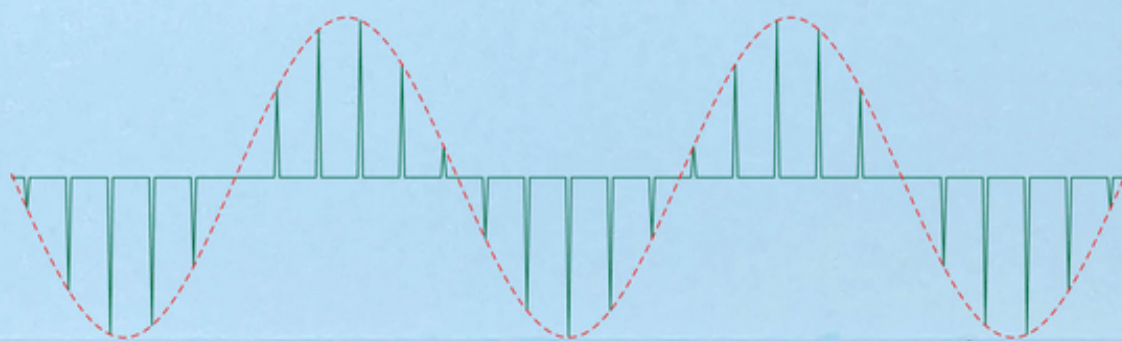


OTFS Modulation

Theory and Applications

Saif Khan Mohammed
Ronny Hadani
Ananthanarayanan Chockalingam



 IEEE Press

WILEY

OTFS Modulation

IEEE Press
445 Hoes Lane
Piscataway, NJ 08854

IEEE Press Editorial Board
Sarah Spurgeon, *Editor-in-Chief*

Moeness Amin
Jón Atli Benediktsson
Adam Drobot
James Duncan

Ekram Hossain
Brian Johnson
Hai Li
James Lyke
Joydeep Mitra

Desineni Subbaram Naidu
Tony Q. S. Quek
Behzad Razavi
Thomas Robertazzi
Diomidis Spinellis

OTFS Modulation

Theory and Applications

Saif Khan Mohammed

Indian Institute of Technology
New Delhi, India

Ronny Hadani

University of Texas
Austin, USA

Ananthanarayanan Chockalingam

Indian Institute of Science
Bangalore, India

 **IEEEPress**

WILEY

Copyright © 2024 by The Institute of Electrical and Electronics Engineers, Inc. All rights reserved.

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.

Published simultaneously in Canada.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning, or otherwise, except as permitted under Section 107 or 108 of the 1976 United States Copyright Act, without either the prior written permission of the Publisher, or authorization through payment of the appropriate per-copy fee to the Copyright Clearance Center, Inc., 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400, fax (978) 750-4470, or on the web at www.copyright.com. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008, or online at <http://www.wiley.com/go/permission>.

Trademarks: Wiley and the Wiley logo are trademarks or registered trademarks of John Wiley & Sons, Inc. and/or its affiliates in the United States and other countries and may not be used without written permission. All other trademarks are the property of their respective owners. John Wiley & Sons, Inc. is not associated with any product or vendor mentioned in this book.

Limit of Liability/Disclaimer of Warranty: While the publisher and author have used their best efforts in preparing this book, they make no representations or warranties with respect to the accuracy or completeness of the contents of this book and specifically disclaim any implied warranties of merchantability or fitness for a particular purpose. No warranty may be created or extended by sales representatives or written sales materials. The advice and strategies contained herein may not be suitable for your situation. You should consult with a professional where appropriate. Further, readers should be aware that websites listed in this work may have changed or disappeared between when this work was written and when it is read. Neither the publisher nor authors shall be liable for any loss of profit or any other commercial damages, including but not limited to special, incidental, consequential, or other damages.

For general information on our other products and services or for technical support, please contact our Customer Care Department within the United States at (800) 762-2974, outside the United States at (317) 572-3993 or fax (317) 572-4002.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic formats. For more information about Wiley products, visit our web site at www.wiley.com.

Library of Congress Cataloging-in-Publication Data:

Names: Mohammed, Saif Khan, author. | Hadani, Ronny, author. | Chockalingam, Ananthanarayanan, author.

Title: OTFS modulation : theory and applications / Saif Khan Mohammed, Ronny Hadani, Ananthanarayanan Chockalingam.

Description: Hoboken, New Jersey : Wiley, [2024] | Includes bibliographical references and index.

Identifiers: LCCN 2024023399 (print) | LCCN 2024023400 (ebook) | ISBN 9781119984184 (hardback) | ISBN 9781119984207 (adobe pdf) | ISBN 9781119984214 (epub)

Subjects: LCSH: Modulation (Electronics) | Wireless communication systems.

Classification: LCC TK5102.5 .M635 2024 (print) | LCC TK5102.5 (ebook) | DDC 621.3815/36–dc23/eng/20240801

LC record available at <https://lcn.loc.gov/2024023399>

LC ebook record available at <https://lcn.loc.gov/2024023400>

Cover Design: Wiley

Cover Images: Courtesy of Shailesh Rao M.

Set in 9.5/12.5pt STIXTwoText by Straive, Chennai, India

To my parents, my wife, and my daughter

– Saif Khan Mohammed

In memory of Noga Porat

– Ronny Hadani

In memory of my elder brother S. A. Anbumani

– Ananthanarayanan Chockalingam

Contents

Preface *xv*

Acknowledgements *xvii*

Acronyms *xxi*

1	Introduction	1
1.1	Cellular Mobile Evolution	2
1.1.1	Moore's Law Drives Rate Increase	4
1.2	Multipath Fading Channels	5
1.2.1	Frequency Domain Characterization	6
1.2.2	Time Domain Characterization	7
1.2.3	Impact of High Dopplers	9
1.2.4	Delay-Doppler Domain Characterization	12
1.2.4.1	Urban Multi-lane Scenario – An Example	14
1.3	Communication Waveforms for Doubly Selective Channels	14
1.3.1	OTFS – A Modulation Waveform for Doubly Selective Channels	16
1.3.2	OTFS Realization – Two Approaches	16
1.4	Waveforms for Radar Sensing	17
1.4.1	OTFS for Integrated Sensing and Communication	18
1.5	Organization of the Book	19
2	Delay-Doppler Signaling and OTFS Modulation	23
2.1	Delay-Doppler Domain	28
2.2	Time and Frequency Domain Modulation	29
2.2.1	Channel Interaction of a TD Pulse	30
2.2.2	Channel Interaction of a FD Pulse	32
2.3	Delay-Doppler (DD) Domain Modulation	34
2.3.1	Origin of Quasi-periodicity	37
2.4	Channel Interaction of a DD Domain Pulse	39
2.4.1	The Channel Interaction is Predictable	40
2.4.2	The Channel Interaction is Non-fading	43
2.4.3	The Channel Interaction is Non-stationary	43
2.5	Time- and Bandwidth-Limited DD Domain Carrier Waveforms	43
2.5.1	Example in Fig. 2.17	47

2.5.2	Orthogonality of Pulsones	49
2.5.3	Optimality as Time- and Bandwidth-limited Signals	50
2.5.4	TDM as a Limiting Case	50
2.5.5	FDM as a Limiting Case	50
2.5.6	TD Pulsones Encode Wireless Channel Dynamics	50
2.5.7	The Fourier Transform as a Composition	51
2.6	Zak-OTFS Modulation and I/O Relation	51
2.6.1	Generalized Transceiver Signal Processing	52
2.6.2	Zak-OTFS Modulation	54
2.6.3	Zak-OTFS Receiver	56
2.6.4	Zak-OTFS I/O Relation	57
2.7	Predictability of the Zak-OTFS I/O Relation in the Crystalline Regime	58
2.7.1	Non-predictability of the Zak-OTFS I/O Relation in the Non-crystalline Regime	61
2.7.2	Crystalline Decomposition	62
2.7.3	Identification of Linear Time-Varying Channels	63
2.7.4	Error in Prediction of the Zak-OTFS I/O Relation	64
2.8	Matrix-vector Description of the I/O Relation	67
2.8.1	Zak-OTFS	67
2.8.2	TDM	68
2.8.3	FDM	69
2.9	Impact of Fading in the Crystalline Regime	70
2.10	Model-free Operation in the Crystalline Regime	73
2.10.1	Model-dependent Operation	73
2.10.2	Model-free Operation	74
2.11	Summary	77
Appendix 2.A	Cascade of Two Doubly Spread Channels and Twisted Convolution	78
2.A.1	Properties of Twisted Convolution	81
Appendix 2.B	Channel Action on DD Domain Signal	83
Appendix 2.C	Zak Transform	84
Appendix 2.D	Derivation of (2.30) and (2.31)	85
Appendix 2.E	Proof of Non-overlapping Received Impulses in a-response	86
Appendix 2.F	Inverse Zak Transforms	87
2.F.1	Inverse Time-Zak Transform	87
2.F.2	Derivation of (2.38)	87
2.F.3	Inverse Frequency-Zak Transform	88
Appendix 2.G	Twisted Convolution Preserves Quasi-periodicity	88
Appendix 2.H	Derivation of TD Pulsone Expression in (2.41)	89
Appendix 2.I	Derivation of FD Pulsone Expression in (2.44)	90
Appendix 2.J	TDM I/O Relation	90
2.J.1	Proof of Theorem 2.5	91
Appendix 2.K	FDM I/O Relation	92
2.K.1	Proof of Theorem 2.6	93
Appendix 2.L	Discrete DD Domain Signals	94
2.L.1	Impulse Signal in Discrete DD Domain	95
Appendix 2.M	Derivation of Zak-OTFS Modulator Architecture	96
Appendix 2.N	Derivation of Zak-OTFS Receiver	97
Appendix 2.O	Proof of Theorem 2.1	99

3	Approximations of OTFS Modulation	101
3.1	Pulsones as a Basis for TD Signals	109
3.2	Generating Pulsones Using the Heisenberg Transform	110
3.3	Generating Time- and Bandwidth-Limited Pulsones	113
3.3.1	Comparing MC Pulsones with Zak Pulsones	117
3.4	Multicarrier OTFS (MC-OTFS) Modulation	118
3.4.1	Two-step MC-OTFS Modulator	119
3.4.2	Zak Transform-Based MC-OTFS Modulator	122
3.5	MC-OTFS Receiver	123
3.5.1	Two-step MC-OTFS Receiver	124
3.5.2	Zak Transform-Based MC-OTFS Receiver	125
3.6	MC-OTFS I/O Relation	126
3.6.1	MC-OTFS I/O for a Two-Step Transceiver	126
3.6.2	MC-OTFS I/O Relation for Zak Transform-Based Transceiver	129
3.7	Comparing MC-OTFS to Zak-OTFS	130
Appendix 3.A	Proof of Theorem 3.1	133
Appendix 3.B	Proof of Corollary 3.14	134
Appendix 3.C	Proof of Theorem 3.2	134
Appendix 3.D	SFT and Inverse SFT	135
Appendix 3.E	Proof of Theorem 3.3	138
Appendix 3.F	Proof of Theorem 3.4	138
Appendix 3.G	Proof of Theorem 3.5	139
Appendix 3.H	Impact of Windowing on a Periodic DD Domain Impulse	140
Appendix 3.I	Proof of Theorem 3.6	141
Appendix 3.J	Proof of Theorem 3.7	142
Appendix 3.K	Proof of Theorem 3.9	143
Appendix 3.L	Proof of Theorem 3.10	144
Appendix 3.M	Proof of Theorem 3.11	145
Appendix 3.N	Derivation of Bi-orthogonality Condition	145
Appendix 3.O	Proof of Corollary 3.5	146
Appendix 3.P	Derivation of I/O Relation for the Two-step MC-OTFS Transceiver	146
Appendix 3.Q	Proof of Theorem 3.13	146
Appendix 3.R	Proof of Corollary 3.6	147
4	Delay-Doppler Diversity in OTFS	149
4.1	Diversity in SISO OTFS	150
4.1.1	System Model	150
4.1.2	Diversity Analysis	154
4.1.3	Full Diversity with Phase Rotation	157
4.2	Diversity in MIMO-OTFS	161
4.2.1	MIMO-OTFS Diversity Analysis	162
4.2.2	MIMO-OTFS with Phase Rotation	163
4.3	Diversity in Space-Time Coded OTFS	165
4.3.1	STC-OTFS Scheme	166
4.3.1.1	Encoding and Decoding	166
4.3.1.2	Diversity Analysis	167
4.3.1.3	Phase Rotation in STC-OTFS	168

4.4	Diversity in OTFS with Antenna Selection	170
4.4.1	MIMO-OTFS with RAS	170
4.4.2	STC-OTFS with RAS	172
4.4.3	Rank of Multi-antenna OTFS Systems	173
4.4.4	Analysis of Multiantenna OTFS with RAS	173
4.4.5	Rank Deficient Multi-antenna OTFS Systems with RAS	178

5 OTFS Signal Detection 183

5.1	Linear Detection	184
5.1.1	Reduced Complexity MMSE Detection	185
5.1.1.1	Complexity Reduction Scheme 1	185
5.1.1.2	Complexity Reduction Scheme 2	187
5.2	Approximate MAP-Based Detection	189
5.2.1	Message Passing Detection	189
5.2.2	Variational Bayes-Based Detection	192
5.2.3	Hybrid MAP and PIC Detection	194
5.2.4	MCMC Sampling-Based Detection	197
5.3	Cross-Domain Detection	198

6 Delay-Doppler Channel Estimation 207

6.1	PN Pilot-Based Estimation	208
6.1.1	Solving for $(\delta_i, \omega_i, \alpha_i)$	209
6.1.2	Performance Results	211
6.2	Exclusive Pilot Approach	212
6.2.1	Impulse-Based Channel Estimation	212
6.2.1.1	Integer DD and Fractional DD	214
6.2.2	DDIPIC-Based Estimation	215
6.2.2.1	Coarse Estimation	217
6.2.2.2	Fine Estimation	217
6.2.2.3	Stopping Criterion	218
6.2.2.4	Refinement of Parameters	218
6.2.2.5	Choice of ϵ	218
6.2.2.6	Complexity	218
6.2.2.7	Performance	219
6.3	Embedded Pilot Approach	221
6.3.1	Embedded Pilot with Guard Symbols	221
6.3.1.1	Threshold-Based Estimation	221
6.3.2	Embedded Pilot Without Guard Symbols	223
6.3.2.1	System Model	223
6.3.2.2	Embedded Frame Structure	224
6.3.2.3	SBL-Based Estimation	224
6.4	Superimposed Pilot Approach	227
6.4.1	System Model	227
6.4.2	SP-NI Scheme	229
6.4.3	SP-I Scheme	230
6.4.4	Performance	231

7	Multuser OTFS	233
7.1	Resource Elements	234
7.2	OTFS Uplink	236
7.2.1	Orthogonal Multiple Access (OMA)	237
7.2.1.1	Guard Band (GB)-Based OMA	237
7.2.1.2	Interleaved Delay-Doppler OMA	240
7.2.1.3	Interleaved Time-Frequency OMA	245
7.2.1.4	Performance of OTFS-Based OMA Methods with Ideal Pulses	249
7.2.1.5	Performance of OTFS-Based OMA Methods with Practical Rectangular Pulses	251
7.2.2	Non-orthogonal Multiple Access	256
7.2.2.1	Multuser MIMO-OTFS Uplink	256
7.2.2.2	Massive MIMO-OTFS	263
7.3	OTFS Downlink	265
7.3.1	Downlink Precoding	266
7.3.1.1	Linear Precoding	267
7.3.1.2	Nonlinear Precoding	270
7.3.2	Massive MIMO-Based OTFS Downlink	271
7.3.2.1	MRT-Based Massive MIMO-OTFS Precoding	271
7.4	Multuser Channel Estimation	276
7.5	Synchronization	277
7.5.1	Frequency Synchronization	277
7.5.2	Timing Synchronization	277
7.5.2.1	Uplink Timing Synchronization in OFDM-Based Systems	278
7.5.2.2	Uplink Timing Synchronization in OTFS-Based Systems	278
7.5.2.3	Downlink Timing Synchronization in OTFS-Based Systems	283
8	Practical Considerations in OTFS Systems	285
8.1	PAPR of MC-OTFS Signals	286
8.1.1	MC-OTFS Transmit Signal	287
8.1.2	Upper Bound on PAPR	288
8.1.3	CCDF of PAPR	289
8.1.4	Results	290
8.1.4.1	Effect of Increasing M and N on the CCDF of PAPR	290
8.1.4.2	Effect of Pulse Shaping on PAPR	291
8.1.4.3	Comparison with GFDM and OFDM	292
8.2	Impact of Phase Noise on OTFS	292
8.2.1	System Model Without Phase Noise	293
8.2.2	System Model with Phase Noise	294
8.2.3	Performance in the Presence of Phase Noise	295
8.3	Impact of IQ Imbalance on OTFS	298
8.3.1	OTFS System Model with IQI	299
8.3.1.1	System Model with Transmitter IQI	299
8.3.1.2	System Model with Receiver IQI	300
8.3.2	Sensitivity of OTFS and OFDM to IQI	300
8.4	Discrete Zak Approach to OTFS	302
8.4.1	DZT-OTFS System Model	302
8.4.2	Vectorized DD Domain I/O Relation	304

8.4.3	Diversity Analysis of DZT-OTFS	305
8.4.4	Performance Results	307
8.4.4.1	Diversity Performance	307
8.4.4.2	BER Performance	309
8.4.4.3	Complexity	310
Appendix 8.A Digital Implementation of Zak-OTFS Modulation		311
8.A.1	Digital Implementation of the Zak-OTFS Receiver	311
8.A.2	Digital Implementation of the Zak-OTFS Transmitter	314
Appendix 8.B Discrete Zak Transform		316
8.B.1	Discrete Delay and Doppler Shifts	318
8.B.2	Discrete Pulsones	319
9	Deep Learning for OTFS Transceiver Design	327
9.1	Learning Framework	328
9.1.1	Classification of Machine Learning Programs	328
9.1.1.1	Classification Based on Training Dataset	328
9.1.1.2	Classification Based on Nature of Objective	329
9.1.2	Traditional Program and Machine Learning Program	329
9.1.3	Deep Learning Using Neural Networks	330
9.1.3.1	Fully Connected Neural Networks	330
9.1.3.2	Recurrent Neural Networks	333
9.1.3.3	Convolutional Neural Networks	333
9.1.4	Common Terminologies in the DL Framework	333
9.2	OTFS Signal Detection Using DNNs	335
9.2.1	Full-DNN Detection Approach	335
9.2.2	Symbol-DNN Detection Approach	336
9.2.2.1	Training and Testing	337
9.2.3	Performance Results	337
9.2.3.1	Static Channel with i.i.d. Gaussian Noise	338
9.2.3.2	Static Channel with Non-Gaussian Noise	339
9.2.3.3	Doppler Channel with Non-Gaussian Noise	342
9.2.3.4	Correlated Gaussian Noise in MIMO-OTFS	342
9.3	IQI Estimation/Compensation Using DNNs	344
9.3.1	DNN-Based OTFS Transceiver	344
9.3.1.1	Tx IQI Compensation	344
9.3.1.2	Rx IQI Compensation	345
9.3.1.3	Channel Training and Detection	346
9.3.2	Performance Results	347
9.3.2.1	Tx IQI Compensation	348
9.3.2.2	Rx IQI Compensation	349
9.3.2.3	Channel Training and Detection	350
9.3.2.4	Combined Performance of All DNNs	351
9.4	DD Channel Estimation Using DNNs	353
9.4.1	Estimation Using Embedded Pilots	355
9.4.1.1	Architecture	356
9.4.1.2	Training Methodology	356
9.4.1.3	Inference from the Network	357

9.4.1.4	Performance Results	358
9.4.2	Estimation Using Interleaved Pilots	360
9.4.2.1	Architecture	362
9.4.2.2	Training Methodology	363
9.4.2.3	Estimation of Delay and Doppler Indices	363
9.4.2.4	Complexity	363
9.4.2.5	Performance Results	364
9.4.3	Estimation Using Superimposed Pilots	365
9.4.3.1	Iterative Scheme	366
9.4.3.2	Performance Results	366
10	OTFS for Radar Sensing	371
10.1	Radar Scene	373
10.2	The Radar Ambiguity Function	374
10.2.1	Radar Resolution	376
10.2.2	Ambiguity Functions for TD and FD Pulses	379
10.3	A Good Radar Waveform: Quasi-localization in TD and FD Using Modulated Pulse Trains	381
10.4	Zak Theoretical Framework for Design of Radar Waveforms	383
10.4.1	Ambiguity Function of the Zak-OTFS Pulsone for Sinc Pulse	388
10.5	Summary	389
Appendix 10.A Proof of (10.7)		390
Appendix 10.B Proof of (10.9)		391
Appendix 10.C Proof of Theorem 10.1		392
Appendix 10.D Proof of (10.21)		394
Appendix 10.E Effect of Time and Bandwidth Limitation on the DD Representation of a Waveform		394
Appendix 10.F Peaks of the Ambiguity Function		395
Appendix 10.G Proof of Theorem 10.2		396
Appendix 10.H Proof of Theorem 10.3		397
Appendix 10.I Proof of Theorem 10.4		398
Appendix 10.J Proof of (10.43) and (10.44)		400
Appendix 10.K Proof of Theorem 10.5		400
References		403
Index		415

Preface

This book is about a new waveform and its associated promising modulation format called orthogonal time frequency and space (OTFS for short) suited for next-generation wireless communication systems such as 6G. Historically, physical layer waveform has been a key differentiator between different generations of wireless systems. Specifically, 2G/3G systems are based on code division multiple access (CDMA), and 4G/5G systems are based on orthogonal frequency division multiplexing (OFDM). Both CDMA and OFDM waveforms enjoy certain beneficial attributes that have resulted in enhanced quality of communications. The increased relevance and use of mobile internet demand a more capable mobile communication infrastructure compared to what is currently deployed. Specifically, the rate, reliability, robustness, and spectral/energy efficiency requirements are envisioned to be an order of magnitude higher in 6G compared to those in 5G. In addition to that, 6G brings in several interesting new use cases that challenge the boundaries of conventional systems. One such use case is high-mobility communication. A fundamental challenge in this context is the rapidly time-varying nature of the underlying channel due to the increased Doppler shifts at high user speeds and high carrier frequencies, rendering waveforms such as OFDM inadequate. Another such use case is intelligent transportation systems where radar sensing capability has become an additional new requirement, which is not adequately accounted for by contemporary waveforms.

The OTFS waveform and its associated modulation format were invented by Ronny Hadani, an applied mathematician (also an author of this book), and were first reported in the literature in 2017. This invention has created wide interest and stimulated a growing body of research in the past few years, mainly due to the demonstrated superior performance of OTFS compared to existing waveforms, particularly under challenging channel conditions involving high Doppler and delay spreads.

Two key aspects of the OTFS communication paradigm are responsible for its superior performance. They are (i) channel representation in the delay-Doppler domain and (ii) signaling of information in the delay-Doppler domain. Both these aspects stand in sharp contrast with the conventional multicarrier communication paradigm where both channel representation and information signaling are carried out in the time-frequency domain. In this regard, OTFS constitutes a fundamental paradigm shift in communication theory, moving it away from traditional time-frequency (TF) signal processing to delay-Doppler (DD) signal processing.

Depending on how the delay-Doppler multiplexed information symbols are converted to time domain for transmission, OTFS research from 2017 until now has evolved in two phases. In the early version of OTFS introduced in 2017, this conversion was done in two steps, namely, DD domain-to-TF domain conversion followed by TF domain-to-time domain conversion. This approach was motivated by compatibility with existing multicarrier modulation in 4G/5G.

The first five years of OTFS research since 2017 has adopted this two-step conversion approach. During this period, our independent research following this approach has resulted in some of the early contributions to OTFS. In the last one year, a more fundamental and promising approach to direct conversion from DD domain to time domain using a Zak theoretic framework has emerged out of our research collaboration, setting a new direction for OTFS research. Two key aspects are central to the Zak approach to OTFS. First, it provides a formal mathematical framework using Zak theory for describing OTFS and studying its fundamental properties, in a manner analogous to how Fourier theory constitutes an appropriate mathematical framework for describing and understanding OFDM. Second, the Zak-based OTFS waveform is more robust to a larger range of delay and Doppler spreads of the channel. This is because the relation between channel input and output in Zak-based OTFS is localized, non-fading, and predictable, even in the presence of significant delay and Doppler spreads, and as a consequence, the channel can be efficiently acquired and equalized.

The academic research on OTFS is growing, and so is the interest from the industry and the community at large, clearly marking the need for a good textbook on OTFS that lays the theory of OTFS ground up and discusses the systems, signal processing, algorithmic, and application aspects. So, when IEEE Press-Wiley evinced interest in a book on new modulation waveforms, we took it as an opportunity to write this book on OTFS. In writing this book, our aim has been to introduce OTFS modulation to the readers in a fundamental, structured, and accessible way emphasizing its key strengths and highlighting its potential for becoming a widely adopted waveform in a variety of use cases and applications. A broader goal is to introduce the delay-Doppler signal processing framework and hint toward its wider applicability in the context of communication theory, radar theory, and signal processing. In this effort, we have been guided by our past research in OTFS, our recent research collaboration on Zak-based OTFS, and our interactions with students, researchers, industry engineers, and the community at large during classroom lectures, tutorial talks, and presentations and discussions in various forums.

The book covers various aspects of OTFS including a thorough discussion of its physical foundation and theoretical underpinnings as well as aspects of system design and algorithms. In this context, the book includes a comprehensive description of the underlying mathematical theory while emphasizing its structural connections to conventional modulation formats and drawing parallels between the Fourier theory of linear time-invariant systems and the more general Zak theory of linear time-varying systems. We hope the book will benefit practicing engineers, researchers, academics, and graduate/senior undergraduate students interested in emerging wireless communication systems and advanced physical layer waveforms/techniques. We also hope the book will be of interest to radar signal processing engineers, as the OTFS waveform and its underlying theory are very well suited for radar applications. People involved in standardization of wireless systems will also find this book useful in their quest to know new waveforms suitable for future wireless systems and standards.

September 2024

Saif Khan Mohammed
New Delhi

Ronny Hadani
Austin

Ananthanarayanan Chockalingam
Bangalore

Acknowledgements

This book is the outcome of our individual and collaborative research on OTFS. It would not have been possible without the active collaboration, support, help, and encouragement we received from many. Our students and research collaborators from both academia and industry were pivotal for sustaining our research interest on OTFS. Our sincere thanks to all of them. Our deep appreciation and special thanks are due to Prof. Robert Calderbank, Duke University, USA, for his inspiring research collaboration on Zak-OTFS that led to the two seminal papers on Zak-OTFS published in IEEE BITS the Information Theory Magazine. We had the opportunity to deliver talks and tutorials on OTFS on several occasions in various forums that served as ideal platforms for interaction with the research community. These interactions have contributed to our deeper understanding and appreciation of OTFS. We thank the hosts and organizers of these events.

We thank Wiley and IEEE for accepting our proposal to write this book. It was a pleasure to work with the Wiley team throughout this book writing project. We thank Michelle Dunckley, Vishal Paduchuru, Elisha Benjamin, and Kavipriya Ramachandran for their help and guidance at various stages of the project that helped us to keep the project on track. Our special thanks are due to Mustaq Ahamed Noorullah and Sundaramoorthy Balasubramani for patiently working with us in the copy editing and proof reading stages of the production process. We appreciate their meticulous efforts.

A brief note on the cover design of the book is due here. We express our sincere thanks to Dr. Shailesh Rao M. for his help in getting the photograph of the crystalline rock formation in St. Mary's island, Udupi, Karnataka state, India, used in the cover design. A noticeable structural similarity between the crystalline rocks in the island and the 3D-rendering of the received delay-Doppler power profile in Zak-OTFS in the crystalline regime (see Fig. 2.37) has been an inspiration for us to use the crystalline rock image in the cover design. Of course, the waveform (the pulsone) responsible for the delay-Doppler crystallization phenomenon has also found its place along with the Arabian sea waves in the cover design.

– Saif Khan Mohammed, Ronny Hadani, and Ananthanarayanan Chockalingam

I thank my students for being a part of my research on OTFS and also for constantly pushing me to write a book on OTFS – Venkatesh Khammammetti, Imran Ali Khan, Muhammad Ubadah, Rahul Kumar Jaiswal, Danish Nisar, Jinu Jayachandran, Alok Kumar Sinha, Saurabh Prakash, and Brijesh Chander Pandey. I would like to thank Prof. Emanuele Viterbo and Prof. Yi Hong at Monash University, Australia, for collaborating in the area of multiuser OTFS precoding and synchronization.

I would like to thank my PhD supervisor Prof. A. Chockalingam, IISc, Bangalore (coauthor of this book), for bringing out the best in me in terms of research and from whom I have inherited the art of perseverance in research. It is Prof. Chockalingam who introduced me to OTFS in 2017.

This book could not have been a reality without the support from Prof. Kishan and Pramila Gupta, Chair Professor position at IIT Delhi. I would specially thank Dr. Surendra Prasad, Honorary Professor, IIT Delhi, for being a mentor and a source of inspiration at the Department of Electrical Engineering, IIT Delhi. I would also like to thank Dr. M. Balakrishnan, Honorary Professor, IIT Delhi, who was my BTech project supervisor when I was an undergraduate student at IIT Delhi and under whom I learnt the basics of digital hardware and microprocessor design. I thank Dr. M. Balakrishnan for introducing me to the world of scientific research and innovation, which is why I chose to be an academic.

I cannot forget to thank my parents for their constant and unconditional support and encouragement and without whom I would not be what I am, my wife and daughter for sacrificing family time so that I could devote more time to book writing and research. I specially thank my wife Shaba for being an unconditional source of support for the last 20 years and also for encouraging me to pursue a career in academics, a decision which I will always be happy about. I also thank my parents-in-law who have always helped me whenever I needed them.

Lastly, I thank Almighty God for enlightening me and my coauthors, which has enabled us to write this book.

– Saif Khan Mohammed

My OTFS journey began when I first met Shlomo Rakib in 2008 during one of my lectures on the canonical basis of eigen vectors of the finite Fourier transform (FFT). We immediately resonated on the personal level, however, only by the end of 2009, we started to work on a project together. The early discussions were conducted on Skype, after working hours, before miraculously converging to an intuitive idea of a communication method based on delay-Doppler shifts. From that point on, over a period of three years, we established the basic principles of OTFS, built a radio implementation, demonstrated its superior performance to a third party and built a company around it called Cohere Technologies. Shlomo has always been a source of knowledge and inspiration for me, and I am grateful for having the opportunity of learning and working side by side with him until today.

I would like to acknowledge the imminent contribution of Clayton Ambrose and Norm Rayes during these early years of development. Clayton, single handedly put together a comprehensive demonstration platform that provided deep insights about the interaction between the OTFS waveform and the wireless channel – confirming our theoretical hypothesis. Jointly with Clayton we performed extensive over-the-air testing under various mobility conditions both indoor and outdoor, at times employing unorthodox methods like driving a car with one hand and holding an antenna with another or holding a transmitter in front of a rotating fan. Norm joined our small team shortly after Clayton and put together the first OTFS radio and conducted a successful over-the-air demonstration to Sprint (telecom company) in their Kansas campus. This facilitated the first substantial Venture capital investment in Cohere. Norm always impressed me as a “renaissance man” – an orchestra of hardware and software engineers distilled in one man!

Although OTFS basic transceiver was in place as early as 2013, it took another several years to reveal its underlying mathematical structure. During this second period of

development, I had the pleasure of working closely, as part of a due diligence process, with John Campbell from Telstra and Giovanni Vannucci from Bell labs. Their insights helped tremendously in explaining the value of OTFS and clarifying its conceptual structure.

Another important stimulator was Cohere's attempt to promote orthogonal time frequency space (OTFS) into the emerging 3GPP 5G standard. Since orthogonal frequency division multiplexing (OFDM) was the waveform of choice at that time, our focus revolved around a multi-carrier variant of the waveform, referred to in this book as MC-OTFS. I would like to thank Anton Monk, the director of the Cohere standards team and my coauthor of the OTFS white paper, who led the audacious effort of promoting OTFS into 3GPP. Anton put together a team of experts and established strategic relations with telecom partners and carriers. The standards team wrote multiple high-quality technical contributions to 3GPP which laid the basis for all future developments. I would like to acknowledge the contribution of Shachar Kons, Yoav Hevron, Michael Tsatsanis, Anthony Ekpenyong, Cristian Ibars, and Paul Harris. Their deep knowledge of theory and vast practical experience combined with a healthy dose of skepticism helped mature OTFS into a complete architecture with comprehensive performance evaluation.

The cherry on top was our conference paper that was published in 2017 and constituted the first peer reviewed publication that exposed OTFS to the academic community. I would like to thank Prof. Andrea Goldsmith, who served as chair of Cohere Technical Advisory Board (TAB), Prof. Andreas Molisch and Prof. Robert Calderbank, both members of Cohere TAB, for putting the time and effort in organizing the writing of this pioneering paper.

In my mind, there were always two parallel derivations of OTFS. One was using the Zak transform giving rise to what we refer to in this book as Zak-OTFS and the other was using the symplectic finite Fourier transform (SFFT), giving rise to MC-OTFS. Until around 2017, I believed that the two derivations are mathematically equivalent and it is a matter of convenience which one to use. However, as often happens in mathematics, beliefs are proved to be false. Eventually, I came to realize that the two derivations are not equivalent and, moreover, got convinced that the Zak-theoretic derivation is the "correct" one while the SFFT derivation is merely an approximation. This realization marks the third period of OTFS development. Most of my work in Cohere on this topic at an early stage was conducted in close collaboration with my former PhD student Jim Delfeld who had numerous key insights and helped develop comprehensive simulations. A later stage collaboration which continues until today is conducted with Shachar Kons who made fundamental contributions to the development and implementation of a complete transceiver structure.

In parallel to the research activity, there was a continuous effort by Cohere to develop a viable product based on OTFS technology. This parallel path turns out to reinforce the research, revealing gaps and inspiring new ideas and directions. The process of building a product is complicated and demanding. It requires collaboration between professionals from multiple disciplines, such as system engineers, communication engineers, hardware and software engineers, testing and quality assurance, and of course program management. I would like to thank Mike Grimwood and Yoav Hevron for leading system engineering, Naveen Rawat for leading software engineering, Norm Reyes for leading hardware engineering, Venu Gopal Reddy Vennapusa for leading the testing and quality assurance, and Ashish Yadav for taking care of program management.

I would also like to convey my sincere admiration to Ray Dolan, Cohere CEO, for carrying the heavy responsibility of navigating our precious "OTFS cargo" through, in times, challenging commercial circumstances, allowing it to mature into what we know it today.

Another important component is the support of operators. I would like to thank Telstra for their continued support to Cohere on numerous fronts including validation trials, standardization efforts, over-the-air testing, and giving us access to people, bandwidth, and facilities. At the same note, I would also like to thank Steven Bye from Sprint, Andrew Ip from Cablevision, Peter Ledl from Deutsche Telekom, and Santiago Tenorio and Francisco Martin Pignatelli from Vodafone for facilitating validation testing. Our work at Cohere would not be possible without continuous financial backing. I would like to extend my gratitude to Chris Schaepe from light speed, Ron Bernal from NEA, Mark Sherman from Telstra Ventures, David Dibble from Cablevision, Sachin Katti from VMware, Caroline Chen from Intel, the venture arm of Koch industries and Bell Canada.

Finally, I would like to thank my wife Noga Porat who supported me over the many years of intense work and emotional struggle, associated with navigating a start-up company through the challenging telecom industry. Turns out, the OTFS meme eventually percolated into our personal lives – we decided to call our son Gal which means wave in Hebrew.

– Ronny Hadani

I got to know about OTFS when it was first presented in a session on *Modulation* in IEEE Wireless Communications and Networking Conference (IEEE WCNC'2017) held in San Francisco in March 2017. Incidentally, I, along with my coauthors Swaroop Jacob and Dr. Lakshmi Narasimhan Theagarajan (now a faculty at IIT Madras), had a paper on *Space-Time Index Modulation* in the same session. Lakshmi Narasimhan presented our paper in the session. The OTFS paper was presented by Dr. Anton Monk, one of the authors of the OTFS paper (also my research collaborator during my post-doc days at UCSD with Prof. Larry Milstein). On his return from the conference, Lakshmi Narasimhan told me that the OTFS paper created big excitement in the session. We felt that the paper talked about something new and interesting. Our journey on OTFS research started from there and I am thankful to many in this exciting journey. I thank all my students who have participated in this journey and have contributed immensely to the advancement of OTFS research. These students had to work with very limited literature on OTFS in the early days, yet they made a big difference through their dedicated efforts and talent. My big thanks are due to my students K. R. Murali, Rosemary Augustine, G. D. Surabhi, Rasheed O. K., Vighnesh Bhat S., Jackson K. F., Ashwitha Naikoti, Sujata Sinha, Sandesh Rao Mattu, Vineetha Yogesh, Fathima Jesbin, Sai Pradeep Muppaneni, Gandhodi Harshavardhan, Anagha V., Jaswanth Bodempudi, Nabarun Roy, Arpan Das, Abhishek Bairwa, Naveed Bin Nazir, and all other students and colleagues for their interest in OTFS and contributions to OTFS research. I have taught parts from early drafts of this book on Zak-OTFS in my graduate level course and I thank the students for attending the course and offering valuable feedback. I gratefully acknowledge the support of the J. C. Bose Fellowship, Department of Science and Technology, Government of India, and the Intel India Faculty Excellence Program. As we conclude writing this book, I am reminded of the immense understanding and support I received from my family all along, and I express my special and sincere thanks to my family.

– Ananthanarayanan Chockalingam

Acronyms

1G	first generation
2D	two-dimensional
2G	second generation
3G	third generation
3GPP	third generation partnership project
4G	fourth generation
5G	fifth generation
5G NR	5G new radio
6G	sixth generation
ACF	auto-correlation function
AI	artificial intelligence
AJ	anti-jamming
AWGN	additive white Gaussian noise
BCE	binary cross entropy
BER	bit error rate
BPCM	basic circulant permutation matrix
BPSK	binary phase shift keying
BS	base station
CCDF	complementary cumulative distribution function
CDMA	code division multiple access
CFO	carrier frequency offset
CLT	central limit theorem
CMOS	complementary metal-oxide semiconductor
CNN	convolutional neural network
CP	cyclic prefix
CP-OFDM	cyclic prefix OFDM
CPSC	cyclic prefix single carrier
CSI	channel state information
CW	continuous wave
DD	delay-Doppler
DDRE	delay-Doppler resource element
DFRC	dual-function radar communication
DFT	discrete Fourier transform
DL	downlink/deep learning
DNN	deep neural network
DZT	discrete Zak transform
EDGE	enhanced data rates for GSM evolution

ELBO	evidence lower bound
EM	expectation maximization
ETU	extended typical urban
FCNN	fully connected neural network
FD	frequency domain
FDM	frequency domain modulation
FDMA	frequency division multiple access
FFT	fast Fourier transform
FLOPS	floating-point operations per second
FMCW	frequency modulated continuous wave
GB	guard band
Gbps	gigabits per second
GFDM	generalized frequency division multiplexing
GHz	gigahertz
GMSK	Gaussian minimum shift keying
GPRS	generalized packet radio service
GPS	global positioning system
GRU	gated recursive unit
HSDPA	high-speed downlink packet access
HSPA	high-speed packet access
I/O	input–output
IAPR	instantaneous-to-average power ratio
ICI	inter-carrier interference
IDD	interleaved delay-Doppler
IDFT	inverse discrete Fourier transform
IF	intermediate frequency
IFFT	inverse fast Fourier transform
IoT	Internet of things
IQ	in-phase quadrature-phase
IQI	IQ imbalance
IS-95	Interim standard-95
ISAC	integrated sensing and communication
ISFFT	inverse symplectic finite Fourier transform
ISFT	inverse symplectic Fourier transform
ISI	inter-symbol interference
ITF	interleaved time-frequency
IZT	inverse Zak transform
kbps	kilobits per second
kHz	kilohertz
LDPC	low density parity check
LHS	left-hand side
LMMSE	linear minimum mean square error
LOS	line-of-sight
LPI	low probability of intercept
LSTM	long short-term memory
LTE	long-term evolution
LTl	linear time-invariant
LTV	linear time-varying
MA	multiple access

MAI	multiple access interference
MAP	maximum a posteriori
Mbps	megabits per second
MC	multicarrier
MCMC	Markov chain Monte Carlo
MHz	megahertz
MIMO	multiple-input multiple-output
ML	maximum-likelihood/machine learning
MLSE	maximum-likelihood sequence estimation
MMSE	minimum mean square error
mmWave	millimeter wave
MP	message passing
MRC	maximal ratio combining
MRT	maximal ratio transmission
MSE	mean square error
MUI	multiuser interference
NLOS	non-line-of-sight
NMSE	normalized mean square error
NOMA	non-orthogonal multiple access
ODC	orthogonal downlink communication
OFDM	orthogonal frequency division multiplexing
OFDMA	orthogonal frequency division multiple access
OMA	orthogonal multiple access
OMP	orthogonal matching pursuit
OOB	out-of-band
OTFS	orthogonal time frequency space
PA	power amplifier
PAPR	peak-to-average power ratio
PDMA	path division multiple access
PDP	power-delay profile
PEP	pairwise error probability
PIC	parallel interference cancellation
PLL	phase lock loop
pmf	probability mass function
PN	pseudo-noise/phase noise
PR	phase rotation
PSD	power spectral density
PSK	phase shift keying
PSS	primary synchronization signal
QAM	quadrature amplitude modulation
QPSK	quadrature phase shift keying
RA	random access
RAS	receive antenna selection
RC	raised cosine
RE	resource element
RF	radio frequency
RHS	right-hand side
RNN	recurrent neural network
RPE	relative prediction error

RRC	root raised cosine
Rx	receiver
SBL	sparse Bayesian learning
SC-FDMA	single-carrier frequency division multiple access
SE	spectral efficiency
SFFT	symplectic finite Fourier transform
SFO	sampling frequency offset
SFT	symplectic Fourier transform
SGD	stochastic gradient descent
SIMO	single-input multiple-output
SINR	signal-to-interference plus noise ratio
SISO	single-input single-output
SMS	short message service
SNR	signal-to-noise ratio
SOC	system-on-chip
SSS	secondary synchronization signal
STC	space-time coding
TA	timing advance
Tbps	terabits per second
TD	time domain
TDL	tapped delay line
TDM	time domain modulation
TDMA	time division multiple access
TEP	timing error probability
TF	time-frequency
TFRE	time-frequency resource element
THP	Tomlinson–Harashima precoding
THz	terahertz
TSNR	transmit SNR
Tx	transmitter
UAV	unmanned aerial vehicle
UE	user equipment
UL	uplink
UT	user terminal
V2I	vehicle-to-infrastructure
V2V	vehicle-to-vehicle
V2X	vehicle-to-everything
VLSI	very large-scale integration
VB	variational Bayes
VP	vector perturbation
WiFi	wireless fidelity
WiMAX	worldwide interoperability for microwave access
ZC	Zadoff–Chu
ZF	zero forcing
ZP	zero padding
ZT	Zak transform

1

Introduction

Cellular mobile communication technology which came into being in the late 1970s has witnessed numerous advances and innovations in the last few decades and has been evolving rapidly. It started with an objective to make large-scale mobile radio service affordable to a sizable segment of the public through efficient use of a limited block of frequency spectrum [1]. A key element that played a crucial role in achieving this objective is the cellular concept, driven by its underlying frequency reuse philosophy [2]. Over these decades, the fraction of global population that has access to mobile communication services has increased tremendously. Present-day cellular mobile communication infrastructure has become a key driving force behind many new-age industries that provide several new services and products. Cellular systems have advanced from being voice service driven in the first generation (1G) in the early 1980s to being data services driven in the fifth generation (5G) now, and likely to be artificial intelligence (AI) driven in the upcoming sixth generation (6G) and beyond in the future. Starting from the early 1980s, the cellular industry has witnessed a new generation of cellular standard replacing its predecessor, once in about every decade. By that count, we are in the midst of a 5G standard which is deployed and operational, and a 6G standard which is in the making for possible deployment in the 2030s.

In the evolution from 1G to 5G, huge strides have been made in enhancing the capabilities of cellular systems through several theoretical advances and technological innovations. This flow of advances has also transformed the teaching of wireless communications [3–8]. Among the several advances and innovations that characterize the evolution of one generation to the next, physical layer waveform design stands out as a crucial enabler to achieve substantially enhanced capabilities across multiple generations. Such capabilities are often in terms of achieving increased data rate, spectral efficiency, energy efficiency, reliability, and robustness to adverse channel characteristics, to name a few. Some of the physical layer advances that distinguished different generations in the past include the following:

- adoption of efficient modulation schemes (e.g., moving from analog modulation to digital modulation),
- use of spread spectrum and multicarrier modulation techniques (to mitigate the effects of adverse channel characteristics),
- use of higher-order modulation (to improve rate and spectral efficiency),
- use of multiple antennas (to improve rate, spectral efficiency, reliability, and energy efficiency),
- use of capacity achieving codes (to achieve ultra-reliable communication),
- use of cooperative relaying (to increase range and improve coverage),
- operation in higher and higher frequency bands (to acquire more bandwidth to increase rate),

and the list continues to grow. We can see several new physical layer techniques emerging in the technological horizon as we look for promising possibilities beyond 5G. Some of them include non-orthogonal multiple access (NOMA), reconfigurable intelligent surfaces (RIS), waveform design for robust communication in rapidly time-varying channels, integrated radar sensing and communication, cell-free communication, and millimeter wave (mmWave) and Terahertz (THz) communications.

1.1 Cellular Mobile Evolution

1G technology, called advanced mobile phone system (AMPS), provided primarily voice services using analog switching of frequency-modulated (FM) voice signals, operating in the 800–900 MHz radio frequency (RF) band. Frequency division multiple access (FDMA) technology was used to support multiple voice calls/users requiring a one-way bandwidth of 30 kHz to support a single voice call. Though a revolutionary telecommunication technology at the time, 1G suffered from inconsistent voice quality and coverage, and fared low on size, energy consumption, and security, leaving a lot of room for improvement, innovation, and growth. Following the success of 1G, second generation (2G) systems were introduced in the early 1990s, incorporating several advancements compared to 1G. Foremost among the advancements is the change in the communication waveform used, viz., a digital modulation waveform replacing the analog counterpart in 1G. The digital nature of the information signal in 2G enabled encrypted voice calls with better security. It also enabled packet transmission for data services. Being the first digital cellular standard, the data rate in 2G was quite modest, viz., about 10 kbps digitally encoded voice or data. Error correcting capability for reliable transmission was added through channel coding of information bits. The way different users are multiplexed was also changed, from FDMA in 1G to a more efficient time division multiple access (TDMA) in 2G. In Global System for Mobile communication (GSM) [3, 4], which was a widely deployed 2G system, signals from eight users were multiplexed in a TDMA frame of eight time slots, at a channel transmission rate of 270.833 kbps using Gaussian minimum shift keying (GMSK) modulation, occupying 200 kHz bandwidth in the 900/1800 MHz frequency bands. In addition to data and other guard/overhead symbols, training symbols are sent in a given user's time slot in a TDMA frame, so that sophisticated equalization techniques can be applied on the received signal to mitigate the effects of multipath fading in the channel. Even at a modest data rate of about 10 kbps, the popularity of 2G voice/data services (talking, SMS texting, downloading) grew immensely, leading to a massive adoption of cell phones by consumers worldwide.

In a related development, 2G brought another multiple access technology for mobile communications, namely, code division multiple access (CDMA), which, for a given system bandwidth, could support more users compared to FDMA and TDMA [5]. CDMA technology is based on a spread spectrum technique, which has its roots in military communications to provide anti-jamming (AJ) and low probability of intercept (LPI) capabilities. In the context of multiuser communication, unlike in FDMA and TDMA where users' transmissions are separated in frequency and time, respectively, CDMA allows different users' transmissions to overlap in both frequency and time. In order to distinguish different users' signals at the base station, different users are assigned different spreading sequences, and each user multiplies its information signal with its assigned spreading sequence. The bandwidth of the spreading signal is kept much larger than that of the information signal. Therefore, the bandwidth of the transmitted signal is also much larger than the information signal bandwidth. The ratio of the bandwidths of the transmitted signal and the information signal is termed the spreading factor. The greater the spreading factor, the greater the number of users

that can be supported. In the CDMA-based 2G standard (IS-95), digitally encoded voice at 9.6 kbps rate is spread using a spreading factor of 128, occupying a transmission bandwidth of 1.25 MHz. Although CDMA had been adopted earlier for civilian applications in satellite communication, there was a technical problem (known as near-far problem) that needed to be solved for its adoption in cellular mobile communication. In a cellular system, since different users can be at different distances from the base station, their received signal powers at the base station receiver will be unequal, which causes the near-far problem. A consequence of the near-far problem is that the performance of a far user can be overwhelmed by the strong received signal from a near user. Left unsolved, this problem will lead to a significant reduction in the number of users supported. The near-far problem was solved using a carefully designed transmit power control loop which made the received powers from all the users to be almost the same at the base station receiver, regardless of their distances from the base station. This technical solution played a key role in making CDMA a feasible technology for cellular mobile communication. In addition, CDMA in 2G took advantage of voice activity-enabled transmission and sectorized cells to reduce interference at the base station receiver. CDMA being an interference-limited system, such ingenious techniques to reduce interference directly resulted in an increase in the number of users supported. Such increase was not possible in FDMA and TDMA because of the orthogonal nature of their transmissions by design. Also, because of the large bandwidth occupied by the transmitted signal, multipaths become more resolvable in CDMA. This was also exploited in 2G CDMA by RAKE combining of multipath signals at the receiver, achieving diversity benefits without having to use multiple antennas. Therefore, when the 3G era started in the early 2000s, TDMA gave way to CDMA for 3G.

As more and more people started using 2G, the demand for data services increased significantly. Therefore, increased data rates and enhanced network capabilities turned out to be key requirements for 3G. 3G systems, therefore, were designed to offer enhanced data rates in the range of hundreds of kbps to several Mbps using variable spreading factor CDMA waveform over a wider bandwidth of 5 MHz compared to 1.25 MHz bandwidth in the fixed spreading factor 2G CDMA. 3G systems operated in the 1.8–2.1 GHz frequency band. The enhanced rates in 3G enabled new data services including video streaming, video conferencing, live video chat, and mobile Internet access (surfing the Internet on mobile) at faster speeds in comparison with 2G. Early versions of smartphones were introduced toward the end of the 3G era, and the popularity of mobile Internet access demanded even higher data rates. Considering the fact that wireless spectrum is at a premium, it is important that rate increase is achieved not by merely increasing the bandwidth, but along with an increase in spectral efficiency (in bps/Hz), and physical layer design plays a key role in achieving this. Rate increase along with spectral efficiency increase, therefore, became an important physical layer design requirement as cellular systems moved from 3G to 4G.

The physical layer waveform designs using binary phase shift keying (BPSK) and GMSK/Quadrature phase shift keying (QPSK) modulation in 2G and 3G systems score low on spectral efficiency (1 or 2 bits per modulation symbol). Higher-order modulation (M -ary quadrature amplitude modulation [QAM]) and multiple-input multiple-output (MIMO) techniques using multiple transmit and receive antennas [9–12] can increase rate without increasing bandwidth. These two techniques along with multicarrier mode of transmission using orthogonal frequency division multiplexing (OFDM) [13, 14] became defining physical layer enhancements in 4G compared to 3G, leaving behind CDMA. Introduced in the 2010s, OFDM-based waveform design in 4G using higher-order QAM (up to 256-QAM), multiple modes of MIMO operation, and capacity achieving turbo coding [15] offered data rates in the tens to hundred Mbps range. 4G, therefore, could provide high-quality video streaming/chat, fast mobile web access, high-definition videos, and online gaming. Multiple users are supported using orthogonal frequency division multiple

access (OFDMA) by assigning different time-frequency (TF) slices to different users [14]. The transition from 4G to 5G was along the lines of increasing the rate to multi-Gbps range, reducing latency to below 10 ms (from tens of ms in 4G), and connecting a massive number of Internet of Things (IoT) devices. Major physical layer advances from 4G to 5G include adoption of massive MIMO technology which uses an array of hundreds of antennas at the base station [16, 17] and use of capacity achieving low density parity check (LDPC) codes and polar codes [18, 19]. Use of a large number of antennas at the base station (e.g., 128 antennas) allows the base station receiver to identify different users' transmissions through their spatial signatures (i.e., channel gains between a given user's transmit antenna and the receive antennas at the base station). While users in CDMA are distinguished in the time domain by their spreading sequences with an associated bandwidth expansion, users in massive MIMO are distinguished in the spatial domain by their spatial signatures toward the base station array without incurring any bandwidth expansion. OFDM continues to be the basic communication waveform in 5G. While 4G used frequency bands in the 2 GHz range, 5G opened up the use of mmWave frequency band (28 GHz) in addition to the use of sub-6 GHz band. The high bandwidth, high speed, and low latency features in 5G are expected to trigger the evolution of augmented reality (AR) and virtual reality (VR) based new-age applications in the mobile world. Expectedly, while 5G is getting rolled out and being used since 2020, we are already in the midst of defining 6G, the next generation cellular system, for possible adoption/roll out by 2030. In the evolution toward 6G, AI technology and immersive technology-driven AR and VR applications are likely to be important drivers from the applications side of things, for which matching physical layer advancements are essential.

1.1.1 Moore's Law Drives Rate Increase

It is interesting to note that the increase in the rate of data transmission in cellular mobile systems has followed Moore's law since early 1990s, i.e., the rate of data transmission has doubled every 18 months. The trajectory of data rate increase across multiple generations is shown in Fig. 1.1. A mere 10 kbps capability in 2G in the early 1990s has grown to a 10 Gbps capability in 5G in the early 2020s, which is a million-fold increase in data rate over three decades (i.e., over 360 months,

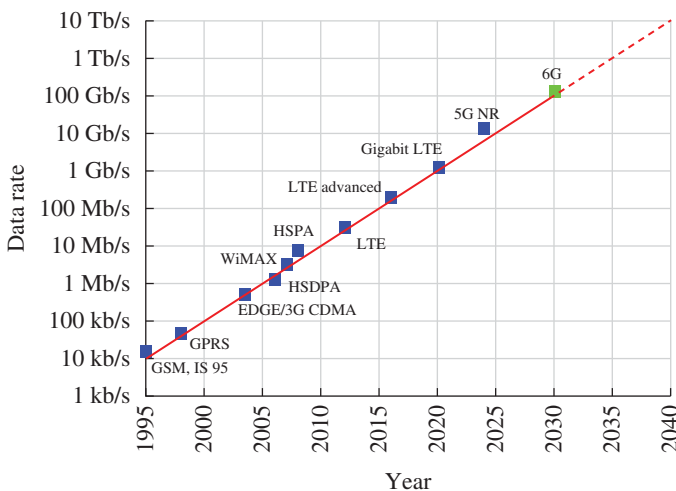


Figure 1.1 Moore's law drives the data rate increase across multiple generations of cellular systems.

which is twenty 18-month slices, and 2^{20} is 1.048576 million). This is indeed a fabulous increase. Now, extrapolating this trajectory, one could envision a 10 Tbps capability by mid/late 2030s, i.e., another thousand-fold increase in rate in the next 15 years. An interesting observation here is that the targeted peak data rate in 6G is in excess of 100 Gbps. Given that the timeline for 6G is around 2030, achieving 10 Tbps capability by around 2040 seems to be a possibility, retaining Moore's law of rate increase.

To achieve the targeted rate in excess of 100 Gbps in 6G, mere incremental changes to previous generation technologies may become inadequate. New waveform designs and new frequency bands of operation will be needed. Communication in the mmWave and Terahertz (THz) bands is promising in this regard because of the availability of abundant bandwidth in these bands. On the other hand, operation in such high frequency bands has a bearing on the waveform design because the channel becomes highly time-varying even in moderate mobility scenarios (e.g., low to moderate user speeds) when operating at these frequencies. In addition, rapidly time-varying channels can manifest in 6G due to another reason, namely, need to support emerging high-mobility use cases, as elaborated below.

Next generation wireless networks are expected to see the emergence of several new use cases and business models driven by the emerging needs of users and system operators. The variety of applications and the corresponding performance requirements in these use cases will be vastly different. Several use case families such as broadband access everywhere (e.g., pervasive video), extreme real-time communications (e.g., tactile internet with sub-millisecond latency), ultra-reliable communications (e.g., e-health services), massive IoT (e.g., sensor networks smart wearables), and high-mobility communications (e.g., high-speed trains, high-speed autonomous/self-driving cars, unmanned aerial vehicles, and airplanes) can be envisioned. In particular, the high-mobility use case family is technologically challenging. Mobility requirements are expressed in terms of the relative speed between the user and the network edge at which consistent user experience must be ensured. For example, in-vehicle mobile broadband service requires a mobility support of up to 500 km/h for bullet trains and 1000 km/h for airplanes. Wireless channels in such high-mobility scenarios vary rapidly in time due to high Doppler spreads. A consequence of rapidly time-varying channels is degradation in communication performance. This is because the channel can vary significantly within one signaling frame time, making signal detection difficult. Also, estimating the channel for signal detection purposes is challenging in rapidly time-varying channels. This is because by the time the channel gets estimated using a received pilot symbol, the channel would have changed. Therefore, waveform designs for next generation wireless systems need to take this into account and ensure robust performance under challenging time-varying channel conditions.

1.2 Multipath Fading Channels

A defining characteristic of wireless channels is the variation of the channel strength over time and frequency, which is termed fading, often classified as large-scale fading and small-scale fading. Large-scale fading is mainly due to path loss which is a function of distance and shadowing. Small-scale fading is due to constructive and destructive addition of multipath components of the transmitted signal at the receiver. Small-scale fading occurs at the spatial scale of the order of carrier wavelength. While large-scale fading is of relevance in issues such as cell-site planning and defining cell radius, small-scale fading is relevant in communication waveform design.

1.2.1 Frequency Domain Characterization

The variation of multipath fading in the frequency domain is characterized by the delay spread and the coherence bandwidth of the channel, denoted by T_d and B_{coh} , respectively. For the purpose of illustration, Fig. 1.2 shows a simple multipath channel with two paths, one direct line-of-sight (LOS) path between the transmitter and receiver and another path reflected by a building, where r_1 , r_2 , and r_3 denote the distances of the paths between the transmitter and receiver, transmitter and reflector, and reflector and receiver, respectively. The path delay of the 1st path between the transmitter and receiver (i.e., LOS path), which is denoted by τ_1 , is $\tau_1 = \frac{r_1}{c}$ seconds, where c is the speed of light. Likewise, the path delay of the 2nd path (reflected path) is $\tau_2 = \frac{r_2 + r_3}{c}$ seconds. The delay difference between the two paths is $\tau_2 - \tau_1$. In a rich scattering environment, there will be a large number of reflected paths with their associated path delays. Now, the delay spread of a channel is defined as the largest magnitude of the delay difference among all possible pairs of paths between the transmitter and receiver, i.e., T_d is given by

$$T_d = \max_{i,j} |\tau_i - \tau_j|, \quad (1.1)$$

where τ_i and τ_j are the delays of the i th and j th paths, respectively. A likely effect of delay spread is that a symbol transmitted in a given signaling interval can spread in to subsequent signaling intervals, causing inter-symbol interference (ISI). Also, the delay spread of a channel defines its frequency coherence which tells how the channel varies in the frequency domain. In this regard, the coherence bandwidth of a channel is defined as the bandwidth over which the channel response remains almost the same. Coherence bandwidth is inversely related to the delay spread, i.e., $B_{\text{coh}} \propto T_d^{-1}$. Typically, B_{coh} is taken to be $\frac{1}{2T_d}$ [6].

The coherence bandwidth of a channel leads to the notion of the channel being characterized as frequency-flat or frequency-selective, depending on whether the coherence bandwidth is larger or smaller than the signaling bandwidth, respectively. Suppose B is the signaling bandwidth. The channel is said to be frequency-flat if $B < B_{\text{coh}}$, and frequency-selective if $B > B_{\text{coh}}$. ISI happens in frequency-selective channels where $B > B_{\text{coh}}$, and hence, frequency-selective channels are also called ISI channels. Frequency selectivity is determined by the number of multipath components that can be resolved within the delay spread. ISI channels are often modeled using a tapped-delay line (TDL) channel model and is shown in Fig. 1.3. The model uses L resolvable channel taps, where each tap corresponds to a path with its associated delay and fade coefficient, $\{\tau_l, h_l\}$, $l = 0, 1, \dots, L - 1$. The delays of the different taps/paths can be arbitrarily spaced. In the figure, the tap delays are shown as equally spaced with T seconds spacing, where $T = \frac{1}{B}$, i.e., the l th path delay is given by $\tau_l = lT$, $l = 0, 1, \dots, L - 1$, and $L = \lceil \frac{T_d}{T} \rceil$. Here, the 1st path is taken to have zero delay (taking the earliest arriving path at the receiver as the reference) and the last path's delay, denoted by τ_{max} , is the delay spread of the channel, i.e., $\tau_{\text{max}} = T_d$. The fade coefficient of the l th tap, h_l , is statistically modeled based on the assumption that a large number of unresolvable multipath

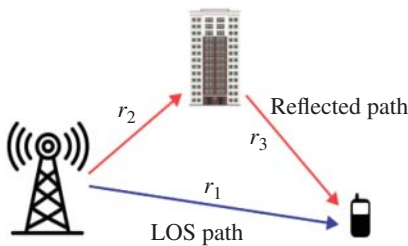


Figure 1.2 Delay spread due to multipath.

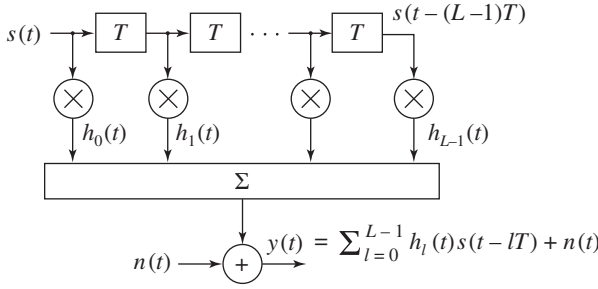


Figure 1.3 Tapped-delay line model of frequency-selective fading channels.

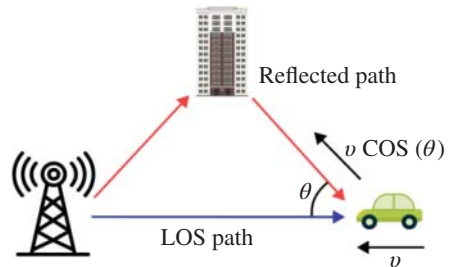
components around the l th path add up, leading to a complex Gaussian distribution of the fade coefficient by central limit theorem (CLT). For the distribution with zero mean, the magnitudes of the fade coefficients are Rayleigh distributed, and for the distribution with non-zero mean, the magnitude distribution becomes Rician.

The average powers in the various taps, given by $\mathbb{E}[|h_l|^2]$, $l = 0, 1, \dots, L-1$, model the strengths of the various paths. The profile of tap delays and their powers is called the power-delay profile (PDP) of the channel. A PDP in which all the taps have equal average powers is called a uniform PDP, which is a widely used model. Another widely used model is the exponential PDP, where the tap powers fall progressively in an exponential fashion for increasing tap indices. While uniform and exponential PDPs are widely followed in academic research, explicit values for the tap delays and powers characterizing a certain operational environment (e.g., urban and rural) are specified in wireless standards. In the standard specification, as in the TDL channel model, the 1st path is taken to have zero delay and the last path's delay $\tau_{\max}$ is the delay spread of channel T_d . It is noted that, for $L = 1$, the TDL channel model described above specializes to a frequency-flat fading model, where the fading effect is only through the multiplicative effect of the fading coefficient h_0 , and there is no ISI.

1.2.2 Time Domain Characterization

The variation of multipath fading in the time domain is characterized by the Doppler spread and coherence time of the channel, denoted by D_s and T_{coh} , respectively. Let us consider the multipath channel with two paths, as shown in Fig. 1.4. The user (car) is moving toward the base station at a velocity of v m/s along the direction of the LOS path (1st path). The Doppler shift of the LOS path, denoted by ν_1 , is given by $\nu_1 = \frac{v}{\lambda} = \frac{f_c v}{c}$, where $\lambda = \frac{c}{f_c}$ is the carrier wavelength, f_c is the carrier frequency, and c is the speed of the light. Larger the carrier frequency and velocity, larger the Doppler shift. The reflected path (2nd path) arrives at angle θ with respect to the direction of the

Figure 1.4 Doppler spread due to multipath.



LOS path. Therefore, the Doppler shift of the reflected path is given by $\nu_2 = \frac{f_c v \cos \theta}{c}$. The Doppler difference between the two paths is given by $\nu_2 - \nu_1$. Now, the Doppler spread of a channel is defined as the largest magnitude of the Doppler difference among all possible pairs of paths between the transmitter and receiver, i.e., D_s is given by

$$D_s = \max_{i,j} |\nu_i - \nu_j|, \quad (1.2)$$

where ν_i and ν_j are the Dopplers of the i th and j th paths, respectively. Larger the Doppler spread, larger is rate at which the channel changes with time. In other words, the Doppler spread of a channel defines its time coherence which tells how the channel varies in the time domain. The coherence time of a channel is defined as the time interval over which the channel response remains almost constant. Coherence time is inversely related to the Doppler spread, i.e., $T_{\text{coh}} \propto D_s^{-1}$. Typically, T_{coh} is taken to be $\frac{1}{4D_s}$ [6]. The coherence time of a channel leads to the notion of the channel being characterized as slow fading or fast fading, depending on whether the coherence time is larger or smaller compared to the signaling duration, respectively. In a block transmission scheme, the signaling duration corresponds to the transmission time of a block/frame of data, where fade variation within a frame duration can influence performance. Suppose T is the signaling/frame duration. The channel is said to be slow fading if $T < T_{\text{coh}}$ (i.e., time-flat fading) and fast fading if $T > T_{\text{coh}}$ (i.e., time-selective fading). Channels which are selective in both time and frequency are called doubly selective channels. Figure 1.5 shows an illustration of time- and frequency-selective channel characteristics, where Fig. 1.5(a) shows the frequency-flat and time-selective fading and Fig. 1.5(b) shows doubly selective fading.

In the TDL channel model in Fig. 1.3, a given realization of the fade coefficient h_l is modeled as complex Gaussian distributed. What is also important is how the tap coefficients vary with time because it impacts communication performance. A statistical quantity that captures this is the auto-correlation function (ACF). An ACF which is a zeroth-order Bessel function of the first kind whose power spectral density (PSD) is a U-shaped spectrum, called Jakes' spectrum, is a widely used model [6, 20]. In this model, the ACF as a function of time lag Δt is given by

$$R(\Delta t) = J_0(2\pi f_m \Delta t), \quad (1.3)$$

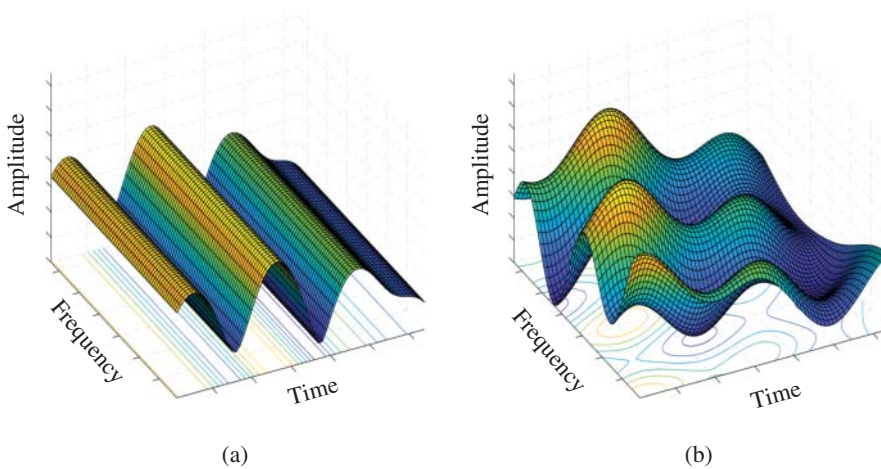


Figure 1.5 Illustration of time- and frequency-selective fading. (a) Frequency-flat, time-selective fading and (b) doubly selective fading.

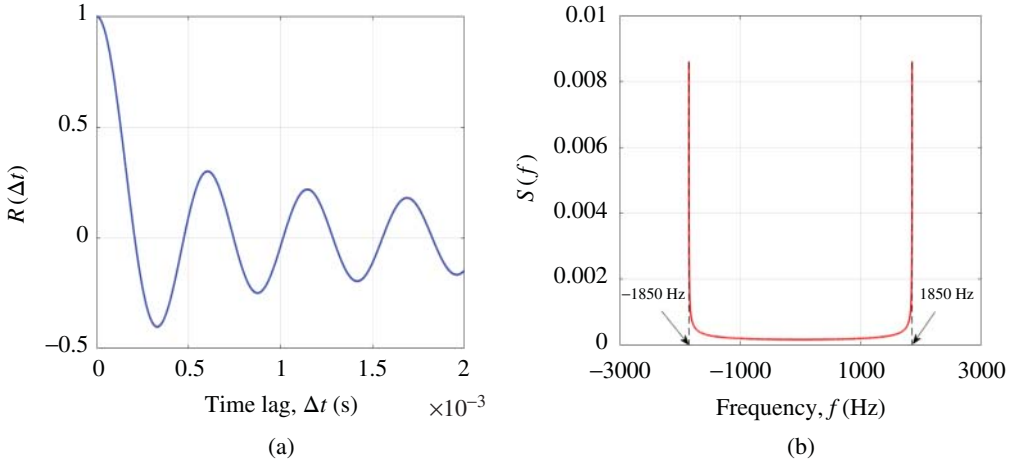


Figure 1.6 ACF and Doppler spectrum for 1.85 kHz maximum Doppler at 4 GHz carrier frequency and 500 km/h velocity. (a) ACF and (b) PSD.

where $f_m = \frac{fcv}{c}$ is the maximum Doppler and $J_0(x) = \frac{1}{\pi} \int_0^\pi e^{-jx \cos \theta} d\theta$. The corresponding PSD is given by

$$S(f) = \frac{1}{\pi f_m} \frac{\text{rect}\left(\frac{f}{2f_m}\right)}{\sqrt{1 - \left(\frac{f}{f_m}\right)^2}}, \quad (1.4)$$

where $\text{rect}(x) = 1$ for $-\frac{1}{2} \leq x \leq \frac{1}{2}$, and 0 otherwise. The ACF and the Doppler spectrum for $f_m = 1.85$ kHz, which corresponds to a carrier frequency of 4 GHz and a velocity of 500 km/h, are shown in Figs. 1.6(a) and (b), respectively. The Doppler spread is the width of the frequency range over which the Doppler spectrum is non-zero. Therefore, the Doppler spread in Fig. 1.6 is $D_s = 2 \times 1.85 = 3.7$ kHz, and the coherence time is $T_{\text{coh}} = \frac{1}{4D_s} = \frac{1}{4 \times 3700} = 0.0675$ ms.

Figure 1.7 shows the Doppler shift as a function of carrier frequency in the range of 0.1–100 GHz for user velocities of 50, 150, and 500 km/h. It can be seen that at 2.1 GHz carrier frequency (a frequency band of operation in 4G), the Doppler is about 290 Hz for a user velocity of 150 km/h. Whereas, at 5.9 GHz carrier frequency (a frequency band of operation in 5G), the same velocity of 150 km/h results in a Doppler of about 830 Hz. At 28 GHz carrier frequency (another frequency band in 5G), the Doppler increases to about 1.3 kHz even at 50 km/h velocity. At 500 km/h velocity (e.g., bullet train speed in high-mobility use cases), a carrier frequency of 4 GHz results in a Doppler of about 1.85 kHz. An observation of interest here is that while the maximum Dopplers in generations up to 4G have remained in the tens to a few hundred Hz range, they have grown to kHz range in 5G and beyond because of higher carrier frequencies and new use cases.

1.2.3 Impact of High Dopplers

High delay spreads and high Doppler spreads impact several aspects of communication. Foremost among them are signal detection and channel estimation aspects, including pilot resources required for channel estimation. As mentioned earlier, high delay spreads cause ISI, which can be viewed as a detrimental factor for signal detection at the receiver. For example, ISI can cause error floors

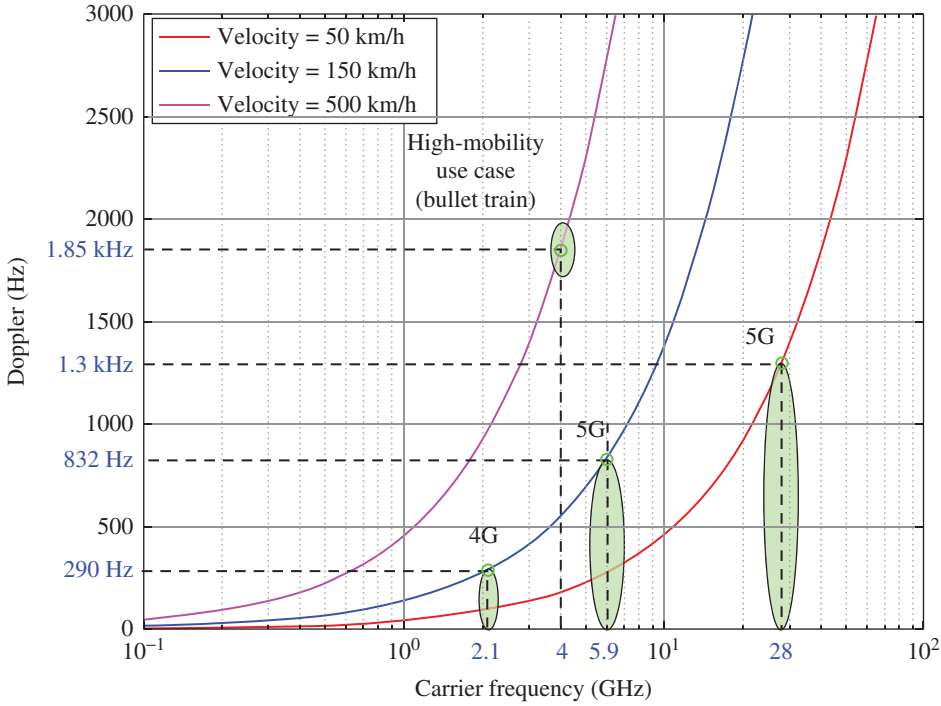


Figure 1.7 Doppler shift as a function of carrier frequency for user velocities of $v = 50, 150, 500$ km/h.

in bit error performance. One way of dealing with ISI has been to adopt a multicarrier (MC) transmission approach (e.g., OFDM), where the signaling bandwidth is divided into multiple orthogonal subcarriers with a much smaller spacing between them compared to the coherence bandwidth of the channel, and multiplexing information symbols on these subcarriers [21–23]. That is, information symbols are multiplexed in the frequency domain. The information symbols in the frequency domain are converted to time domain for transmission using inverse discrete Fourier transform (IDFT). The received time domain signal is converted back to the frequency domain using discrete Fourier transform (DFT), and signal detection is carried out in the frequency domain. By doing so, the channel seen by a given symbol on its subcarrier becomes frequency-flat, thereby a simple one-tap equalizer in the frequency domain is adequate for signal detection. An alternate way to view ISI is that the delay spread provides multiple replicas (echoes) of a transmitted symbol at the receiver, providing the opportunity to harness diversity benefits by combining the replicas. For this, sophisticated signal processing (e.g., equalization/interference cancellation) is needed at the receiver, which comes with its added complexity. This approach, called the single carrier (SC) approach [21, 24], achieves better performance compared to the MC approach at the cost of increased equalization complexity. The SC scheme can be implemented as a DFT-precoded MC scheme, where a DFT precoder precedes the IDFT operation in the MC transmitter. The SC waveform has better peak-to-average power ratio (PAPR) characteristics compared to the MC waveform. Both MC and SC approaches have been adopted in the standards, e.g., 4G has used MC and SC approaches on the downlink and uplink, respectively. Figure 1.8(a) shows the uncoded bit error rate (BER) performance of OFDM and cyclic prefixed single carrier (CPSC) schemes. The following system parameters are considered: $M = 256$ subcarriers, subcarrier spacing $\Delta f = 3.75$ kHz, channel model with $L = 5$ delay taps, uniform PDP, and zero Doppler. BPSK modulation and perfect channel

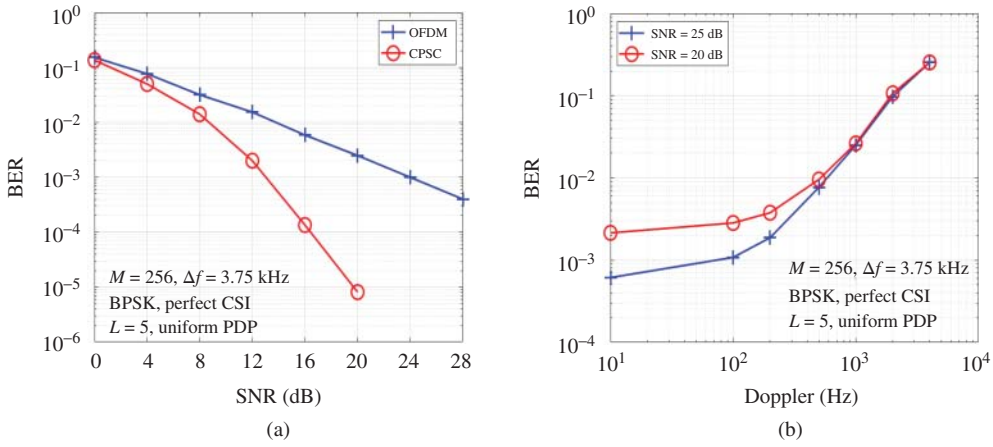


Figure 1.8 (a) BER performance of OFDM and CPSC. (b) BER performance of OFDM as a function of Doppler.

state information (CSI) at the receiver are assumed. A single-tap frequency domain equalizer is used for OFDM, and a minimum mean square error (MMSE) time domain equalizer is used for CPSC. As mentioned before, CPSC achieves better BER performance compared to OFDM at the expense of increased equalizer complexity.

While high delay spreads in communication channels manifest as ISI, high Doppler spreads manifest as inter-carrier interference (ICI) in MC schemes like OFDM. An important and essential attribute in OFDM signaling is the orthogonality among subcarriers, which has to be retained at both the transmitter and receiver for good performance. The orthogonality among subcarriers at the receiver gets compromised due to ICI caused by non-zero Doppler spreads [25, 26]. More the Doppler spread, more the loss of subcarrier orthogonality and the consequent loss in performance. As long as the Doppler spread is much smaller compared to the subcarrier spacing, the loss in orthogonality and performance is not appreciable. Practical systems are operated in this regime. However, with higher Doppler spreads, the performance loss becomes severe, causing error floors. This is illustrated in Fig. 1.8(b) which shows the uncoded BER performance of OFDM as a function of Doppler for 20 and 25 dB signal-to-noise ratios (SNRs) in an OFDM system with 256 subcarriers, 3.75 kHz subcarrier spacing, 5-tap channel with uniform PDP, BPSK modulation, perfect CSI at the receiver, and a single-tap frequency domain equalizer. Increased degradation in BER is observed for increased Doppler. For 800 Hz Doppler and above, increasing the SNR from 20–25 dB does not decrease the BER, meaning the BER has hit an error floor. This is due to the ICI due to non-zero Doppler.

The signal detection performance shown in Figs. 1.8(a) and (b) is under the assumption of perfect channel knowledge at the receiver. In practice, the channel is estimated at the receiver using transmitted pilot symbols and the estimated channel is used for signal detection. In order to account for the time variations in the channel, the channel needs to be estimated at least once every coherence time. This requires pilot symbols to be sent periodically, once every coherence time. Therefore, pilot symbols need to be sent more often in high-Doppler channels because of the low coherence times. This increases pilot overhead. Alternately viewed, for a fixed rate of pilot symbol transmission, high Dopplers (with low coherence times compared to the inter-pilot period) will lead to outdated channel estimates to be used in signal detection. This will severely degrade performance. Therefore, with an increasing need for future wireless systems to operate in high-Doppler channels, new

communication waveforms that can offer robust performance in rapidly time-varying channels are becoming important. Toward this, recent research has shown that viewing doubly selective channels in the delay-Doppler (DD) domain (rather than viewing them in the traditional time-frequency domain as described above) has a beneficial role in waveform design for time-varying channels. Characterization of doubly selective wireless channels in the DD domain is introduced in the following Section 1.2.4.

1.2.4 Delay-Doppler Domain Characterization

When a waveform is transmitted, the wireless channel delays it in time (delay shift) and shifts its frequency contents (Doppler shift), and a DD-shifted waveform is received. A linear time-varying multipath channel can be represented in different ways, depending upon the domain in which the channel response is characterized, namely, combinations of time (t), frequency (f), delay (τ), and Doppler (ν), including time-frequency representation $H(t, f)$, time-delay representation $g(t, \tau)$, DD representation $h(\tau, \nu)$, and Doppler-frequency representation $B(\nu, f)$ [27]. Each of these representations has a transform relation to other representations. For example, as illustrated in Fig. 1.9, the Fourier transform of time-delay representation $g(t, \tau)$ along delay gives time-frequency (TF) representation $H(t, f)$. Likewise, the Fourier transform of $g(t, \tau)$ along time gives $h(\tau, \nu)$. Similarly, $H(t, f)$ has a Fourier relation to $B(\nu, f)$ along time, and $h(\tau, \nu)$ has a Fourier relation to $B(\nu, f)$ along delay. $H(t, f)$ and $h(\tau, \nu)$ are related through a two-dimensional transform, namely, symplectic finite Fourier transform (SFFT).

In representations with time variable in them, namely, $H(t, f)$ and $g(t, \tau)$, the channel coefficients vary with time at a rate inversely proportional to coherence time, which depends on the user's relative velocity and the carrier frequency of operation as described before. High mobility or high frequency of operation would cause the channel to vary rapidly, making channel estimation using these representations difficult. On the other hand, in representations with no time variable in them, the channel responses can be almost time invariant. For example, the channel taps in the DD representation correspond to a group of reflectors, each with an associated delay (depending on the reflector's relative distance), a Doppler (depending on the reflector's relative velocity), and a channel coefficient (see Fig. 1.10). Letting τ_i , ν_i , and h_i to denote the i th reflector path's delay, Doppler, and channel coefficient, respectively, the DD representation of the channel is given by

$$h(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (1.5)$$

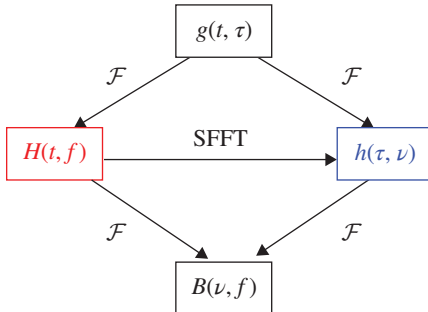


Figure 1.9 Different representations of linear time-varying channels.

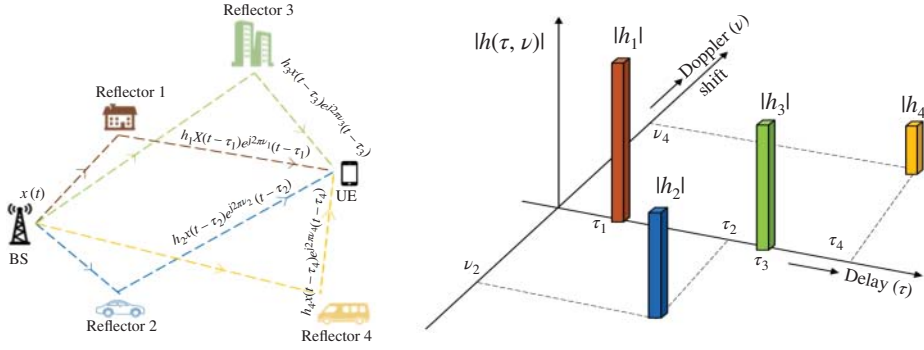


Figure 1.10 Time-varying channel response in the delay-Doppler domain, $h(\tau, \nu)$.

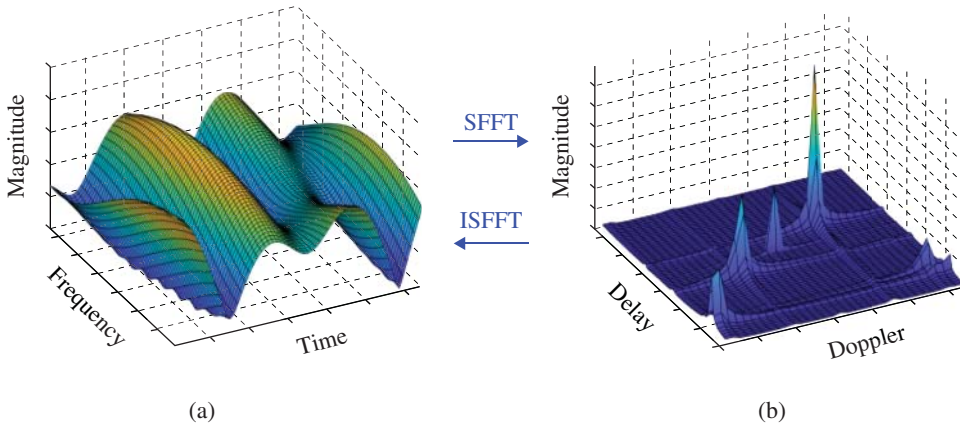


Figure 1.11 Time-varying channel response in time-frequency $H(t, f)$ and delay-Doppler $h(\tau, \nu)$ representations. (a) TF representation and (b) DD representation.

where P is the number of paths (in Fig. 1.10, $P = 4$), and the received signal at the receiver for a input signal $x(t)$ is given by

$$y(t) = \int_{\nu} \int_{\tau} h(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n(t), \quad (1.6)$$

where $n(t)$ is the additive noise at the receiver. In the DD representation, the channel changes only at the speed of the underlying physics of the reflectors. The relative velocity and the distance of the reflectors remain roughly constant for at least a few milliseconds. Hence, the DD representation $h(\tau, \nu)$ appears time invariant for a longer observation duration compared to that in the time-frequency representation $H(t, f)$. Besides time invariance, the DD representation results in a sparse representation of the channel because the wireless channel response is determined by a small number of dominant reflectors. This means that only fewer channel parameters are required to be estimated. This is illustrated in Fig. 1.11 which shows the time-varying channel response viewed in both time-frequency and DD representations. This is further illustrated through an example below.

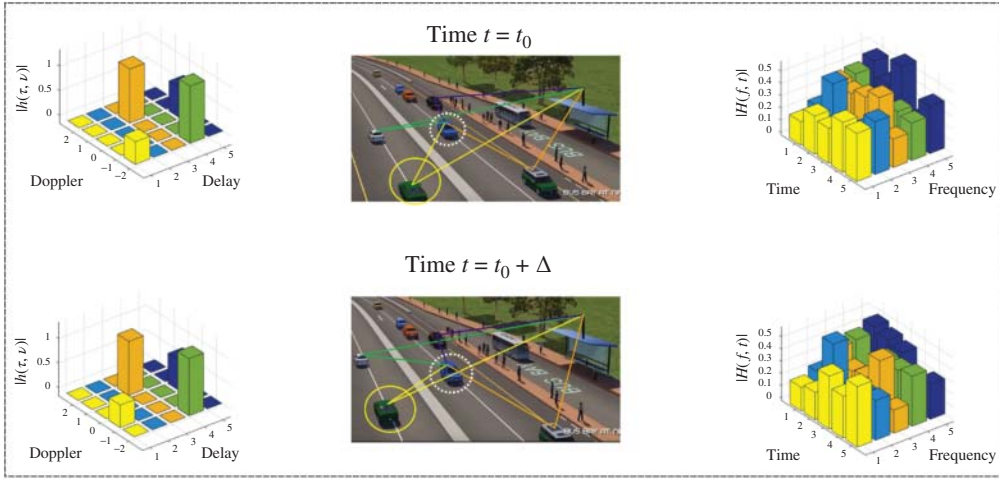


Figure 1.12 Example of a wireless channel in an urban multi-lane scenario illustrating the sparsity and slow variability of the channel in the delay-Doppler representation.

1.2.4.1 Urban Multi-lane Scenario – An Example

Figure 1.12 shows an example scenario encountered in urban multi-lanes. In this figure, the transmitter mounted on top of the bus bay transmits a signal that is intended to be received by one of the moving cars (car in the dotted line circle). The transmitted signal gets reflected by multiple mobile reflectors (other cars) and reaches the receiver. For the sake of illustration, let us consider four paths corresponding to four reflectors. The impulse response of the wireless channel in this scenario is illustrated in both the DD domain and time-frequency domain in the same figure at time $t = t_0$ and $t = t_0 + \Delta$. From the figure, it can be seen that the channel representation in the DD domain is sparse compared to that in the time-frequency representation. Also, since the channel coefficients in the DD representation correspond to a group of reflectors with a particular delay and Doppler shift, we see four non-zero channel coefficients corresponding to the four reflectors in the figure. This illustrates that the DD representation of the channel response is sparse and captures the geometry of the wireless environment. The figure also illustrates the channel at time $t = t_0 + \Delta$, in which one of the reflectors (car in the thick line circle) has decelerated. Due to the deceleration (decrease in the speed) of one of the reflectors, the coefficient corresponding to this reflector now falls in a different bin on the Doppler axis (see the bar in the delay bin 1) in the DD grid, whereas other coefficients remain in their bins at time t_0 . However, the time-frequency representation of the channel shows significant variation at times t_0 and $t_0 + \Delta$. This illustrates the slow variability and sparse nature of the channel in the DD representation compared to the time-frequency representation. The time-invariant and sparse nature in the DD representation simplifies the channel estimation task in rapidly time-varying channels.

1.3 Communication Waveforms for Doubly Selective Channels

Enhancements to mobile communication services across multiple generations have been enabled by judicious allocation of time and frequency resources. To date, these enhancements have been achieved primarily within the framework of time domain modulation (TDM) or frequency domain modulation (FDM). In TDM-based schemes, information is carried by a narrow time domain (TD)

pulse, and an information packet is a stream of such TD pulses. In TDM, the signal is localized in the TD but not in the frequency domain (FD). As a consequence, the interaction between a TD pulse and the environment leads to time-selective fading. Likewise, in FDM-based schemes, information is carried by a narrow FD pulse, which is essentially a sinusoid in the time domain. The FDM signal is localized in the FD but not in the TD, and this lack of localization in the TD causes frequency-selective fading. The lack of time-frequency (TF) localization in TDM/FDM signals also makes the relation between the channel input and channel output less predictable. Fading and lack of predictability degrade the performance of TDM/FDM on doubly spread channels, i.e., channels that are selective in both time and frequency [27]. As carrier frequencies increase and high-speed use cases emerge, we encounter doubly spread channels that are more extreme. TDM/FDM-based waveforms have served well in wireless generations to date, but better TF localization may be required in the wireless generations to come.

A TD pulse is an ideal waveform for pure delay channels (channels with non-zero delay and zero Doppler) as it is possible to separate reflections according to their range. Similarly, an FD pulse is an ideal waveform for pure Doppler channels (channels with non-zero Doppler and zero delay) as it is possible to separate reflections according to their velocity. Recent research has shown that a pulse in the DD domain is a natural and ideal waveform for doubly spread channels with non-zero delays and Dopplers comprising reflections of various ranges and velocities [28–34]. A DD pulse, when viewed in the time domain, exhibits the characteristics of both the pulse and tone. Specifically, the time domain realization of a DD pulse consists of a pulse train modulated by a tone. Hence, a DD pulse in the time domain is called a *pulsone*. An illustration of a pulse in the DD domain and its time domain realization (pulsone) are shown in Fig. 1.13. The pulse rate and the tone frequency are determined by the delay and Doppler periods of the pulsone. Traditional TD and FD pulse waveforms manifest as special/limiting cases of a pulsone. Specifically, a TDM waveform is a pulsone with infinite delay period and zero Doppler period. Such a waveform perfectly matches to delay-only propagation environments. Reciprocally, an FDM waveform is a pulsone with infinite Doppler period and zero delay period. Such a waveform perfectly matches to Doppler-only propagation environments. Therefore, using pulsones (i.e., DD pulses) as modulation carriers provides us with a generalized DD domain modulation framework in which TDM and FDM become special cases. Orthogonal time frequency space (OTFS) waveform, which is the subject matter in this book, belongs to this family of DD domain modulation waveforms. The OTFS waveform is a localized pulse in the DD domain and is a natural modulation waveform for doubly spread channels.

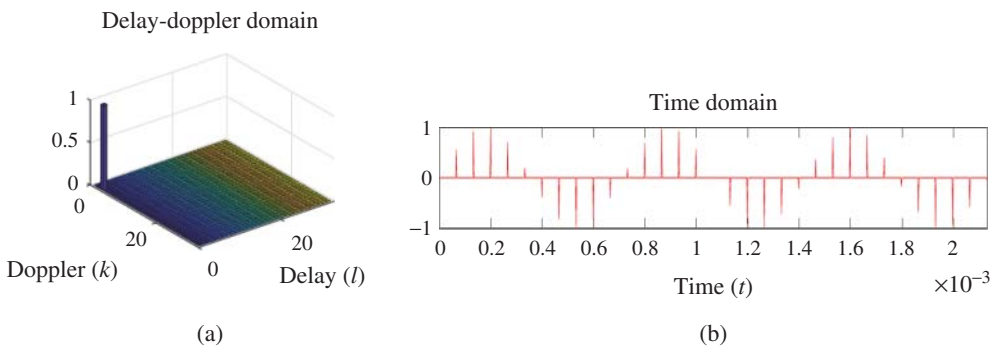


Figure 1.13 A DD domain pulse and its time domain realization (pulsone). (a) A DD pulse and (b) corresponding pulsone.

1.3.1 OTFS – A Modulation Waveform for Doubly Selective Channels

The OTFS waveform has been introduced in [28] by Hadani et al. in 2017 as a modulation waveform for robust communication in doubly selective channels. OTFS converts a doubly selective channel into an almost time-invariant and non-fading channel, which gives impressive resilience to large delay and Doppler spreads. This is an important attribute of the OTFS waveform which other TDM/FDM-based waveforms do not possess. Two basic features are key to the good performance and growing popularity of OTFS. They include the following:

- viewing/representing the channel in the DD domain, and
- multiplexing the information symbols in the DD domain.

Both these features stand in sharp contrast with conventional multicarrier communication paradigm where both channel representation and information signaling are carried out in the time-frequency domain. In this regard, OTFS constitutes a fundamental paradigm shift in communication theory, moving it away from traditional time-frequency signal processing to DD signal processing. The first feature is a compact and sparse DD domain parameterization of the wireless channel, where the parameters map directly to physical attributes of the reflectors that comprise the scattering environment, and the second feature is a novel waveform/modulation technique, matched to the DD channel model, that embeds information symbols in the DD domain. These features make the relation between the channel input and output to be localized, non-fading, and predictable, even in the presence of significant delay and Doppler spread. As a consequence, the channel can be efficiently acquired and equalized. By avoiding fading, the post-equalization SNR remains almost constant across all information symbols in a frame. This, in turn, leads to superior bit error performance compared to contemporary multicarrier waveforms. Furthermore, being a localized pulse in the DD domain, the OTFS waveform makes it possible to separate reflectors along both delay and Doppler domains simultaneously and to achieve a high-resolution DD radar image of the environment. In other words, the DD parameterization provides a common mathematical framework for both communication and radar. This is another key feature of the OTFS framework, which makes OTFS suitable for integrated communication and radar sensing in applications such as intelligent transportation systems.

1.3.2 OTFS Realization – Two Approaches

The basic premise of OTFS modulation is the multiplexing of information symbols in the DD domain and converting them into the time domain for transmission. This DD domain to time domain conversion can be achieved in two ways. The first way is to carry out this conversion in two steps, namely, the DD domain to TF domain conversion followed by the TF domain to time domain conversion. This two-step approach is shown in Fig. 1.14(a). The DD domain to TF domain conversion is done using inverse symplectic finite Fourier transform (ISFFT) and the TF domain to time domain conversion is done using Heisenberg transform. Corresponding inverse transforms, namely, Wigner transform and SFFT, respectively, are carried out for time domain to TF domain conversion and TF domain to DD domain conversion, and the symbol detection is carried out in the DD domain. It is noted that the inner Heisenberg–Wigner block is nothing but a multicarrier modulation block on which an outer ISFFT-SFFT block is overlaid. Therefore, this approach has the benefit of realizing OTFS as an overlay on existing multicarrier modulation schemes (e.g., OFDM in 4G/5G). This approach is, therefore, called the multicarrier OTFS (MC-OTFS) approach. OTFS was first introduced using this MC approach [28], motivated mainly by its backward compatibility

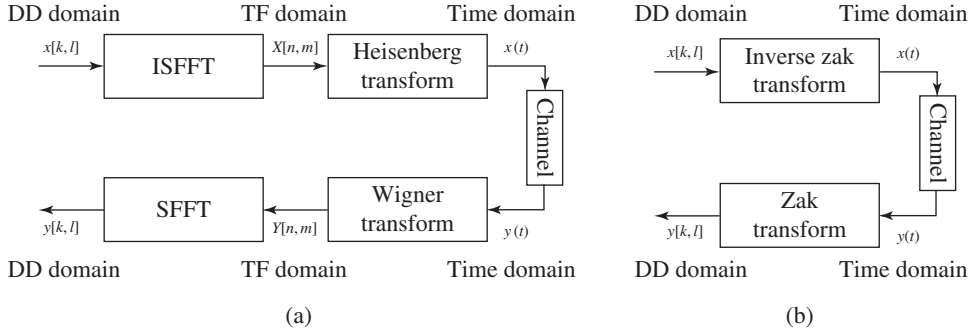


Figure 1.14 Two approaches of realizing OTFS. (a) MC-OTFS approach and (b) Zak-OTFS approach.

with the legacy multicarrier approach. Since then, most research on OTFS to date have adopted this approach [35–47]. These studies have shown that MC-OTFS achieves superior performance compared to legacy multicarrier systems, particularly in high-Doppler channels.

The second approach to realize OTFS is to carry out the DD domain to time domain conversion in a single step using the Zak transform approach, as shown in Fig. 1.14(b). In this approach, the information symbols multiplexed in the DD domain are directly converted to the time domain using inverse Zak transform (IZT) at the transmitter. At the receiver, the received time domain signal is converted back to the DD domain using Zak transform (ZT) for symbol detection. This approach, referred to as the Zak OTFS (Zak-OTFS) approach, has been introduced more recently [33, 34]. The Zak-OTFS approach becomes important on two counts: (i) Zak-OTFS is more robust to large DD spreads compared to MC-OTFS, and (ii) Zak-OTFS is backed by the formal mathematical framework of Zak theory. This book deals with both the MC and Zak approaches of OTFS. Chapters 2, 3, and 10 deal with Zak-OTFS, and the remaining Chapters 4–9 deal with MC-OTFS. Chapter 2 introduces Zak-OTFS and its foundational aspects. Chapter 3 deals with the beneficial attributes of Zak-OTFS such as predictability of input–output relation and non-fading, and the origins of these attributes. It describes how the MC-OTFS approach can be cast in the framework of the Zak transform approach, which formally brings out the basic structural differences between MC-OTFS and Zak-OTFS approaches. It also presents the performance aspects of Zak-OTFS in comparison with that of MC-OTFS. Chapter 10 deals with the radar sensing aspects of Zak-OTFS.

1.4 Waveforms for Radar Sensing

In the emerging intelligent transportation systems involving autonomous vehicles/self-driving cars, the ability of a vehicle to continuously sense and track its dynamically changing environment and react accordingly is important. The sensed information and the data communicated from neighboring vehicles (that also sense their environment) and roadside units are essential to avoid obstacles, select routes, and comply with traffic regulations, all in real time. Therefore, in addition to communication functionality, radar sensing functionality becomes an important additional requirement in order to meet the enhanced road safety objectives in such environments [48]. Radar sensing helps in recognizing and responding to potential hazards in fast-moving environments. Accurate radar imaging of the environment in high resolution is essential for this. The radar sensor should be able to reliably detect all objects inside the observation area and to accurately estimate range, radial velocity, and azimuth angle simultaneously in multi-target situations. This

is challenging in automotive radars because while conventional radars, such as those used in military applications and air traffic control, are required to detect a relatively small number of targets in ranges on the order of tens or hundreds of kilometers, automotive radars must detect a larger number of objects in short ranges on the order of a few tens of meters. Furthermore, simultaneously achieving accurate estimates of range, velocity, and azimuth puts conflicting requirements on the sensing waveform used. A waveform inherently good for range estimation (delay estimation) due to its good localization in time may not be good for velocity estimation (Doppler estimation) due to its poor localization in frequency. This is because a pulse/waveform cannot be simultaneously localized in both time and frequency due to Heisenberg's uncertainty principle. Pulse-based waveforms (using TD pulses which are well localized in time) and tone-based waveforms (using FD pulses which are well localized in frequency) are good for range and velocity estimation, respectively, and not for both simultaneously. Also, the ability of a radar waveform to resolve multiple targets unambiguously is determined by its ambiguity function in the DD domain. The ambiguity function of a TD pulse is narrow along the delay domain but wide along the Doppler domain, whereas the ambiguity function of a FD pulse is narrow along the Doppler domain but wide along the delay domain. Commonly used radar waveforms include pulsed wave, continuous wave (CW), frequency-modulated continuous wave (FMCW), and phase-coded wave. Communication waveforms such as OFDM and CPSC waveforms have also been considered for radar sensing [48, 49], and so are multiple-input multiple-output (MIMO) radar techniques to achieve spatial diversity to improve target detection and discrimination capability [50, 51]. Waveform designs with good TF localization that can provide a good trade-off in range and velocity estimation performance simultaneously have been of interest in radar sensing.

Looking back to 1953, only five years after Claude Shannon created information theory, Philip Woodward described how to think of radar in information theoretic terms [52]. He suggested that we view the radar scene as an unknown operator parameterized by delay and Doppler and radar waveforms as questions that we ask the operator. Woodward proposed to define a good question in terms of lack of ambiguity in the answer, and he sought questions with good localization in delay and Doppler. That is, good radar waveforms are those with good localization in delay and Doppler. This observation suggests that the OTFS waveform is a naturally good waveform for radar sensing for the following reasons: (i) being a pulse train modulated by a tone, the OTFS waveform (pulsone) inherently possesses the characteristics of both the pulse and tone and (ii) pulsones have good DD localization, an attribute suggested by Woodward for a waveform to be good for radar sensing. Despite its potential as a promising waveform for radar sensing, research on OTFS in the context of radar sensing is yet to take off in a big way. This book presents some of the foundational aspects of the OTFS waveform for radar sensing applications (in Chapter 10), providing a basis for new research in this direction.

1.4.1 OTFS for Integrated Sensing and Communication

Both communication and radar sensing applications have witnessed significant growth in recent years. In addition to traditional application domains such as military radar and air-traffic control, radar sensing has found application in new domains such as automotive radar, localization, earth geophysical monitoring, and security. For example, autonomous vehicles are equipped with radar transceivers, in addition to other sensing technologies including cameras, lidar, laser-based sensors, and the global positioning system (GPS). Radar technology is more robust to weather conditions and poor visibility compared to other sensing technologies. Because of

the high-volume need for radar transceivers in these new application domains, the size, power, and cost considerations become important. Also, increasing radar sensing applications put increased demand on the wireless spectrum. Demand for increased wireless spectrum has also been the norm in communication applications. For example, AR/VR technology is expected to drive several new high data rate applications in 6G and beyond, and this demands increased bandwidth resources. This has pushed the communication carrier frequencies toward bands traditionally assigned to radar systems. Therefore, coexistence of communication and radar systems using overlapping bandwidths becomes important [53]. Interestingly, the RF front-end hardware, signal processing, and waveform design for both radar and wireless communication tend to be similar, making their co-existence viable. One way to realize coexistence is to use radar to perform communication by embedding information into radar waveforms, a technique known as dual-function radar communication (DFRC). Another way is to exploit the waveforms transmitted by a communication network to perform radar sensing functions. Either way, jointly implementing radar and communication can contribute to reducing the system size and cost as well as alleviating electromagnetic interference and spectrum-related concerns [54]. Hence, there has been a growing interest and opportunity to take a fresh look at new waveform designs that can jointly address the communication and radar sensing requirements. Since the OTFS carrier waveform is a localized pulse in the DD domain, it is possible to separate targets (reflectors) along both delay and Doppler simultaneously and achieve a high-resolution DD radar image of the environment. In addition to being a good waveform to provide predictable and non-fading communication in doubly selective channels, the OTFS waveform also serves as a good radar waveform, making it a promising waveform for integrated sensing and communications (ISAC).

1.5 Organization of the Book

The key objective of this book is to introduce OTFS modulation to the readers in a fundamental, structured, and accessible way, emphasizing its key strengths and highlighting its potential for becoming a widely adopted waveform in a variety of use cases. Our broader goal is to introduce the DD signal processing framework and hint toward its wider applicability in the context of communication theory, radar theory, and signal processing. The book covers various aspects of OTFS including a thorough discussion of its physical foundation and theoretical underpinnings and also covering aspects of system design and algorithms. In this context, the book includes a comprehensive description of the underlying mathematical theory while emphasizing its structural connections to conventional modulation formats like OFDM. In accordance with this line of development, the OTFS modulation is explained in a manner analogous to the well-understood multicarrier OFDM modulation, drawing parallels between Fourier theory of linear time-invariant systems to the more general Zak theory of linear time-varying systems. The development emphasizes the manner in which Zak theory constitutes an appropriate mathematical framework for describing OTFS and studying its fundamental properties, in analogy to the manner in which Fourier theory constitutes an appropriate mathematical framework for describing and understanding OFDM. Having laid down the mathematical framework, the book proceeds to explain how the theory is applied for the analysis and development of the algorithmic and signal processing aspects of OTFS. As mentioned earlier, OTFS can be realized using a multicarrier-based approach (MC-OTFS) as well as a Zak transform-based approach (Zak-OTFS). While OTFS was first introduced and researched in the multicarrier framework [28, 35], a formal and detailed articulation of OTFS in the Zak theoretic framework and its benefits is a more recent development [33, 34]. This book covers both the

Zak and multicarrier approaches to OTFS, highlighting their theoretical underpinnings, signal processing aspects, and performance. Chapters 2, 3, and 10 deal with Zak-OTFS, and the remaining chapters (Chapters 4–9) deal with MC-OTFS. Also, Chapter 3 describes how MC-OTFS is different from Zak-OTFS and how MC-OTFS can be viewed in the Zak framework, and Chapter 8 covers a discrete Zak transform-based implementation of OTFS.

Chapter 2 introduces the DD representation of signals as another representation in addition to the already known TD and FD domain signal representations. Just as the Fourier transform of the TD representation gives the FD representation, the Zak transform of the TD representation gives the DD representation. The DD representation of any TD signal is a quasi-periodic function in the DD domain whose period along the delay and Doppler domain is inverse of each other. TD pulses are ideally suited for carrying information in delay-only channels as the input–output (I/O) relation is predictable, i.e., the TD output to a TD pulse at some time t_2 can be predicted from the channel response to another TD pulse at some other time t_1 . Similarly, FD pulses are suited for Doppler-only channels. However, in doubly spread channels where channel paths induce both delay and Doppler shifts, the I/O relation for TD modulation (based on TD pulses) and FD modulation (based on FD pulses) is not predictable. Predictability in doubly spread channels requires the information carrier to be localized in both time and frequency. Heisenberg’s uncertainty principle says that simultaneous TD and FD localization is not possible. However, there exist signals that are quasi-localized in TD, FD, and DD domains, i.e., although these signals are not localized in the entire domain, they are still localized within periodic intervals. Quasi-periodic pulses in the DD domain are quasi-localized in the DD domain, and their TD and FD realizations consist of a train of pulses spaced regularly apart (i.e., within each periodic interval, there is exactly one localized pulse). This chapter introduces Zak-OTFS modulation where information is carried by quasi-periodic DD pulses. It is shown that the I/O relation is simple, i.e., the received DD domain symbols are given by twisted convolution between the transmitted DD domain symbols and the effective DD domain channel response (also referred to as the effective discrete DD domain channel filter). It is further shown that even in doubly spread channels, the I/O relation of Zak-OTFS modulation is predictable when the delay and Doppler spread of the effective DD domain channel filter are, respectively, less than the delay and Doppler period of the quasi-periodic DD pulses. This condition on the delay and Doppler period of the Zak-OTFS carriers is referred to as the crystallization condition. Predictability implies that the channel response to a DD pulse at a particular DD location can be predicted from the known response to a DD pulse at some other location. In other words, predictability implies that the effective DD domain channel filter can be accurately estimated in the crystalline regime (i.e., when the delay and Doppler periods satisfy the crystallization condition). This enables a model-free mode of operation where the receiver acquires the discrete taps of the effective DD domain channel filter instead of estimating the path delay and Doppler shifts of the underlying physical channel. The error rate performance of Zak-OTFS-based model-free communication in the crystalline regime is significantly more reliable when compared to that of TD and FD pulse-based modulation, especially in high delay and Doppler spread scenarios.

Chapter 3 introduces a well-known approximation to Zak-OTFS, known as multicarrier OTFS (MC-OTFS). The first research article on OTFS modulation [28] (2017 research paper by Hadani et al.) introduced MC-OTFS, and a majority of the subsequent research has been on MC-OTFS. Zak-OTFS modulation is more recent and has been introduced as a better and natural fit for doubly spread channels in [33, 34]. In this chapter, it is shown that MC-OTFS modulation arises when attempt is made to generate Zak-OTFS pulses (carriers of Zak-OTFS modulation) using a multicarrier modulator (e.g., OFDM modulator) in existing 4G/5G modems. In other words, MC-OTFS is an approximation to Zak-OTFS modulation under a constraint that the modem is compatible

with existing 4G/5G standards. In this chapter, the MC-OTFS modulator and receiver are derived from first principles. The MC-OTFS I/O relation is derived and is shown to be not as simple as that of Zak-OTFS, which makes acquiring of the MC-OTFS I/O relation much more challenging than acquiring of the Zak-OTFS I/O relation where it suffices to simply estimate the taps of the effective DD domain channel filter from the received response to a pilot. Therefore, simple DD domain pilot-based acquisition of the MC-OTFS I/O relation is not as accurate as that of the Zak-OTFS I/O relation, specially in high delay and Doppler spread scenarios. In other words, for the same pilot overhead, the Zak-OTFS I/O relation appears to be more predictable than the MC-OTFS I/O relation. The bit error performance comparison between Zak-OTFS and MC-OTFS reveals the significantly superior performance of Zak-OTFS, especially at high Doppler spreads.

Chapter 4 deals with the analytical characterization of the diversity performance of MC-OTFS, which is a measure of the bit error performance in the high SNR regime. Since the channel representation and channel estimation are in the DD domain, OTFS can achieve DD domain diversity. A maximum DD diversity order equal to the number of DD domain paths is possible. Phase-rotated OTFS is shown to achieve the full DD diversity in the system. The diversity analysis for single-input single-output (SISO), MIMO, and space-time-coded OTFS are presented. Diversity analysis of OTFS with receive antenna selection (RAS) is also presented.

In Chapter 5, algorithms suitable for OTFS signal detection at the receiver are presented. Since information symbols are multiplexed in the DD domain at the transmitter, the signal detection function at the receiver is also carried out in the DD domain. Several algorithms have been proposed in the literature for this purpose. This chapter classifies the detection algorithms into three categories of algorithms based on the type of signal processing carried out and presents different algorithms in these categories. The three categories are as follows: (i) linear detection algorithms such as minimum mean square error (MMSE) and zero-forcing (ZF) based detection, and their low complexity implementations, (ii) approximate maximum a posteriori probability (MAP) detection algorithms such as message passing (MP)-based detection and variants, variational Bayes-based detection, hybrid MAP and parallel interference cancellation (PIC)-based detection, and Markov chain Monte Carlo (MCMC) sampling-based detection, and (iii) cross-domain-based detection which involves iterative processing between the time domain and DD domain.

Chapter 6 addresses the problem of DD channel estimation in OTFS. OTFS signal detection at the receiver requires the knowledge of the DD channel. In presenting the signal detection algorithms in Chapter 5, perfect knowledge of the DD channel is assumed. In practice, the channel needs to be estimated and the estimated channel is used for signal detection. Channel estimation is done using pilot symbols sent by the transmitter. The chapter classifies the estimation methods into three categories depending on how the pilot symbols are organized in an OTFS frame. They include the following: (i) exclusive pilot frame, (ii) embedded pilot frame, and (iii) superimposed pilot frame. Exclusive pilot frames consist of only pilot symbol(s) and no data symbols. Embedded pilot frames consist of both pilot and data symbols with guard symbols in between. Superimposed pilot frames consist of pilot and data symbols superimposed on each other. Channel estimation algorithms for each of the above categories of pilot arrangement are presented.

Chapter 7 presents OTFS-based multiuser communication in both the uplink and downlink. Both orthogonal and non-orthogonal multiple access are considered. In orthogonal multiple access, users are allocated distinct physical resource in either the DD or the TF domain. Different resource allocation strategies are discussed, and their performances are compared. In NOMA, multiple users can communicate on the same physical resource, thereby increasing the overall spectral efficiency. Multiple antennas are also considered, and they are shown to improve multiuser performance significantly. Multiuser timing and frequency synchronization are also discussed.

Chapter 8 is concerned with some of the practical considerations involved in OTFS signals and systems. It first considers the PAPR characteristics of MC-OTFS signals and compares with those of OFDM and generalized frequency division multiplexing (GFDM). It then considers the effect of RF hardware impairments such as oscillator phase noise and in-phase quadrature-phase (IQ) imbalances in RF transmit and receive chains on the performance of OTFS transceivers. This chapter also introduces a discrete Zak transform (DZT)-based implementation of Zak OTFS, which has the advantages of implementation simplicity and good performance.

Chapter 9 deals with the use of deep learning tools and techniques for the design of efficient OTFS transceivers. Machine learning and AI algorithms have been widely adopted in various application domains. They have also been adopted for physical layer design and optimization in wireless communications. The compactness and predictability of the OTFS I/O relation make it a natural fit for machine learning and AI algorithms. This chapter presents deep neural network (DNN)-based design of OTFS transceiver functionalities such as OTFS signal detection, DD channel estimation, and IQ imbalance (IQI) compensation, and their performance.

Chapter 10 considers the OTFS waveform for radar sensing applications and studies a Zak theory-based framework for the design of good waveforms for radar sensing. It highlights the similarity between the structure of the carrier waveform proposed by Woodward in his influential text [52] (a train of narrow TD Gaussian pulses modulated with a broad Gaussian envelope) and the Zak-OTFS pulsed carrier waveform (a train of narrow impulses modulated by a sinusoid). This chapter derives the radar ambiguity function of the Zak-OTFS carrier waveform and illustrates that, with the proper choice of the delay and Doppler periods of the pulsed carrier, unambiguous DD estimation is possible using Zak-OTFS. Radar resolution depends on the shape of the ambiguity function. For Zak-OTFS-based radar sensing, the ambiguity function is given by a precise and simple analytical expression of the DD domain pulse shaping function/filter used for limiting the time duration and bandwidth of the Zak-OTFS pulsed carrier. The Zak theoretical framework therefore simplifies the design of radar waveforms to that of designing pulse shaping filters. This is a paradigm shift in the manner radar waveforms have been designed so far.

2

Delay-Doppler Signaling and OTFS Modulation

This chapter introduces signaling in the delay-Doppler (DD) domain for doubly spread channels, where the channel paths induce both delay and Doppler shifts. The main plot of this chapter is summarized in Fig. 2.1. In traditional time domain modulation (TDM), time domain (TD) pulses carry information and are suitable for delay-only channels (where the path Doppler shift is zero), for which the TD output is given by linear convolution of the TD information input with the TD impulse response of the channel which does not vary with time. Therefore, in delay-only channels, the channel impulse response can be estimated and the energy of the received channel response to a TD pulse is invariant of the time at which the pulse is transmitted.

However, channel path Doppler shift results in a time-varying channel impulse response due to which the channel response to an impulse at time t_1 cannot be predicted/estimated from the response to an impulse at some other time t_0 , i.e., the interaction between the channel and a TD pulse is *non-predictable* in doubly spread channels. Also, in the presence of channel path Doppler shift, the energy of the received channel response to a TD pulse varies with the time at which the pulse is transmitted, i.e., the interaction between the channel and a TD pulse is *fading*.

Similarly, traditional frequency domain modulation (FDM), where information is carried by pulses in the frequency domain (FD), is ideally suited for Doppler-only channels, i.e., the interaction between an FD pulse and a Doppler-only channel is predictable and non-fading. However, in the presence of channel path delay, the interaction between an FD pulse and a doubly spread channel is non-predictable and fading. The non-predictable and fading channel interaction of TD and FD pulses with a doubly spread channel is because TD and FD pulses cannot be simultaneously localized in both TD and FD.

Pulses in the DD domain can, however, eliminate this obstruction and this is why DD domain signaling is suited for doubly spread channels. Most of this chapter is devoted toward demonstrating that the interaction between a DD domain information signal (linear combination of DD pulses) and a doubly spread channel is predictable and non-fading under certain condition, which is referred to as the *crystallization condition* [33, 34].

The cascade of two delay-only (Doppler-only) channels is another delay-only (Doppler-only) channel, whose TD (FD) impulse response is given by the linear convolution of the TD (FD) impulse responses of those two channels. In TDM and FDM, the action of a delay-only (Doppler-only) channel on a TD (FD) input is through TD (FD) linear convolution, which is the same operation that describes the impulse response of a cascade of two delay-only (Doppler-only) channels. Similarly, the cascade of two doubly spread channels is another doubly spread channel whose DD domain representation (known as the DD channel spreading function) is given by the *twisted convolution* of the DD spreading functions of those two channels. Therefore, by analogy with TDM/FDM, the

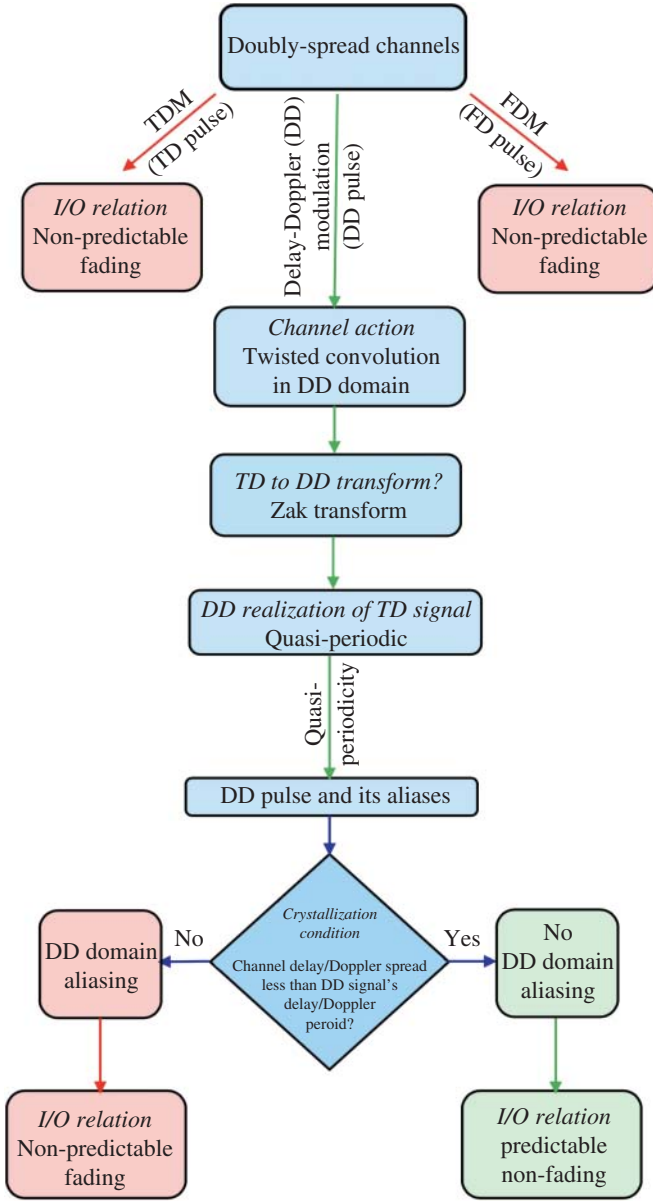


Figure 2.1 Delay-Doppler signaling in doubly spread channels.

action of a doubly spread channel on a DD domain input signal should be through twisted convolution, i.e., the DD domain output is given by twisted convolution between the channel DD spreading function and the DD domain input signal.

Since the transmit signal on the channel is a TD signal, the DD domain information signal needs to be transformed to a TD signal at the transmitter. At the receiver, the received TD signal needs to be transformed back to the DD domain. However, this transform should be such that the received DD domain signal (i.e., DD domain output) and the DD domain information signal (i.e., DD domain

input) are related through the twisted convolution operation. The unique transform which satisfies this criterion is the *Zak transform*.

The Zak transform is parameterized by two parameters (τ_p, ν_p) , where $\nu_p \tau_p = 1$. The Zak transform of a TD signal gives its DD domain representation which is always *quasi-periodic* in the DD domain, viz., periodic along the Doppler domain with period ν_p and quasi-periodic along the delay domain with period τ_p . In other words, for a DD domain signal to have a TD representation, it must be quasi-periodic in the DD domain. Due to this quasi-periodicity constraint, a DD pulse has infinitely many aliases that are separated from each other along the delay domain by integer multiples of the delay period τ_p and along the Doppler domain by integer multiples of the Doppler period ν_p . Therefore, in a way, a DD pulse is simultaneously localized within the fundamental DD period $((\tau, \nu) \in [0, \tau_p) \times [0, \nu_p))$ along both delay and Doppler domains.

The aliases lead to an interesting phenomenon in DD signal processing. When the delay spread and/or Doppler spread of the doubly spread channel exceed the delay period (τ_p) and Doppler period (ν_p), respectively, then the channel response to the aliases start superimposing with each other. This phenomenon is referred to as *DD domain aliasing*. Due to DD domain aliasing, the interaction between a DD pulse and a doubly spread channel is non-predictable and fading. However, when the delay period and Doppler period are chosen to be greater than the delay spread and Doppler spread, respectively, there is no DD domain aliasing, and the interaction between a DD pulse and a doubly spread channel is predictable and non-fading. In this scenario, the received signal power in the DD domain does not depend on the DD location of the transmitted pulse, and, therefore, the received power profile is flat just like the surface of crystalline solids. This is why the condition where the delay and Doppler periods are greater than the respective channel spreads is referred to as the *crystallization condition*.

In Section 2.1, the DD domain representation of a doubly spread channel is introduced. Section 2.2 explains the non-predictable and fading interaction between a doubly spread channel and TD/FD pulses. In Section 2.3, the fundamentals of DD domain signaling are introduced, where it is shown that a doubly spread channel acts on a DD domain input signal through twisted convolution, and the unique transform which converts a TD signal to its DD representation is the Zak transform. It is also shown that the DD representation of any TD signal is a quasi-periodic function in the DD domain. Therefore, a DD pulse consists of infinitely many aliases. In Section 2.4, the interaction between a quasi-periodic DD pulse and a doubly spread channel is studied. It is shown that this interaction is predictable and non-fading if the crystallization condition is satisfied.

In Section 2.5, DD domain pulse shaping is introduced for limiting the time duration and bandwidth of the TD representation of a DD pulse. The expression for the time- and bandwidth-limited TD/FD realization of a DD pulse is derived, and it is named as TD/FD *pulsone* since it appears as a periodic pulse train modulated by a sinusoid/tone. It is observed that the time duration T of a TD pulsone is limited by the inverse Doppler domain spread of its DD realization, and the bandwidth B is limited by the inverse delay domain spread of the DD realization. The pulses in the TD pulsone pulse train are separated by the delay period τ_p . Therefore, with τ_p becoming very large, the TD pulsone consists of only a single TD pulse, i.e., it becomes a TDM carrier. Similarly, the pulses in the pulse train of an FD pulsone are separated by the Doppler period ν_p . Therefore, with ν_p becoming very large, the FD pulsone consists of only a single FD pulse, i.e., it becomes an FDM carrier. Therefore, TDM and FDM are special cases of DD domain modulation. Also, this section introduces the inverse time-Zak and the inverse frequency-Zak transforms that convert a quasi-periodic DD domain signal to its TD and FD representations, respectively. It turns out that the Fourier transform is the composition of the time-Zak and inverse frequency-Zak transforms.

With time and bandwidth constraints of T and B , respectively, an information carrying DD pulse has a spread of $1/B$ along the delay domain and a spread of $1/T$ along the Doppler domain. Therefore, distinct information carrying DD pulses can be placed at intervals of $1/B$ along the delay domain and $1/T$ along the Doppler domain. Since DD pulses are quasi-periodic, there are at most $M = \tau_p/(1/B) = B\tau_p$ distinct DD pulses along the delay domain and at most $N = \nu_p/(1/T) = T\nu_p$ distinct pulses along the Doppler domain. In other words, totally $M \times N = BT$ distinct information symbols can be communicated. This gives rise to *Zak-orthogonal time frequency space (OTFS)* modulation, which is discussed in Section 2.6.

In Zak-OTFS modulation, at the transmitter, MN information symbols are mounted on MN DD pulses residing on the DD domain information grid/lattice, where the delay and Doppler co-ordinates of the grid points are integer multiples of $1/B$ and integer multiples of $1/T$, respectively. At the receiver, the received TD signal is firstly converted to the DD domain through the time-Zak transform, followed by DD domain match-filtering. The resulting match-filtered DD domain signal is then sampled on the information grid to obtain a discrete DD domain output signal. In Section 2.6, the relation between the information-carrying discrete DD domain input signal at the transmitter and the discrete DD domain output signal at the receiver is derived (referred to as the Zak-OTFS I/O relation).

In Section 2.7, using the Zak-OTFS I/O relation, it is shown that the I/O relation is predictable and non-fading when the crystallization condition is satisfied. In Section 2.8, the I/O relation for TDM/FDM and Zak-OTFS modulation is expressed in a matrix-vector form which facilitates the performance evaluation study. In Section 2.9, the bit error rate (BER) performance of TDM/FDM/Zak-OTFS is compared for the scenario where the receiver has perfect knowledge of the I/O relation. It is shown that in doubly spread channels, Zak-OTFS achieves a superior BER performance compared to TDM and FDM when the crystallization condition is satisfied. This is in conformance with the fact that the Zak-OTFS I/O relation is non-fading when the crystallization condition is satisfied.

In Section 2.10, the Zak-OTFS BER performance is studied when the I/O relation is not known perfectly at the receiver. In such a scenario, there are two possible approaches: The first approach is a *model-dependent* approach where the receiver assumes an underlying channel model and estimates its parameters from the received response to a DD pilot signal. An estimate of the I/O relation is then constructed using the acquired estimates of the parameters of the underlying channel. The second approach is a *model-free* approach where the receiver only estimates the I/O relation and not the parameters of some underlying channel model. The estimated I/O relation is then used to detect the information symbols from the received discrete DD domain output signal. Simulations reveal that the model-free approach achieves better BER performance than the model-dependent approach in scenarios where the underlying channel model parameters cannot be estimated accurately (for example, when delay resolution $1/B$ and Doppler resolution $1/T$ are not fine enough to separate the channel paths). Even with a simple impulse pilot-based estimation of the I/O relation, the model-free approach achieves good BER performance which is almost invariant of the channel Doppler spread as long as the system operates deep within the crystalline regime (i.e., the delay spread is sufficiently less than the delay period, and the Doppler spread is sufficiently less than the Doppler period).

Appendices at the end of this chapter contain proofs/derivations of important results. For a tutorial introduction to Zak-OTFS modulation, see [33, 34]. Table 2.1 collects the terminologies and notations used in the chapter.

Table 2.1 Glossary.

Notation	Description	Reference
$*_{\sigma}$	Twisted convolution	See (2.20)
$\star$	Linear convolution	
$\mathcal{Z}_t$	Zak transform	See (2.23) and Appendix 2.C
$\mathcal{Z}_t^{-1}$	Inverse Zak transform	See Appendix 2.F
τ_p, ν_p	Delay and Doppler period of Zak-OTFS modulation	See Figs. 2.11 and 2.22
D_0	Fundamental DD period	See (2.18)
Quasi-periodic	Attribute of DD domain	See (2.25)
Quasi-periodic DD pulse at (τ_a, ν_a) (continuous)	DD domain Dirac impulses at $(\tau_a + n\tau_p, \nu_a + m\nu_p), n, m \in \mathbb{Z}$	$x_{\text{dd}}(\tau, \nu; \tau_a, \nu_a)$ in (2.28)
$w_{\text{tx}}(\tau, \nu)$	Transmit pulse shaping filter	See (2.39)
$x_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu; \tau_0, \nu_0)$	Filtered quasi-periodic DD pulse	See (2.39)
$w_{\text{rx}}(\tau, \nu)$	Receive pulse shaping filter	See (2.62)
Zak-OTFS pulsone	TD realization of filtered quasi-periodic DD pulse $x_{\text{dd}}(\tau, \nu)$	$\mathcal{Z}_t^{-1}(w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu))$ in (2.41)
B and T	Bandwidth and time duration	See Fig. 2.17
$M = B\tau_p$	Delay domain period of discrete DD signals	Para. after (2.51)
$N = T\nu_p$	Doppler domain period of discrete DD signals	Para. after (2.51)
Quasi-periodic DD pulse (discrete)	$x^{(b)}[k, l]$, DD pulse at $(k^{(b)}, l^{(b)})$	See (2.68)
$h_{\text{phy}}(\tau, \nu)$	DD spreading function of underlying channel	See (2.1)
$h_{\text{eff}}(\tau, \nu)$	Effective continuous DD domain channel filter	See (2.64)
τ_{eff} and ν_{eff}	Effective delay and Doppler spread of $h_{\text{eff}}(\tau, \nu)$	See (2.67)
$h_{\text{eff}}[k, l]$	Effective discrete DD domain channel filter	See (2.65)
Crystallization condition	Delay and Doppler periods exceed respective effective channel spreads	See (2.67)
Crystalline regime	(τ_p, ν_p) values which satisfy crystallization condition	
Λ_{dd}	Zak-OTFS information grid	See (2.50)
Predictability of I/O relation	Channel response to any input can be predicted from the response to a pulse in modulation domain	Section 2.7

2.1 Delay-Doppler Domain

A typical wireless channel between a transmitter and a receiver is determined by a relatively small number of dominant propagation paths. Figure 2.2 shows a TD signal $x(t)$ transmitted by a base station (BS) and received at some user equipment (UE) through four propagation paths. Each path is characterized by the *path delay*, which is the time taken by the signal to propagate along the path, and by the *path Doppler shift*, which is the frequency shift induced by the relative motion of the transmitter, reflector, and receiver. In Fig. 2.2, it is assumed that the BS and UE are stationary so that only reflector motion determines the path Doppler shift. The signal copy received at UE along the i th path is given by $h_i x(t - \tau_i) e^{j2\pi\nu_i(t - \tau_i)}$, where h_i is the complex channel path gain, τ_i seconds is the path delay, and ν_i Hz is the path Doppler shift.

Wireless channels are inherently *doubly spread*, characterized by the delay, Doppler shift, and complex channel gain of each channel path. The action of the wireless channel on the transmitted signal $x(t)$ is specified by the *DD spreading function* $h_{\text{phy}}(\tau, \nu)$, where $\tau \in \mathbb{R}$ and $\nu \in \mathbb{R}$ are the delay and Doppler variables, respectively [55]. In Fig. 2.2,

$$h_{\text{phy}}(\tau, \nu) = \sum_{i=1}^4 h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (2.1)$$

where $\delta(\cdot)$ denotes the Dirac-delta impulse function.¹ The noise-free TD signal² received at the UE is given by [55]

$$y(t) = \iint h_{\text{phy}}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu. \quad (2.2)$$

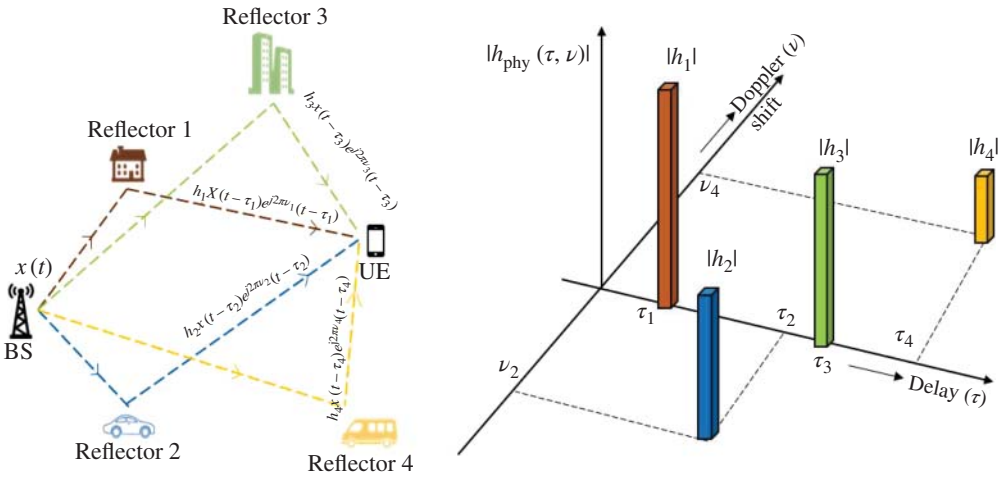


Figure 2.2 Delay-Doppler spreading function $h_{\text{phy}}(\tau, \nu)$ of a doubly spread wireless channel. Four dominant paths between the base station (BS) and the user equipment (UE) result in a function $h_{\text{phy}}(\tau, \nu)$ comprising impulses at $(\tau, \nu) = (\tau_i, \nu_i)$, where τ_i is the i th path delay and ν_i is the i th path Doppler shift. Note that the first and third reflectors do not introduce Doppler shifts since they are stationary. Also, the Doppler shift induced by the second and fourth reflectors have opposite polarity as these mobile reflectors are travelling in opposite directions. Source: Mohammed et al. [33] IEEE.

¹ The same notation δ is used for both discrete and continuous Dirac-delta functions, $\delta[\cdot]$ for the discrete case and $\delta(\cdot)$ for the continuous case.

² In this chapter, the missing lower limit in integration must be interpreted as $-\infty$ and the missing upper limit must be interpreted as ∞ .

In Fig. 2.2,

$$y(t) = \sum_{i=1}^4 h_i x(t - \tau_i) e^{j2\pi v_i(t - \tau_i)}, \quad (2.3)$$

which is the sum of the signal copies received along the four paths.

2.2 Time and Frequency Domain Modulation

In TDM, information is carried by a *narrow TD pulse*, and an information packet is the superposition of distinct TD pulses, each modulated by an information symbol. In FDM, information is carried by a *narrow FD pulse*, which is essentially a sinusoid in TD. In FDM, an information packet is a TD signal, which is the inverse Fourier transform of a superposition of distinct FD pulses, each modulated by an information symbol.

The information carrier in TDM is localized in the TD but not in the FD. Similarly, the information carrier in FDM is localized in the FD but not in the TD (the sinusoid $e^{j2\pi f_0 t}$ shown in Fig. 2.3 is spread in time, but localized at $f = f_0$ in FD). As will be explained in the sequel, the implication of this lack of localization is that the TDM/FDM I/O relation under doubly spread channel witnesses fading and non-predictability (see Sections 2.2.1 and 2.2.2). The fading attribute implies a degradation in the BER performance, while the non-predictability attribute implies frequent acquisition of the effective channel response in order to maintain good performance.

Next, in this section, the example shown in Fig. 2.4 is used to illustrate how a doubly spread wireless channel interacts with the carrier waveforms for TDM and FDM. The doubly spread wireless channel considered in Fig. 2.4 comprises four paths. The first and third paths are due to reflection from the stationary building, and, therefore, these paths do not result in any Doppler shift. The second and fourth paths are due to reflection from moving vehicles, and these therefore induce Doppler shift. To highlight the phenomenon of fading and non-predictability of channel interaction in TDM, let us consider the first and second paths to have the same delay so that the TD pulses received along these two paths superimpose in a time-varying manner (due to the Doppler shift of the second path). Similarly, choosing the first and third paths to have the same Doppler shift (i.e., zero) but different delay highlights the phenomenon of fading and non-predictability of

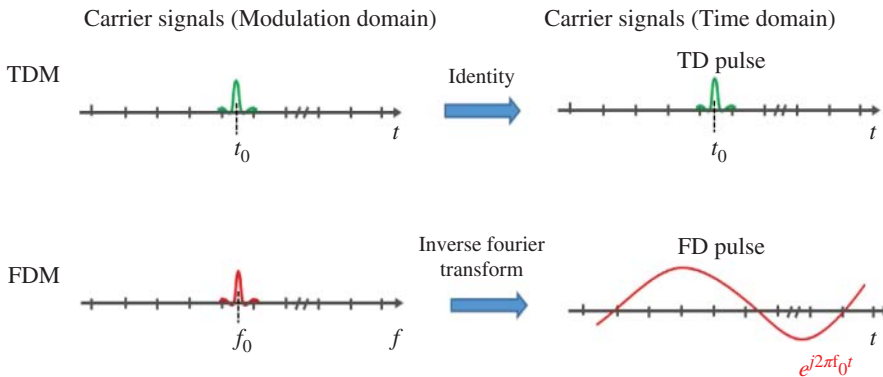


Figure 2.3 Information carriers for TDM/FDM in the modulation domain and time domain. Traditional TDM and FDM carriers are narrow pulses in TD and FD but spread in FD and TD, respectively, manifesting the fundamental obstruction for TF localization. Source: Mohammed et al. [33] IEEE.

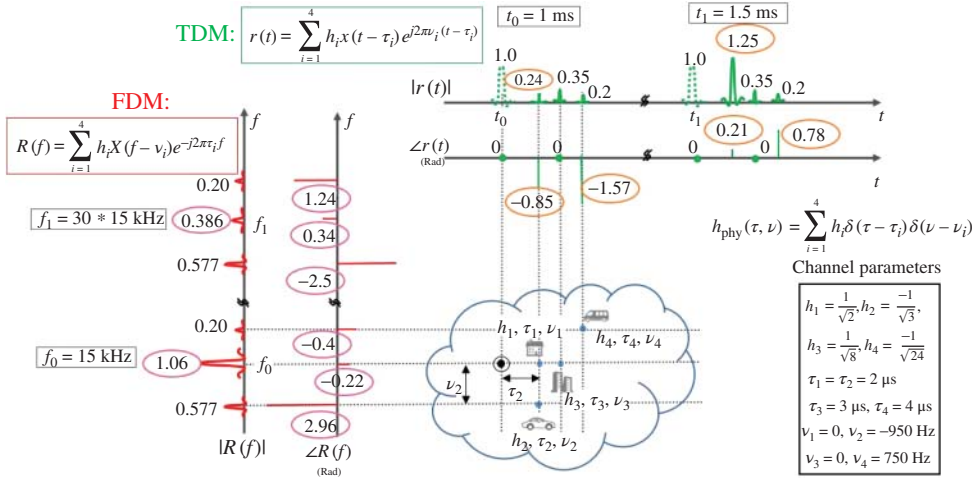


Figure 2.4 Interaction of a doubly spread channel with the TDM/FDM carrier waveform. The channel gain, delay, and Doppler shifts of the four paths are given in the channel parameter box. Two TD pulses are transmitted at $t = t_0$ and $t = t_1$. The channel interaction is fading and non-stationary. For example, the magnitude and phase of the pulse received at $(t_0 + 2 \mu\text{s})$ are 0.24 and -0.85 radians, respectively, whereas those for the pulse received at $(t_1 + 2 \mu\text{s})$ are different, i.e., 1.25 and 0.21 radians, respectively. Similarly, in FDM, two FD pulses are transmitted at $f = f_0$ and $f = f_1$. The channel interaction is fading and non-stationary. For example, the magnitude and phase of the FD pulse received at f_0 are 1.06 and -0.22 radians, respectively, whereas those for the pulse received at f_1 are 0.386 and 0.34 radians, respectively. Source: Mohammed et al. [33] IEEE.

channel interaction in FDM. This is discussed in detail in Section 2.2.1 for TDM and Section 2.2.2 for FDM.

2.2.1 Channel Interaction of a TD Pulse

Figure 2.4 illustrates the TD response to a transmitted signal,

$$x(t) = \delta(t - t_0) + \delta(t - t_1), \quad (2.4)$$

comprising two narrow TD pulses³ transmitted at $t = t_0 = 1 \text{ ms}$ and $t = t_1 = 1.5 \text{ ms}$. A stationary doubly spread channel is considered comprising four channel paths with parameters h_i , τ_i , and ν_i listed in the channel parameter box. Let $h(t; t_i)$, $i = 0, 1$, denote the TD impulse response of the channel to a TD pulse transmitted at time t_i . The channel response to the input impulse at $t = t_i$ is then given by $y_i(t) = h(t - t_i; t_i)$, $i = 0, 1$. From (2.2), the expression for the noise-free channel response to the impulse at $t = t_i$ is given by

$$\begin{aligned} y_i(t) &= \iint h_{\text{phy}}(\tau, \nu) \delta(t - \tau - t_i) e^{j2\pi\nu(t - \tau)} d\tau d\nu \\ &= \int h_{\text{phy}}(t - t_i, \nu) e^{j2\pi\nu t_i} d\nu. \end{aligned} \quad (2.5)$$

³ Narrow pulses are represented by Dirac-delta impulses since the width of a pulse is much smaller than the channel path delays and the inverse of the maximum Doppler shift.

In this example, since $v_1 = v_3 = 0$ Hz,

$$\begin{aligned} h_{\text{phy}}(\tau, \nu) &= \sum_{i=1}^4 h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i) \\ &= h_1 \delta(\tau - 2\mu\text{s}) \delta(\nu) + h_2 \delta(\tau - 2\mu\text{s}) \delta(\nu - \nu_2) \\ &\quad + h_3 \delta(\tau - 3\mu\text{s}) \delta(\nu) + h_4 \delta(\tau - 4\mu\text{s}) \delta(\nu - \nu_4), \end{aligned} \quad (2.6)$$

and, therefore,

$$\begin{aligned} h_{\text{phy}}(t - t_i, \nu) &= h_1 \delta(t - t_i - 2\mu\text{s}) \delta(\nu) + h_2 \delta(t - t_i - 2\mu\text{s}) \delta(\nu - \nu_2) \\ &\quad + h_3 \delta(t - t_i - 3\mu\text{s}) \delta(\nu) + h_4 \delta(t - t_i - 4\mu\text{s}) \delta(\nu - \nu_4). \end{aligned} \quad (2.7)$$

Using this in (2.5), it follows that

$$\begin{aligned} y_i(t) &= h_1 \delta(t - t_i - 2\mu\text{s}) + h_2 \delta(t - t_i - 2\mu\text{s}) e^{j2\pi\nu_2 t_i} \\ &\quad + h_3 \delta(t - t_i - 3\mu\text{s}) + h_4 \delta(t - t_i - 4\mu\text{s}) e^{j2\pi\nu_4 t_i}. \end{aligned} \quad (2.8)$$

Since $y_i(t) = h(t - t_i; t_i)$, $i = 0, 1$, it follows from (2.8) that

$$\begin{aligned} h(t; t_0) &= (h_1 + h_2 e^{j2\pi\nu_2 t_0}) \delta(t - 2\mu\text{s}) + h_3 \delta(t - 3\mu\text{s}) \\ &\quad + h_4 e^{j2\pi\nu_4 t_0} \delta(t - 4\mu\text{s}) \end{aligned} \quad (2.9)$$

and

$$\begin{aligned} h(t; t_1) &= (h_1 + h_2 e^{j2\pi\nu_2 t_1}) \delta(t - 2\mu\text{s}) + h_3 \delta(t - 3\mu\text{s}) \\ &\quad + h_4 e^{j2\pi\nu_4 t_1} \delta(t - 4\mu\text{s}). \end{aligned} \quad (2.10)$$

The important question now is if the response to the impulse at $t = t_1$, i.e., $y_1(t) = h(t - t_1; t_1)$, can be *predicted* from the response to the impulse at $t = t_0$, i.e., $y_0(t) = h(t - t_0; t_0)$. This prediction is possible if and only if $h(t; t_1)$ can be predicted from the knowledge of $h(t; t_0)$.

After estimating the path delays from $h(t; t_0)$, it can be predicted that $h(t; t_1)$ will involve path delays of 2, 3, and 4 μs . In the following, the predictability for each of these three paths is examined.

2.2.1.1 $h_3 \delta(t - 3\mu\text{s})$

This term due to the third channel path is common to $h(t; t_0)$ and $h(t; t_1)$. *Channel paths which do not induce Doppler shift and whose delay is distinct from that of other paths result in a predictable channel interaction.* The complex gain h_3 in this term does not depend on when the pulse was transmitted, so the interaction of the pulse with this path is stationary and non-fading (received pulse magnitude $|h_3|$ is independent of when the pulse was transmitted). The response of such a channel path to any transmitted pulse can therefore be *predicted*.

2.2.1.2 $h_4 e^{j2\pi\nu_4 t_0} \delta(t - 4\mu\text{s})$ and $h_4 e^{j2\pi\nu_4 t_1} \delta(t - 4\mu\text{s})$

These terms are due to the fourth channel path, whose Doppler shift ν_4 interacts with the time at which the pulse was transmitted. As it is not possible to simultaneously estimate h_4 and ν_4 from $h_4 e^{j2\pi\nu_4 t_0}$, the complex gain $h_4 e^{j2\pi\nu_4 t_1}$ of the corresponding term in $h(t; t_1)$ cannot be predicted. The received signal power is independent of the time at which the pulse is transmitted since $|h_4 e^{j2\pi\nu_4 t_i}|^2 = |h_4|^2$, but the phase is time dependent. *Channel paths which induce Doppler shift and whose delay is different from that of other paths result in a non-predictable channel interaction that is non-stationary and non-fading.*

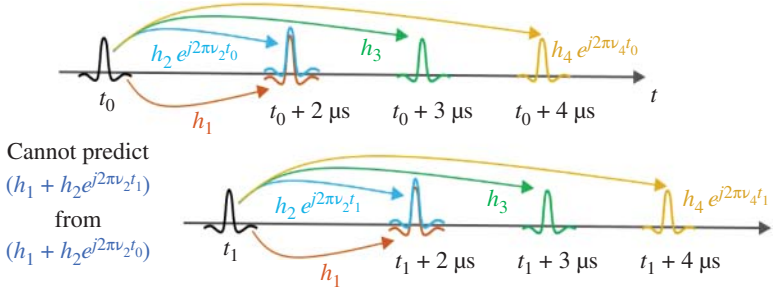


Figure 2.5 Since the first and second paths have the same delay, the response received along these two paths is not separable at the receiver. This results in a non-predictable and fading channel interaction.

2.2.1.3 $(h_1 + h_2 e^{j2\pi v_2 t_0}) \delta(t - 2 \mu s)$ and $(h_1 + h_2 e^{j2\pi v_2 t_1}) \delta(t - 2 \mu s)$

The first and second paths have the same delay, but only the second path introduces a Doppler shift. It is not possible to predict $(h_1 + h_2 e^{j2\pi v_2 t_1}) \delta(t - 2 \mu s)$ from $(h_1 + h_2 e^{j2\pi v_2 t_0}) \delta(t - 2 \mu s)$ since it is not possible to separately estimate h_1 , h_2 , and v_2 from $(h_1 + h_2 e^{j2\pi v_2 t_0}) \delta(t - 2 \mu s)$. Note that, in this case, both the complex gain and received signal power are time dependent (i.e., they depend on the time at which the pulse is transmitted). *Two or more channel paths which have the same path delay and at least one of which induces a Doppler shift which is different from that induced by the other paths result in a non-predictable channel interaction that is fading and non-stationary.*

Therefore, in general, the overall interaction of a doubly spread channel with a TDM pulse is non-predictable and fading. This is primarily because, due to its TD localization attribute, the TDM carrier waveform cannot separate the response received from two paths having the same delay but different Doppler shifts. For the example channel in Fig. 2.4, we illustrate this in Fig. 2.5.

2.2.2 Channel Interaction of a FD Pulse

The same doubly spread channel illustrated in Fig. 2.4 is considered. Let us consider the FD channel response to a transmitted FD signal given by

$$X(f) = \delta(f - f_0) + \delta(f - f_1), \quad (2.11)$$

comprising two FD pulses transmitted at $f = f_0 = 15$ kHz and $f = f_1 = 450$ kHz. Let $H(f; f_i)$ denote the FD impulse response of the channel to a FD impulse transmitted at frequency f_i . The transmitted TD signal is then given by

$$x(t) = e^{j2\pi f_0 t} + e^{j2\pi f_1 t}. \quad (2.12)$$

Using (2.2), the channel output is given by

$$y(t) = \iint h_{\text{phy}}(\tau, \nu) [e^{j2\pi f_0(t-\tau)} + e^{j2\pi f_1(t-\tau)}] e^{j2\pi \nu(t-\tau)} d\tau d\nu. \quad (2.13)$$

From the expression of $h(\tau, \nu)$ in (2.6), it follows that

$$y(t) = \sum_{i=1}^4 h_i e^{j2\pi(f_0 + \nu_i)(t-\tau_i)} + \sum_{i=1}^4 h_i e^{j2\pi(f_1 + \nu_i)(t-\tau_i)}. \quad (2.14)$$

The Fourier transform of the TD response $y(t)$ gives the FD channel response

$$Y(f) = \underbrace{\sum_{i=1}^4 h_i e^{-j2\pi(f_0+v_i)\tau_i} \delta(f - (f_0 + v_i))}_{H(f-f_0;f_0)} + \underbrace{\sum_{i=1}^4 h_i e^{-j2\pi(f_1+v_i)\tau_i} \delta(f - (f_1 + v_i))}_{H(f-f_1;f_1)}, \quad (2.15)$$

where $H(f-f_0;f_0)$ and $H(f-f_1;f_1)$ are the FD channel responses to the FD pulses transmitted at $f=f_0$ and $f=f_1$, respectively. Again, the important question is whether the response $H(f-f_1;f_1)$ can be predicted from the response $H(f-f_0;f_0)$. This is equivalent to the question as to whether $H(f;f_1)$ can be predicted from $H(f;f_0)$. Toward this end, from the expression of $H(f-f_0;f_0)$ and $H(f-f_1;f_1)$ in (2.15) and the values of path delays and Doppler shifts in Fig. 2.4, it follows that

$$H(f;f_0) = [h_1 e^{-j2\pi f_0 \tau_1} + h_3 e^{-j2\pi f_0 \tau_3}] \delta(f) + h_2 e^{-j2\pi(f_0+v_2)\tau_2} \delta(f+950) + h_4 e^{-j2\pi(f_0+v_4)\tau_4} \delta(f-750) \quad (2.16)$$

and

$$H(f;f_1) = [h_1 e^{-j2\pi f_1 \tau_1} + h_3 e^{-j2\pi f_1 \tau_3}] \delta(f) + h_2 e^{-j2\pi(f_1+v_2)\tau_2} \delta(f+950) + h_4 e^{-j2\pi(f_1+v_4)\tau_4} \delta(f-750). \quad (2.17)$$

From the expression of $H(f;f_0)$, it is clear that it comprises impulses at $f=v_1=v_3=0$ Hz, $f=v_2=-950$ Hz, and $f=v_4=750$ Hz. Let us see if the FD channel gain coefficients at these Doppler shifts in $H(f;f_0)$ can be used to predict the corresponding coefficients in $H(f;f_1)$.

2.2.2.1 $h_2 e^{-j2\pi(f_0+v_2)\tau_2} \delta(f+950\text{Hz})$

The second path interacts with the FD pulse transmitted at $f=f_0$, resulting in a received pulse at $f=f_0+v_2$ having complex gain $h_2 e^{-j2\pi(f_0+v_2)\tau_2}$. From this complex gain, the complex gain $h_2 e^{-j2\pi(f_1+v_2)\tau_2}$ of the corresponding term in $H(f;f_1)$ cannot be predicted since it is not possible to estimate h_2 and τ_2 from $h_2 e^{-j2\pi(f_0+v_2)\tau_2}$. Since there is no other path with the same Doppler shift, the magnitude of the received FD pulse along the second path is $|h_2 e^{-j2\pi(f_1+v_2)\tau_2}| = |h_2|$. There is no FD fading since this magnitude does not depend on the FD location of the transmitted pulse. The same analysis applies to the interaction of the fourth channel path with the FD pulses. *Channel paths whose Doppler shift is distinct from that of other paths result in a non-predictable, non-fading, and non-stationary interaction between the channel and the FD pulse.*

2.2.2.2 $(h_1 e^{-j2\pi f_0 \tau_1} + h_3 e^{-j2\pi f_0 \tau_3}) \delta(f)$

The first and third paths do not induce any Doppler shift and have different path delays. The interaction of these two paths with the pulse transmitted at $f=f_0$ results in a received pulse at $f=f_0$ having complex gain $(h_1 e^{-j2\pi f_0 \tau_1} + h_3 e^{-j2\pi f_0 \tau_3})$ as both paths induce the same Doppler shift. From this complex gain, it is not possible to predict the complex gain $(h_1 e^{-j2\pi f_1 \tau_1} + h_3 e^{-j2\pi f_1 \tau_3})$ of the pulse received due to the interaction of the first and third paths with the pulse transmitted at $f=f_1$. This is because it is not possible to separately estimate h_1, h_3, τ_1 , and τ_3 from $(h_1 e^{-j2\pi f_0 \tau_1} + h_3 e^{-j2\pi f_0 \tau_3})$. Also, the magnitude of the superposition of the received pulse along the first and third paths depends on

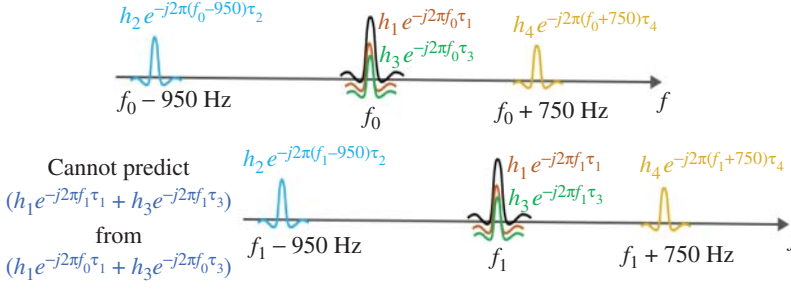


Figure 2.6 Since the first and third paths induce the same Doppler shift, the response received along these two paths is not separable at the receiver. This results in a non-predictable and fading channel interaction.

the location of the transmit pulse, i.e., $|h_1 e^{-j2\pi f_i \tau_1} + h_3 e^{-j2\pi f_i \tau_3}|$ depends on f_i , and therefore, in general, $|h_1 e^{-j2\pi f_i \tau_1} + h_3 e^{-j2\pi f_i \tau_3}| \neq |h_1 e^{-j2\pi f_0 \tau_1} + h_3 e^{-j2\pi f_0 \tau_3}|$. Hence, this interaction is non-stationary and fading. *Two or more channel paths which induce the same Doppler shift, and at least one of which has a path delay which is different from those of the other paths result in a non-predictable channel interaction that is fading and non-stationary.*

Therefore, in general, the overall interaction of a doubly spread channel with a FDM pulse is non-predictable and fading. This is primarily because, due to its FD localization attribute, the FDM carrier waveform cannot separate the response received from two paths having same Doppler shifts but different delays. For the example channel in Fig. 2.4, we illustrate this in Fig. 2.6.

2.3 Delay-Doppler (DD) Domain Modulation

The interaction of a doubly spread channel with a TD/FD pulse is, in general, non-predictable and fading. This is because both TD and FD pulses cannot be simultaneously localized in TD and FD due to Heisenberg's uncertainty principle. However, the obstruction of simultaneous TD and FD localization can be effectively avoided if there are DD domain pulses which are localized in a 2D interval

$$\mathcal{D}_0 \triangleq \{(\tau, \nu) \mid 0 \leq \tau < \tau_p, 0 \leq \nu < \nu_p\}, \quad (2.18)$$

of the DD domain, and which repeat along both the delay and Doppler domains with periods τ_p (delay period) and $\nu_p = 1/\tau_p$ (Doppler period), respectively. Then, the pulses are really not localized since they repeat, and at the same time, they are localized within the interval $\mathcal{D}_0$ which is referred to as the *fundamental DD period*.

In TDM and FDM, information is carried by TD and FD pulses. In DD domain modulation, information is carried by pulses in the DD domain. In Section 2.4, the interaction of a DD domain pulse with a doubly spread channel is studied and it is shown that this interaction is predictable and non-fading when the channel delay and Doppler spreads are less than the delay period and Doppler period, respectively. However, in order to understand this interaction, a mathematical framework for DD domain signals, their TD representation, and channel interaction is required. In this section, this framework is developed along the same lines as that for TDM and FDM.

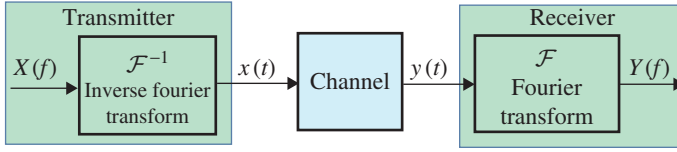


Figure 2.7 In FDM, the FD domain information signal $X(f)$ is converted to the TD transmit signal $x(t)$ using the inverse Fourier transform $\mathcal{F}^{-1}$. At the receiver, the received signal $y(t)$ is then converted to the FD domain signal $Y(f)$ using the Fourier transform. Information is decoded from $Y(f)$.

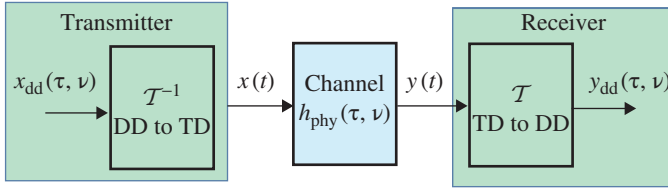


Figure 2.8 In DD domain modulation, the DD domain information signal $x_{dd}(\tau, \nu)$ is converted to the TD transmit signal $x(t)$ using the transform $\mathcal{T}^{-1}$. At the receiver, the received TD signal $y(t)$ is then converted to the DD domain signal $y_{dd}(\tau, \nu)$ using the transform $\mathcal{T}$. Information is decoded from $y_{dd}(\tau, \nu)$.

In FDM, the FD information signal is transformed to the transmit TD signal using the inverse Fourier transform and the received TD signal is converted back to FD using the Fourier transform (see Fig. 2.7). Similarly, in DD domain modulation, the information signal $x_{dd}(\tau, \nu)$ is converted to the transmit TD signal $x(t)$ using the transform $\mathcal{T}^{-1}$, and at the receiver, the received TD signal is converted back to the DD domain using the transform $\mathcal{T}$ (see Fig. 2.8), i.e.,

$$\begin{aligned}
 y_{dd}(\tau, \nu) &= \mathcal{T}(y(t)), \\
 y(t) &= \iint h_{\text{phy}}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu, \\
 x(t) &= \mathcal{T}^{-1}(x_{dd}(\tau, \nu)).
 \end{aligned} \tag{2.19}$$

The next immediate questions that arise are, (i) what is the transform $\mathcal{T}$ and (ii) what is the channel action, i.e., what operation between the DD channel representation $h_{\text{phy}}(\tau, \nu)$ and the DD domain channel input $x_{dd}(\tau, \nu)$ would give the DD domain channel output $y_{dd}(\tau, \nu)$. To answer these questions, again clues are taken from TDM and FDM.

In both TDM (delay-only channels) and FDM (Doppler-only channels), the effective channel impulse response of a cascade of two channels is linear convolution of the impulse responses of those two channels. Also, in both cases, the channel action on an input signal is also described by the same operation, which is linear convolution, i.e., the channel output in TDM (FDM) is the linear convolution of the TD (FD) channel impulse response with the channel input.

In a similar way, the cascade of two doubly spread channels (having DD spreading functions $h_1(\tau, \nu)$ and $h_2(\tau, \nu)$) is a doubly spread channel whose DD spreading function $h_{\text{phy}}(\tau, \nu)$ is given by the *twisted convolution* (denoted by $*_{\sigma}$) of $h_1(\tau, \nu)$ and $h_2(\tau, \nu)$ (see Fig. 2.9), i.e.,

$$\begin{aligned}
 h_{\text{phy}}(\tau, \nu) &= h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu) \\
 &= \iint h_2(\tau', \nu') h_1(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu',
 \end{aligned} \tag{2.20}$$

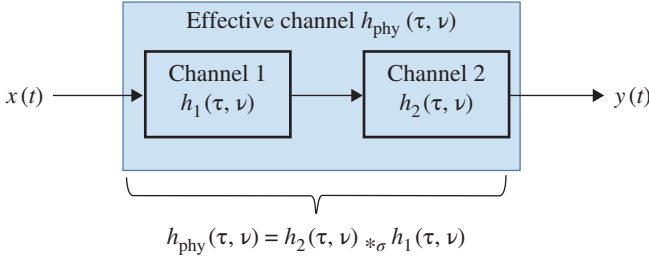


Figure 2.9 Cascade of two doubly spread channels.

which is shown formally in Appendix 2.A. In TDM and FDM, the operation describing a cascade of two channels is the same as the operation describing the channel action on an input information signal, and, therefore, it is natural to require that the interaction between a DD domain information signal and a doubly spread channel will be given by twisted convolution (this is in fact the case, as is argued in Appendix 2.B). In other words, the DD domain representation $y_{\text{dd}}(\tau, \nu)$ of the received TD signal $y(t)$ is given by the twisted convolution of the channel DD spreading function and the DD domain representation $x_{\text{dd}}(\tau, \nu)$ of the transmitted TD signal $x(t)$, i.e.,

$$\begin{aligned} \mathcal{T}(y(t)) &= y_{\text{dd}}(\tau, \nu) = h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu) \\ &= h_{\text{phy}}(\tau, \nu) *_{\sigma} \mathcal{T}(x(t)). \end{aligned} \quad (2.21)$$

The next question is, what is the transform $\mathcal{T}$ such that the channel action is twisted convolution, i.e., for any transmit TD signal $x(t)$, $\mathcal{T}(y(t)) = h_{\text{phy}}(\tau, \nu) *_{\sigma} \mathcal{T}(x(t))$. For FDM (in Doppler-only channels), this is the same as trying to find a FD to TD transform $\mathcal{F}$ for which the channel action is FD linear convolution, i.e., for any transmit TD signal $x(t)$, $\mathcal{F}(y(t)) = H(f) \star \mathcal{F}(x(t))$, where $\star$ denotes linear convolution and $H(f)$ is the FD channel impulse response. For FDM, it is known that this transform $\mathcal{F}$ is nothing but the *Fourier* transform. For DD domain modulation, it can be shown that the TD to DD transform for which the channel action is twisted convolution is the *Zak* transform (see Appendix 2.C). The Zak transform (denoted by $\mathcal{Z}_t$) is parameterized by parameters (τ_p, ν_p) , with $\nu_p = 1/\tau_p$ and the associated period lattice

$$\Lambda_p \triangleq \{(n\tau_p, m\nu_p) \mid n, m \in \mathbb{Z}\}. \quad (2.22)$$

The Zak transform of a TD signal $x(t)$ gives its DD domain representation.⁴

$$x_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(x(t)) \triangleq \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} x(\tau + k\tau_p) e^{-j2\pi\nu k\tau_p}. \quad (2.23)$$

Note that the channel action is twisted convolution for any choice of the parameter (τ_p, ν_p) . The channel action on a DD domain signal is depicted in Fig. 2.10, i.e.,

$$y_{\text{dd}}(\tau, \nu) = h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu). \quad (2.24)$$

⁴ The Zak transform appears in the signal processing literature as a tool for analyzing signals (see [56]), and not as a modulation technique. This was also true of the Fourier transform until the introduction of orthogonal frequency division multiplexing (OFDM) in the late 1960s. In a similar context, the introduction and treatment of Zak transform for modulation purposes in [33, 34] and in this book can be viewed as new.

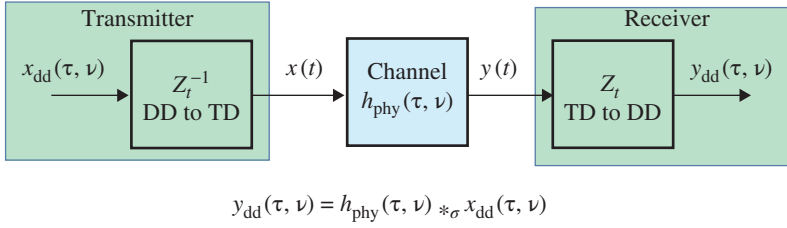


Figure 2.10 Channel action on a DD domain signal.

2.3.1 Origin of Quasi-periodicity

Observe from (2.23) that, for any $n, m \in \mathbb{Z}$, $x_{dd}(\tau, \nu)$ satisfies

$$\begin{aligned}
 x_{dd}(\tau + n\tau_p, \nu + m\nu_p) &= \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} x(\tau + (n+k)\tau_p) e^{-j2\pi(\nu + m\nu_p)k\tau_p} \\
 &= \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} x(\tau + (n+k)\tau_p) e^{-j2\pi\nu(n+k-n)\tau_p} \\
 &= e^{j2\pi n\nu\tau_p} \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} x(\tau + (n+k)\tau_p) e^{-j2\pi\nu(n+k)\tau_p} \\
 &= e^{j2\pi n\nu\tau_p} x_{dd}(\tau, \nu),
 \end{aligned} \tag{2.25}$$

where the second step follows from the fact that $\tau_p \nu_p = 1$, and the third step follows from the fact that the infinite summation on the RHS is $x_{dd}(\tau, \nu)$ (see (2.23)). Substituting $n = 0$ in (2.25), it follows that, for any $m \in \mathbb{Z}$, $x_{dd}(\tau, \nu + m\nu_p) = x_{dd}(\tau, \nu)$, i.e., $x_{dd}(\tau, \nu)$ is periodic along the Doppler domain with period ν_p . Substituting $m = 0$ in (2.25), it follows that, for any $n \in \mathbb{Z}$, $x_{dd}(\tau + n\tau_p, \nu) = e^{j2\pi n\nu\tau_p} x_{dd}(\tau, \nu)$, i.e., a shift along the delay domain by an integer multiple of τ_p introduces a multiplicative phase term $e^{j2\pi n\nu\tau_p}$. Therefore, any DD domain signal (which is the Zak transform of some TD signal) is *quasi-periodic* in the DD domain with period τ_p and ν_p along the delay and Doppler domain, respectively. This also implies that TD realization may not exist for any arbitrary DD function $g(\tau, \nu)$. The TD realization of a DD function exists if and only if it is quasi-periodic. Such DD functions are referred to as DD domain signals.

Table 2.2 summarizes the channel action, transform, and information carrier in TDM, FDM, and DD domain modulation.

In DD domain modulation, the information carrier is a pulse in the DD domain, similar to a FD pulse in FDM and a TD pulse in TDM. A Dirac-delta TD pulse at $t = t_0$ is given by $\delta(t - t_0)$, and a Dirac-delta FD pulse at $f = f_0$ is given by $\delta(f - f_0)$. Therefore, a DD domain signal playing the role of a Dirac-delta pulse at a DD point (τ_0, ν_0) is a quasi-periodic extension of $\delta(\tau - \tau_0)\delta(\nu - \nu_0)$.

Table 2.2 TDM, FDM, and DD domain modulation characteristics.

Modulation	Channel	Domain	Transform	Channel action	Information carrier
TDM	Delay-only	TD	Identity	Linear convolution	TD pulse
FDM	Doppler-only	FD	Fourier	Linear convolution	FD pulse
DD domain modulation	Doubly spread	DD	Zak	Twisted convolution	DD pulse

Hence, Dirac-delta impulses should also be located at $(\tau_0 + n\tau_p, \nu_0 + m\nu_p)$ for all $n, m \in \mathbb{Z}$. This implies that there would also be a single Dirac-delta impulse inside the fundamental DD period D_0 . All other impulses not inside D_0 are called as *aliases* of the impulse inside D_0 (see Fig. 2.11). The expression for a DD pulse at $(\tau_0, \nu_0) \in D_0$ is therefore given by

$$x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0) \triangleq \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n\nu_0\tau_p} \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p). \quad (2.26)$$

A quasi-periodic DD pulse is depicted in Fig. 2.11. In Section 2.4, the interaction of a DD domain pulse with a doubly spread channel is studied. Toward this, the following result is very useful. The action of a channel path (inducing delay τ_1 and Doppler shift ν_1) on a DD domain signal $x_{\text{dd}}(\tau, \nu)$ is given by

$$[\delta(\tau - \tau_1) \delta(\nu - \nu_1)] *_{\sigma} x_{\text{dd}}(\tau, \nu) = e^{j2\pi\nu_1(\tau - \tau_1)} x_{\text{dd}}(\tau - \tau_1, \nu - \nu_1), \quad (2.27)$$

i.e., the signal is geometrically shifted by (τ_1, ν_1) in the DD domain along with multiplication with the *twist* term $e^{j2\pi\nu_1(\tau - \tau_1)}$ (see Fig. 2.12).

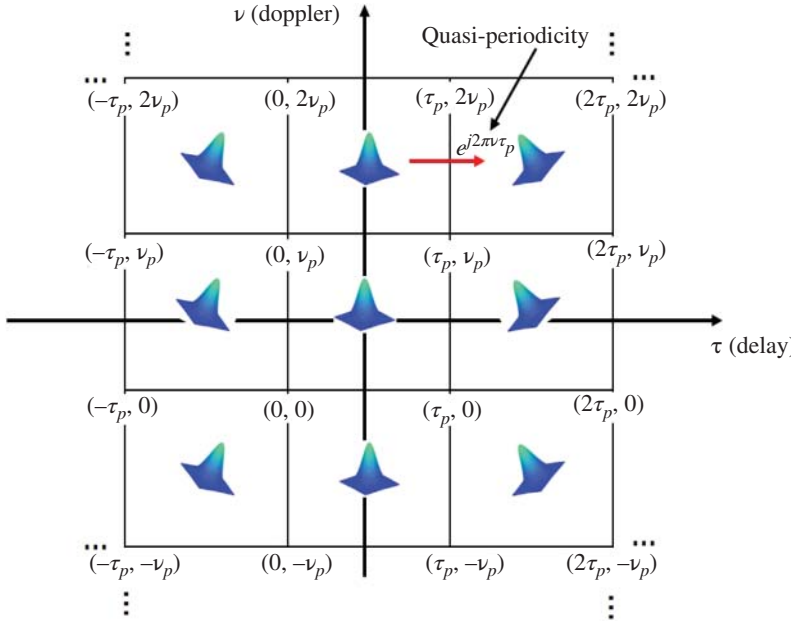
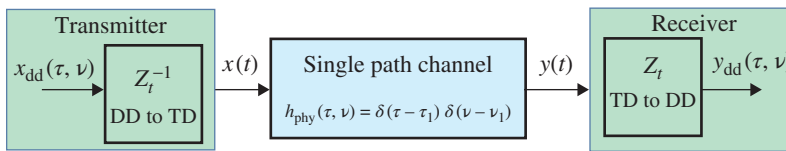


Figure 2.11 A DD domain pulse is localized only within the fundamental DD domain period D_0 as it repeats infinitely many times in a quasi-periodic manner. Source: Mohammed et al. [33] IEEE.



$$y_{\text{dd}}(\tau, \nu) = [\delta(\tau - \tau_1) \delta(\nu - \nu_1)] *_{\sigma} x_{\text{dd}}(\tau, \nu) = e^{j2\pi\nu_1(\tau - \tau_1)} x_{\text{dd}}(\tau - \tau_1, \nu - \nu_1)$$

Figure 2.12 Action of a single path on a DD domain signal.

2.4 Channel Interaction of a DD Domain Pulse

In this section, let us study the interaction of the doubly spread channel in Fig. 2.4 with two DD domain pulses transmitted at $(\tau, \nu) = (\tau_a, \nu_a)$ and $(\tau, \nu) = (\tau_b, \nu_b)$, $0 \leq \tau_a, \tau_b < \tau_p$, $0 \leq \nu_a, \nu_b < \nu_p$. The expression for a quasi-periodic DD domain impulse signal at (τ_0, ν_0) is given by (2.26). Therefore, the quasi-periodic DD domain signal consisting of two impulses at (τ_a, ν_a) and (τ_b, ν_b) is given by

$$x_{dd}(\tau, \nu) = \underbrace{\sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} e^{j2\pi n \nu_a \tau_p} \delta(\tau - n\tau_p - \tau_a) \delta(\nu - m\nu_p - \nu_a)}_{x_{dd}(\tau, \nu; \tau_a, \nu_a)} + \underbrace{\sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} e^{j2\pi n \nu_b \tau_p} \delta(\tau - n\tau_p - \tau_b) \delta(\nu - m\nu_p - \nu_b)}_{x_{dd}(\tau, \nu; \tau_b, \nu_b)}, \quad (2.28)$$

where $x_{dd}(\tau, \nu; \tau_a, \nu_a)$ and $x_{dd}(\tau, \nu; \tau_b, \nu_b)$ denote the DD domain impulse signals at (τ_a, ν_a) and (τ_b, ν_b) , respectively. The transmit TD signal is then given by the inverse time-Zak transform $\mathcal{Z}_t^{-1}$ (see Fig. 2.10) of $x_{dd}(\tau, \nu)$. At the receiver, the received TD signal is transformed to the DD domain through the Zak transform $\mathcal{Z}_t$ (as in (2.23)), resulting in the DD domain signal

$$y_{dd}(\tau, \nu) = y_{dd}(\tau, \nu; \tau_a, \nu_a) + y_{dd}(\tau, \nu; \tau_b, \nu_b), \quad (2.29)$$

where $y_{dd}(\tau, \nu; \tau_a, \nu_a)$ and $y_{dd}(\tau, \nu; \tau_b, \nu_b)$ denote the DD domain channel response to the DD domain impulses at (τ_a, ν_a) and (τ_b, ν_b) , respectively. For the doubly spread channel example in (2.6), the expressions for these response signals are derived in Appendix 2.D and are given by the a -response

$$y_{dd}(\tau, \nu; \tau_a, \nu_a) = \sum_{i=1}^4 h_i e^{j2\pi \nu_i \tau_a} \underbrace{\left[\sum_{(n,m) \in \mathbb{Z}} e^{j2\pi n(\nu_a + \nu_i) \tau_p} \delta(\tau - (\tau_a + \tau_i) - n\tau_p) \delta(\nu - (\nu_a + \nu_i) - m\nu_p) \right]}_{x_{dd}(\tau, \nu; \tau_a + \tau_i, \nu_a + \nu_i)} \quad (2.30)$$

and the b -response

$$y_{dd}(\tau, \nu; \tau_b, \nu_b) = \sum_{i=1}^4 h_i e^{j2\pi \nu_i \tau_b} \underbrace{\left[\sum_{(n,m) \in \mathbb{Z}} e^{j2\pi n(\nu_b + \nu_i) \tau_p} \delta(\tau - (\tau_b + \tau_i) - n\tau_p) \delta(\nu - (\nu_b + \nu_i) - m\nu_p) \right]}_{x_{dd}(\tau, \nu; \tau_b + \tau_i, \nu_b + \nu_i)}. \quad (2.31)$$

From (2.27), it is known that channel path delay shifts the DD domain pulse along the delay domain and that channel path Doppler shifts the DD domain pulse along the Doppler domain. From (2.30), it follows that the a -response includes four *distinct* pulses received at $(\tau_a + \tau_i, \nu_a + \nu_i)$, and from (2.31), it is observed that the b -response includes four distinct pulses received at $(\tau_b + \tau_i, \nu_b + \nu_i)$, where $i = 1, 2, 3, 4$. Figure 2.13 illustrates that the received pulse along each path is observed *separately* in the DD domain.

Again, the main question is whether the b -response $y_{dd}(\tau, \nu; \tau_b, \nu_b)$ can be predicted from the a -response $y_{dd}(\tau, \nu; \tau_a, \nu_a)$.

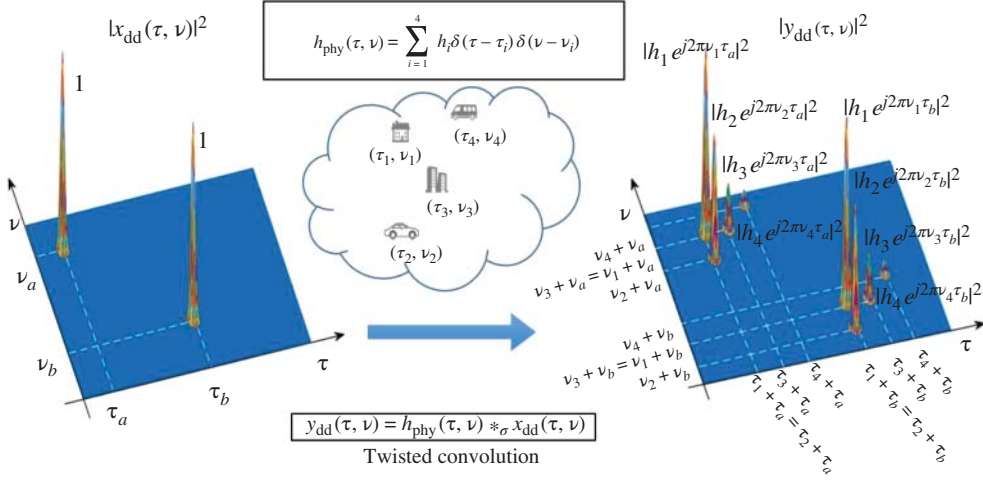


Figure 2.13 The transmit DD pulse at (τ_a, ν_a) is “effectively” localized along both the delay and Doppler domains, and, therefore, DD pulses are received along each path at distinct DD locations $(\tau_a + \tau_i, \nu_a + \nu_i)$, $i = 1, 2, 3, 4$, as compared to TDM and FDM (see Figs. 2.5 and 2.6, where only three paths are detected at the receiver). Due to quasi-periodicity, the transmit DD pulse is repeated along the delay and Doppler domain with period τ_p and ν_p , respectively. These repetitions/aliases also interact with the channel. If τ_p is less than the channel delay spread and/or if ν_p is less than the channel Doppler spread (i.e., (τ_p, ν_p) does not satisfy (2.32)), then the received response to the quasi-periodic aliases could *overlap/superimpose/alias* with the received response due to the DD impulse transmitted at $(\tau_a, \nu_a) \in \mathcal{D}_0$, thereby making channel prediction non-simple and difficult. However, if (τ_p, ν_p) is properly chosen to satisfy (2.32), then the channel response to a DD pulse transmitted at some other DD location (e.g., (τ_b, ν_b)) can be predicted from perfect knowledge of the channel response to the DD pulse transmitted at (τ_a, ν_a) . Also, when (τ_p, ν_p) satisfies (2.32), the total energy of the received channel response is invariant of the location of the transmitted DD pulse. Therefore, the interaction of a DD pulse with the channel is *predictable and non-fading* if (2.32) is satisfied. Source: Mohammed et al. [33] IEEE.

2.4.1 The Channel Interaction is Predictable

The channel interaction is predictable when the delay period τ_p is greater than the channel delay spread and the Doppler period ν_p is greater than the channel Doppler spread, i.e.,

$$\tau_p > (\max_i \tau_i - \min_i \tau_i), \text{ and } \nu_p > (\max_i \nu_i - \min_i \nu_i). \quad (2.32)$$

Condition (2.32) is called the *crystallization condition*. When the crystallization condition holds, the b -response can be predicted from the a -response. From (2.30), it is clear that the received a -response along the i th path consists of infinitely many impulses at $(\tau_a + \tau_i + n\tau_p, \nu_a + \nu_i + m\nu_p)$, $n, m \in \mathbb{Z}$ and $i = 1, 2, 3, 4$. In Appendix 2.E, it is shown that when the crystallization condition is satisfied, *none of these pulses overlap*, irrespective of the path along which they are received.

When the crystallization condition is satisfied, the path delay and Doppler spreads are less than τ_p and ν_p , respectively. Therefore, in general, let the path delays and Doppler shifts lie within the interval $[\tau_{\min}, \tau_{\max})$ and $[\nu_{\min}, \nu_{\max})$, respectively, where $0 < (\tau_{\max} - \tau_{\min}) < \tau_p$ and $0 < (\nu_{\max} - \nu_{\min}) < \nu_p$. These intervals are usually known at the receiver. The (n, m) th term in the a -response (see (2.30)), i.e.,

$$\sum_{i=1}^4 h_i e^{j2\pi\nu_i\tau_a} e^{j2\pi n(\nu_a + \nu_i)\tau_p} \delta(\tau - \tau_a - \tau_i - n\tau_p) \delta(\nu - \nu_a - \nu_i - m\nu_p), \quad (2.33)$$

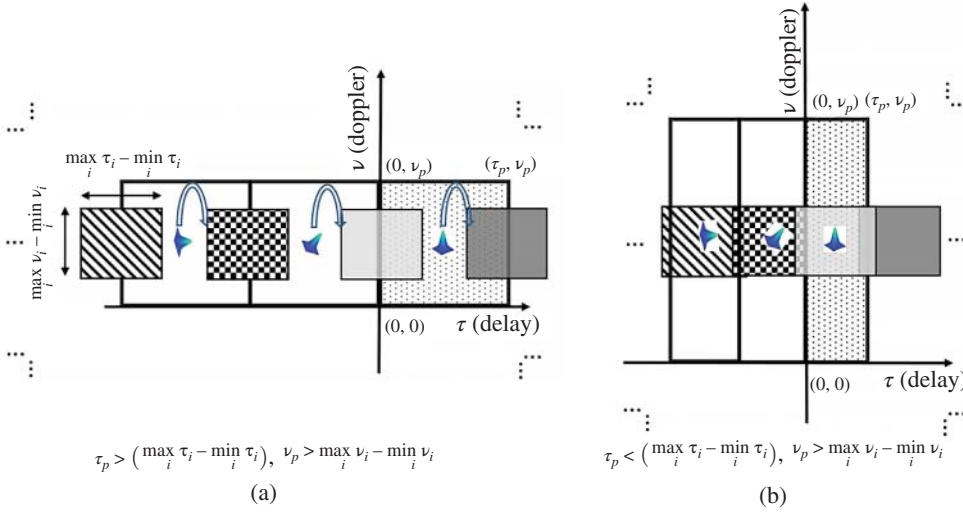


Figure 2.14 A DD domain pulse is transmitted in the fundamental DD domain period D_0 . Rectangular DD domain regions containing the received DD domain pulses corresponding to the transmitted DD pulse in D_0 and its quasi-periodic aliases are depicted as shaded rectangles. The size of each such rectangular region is invariant of the delay and Doppler domain period, i.e., they have length equal to the channel delay spread $(\max_i \tau_i - \min_i \tau_i)$ along the delay domain and equal to the channel Doppler spread $(\max_i \nu_i - \min_i \nu_i)$ along the Doppler domain. Two choices of the delay and Doppler domain period are considered, choice (a) the delay and Doppler domain periods satisfy the crystallization condition in (2.32), and choice (b) Doppler domain period is greater than the channel Doppler spread, but the delay domain period is less than the channel delay spread. The fundamental DD domain period D_0 is depicted by a dotted rectangle. For choice A, DD pulses received in D_0 are within the light gray and the dark gray shaded rectangles which do not overlap and therefore there is no aliasing. However, in choice B, due to small τ_p and the invariance of the size of the shaded rectangles, the light gray rectangle overlaps with the dark gray and the checkered rectangles resulting in delay domain aliasing. Similarly, there would be aliasing along the Doppler domain if the Doppler domain period is chosen to be smaller than the channel Doppler spread. Source: Mohammed et al. [33] IEEE.

consists of impulses which lie within the DD domain rectangles/intervals $\mathcal{R}_{a,n,m} \triangleq [\tau_a + \tau_{\min} + n\tau_p, \tau_a + \tau_{\max} + n\tau_p] \times [\nu_a + \nu_{\min} + m\nu_p, \nu_a + \nu_{\max} + m\nu_p]$, $n, m \in \mathbb{Z}$. The boundary of this rectangle is known at the receiver (see Fig. 2.14 for an illustration of these rectangles along the delay domain). When the crystallization condition is satisfied, no two impulses in the a -response overlap, i.e., the received impulses in $\mathcal{R}_{a,n,m}$ can be separated out from the a -response.

Similarly, the (n, m) th term in the b -response (see (2.31)) is given by

$$\sum_{i=1}^4 h_i e^{j2\pi \nu_i \tau_b} e^{j2\pi n(\nu_b + \nu_i) \tau_p} \delta(\tau - \tau_b - \tau_i - n\tau_p) \delta(\nu - \nu_b - \nu_i - m\nu_p), \quad (2.34)$$

which consists of impulses within the known rectangles $\mathcal{R}_{b,n,m} = [\tau_b + \tau_{\min} + n\tau_p, \tau_b + \tau_{\max} + n\tau_p] \times [\nu_b + \nu_{\min} + m\nu_p, \nu_b + \nu_{\max} + m\nu_p]$, $n, m \in \mathbb{Z}$. It is next shown that from the knowledge of the impulses in the a -response which lie within $\mathcal{R}_{a,n,m}$, the impulses in the b -response which lie within $\mathcal{R}_{b,n,m}$ can be predicted. For example, the i th impulse in $\mathcal{R}_{a,n,m}$ is located at $(\tau_a + \tau_i + n\tau_p, \nu_a + \nu_i + m\nu_p)$ and has complex gain $h_i e^{j2\pi \nu_i \tau_a} e^{j2\pi n(\nu_a + \nu_i) \tau_p}$. Shifting the location of this impulse by $(\tau_b - \tau_a)$ and $(\nu_b - \nu_a)$ along the delay and Doppler domain, respectively, gives the location of the i th impulse in $\mathcal{R}_{b,n,m}$, and multiplying its gain by $e^{j2\pi \nu_i (\tau_b - \tau_a)} e^{j2\pi n(\nu_b - \nu_a) \tau_p}$ gives the

complex gain of the i th impulse in $\mathcal{R}_{b,n,m}$, i.e.,

$$h_i e^{j2\pi v_i \tau_a} e^{j2\pi n(v_a + v_i) \tau_p} \times [e^{j2\pi v_i(\tau_b - \tau_a)} e^{j2\pi n(v_b - v_a) \tau_p}] = h_i e^{j2\pi v_i \tau_b} e^{j2\pi n(v_b + v_i) \tau_p}. \quad (2.35)$$

Note that the multiplication factor $e^{j2\pi v_i(\tau_b - \tau_a)} e^{j2\pi n(v_b - v_a) \tau_p}$ in the above is known to the receiver as it knows (τ_a, v_a) , (τ_b, v_b) , and it can estimate v_i from the *difference* between the Doppler domain location of the i th received impulse in $\mathcal{R}_{a,n,m}$ (i.e., $(v_a + v_i + m v_p)$) and the Doppler domain location of the transmitted pulse at $(v_a + m v_p)$. This prediction is illustrated in Fig. 2.13 for $(n, m) = (0, 0)$, i.e., the impulses received in $\mathcal{R}_{b,0,0}$ are predicted from the impulses received in $\mathcal{R}_{a,0,0}$. Since each received impulse in the b -response lies in $\mathcal{R}_{b,n,m}$ for some (n, m) , it can be predicted from the corresponding impulse in $\mathcal{R}_{a,n,m}$.

The predictive relation (2.35) *breaks down* when the crystallization condition does not hold. This is because when the crystallization condition does not hold, impulses received along two different paths may overlap. This happens because a DD domain impulse signal comprises infinitely many impulses (referred to as aliases of the impulse within the fundamental period) and the response associated with these aliases overlaps/interacts with the response associated with the impulse within the fundamental period.⁵ This phenomenon is referred to as *DD domain aliasing* and is illustrated in Fig. 2.14(b).

To understand why the predictive relation (2.35) may breakdown when the crystallization condition does not hold, let us consider that in our four path example, $\tau_4 > \tau_1$, $v_4 > v_1$, and $(\tau_4 - \tau_1)(v_4 - v_1) = 1$. With $\tau_p = \tau_4 - \tau_1$ and $v_p = v_4 - v_1$, it is clear that the crystallization condition in (2.32) is not satisfied. In this situation, the responses associated with the pulse transmitted at (τ_a, v_a) and its alias at $(\tau_a - \tau_p, v_a - v_p)$ are both received at the same location $(\tau_a + \tau_1, v_a + v_1)$ along the 1st and the 4th channel paths, respectively (for the alias at $(\tau_a - \tau_p, v_a - v_p)$, the impulse received along the fourth path is located at $(\tau_a - \tau_p + \tau_4, v_a - v_p + v_4) = (\tau_a + \tau_1, v_a + v_1)$) (see Fig. 2.15). From (2.30), it is clear that the impulse received at $(\tau_a + \tau_1, v_a + v_1)$ along the $i = 1$ st path is due to the $(n, m) = (0, 0)$ th term (with $i = 1$), i.e., $h_1 e^{j2\pi v_1 \tau_a} \delta(\tau - \tau_1 - \tau_a) \delta(v - v_1 - v_a)$ and that received at the same location along the $i = 4$ th path is due to the $(n, m) = (-1, -1)$ th term (with $i = 4$), i.e., $h_4 e^{j2\pi v_4 \tau_a} e^{-j2\pi \tau_p(v_a + v_4)} \delta(\tau - \tau_1 - \tau_a) \delta(v - v_1 - v_a)$. Since these two received impulses overlap, the received a -response contains their superposition

$$(h_1 e^{j2\pi v_1 \tau_a} + h_4 e^{j2\pi v_4 \tau_a} e^{-j2\pi(v_a + v_1)(\tau_4 - \tau_1)}) \delta(\tau - \tau_1 - \tau_a) \delta(v - v_1 - v_a), \quad (2.36)$$

from which it is *difficult to separate them apart*.

Therefore, the complex gain $(h_1 e^{j2\pi v_1 \tau_b} + h_4 e^{j2\pi v_4 \tau_b} e^{-j2\pi(v_b + v_1)(\tau_4 - \tau_1)})$ of the received pulse at $(\tau_b + \tau_1, v_b + v_1)$ in the b -response cannot be simply predicted from the complex gain of the received pulse at $(\tau_a + \tau_1, v_a + v_1)$ in the a -response. Hence, *DD domain aliasing* results in a *non-predictable* interaction between a DD domain pulse and a doubly spread channel.

Condition (2.32) precludes DD domain aliasing. Since $\tau_p v_p = 1$, condition (2.32) requires that

$$\left(\max_i \tau_i - \min_i \tau_i \right) \times \left(\max_i v_i - \min_i v_i \right) < 1, \quad (2.37)$$

that is, the product of the channel delay spread and Doppler spread is less than one. This condition is generally satisfied for most doubly spread channels of practical interest. For example, in a typical cellular wireless system with channel delay spread 5 μ s and Doppler spread 1000 Hz, the product of the delay and Doppler spread is only 5×10^{-3} .

⁵ In the a -response, the $(n, m) = (0, 0)$ th term $\sum_{i=1}^4 h_i e^{j2\pi v_i \tau_a} \delta(\tau - \tau_a - \tau_i) \delta(v - v_a - v_i)$ corresponds to the response associated with the impulse located at (τ_a, v_a) within the fundamental DD period and the other terms in the a -response for $(n, m) \neq (0, 0)$ correspond to the response associated with impulse aliases at $(\tau_a + n\tau_p, v_a + m v_p)$.

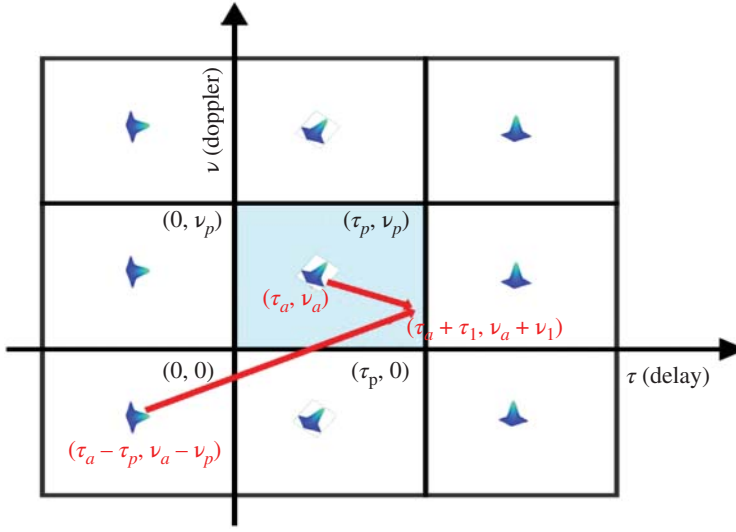


Figure 2.15 DD domain aliasing ($\tau_4 > \tau_1$, $\nu_4 > \nu_1$, and $(\tau_4 - \tau_1)(\nu_4 - \nu_1) = 1$). $\tau_p = (\tau_4 - \tau_1)$ and $\nu_p = (\nu_4 - \nu_1)$. The responses to the impulses at (τ_a, ν_a) and $(\tau_a - \tau_p, \nu_a - \nu_p)$ overlap at $(\tau_a + \tau_1, \nu_a + \nu_1)$, resulting in non-predictability and fading.

2.4.2 The Channel Interaction is Non-fading

The channel interaction is non-fading when the crystallization condition (2.32) holds, in the sense that the amplitude of the response does not depend on the location of the transmitted pulse. Specifically, in our example, when the crystallization condition holds, the power distribution of the b -response is equal to the power distribution of the a -response. In Fig. 2.13, for both the a -response and b -response, the amplitude of the pulse received along the i th path is $|h_i|$. When the crystallization condition does not hold, then due to aliasing, the power distribution in general depends on the location of the transmit pulse. For the example in Fig. 2.15, the amplitude of the pulse received at $(\tau_a + \tau_1, \nu_a + \nu_1)$ is $|h_1 e^{j2\pi\nu_1\tau_a} + h_4 e^{j2\pi\nu_4\tau_a} e^{-j2\pi(\nu_a + \nu_1)(\tau_4 - \tau_1)}|$. It can be verified that this expression in general depends on the location (τ_a, ν_a) of the transmitted pulse.

2.4.3 The Channel Interaction is Non-stationary

It is noted that even when the crystallization condition holds, the actual phases in the response depend on the location of the transmitted pulse. In our example, the complex gain of the impulse received at $(\tau_a + \tau_i, \nu_a + \nu_i)$ along the i th path is $h_i e^{j2\pi\nu_i\tau_a}$ in the a -response and is $h_i e^{j2\pi\nu_i\tau_b}$ in the b -response. However, non-stationarity is not a major issue as long as predictability (2.35) is maintained.

2.5 Time- and Bandwidth-Limited DD Domain Carrier Waveforms

In TDM, the TD carrier waveform $\delta(t - t_0)$ has infinite bandwidth, and, in FDM, the TD carrier waveform $e^{j2\pi f_0 t}$ (i.e., inverse Fourier transform of FD pulse $\delta(f - f_0)$) has infinite time duration. In DD domain modulation, the TD carrier waveform is given by the inverse Zak transform of a

DD pulse. A DD pulse at (τ_0, ν_0) is given by (2.26). Its TD realization is given by its inverse Zak transform,⁶ i.e.,

$$\begin{aligned} x(t) &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{j2\pi n \nu_0 \tau_p} \delta(t - \tau_0 - n\tau_p) \\ &= \sqrt{\tau_p} e^{j2\pi \nu_0 (t - \tau_0)} \left[\sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) \right], \end{aligned} \quad (2.38)$$

which is a train of TD impulses modulated by a tone $e^{j2\pi \nu_0 (t - \tau_0)}$ of frequency ν_0 . The impulses in the pulse train are separated by τ_p seconds, and the n th impulse is located at $t = (\tau_0 + n\tau_p)$. Therefore, the TD realization of a DD pulse is appropriately called a *pulsone* (a pulse modulated by a tone). However, the TD pulsone in (2.38) clearly has infinite time duration and bandwidth.

In order to have a time- and bandwidth-limited carrier, in TDM, the carrier $\delta(t - t_0)$ is *linearly convolved* (filtered) with a time- and bandwidth-limited TD pulse $p(t)$. Similarly, in FDM, the FD carrier $\delta(f - f_0)$ is *linearly convolved* with a time- and bandwidth-limited FD pulse $Q(f)$. In TDM (delay-only channels) and FDM (Doppler-only channels), linear convolution is the preferred operation to ensure limiting time and bandwidth as it is also how the channel interacts with the channel input. Since linear convolution is associative, the overall effective I/O relation is also described by linear convolution which guarantees predictability and no fading.⁷

Similarly, for DD domain modulation, the preferred filtering operation at the transmitter is twisted convolution since the effective DD domain I/O relation would be described by twisted convolution⁸ for which in Section 2.4 it has already been shown that the channel interaction would be predictable and non-fading (with (τ_p, ν_p) chosen to satisfy the crystallization condition in (2.32)). For a DD pulse $x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0)$ at (τ_0, ν_0) , the filtered pulse is given by

$$\begin{aligned} x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) &= w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0) \\ &= \sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} w_{\text{tx}}(\tau - n\tau_p - \tau_0, \nu - m\nu_p - \nu_0) e^{j2\pi \nu (n\tau_p + \tau_0)} e^{-j2\pi \tau_0 (\nu_0 + m\nu_p)}, \end{aligned} \quad (2.39)$$

where $w_{\text{tx}}(\tau, \nu)$ is the DD domain pulse shaping waveform at the transmitter. The last step follows in a straightforward manner by evaluating the twisted convolution integral. Note that the filtered DD pulse $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ consists of the pulse $w_{\text{tx}}(\tau - \tau_0, \nu - \nu_0) e^{j2\pi (\nu - \nu_0) \tau_0}$ at (τ_0, ν_0) in the fundamental DD period (see the $(n, m) = (0, 0)$ th term in (2.39)), and its aliases at $(\tau_0 + n\tau_p, \nu_0 + m\nu_p)$, $n, m \in \mathbb{Z}$ (see the terms in (2.39) corresponding to $(n, m) \neq (0, 0)$).

The TD transmit signal $x(t)$ is now given by the inverse Zak transform of $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ (see Fig. 2.16). However, for a TD realization to exist, $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ must be quasi-periodic (see (2.25)). Interestingly, twisted convolution conserves quasi-periodicity, i.e., the twisted convolution of a quasi-periodic DD domain signal with any arbitrary DD domain function (which need not be quasi-periodic) is also quasi-periodic (see Appendix 2.G). At the receiver, the Zak transform of the received TD signal $y(t)$ gives the DD domain output signal $y_{\text{dd}}(\tau, \nu)$. Since the channel action on a DD domain input signal is through twisted convolution with the channel DD spreading function

6 See Appendix 2.F.1 for a discussion on the inverse Zak transform and Appendix 2.F.2 for the derivation of (2.38).

7 With linear convolution, the effective TD channel response for TDM (delay-only channel) is $h(t) \star p(t)$, and the effective FD channel response for FDM (Doppler-only channel) is $G(f) \star Q(f)$, where $h(t)$ is the TD impulse response of the delay-only channel and $G(f)$ is the FD impulse response of the Doppler-only channel.

8 This follows from the fact that the cascade of twisted convolution/filtering operations is equivalent to a single twisted convolution with an effective DD filter (see Appendix 2.A).

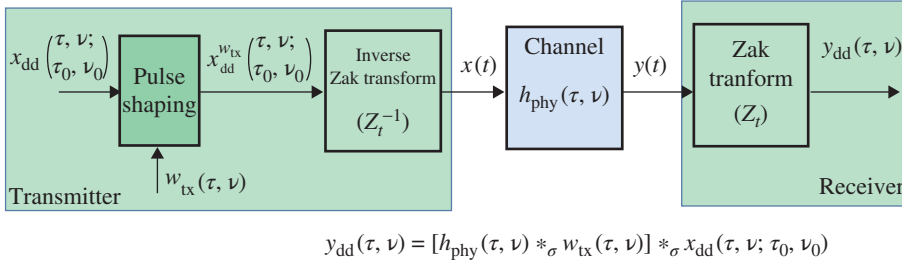


Figure 2.16 Pulse shaping of a DD domain pulse.

$h_{\text{phy}}(\tau, \nu)$ (see Appendix 2.C), it follows that

$$\begin{aligned}
 y_{\text{dd}}(\tau, \nu) &= h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0) \\
 &= h_{\text{phy}}(\tau, \nu) *_{\sigma} (w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0)) \\
 &= \underbrace{(h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu))}_{\text{Effective DD spreading function}} *_{\sigma} x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0),
 \end{aligned} \tag{2.40}$$

where the second step follows from (2.39) and the third step follows from the associativity of twisted convolution.

The next immediate question is how to choose $w_{\text{tx}}(\tau, \nu)$ in order that the TD carrier $x(t) = \mathcal{Z}_t^{-1}(x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0))$ is approximately time and bandwidth limited to T seconds and B Hz, respectively. In the following, it is shown that the TD carrier $x(t)$ is approximately time and bandwidth limited, respectively, to the inverse of the Doppler domain and delay domain spreads of the pulse shaping waveform $w_{\text{tx}}(\tau, \nu)$.

For a factorizable DD domain function $w_{\text{tx}}(\tau, \nu) = W_1(\tau)W_2(\nu)$,

$$\begin{aligned}
 x(t) &= \mathcal{Z}_t^{-1}(x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0)) \\
 &= \sqrt{\tau_p} \sum_{n=-\infty}^{\infty} g(n\tau_p + \tau_0) W_1(t - n\tau_p - \tau_0),
 \end{aligned} \tag{2.41}$$

where $g(t) \triangleq w_2(t) e^{j2\pi\nu_0(t-\tau_0)}$ and

$$w_2(t) \triangleq \mathcal{F}^{-1}(W_2(f)) = \int W_2(f) e^{j2\pi ft} df, \tag{2.42}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and its inverse (see Appendix 2.H for the derivation of (2.41)), respectively. From (2.41), it is clear that $x(t)$ consists of a train of pulses, where the n th pulse located at $(n\tau_p + \tau_0)$ is a replica of $W_1(t)$ time shifted by $(n\tau_p + \tau_0)$. The n th pulse is also multiplied by the value of $g(t)$ at the pulse location. The waveform $g(t)$ is a complex TD sinusoid of frequency ν_0 modulated by the waveform $w_2(t)$ (which is the inverse Fourier transform of $W_2(\cdot)$). Therefore, $x(t)$ is time limited by the duration of $w_2(t)$, which is approximately the inverse of the Doppler domain spread of $W_2(\nu)$. Since the Doppler domain spread of $W_2(\nu)$ is the same as that of $w_{\text{tx}}(\tau, \nu)$, it is clear that the time duration of $x(t)$ is approximately the inverse of the Doppler domain spread of $w_{\text{tx}}(\tau, \nu)$. Furthermore, from (2.41), it is also clear that the pulses in $x(t)$ are spaced apart by the delay period τ_p and the TD spread of each pulse is the same as the delay domain spread of $W_1(\tau)$, which is the same as the delay domain spread of $w_{\text{tx}}(\tau, \nu)$. Since $x(t)$ is a pulse train modulated by a complex sinusoid/tone (of frequency ν_0), it is referred to as a TD pulsone (see Fig. 2.17).

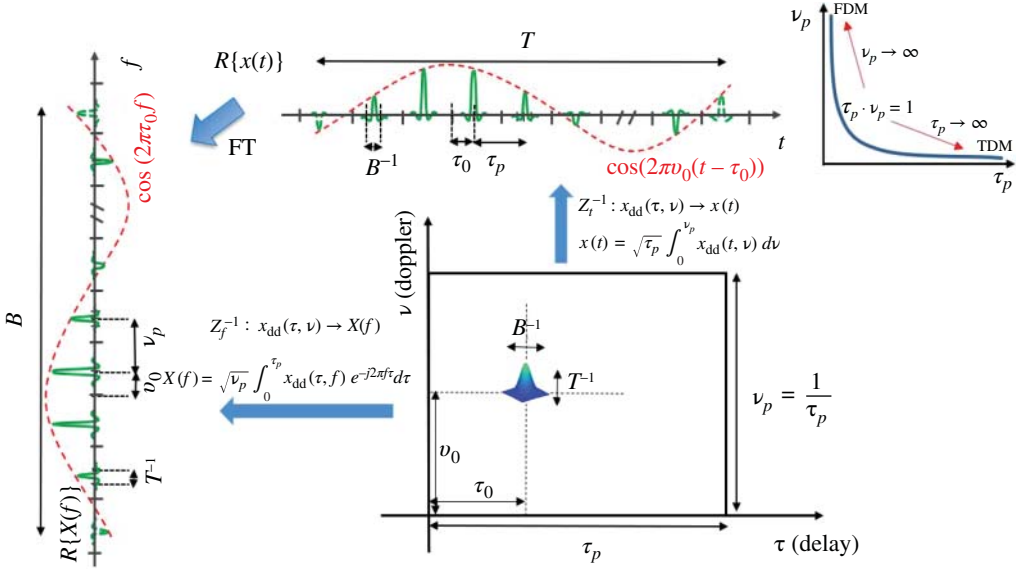


Figure 2.17 DD domain pulse and its TD (FD) realization, referred to as TD (FD) pulsone. Pulse shaping waveform is $w_{\text{tx}}(\tau, \nu) = \sqrt{BT} \text{sinc}(B\tau) \text{sinc}(T\nu)$. The TD pulsone comprises a finite duration pulse train modulated by a TD tone. The FD pulsone comprises a finite bandwidth pulse train modulated by a FD tone. The location of the pulses in the TD/FD pulse train and the frequency of the modulating TD/FD tone are determined by the location of the DD domain pulse (τ_0, ν_0) . The time duration and bandwidth of a pulsone are inversely proportional to the characteristic width of the DD domain pulse along the Doppler domain and the delay domain, respectively. The number of non-overlapping DD pulses, each spread over an area $B^{-1}T^{-1}$, inside the fundamental period D_0 (which has unit area) is equal to the time–bandwidth product BT , and the corresponding pulsones are orthogonal to one another. As $\nu_p \rightarrow \infty$, the FD pulsone approaches a single FD pulse, which is the FDM carrier. Similarly, as $\tau_p \rightarrow \infty$, the TD pulsone approaches a single TD pulse, which is the TDM carrier. Therefore, DD domain pulses give rise to a family of carrier waveforms parameterized by τ_p that interpolates between the TDM and FDM carrier waveforms. Source: Mohammed et al. [33] IEEE.

Next, in order to understand the bandwidth of a DD pulse, the FD realization of the DD pulse $x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0)$ is analyzed. Its expression is obtained by applying the inverse frequency-Zak transform $\mathcal{Z}_f^{-1}$, given by⁹ [57],[56]

$$X(f) = \mathcal{F} \left(\mathcal{Z}_t^{-1} \left(x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0) \right) \right) = \mathcal{Z}_f^{-1} \left(x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0) \right). \quad (2.43)$$

For a factorizable DD domain function $w_{\text{tx}}(\tau, \nu) = W_1(\tau)W_2(\nu)$, the FD realization for the DD domain pulse $x_{\text{dd}}^w(\tau, \nu; \tau_0, \nu_0)$ is given by (see Appendix 2.I)

$$X(f) = \left[\sqrt{\nu_p} \sum_{m=-\infty}^{\infty} \tilde{G}(m\nu_p + \nu_0) W_2(f - m\nu_p - \nu_0) \right] w_1(f), \quad (2.44)$$

where $\tilde{G}(f) \triangleq e^{-j2\pi\tau_0 f}$ and

$$w_1(f) \triangleq \mathcal{F}(W_1(t)) = \int W_1(t) e^{-j2\pi f t} dt. \quad (2.45)$$

From (2.44), it is clear that $X(f)$ consists of a train of pulses, where the m th pulse located at $(m\nu_p + \nu_0)$ is a replica of $W_2(f)$ frequency shifted by $(m\nu_p + \nu_0)$. The pulses in the pulse train are separated by ν_p Hz. Each pulse is also multiplied with the value of $\tilde{G}(f)$ at the pulse location. The

⁹ See Appendix 2.F.3 for a discussion on the inverse frequency-Zak transform.

FD waveform $\tilde{G}(f)$ is a complex FD sinusoid/tone of frequency τ_0 , and, therefore, $X(f)$ is appropriately referred to as a *FD pulsone*. Therefore, the co-ordinates τ_0 and ν_0 of the location of the DD pulse $x_{\text{dd}}^{w_{\text{ix}}}(\tau, \nu; \tau_0, \nu_0)$ inside the fundamental DD domain are nothing but the frequencies of the modulating tone in the FD and TD realizations of the DD pulse. The infinite FD pulse train $\left[\sqrt{\nu_p} \sum_{m=-\infty}^{\infty} \tilde{G}(m\nu_p + \nu_0) W_2(f - m\nu_p - \nu_0) \right]$ is modulated by the FD waveform $w_1(f)$. Therefore, the bandwidth of $x_{\text{dd}}^{w_{\text{ix}}}(\tau, \nu; \tau_0, \nu_0)$ is the same as the spread of $w_1(f)$. Since $w_1(f)$ is the Fourier transform of $W_1(t)$, its spread is approximately the inverse of the spread of $W_1(t)$. Since the spread of $W_1(\cdot)$ is the same as the delay domain spread of $w_{\text{ix}}(\tau, \nu) = W_1(\tau) W_2(\nu)$, it follows that the bandwidth of the DD pulse $x_{\text{dd}}^{w_{\text{ix}}}(\tau, \nu; \tau_0, \nu_0)$ is approximately the inverse of the delay domain spread of $w_{\text{ix}}(\tau, \nu)$ (see Fig. 2.17).¹⁰

2.5.1 Example in Fig. 2.17

In Fig. 2.17, we consider the factorizable pulse shaping waveform $w_{\text{ix}}(\tau, \nu) = W_1(\tau) W_2(\nu)$, where $W_1(\tau) = \sqrt{B} \text{sinc}(B\tau)$ and $W_2(\nu) = \sqrt{T} \text{sinc}(T\nu)$. From the TD pulsone expression in (2.41), it follows that

$$w_2(t) = \begin{cases} \frac{1}{\sqrt{T}}, & -\frac{T}{2} \leq t < \frac{T}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (2.46)$$

Using this in (2.41), the TD pulsone expression is given by

$$x(t) = \sqrt{\frac{B\tau_p}{T}} \sum_{n \in \mathbb{Z} \mid -\frac{T}{2} \leq (n\tau_p + \tau_0) < \frac{T}{2}} e^{j2\pi n\nu_0\tau_p} \text{sinc}(B(t - n\tau_p - \tau_0)). \quad (2.47)$$

It is clear that the TD pulsone will consists of pulses $\text{sinc}(B(t - n\tau_p - \tau_0))$ for only those integer n for which the pulse location $(n\tau_p + \tau_0)$ lies within the TD interval $\left[-\frac{T}{2}, \frac{T}{2}\right)$. Since each narrow sinc pulse has a main lobe width of $2/B$ and the interval $\left[-\frac{T}{2}, \frac{T}{2}\right)$ is T seconds wide, the TD pulsone is time limited to roughly $\left(T + \frac{2}{B}\right)$ seconds.

For this example, the FD pulsone expression in (2.44) is given by

$$X(f) = \left[\sqrt{T\nu_p} \sum_{m=-\infty}^{\infty} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \text{sinc}(T(f - m\nu_p - \nu_0)) \right] w_1(f), \quad (2.48)$$

where

$$w_1(f) = \begin{cases} \frac{1}{\sqrt{B}}, & -\frac{B}{2} \leq f < \frac{B}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (2.49)$$

Since $w_1(f)$ is limited to the FD interval $\left[-\frac{B}{2}, \frac{B}{2}\right)$, from (2.48), it is clear that the FD pulsone $X(f)$ is also limited to the same interval, and, therefore, the pulsone bandwidth is B Hz.

¹⁰ Here, the inverse dependence between the time (bandwidth) of a pulsone and the Doppler (delay) spread of $w_{\text{ix}}(\tau, \nu)$ has been shown for a factorizable $w_{\text{ix}}(\tau, \nu)$. However, this fact holds for general non-factorizable pulse shaping waveforms also.

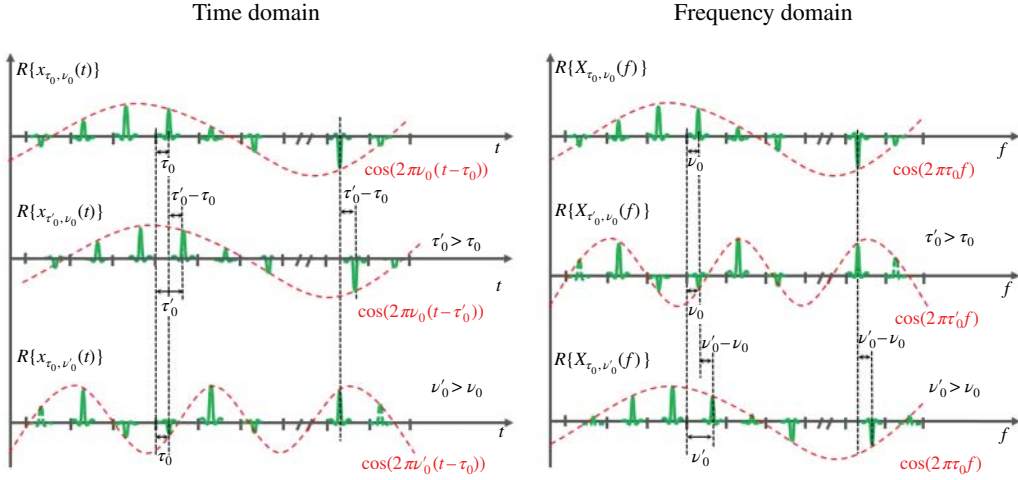


Figure 2.18 Impact of the DD domain pulse location on the TD/FD pulsed characteristics. The figure consists of six plots organized into three rows and two columns (one column for TD and the other for FD). The TD plot in the second row of the figure shows that a shift in the delay domain location of a DD domain pulse from τ_0 to τ'_0 translates to a time displacement of the TD pulsed by $(\tau'_0 - \tau_0)$ seconds. The effect of this shift on the corresponding FD pulsed (shown on the right in the same row) is that the modulating FD tone changes from $e^{-j2\pi\tau_0 f}$ to $e^{-j2\pi\tau'_0 f}$ (only the real part of the TD/FD pulsed is plotted). The FD plot in the third row of the figure shows that a shift in the Doppler domain location of a DD domain pulse from ν_0 to ν'_0 translates to a frequency displacement of the FD pulsed by $(\nu'_0 - \nu_0)$ Hz. The effect of this shift on the corresponding TD pulsed (shown on the left in the same row) is that the modulating TD tone changes from $e^{j2\pi\nu_0(t-\tau_0)}$ to $e^{j2\pi\nu'_0(t-\tau_0)}$. Source: Mohammed et al. [33] IEEE.

Next, through Fig. 2.18, the effect of shift in the location of the DD domain pulse on the structure of the corresponding TD and FD pulsed is studied. In a nutshell, a shift of the DD domain pulse along the delay domain translates to a time displacement of the TD pulsed, and a shift of the DD domain pulse along the Doppler domain translates to a frequency displacement of the FD pulsed. This is also clear from the TD and FD pulsed expressions in (2.41) and (2.44). For the example, with the *sinc* pulse shaping waveform, from (2.47), it is clear that shifting the DD domain pulse location along the delay domain from (τ_0, ν_0) to (τ'_0, ν_0) will shift the TD pulsed by $(\tau'_0 - \tau_0)$ since the expression for the n th sinc pulse in the TD pulsed changes from $\text{sinc}(B(t - n\tau_p - \tau_0))$ to $\text{sinc}(B(t - n\tau_p - \tau'_0))$. Similarly, from the FD pulsed expression in (2.48), it is clear that shifting the DD domain pulse location along the Doppler domain from (τ_0, ν_0) to (τ_0, ν'_0) will shift the FD pulsed by $(\nu'_0 - \nu_0)$ since the expression for the m th sinc pulse in the FD pulsed changes from $\text{sinc}(T(f - m\nu_p - \nu_0))$ to $\text{sinc}(T(f - m\nu_p - \nu'_0))$.

Figures 2.19 and 2.20 depict the effect of increasing the width of the DD domain pulse on the structure of the corresponding TD and FD pulsed. In a nutshell, Fig. 2.19 shows that an increase in the width of the DD domain pulse along the delay domain translates to a reduction of the FD pulsed bandwidth, and Fig. 2.20 shows that an increase in the width of the DD domain pulse along the Doppler domain translates to a reduction of the TD pulsed time duration.

Next, global properties of pulsed are derived from the local properties described above.

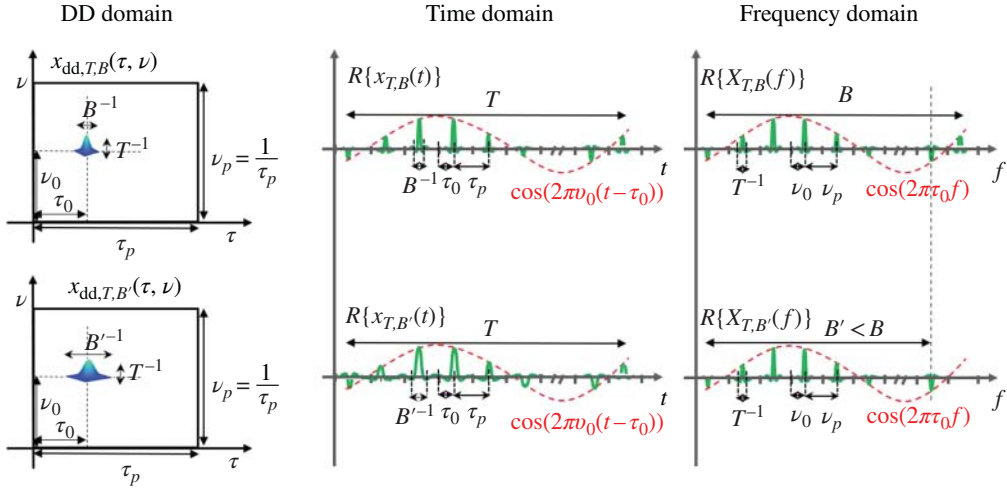


Figure 2.19 Impact of the DD domain pulse width along delay domain on the TD/FD pulsone characteristics. Increasing the width of the DD domain pulse along the delay domain from $\frac{1}{B}$ to $\frac{1}{B'}$ translates to a reduction of the FD pulsone bandwidth from B to B' and an increase in the width of each TD pulse in the corresponding TD pulsone from $\frac{1}{B}$ to $\frac{1}{B'}$. Source: Mohammed et al. [33] IEEE.

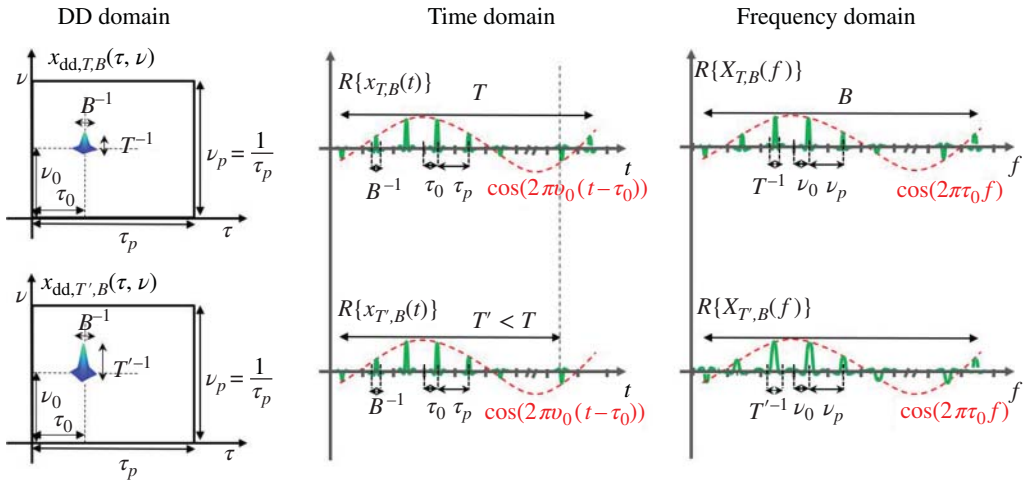


Figure 2.20 Impact of the DD domain pulse width along the Doppler domain on the TD/FD pulsone characteristics. Increasing the width of the DD domain pulse along the Doppler domain from $\frac{1}{T}$ to $\frac{1}{T'}$ translates to a reduction of the TD pulsone time duration from T to T' and an increase in the width of each FD pulse in the corresponding FD pulsone from $\frac{1}{T}$ to $\frac{1}{T'}$. Source: Mohammed et al. [33] IEEE.

2.5.2 Orthogonality of Pulsones

From Fig. 2.17, it is observed that a DD domain pulse is localized within a rectangular DD domain region which is $1/B$ seconds wide along the delay domain and $1/T$ Hz wide along the Doppler domain. Geometrically, this implies almost no overlap between two DD pulses whose delay domain

locations differ by $1/B$ or whose Doppler domain locations differ by $1/T$, i.e., such pulses are almost orthogonal.

The same observation can be made by separately observing the TD and FD realizations of a DD pulse. From Fig. 2.18, it is clear that shifting the location of the DD domain pulse by $1/B$ along the delay domain induces a time displacement of the TD pulsones by $1/B$ seconds. From Fig. 2.17, it is also known that each pulse in the TD pulsones has time duration $1/B$ and that consecutive pulses are separated by τ_p . When a DD domain pulse is shifted by $1/B$ along the delay domain, the corresponding TD pulsones do not overlap. Similarly, when a DD domain pulse is shifted by $1/T$ Hz along the Doppler domain, the corresponding FD pulsones do not overlap. Delay shifts by integer multiples of $1/B$ and Doppler shifts by integer multiples of $1/T$ lead to TD/FD pulsones that are almost orthogonal. Here, the impact of delay and Doppler shifts on the orthogonality between pulsones is understood *separately* through their TD and FD realizations, respectively. The geometric interpretation of orthogonality in the previous paragraph is however much *simpler* since the pulsones are viewed as a pulse in the DD domain.

2.5.3 Optimality as Time- and Bandwidth-limited Signals

In [58, 59], Slepian, Landau, and Pollack introduced the family of prolate spheroidal waveforms to measure the space of essentially time- and bandwidth-limited signals. For signals limited approximately to a time duration of T seconds and bandwidth B Hz, they showed that the number of orthogonal carrier waveforms is approximately equal to the time–bandwidth product BT . Pulsones with time duration T and bandwidth B also span this same space. Recall that pulsones are almost orthogonal if the location of the corresponding DD domain pulses differs by integer multiples of $1/B$ along the delay domain and by integer multiples of $1/T$ along the Doppler domain. The number of approximately orthogonal carrier waveforms is essentially $\frac{\tau_p v_p}{\frac{1}{B} \frac{1}{T}} = BT$, which is the time–bandwidth product.

2.5.4 TDM as a Limiting Case

In the TD pulsones shown in Fig. 2.17, as $\tau_p \rightarrow \infty$, the TD pulses located at $t = \tau_0 + n\tau_p$, $n \in \mathbb{Z}$, $n \neq 0$, move toward $\pm\infty$ and only the TD pulse at $t = \tau_0$ remains, i.e., the TD pulsones approach a TD pulse, which is the TDM carrier. In other words, as the Doppler domain collapses, the TD pulsones tend to the TDM carrier.

2.5.5 FDM as a Limiting Case

In the FD pulsones shown in Fig. 2.17, as $v_p \rightarrow \infty$ (i.e., $\tau_p = (1/v_p) \rightarrow 0$), the pulses located at $f = v_0 + m v_p$, $m \in \mathbb{Z}$, $m \neq 0$, move toward $\pm\infty$ and only the FD pulse at $f = v_0$ remains, i.e., the FD pulsones approach a FD pulse, which is the FDM carrier. In other words, as the delay domain collapses, the FD pulsones tend to the FDM carrier.

DD domain pulses therefore give rise to a family of carrier waveforms (pulsones) parameterized by τ_p that interpolates between TDM and FDM (see the hyperbola $\tau_p v_p = 1$ in the top right corner of Fig. 2.17).

2.5.6 TD Pulsones Encode Wireless Channel Dynamics

TD pulsones are engineered to mirror the dynamics of the wireless channel. The effect of channel path delay on a DD domain pulse is to simply shift the pulse along the delay domain by an amount

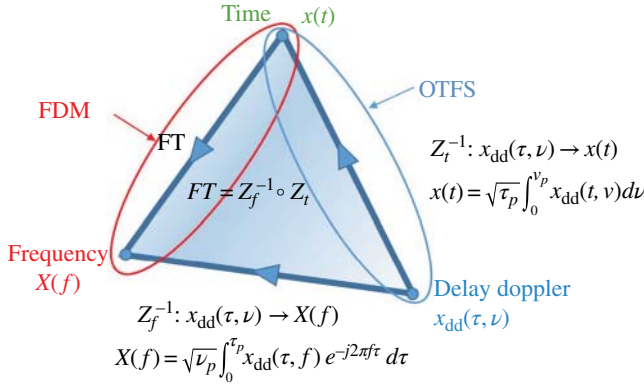


Figure 2.21 Three different basic signal representations in the TD, FD, and DD domain. Signal representations in these domains are related through transforms. The well-known Fourier transform is in fact a composition of the time-Zak transform Z_t and the inverse frequency-Zak transform Z_f^{-1} . Source: Mohammed et al. [33] IEEE.

equal to the path delay. The effect of a channel path Doppler shift on a DD domain pulse is to simply shift the pulse along the Doppler domain by an amount equal to the path Doppler shift.

2.5.7 The Fourier Transform as a Composition

Figure 2.17 illustrates that a TD signal can be mapped to its FD realization by first applying the Zak transform Z_t from the TD to the DD domain and then applying the inverse frequency-Zak transform Z_f^{-1} from the DD domain to the FD. In other words, the Fourier transform is the composition of Z_t and Z_f^{-1} . Figure 2.21 represents the three signal representations (TD, FD, and DD domains) as the three vertices of a triangle and labels the edges between vertices by the transforms between signal representations.

2.6 Zak-OTFS Modulation and I/O Relation

TD pulsones are time- and bandwidth-limited TD carrier waveforms for DD domain modulation. From Section 2.5, it is known that the DD pulse corresponding to a TD pulsone (time and bandwidth limited to T and B , respectively) has a delay and Doppler domain spread of approximately $1/B$ and $1/T$, respectively. Therefore, DD pulses located at DD points separated by at least $1/T$ along the Doppler domain and at least $1/B$ along the delay domain overlap/interfere minimally, and, hence, distinct information symbols can be carried by DD pulses which are separated by exactly $1/T$ along the Doppler domain and by $1/B$ along the delay domain. Accordingly, one possible choice for the DD location of these pulses is given by the set of DD points,

$$\Lambda_{\text{dd}} \triangleq \left\{ \left(\frac{k}{B}, \frac{l}{T} \right) \mid k, l \in \mathbb{Z} \right\}, \quad (2.50)$$

which is referred to as the *information grid* or the *information lattice*. An important sub-lattice of the information lattice is the period lattice, which defines the delay and Doppler period of quasi-periodic DD functions, and is given by

$$\Lambda_p \triangleq \{ (n\tau_p, m\nu_p) \mid n, m \in \mathbb{Z} \}. \quad (2.51)$$

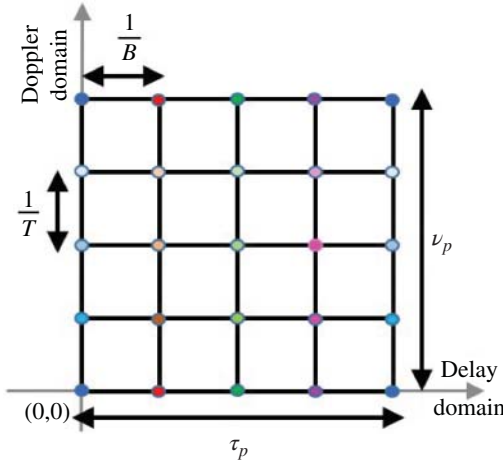


Figure 2.22 Zak-OTFS modulation. Information symbols modulate DD pulses located on the information grid.

Since the DD pulses are quasi-periodic with delay and Doppler period τ_p and ν_p , respectively, the number of distinct pulse locations along the delay domain is $M \triangleq \tau_p/(1/B) = B\tau_p$. Similarly, the number of distinct pulse locations along the Doppler domain is $N \triangleq \nu_p/(1/T) = T\nu_p$. One canonical choice of such delay and Doppler locations, which lie within the fundamental DD period, is given by $\left\{\frac{k}{B} \mid k = 0, 1, \dots, M-1\right\}$ and $\left\{\frac{l}{T} \mid l = 0, 1, \dots, N-1\right\}$, respectively. Therefore, the number of distinct DD domain information symbols which can be communicated is MN . The (k, l) th information symbol $x[k, l]$ is carried by the DD pulse located at $\left(\frac{k}{B}, \frac{l}{T}\right)$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$. An illustration is shown in Fig. 2.22, where the DD pulse locations are highlighted as dots on the information grid.

This modulation method where information symbols are carried by DD domain pulses is referred to as *Zak-OTFS* modulation. Zak-OTFS modulation is similar to TDM and FDM, in the sense that all three follow the generalized transceiver signal processing template, as shown in Fig. 2.23.

2.6.1 Generalized Transceiver Signal Processing

In Fig. 2.23, a packet of BT information symbols $x[k]$, $k = 0, 1, \dots, (BT-1)$ is to be communicated using a TD signal time and bandwidth limited to T and B , respectively. At the transmitter, the information symbols are embedded into a discrete information signal $x_{\mathcal{M}}[\cdot]$ in the *modulation domain* $\mathcal{M}$. For TDM, FDM, and Zak-OTFS modulation, the modulation domains are TD, FD, and DD, respectively (i.e., $\mathcal{M}$ is td , fd , and dd , respectively). As shown in Fig. 2.23, $x_{\mathcal{M}}[\cdot]$ is then lifted to a continuous modulation domain signal $x_{\mathcal{M}}(\cdot)$ by modulating Dirac-delta functions on the information grid $\Lambda_{\mathcal{M}}$. For the transmit TD signal to satisfy the time and bandwidth constraints, $x_{\mathcal{M}}(\cdot)$ is then filtered with a pulse shaping waveform $w_{\text{tx}}(\cdot)$ in the modulation domain, resulting in the filtered signal $x_{\mathcal{M}}^{w_{\text{tx}}}(\cdot)$. The TD realization of the filtered signal $x_{\mathcal{M}}^{w_{\text{tx}}}(\cdot)$ is the TD transmit signal $s_{\text{td}}(t)$, which is computed using a modulation domain to TD transform.

The signal $s_{\text{td}}(t)$ interacts with the doubly spread channel to provide a received TD signal $r_{\text{td}}(t)$, given by [55]

$$r_{\text{td}}(t) = \iint h_{\text{phy}}(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n_{\text{td}}(t), \quad (2.52)$$

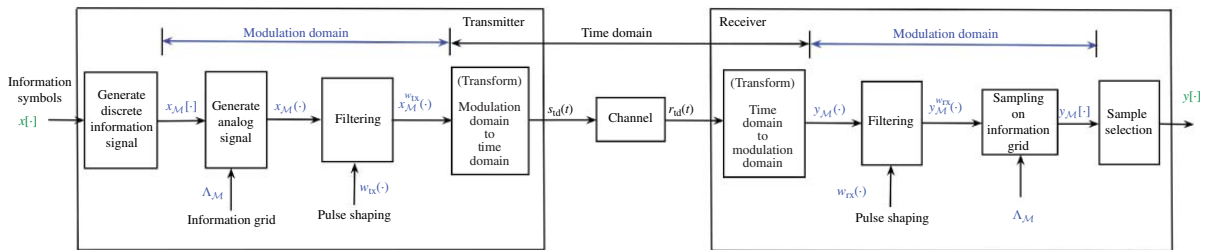


Figure 2.23 Transceiver signal processing. Source: Mohammed et al. [33] IEEE.

Table 2.3 Signal processing steps in TDM and FDM.

Transceiver operation	TDM	FDM
Generating the discrete information signal	$x_{\text{td}}[k] \triangleq \begin{cases} x[k], & k = 0, 1, \dots, (BT - 1) \\ 0, & \text{otherwise} \end{cases}$	$x_{\text{fd}}[k] \triangleq \begin{cases} x[k], & k = 0, 1, \dots, (BT - 1) \\ 0, & \text{otherwise} \end{cases}$
Generating the analog signal on the information grid	$\Lambda_{\text{td}} = \{q/B \mid q \in \mathbb{Z}\}$ $x_{\text{td}}(t) = \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] \delta(t - k/B)$	$\Lambda_{\text{fd}} = \{q/T \mid q \in \mathbb{Z}\}$ $x_{\text{fd}}(f) = \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \delta(f - k/T)$
Shaping the pulse at the transmitter	$x_{\text{td}}^{w_{\text{tx}}}(t) = w_{\text{tx}}(t) \star x_{\text{td}}(t)$	$x_{\text{fd}}^{w_{\text{tx}}}(f) = w_{\text{tx}}(f) \star x_{\text{fd}}(f)$
Converting from the modulation domain to the time domain	$s_{\text{td}}(t) = x_{\text{td}}^{w_{\text{tx}}}(t)$	$s_{\text{td}}(t) = \int x_{\text{fd}}^{w_{\text{tx}}}(f) e^{j2\pi ft} df$
Applying the channel $h(\tau, \nu)$	$r_{\text{td}}(t)$ is given by (2.52)	$r_{\text{td}}(t)$ is given by (2.52)
Converting from the time domain to modulation domain	$y_{\text{td}}(t) = r_{\text{td}}(t)$	$y_{\text{fd}}(f) = \int r_{\text{td}}(t) e^{-j2\pi ft} dt$
Shaping the pulse at the receiver	$y_{\text{td}}^{w_{\text{rx}}}(t) = w_{\text{rx}}(t) \star y_{\text{td}}(t)$	$y_{\text{fd}}^{w_{\text{rx}}}(f) = w_{\text{rx}}(f) \star y_{\text{fd}}(f)$
Sampling on the information grid	$y_{\text{td}}[k] = y_{\text{td}}^{w_{\text{rx}}}(t = k/B), k \in \mathbb{Z}.$	$y_{\text{fd}}[k] = y_{\text{fd}}^{w_{\text{rx}}}(f = k/T), k \in \mathbb{Z}.$

Source: Mohammed et al. [33] IEEE.

where $h_{\text{phy}}(\tau, \nu)$ is the DD spreading function of the channel and $n_{\text{td}}(t)$ is additive white Gaussian noise (AWGN). This signal is converted to the appropriate modulation domain, using the inverse of the transform used at the transmitter, to provide a signal $y_{\mathcal{M}}(\cdot)$. Then, matched filtering at the receiver with the receive filter $w_{\text{rx}}(\cdot)$ produces a modulation domain analog signal $y_{\mathcal{M}}^{w_{\text{rx}}}(\cdot)$. This signal is then sampled on the information grid to provide the discrete received modulation domain signal $y_{\mathcal{M}}[\cdot]$, which is processed to recover the information symbols. The modulation domain signals $x_{\mathcal{M}}[\cdot]$ and $y_{\mathcal{M}}[\cdot]$ are said to be the “input” and “output,” respectively. The output signal can be expressed in terms of the input signal, and this expression is referred to as the input–output relation.

The transceiver signal processing steps in TDM and FDM are listed in Table 2.3. The input–output relation for TDM and FDM in a doubly spread channel is given in Appendices 2.J and 2.K, respectively.

2.6.2 Zak-OTFS Modulation

As discussed earlier, in Zak-OTFS modulation, the $BT = MN$ information symbols are carried by quasi-periodic DD pulses at locations of the information lattice, which are inside the fundamental period D_0 . Let $x[k, l]$, $k = 0, 1, \dots, M - 1$, $l = 0, 1, \dots, N - 1$ denote the information symbol carried by the (k, l) th DD pulse located at $(\tau = k\tau_p/M, \nu = l\nu_p/N)$. The TD carrier waveform that carries $x[k, l]$ is, therefore, the TD realization of the (k, l) th DD pulse, i.e., the (k, l) th TD pulsone is given

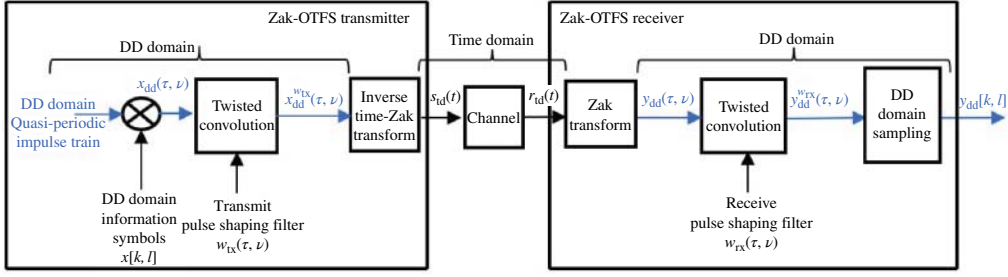


Figure 2.24 Zak-OTFS modulation. Source: Mohammed et al. [34] IEEE.

by (see (2.41) for the expression of the TD pulsone corresponding to a quasi-periodic DD pulse at (τ_0, ν_0))

$$x_{\mathcal{Z}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{tx}}(t) \triangleq \mathcal{Z}_t^{-1} \left(x_{dd}^{w_{tx}}(\tau, \nu; k\tau_p/M, l\nu_p/N) \right). \quad (2.53)$$

Therefore, the Zak-OTFS modulated signal that carries all MN information symbols is given by

$$\begin{aligned} s_{td}(t) &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{\mathcal{Z}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{tx}}(t) \\ &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \mathcal{Z}_t^{-1} \left(x_{dd}^{w_{tx}}(\tau, \nu; k\tau_p/M, l\nu_p/N) \right). \end{aligned} \quad (2.54)$$

In Appendix 2.M, it is shown that Zak-OTFS modulation can be implemented using the sequence of signal processing operations illustrated in the transmitter module in Fig. 2.24 and listed in Table 2.4. This sequence of operations is described in the following.

Table 2.4 Signal processing steps in Zak-OTFS modulation.

Transceiver operation	Zak-OTFS
Generating the discrete information signal	$x_{dd}[k + nM, l + mN] \triangleq x[k, l] e^{j2\pi n \frac{l}{N}}.$
Generating the analog signal on the information grid	$\Lambda_{dd} = \left\{ \left(k \frac{\tau_p}{M}, l \frac{\nu_p}{N} \right) \mid k, l \in \mathbb{Z} \right\}$ $x_{dd}(\tau, \nu) = \sum_{k, l \in \mathbb{Z}} x_{dd}[k, l] \delta \left(\tau - k \frac{\tau_p}{M} \right) \delta \left(\nu - l \frac{\nu_p}{N} \right)$
Shaping the pulse at the transmitter	$x_{dd}^{w_{tx}}(\tau, \nu) = w_{tx}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu)$
Converting from the modulation domain to the time domain	$s_{td}(t) = \mathcal{Z}_t^{-1} \left(x_{dd}^{w_{tx}}(\tau, \nu) \right)$
Applying the channel $h(\tau, \nu)$	$r_{td}(t) = \iint h_{\text{phy}}(\tau, \nu) s_{td}(t - \tau) e^{j2\pi \nu(t - \tau)} d\tau d\nu$
Converting from the time domain to modulation domain	$y_{dd}(\tau, \nu) = \mathcal{Z}_t \left(r_{td}(t) \right)$
Shaping the pulse at the receiver	$y_{dd}^{w_{rx}}(\tau, \nu) = w_{rx}(\tau, \nu) *_{\sigma} y_{dd}(\tau, \nu)$
Sampling on the information grid	$y_{dd}[k', l'] = y_{dd}^{w_{rx}} \left(\tau = k' \frac{\tau_p}{M}, \nu = l' \frac{\nu_p}{N} \right), \quad k', l' \in \mathbb{Z}$

Source: Mohammed et al. [33] IEEE.

The information symbols $x[k, l]$ are firstly embedded in a discrete DD domain information signal $x_{\text{dd}}[k, l]$, $k, l \in \mathbb{Z}$, defined as follows (see Appendix 2.L for a discussion on discrete DD domain information signals). For all $k = 0, 1, \dots, M-1$ and all $l = 0, 1, \dots, N-1$,

$$x_{\text{dd}}[k + nM, l + mN] \triangleq x[k, l] e^{j2\pi n \frac{l}{N}}, \quad m, n \in \mathbb{Z}. \quad (2.55)$$

Clearly, it follows that for all $k, l, n, m \in \mathbb{Z}$,

$$x_{\text{dd}}[k + nM, l + mN] = x_{\text{dd}}[k, l] e^{j2\pi n \frac{l}{N}}. \quad (2.56)$$

The discrete DD domain information signal $x_{\text{dd}}[k, l]$ is then lifted to the information grid Λ_{dd} (see (2.50)), resulting in the continuous DD domain analog information signal $x_{\text{dd}}(\tau, \nu)$, which is given by

$$x_{\text{dd}}(\tau, \nu) \triangleq \sum_{k, l \in \mathbb{Z}} x_{\text{dd}}[k, l] \delta\left(\tau - k \frac{\tau_p}{M}\right) \delta\left(\nu - l \frac{\nu_p}{N}\right). \quad (2.57)$$

Since $x_{\text{dd}}[\cdot, \cdot]$ satisfies (2.56), for any arbitrary integers $m, n \in \mathbb{Z}$, it follows that $x_{\text{dd}}(\tau, \nu)$ is quasi-periodic (see Theorem 2.8 in Appendix 2.L), i.e., for all $n, m \in \mathbb{Z}$, it satisfies (2.25), i.e.,

$$x_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) = e^{j2\pi n \nu \tau_p} x_{\text{dd}}(\tau, \nu). \quad (2.58)$$

Since the lifted analog DD domain signal is quasi-periodic, any discrete DD domain signal satisfying (2.56) is said to be quasi-periodic in the discrete DD domain with period M along the delay domain and period N along the Doppler domain.

In order to satisfy the time and bandwidth constraints for the transmit TD signal, from Section 2.5 (see (2.39)), we know that $x_{\text{dd}}(\tau, \nu)$ must be filtered (twisted convolved) in the DD domain with a DD domain filter $w_{\text{tx}}(\tau, \nu)$, resulting in

$$x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu) \triangleq w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu). \quad (2.59)$$

Since $x_{\text{dd}}(\tau, \nu)$ is quasi-periodic and twisted convolution preserves quasi-periodicity (see Appendix 2.G), it follows that $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu)$ is also quasi-periodic. The inverse Zak transform (see (2.F.1) in Appendix 2.F) of this filtered signal gives the transmit TD signal $s_{\text{id}}(t)$, i.e.,

$$s_{\text{id}}(t) \triangleq \mathcal{Z}_t^{-1} \left(x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu) \right). \quad (2.60)$$

Comparing Figs. 2.23 and 2.24, it is clear that Zak-OTFS modulation is similar to TDM/FDM, in the sense that it follows the same generalized modulator template. The differences are that, in Zak-OTFS, the modulation domain is the 2D DD domain, the transform between the modulation domain and TD is Zak transform, and filtering in the modulation domain takes place through twisted convolution (see Table 2.2).

2.6.3 Zak-OTFS Receiver

Zak-OTFS receiver signal processing is similar to that in TDM/FDM and follows the same generalized template, as illustrated through the receiver shown in Fig. 2.23. Receiver signal processing for Zak-OTFS modulation is shown in Fig. 2.24 and listed in Table 2.4. A detailed derivation of the Zak-OTFS receiver from first principles is given in Appendix 2.N. In the following, the Zak-OTFS receiver signal processing steps are described.

The received TD signal $r_{\text{id}}(t)$ is given by (2.52). At the receiver, Zak transform (2.23) converts the received TD signal to a DD domain signal $y_{\text{dd}}(\tau, \nu)$, i.e.,

$$y_{\text{dd}}(\tau, \nu) \triangleq \mathcal{Z}_t \left(r_{\text{id}}(t) \right). \quad (2.61)$$

The received DD domain signal is then match filtered (i.e., twisted convolved) with the receive pulse shaping filter $w_{\text{rx}}(\tau, \nu)$, resulting in the received filtered DD domain signal¹¹ $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$, i.e.,

$$y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu). \quad (2.62)$$

This filtered received DD domain signal is then sampled on the DD domain information grid Λ_{dd} , resulting in the discrete DD domain output signal $y_{\text{dd}}[k, l]$, $k, l \in \mathbb{Z}$,

$$y_{\text{dd}}[k, l] = y_{\text{dd}}^{w_{\text{rx}}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right). \quad (2.63)$$

2.6.4 Zak-OTFS I/O Relation

The following theorem gives the relation between the Zak-OTFS output signal $y_{\text{dd}}[k, l]$ and the input information signal $x_{\text{dd}}[k, l]$.

Theorem 2.1 The Zak-OTFS output signal $y_{\text{dd}}[k, l]$ is given by

$$y_{\text{dd}}[k, l] = h_{\text{eff}}[k, l] *_{\sigma} x_{\text{dd}}[k, l] + n_{\text{dd}}[k, l], \quad (2.64)$$

where $h_{\text{eff}}[k, l]$ and $n_{\text{dd}}[k, l]$ are the effective discrete DD domain channel filter and the discrete DD domain noise, respectively, which are given by

$$\begin{aligned} h_{\text{eff}}[k, l] &= h_{\text{eff}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right), \\ h_{\text{eff}}(\tau, \nu) &= w_{\text{rx}}(\tau, \nu) *_{\sigma} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu), \\ n_{\text{dd}}[k, l] &= n_{\text{dd}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right), \\ n_{\text{dd}}(\tau, \nu) &= w_{\text{rx}}(\tau, \nu) *_{\sigma} \mathcal{Z}_t(n_{\text{td}}(t)). \end{aligned}$$

In (2.64), the twisted convolution between any two discrete DD functions $h_{\text{eff}}[k, l]$ and $x_{\text{dd}}[k, l]$ is given by

$$\begin{aligned} h_{\text{eff}}[k, l] *_{\sigma} x_{\text{dd}}[k, l] &= \sum_{k', l' \in \mathbb{Z}} h_{\text{eff}}[k', l'] x_{\text{dd}}[k - k', l - l'] e^{j2\pi \frac{l'(k-k')}{MN}} \\ &= \sum_{k', l' \in \mathbb{Z}} h_{\text{eff}}[k - k', l - l'] x_{\text{dd}}[k', l'] e^{j2\pi \left(\frac{l-l'}{N} \right) \frac{k'}{M}}. \end{aligned} \quad (2.65)$$

Proof: See Appendix 2.O.

11 In TDM (delay-only channel), the channel acts on an input TD signal through linear convolution. At the receiver, it suffices to filter (i.e., linearly convolve) the received TD signal with a filter having impulse response $w_{\text{rx}}(t)$ that is *matched* to the combined response $h_w(t) = h(t) \star w_{\text{tx}}(t)$ of the channel (impulse response $h(t)$) and the transmit pulse $w_{\text{tx}}(t)$, i.e., $w_{\text{rx}}(t) = \overline{h_w}(t) \triangleq h_w^*(-t)$, followed by TD sampling on the information grid, i.e., at integer multiples of $1/B$. Furthermore, it holds that $\overline{h_w}(t) = \overline{w_{\text{tx}}}(t) \star \overline{h}(t)$. Along the same lines, in a Zak-OTFS receiver, it suffices to filter (i.e., twisted convolve) the received DD domain signal with a DD domain pulse $w_{\text{rx}}(\tau, \nu)$ that is *matched* to the combined DD domain response $h_w(\tau, \nu) = h(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu)$ of the channel and the transmit pulse, i.e., $w_{\text{rx}}(\tau, \nu) = \overline{h_w}(\tau, \nu) \triangleq h_w^*(-\tau, -\nu) e^{j2\pi \tau \nu}$, followed by DD domain sampling on the DD domain information grid Λ_{dd} (see (2.N.5) in Appendix 2.N). Furthermore, it holds that $\overline{h_w}(\tau, \nu) = \overline{w_{\text{tx}}}(\tau, \nu) *_{\sigma} \overline{h}(\tau, \nu)$.

Note that the RHS of (2.65) is a weighted double sum of terms, where the (k', l') th term is given by (see the second line in (2.65))

$$h_{\text{eff}}[k, l] *_{\sigma} [\delta[k - k'] \delta[l - l']] , \quad k, l \in \mathbb{Z}, \quad (2.66)$$

multiplied by the weight $x_{\text{dd}}[k', l']$.

The Zak-OTFS I/O relation is valid, whether the crystallization condition is satisfied or not. This is because the channel action (i.e., twisted convolution in the DD domain) is independent of whether the delay and Doppler period parameters of Zak-OTFS modulation are greater than the respective channel spreads.

The Zak-OTFS I/O relation (2.64) for doubly spread channels is a generalization of the TDM I/O relation for linear time-invariant (LTI) channels. In an LTI (i.e., delay-only) channel, the match-filtered received continuous TD signal is given by $y(t) = w_{\text{rx}}(t) \star h(t) \star w_{\text{tx}}(t) \star \left(\sum_k x[k] \delta(t - k/B) \right)$, where $h(t)$ is the channel impulse response, $w_{\text{tx}}(t)$ and $w_{\text{rx}}(t)$ are the transmit and receive filters, $x[k]$ is the discrete information signal, and $\star$ denotes linear convolution. Sampling $y(t)$ on the TDM information grid (time instances which are integer multiples of $1/B$) gives the discrete output $y[k] = h[k] \star x[k]$, where $h[k]$ is the discrete effective channel filter, obtained by sampling $w_{\text{rx}}(t) \star h(t) \star w_{\text{tx}}(t)$ on the TDM information grid. Similarly, for Zak-OTFS modulation on a doubly spread channel, $h_{\text{eff}}[k, l]$ in (2.64) is the discrete DD domain effective channel filter, and $y_{\text{dd}}[k, l]$ and $x_{\text{dd}}[k, l]$ are the discrete DD domain channel output and input, respectively. This structural similarity between TDM and Zak-OTFS is captured in Fig. 2.23 and in Tables 2.3 and 2.4.

An important difference between an LTI channel and a linear time varying (LTV) channel is the following. An LTI channel has TD sinusoids as eigen-modes/eigen-functions/eigen-waveforms, i.e., a single frequency sinusoidal input to an LTI system results in a sinusoidal output of the same frequency. Eigen-waveforms for LTV channels do not exist. However, LTV channels have geometric modes/geometric functions. These geometric modes are Zak-OTFS TD pulsones, which are quasi-periodic pulses in the DD domain, and the action of a channel-induced delay and Doppler shift is to *geometrically* translate the DD pulse by the same shifts along the respective domains.

2.7 Predictability of the Zak-OTFS I/O Relation in the Crystalline Regime

In this section, it is shown that the Zak-OTFS I/O relation is predictable in the crystalline regime, i.e., when (τ_p, ν_p) with $\tau_p \nu_p = 1$ satisfy the crystallization condition

$$\tau_p > \tau_{\text{eff}}, \quad \nu_p > \nu_{\text{eff}}, \quad (2.67)$$

where τ_{eff} and ν_{eff} denote the spread of the support of $h_{\text{eff}}(\tau, \nu)$ along the delay axis and Doppler axis, respectively. Specifically, it is shown that when the crystallization condition holds, the channel response to a *light gray* pilot located at $(k^{(g)}, l^{(g)})$ can be accurately estimated from the channel response to a *dark gray* pilot located at $(k^{(b)}, l^{(b)})$ in Fig. 2.25. A special case is a delay-only channel (i.e., LTI channel), where we know that the response to a particular impulse input can be used to predict/estimate the channel response to any other impulse input. In Fig. 2.25, the dark gray pilot is a discrete DD domain impulse signal (see (2.L.9) in Appendix 2.L.1), given by

$$x^{(b)}[k, l] = \sum_{m, n \in \mathbb{Z}} e^{j2\pi \frac{nl^{(b)}}{N}} \delta[k - (k^{(b)} + nM)] \delta[l - (l^{(b)} + mN)]. \quad (2.68)$$

Similarly, the light gray pilot is a discrete DD domain impulse signal, given by

$$x^{(g)}[k, l] = \sum_{m, n \in \mathbb{Z}} e^{j2\pi \frac{nl^{(g)}}{N}} \delta[k - (k^{(g)} + nM)] \delta[l - (l^{(g)} + mN)]. \quad (2.69)$$

From Theorem 2.1, the discrete DD domain channel response to the dark gray pilot is given by¹²

$$\begin{aligned} y^{(b)}[k, l] &= h_{\text{eff}}[k, l] *_{\sigma} x^{(b)}[k, l] \\ &= \sum_{n, m \in \mathbb{Z}} \left[h_{\text{eff}}[k - (k^{(b)} + nM), l - (l^{(b)} + mN)] \right. \\ &\quad \left. e^{j2\pi \frac{nl^{(b)}}{N}} e^{j2\pi \frac{(l - l^{(b)} - mN)(k^{(b)} + nM)}{M}} \right]. \end{aligned} \quad (2.70)$$

From (2.70), it is observed that the total response to the dark gray pilot is a sum of local responses to its constituent impulses, where the response to the (n, m) th constituent impulse (i.e., the (n, m) th term $e^{j2\pi \frac{nl^{(b)}}{N}} \delta[k - (k^{(b)} + nM)] \delta[l - (l^{(b)} + mN)]$ in (2.68)) is given by the (n, m) th term in (2.70), i.e.,

$$\begin{aligned} y_{n,m}^{(b)}[k, l] &= h_{\text{eff}}[k - (k^{(b)} + nM), l - (l^{(b)} + mN)] \\ &\quad e^{j2\pi \frac{nl^{(b)}}{N}} e^{j2\pi \frac{(l - l^{(b)} - mN)(k^{(b)} + nM)}{M}}. \end{aligned} \quad (2.71)$$

Observe that the (n, m) th response coincides up to multiplicative phases with the effective discrete DD domain channel filter $h_{\text{eff}}[k, l]$ shifted by $(k^{(b)} + nM, l^{(b)} + mN)$. Similarly, the response to the light gray pilot is a sum of local responses, where the (n, m) th response is given by

$$\begin{aligned} y_{n,m}^{(g)}[k, l] &= \left[h_{\text{eff}}[k - (k^{(g)} + nM), l - (l^{(g)} + mN)] \right. \\ &\quad \left. e^{j2\pi \frac{nl^{(g)}}{N}} e^{j2\pi \frac{(l - l^{(g)} - mN)(k^{(g)} + nM)}{M}} \right]. \end{aligned} \quad (2.72)$$

In Fig. 2.25, the local response components of the dark gray (light gray) pilot are depicted as dark gray (light gray) ellipses.¹³ Observe that the (n, m) th light gray response can always be calculated from the (n, m) th dark gray response according to the following rule:

$$\begin{aligned} y_{n,m}^{(g)}[k, l] &= y_{n,m}^{(b)}[k - (k^{(g)} - k^{(b)}), l - (l^{(g)} - l^{(b)})] \\ &\quad e^{j2\pi n \frac{(l^{(g)} - l^{(b)})}{N}} e^{j2\pi \frac{(l - l^{(g)} - mN)(k^{(g)} - k^{(b)})}{M}}, \end{aligned} \quad (2.73)$$

which follows from (2.71) and (2.72).¹⁴ The individual local responses $y_{n,m}^{(b)}[k, l]$, $n, m \in \mathbb{Z}$, are superimposed at the receiver, and, in general, cannot be separated from one another in the received signal $y^{(b)}[k, l]$. However, when the dark gray (light gray) ellipses do not overlap with each other, the local responses do not superimpose/interact. As a consequence, in this situation, the local responses in the response to the dark gray pilot can be separated from $y^{(b)}[k, l]$ and the corresponding local response components in the channel response to the light gray pilot can be *predicted* separately

¹² Since the objective in this section is to study the predictability attribute of the Zak-OTFS I/O relation, an AWGN-free received signal is considered here.

¹³ Ellipses are used only as visual metaphor and the discussion here is valid for any arbitrary shape of the support set of $h_{\text{eff}}[k, l]$.

¹⁴ To obtain the (n, m) th response component to the light gray pilot from that of the dark gray pilot, supports are first aligned by translating the dark gray ellipses by $(k^{(g)} - k^{(b)})$ along delay and by $(l^{(g)} - l^{(b)})$ along Doppler. This is followed by multiplication with a deterministic factor that is independent of the channel.

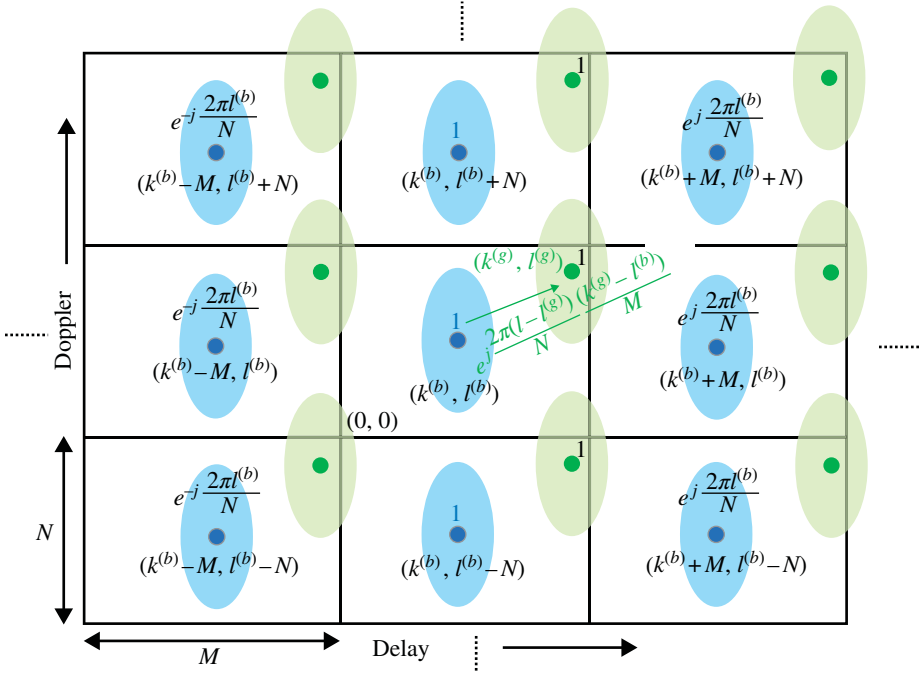


Figure 2.25 Prediction in the crystalline regime. In the crystalline regime, the channel response to the light gray DD pilot signal (impulse locations depicted as light gray dots) can be predicted from the channel response to the dark gray DD pilot signal (impulse locations depicted as dark gray dots). Source: Mohammed et al. [34] IEEE.

using (2.73), adding all of which gives the total light gray response $y^{(g)}[k, l]$, i.e., the Zak-OTFS I/O relation is predictable.

Therefore, with time- and bandwidth-limited waveforms, the crystallization condition is said to be satisfied if and only if the local response components do not overlap/interact. The support set of the (n, m) th local response $h_{\text{eff}}[k - (k^{(b)} + nM), l - (l^{(b)} + mN)]$ is simply the support set of the channel filter $h_{\text{eff}}[k, l]$ shifted by $(k^{(b)} + nM)$ along the discrete delay domain and by $(l^{(b)} + mN)$ along the discrete Doppler domain. Since the delay and Doppler spread of the support set of the channel filter is invariant to the delay and Doppler shift, it follows that the delay and Doppler spread of the support set of each local response component is the same as that for the channel filter. As the shifts for the local responses differ by integer multiples of M along the delay domain and by integer multiples of N along the Doppler domain, these local responses do not superimpose/overlap/interact if and only if the *delay spread and Doppler spread of the channel filter $h_{\text{eff}}[k, l]$ are less than the delay period M and Doppler period N , respectively*. Since the Zak-OTFS I/O relation is predictable when this condition is satisfied, it is referred to as the crystallization condition with time- and bandwidth-limited waveforms.

From Theorem 2.1, it is known that $h_{\text{eff}}[k, l]$ is obtained by sampling the continuous DD domain channel filter $h_{\text{eff}}(\tau, \nu)$ on the information grid Λ_{dd} .¹⁵ Therefore, the crystallization condition with time- and bandwidth-limited waveforms is equivalently that the delay spread and Doppler spread of the effective continuous DD domain filter $h_{\text{eff}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu)$ are less than the

¹⁵ The DD domain function $h_{\text{eff}}(\tau, \nu)$ is the effective continuous DD domain channel filter since the AWGN-free received filtered continuous DD domain signal is given by $h_{\text{eff}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu)$ (see (2.O.2) in Appendix 2.O).

delay period (τ_p) and Doppler period (ν_p), respectively. From Corollary 2.4 in Appendix 2.A.1, it follows that the delay (Doppler) spread of $h_{\text{eff}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu)$ is the sum of the delay (Doppler) spreads of the pulse shaping waveforms $w_{\text{rx}}(\tau, \nu)$ and $w_{\text{tx}}(\tau, \nu)$, and the underlying channel DD spreading function $h_{\text{phy}}(\tau, \nu)$. Since the delay spread and Doppler spread of the pulse shaping waveforms are approximately $2/B$ and $2/T$, respectively, the *crystallization condition for time- and bandwidth-limited Zak-OTFS waveforms* is given by

$$\tau_p > \frac{2}{B} + \left(\max_i \tau_i - \min_i \tau_i \right) \quad \text{and} \quad \nu_p > \frac{2}{T} + \left(\max_i \nu_i - \min_i \nu_i \right). \quad (2.74)$$

The energy profile of the local response components is independent of the position of the pilot, i.e.,

$$\begin{aligned} |y_{n,m}^{(g)}[k, l]| &\stackrel{(a)}{=} |y_{n,m}^{(b)}[k - (k^{(g)} - k^{(b)}), l - (l^{(g)} - l^{(b)})]| \\ &\quad e^{j2\pi n \frac{(l^{(g)} - l^{(b)})}{N}} e^{j2\pi \frac{(l - l^{(g)} - mN)}{N} \frac{(k^{(g)} - k^{(b)})}{M}} \\ &= |y_{n,m}^{(b)}[k - (k^{(g)} - k^{(b)}), l - (l^{(g)} - l^{(b)})]|, \end{aligned} \quad (2.75)$$

where step (a) follows from (2.73). When the crystallization condition is satisfied, the local response components do not superimpose/overlap/interact, and, therefore, the energy profile of the received pilot is the same irrespective of whether the dark gray or the light gray pilot was transmitted, i.e., the energy profile is invariant of the pilot position. In other words, in the crystalline regime, the Zak-OTFS I/O relation is *non-fading*.

Some recent studies on OTFS are based on the Zak transform [60–65]. However, none of these studies investigate the predictability of the I/O relation in the DD domain. In particular, none of these emphasize the important fact that only Zak-OTFS (where the information bits are encoded as a discrete quasi-periodic function and filtering is applied through twisted convolution) *admits* a predictable I/O relation if the crystallization condition in (2.74) holds.

2.7.1 Non-predictability of the Zak-OTFS I/O Relation in the Non-crystalline Regime

When the crystallization condition fails to hold, the local channel responses interact (i.e., the ellipses overlap with one another). This phenomenon is referred to as *DD domain aliasing*. It is now explained as to how non-predictability arises from DD domain aliasing.

The aliasing phenomenon is illustrated through Fig. 2.26. Again, the objective is to predict the response to a light gray pilot from the response to a dark gray pilot. Let us now consider a situation where the Doppler spread of the effective channel filter is greater than the Doppler period. Under this assumption, local responses interact. Specifically, in the example considered here, the (n, m) th response interacts with the $(n, m - 1)$ th response. The interaction is depicted in Fig. 2.26 as overlap between ellipses. Consider a generic point (k, l) residing in the overlap of the $(0, 0)$ and $(0, -1)$ dark gray ellipses, depicted in Fig. 2.26 as a square with a black border. The response received at this point is the superposition of two local responses $y_{0,0}^{(b)}[k, l] + y_{0,-1}^{(b)}[k, l]$, which is equal to

$$h_{\text{eff}}[\Delta k, \Delta l] e^{j2\pi \frac{\Delta l}{N} \frac{k^{(b)}}{M}} + h_{\text{eff}}[\Delta k, \Delta l + N] e^{j2\pi \frac{\Delta l + N}{N} \frac{k^{(b)}}{M}}, \quad (2.76)$$

where $\Delta k = k - k^{(b)}$ and $\Delta l = l - l^{(b)}$. Consider the parallel point (k', l') residing in the overlap of the $(0, 0)$ and $(0, -1)$ light gray ellipses, depicted in Fig. 2.26 as a square with a light gray border. Here, $k' = k + (k^{(g)} - k^{(b)})$ and $l' = l + (l^{(g)} - l^{(b)})$. The response at this point is a superposition of two local responses $y_{0,0}^{(g)}[k', l'] + y_{0,-1}^{(g)}[k', l']$, which is explicitly equal to

$$h_{\text{eff}}[\Delta k, \Delta l] e^{j2\pi \frac{\Delta l}{N} \frac{k^{(g)}}{M}} + h_{\text{eff}}[\Delta k, \Delta l + N] e^{j2\pi \frac{\Delta l + N}{N} \frac{k^{(g)}}{M}}. \quad (2.77)$$

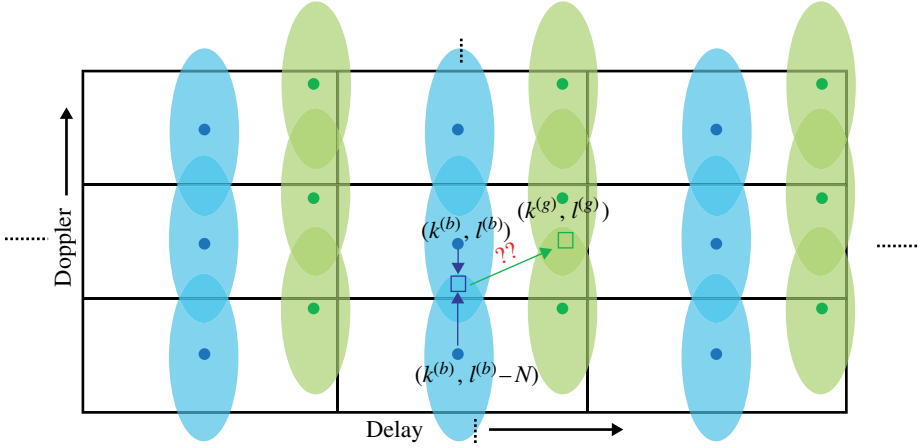


Figure 2.26 Non-crystalline regime: Partially predictable Zak-OTFS I/O relation. The response to the light gray DD pilot signal can be predicted from the response to the dark gray DD pilot signal only for the part of the response which is free from any overlap/aliasing/self-interaction. Source: Mohammed et al. [34] IEEE.

Observe that unless both the individual local terms in (2.76) are separately known, the light gray response in (2.77) cannot be predicted from the total dark gray response.

A consequence of DD domain aliasing is that the energy profile of the received response depends on the DD location of the transmitted impulse pilot. In our example, the energy profile of the received response to the dark gray pilot will, in general, be different from that for the light gray pilot. This is because the linear combination in (2.76) can be different from that in (2.77), depending on the values of the phase coefficients in each case.

2.7.2 Crystalline Decomposition

The previous discussion reveals that the channel response is non-predictable at points inside the overlap region between interacting ellipses. Outside this region, the response is in fact predictable. This observation suggests a *decomposition* of the effective channel filter into a predictable and non-predictable component.

To see this, consider a DD domain pilot at the origin (i.e., discrete DD domain impulse signal at $(0, 0)$, see (2.L.9) in Appendix 2.L.1). From Theorem 2.1, the AWGN-free channel response to this pilot is given by

$$\begin{aligned}
 h_{\text{eff}}[k, l] *_{\sigma} & \overbrace{\left(\sum_{n, m \in \mathbb{Z}} \delta[k - nM] \delta[l - mN] \right)}^{\text{Impulse at origin}} \\
 &= h_{\text{eff}}[k, l] + \sum_{n, m \in \mathbb{Z}, (n, m) \neq (0, 0)} h_{\text{eff}}[k - nM, l - mN] e^{j2\pi \frac{nl}{N}}, \quad (2.78)
 \end{aligned}$$

where the first term is the local response to the impulse at the origin and the remaining terms are local responses to other impulses. The main observation is that if (k, l) satisfies $h_{\text{eff}}[k, l] \neq 0$ and $h_{\text{eff}}[k - nM, l - mN] \neq 0$ for some $(n, m) \neq (0, 0)$, then the response at (k, l) is a *superposition* of at least two local responses, viz., the local response to the $(0, 0)$ th impulse (which is $h_{\text{eff}}[k, l]$ itself) and the local response to the (n, m) th alias impulse. Consequently, the channel filter tap at

(k, l) (i.e., $h_{\text{eff}}[k, l]$) cannot be estimated from the received response. Such taps are referred to as *non-predictable* taps. Alternatively, it is easy to see that all other taps in the support of $h_{\text{eff}}[k, l]$ can be estimated from the channel response. These taps are referred to as *predictable* taps. The support set of all predictable taps is denoted by $\mathcal{P}$, and the complement support set of all non-predictable taps by $\mathcal{P}^c$.

Recall that given a set S , the indicator function $\chi_S[k, l]$ takes the value 1 for $(k, l) \in S$, and the value 0 otherwise. The *crystalline decomposition* of the effective channel filter is then given by

$$h_{\text{eff}}[k, l] = \chi_{\mathcal{P}}[k, l] h_{\text{eff}}[k, l] + \chi_{\mathcal{P}^c}[k, l] h_{\text{eff}}[k, l], \quad (2.79)$$

where the first term is the predictable component and the second term is the non-predictable component. A mathematically exact definition of the crystallization condition is that the non-predictable component $\mathcal{P}^c$ vanishes, and this condition is weaker than (2.67). However, we shall focus on the stricter condition given by (2.67) since it relates directly to delay and Doppler spread, channel characteristics that are important in wireless systems engineering.

Remark We can view the inability to predict the Zak-OTFS I/O relation from a single pilot as the infeasibility of inverting a certain linear system. We send a quasi-periodic pilot at the origin and then make MN measurements by sampling the channel DD response on the information grid (M taps along the delay axis and N taps along the Doppler axis). Predictability requires that we are able to calculate the effective channel taps $h_{\text{eff}}[k, l]$ from the MN measurements. This is not possible, for example, when the number of taps to be estimated exceeds the number of measurements.

2.7.3 Identification of Linear Time-Varying Channels

Acquisition of the I/O relation is equivalent to channel identification. In [66], Bello conjectured that a LTV channel is identifiable if and only if the channel is *underspread*, i.e., the support area A of the channel DD spreading function is less than one. This conjecture was proved in [67] and extended in [68]. We note that the method developed in [68] is to use a weighted train of Dirac impulses as a probe signal then to use the Zak transform at the receiver to derive a system of linear equations that specifies the channel.

The channel estimation algorithm proposed in [66–68] involves solving a linear system of equations with a large number of unknown channel parameters (roughly equal to BTA, where BT is the time–bandwidth product of the communication system and A is the support area of the channel DD spreading function). This requires inversion of large matrices, which makes practical implementation challenging.

In contrast, a single pulsone is used as a probing signal in Zak-OTFS, e.g., the TD realization of a quasi-periodic DD pulse at the origin. The channel response to such a probing/pilot signal is given by (2.78). When the crystallization conditions in (2.67) hold, the terms $h_{\text{eff}}[k - nM, l - mN] e^{j2\pi \frac{nl}{N}}$, $(n, m) \neq (0, 0)$, in (2.78) have support that does not overlap with the support of the effective channel filter $h_{\text{eff}}[k, l]$, and, hence, its taps can be read off directly from the received response. Therefore, probing a linear time-varying channel with a Zak-OTFS pulsone significantly simplifies the acquisition of the I/O relation in comparison to other probing signals used in [66–68]. The underspread condition is an intrinsic property of the channel, whereas the crystallization condition is a property of the channel in relation to the (τ_p, ν_p) parameters of Zak-OTFS modulation. Note that we use the two-dimensional period lattice (with generators $(\tau_p, 0)$ and $(0, \nu_p)$) to define Zak-OTFS modulation, and that it is the combination of underspread channel and Zak-OTFS modulation that reduces the complexity of channel estimation.

If we ignore modulation and simply focus on channel support, the crystallization condition (2.67) is *not the same* as the channel underspread condition $A < 1$. The crystallization condition requires the delay spread to be less than the delay period (τ_p) and the Doppler spread to be less than the Doppler period (ν_p). Since $\tau_p \nu_p = 1$, the crystallization condition is more *restrictive* than the channel underspread condition, i.e., there are channels which are underspread ($A < 1$), but for which it is not possible to choose (τ_p, ν_p) , $\tau_p \nu_p = 1$, such that the delay and Doppler spreads are less than the respective periods. Being more restrictive, however, allows *simple* channel estimation (as discussed above).

2.7.4 Error in Prediction of the Zak-OTFS I/O Relation

In this section, the prediction accuracy is analyzed for different choices of the DD period and shaping filter. The analysis is carried through simulations of a simple two-path channel, where the first path has a delay of $\tau_1 = 0 \mu\text{s}$ and a Doppler shift of $\nu_1 = 815 \text{ Hz}$, and the second path has a delay of $\tau_2 = 5 \mu\text{s}$ and a Doppler shift of $\nu_2 = -815 \text{ Hz}$. The normalized channel gains for the two paths are $h_1 = h_2 = 1/\sqrt{2}$. The system bandwidth is $B = 0.96 \text{ MHz}$, and the frame duration is $T = 1.6 \text{ ms}$. Two choices for the transmit/receive shaping filters are considered. The first choice is a sinc filter, given by

$$w_{\text{rx}}(\tau, \nu) = w_{\text{tx}}(\tau, \nu) = \sqrt{BT} \text{sinc}(B\tau) \text{sinc}(\nu T). \quad (2.80)$$

Observe that the slow decay of the sinc function amplifies DD domain aliasing, thereby reducing predictability. The second choice is a root raised cosine (RRC) filter, given by

$$w_{\text{rx}}(\tau, \nu) = w_{\text{tx}}(\tau, \nu) = \sqrt{BT} \text{rrc}_{\beta_\tau}(B\tau) \text{rrc}_{\beta_\nu}(T\nu), \quad (2.81)$$

where the function $\text{rrc}_\beta(\cdot)$ for $0 \leq \beta \leq 1$ is given by [69]

$$\text{rrc}_\beta(x) = \frac{\sin(\pi x(1 - \beta)) + 4\beta x \cos(\pi x(1 + \beta))}{\pi x (1 - (4\beta x)^2)}. \quad (2.82)$$

Observe that the RRC function has a faster decay than the sinc function, causing less DD domain aliasing, thereby improving predictability. However, reduced aliasing in this manner comes at the cost of an increase in frame duration and bandwidth. In the simulation, $\beta_\tau = 0.1$ and $\beta_\nu = 0.2$, corresponding to a 10% increase in bandwidth and a 20% increase in duration. The energy of the transmit/receive filters is normalized by setting

$$\iint |w_{\text{tx}}(\tau, \nu)|^2 d\tau d\nu = \iint |w_{\text{rx}}(\tau, \nu)|^2 d\tau d\nu = 1. \quad (2.83)$$

A pilot signal at $(k^{(b)}, l^{(b)}) = \left(\frac{M}{2}, \frac{N}{2}\right)$ is used to estimate the effective channel filter $h_{\text{eff}}[k, l]$. Recall that, in the crystalline regime, local responses do not interact, and, therefore, the total response in the fundamental period coincides with the $(0, 0)$ th local response $y_{0,0}^{(b)}[k, l]$, which is given by

$$y^{(b)}[k, l] = h_{\text{eff}}[k - M/2, l - N/2] e^{j\pi \frac{(l-N/2)}{N}} \quad (2.84)$$

for $0 \leq k < M$ and $0 \leq l < N$. From this, it follows that

$$h_{\text{eff}}[k, l] = y^{(b)}[k + M/2, l + N/2] e^{-j\pi \frac{l}{N}} \quad (2.85)$$

for $-\frac{M}{2} \leq k < \frac{M}{2}$ and $-\frac{N}{2} \leq l < \frac{N}{2}$. Set the channel filter estimate to be

$$\hat{h}_{\text{eff}}[k, l] = \begin{cases} y^{(b)} \left[k + \frac{M}{2}, l + \frac{N}{2} \right] e^{-j\pi \frac{l}{N}}, & -\frac{M}{2} \leq k < \frac{M}{2} \\ & -\frac{N}{2} \leq l < \frac{N}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (2.86)$$

Using (2.86), the effective channel response to any pilot can be predicted. The predicted channel response to a pilot located at $(k^{(g)}, l^{(g)})$ is given by

$$\hat{y}^{(g)}[k, l] = \hat{h}_{\text{eff}}[k, l] *_{\sigma} x^{(g)}[k, l]. \quad (2.87)$$

The predicted response should be compared to the true channel response

$$y^{(g)}[k, l] = h_{\text{eff}}[k, l] *_{\sigma} x^{(g)}[k, l]. \quad (2.88)$$

The relative prediction error (RPE) is given by

$$E(k^{(g)}, l^{(g)}) \triangleq \frac{\sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} |\hat{y}^{(g)}[k, l] - y^{(g)}[k, l]|^2}{\sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} |y^{(g)}[k, l]|^2}. \quad (2.89)$$

Figures 2.27 and 2.28 depict the RPE as a two-dimensional heatmap for three different points on the period curve: the point $\nu_p = 1.25$ kHz representing a TDM approximation, $\nu_p = 30$ kHz representing a point in the crystalline regime, and $\nu_p = 240$ kHz representing the FDM approximation¹⁶. Figure 2.27 assumes sinc transmit/receive shaping filters, whereas Fig. 2.28 assumes RRC transmit/receive filters.

When $\nu_p = 1.25$ kHz ($\tau_p = 800$ μ s), the Doppler period is smaller than the channel Doppler spread (1.63 kHz), causing aliasing along the Doppler dimension which, in turn, creates fading along delay and non-predictability. In this situation, the RPE is minimized at the pilot location and it increases rapidly toward 0 dB as we move away along the delay domain.

When $\nu_p = 240$ kHz, ($\tau_p = 4.16$ μ s), the delay period is smaller than the channel delay spread (5 μ s) causing aliasing along the delay dimension, which, in turn, creates fading along Doppler and non-predictability. In this situation, the RPE is minimized at the pilot location and it now increases rapidly toward 0 dB as we move away from the pilot location along the Doppler domain.

Finally, when $\nu_p = 30$ kHz, ($\tau_p = 33.3$ μ s), the crystallization condition holds and predictability is maintained. In this situation, the RPE is small.

Comparing Figs. 2.27 and 2.28, it is observed that the choice of transmit/receive shaping filter makes a significant difference. Using RRC filters lead to a RPE of roughly -50 dB, whereas using sinc filters lead to a RPE of roughly -20 dB.

¹⁶ Zak-OTFS is a family of modulations parameterized by (τ_p, ν_p) (see Fig. 2.17). As the Doppler period ν_p tends to infinity and the delay period $\tau_p = 1/\nu_p$ tends to zero, the proposed Zak-OTFS modulation behaves like FDM, i.e., Zak-OTFS is approximately FDM. Similarly, when the delay period tends to infinity and the Doppler period tends to zero, the proposed Zak-OTFS is approximately TDM.

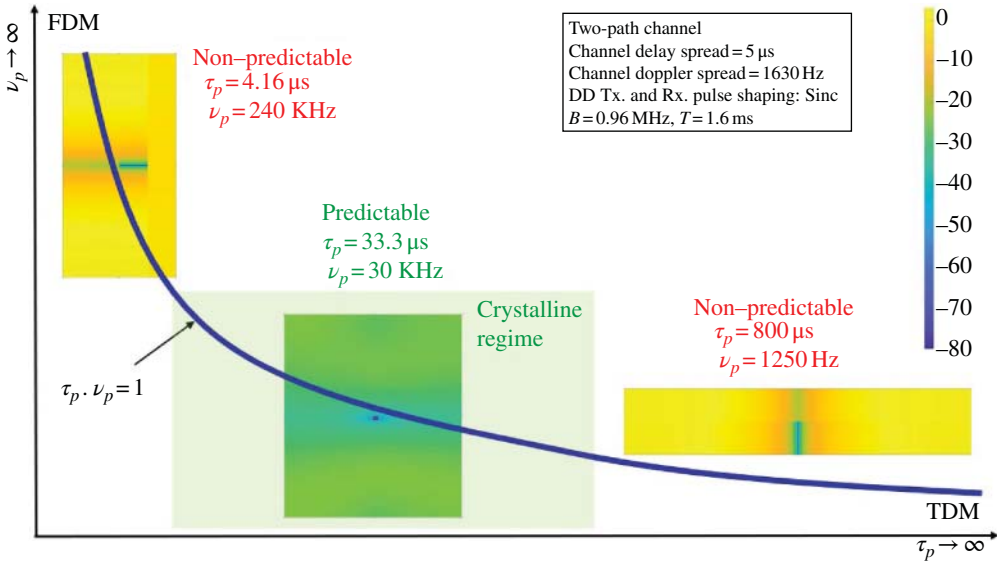


Figure 2.27 Heatmap showing relative prediction error (RPE), in dB, as a function of delay (horizontal axis) and Doppler (vertical axis) with sinc pulse shaping filters. RPE is significantly smaller in the crystalline regime when compared to that in the non-crystalline regime. Source: Mohammed et al. [34] IEEE.

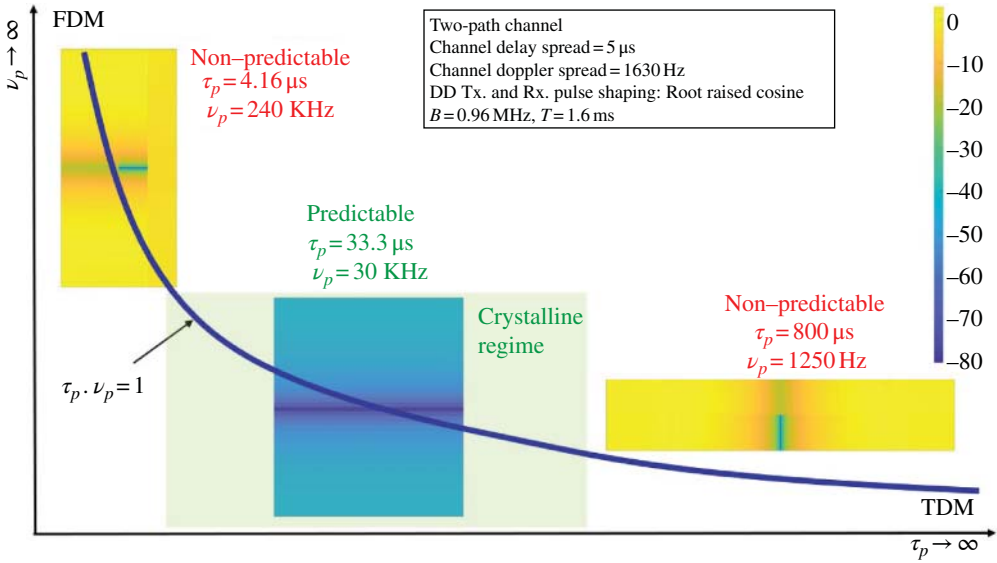


Figure 2.28 Heatmap showing RPE, in dB, as a function of delay (horizontal axis) and Doppler (vertical axis) with RRC pulse shaping filters. In the crystalline regime, RRC pulse results in significantly smaller RPE when compared to sinc pulses, although at the cost of a higher OTFS frame duration and bandwidth. Source: Mohammed et al. [34] IEEE.

2.8 Matrix-vector Description of the I/O Relation

The channel equation relating the received and transmit TD signals is given by [55]

$$r_{\text{td}}(t) = \iint h_{\text{phy}}(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n_{\text{td}}(t), \quad (2.90)$$

where $h_{\text{phy}}(\tau, \nu)$ is the DD channel representation/spreading function and $n_{\text{td}}(t)$ is the DD domain AWGN signal. This equation induces a relation between the input and output signals for TDM, FDM, and Zak-OTFS modulation (see Appendix 2.J for TDM I/O relation, Appendix 2.K for FDM I/O relation, and Theorem 2.1 in Section 2.6.4 for Zak-OTFS I/O relation). This section presents the equivalent matrix-vector representation of the I/O relation for Zak-OTFS, TDM, and FDM.

Although there are other equivalent channel representations (e.g., the time-delay representation $h_{\text{phy}}(t, \tau)$), we consider the DD representation here since the Zak-OTFS I/O relation in the DD domain is described simply in terms of twisted convolution of the DD domain channel input with a discrete DD domain channel filter $h_{\text{eff}}[k, l]$. The channel filter $h_{\text{eff}}[k, l]$ can be acquired efficiently using a single pulsed probe signal, making it possible (in the crystalline regime) to predict the channel response to any Zak-OTFS input (see Theorem 2.1 and Section 2.7).

2.8.1 Zak-OTFS

From Theorem 2.1 in Section 2.6.4, the Zak-OTFS I/O relation is given by

$$y_{\text{dd}}[k, l] = \sum_{k', l' \in \mathbb{Z}} h_{\text{eff}}[k - k', l - l'] x_{\text{dd}}[k', l'] e^{j2\pi \frac{(l-l')}{N} \frac{k'}{M}} + n_{\text{dd}}[k, l], \quad (2.91)$$

where $h_{\text{eff}}[k, l]$ is the effective DD domain channel filter and $n_{\text{dd}}[k, l]$ is the discrete DD domain AWGN signal obtained by sampling

$$n_{\text{dd}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} \mathcal{Z}_t(n_{\text{td}}(t)).$$

Since $y_{\text{dd}}[k, l]$ is obtained by sampling the quasi-periodic received filtered DD domain signal $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$ on the information grid (see Fig. 2.24), from Theorem 2.7 in Appendix 2.L, it follows that $y_{\text{dd}}[k, l]$ is a quasi-periodic discrete DD domain signal, i.e., it satisfies (see (2.L.2) in Appendix 2.L)¹⁷

$$y_{\text{dd}}[k + nM, l + mN] = e^{j2\pi \frac{nl}{N}} y_{\text{dd}}[k, l] \quad (2.92)$$

for all $k, l, n, m \in \mathbb{Z}$.¹⁸

The quasi-periodicity of $y_{\text{dd}}[k, l]$ implies that the MN received samples in the fundamental DD period ($0 \leq k < M, 0 \leq l < N$) constitute a sufficient statistic for the detection of the transmitted information symbols $x[k, l]$. This is illustrated through an example in Fig. 2.29, where a part of the response to a DD pilot received within the fundamental DD period (union of regions I', II', III, and IV' in Fig. 2.29) can be used to reconstruct any local response component (e.g., union of regions I, II, III, and IV in Fig. 2.29). Hence, the I/O relation (2.91) can be reduced to a finite matrix form relating the $MN \times 1$ vector of received samples $y_{\text{dd}}[k, l]$ to the $MN \times 1$ vector of transmitted symbols $x[k, l]$,

¹⁷ The quasi-periodicity of $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu)$ follows from the fact that, (i) $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(r_{\text{td}}(t))$ is quasi-periodic as it is the DD representation of a TD signal (see (2.25)), and (ii) $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$ is the twisted convolution of a quasi-periodic signal $y_{\text{dd}}(\tau, \nu)$ with the filter $w_{\text{rx}}(\tau, \nu)$ and twisted convolution *preserves* quasi-periodicity (see Appendix 2.G).

¹⁸ Note that $y_{\text{dd}}[k, l]$ is quasi-periodic irrespective of whether the crystallization condition is satisfied or not. This is because quasi-periodicity is an intrinsic property of the Zak transform (i.e., the Zak transform of a TD signal is quasi-periodic), which is independent of the channel.

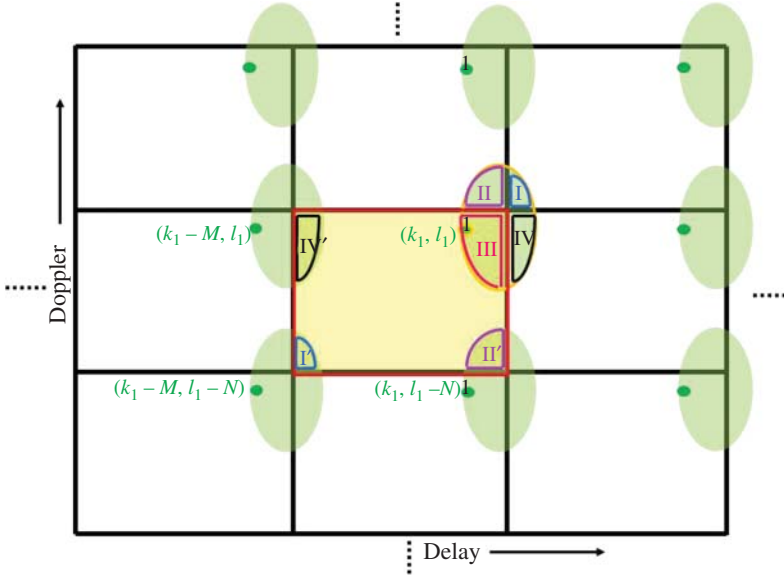


Figure 2.29 Exploiting quasi-periodicity: Generating complete response from response received in the fundamental DD period. Source: Mohammed et al. [34] IEEE.

where on both sides $k = 0, 1, \dots, M - 1$ and $l = 0, 1, \dots, N - 1$. In more detail, define the $MN \times 1$ vectors $\mathbf{y}_{dd}$, $\mathbf{x}_{dd}$, and $\mathbf{n}_{dd}$, such that

$$\begin{aligned} (\mathbf{y}_{dd})_{kN+l+1} &= y_{dd}[k, l], \\ (\mathbf{x}_{dd})_{kN+l+1} &= x_{dd}[k, l], \\ (\mathbf{n}_{dd})_{kN+l+1} &= n_{dd}[k, l]. \end{aligned} \quad (2.93)$$

In addition, define the $MN \times MN$ matrix $\mathbf{H}_{dd}$, such that

$$(\mathbf{H}_{dd})_{k'N+l'+1, kN+l+1} = H_{dd}[k'N + l', kN + l], \quad (2.94)$$

where, using (2.91), the RHS in the above equation is given by

$$\begin{aligned} H_{dd}[k'N + l', kN + l] &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} h_{\text{eff}}[k' - k - nM, l' - l - mN] e^{j2\pi nl/N} e^{j2\pi \frac{l' - l - mN}{N} \frac{k + nM}{M}}, \\ &k', k = 0, 1, \dots, M - 1, \quad l', l = 0, 1, \dots, N - 1. \end{aligned} \quad (2.95)$$

The matrix-vector formulation of the DD domain I/O relation (2.91) can then be written in the form

$$\mathbf{y}_{dd} = \mathbf{H}_{dd} \mathbf{x}_{dd} + \mathbf{n}_{dd}. \quad (2.96)$$

Since the I/O relation in (2.91) is valid irrespective of whether we operate in the crystalline regime or not, the same holds for the corresponding matrix-vector formulation in (2.96).

2.8.2 TDM

From the TDM discrete I/O relation given by Theorem 2.5 in Appendix 2.J, it follows that

$$y_{td}[k] = \sum_{k' \in \mathbb{Z}} x_{td}[k'] h_{td}[k - k'; k'] + n_{td}[k], \quad (2.97)$$

where $h_{\text{td}}[n; k]$ is the effective TD channel filter and the noise term $n_{\text{td}}[k]$ is given by sampling the continuous function $w_{\text{rx}}(t) \star n_{\text{td}}(t)$ at $t = k/B$, where $w_{\text{rx}}(t)$ is the matched filter at the receiver and $\star$ denotes linear convolution. Equation (2.97) can be expressed in a matrix-vector form as follows. Arrange the transmitted information symbols as a BT -length column vector $\mathbf{x}_{\text{td}}$, where

$$(\mathbf{x}_{\text{td}})_{k+1} = x_{\text{td}}[k] = x[k], \quad (2.98)$$

for $k = 0, 1, \dots, BT - 1$, and arrange the received TD samples as a $(BT + K_1 + K_2)$ -length column vector $\mathbf{y}_{\text{td}}$, where

$$(\mathbf{y}_{\text{td}})_{k+1+K_1} = y_{\text{td}}[k] \quad (2.99)$$

for $k = -K_1, \dots, BT - 1 + K_2$, where the constants $K_1, K_2 \in \mathbb{Z}$ are determined by the channel delay spread. Similarly, arrange the received sampled noise as a $(BT + K_1 + K_2)$ -length vector $\mathbf{n}_{\text{td}}$, where

$$(\mathbf{n}_{\text{td}})_{k+1+K_1} = n_{\text{td}}[k] \quad (2.100)$$

for $k = -K_1, \dots, BT - 1 + K_2$. Finally, arrange the effective channel filter as a $(BT + K_1 + K_2) \times BT$ matrix $\mathbf{H}_{\text{td}}$, where

$$(\mathbf{H}_{\text{td}})_{k'+1, k+1} = h_{\text{td}}[k' - k - K_1; k] \quad (2.101)$$

for $k' = 0, 1, \dots, (BT + K_1 + K_2 - 1)$, $k = 0, 1, \dots, (BT - 1)$. Using (2.97), the I/O relation in matrix-vector form is then given by

$$\mathbf{y}_{\text{td}} = \mathbf{H}_{\text{td}} \mathbf{x}_{\text{td}} + \mathbf{n}_{\text{td}}. \quad (2.102)$$

2.8.3 FDM

From Theorem 2.6 in Appendix 2.K, the FDM discrete I/O relation in a doubly spread channel is given by

$$y_{\text{fd}}[k] = \sum_{k' \in \mathbb{Z}} x_{\text{fd}}[k'] h_{\text{fd}}[k - k'; k'] + n_{\text{fd}}[k], \quad (2.103)$$

where $h_{\text{fd}}[n; k]$ is the effective FD channel filter and the noise term $n_{\text{fd}}[k]$ is obtained by sampling the continuous function $w_{\text{rx}}(f) \star n_{\text{fd}}(f)$ at $f = k/T$, where

$$n_{\text{fd}}(f) = \int n_{\text{td}}(t) e^{-j2\pi ft} dt.$$

Arrange the transmitted information symbols as a BT -length column vector $\mathbf{x}_{\text{fd}}$, such that

$$(\mathbf{x}_{\text{fd}})_{k+1} = x_{\text{fd}}[k] = x[k] \quad (2.104)$$

for $k = 0, 1, \dots, BT - 1$. Arrange the received samples as a $(BT + L_1 + L_2)$ -length column vector $\mathbf{y}_{\text{fd}}$, such that

$$(\mathbf{y}_{\text{fd}})_{k+1+L_1} = y_{\text{fd}}[k] \quad (2.105)$$

for $k = -L_1, \dots, BT - 1 + L_2$, where the constants $L_1, L_2 \in \mathbb{Z}$ are determined by the channel Doppler spread. Similarly, arrange the received noise samples as a $(BT + L_1 + L_2)$ -length column vector $\mathbf{n}_{\text{fd}}$, such that

$$(\mathbf{n}_{\text{fd}})_{k+1+L_1} = n_{\text{fd}}[k] \quad (2.106)$$

for $k = -L_1, \dots, BT - 1 + L_2$. Finally, arrange the FD channel filter as a $(BT + L_1 + L_2) \times BT$ matrix $\mathbf{H}_{\text{fd}}$, such that

$$(\mathbf{H}_{\text{fd}})_{k'+1, k+1} = h_{\text{fd}}[k' - k - L_1; k] \quad (2.107)$$

for $k' = 0, 1, \dots, BT + L_1 + L_2 - 1$ and $k = 0, 1, \dots, BT - 1$. Putting everything together, the matrix-vector relation is given by

$$\mathbf{y}_{\text{fd}} = \mathbf{H}_{\text{fd}} \mathbf{x}_{\text{fd}} + \mathbf{n}_{\text{fd}}. \quad (2.108)$$

2.9 Impact of Fading in the Crystalline Regime

In this section, the uncoded BER performance of Zak-OTFS, TDM, and FDM is compared for the Veh-A channel model in [70]. Since the impact of fading is to be studied, the I/O relation is assumed to be known perfectly. The performance of Zak-OTFS is studied as we move along the hyperbola $\tau_p \nu_p = 1$ by choosing different Doppler periods ν_p . It is demonstrated that (i) the performance of Zak-OTFS is superior in the crystalline regime, (ii) the performance approaches the TDM performance as the Doppler period shrinks, and (iii) the performance approaches the FDM performance as the delay period shrinks.

The Veh-A channel model consists of six channel paths, and the DD spreading function is given by

$$h_{\text{phy}}(\tau, \nu) = \sum_{i=1}^6 h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (2.109)$$

where h_i , τ_i , and ν_i denote the gain, delay, and Doppler shift of the i th channel path, respectively. Table 2.5 lists the power-delay profile (PDP) for the six channel paths. The maximum Doppler shift is $\nu_{\text{max}} = 815$ Hz, the Doppler spread is 1.63 kHz, and the delay spread is $\tau_{\text{max}} \triangleq \max_i \tau_i - \min_i \tau_i = 2.5 \mu\text{s}$. The Doppler shift of the i th path is modeled as $\nu_i = \nu_{\text{max}} \cos(\theta_i)$, where the variables $\theta_i, i = 1, 2, \dots, 6$, are independent and distributed uniformly in the interval $[0, 2\pi)$. The time duration T of a data frame is 1.6 ms, and the bandwidth B is 0.96 MHz.

The transmit and receive filters are specified next. For TDM, the pulse shaping filter at the transmitter and the matched filter at the receiver are both narrow TD sinc pulses of bandwidth B and TD pulse width proportional to $1/B$, given by

$$w_{\text{tx}}(t) = w_{\text{rx}}(t) = \sqrt{B} \text{sinc}(Bt). \quad (2.110)$$

For FDM, the transmit and receive filters are narrow FD sinc pulses with width proportional to $1/T$ so that the time realization has duration T , given by

$$w_{\text{tx}}(f) = w_{\text{rx}}(f) = \sqrt{T} \text{sinc}(fT). \quad (2.111)$$

Table 2.5 Power-delay profile of the doubly spread Veh-A channel.

Path no. i	1	2	3	4	5	6
Rel. Delay τ_i (μs)	0	0.31	0.71	1.09	1.73	2.51
Rel. Power $\frac{\mathbb{E}[h_i ^2]}{\mathbb{E}[h_1 ^2]}$ (dB)	0	-1	-9	-10	-15	-20

Source: Mohammed et al. [34] IEEE.

For Zak-OTFS, the DD domain transmit and receive filters are the product of a narrow pulse in the delay domain with width proportional to $1/B$ and a narrow pulse in the Doppler domain with width proportional to $1/T$, given by

$$w_{\text{tx}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) = \sqrt{BT} \text{sinc}(B\tau) \text{sinc}(T\nu). \quad (2.112)$$

The BER performance is measured as a function of the received signal-to-noise ratio (SNR), which is the ratio of the power of the information carrying signal to the power of the AWGN in the received TD signal. The complex channel gains are normalized by setting $\sum_{i=1}^6 \mathbb{E}[|h_i|^2] = 1$. The transmitted signal power P_T is defined to be the ratio of the average energy of the transmitted signal to the frame duration T as

$$P_T \triangleq \frac{\mathbb{E} \left[\int |s_{\text{td}}(t)|^2 dt \right]}{T}. \quad (2.113)$$

If N_0 is the noise power spectral density of the AWGN $n_{\text{td}}(t)$, then the noise power at the receiver is $N_0 B$, and the SNR is given by

$$\gamma \triangleq \frac{P_T}{N_0 B}. \quad (2.114)$$

It is assumed that the Zak-OTFS receiver knows the effective channel matrix $\mathbf{H}_{\text{dd}}$ perfectly, the TDM receiver knows $\mathbf{H}_{\text{td}}$ perfectly, and the FDM receiver knows $\mathbf{H}_{\text{fd}}$ perfectly. *Note that perfect knowledge of the I/O relation does not imply perfect knowledge of the channel DD spreading function.* The matrix-vector I/O relations given by (2.96), (2.102), and (2.108) have the same structure as those of a multiple input multiple output (MIMO) system. A linear minimum mean squared error (LMMSE) equalizer is therefore used to detect the transmitted 4-QAM information symbols.

Figure 2.30 illustrates how the BER performance of Zak-OTFS changes as we move along the hyperbola $\tau_p \nu_p = 1$, choosing Doppler periods $\nu_p = 0.625, 1.25, 15, 480$, and 960 kHz.

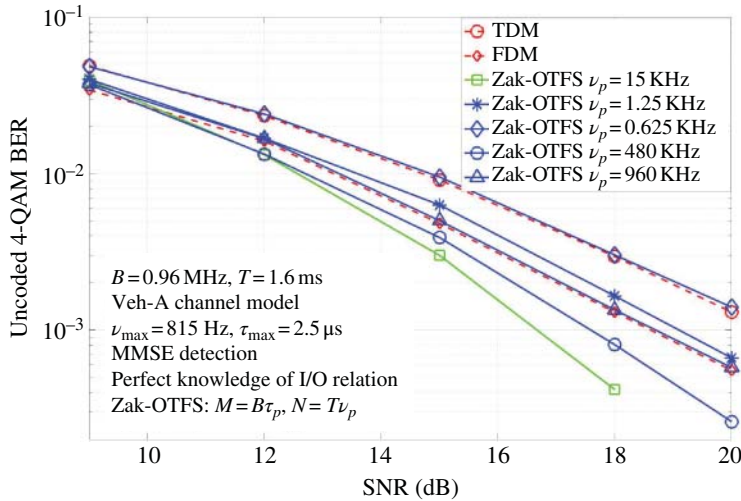


Figure 2.30 BER performance of Zak-OTFS, TDM, and FDM on a doubly spread Veh-A channel as we traverse the hyperbola $\tau_p \nu_p = 1$. The performance of Zak-OTFS is superior in the crystalline regime ($\nu_p = 15$ kHz), performance approaches TDM as the delay period grows ($\nu_p = 1.25$ and 0.625 kHz), and performance approaches FDM as the Doppler period grows ($\nu_p = 480$ and 960 kHz). Source: Mohammed et al. [34] IEEE.

When $\nu_p = 15$ kHz, the crystallization conditions hold, the I/O relation is non-fading (see (2.75)), and the BER performance is superior to those of both TDM and FDM. This is because the I/O relations for TDM and FDM exhibit fading on doubly spread channels.

As the Doppler period increases, first to 480 kHz and then to 960 kHz, the delay spread is no longer less than the delay period, and Zak-OTFS operates outside the crystalline regime. Aliasing occurs along the delay domain, and the received symbol energy varies (fades) along the Doppler domain. When $\nu_p = 960$ kHz, the BER performance of Zak-OTFS coincides with FDM performance, which is expected, given that FDM coincides with Zak-OTFS in the limit of vanishing delay period.

As the Doppler period decreases, first to 1.25 kHz and then to 0.625 kHz, the Doppler spread is no longer less than the Doppler period, and Zak-OTFS operates outside the crystalline regime. Aliasing now occurs along the Doppler domain, and the received symbol energy varies (fades) along the delay domain. When $\nu_p = 0.625$ kHz, the BER performance of Zak-OTFS almost coincides with TDM performance, which is expected, given that TDM coincides with Zak-OTFS in the limit of vanishing Doppler period.

A *delay-only* Veh-A channel is matched to TDM and mismatched to FDM. The channel is frequency-selective, not time-selective; hence, FDM exhibits fading and TDM does not. Figure 2.31 compares the BER performance of TDM and FDM with that of Zak-OTFS in a delay-only Veh-A channel (i.e., $\nu_i = 0, i = 1, 2, \dots, 6$) for Doppler periods $\nu_p = 0.625, 15$, and 960 kHz. When $\nu_p = 0.625$ kHz, the delay period $\tau_p = 1.6$ ms is much larger than the delay spread ($2.5 \mu\text{s}$), and the BER performance is essentially the same as that of TDM. While operating in the crystalline regime (for example, $\nu_p = 15$ kHz and $\tau_p = 66.66 \mu\text{s}$), there is no fading, and the BER performance changes very little with change in ν_p . Outside the crystalline regime, when the Doppler period is large and the delay period is smaller than the delay spread (for example, $\nu_p = 960$ kHz and $\tau_p = 1.04 \mu\text{s}$), the BER performance degrades considerably.

In contrast, a *Doppler-only* Veh-A channel is matched to FDM and mismatched to TDM. The channel is time-selective, not frequency-selective; hence, TDM exhibits fading and FDM does

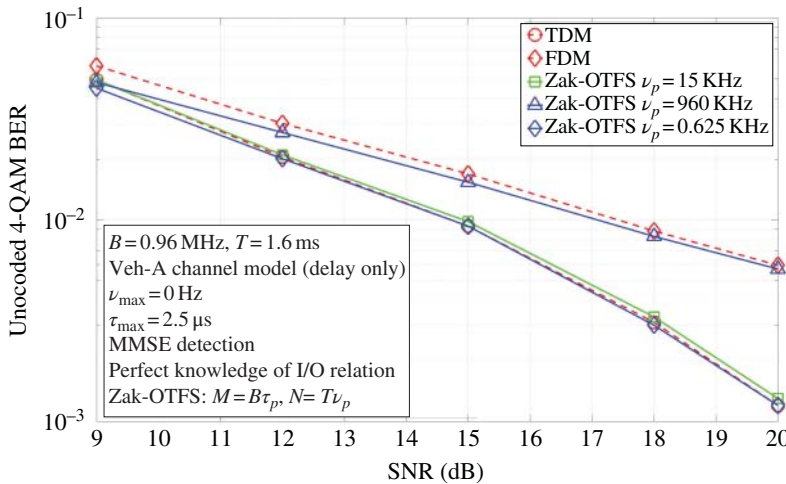


Figure 2.31 BER performance of Zak-OTFS, TDM, and FDM on a delay-only Veh-A channel. The I/O relation for Zak-OTFS in the crystalline regime ($\nu_p = 15$ kHz) is non-fading; hence, the BER performance matches that of TDM. Frequency-selective fading degrades the BER performance of FDM. Source: Mohammed et al. [34] IEEE.

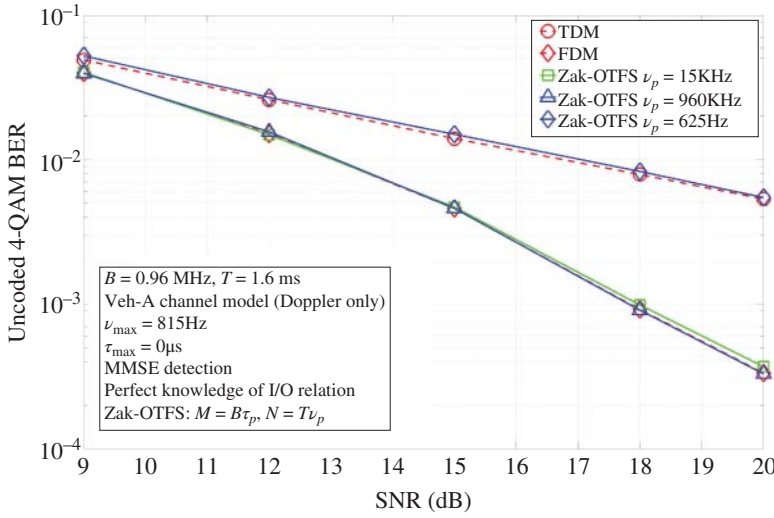


Figure 2.32 BER performance of Zak-OTFS, TDM, and FDM on a Doppler-only Veh-A channel. The I/O relation for Zak-OTFS in the crystalline regime ($\nu_p = 15$ kHz) is non-fading; hence, the BER performance matches that of FDM. Time-selective fading degrades the BER performance of TDM. Source: Mohammed et al. [34] IEEE.

not. Figure 2.32 compares the BER performance of TDM and FDM with that of Zak-OTFS in a Doppler-only Veh-A channel (i.e., $\tau_{\max} = 0$), for Doppler periods $\nu_p = 0.625, 15$, and 960 kHz. When $\nu_p = 960$ kHz, the Doppler period is much larger than the Doppler spread (1.63 kHz), and the BER performance is essentially the same as that of FDM. Again, while operating in the crystalline regime (for example, $\nu_p = 15$ kHz and $\tau_p = 66.66 \mu\text{s}$), there is no fading, and the BER performance changes very little with change in ν_p . Outside the crystalline regime (for example, $\nu_p = 0.625$ kHz), the BER performance degrades considerably.

2.10 Model-free Operation in the Crystalline Regime

In this section, the uncoded BER performance of Zak-OTFS, TDM, and FDM is compared in the scenario considered above, but relaxing the perfect I/O knowledge assumption. The effective channel matrix can be acquired from the I/O relation (see Section 2.8), which, in turn, depends on the DD spreading function $h_{\text{phy}}(\tau, \nu)$. The *model-dependent* and *model-free* modes of operation correspond to different approaches to estimating the effective channel matrix.

2.10.1 Model-dependent Operation

Here, a model is imposed on the DD spreading function $h_{\text{phy}}(\tau, \nu)$, typically by prescribing a finite number of paths and constraining their delay and Doppler shifts. Given this model, the receiver estimates $h_{\text{phy}}(\tau, \nu)$ and then forms an estimate of the effective channel matrix. The accuracy of this estimate is limited by time and bandwidth constraints on the pilot signal and by any mismatch between the channel model and the physical channel.

2.10.2 Model-free Operation

Here, the receiver estimates the effective DD domain channel filter taps $h_{\text{eff}}[k, l]$ directly, without reference to any model for the DD spreading function $h_{\text{phy}}(\tau, \nu)$. Although model-free operation can be chosen for both the crystalline and non-crystalline regimes, communication reliability degrades in the non-crystalline regime since the effective channel filter taps cannot be estimated accurately (see Section 2.7.1). On the other hand, in the crystalline regime, the effective channel filter taps $h_{\text{eff}}[k, l]$ can be estimated from the response to a single pilot symbol (see Section 2.7).¹⁹

The model-dependent mode of operation is a challenge for the Veh-A channel introduced in Section 2.9 (with sinc pulse shaping filters). Since channel bandwidth $B = 0.96$ MHz, the delay domain resolution is $1/B \approx 1.04 \mu\text{s}$ and the first three paths introduce delay shifts in the interval $[0, 0.71] \mu\text{s}$ (see Table 2.5). Therefore, these paths are not separable and hence cannot be estimated accurately. However, model-free operation is still possible for TDM, FDM, and Zak-OTFS in the crystalline regime (e.g., $\nu_p = 15$ kHz). The effective channel filter taps are estimated from the response to a pilot frame with a single high-energy pilot and no information symbols. To begin with, it is supposed that the received pilot is not subject to AWGN. For Zak-OTFS, the single pilot is located at the center $(M/2, N/2)$ of the fundamental period, and (2.86) is used to estimate the effective filter taps. For TDM and FDM, a single pilot located at $k = BT/2$ is used to estimate the effective channel filter taps $h_{\text{td/fd}}[n; k = BT/2]$, $n \in \mathbb{Z}$, and this estimate is simply reused for all other taps $h_{\text{td/fd}}[n; k]$, $n \in \mathbb{Z}$, $k = 0, 1, \dots, BT - 1$, $k \neq BT/2$.

For model-dependent operation, the pilots as described above are used to estimate the complex channel gain, delay, and Doppler shift of each channel path. At the receiver, DD points with significant received energy are interpreted as the locations of channel path delays and Doppler shifts. Given the locations of delay and Doppler shifts, the received pilot signal depends linearly on the channel path gains, and, therefore, least squares is used to estimate the vector of complex channel gains and then reconstruct $\mathbf{H}_{\text{dd}}$ using (2.95).

Figure 2.33 compares the BER performance of the model-dependent and model-free modes of Zak-OTFS in the crystalline regime. It also includes the BER performance of TDM and FDM, which is poor, and which does not improve with increasing SNR. Since the channel is doubly spread, the I/O relation for TDM/FDM is non-stationary in the TD/FD and it cannot be accurately predicted from the response to a single pilot at $k = BT/2$. Hence, the estimate of the effective channel matrix is inaccurate, and the probability of misdetection is high. The BER performance of the model-dependent mode of Zak-OTFS is slightly better, but it also exhibits a high error floor because insufficient frame bandwidth and duration preclude the accurate estimation of channel path gains, delays, and Doppler shifts.²⁰ The BER performance of the model-free mode of Zak-OTFS is considerably better, which is only slightly worse than the performance with perfect knowledge of the I/O relation.

Next, through Fig. 2.34, we explore what we lose when it is possible to learn the channel, but we choose to operate model-free. Again, the time duration T and bandwidth B of the OTFS frame are fixed to 1.6 ms and 0.96 MHz, respectively. There are $M = 64$ delay bins and $N = 24$ Doppler bins.

¹⁹ The model-free approach considered here is a LTV channel generalization of the approach taken in LTI channels where only the taps of the effective discrete-time channel filter are estimated instead of estimating the continuous time impulse response of the underlying physical channel.

²⁰ For example, for the Veh-A channel model, the first three paths introduce delay shifts which lie in the interval $[0, 0.71] \mu\text{s}$. With a channel bandwidth of $B = 0.96$ MHz, the delay domain resolution is $1/B \approx 1.04 \mu\text{s}$, which is more than $0.71 \mu\text{s}$, and, therefore, the first three dominant paths are not separable/resolvable. These non-separable paths cannot be estimated accurately.

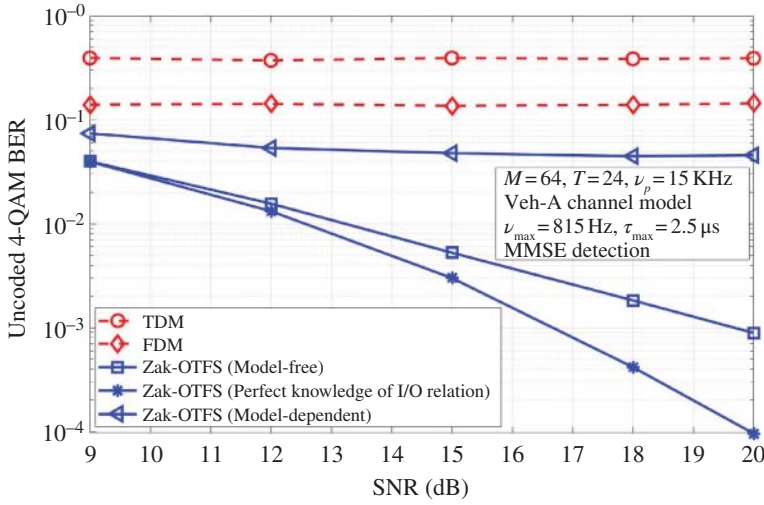


Figure 2.33 BER performance of Zak-OTFS, TDM, and FDM on the Veh-A channel model, where the signal bandwidth and duration are insufficient to estimate channel path delays and Doppler shifts. In the crystalline regime ($\nu_p = 15$ kHz), the BER performance of the model-free mode of Zak-OTFS is only slightly worse than the performance with perfect knowledge of the I/O relation. Source: Mohammed et al. [34] IEEE.

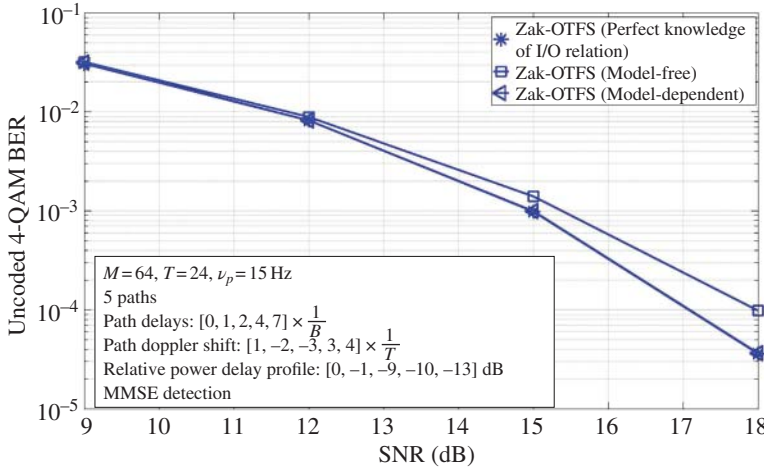


Figure 2.34 BER performance of the model-dependent and model-free mode of Zak-OTFS for a five-path resolvable channel, where the delay shifts are integer multiples of the delay resolution $1/B$, and the Doppler shifts are integer multiples of the Doppler resolution $1/T$. The model-dependent performance coincides with that of performance with perfect knowledge of the I/O relation. The model-free performance is only slightly inferior. Source: Mohammed et al. [34] IEEE.

Again, the delay spread is roughly $2.5\mu\text{s}$ and the Doppler spread is 1.63 kHz. A five-path resolvable channel is considered, where the delay shifts are integer multiples of the delay resolution $1/B$, and the Doppler shifts are integer multiples of the Doppler resolution $1/T$. To be precise, the path delays are $[0, 1, 2, 4, 7] \times 1/B$, the Doppler shifts are $[1, -2, -3, 3, 4] \times 1/T$, and the relative average power of the paths is $[0, -1, -9, -10, -13]$ dB. Again, consider Zak-OTFS operating in the crystalline regime ($\nu_p = 15$ kHz).

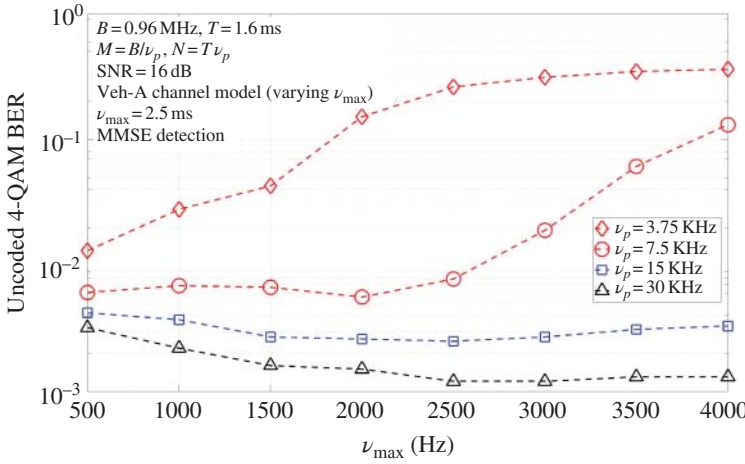


Figure 2.35 Model-free performance as a function of $\nu_{\max}$ for the Veh-A channel model introduced in Section 2.9. When the Doppler spread $2\nu_{\max}$ is significantly less than ν_p , the performance does not degrade as $\nu_{\max}$ increases. When the Doppler spread $2\nu_{\max}$ is close to ν_p , the performance degrades because of Doppler domain aliasing. When operating deep in the crystalline regime, the BER performance is consistently very good over a wide range of Dopplers. Source: Mohammed et al. [34] IEEE.

Figure 2.34 illustrates that for this resolvable channel, the model-dependent performance almost coincides with that of performance with perfect knowledge of the I/O relation. This is expected since the estimation of the channel spreading function is accurate when paths are resolvable. Furthermore, the model-free performance is only slightly inferior, despite the high Doppler spread of 4.375 kHz. Why the small degradation? Pulse shaping filters cause self-interaction/aliasing between the received pilot in the fundamental period and its quasi-periodic aliases. Though this aliasing is small in the crystalline regime, it is non-zero²¹.

Next, through Fig. 2.35, the model-free performance in the crystalline regime is explored as we traverse the hyperbola $\tau_p \nu_p = 1$, moving toward FDM. The Veh-A channel introduced in Section 2.9 is considered, fixing the SNR at 16 dB. The performance for Doppler periods $\nu_p = 3.75, 7.5, 15$, and 30 kHz is studied as $\nu_{\max}$ varies between 500 Hz and 4 kHz. For $\nu_p = 15, 30$ kHz, the system operates deep in the crystalline regime since the Doppler period is significantly larger than the Doppler spread (which is at most 8 kHz), and the corresponding delay period (66.66, 33.33 μ s) is significantly larger than the delay spread (2.5 μ s). The spacing between quasi-periodic aliases limits self-interaction even with sinc pulse shaping filters. The Zak-OTFS I/O relation is predictable, and, therefore, the taps of the effective channel filter can be accurately estimated. Figure 2.35 illustrates that increasing the spacing from 15 to 30 kHz improves performance slightly. For both values of Doppler periods, the BER performance is very good and remains almost invariant to increasing Doppler spread. The BER performance changes when the Doppler period is reduced to 7.5 kHz. When the Doppler spread $2\nu_{\max}$ is less than 4 kHz, the interaction between the received DD pilot in the fundamental period and its quasi-periodic aliases is not significant. As the Doppler spread increases beyond 4 kHz, BER increases steadily. When the Doppler spread is the same as

²¹ From Corollary 2.4 in Appendix 2.A.1, it follows that the DD domain spread of the effective continuous DD domain channel filter $h_{\text{eff}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu)$ is the sum of the DD spread of $h_{\text{phy}}(\tau, \nu)$ and the DD spread of the transmit and receive DD pulse shaping filters $w_{\text{tx}}(\tau, \nu)$ and $w_{\text{rx}}(\tau, \nu)$. Hence, the maximum effective DD spread of $h_{\text{eff}}(\tau, \nu)$ is more than $\tau_{\max}$ along the delay domain and more than $2\nu_{\max}$ along the Doppler domain (see (2.74)).

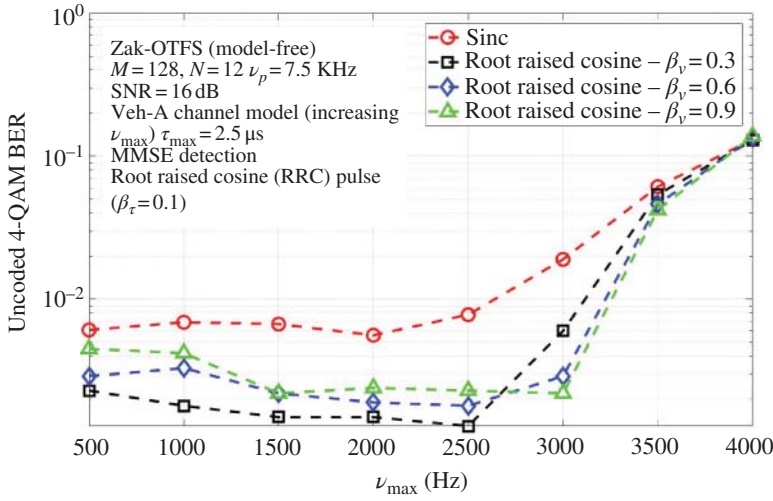


Figure 2.36 Better pulse shaping filters extend the region of reliable model-free operation. BER performance with sinc pulse and RRC pulses with $\beta_r = 0.1$, and $\beta_v = 0.3, 0.6$, and 0.9 as $\nu_{\max}$ varies between 500 Hz and 4 kHz. Source: Mohammed et al. [34] IEEE.

the Doppler period ($\nu_{\max} = 3.75$ kHz), the performance degradation becomes quite severe. This is because the sinc pulse shaping filters leak energy outside their ideal delay width ($1/B = 1.04 \mu\text{s}$) and Doppler width ($1/T = 625$ Hz). On further reducing the Doppler period to 3.75 kHz, the BER performance starts to degrade at a smaller Doppler shift of $\nu_{\max} = 500$ Hz.

Next, through Fig. 2.36, it is shown that better pulse shaping filters extend the region of reliable model-free operation by reducing aliasing. DD domain RRC pulses (specified in (2.81)) are considered for the pulse shaping filters, parameterized by a roll-off parameter β_r that controls localization in delay and a parameter β_v that controls localization in Doppler. The sinc pulse corresponds to $\beta_r = \beta_v = 0$, and localization improves as the parameters β_r and β_v increase from 0 to 1. Better localization implies lesser DD domain aliasing and, therefore, accurate prediction/estimation of the I/O relation for a higher Doppler spread. Fix $\beta_r = 0.1$, and for $\beta_v = 0.3, 0.6$, and 0.9 , the BER performance over the Veh-A channel is measured as $\nu_{\max}$ varies between 500 Hz and 4 kHz (fixed SNR of 16 dB, $M = 128, N = 12, \nu_p = 7.5$ KHz). Figure 2.36 illustrates that increasing β_v from 0 (a sinc pulse) to 0.3 extends the range of Doppler spreads for which the BER performance is flat (from 4 to 5 kHz). Increasing β_v further to 0.6 and 0.9 extends the range still further. The cost of introducing better filters is a reduction in spectral efficiency. When the sinc filter is replaced by a RRC filter, the frame duration is increased by a factor $(1 + \beta_v)$ and the frame bandwidth is increased by a factor $(1 + \beta_r)$.

2.11 Summary

In doubly spread channels, TDM and FDM exhibit non-predictable and fading I/O relation. This is because TDM and FDM are suited to delay-only and Doppler-only channels, respectively, but not to doubly spread channels. Zak-OTFS modulation, on the other hand, is naturally suited for doubly spread channels. Zak-OTFS modulation is parameterized by a delay domain period and a Doppler

Table 2.6 Attributes of TDM/FDM/Zak-OTFS I/O relation.

Attribute	Channel type	TDM	FDM	Zak-OTFS
Non-fading	Delay spread only	✓	×	✓
	Doppler spread only	×	✓	✓
	Doubly spread	×	×	✓
Predictable	Delay spread only	✓	×	✓
	Doppler spread only	×	✓	✓
	Doubly spread	×	×	✓

Source: Mohammed et al. [33] IEEE.

domain period. When operated in the crystalline regime (delay period and Doppler period are sufficiently larger than the delay spread and Doppler spread, respectively, of the *effective* channel), Zak-OTFS exhibits a predictable and non-fading I/O relation (see Table 2.6).

The predictable and non-fading attribute of the Zak-OTFS I/O relation is depicted in Fig. 2.37 as we move along the hyperbola $\tau_p \nu_p = 1$. The average received power profile is flat (i.e., no-fading), and the I/O relation is predictable when the crystallization condition is satisfied (i.e., when the delay and Doppler periods are sufficiently larger than the respective channel spreads). When we move out of the crystalline regime, the Zak-OTFS I/O relation is non-predictable and fading (i.e., the average received power profile is not flat).

Appendix 2.A Cascade of Two Doubly Spread Channels and Twisted Convolution

Theorem 2.2 For the cascade of two doubly spread channels, as shown in Fig. 2.9, the cascaded channel is also doubly spread and its effective DD spreading function $h_{\text{phy}}(\tau, \nu)$ is given by the twisted convolution of the DD spreading functions of the two channels in the cascade, i.e.,

$$h_{\text{phy}}(\tau, \nu) = h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu). \quad (2.A.1)$$

Proof: Consider the cascade of two doubly spread channels, as shown in Fig. 2.9. Let $y_1(t)$ denote the TD output of channel 1 having DD spreading function $h_1(\tau, \nu)$. Then, from (2.2), it follows that

$$y_1(t) = \iint h_1(\tau'', \nu'') x(t - \tau'') e^{j2\pi\nu''(t-\tau'')} d\tau'' d\nu''. \quad (2.A.2)$$

The output of channel 1 is input to channel 2 having DD spreading function $h_2(\tau, \nu)$. The output of channel 2 is, therefore, given by

$$y(t) = \iint h_2(\tau', \nu') y_1(t - \tau') e^{j2\pi\nu'(t-\tau')} d\tau' d\nu'. \quad (2.A.3)$$

Using (2.A.2) in (2.A.3) gives

$$y(t) = \iiint \iint h_2(\tau', \nu') h_1(\tau'', \nu'') x(t - \tau' - \tau'') e^{j2\pi\nu''(t-\tau'-\tau'')} e^{j2\pi\nu'(t-\tau')} d\tau' d\nu' d\tau'' d\nu''. \quad (2.A.4)$$

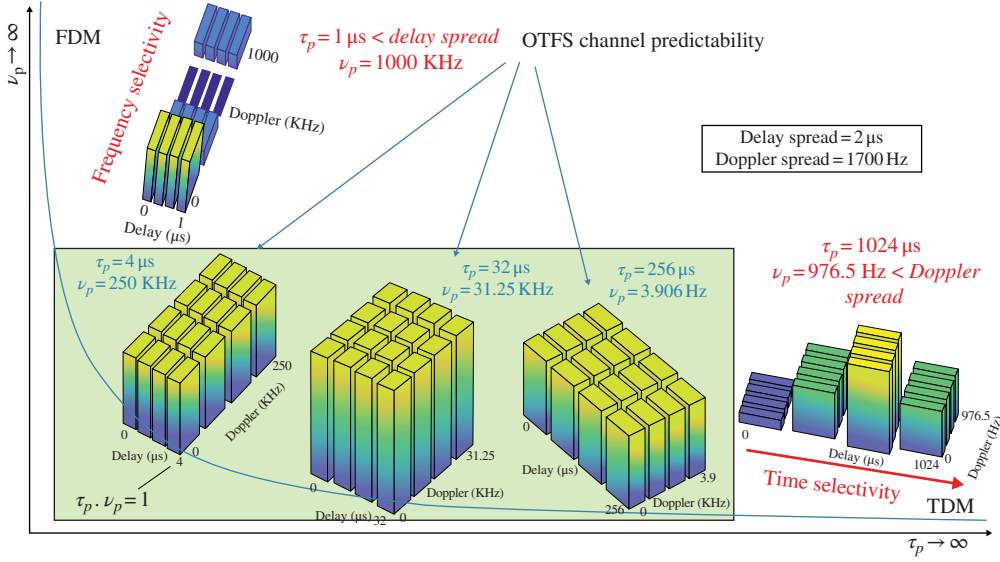


Figure 2.37 For a fixed $M = N = 4$, the average power of the received discrete DD domain signal is plotted for varying delay and Doppler domain period (τ_p, ν_p) under the constraint $\tau_p \nu_p = 1$. The channel considered is that in Fig. 2.4. As $\tau_p \rightarrow 0$, the delay period becomes smaller than the effective channel delay spread due to which there is aliasing along the delay domain, resulting in variation in the received power, i.e., fading along the Doppler domain. Zak-OTFS $\rightarrow$ FDM as $\tau_p \rightarrow 0$, which explains frequency selectivity in FDM. On the other hand, as $\tau_p \rightarrow \infty$, $\nu_p = (1/\tau_p) \rightarrow 0$, and the Doppler period becomes smaller than the effective channel Doppler spread due to which there is aliasing along the Doppler domain, resulting in fading along the delay domain. Zak-OTFS $\rightarrow$ TDM as $\tau_p \rightarrow \infty$, which explains time selectivity in TDM. However, when the delay period and Doppler period are greater than the effective channel delay spread and Doppler spread, respectively, there is no aliasing and the Zak-OTFS I/O relation is predictable and exhibits no fading (the average received power profile is flat and appears like the surface of *crystalline solids*). Source: Mohammed et al. [33] IEEE.

Substituting the integration variable pair (τ'', τ') with the pair (τ, τ') , $\tau = \tau' + \tau''$, and, similarly, substituting (ν'', ν') with the pair (ν, ν') , $\nu = \nu' + \nu''$, it follows that

$$\begin{aligned}
 y(t) &= \iiint h_2(\tau', \nu') h_1(\tau - \tau', \nu - \nu') x(t - \tau) e^{j2\pi(\nu - \nu')(t - \tau)} e^{j2\pi\nu'(t - \tau + (\tau - \tau'))} d\tau' d\nu' d\tau \\
 &= \iint x(t - \tau) e^{j2\pi\nu(t - \tau)} \underbrace{\left[\iint h_2(\tau', \nu') h_1(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \right]}_{h_{\text{phy}}(\tau, \nu)} d\tau d\nu \\
 &= \iint x(t - \tau) e^{j2\pi\nu(t - \tau)} h_{\text{phy}}(\tau, \nu) d\tau d\nu,
 \end{aligned} \tag{2.A.5}$$

which is the same as the input–output relation of a doubly spread channel (see (2.2)) with the DD spreading function

$$h_{\text{phy}}(\tau, \nu) = \iint h_2(\tau', \nu') h_1(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu'. \tag{2.A.6}$$

The above operation between the DD domain functions $h_2(\tau, \nu)$ and $h_1(\tau, \nu)$ is referred to as *twisted convolution* and is denoted by $*_{\sigma}$, i.e.,

$$h_{\text{phy}}(\tau, \nu) = h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu). \quad (2.A.7)$$

The nomenclature “twisted” is primarily due to the term $e^{j2\pi\nu'(\tau-\tau')}$ in the integrand in (2.A.6), without which the RHS would simply be linear 2D convolution. The DD spreading function for a cascade of two doubly spread channels cannot be described by a simple 2D linear convolution of the DD spreading functions of the cascaded channels, as is revealed by the following example.

Example 2.1 Consider the cascade of a delay-only and Doppler-only channel. The cascade input and output are $x(t)$ and $y(t)$, respectively. The DD spreading function of the delay-only and Doppler-only channels is given by $h_{\text{dly}}(\tau, \nu) = \delta(\tau - \tau_1)\delta(\nu)$ and $h_{\text{dplr}}(\tau, \nu) = \delta(\tau)\delta(\nu - \nu_1)$, respectively. Here, τ_1 and ν_1 are the single-path delay and Doppler induced by the delay-only channel and the Doppler-only channel, respectively. When the cascade order is such that the delay-only channel output (i.e., $y_1(t)$) is input to the Doppler-only channel, the cascade output is given by

$$\begin{aligned} y(t) &= \iint h_{\text{dplr}}(\tau, \nu) y_1(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \\ &= \iint \delta(\tau)\delta(\nu - \nu_1) y_1(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \\ &= y_1(t) e^{j2\pi\nu_1 t}, \end{aligned} \quad (2.A.8)$$

where $y_1(t)$ is the output of the delay-only channel, which is given by

$$\begin{aligned} y_1(t) &= \iint h_{\text{dly}}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \\ &= \iint \delta(\nu)\delta(\tau - \tau_1) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \\ &= x(t - \tau_1). \end{aligned} \quad (2.A.9)$$

From (2.A.8) and (2.A.9), it follows that for this cascade order (the delay-only channel followed by the Doppler-only channel), the I/O relation is

$$y(t) = x(t - \tau_1) e^{j2\pi\nu_1 t}. \quad (2.A.10)$$

If the effective DD spreading function of the cascade of two channels were to be linear convolution of their DD spreading functions, then the effective $h(\tau, \nu)$ for the cascade in this example would have been

$$\begin{aligned} &\iint h_{\text{dplr}}(\tau', \nu') h_{\text{dly}}(\tau - \tau', \nu - \nu') d\tau' d\nu' \\ &= \iint \delta(\tau') \delta(\nu' - \nu_1) \delta(\tau - \tau' - \tau_1) \delta(\nu - \nu') d\tau' d\nu' \\ &= \delta(\tau - \tau_1) \delta(\nu - \nu_1), \end{aligned} \quad (2.A.11)$$

for which the cascade output would be

$$\iint \delta(\tau - \tau_1) \delta(\nu - \nu_1) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu = x(t - \tau_1) e^{j2\pi\nu_1(t-\tau_1)}. \quad (2.A.12)$$

This, however, is not the same as the correct cascade output $x(t - \tau_1) e^{j2\pi\nu_1 t}$ (see (2.A.10)). This shows that the effective DD spreading function of the cascade of two doubly spread channels cannot be described by a linear convolution of the DD spreading function of the cascaded channels.

2.A.1 Properties of Twisted Convolution

Corollary 2.1 Twisted convolution is associative but non-commutative, i.e., for any DD domain functions $h_1(\tau, \nu)$, $h_2(\tau, \nu)$, and $h_3(\tau, \nu)$,

$$\begin{aligned} h_3(\tau, \nu) *_{\sigma} (h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu)) &= (h_3(\tau, \nu) *_{\sigma} h_2(\tau, \nu)) *_{\sigma} h_1(\tau, \nu) \\ &= h_3(\tau, \nu) *_{\sigma} h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu). \end{aligned} \quad (2.A.13)$$

Also, in general,

$$h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu) \neq h_1(\tau, \nu) *_{\sigma} h_2(\tau, \nu). \quad (2.A.14)$$

Proof: Associativity of twisted convolution follows directly by explicit evaluation of the LHS and RHS in (2.A.13), by using the definition of twisted convolution in (2.A.6).

Non-commutativity of twisted convolution can be seen from the above example of the cascade of a delay-only and Doppler-only channel. As seen above, when the cascade consists of the delay-only channel first followed by the Doppler only channel, the cascade output is $x(t - \tau_1) e^{j2\pi\nu_1 t}$ and the effective DD spreading function of the cascade is $h_{\text{dplr}}(\tau, \nu) *_{\sigma} h_{\text{dly}}(\tau, \nu)$. However, when the order of the channels in the cascade is reversed, it is seen that the output is $x(t - \tau_1) e^{j2\pi\nu_1(t - \tau_1)}$ which is different. Now the effective spreading function is $h_{\text{dly}}(\tau, \nu) *_{\sigma} h_{\text{dplr}}(\tau, \nu)$. For the same cascade input $x(t)$, the cascade output $y(t)$ is different for different cascading order, and, therefore, $h_{\text{dplr}}(\tau, \nu) *_{\sigma} h_{\text{dly}}(\tau, \nu) \neq h_{\text{dly}}(\tau, \nu) *_{\sigma} h_{\text{dplr}}(\tau, \nu)$.

Corollary 2.2 Delay-only and Doppler-only channels are special instances of a doubly spread channel. For a general delay-only channel, the DD spreading function is $a(\tau)\delta(\nu)$, where $a(t)$ is TD channel impulse response. Similarly, for a general Doppler-only channel, the DD spreading function is $\delta(\tau)b(\nu)$, where $b(f)$ is the FD channel impulse response of the Doppler-only channel.

Proof: For a delay-only channel, for TD channel input $x(t)$, the TD channel output is

$$\begin{aligned} y(t) &= \iint h_{\text{phy}}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu \\ &= \iint a(\tau)\delta(\nu) x(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu \\ &= \int a(\tau) x(t - \tau) d\tau, \end{aligned} \quad (2.A.15)$$

which is simply TD linear convolution and $a(t)$ is the TD channel impulse response. For a Doppler-only channel, with FD channel input $X(f)$ and TD channel input $x(t) = \int X(f) e^{j2\pi ft} df$, the TD channel output is given by

$$\begin{aligned} y(t) &= \iint h_{\text{phy}}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu \\ &= \iint \delta(\tau) b(\nu) x(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu \\ &= x(t) \int b(\nu) e^{j2\pi\nu t} d\nu. \end{aligned} \quad (2.A.16)$$

The FD channel output is therefore given by

$$\begin{aligned}
 Y(f) &= \int y(t) e^{-j2\pi ft} dt \\
 &= \int b(\nu) \underbrace{\left(\int x(t) e^{-j2\pi(f-\nu)t} dt \right)}_{=X(f-\nu)} d\nu \\
 &= \int b(\nu) X(f-\nu) d\nu,
 \end{aligned} \tag{2.A.17}$$

which is simply FD linear convolution and $b(f)$ is the FD channel impulse response.

The following result is on the support set of the twisted convolution of two DD domain functions $a(\tau, \nu)$ and $b(\tau, \nu)$ having finite support.

Corollary 2.3 Let $a(\tau, \nu)$ and $b(\tau, \nu)$ be two DD domain functions having DD domain support sets $\mathcal{A}$ and $\mathcal{B}$, respectively, i.e., $|a(\tau, \nu)|$ and $|b(\tau, \nu)|$ are non-zero only for $(\tau, \nu) \in \mathcal{A}$ and $(\tau, \nu) \in \mathcal{B}$, respectively. Then, the support set of $a(\tau, \nu) *_{\sigma} b(\tau, \nu)$ and $b(\tau, \nu) *_{\sigma} a(\tau, \nu)$ are both subsets of the direct sum of the support sets $\mathcal{A}$ and $\mathcal{B}$, i.e.,

$$\begin{aligned}
 \text{Support}(a(\tau, \nu) *_{\sigma} b(\tau, \nu)) &\subseteq \mathcal{A} \oplus \mathcal{B}, \\
 \text{Support}(b(\tau, \nu) *_{\sigma} a(\tau, \nu)) &\subseteq \mathcal{A} \oplus \mathcal{B}, \\
 \text{where } \mathcal{A} \oplus \mathcal{B} &\triangleq \{(\tau_a + \tau_b, \nu_a + \nu_b) \mid (\tau_a, \nu_a) \in \mathcal{A}, (\tau_b, \nu_b) \in \mathcal{B}\}.
 \end{aligned} \tag{2.A.18}$$

Proof: From (2.A.6), it follows that

$$\begin{aligned}
 c(\tau, \nu) &\stackrel{\Delta}{=} a(\tau, \nu) *_{\sigma} b(\tau, \nu) \\
 &= \iint_{(\tau', \nu') \in \mathcal{A}} a(\tau', \nu') b(\tau - \tau', \nu - \nu') e^{j2\pi \nu'(\tau - \tau')} d\tau' d\nu'.
 \end{aligned} \tag{2.A.19}$$

Clearly, for a given (τ, ν) , $c(\tau, \nu)$ is zero if for all $(\tau', \nu') \in \mathcal{A}$, $b(\tau - \tau', \nu - \nu') = 0$, i.e., $(\tau - \tau', \nu - \nu') \notin \mathcal{B}$, equivalently, $(\tau, \nu) \notin \{(\tau' + \tau_b, \nu' + \nu_b) \mid (\tau_b, \nu_b) \in \mathcal{B}\}$. In other words, $c(\tau, \nu)$ is zero whenever (τ, ν) does not belong to the set

$$\bigcup_{(\tau', \nu') \in \mathcal{A}} \{(\tau' + \tau_b, \nu' + \nu_b) \mid (\tau_b, \nu_b) \in \mathcal{B}\}. \tag{2.A.20}$$

However, $c(\tau, \nu)$ could be zero even when (τ, ν) belongs to this set. A similar argument also holds for the support set of $b(\tau, \nu) *_{\sigma} a(\tau, \nu)$. This set in (2.A.20) is, therefore, a super set of the support set of both $a(\tau, \nu) *_{\sigma} b(\tau, \nu)$ and $b(\tau, \nu) *_{\sigma} a(\tau, \nu)$. This set is in fact $\mathcal{A} \oplus \mathcal{B}$, which completes the proof.

The following result shows that the delay and Doppler spread of the twisted convolution of two DD domain functions/filters is upper bounded by the sum of the respective delay and Doppler spreads.

Corollary 2.4 Let $a(\tau, \nu)$ be a continuous DD domain function/filter having finite support set $\mathcal{A}$ (i.e., $\iint_{(\tau, \nu) \in \mathcal{A}} d\tau d\nu < \infty$). The delay and Doppler spread of $a(\tau, \nu)$ is given by

$$\begin{aligned}
 \Delta\tau_a &\stackrel{\Delta}{=} \max_{(\tau_1, \nu_1), (\tau_2, \nu_2) \in \mathcal{A}} |\tau_2 - \tau_1|, \\
 \Delta\nu_a &\stackrel{\Delta}{=} \max_{(\tau_1, \nu_1), (\tau_2, \nu_2) \in \mathcal{A}} |\nu_2 - \nu_1|.
 \end{aligned} \tag{2.A.21}$$

Let $b(\tau, \nu)$ be another continuous DD domain function/filter having finite support set $\mathcal{B}$ and delay (Doppler) spread denoted by $\Delta\tau_b$ ($\Delta\nu_b$). The delay and Doppler spread of their twisted convolution is then upper bounded by $(\Delta\tau_a + \Delta\tau_b)$ and $(\Delta\nu_a + \Delta\nu_b)$, respectively.

Proof: From Corollary 2.3, it is known that the support set of $c(\tau, \nu) \stackrel{\Delta}{=} a(\tau, \nu) *_\sigma b(\tau, \nu)$ is a subset of $\mathcal{A} \oplus \mathcal{B}$. Therefore, the delay spread of $c(\tau, \nu)$ satisfies

$$\begin{aligned}
 \Delta\tau_c &\leq \max_{(\tau_1, \nu_1), (\tau_2, \nu_2) \in \mathcal{A} \oplus \mathcal{B}} |\tau_2 - \tau_1|, \\
 &= \max_{(\tau_{a,1}, \nu_{a,1}), (\tau_{a,2}, \nu_{a,2}) \in \mathcal{A}, (\tau_{b,1}, \nu_{b,1}), (\tau_{b,2}, \nu_{b,2}) \in \mathcal{B}} |(\tau_{a,2} + \tau_{b,2}) - (\tau_{a,1} + \tau_{b,1})| \\
 &= \max_{(\tau_{a,1}, \nu_{a,1}), (\tau_{a,2}, \nu_{a,2}) \in \mathcal{A}, (\tau_{b,1}, \nu_{b,1}), (\tau_{b,2}, \nu_{b,2}) \in \mathcal{B}} |(\tau_{a,2} - \tau_{a,1}) + (\tau_{b,2} - \tau_{b,1})| \\
 &\leq \max_{(\tau_{a,1}, \nu_{a,1}), (\tau_{a,2}, \nu_{a,2}) \in \mathcal{A}, (\tau_{b,1}, \nu_{b,1}), (\tau_{b,2}, \nu_{b,2}) \in \mathcal{B}} [|\tau_{a,2} - \tau_{a,1}| + |\tau_{b,2} - \tau_{b,1}|] \\
 &= \max_{(\tau_{a,1}, \nu_{a,1}), (\tau_{a,2}, \nu_{a,2}) \in \mathcal{A}} |\tau_{a,2} - \tau_{a,1}| + \max_{(\tau_{b,1}, \nu_{b,1}), (\tau_{b,2}, \nu_{b,2}) \in \mathcal{B}} |\tau_{b,2} - \tau_{b,1}| \\
 &= \Delta\tau_a + \Delta\tau_b,
 \end{aligned} \tag{2.A.22}$$

where the second step follows from the definition of the set $\mathcal{A} \oplus \mathcal{B}$ in (2.A.18). The fourth step follows from the triangular inequality, and the last step follows from the definition of $\Delta\tau_a$ and $\Delta\tau_b$. Similarly, it can be shown that the Doppler spread of the set $\mathcal{A} \oplus \mathcal{B}$ is less than equal to $(\Delta\nu_a + \Delta\nu_b)$.

Appendix 2.B Channel Action on DD Domain Signal

In TDM (delay-only channel) and FDM (Doppler-only channel), the channel acts on the TD/FD information signal through TD/FD linear convolution (see Corollary 2.2 in Appendix 2.A.1). Similarly, in DD domain modulation, the DD domain realization of the received TD signal is expected to be given by some operation ($\otimes$) between the DD representation of the channel, i.e., the channel DD spreading function, and the DD realization of the transmitted TD signal,²² i.e.,

$$\begin{aligned}
 y_{dd}(\tau, \nu) &= \mathcal{T}(y(t)) = h_{\text{phy}}(\tau, \nu) \otimes \mathcal{T}(x(t)) \\
 &= h_{\text{phy}}(\tau, \nu) \otimes x_{dd}(\tau, \nu).
 \end{aligned} \tag{2.B.1}$$

In the following, it is argued that the operation $\otimes$ is *twisted convolution*, i.e., the channel action on a DD domain signal and the cascade of two doubly spread channels are described by the same operation, which is twisted convolution. To appreciate this, consider the channel cascade in Fig. 2.9. An equivalent representation for this channel cascade is shown in Fig. 2.B.1, whose input and output are the same as those in Fig. 2.9.

From Fig. 2.9, it follows that

$$y_{dd}(\tau, \nu) = (h_2(\tau, \nu) *_\sigma h_1(\tau, \nu)) \otimes x_{dd}(\tau, \nu). \tag{2.B.2}$$

From the equivalent representation in Fig. 2.B.1, it follows that

$$\begin{aligned}
 y_{dd}(\tau, \nu) &= h_2(\tau, \nu) \otimes y_{dd,1}(\tau, \nu) \\
 &= h_2(\tau, \nu) \otimes (h_1(\tau, \nu) \otimes x_{dd}(\tau, \nu)).
 \end{aligned} \tag{2.B.3}$$

²² Note that $x(t) = \mathcal{T}^{-1}(x_{dd}(\tau, \nu))$, and, therefore, $\mathcal{T}(x(t)) = x_{dd}(\tau, \nu)$.

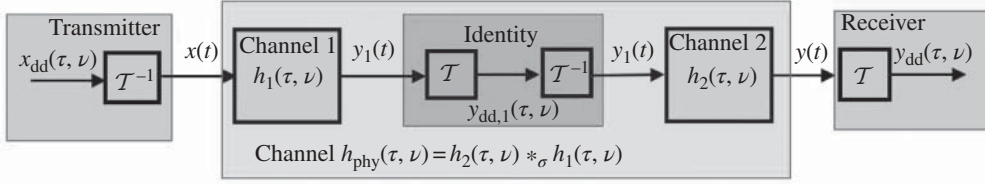


Figure 2.B.1 Equivalent channel representation for the channel cascade in Fig. 2.9.

From the expressions in (2.B.2) and (2.B.3), it follows that the operation $\circledast$ needs to be such that for any $h_1(\tau, \nu)$, $h_2(\tau, \nu)$, and $x_{dd}(\tau, \nu)$,

$$(h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu)) \circledast x_{dd}(\tau, \nu) = h_2(\tau, \nu) \circledast (h_1(\tau, \nu) \circledast x_{dd}(\tau, \nu)). \quad (2.B.4)$$

Since twisted convolution is associative, i.e., $(h_2(\tau, \nu) *_{\sigma} h_1(\tau, \nu)) *_{\sigma} x_{dd}(\tau, \nu) = h_2(\tau, \nu) *_{\sigma} (h_1(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu))$ (see Corollary 2.1), a good choice for the operation $\circledast$ which satisfies (2.B.4) is the twisted convolution operation.

Appendix 2.C Zak Transform

It is shown that with Zak transform as the transform $\mathcal{T}$ for TD to DD conversion, the action of a doubly spread channel is described by twisted convolution.

Theorem 2.3 In Fig. 2.10, the DD domain representation $y_{dd}(\tau, \nu) = \mathcal{Z}_t(y(t))$ of the channel TD output $y(t)$ is given by

$$\begin{aligned} y_{dd}(\tau, \nu) &= \mathcal{Z}_t(y(t)) = h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu), \\ x_{dd}(\tau, \nu) &\triangleq \mathcal{Z}_t(x(t)). \end{aligned} \quad (2.C.1)$$

Proof:

$$\begin{aligned} y_{dd}(\tau, \nu) &= \mathcal{Z}_t(y(t)) \\ &\stackrel{(a)}{=} \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} y(\tau + k\tau_p) e^{-j2\pi\nu k\tau_p} \\ &= \sqrt{\tau_p} \sum_{k=-\infty}^{\infty} \left[\iint h_{\text{phy}}(\tau', \nu') x(\tau + k\tau_p - \tau') e^{j2\pi\nu'(\tau + k\tau_p - \tau')} d\tau' d\nu' \right] e^{-j2\pi\nu k\tau_p} \\ &\quad \underbrace{= y(\tau + k\tau_p) \text{ (see (2.2))}} \\ &\stackrel{(b)}{=} \iint h_{\text{phy}}(\tau', \nu') \left[\underbrace{\sqrt{\tau_p} \sum_{k=-\infty}^{\infty} x(\tau - \tau' + k\tau_p) e^{-j2\pi k\tau_p(\nu - \nu')}}_{= x_{dd}(\tau - \tau', \nu - \nu'), \text{ see (2.23)}} \right] e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \\ &= \iint h_{\text{phy}}(\tau', \nu') x_{dd}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \\ &= h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu), \end{aligned} \quad (2.C.2)$$

where step (a) follows from (2.23). In step (b), note that the term within square brackets is the Zak transform of $x(t)$ evaluated at $(\tau - \tau', \nu - \nu')$. Since $x(t)$ is the inverse time-Zak transform of $x_{dd}(\tau, \nu)$,

it follows that the term within the square brackets is $x_{\text{dd}}(\tau - \tau', \nu - \nu')$. The last step follows from the definition of the twisted convolution operation in (2.20).

The next theorem shows that the inner product between two TD signals is conserved in the DD domain.

Theorem 2.4 Given two TD signals $x(t)$ and $y(t)$ and their respective DD domain realizations $x_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(x(t))$ and $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(y(t))$, the inner product between $x(t)$ and $y(t)$ is conserved in the DD domain, i.e.,

$$\int_{-\infty}^{\infty} x(t) y^*(t) dt = \int_0^{\tau_p} \int_0^{\nu_p} x_{\text{dd}}(\tau, \nu) y_{\text{dd}}^*(\tau, \nu) d\nu d\tau. \quad (2.C.3)$$

Proof: Using (2.23), the DD domain representations of $x(t)$ and $y(t)$ are given by

$$\begin{aligned} x_{\text{dd}}(\tau, \nu) &= \sqrt{\tau_p} \sum_{k_1 \in \mathbb{Z}} x(\tau + k_1 \tau_p) e^{-j2\pi k_1 \nu \tau_p}, \\ y_{\text{dd}}(\tau, \nu) &= \sqrt{\tau_p} \sum_{k_2 \in \mathbb{Z}} y(\tau + k_2 \tau_p) e^{-j2\pi k_2 \nu \tau_p}. \end{aligned} \quad (2.C.4)$$

Using these expressions on the RHS in (2.C.3), it follows that

$$\begin{aligned} \int_0^{\tau_p} \int_0^{\nu_p} x_{\text{dd}}(\tau, \nu) y_{\text{dd}}^*(\tau, \nu) d\nu d\tau &= \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} \int_0^{\tau_p} \int_0^{\nu_p} x(\tau + k_1 \tau_p) y^*(\tau + k_2 \tau_p) e^{j2\pi \nu (k_2 - k_1) \tau_p} d\nu d\tau \\ &= \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} \int_0^{\tau_p} x(\tau + k_1 \tau_p) y^*(\tau + k_2 \tau_p) \left[\tau_p \int_0^{\nu_p} e^{j2\pi \nu (k_2 - k_1) \tau_p} d\nu \right] d\tau \\ &= \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} \int_0^{\tau_p} x(\tau + k_1 \tau_p) y^*(\tau + k_2 \tau_p) \delta[k_2 - k_1] d\tau \\ &= \sum_{k_1 \in \mathbb{Z}} \int_0^{\tau_p} x(\tau + k_1 \tau_p) y^*(\tau + k_1 \tau_p) d\tau = \int_{-\infty}^{\infty} x(t) y^*(t) dt. \end{aligned} \quad (2.C.5)$$

Appendix 2.D Derivation of (2.30) and (2.31)

From (2.27), it is clear that the DD response of only the i th channel path with gain h_i , delay τ_i , and Doppler ν_i to a DD domain signal $x_{\text{dd}}(\tau, \nu)$ is

$$h_i e^{j2\pi \nu_i (\tau - \tau_i)} x_{\text{dd}}(\tau - \tau_i, \nu - \nu_i).$$

Since the overall received signal is a superposition of the signal received along all the four paths, the DD channel response to the input DD signal $x_{\text{dd}}(\tau, \nu)$ is

$$\sum_{i=1}^4 h_i e^{j2\pi \nu_i (\tau - \tau_i)} x_{\text{dd}}(\tau - \tau_i, \nu - \nu_i).$$

Therefore, the DD domain channel response (i.e., a -response) to the DD impulse signal at (τ_a, ν_a) is obtained by substituting $x_{\text{dd}}(\tau, \nu)$ in the above equation with $x_{\text{dd}}(\tau, \nu; \tau_a, \nu_a)$. Furthermore, substituting $x_{\text{dd}}(\tau, \nu; \tau_a, \nu_a)$ with its expression from (2.28), the expression for the a -response is given by

$$\begin{aligned}
 y_{\text{dd}}(\tau, \nu; \tau_a, \nu_a) &= \sum_{i=1}^4 h_i e^{j2\pi\nu_i(\tau-\tau_i)} x_{\text{dd}}(\tau - \tau_i, \nu - \nu_i; \tau_a, \nu_a) \\
 &= \sum_{i=1}^4 h_i \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi\nu_i(\tau-\tau_i)} e^{j2\pi n\nu_a\tau_p} \delta(\tau - n\tau_p - \tau_a - \tau_i) \delta(\nu - m\nu_p - \nu_a - \nu_i) \\
 &= \sum_{i=1}^4 h_i \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi\nu_i(n\tau_p+\tau_a)} e^{j2\pi n\nu_a\tau_p} \delta(\tau - n\tau_p - \tau_a - \tau_i) \delta(\nu - m\nu_p - \nu_a - \nu_i) \\
 &= \sum_{i=1}^4 h_i e^{j2\pi\nu_i\tau_a} \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n\tau_p(\nu_a+\nu_i)} \delta(\tau - n\tau_p - (\tau_a + \tau_i)) \delta(\nu - m\nu_p - (\nu_a + \nu_i)) \\
 &= \sum_{i=1}^4 h_i e^{j2\pi\nu_i\tau_a} x_{\text{dd}}(\tau, \nu; \tau_a + \tau_i, \nu_a + \nu_i), \tag{2.D.1}
 \end{aligned}$$

where the last equality follows directly by substituting (τ_a, ν_a) with $(\tau_a + \tau_i, \nu_a + \nu_i)$ in the expression for $x_{\text{dd}}(\tau, \nu; \tau_a, \nu_a)$ (see (2.28)). The expression for $y_{\text{dd}}(\tau, \nu; \tau_b, \nu_b)$ is similar but with (τ_a, ν_a) replaced by (τ_b, ν_b) .

Appendix 2.E Proof of Non-overlapping Received Impulses in a -response

Constituent impulses in the a -response, which are received along the same path, do not overlap irrespective of whether the crystallization condition is satisfied or not (in (2.30) the impulses received along the i th path are located at $\tau_a + \tau_i + n\tau_p, \nu_a + \nu_i + m\nu_p, n, m \in \mathbb{Z}$. When the crystallization condition in (2.32) is satisfied, any two impulses in the a -response received along different paths do not overlap. The condition for such an overlap is that, for some $i \neq k, i, k = 1, 2, 3, 4$, and some $n_1, n_2, m_1, m_2 \in \mathbb{Z}$,

$$\begin{aligned}
 \tau_a + \tau_i + n_1\tau_p &= \tau_a + \tau_k + n_2\tau_p, \\
 \nu_a + \nu_i + m_1\nu_p &= \nu_a + \nu_k + m_2\nu_p. \tag{2.E.1}
 \end{aligned}$$

Next, it is shown that the overlapping condition in (2.E.1) can *never* be satisfied when the crystallization condition (2.32) holds. The condition in (2.E.1) is equivalent to the fact that

$$(\tau_i - \tau_k) = (n_2 - n_1)\tau_p, \quad (\nu_i - \nu_k) = (m_2 - m_1)\nu_p \tag{2.E.2}$$

for some $i \neq k$ and $n_1, n_2, m_1, m_2 \in \mathbb{Z}$. In other words, this overlapping condition is only satisfied if the difference between the delay and Doppler shifts of two distinct paths is integer multiple of τ_p and ν_p , respectively. However, when the crystallization condition holds, for any $i, k = 1, 2, 3, 4$, $|\tau_i - \tau_k| < \tau_p$ and $|\nu_i - \nu_k| < \nu_p$. Therefore, the only possible solution to (2.E.2) is $(n_2 - n_1) = (m_2 - m_1) = 0$. This, however, implies that $\tau_i = \tau_k$ and $\nu_i = \nu_k$ which is not possible as the paths are distinct.

Appendix 2.F Inverse Zak Transforms

Here, Appendix 2.F.1 describes the inverse time-Zak transform, which gives the TD realization of a quasi-periodic DD domain signal. Using this, in Appendix 2.F.2, the expression for the TD realization of the quasi-periodic DD pulse $\sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n v_0 \tau_p} \delta(\tau - \tau_0 - n\tau_p) \delta(v - v_0 - m v_p)$ at $(\tau_0, v_0) \in \mathcal{D}_0$ is derived (see (2.26)). In Appendix 2.F.3, the inverse frequency-Zak transform is described.

2.F.1 Inverse Time-Zak Transform

The TD realization of a quasi-periodic DD signal $x_{\text{dd}}(\tau, v)$ is given by [56]

$$x(t) = \mathcal{Z}_t^{-1}(x_{\text{dd}}(\tau, v)) \stackrel{\Delta}{=} \sqrt{\tau_p} \int_0^{v_p} x_{\text{dd}}(t, v) dv. \quad (2.F.1)$$

To show this, start with the RHS above and substitute the expression for $x_{\text{dd}}(\cdot, \cdot)$ inside the integrand with the Zak transform expression in (2.23) as

$$\begin{aligned} \sqrt{\tau_p} \int_0^{v_p} x_{\text{dd}}(t, v) dv &= \sum_{k=-\infty}^{\infty} \tau_p \int_0^{v_p} x(t + k\tau_p) e^{-j2\pi v k \tau_p} dv \\ &= \sum_{k=-\infty}^{\infty} x(t + k\tau_p) \left(\tau_p \int_0^{v_p} e^{-j2\pi v k \tau_p} dv \right). \end{aligned} \quad (2.F.2)$$

Note that

$$\tau_p \int_0^{v_p} e^{-j2\pi v k \tau_p} dv = \begin{cases} 0, & k \neq 0 \\ 1, & k = 0. \end{cases} \quad (2.F.3)$$

Using (2.F.3) in (2.F.2), it follows that

$$\sqrt{\tau_p} \int_0^{v_p} x_{\text{dd}}(t, v) dv = x(t), \quad (2.F.4)$$

which is the LHS of (2.F.1).

2.F.2 Derivation of (2.38)

Using the inverse time-Zak transform in (2.F.1), the TD realization of the DD pulse at (τ_0, v_0) is given by

$$\begin{aligned} x(t) &= \sqrt{\tau_p} \int_0^{v_p} \left[\sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n v_0 \tau_p} \delta(t - \tau_0 - n\tau_p) \delta(v - v_0 - m v_p) \right] dv \\ &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{j2\pi n v_0 \tau_p} \delta(t - \tau_0 - n\tau_p) \left[\sum_{m \in \mathbb{Z}} \int_0^{v_p} \delta(v - v_0 - m v_p) dv \right], \end{aligned} \quad (2.F.5)$$

where

$$\begin{aligned} \sum_{m \in \mathbb{Z}} \int_0^{\nu_p} \delta(\nu - \nu_0 - m\nu_p) d\nu &= \sum_{m \in \mathbb{Z}} \int_{-m\nu_p}^{-(m-1)\nu_p} \delta(\nu - \nu_0) d\nu \\ &= \int_{-\infty}^{\infty} \delta(\nu - \nu_0) d\nu = 1. \end{aligned} \quad (2.F.6)$$

Using (2.F.6) in (2.F.5) gives (2.38).

2.F.3 Inverse Frequency-Zak Transform

The inverse frequency-Zak transform gives the FD realization of a quasi-periodic DD domain signal $x_{\text{dd}}(\tau, \nu)$ and is given by [56]

$$X(f) = \mathcal{Z}_f^{-1}(x_{\text{dd}}(\tau, \nu)) \triangleq \sqrt{\nu_p} \int_0^{\tau_p} x_{\text{dd}}(\tau, f) e^{-j2\pi f \tau} d\tau. \quad (2.F.7)$$

To show this, start with the RHS above and substitute the expression for $x_{\text{dd}}(\cdot, \cdot)$ inside the integrand with the Zak transform expression in (2.23) as

$$\begin{aligned} \sqrt{\nu_p} \int_0^{\tau_p} x_{\text{dd}}(\tau, f) e^{-j2\pi f \tau} d\tau &= \sum_{k \in \mathbb{Z}} \left[\int_0^{\tau_p} x(\tau + k\tau_p) e^{-j2\pi k f \tau_p} e^{-j2\pi f \tau} d\tau \right] \\ &= \sum_{k \in \mathbb{Z}} \left[\int_0^{\tau_p} x(\tau + k\tau_p) e^{-j2\pi f(\tau + k\tau_p)} d\tau \right] \\ &= \sum_{k \in \mathbb{Z}} \left[\int_{k\tau_p}^{(k+1)\tau_p} x(t) e^{-j2\pi f t} dt \right] \\ &= \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt = X(f), \end{aligned} \quad (2.F.8)$$

where the third step follows from substituting the integration variable τ by $t = (\tau + k\tau_p)$ for the k th integral.

Appendix 2.G Twisted Convolution Preserves Quasi-periodicity

In the following, it is shown that the twisted convolution of a quasi-periodic signal $b_{\text{dd}}(\tau, \nu)$ with any arbitrary DD domain filter $a(\tau, \nu)$ results in a DD domain signal $c(\tau, \nu) = a(\tau, \nu) *_{\sigma} b_{\text{dd}}(\tau, \nu)$, which is also quasi-periodic. This then implies the quasi-periodicity of $x_{\text{dd}}^{\text{wix}}(\tau, \nu; \tau_0, \nu_0)$ in (2.39). From the definition of twisted convolution, it follows that

$$\begin{aligned} c(\tau, \nu) &= a(\tau, \nu) *_{\sigma} b_{\text{dd}}(\tau, \nu) \\ &= \iint a(\tau', \nu') b_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi \nu'(\tau - \tau')} d\tau' d\nu'. \end{aligned} \quad (2.G.1)$$

For arbitrary $m, n \in \mathbb{Z}$, it follows that

$$\begin{aligned}
 c(\tau + n\tau_p, \nu + m\nu_p) &\stackrel{(a)}{=} \iint a(\tau', \nu') b_{\text{dd}}(\tau + n\tau_p - \tau', \nu + m\nu_p - \nu') e^{j2\pi\nu'(\tau + n\tau_p - \tau')} d\tau' d\nu' \\
 &\stackrel{(b)}{=} \iint a(\tau', \nu') b_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi n\tau_p(\nu - \nu')} e^{j2\pi\nu'(\tau + n\tau_p - \tau')} d\tau' d\nu' \\
 &= e^{j2\pi n\tau_p\nu} \iint a(\tau', \nu') b_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \\
 &\stackrel{(c)}{=} e^{j2\pi n\tau_p\nu} c(\tau, \nu),
 \end{aligned} \tag{2.G.2}$$

where steps (a) and (c) follow from (2.G.1) and step (b) follows from the quasi-periodicity of $b_{\text{dd}}(\tau, \nu)$ (see (2.25)). Since $c(\tau + n\tau_p, \nu + m\nu_p) = e^{j2\pi n\tau_p\nu} c(\tau, \nu)$ for any arbitrary $m, n \in \mathbb{Z}$, $c(\tau, \nu)$ satisfies the quasi-periodicity condition in (2.25), and, therefore, it is quasi-periodic.

Appendix 2.H Derivation of TD Pulsone Expression in (2.41)

Substituting the expression for $x_{\text{dd}}^{w_{\text{ix}}}(\tau, \nu; \tau_0, \nu_0)$ from (2.39) into the inverse time-Zak transform in (2.F.1) (see Appendix 2.F.1), it follows that

$$\begin{aligned}
 x(t) &= \sqrt{\tau_p} \sum_{m, n=-\infty}^{\infty} \int_0^{\nu_p} w_{\text{ix}}(t - n\tau_p - \tau_0, \nu - m\nu_p - \nu_0) e^{j2\pi\nu(n\tau_p + \tau_0)} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} d\nu \\
 &= \sqrt{\tau_p} \sum_{m, n=-\infty}^{\infty} \int_0^{\nu_p} W_1(t - n\tau_p - \tau_0) W_2(\nu - m\nu_p - \nu_0) e^{j2\pi\nu(n\tau_p + \tau_0)} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} d\nu \\
 &= \sqrt{\tau_p} \sum_{n=-\infty}^{\infty} W_1(t - n\tau_p - \tau_0) \left[\sum_{m=-\infty}^{\infty} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \int_0^{\nu_p} e^{j2\pi\nu(n\tau_p + \tau_0)} W_2(\nu - m\nu_p - \nu_0) d\nu \right].
 \end{aligned} \tag{2.H.1}$$

In the m th term of the inner summation, replacing the integration variable ν with $\nu' = \nu - m\nu_p - \nu_0$, it follows that

$$\begin{aligned}
 e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \int_0^{\nu_p} e^{j2\pi\nu(n\tau_p + \tau_0)} W_2(\nu - m\nu_p - \nu_0) d\nu \\
 = e^{j2\pi\nu_0 n\tau_p} \int_{-m\nu_p - \nu_0}^{-(m-1)\nu_p - \nu_0} e^{j2\pi\nu'(\tau_0 + n\tau_p)} W_2(\nu') d\nu'.
 \end{aligned} \tag{2.H.2}$$

Substituting the expression on the RHS of (2.H.2) in (2.H.1), it follows that

$$x(t) = \sqrt{\tau_p} \sum_{n=-\infty}^{\infty} W_1(t - n\tau_p - \tau_0) e^{j2\pi\nu_0 n\tau_p} \left[\sum_{m=-\infty}^{\infty} \int_{-m\nu_p - \nu_0}^{-(m-1)\nu_p - \nu_0} e^{j2\pi\nu'(\tau_0 + n\tau_p)} W_2(\nu') d\nu' \right]. \tag{2.H.3}$$

Substituting integration variable ν' with f , it then follows that

$$x(t) = \sqrt{\tau_p} \sum_{n=-\infty}^{\infty} W_1(t - n\tau_p - \tau_0) e^{j2\pi\nu_0 n\tau_p} \left[\int_{-\infty}^{\infty} W_2(f) e^{j2\pi f(\tau_0 + n\tau_p)} df \right]. \tag{2.H.4}$$

This completes the derivation.

Appendix 2.I Derivation of FD Pulsone Expression in (2.44)

Substituting the expression for $x_{\text{dd}}^{u_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ from (2.39) into the inverse frequency-Zak transform in (2.F.7) (see Appendix 2.F.3), it follows that

$$\begin{aligned}
 X(f) &= \sqrt{v_p} \sum_{m,n=-\infty}^{\infty} \int_0^{\tau_p} w_{\text{tx}}(\tau - n\tau_p - \tau_0, f - m\nu_p - \nu_0) e^{j2\pi f(n\tau_p + \tau_0)} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} e^{-j2\pi\tau f} d\tau \\
 &= \sqrt{v_p} \sum_{m,n=-\infty}^{\infty} \int_0^{\tau_p} W_1(\tau - n\tau_p - \tau_0) W_2(f - m\nu_p - \nu_0) e^{-j2\pi f(\tau - n\tau_p - \tau_0)} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} d\tau \\
 &= \sqrt{v_p} \sum_{m=-\infty}^{\infty} W_2(f - m\nu_p - \nu_0) e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \left[\sum_{n=-\infty}^{\infty} \int_0^{\tau_p} W_1(\tau - n\tau_p - \tau_0) e^{-j2\pi f(\tau - n\tau_p - \tau_0)} d\tau \right].
 \end{aligned} \tag{2.I.1}$$

In the n th term of the inner summation, replacing the integration variable τ with $\tau' = \tau - n\tau_p - \tau_0$, it follows that

$$\sum_{n=-\infty}^{\infty} \int_0^{\tau_p} W_1(\tau - n\tau_p - \tau_0) e^{-j2\pi f(\tau - n\tau_p - \tau_0)} d\tau = \sum_{n=-\infty}^{\infty} \int_{-n\tau_p - \tau_0}^{-(n-1)\tau_p - \tau_0} W_1(\tau') e^{-j2\pi f\tau'} d\tau'. \tag{2.I.2}$$

Substituting the expression on the RHS of (2.I.2) in (2.I.1), it follows that

$$X(f) = \sqrt{v_p} \sum_{m=-\infty}^{\infty} W_2(f - m\nu_p - \nu_0) e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \left[\sum_{n=-\infty}^{\infty} \int_{-n\tau_p - \tau_0}^{-(n-1)\tau_p - \tau_0} W_1(\tau') e^{-j2\pi f\tau'} d\tau' \right]. \tag{2.I.3}$$

Substituting integration variable τ' with t , it follows that

$$X(f) = \sqrt{v_p} \sum_{m=-\infty}^{\infty} W_2(f - m\nu_p - \nu_0) e^{-j2\pi\tau_0(\nu_0 + m\nu_p)} \left[\int_{-\infty}^{\infty} W_1(t) e^{-j2\pi ft} dt \right]. \tag{2.I.4}$$

This completes the derivation.

Appendix 2.J TDM I/O Relation

The following theorem states the TDM I/O relation.

Theorem 2.5 It follows from Table 2.3 that the output signal $y_{\text{td}}[k], k \in \mathbb{Z}$ is obtained from the input signal $x_{\text{td}}[k], k \in \mathbb{Z}$ by a discrete time convolution

$$\begin{aligned}
 y_{\text{td}}[k'] &= \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] h_{\text{td}}[k' - k; k] + n_{\text{td}}[k'], \text{ where, for } n \in \mathbb{Z}, \\
 h_{\text{td}}[n; k] &\triangleq \iint \left[e^{j2\pi\nu \frac{k}{B}} h_{\text{phy}}(\tau, \nu) w_{\text{rx}}(\tau') e^{j2\pi\nu(\frac{n}{B} - \tau - \tau')} w_{\text{tx}}\left(\frac{n}{B} - \tau - \tau'\right) \right] d\tau d\nu d\tau', \\
 n_{\text{td}}[k'] &\triangleq \int w_{\text{rx}}(\tau') n_{\text{td}}\left(\frac{k'}{B} - \tau'\right) d\tau'.
 \end{aligned} \tag{2.J.1}$$

Here, $n_{\text{td}}(t)$ is the AWGN at the receiver input.

Proof: See Appendix 2.J.1.

The discrete TD filter $h_{\text{td}}[\cdot; k]$ represents the effective discrete TD channel response to the k th discrete input symbol $x_{\text{td}}[k]$. The dependence of $h_{\text{td}}[\cdot; k]$ on k is not simple, and knowledge of this filter $h_{\text{td}}[\cdot; q]$ for some integer q is not sufficient to accurately predict $h_{\text{td}}[\cdot; k]$ for all $k \in \mathbb{Z}$. Also, the total received signal energy for the k th input $x_{\text{td}}[k]$ is proportional to $\sum_{k' \in \mathbb{Z}} |h_{\text{td}}[k' - k; k]|^2$, which depends on k , i.e., the received signal energy is not constant but varies. Therefore, for a generic doubly spread channel, the TDM input–output relation is *non-predictable*, *fading*, and *non-stationary* (cf. Section 2.2.1).

If there is no Doppler shift, then $h_{\text{phy}}(\tau, \nu) = g(\tau)\delta(\nu)$. Let $h_{\text{td}}(t) \triangleq w_{\text{rx}}(t) \star g(t) \star w_{\text{tx}}(t)$. By substituting $h_{\text{phy}}(\tau, \nu) = g(\tau)\delta(\nu)$ in (2.J.1), we get $h_{\text{td}}[n; k] = h_{\text{td}}(t = n/B)$, which is independent of k . In this case, the TDM input–output relation is predictable, non-fading, and stationary.

2.J.1 Proof of Theorem 2.5

From Table 2.3, it follows that

$$x_{\text{td}}^{w_{\text{tx}}}(t) = w_{\text{tx}}(t) \star x_{\text{td}}(t) = \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] w_{\text{tx}}(t - k/B). \quad (2.J.2)$$

Since the TD transmit signal $s_{\text{td}}(t) = x_{\text{td}}^{w_{\text{tx}}}(t)$, the received signal is given by

$$\begin{aligned} r_{\text{td}}(t) &= \iint h_{\text{phy}}(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n_{\text{td}}(t) \\ &\stackrel{(a)}{=} \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] \iint h_{\text{phy}}(\tau, \nu) w_{\text{tx}}(t - \tau - k/B) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n_{\text{td}}(t), \end{aligned} \quad (2.J.3)$$

where $n_{\text{td}}(t)$ is AWGN at the receiver and step (a) follows from (2.J.2). Next, the received TD signal after pulse shaping is given by

$$\begin{aligned} y_{\text{td}}^{w_{\text{rx}}}(t) &= w_{\text{rx}}(t) \star r_{\text{td}}(t) = \int w_{\text{rx}}(\tau') r_{\text{td}}(t - \tau') d\tau' \\ &\stackrel{(a)}{=} \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] \iiint w_{\text{rx}}(\tau') h_{\text{phy}}(\tau, \nu) w_{\text{tx}}(t - \tau - \tau' - k/B) e^{j2\pi\nu(t-\tau-\tau')} d\tau d\nu d\tau' \\ &\quad + \int w_{\text{rx}}(\tau') n_{\text{td}}(t - \tau') d\tau', \end{aligned} \quad (2.J.4)$$

where step (a) follows from the expression of $r_{\text{td}}(t)$ in (2.J.3). The received pulse-shaped TD signal is then sampled at integer multiples of $1/B$, resulting in the discrete time received signal $y_{\text{td}}[k']$, $k' \in \mathbb{Z}$, given by

$$\begin{aligned} y_{\text{td}}[k'] &= y_{\text{td}}^{w_{\text{rx}}}(t = k'/B) \\ &\stackrel{(a)}{=} \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] \iiint w_{\text{rx}}(\tau') h_{\text{phy}}(\tau, \nu) w_{\text{tx}}\left(\frac{k'}{B} - \tau - \tau' - \frac{k}{B}\right) e^{j2\pi\nu\left(\frac{k'}{B} - \tau - \tau'\right)} d\tau d\nu d\tau' \\ &\quad + \int w_{\text{rx}}(\tau') n_{\text{td}}\left(\frac{k'}{B} - \tau'\right) d\tau' \\ &= \sum_{k \in \mathbb{Z}} x_{\text{td}}[k] h_{\text{td}}[k' - k; k] + n_{\text{td}}[k'], \end{aligned} \quad (2.J.5)$$

where

$$h_{\text{fd}}[n; k] \triangleq \iiint w_{\text{rx}}(\tau') h_{\text{phy}}(\tau, \nu) w_{\text{tx}}\left(\frac{n}{B} - \tau - \tau'\right) e^{j2\pi\nu\frac{k}{B}} e^{j2\pi\nu(\frac{n}{B} - \tau - \tau')} d\tau d\nu d\tau', \quad (2.J.6)$$

$$n_{\text{fd}}[k'] \triangleq \int w_{\text{rx}}(\tau') n_{\text{td}}\left(\frac{k'}{B} - \tau'\right) d\tau', \quad (2.J.7)$$

and step (a) follows from (2.J.4).

Appendix 2.K FDM I/O Relation

The following theorem states the FDM input–output relation.

Theorem 2.6 It follows from Table 2.3 that the output signal $y_{\text{fd}}[k], k \in \mathbb{Z}$, is obtained from the input signal $x_{\text{fd}}[k], k \in \mathbb{Z}$, by the discrete FD convolution as

$$y_{\text{fd}}[k'] = \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] h_{\text{fd}}[k' - k; k] + n_{\text{fd}}[k'], \quad (2.K.1)$$

where

$$h_{\text{fd}}[n; k] = \iiint \left[e^{-j2\pi\tau\frac{k}{T}} h_{\text{phy}}(\tau, \nu) w_{\text{rx}}(f') e^{-j2\pi\tau(\frac{n}{T} - f')} w_{\text{tx}}\left(\frac{n}{T} - \nu - f'\right) \right] d\tau d\nu df', \quad (2.K.2)$$

$$n_{\text{fd}}[k'] \triangleq \iint w_{\text{rx}}(f') n_{\text{td}}(t) e^{-j2\pi(\frac{k'}{T} - f')t} dt df'. \quad (2.K.3)$$

Proof: See Appendix 2.K.1.

The discrete FD filter $h_{\text{fd}}[\cdot; k]$ represents the effective discrete FD channel response to the k th discrete input symbol $x_{\text{fd}}[k]$. The dependence of $h_{\text{fd}}[\cdot; k]$ on k is not simple, and knowledge of this filter $h_{\text{fd}}[\cdot; q]$ for some integer q is not sufficient to accurately predict $h_{\text{fd}}[\cdot; k]$ for all $k \in \mathbb{Z}$. Also, the total received signal energy for the k th input $x_{\text{fd}}[k]$ is proportional to $\sum_{k' \in \mathbb{Z}} |h_{\text{fd}}[k' - k; k]|^2$, which depends on k , i.e., the received signal energy is not constant but varies. Therefore, for a generic doubly spread channel, the FDM input–output relation is *non-predictable*, *fading*, and *non-stationary* (cf. Section 2.2.2).

If there are no path delays (Doppler-only channel), then $h_{\text{phy}}(\tau, \nu) = \delta(\tau)g(\nu)$. Let $h_{\text{fd}}(f) \triangleq w_{\text{rx}}(f) \star g(f) \star w_{\text{tx}}(f)$. By substituting $h_{\text{phy}}(\tau, \nu) = \delta(\tau)g(\nu)$ in (2.K.2), we get $h_{\text{fd}}[n; k] = h_{\text{fd}}(f = n/T)$, which is independent of k . In this special case, the FDM I/O relation is predictable, non-fading, and stationary.

2.K.1 Proof of Theorem 2.6

From Table 2.3, it follows that

$$x_{\text{fd}}^{w_{\text{tx}}}(f) = w_{\text{tx}}(f) \star x_{\text{fd}}(f) = \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] w_{\text{tx}}\left(f - \frac{k}{T}\right). \quad (2.K.4)$$

The TD transmit signal is then given by

$$\begin{aligned} s_{\text{td}}(t) &= \int x_{\text{fd}}^{w_{\text{tx}}}(f) e^{j2\pi ft} df \stackrel{(a)}{=} \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \int w_{\text{tx}}\left(f - \frac{k}{T}\right) e^{j2\pi ft} df \\ &= \left(\sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] e^{j2\pi \frac{k}{T} t} \right) w_{\text{tx,td}}(t), \end{aligned} \quad (2.K.5)$$

where $w_{\text{tx,td}}(t) \stackrel{\Delta}{=} \int w_{\text{tx}}(f) e^{j2\pi ft} df$ is the TD realization of the FD transmit pulse $w_{\text{tx}}(f)$. In step (a), the expression of $x_{\text{fd}}^{w_{\text{tx}}}(f)$ has been used from (2.K.4). The received TD signal is then given by

$$\begin{aligned} r_{\text{td}}(t) &= \iint h_{\text{phy}}(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi \nu(t - \tau)} d\tau d\nu + n_{\text{td}}(t) \\ &\stackrel{(a)}{=} \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \iint h_{\text{phy}}(\tau, \nu) e^{j2\pi(t - \tau)\frac{k}{T}} w_{\text{tx,td}}(t - \tau) e^{j2\pi \nu(t - \tau)} d\tau d\nu + n_{\text{td}}(t), \end{aligned} \quad (2.K.6)$$

where step (a) follows from (2.K.5). The received FD signal is then given by

$$\begin{aligned} y_{\text{fd}}(f) &\stackrel{(a)}{=} \int r_{\text{td}}(t) e^{-j2\pi ft} dt \\ &\stackrel{(b)}{=} \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \iint h_{\text{phy}}(\tau, \nu) e^{-j2\pi f\tau} \left[\int w_{\text{tx,td}}(t - \tau) e^{-j2\pi\left(f - \nu - \frac{k}{T}\right)(t - \tau)} dt \right] d\tau d\nu \\ &\quad + \int n_{\text{td}}(t) e^{-j2\pi ft} dt \\ &\stackrel{(c)}{=} \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \iint h_{\text{phy}}(\tau, \nu) e^{-j2\pi f\tau} w_{\text{tx}}\left(f - \nu - \frac{k}{T}\right) d\tau d\nu + \int n_{\text{td}}(t) e^{-j2\pi ft} dt, \end{aligned} \quad (2.K.7)$$

where in step (a), the expression of $r_{\text{td}}(t)$ from (2.K.6) is substituted to arrive at step (b). Step (c) follows from the fact that, in step (b), the integral $\left[\int w_{\text{tx,td}}(t - \tau) e^{-j2\pi\left(f - \nu - \frac{k}{T}\right)(t - \tau)} dt \right]$ is simply the FD pulse $w_{\text{tx}}(f)$ shifted by $(\nu + k/T)$. Next, the received FD signal $y_{\text{fd}}(f)$ is filtered with a FD pulse, resulting in the pulse-shaped received FD signal

$$\begin{aligned} y_{\text{fd}}^{w_{\text{rx}}}(f) &\stackrel{(a)}{=} w_{\text{rx}}(f) \star y_{\text{fd}}(f) = \int w_{\text{rx}}(f') y_{\text{fd}}(f - f') df' \\ &\stackrel{(b)}{=} \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \iiint h_{\text{phy}}(\tau, \nu) e^{-j2\pi(f - f')\tau} w_{\text{tx}}\left(f - f' - \nu - \frac{k}{T}\right) w_{\text{rx}}(f') d\tau d\nu df' \\ &\quad + \iint w_{\text{rx}}(f') n_{\text{td}}(t) e^{-j2\pi(f - f')t} dt df', \end{aligned} \quad (2.K.8)$$

where in step (a), the expression for $y_{\text{fd}}(f)$ from (2.K.7) has been substituted. The received pulse-shaped FD signal $y_{\text{fd}}^{w_{\text{rx}}}(f)$ is then sampled at integer multiples of $1/T$ in FD, resulting in the discrete FD received signal

$$\begin{aligned}
 y_{\text{fd}}[k'] &= y_{\text{fd}}^{w_{\text{rx}}} \left(f = \frac{k'}{T} \right) \\
 &= \sum_{k \in \mathbb{Z}} x_{\text{fd}}[k] \underbrace{\int \int \int e^{-j2\pi \frac{k}{T} \tau} h_{\text{phy}}(\tau, \nu) e^{-j2\pi \left(\frac{k'-k}{T} - f' \right) \tau} w_{\text{rx}}(f') w_{\text{tx}} \left(\frac{(k'-k)}{T} - \nu - f' \right) d\tau d\nu df'}_{\triangleq h_{\text{fd}}[k'-k; k]} \\
 &\quad + \underbrace{\int \int w_{\text{rx}}(f') n_{\text{td}}(t) e^{-j2\pi \left(\frac{k'}{T} - f' \right) t} dt df'}_{\triangleq n_{\text{fd}}[k']}, \tag{2.K.9}
 \end{aligned}$$

where the expression of $y_{\text{fd}}^{w_{\text{rx}}}(f)$ is from (2.K.8).

Appendix 2.L Discrete DD Domain Signals

A discrete DD domain signal arises when a quasi-periodic signal on the continuous DD domain is sampled on the information grid Λ_{dd} . Therefore, if $x_{\text{dd}}(\tau, \nu)$ is a quasi-periodic DD domain signal, the corresponding DD domain signal $x_{\text{dd}}[k, l]$, $k, l \in \mathbb{Z}$, is given by

$$x_{\text{dd}}[k, l] \triangleq x_{\text{dd}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right). \tag{2.L.1}$$

The following theorem shows that $x_{\text{dd}}[k, l]$ satisfies a quasi-periodicity condition in the discrete DD domain.

Theorem 2.7 The discrete DD domain signal $x_{\text{dd}}[k, l]$ obtained by sampling a continuous DD domain quasi-periodic signal $x_{\text{dd}}(\tau, \nu)$ on the information grid Λ_{dd} is quasi-periodic in the discrete DD domain with period M along the discrete delay domain and period N along the discrete Doppler domain, i.e., for all $k, l, n, m \in \mathbb{Z}$, it satisfies

$$x_{\text{dd}}[k + nM, l + mN] = e^{j2\pi \frac{nl}{N}} x_{\text{dd}}[k, l]. \tag{2.L.2}$$

Proof: From (2.L.1), it follows that

$$\begin{aligned}
 x_{\text{dd}}[k + nM, l + mN] &= x_{\text{dd}} \left(\tau = (k + nM) \frac{\tau_p}{M}, \nu = (l + mN) \frac{\nu_p}{N} \right) \\
 &= x_{\text{dd}} \left(n\tau_p + k \frac{\tau_p}{M}, m\nu_p + l \frac{\nu_p}{N} \right). \tag{2.L.3}
 \end{aligned}$$

It is given that $x_{\text{dd}}(\tau, \nu)$ is quasi-periodic, i.e., it satisfies (2.25), and, therefore,

$$\begin{aligned}
 x_{\text{dd}} \left(n\tau_p + k \frac{\tau_p}{M}, m\nu_p + l \frac{\nu_p}{N} \right) &= e^{j2\pi n\tau_p \left(l \frac{\nu_p}{N} \right)} \underbrace{x_{\text{dd}} \left(k \frac{\tau_p}{M}, l \frac{\nu_p}{N} \right)}_{=x_{\text{dd}}[k, l]} \\
 &= e^{j2\pi \frac{nl}{N}} x_{\text{dd}}[k, l] \tag{2.L.4}
 \end{aligned}$$

since $\tau_p \nu_p = 1$, and the last step follows from (2.L.1). Using (2.L.4) in (2.L.3) completes the proof.

Also, starting with any arbitrary discrete DD domain quasi-periodic signal $x_{\text{dd}}[k, l]$ satisfying (2.L.2), the corresponding continuous DD domain function $x_{\text{dd}}(\tau, \nu)$ is obtained by lifting $x_{\text{dd}}[k, l]$ to the information grid Λ_{dd} , i.e.,

$$x_{\text{dd}}(\tau, \nu) = \sum_{k, l \in \mathbb{Z}} x_{\text{dd}}[k, l] \delta \left(\tau - k \frac{\tau_p}{M} \right) \delta \left(\nu - l \frac{\nu_p}{N} \right). \quad (2.L.5)$$

The following is the converse of Theorem 2.7.

Theorem 2.8 If a discrete DD domain function $x_{\text{dd}}[k, l]$ satisfies the quasi-periodicity condition in (2.L.2), then the continuous DD domain function $x_{\text{dd}}(\tau, \nu)$ obtained by lifting $x_{\text{dd}}[k, l]$ to the information grid Λ_{dd} (as in (2.L.5)) is quasi-periodic, i.e., it satisfies (2.25).

Proof: For any $n, m \in \mathbb{Z}$,

$$\begin{aligned} x_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) &= \sum_{k, l \in \mathbb{Z}} x_{\text{dd}}[k, l] \delta \left(\tau + n\tau_p - k \frac{\tau_p}{M} \right) \delta \left(\nu + m\nu_p - l \frac{\nu_p}{N} \right) \\ &= \sum_{k, l \in \mathbb{Z}} x_{\text{dd}}[k, l] \delta \left(\tau - (k - nM) \frac{\tau_p}{M} \right) \delta \left(\nu - (l - mN) \frac{\nu_p}{N} \right) \\ &\stackrel{(a)}{=} \sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k' + nM, l' + mN] \delta \left(\tau - k' \frac{\tau_p}{M} \right) \delta \left(\nu - l' \frac{\nu_p}{N} \right) \\ &\stackrel{(b)}{=} \sum_{k', l' \in \mathbb{Z}} e^{j2\pi n \frac{l'}{N}} x_{\text{dd}}[k', l'] \delta \left(\tau - k' \frac{\tau_p}{M} \right) \delta \left(\nu - l' \frac{\nu_p}{N} \right) \\ &\stackrel{(c)}{=} \sum_{k', l' \in \mathbb{Z}} e^{j2\pi n \nu \tau_p} x_{\text{dd}}[k', l'] \delta \left(\tau - k' \frac{\tau_p}{M} \right) \delta \left(\nu - l' \frac{\nu_p}{N} \right) \\ &= e^{j2\pi n \nu \tau_p} x_{\text{dd}}(\tau, \nu), \end{aligned} \quad (2.L.6)$$

where in step (a), the summation variable k is substituted with $k' = k - nM$ and l is substituted with $l' = l - mN$. Step (b) follows from (2.L.2). Step (c) follows from the fact that the l' th term comprises a Dirac-delta at $\nu = l' \nu_p / N$. The last step follows from (2.L.5).

2.L.1 Impulse Signal in Discrete DD Domain

A continuous DD domain quasi-periodic impulse at $\left(\frac{k_1 \tau_p}{M}, \frac{l_1 \nu_p}{N} \right)$ when sampled on the information grid Λ_{dd} gives a quasi-periodic impulse signal in the discrete DD domain at (k_1, l_1) . Also, when a quasi-periodic impulse signal in the discrete DD domain at (k_1, l_1) is lifted to the continuous DD domain on the information grid, we get a continuous DD domain quasi-periodic impulse at $\left(\frac{k_1 \tau_p}{M}, \frac{l_1 \nu_p}{N} \right)$ (see Theorems 2.7 and 2.8 in Appendix 2.L). From (2.26), it follows that a quasi-periodic impulse in the continuous DD domain at $\left(\frac{k_1 \tau_p}{M}, \frac{l_1 \nu_p}{N} \right)$ is given by

$$\sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n \frac{l_1 \nu_p}{N} \tau_p} \delta \left(\tau - \frac{k_1 \tau_p}{M} - n\tau_p \right) \delta \left(\nu - \frac{l_1 \nu_p}{N} - m\nu_p \right). \quad (2.L.7)$$

When sampled on the information grid, i.e., $(\tau, \nu) = \left(\frac{k \tau_p}{M}, \frac{l \nu_p}{N} \right)$, we get

$$\sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi n \frac{l_1 \nu_p}{N}} \delta \left((k - k_1 - nM) \frac{\tau_p}{M} \right) \delta \left((l - l_1 - mN) \frac{\nu_p}{N} \right), \quad (2.L.8)$$

where $k, l \in \mathbb{Z}$. Since $k, l \in \mathbb{Z}$, the above signal is a discrete DD domain signal denoted by

$$x_{\text{dd}}[k, l] = \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \delta[k - (k_1 + nM)] \delta[l - (l_1 + mN)], \quad (2.L.9)$$

where $k, l \in \mathbb{Z}$. Note that this discrete DD domain signal consists of infinitely many discrete DD domain Dirac-delta impulses at $(k_1 + nM, l_1 + mN)$, $n, m \in \mathbb{Z}$, with any two Dirac-delta impulses separated by an integer multiple of M along the discrete delay domain and an integer multiple of N along the discrete Doppler domain (note that M and N are the discrete DD domain periods). Therefore, the discrete DD domain signal in (2.L.9) is referred to as the discrete DD domain impulse signal at (k_1, l_1) .

Appendix 2.M Derivation of Zak-OTFS Modulator Architecture

Starting from (2.54), we derive the modulator architecture in Fig. 2.24. From the linearity of the Zak and Inverse Zak transform, (2.54) is equivalent to

$$s_{\text{td}}(t) = \mathcal{Z}_t^{-1} \left(\sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} x[k, l] x_{\text{dd}}^{\text{wtx}}(\tau, \nu; k\tau_p/M, lv_p/N) \right). \quad (2.M.1)$$

Therefore,

$$\begin{aligned} \mathcal{Z}_t(s_{\text{td}}(t)) &= \sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} x[k, l] x_{\text{dd}}^{\text{wtx}}(\tau, \nu; k\tau_p/M, lv_p/N) \\ &\stackrel{(a)}{=} \sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} x[k, l] (w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu; k\tau_p/M, lv_p/N)) \\ &= w_{\text{tx}}(\tau, \nu) *_{\sigma} \left(\sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} x[k, l] x_{\text{dd}}(\tau, \nu; k\tau_p/M, lv_p/N) \right) \\ &\stackrel{(b)}{=} w_{\text{tx}}(\tau, \nu) *_{\sigma} \left(\sum_{n, m \in \mathbb{Z}} \sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} \underbrace{x[k, l] e^{j2\pi \frac{nl}{N}}}_{=x_{\text{dd}}[k+nM, l+mN]} \delta(\tau - n\tau_p - k\tau_p/M) \delta(\nu - m\nu_p - lv_p/N) \right) \\ &\stackrel{(c)}{=} w_{\text{tx}}(\tau, \nu) *_{\sigma} \left(\sum_{n, m \in \mathbb{Z}} \sum_{k=0}^{M-1N-1} \sum_{l=0}^{M-1N-1} x_{\text{dd}}[k+nM, l+mN] \delta\left(\tau - (k+nM)\frac{\tau_p}{M}\right) \right. \\ &\quad \left. \delta\left(\nu - (l+mN)\frac{\nu_p}{N}\right) \right) \\ &\stackrel{(d)}{=} w_{\text{tx}}(\tau, \nu) *_{\sigma} \underbrace{\left(\sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k', l'] \delta\left(\tau - k'\frac{\tau_p}{M}\right) \delta\left(\nu - l'\frac{\nu_p}{N}\right) \right)}_{=x_{\text{dd}}(\tau, \nu)} \\ &= w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu), \end{aligned} \quad (2.M.2)$$

where step (a) follows from (2.39) and step (b) follows from the general expression of $x_{\text{dd}}(\tau, \nu; \tau_0, \nu_0)$ in (2.26). Step (c) follows from the definition of the discrete DD domain information signal $x_{\text{dd}}[k, l]$ in (2.56) (see also “Generating the discrete information signal” step in Table 2.4). In step (c), $(k +$

nM) and $(l + mN)$ are substituted with k' and l' , respectively, thereby getting the expression in step (d). The bracketed term in step (d) is the continuous DD domain information signal $x_{\text{dd}}(\tau, \nu)$ defined in (2.57) (see also “Generating the information signal on the information grid” in Table 2.4). Hence, the TD Zak-OTFS modulated signal is equivalently given by

$$s_{\text{td}}(t) = \mathcal{Z}_t^{-1} \left(\underbrace{w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu)}_{=x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu)} \right) = \mathcal{Z}_t^{-1} (x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu)), \quad (2.M.3)$$

where the bracketed term is the filtered/pulse-shaped continuous DD domain information signal $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu)$ defined in (2.59) (see also “Shaping the pulse at the transmitter” in Table 2.4).

The modulator architecture shown in Fig. 2.24 follows from (2.M.2) and (2.M.3).

Appendix 2.N Derivation of Zak-OTFS Receiver

Since the noise at the receiver is AWGN, at the receiver, it suffices to compute the projections of the received TD signal $r_{\text{td}}(t)$ onto the MN carrier waveforms/TD pulsones $x_{\mathcal{Z}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{tx}}}(t)$, $k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$ (see (2.53) and (2.54)). The projection of $r_{\text{td}}(t)$ onto the (k, l) th TD pulson is

$$\begin{aligned} y_{\text{dd}}[k, l] &\triangleq \int r_{\text{td}}(t) \left(x_{\mathcal{Z}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{tx}}}(t) \right)^* dt \\ &\stackrel{(a)}{=} \int_0^{\tau_p} \int_0^{\nu_p} \mathcal{Z}_t(r_{\text{td}}(t)) \underbrace{\left[\mathcal{Z}_t \left(x_{\mathcal{Z}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{tx}}}(t) \right) \right]^*}_{=x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; k\tau_p/M, l\nu_p/N)} d\tau d\nu \\ &\stackrel{(b)}{=} \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) (x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; k\tau_p/M, l\nu_p/N))^* d\tau d\nu, \end{aligned} \quad (2.N.1)$$

where step (a) follows from Theorem 2.4 in Appendix 2.C and step (b) follows from (2.53). Step (b) also uses the fact that $y_{\text{dd}}(\tau, \nu)$ is the Zak transform of $r_{\text{td}}(t)$ (see (2.61)). Using the general expression of $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ in (2.39) for $(\tau_0, \nu_0) = (k\tau_p/M, l\nu_p/N)$ and substituting the resulting expression of $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; k\tau_p/M, l\nu_p/N)$ on the RHS of (2.N.1), it follows that

$$\begin{aligned} y_{\text{dd}}[k, l] &= \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) \left[\sum_{n, m \in \mathbb{Z}} w_{\text{tx}}^* \left(\tau - n\tau_p - k\frac{\tau_p}{M}, \nu - m\nu_p - l\frac{\nu_p}{N} \right) \right. \\ &\quad \left. e^{-j2\pi\nu(n\tau_p + k\frac{\tau_p}{M})} e^{j2\pi k\frac{\tau_p}{M}(l\frac{\nu_p}{N} + m\nu_p)} \right] d\tau d\nu \\ &\stackrel{(a)}{=} \sum_{n, m \in \mathbb{Z}} \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) w_{\text{tx}}^* \left(\tau - n\tau_p - k\frac{\tau_p}{M}, \nu - m\nu_p - l\frac{\nu_p}{N} \right) \\ &\quad e^{-j2\pi\nu(n\tau_p + k\frac{\tau_p}{M})} e^{j2\pi k\frac{\tau_p}{M}(l\frac{\nu_p}{N} + m\nu_p)} d\tau d\nu \end{aligned}$$

$$\begin{aligned}
&\stackrel{(b)}{=} \sum_{n,m \in \mathbb{Z}} \int_{(n-1)\tau_p + k\tau_p/M}^{n\tau_p + k\tau_p/M} \int_{(m-1)v_p + lv_p/N}^{mv_p + lv_p/N} \left[w_{\text{tx}}^*(-\tau', -\nu') y_{\text{dd}}(n\tau_p + k\tau_p/M - \tau', mv_p + lv_p/N - \nu') \right. \\
&\quad \left. e^{j2\pi k \frac{\tau_p}{M} \nu'} e^{j2\pi n\tau_p(\nu' - mv_p - lv_p/N)} \right] d\tau' d\nu',
\end{aligned} \tag{2.N.2}$$

where in step (a), for the integral in the (n, m) th term on the RHS, the integration variables τ and ν are substituted with $\tau' = (n\tau_p + k\tau_p/M - \tau)$ and $\nu' = (mv_p + lv_p/N - \nu)$, which gives the expression in step (b). Since $y_{\text{dd}}(\tau, \nu)$ is the Zak transform/DD representation of $r_{\text{id}}(t)$, it is a quasi-periodic DD function (see (2.25)), and, therefore,

$$\begin{aligned}
&y_{\text{dd}}\left(n\tau_p + k\frac{\tau_p}{M} - \tau', mv_p + l\frac{v_p}{N} - \nu'\right) \\
&= e^{j2\pi n\tau_p(mv_p + l\frac{v_p}{N} - \nu')} y_{\text{dd}}\left(k\frac{\tau_p}{M} - \tau', l\frac{v_p}{N} - \nu'\right) \\
&= e^{j2\pi \frac{nl}{N}} e^{-j2\pi n\tau_p \nu'} y_{\text{dd}}\left(k\frac{\tau_p}{M} - \tau', l\frac{v_p}{N} - \nu'\right).
\end{aligned} \tag{2.N.3}$$

Using (2.N.3) in (2.N.2), it follows that

$$\begin{aligned}
y_{\text{dd}}[k, l] &= \sum_{n,m \in \mathbb{Z}} \int_{(n-1)\tau_p + k\tau_p/M}^{n\tau_p + k\tau_p/M} \int_{(m-1)v_p + lv_p/N}^{mv_p + lv_p/N} \left[w_{\text{tx}}^*(-\tau', -\nu') y_{\text{dd}}(k\tau_p/M - \tau', lv_p/N - \nu') \right. \\
&\quad \left. e^{j2\pi k \frac{\tau_p}{M} \nu'} \right] d\tau' d\nu' \\
&\stackrel{(a)}{=} \iint w_{\text{tx}}^*(-\tau', -\nu') y_{\text{dd}}(k\tau_p/M - \tau', lv_p/N - \nu') e^{j2\pi k \frac{\tau_p}{M} \nu'} d\tau' d\nu' \\
&= \iint w_{\text{tx}}^*(-\tau', -\nu') e^{j2\pi \nu' \tau'} y_{\text{dd}}(k\tau_p/M - \tau', lv_p/N - \nu') e^{j2\pi \nu' (k\frac{\tau_p}{M} - \tau')} d\tau' d\nu',
\end{aligned} \tag{2.N.4}$$

where step (a) follows from the fact that the integrands are independent of (n, m) and that the union of the integral limit intervals for all $n, m \in \mathbb{Z}$ is the entire DD domain. Let

$$w_{\text{rx}}^{\text{mf}}(\tau, \nu) \triangleq w_{\text{tx}}^*(-\tau, -\nu) e^{j2\pi \nu \tau} \tag{2.N.5}$$

denote the DD domain matched-filter at the receiver corresponding to the transmit filter $w_{\text{tx}}(\tau, \nu)$. Using this in (2.N.4) gives

$$y_{\text{dd}}[k, l] = \iint w_{\text{rx}}^{\text{mf}}(\tau', \nu') y_{\text{dd}}\left(k\frac{\tau_p}{M} - \tau', l\frac{v_p}{N} - \nu'\right) e^{j2\pi \nu' (k\frac{\tau_p}{M} - \tau')} d\tau' d\nu'. \tag{2.N.6}$$

The twisted convolution of the matched-filter with the received DD domain signal $y_{\text{dd}}(\tau, \nu)$ is given by (see also (2.62))

$$\begin{aligned}
y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) &\triangleq w_{\text{rx}}^{\text{mf}}(\tau, \nu) *_\sigma y_{\text{dd}}(\tau, \nu) \\
&\stackrel{(a)}{=} \iint w_{\text{rx}}^{\text{mf}}(\tau', \nu') y_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi \nu' (\tau - \tau')} d\tau' d\nu',
\end{aligned} \tag{2.N.7}$$

where step (a) follows from the definition of twisted convolution in (2.20). Comparing the integrals on the RHS of (2.N.6) and (2.N.7), it follows that the projection of the (k, l) th TD pulsone on the received TD signal is equal to the DD domain symbol obtained by sampling the filtered received DD domain signal $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$ at the (k, l) th information lattice/grid point $(\tau, \nu) = \left(k \frac{\tau_p}{M}, l \frac{\nu_p}{N}\right)$, i.e.,

$$y_{\text{dd}}[k, l] = y_{\text{dd}}^{w_{\text{rx}}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right). \quad (2.N.8)$$

Appendix 2.O Proof of Theorem 2.1

From (2.24), it is known that a doubly spread channel acts on a DD domain input signal through twisted convolution (see Fig. 2.10). Therefore, from the Zak-OTFS transceiver signal processing in Fig. 2.24, it follows that

$$\begin{aligned} y_{\text{dd}}(\tau, \nu) &= \mathcal{Z}_t \left(r_{\text{td}}(t) \right) \\ &= \mathcal{Z}_t \left(\iint h(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu + n_{\text{td}}(t) \right) \\ &= \mathcal{Z}_t \left(\iint h(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu \right) + \mathcal{Z}_t \left(n_{\text{td}}(t) \right) \\ &= h_{\text{phy}}(\tau, \nu) *_{\sigma} \mathcal{Z}_t \left(s_{\text{td}}(t) \right) + \mathcal{Z}_t \left(n_{\text{td}}(t) \right) \\ &= h_{\text{phy}}(\tau, \nu) *_{\sigma} x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu) + \mathcal{Z}_t \left(n_{\text{td}}(t) \right). \end{aligned} \quad (2.O.1)$$

The last step follows from the fact that the transmit TD signal $s_{\text{td}}(t)$ is the inverse time-Zak transform of $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu)$. Using (2.O.1) in (2.62), it follows that

$$\begin{aligned} y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) &= w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu) \\ &\stackrel{(a)}{=} \left(w_{\text{rx}}(\tau, \nu) *_{\sigma} \left[h_{\text{phy}}(\tau, \nu) *_{\sigma} \left(w_{\text{tx}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu) \right) \right] \right) \\ &\quad + \underbrace{w_{\text{rx}}(\tau, \nu) *_{\sigma} \mathcal{Z}_t \left(n_{\text{td}}(t) \right)}_{\triangleq n_{\text{dd}}(\tau, \nu)} \\ &\stackrel{(b)}{=} \underbrace{\left(w_{\text{rx}}(\tau, \nu) *_{\sigma} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu) \right)}_{\triangleq h_{\text{eff}}(\tau, \nu)} *_{\sigma} x_{\text{dd}}(\tau, \nu) + n_{\text{dd}}(\tau, \nu), \end{aligned} \quad (2.O.2)$$

where step (a) follows from (2.59). Step (b) follows from the associativity of the twisted convolution operation (see Corollary 2.1 in Appendix 2.A.1). Furthermore,

$$\begin{aligned} h_{\text{eff}}(\tau, \nu) *_{\sigma} x_{\text{dd}}(\tau, \nu) &= \int \int h_{\text{eff}}(\tau', \nu') x_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau-\tau')} d\tau' d\nu' \\ &\stackrel{(a)}{=} \sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k', l'] \int \int h_{\text{eff}}(\tau', \nu') \delta \left(\tau - \tau' - \frac{k' \tau_p}{M} \right) \delta \left(\nu - \nu' - \frac{l' \nu_p}{N} \right) e^{j2\pi\nu'(\tau-\tau')} d\tau' d\nu' \\ &= \sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k', l'] h_{\text{eff}} \left(\tau - \frac{k' \tau_p}{M}, \nu - \frac{l' \nu_p}{N} \right) e^{j2\pi \left(\nu - \frac{l' \nu_p}{N} \right) \frac{k' \tau_p}{M}}, \end{aligned} \quad (2.O.3)$$

where step (a) follows from the expression of $x_{\text{dd}}(\tau, \nu)$ in (2.57). Using (2.O.3) in (2.O.2), it follows that

$$y_{\text{dd}}^{\text{wrx}}(\tau, \nu) = \sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k', l'] h_{\text{eff}}\left(\tau - \frac{k' \tau_p}{M}, \nu - \frac{l' \nu_p}{N}\right) e^{j2\pi\left(\nu - \frac{l' \nu_p}{N}\right) \frac{k' \tau_p}{M}} + n_{\text{dd}}(\tau, \nu). \quad (2.O.4)$$

Using (2.O.4) in (2.N.8), it follows that

$$\begin{aligned} y_{\text{dd}}[k, l] &= y_{\text{dd}}^{\text{wrx}}\left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N}\right) \\ &\stackrel{\Delta}{=} \overbrace{h_{\text{eff}}[k - k', l - l']} \\ &= \sum_{k', l' \in \mathbb{Z}} x_{\text{dd}}[k', l'] h_{\text{eff}}\left(\frac{k \tau_p}{M} - \frac{k' \tau_p}{M}, \frac{l \nu_p}{N} - \frac{l' \nu_p}{N}\right) e^{j2\pi\left(\frac{l \nu_p}{N} - \frac{l' \nu_p}{N}\right) \frac{k' \tau_p}{M}} \\ &\quad + \underbrace{n_{\text{dd}}\left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N}\right)}_{\stackrel{\Delta}{=} n_{\text{dd}}[k, l]} \\ &= \sum_{k', l' \in \mathbb{Z}} h_{\text{eff}}[k - k', l - l'] x_{\text{dd}}[k', l'] e^{j2\pi \frac{(l - l')}{N} \frac{k'}{M}} + n_{\text{dd}}[k, l] \\ &= \sum_{k', l' \in \mathbb{Z}} h_{\text{eff}}[k', l'] x_{\text{dd}}[k - k', l - l'] e^{j2\pi \frac{l'}{N} \frac{(k - k')}{M}} + n_{\text{dd}}[k, l], \end{aligned} \quad (2.O.5)$$

where the third step follows from the fact that $\tau_p \nu_p = 1$. The last step follows from substituting the summation variables (k', l') with $(k - k', l - l')$.

3

Approximations of OTFS Modulation

In Zak-orthogonal time frequency space (OTFS) modulation discussed in Chapter 2, information symbols are carried by time domain (TD) pulsones which are quasi-periodic pulses in the delay-Doppler (DD) domain. A TD pulsone is generated by computing the inverse Zak transform of a quasi-periodic DD domain pulse filtered (twisted convolution) with a transmit pulse. At the Zak-OTFS receiver, the received TD signal is converted to DD domain using Zak transform. This DD domain signal is then matched-filtered (twisted convolution) with a receive pulse followed by sampling on the information grid Λ_{dd} .

Transceivers designed for orthogonal frequency division multiplexing (OFDM)-based 4G/5G systems consist of an OFDM modulator and demodulator, which are *incompatible* with Zak/inverse Zak transform. In this chapter, a variant of OTFS modulation, called multicarrier OTFS (MC-OTFS) is derived, which is compatible with existing 4G/5G transceivers. In the history of OTFS, it was MC-OTFS which appeared first in [28] by Hadani et al., followed by Zak-OTFS later in [33, 34]. In [28], the MC-OTFS modulation equations were given without any derivation or insights into the predictability attribute. In this chapter, MC-OTFS modulation is derived naturally in an attempt to generate pulsones using an OFDM modulator. For this derivation, two approaches are followed.

The first approach is based on the Heisenberg transform (which is a generalization of OFDM), which maps a time-frequency (TF) function to a TD signal using a multicarrier (MC) signal generator. It is observed that due to the OFDM compatibility constraint, the pulsones generated are different from the Zak-OTFS pulsones in Chapter 2 and are called multicarrier (MC) pulsones. The second approach is based on Zak theory (introduced in Chapter 2), where the MC pulsones are generated by applying the inverse Zak transform to its DD domain realization. The quasi-periodic DD domain realization of a MC pulsone is a periodic DD domain pulse linearly convolved with a transmit pulse and then multiplied with the Zak transform of the MC generator. Clearly, this is different from a Zak-OTFS pulsone, whose DD realization is a quasi-periodic DD domain pulse twisted convolved with a transmit pulse.

From Chapter 2, it is clear that Zak-OTFS pulsones are matched to a doubly spread channel, i.e., they are generated using twisted convolution which is also how the channel acts on its input, and this is why the effective DD domain I/O relation is described by twisted convolution, i.e., the DD domain output at the receiver is given by twisted convolution between the DD domain information input and the effective DD domain channel filter. However, for MC pulsones that correspond to periodic DD domain pulses (and not quasi-periodic DD domain pulses) and are realized through linear convolution (and not twisted convolution), the effective DD domain I/O relation is not matched to the channel action. In other words, the MC-OTFS I/O relation is not described by twisted convolution with some effective DD domain filter, but it is instead a complex mixture of linear convolution with DD domain pulses, multiplication with MC generators, and

twisted convolution with the channel spreading function. This is why it is challenging to acquire the MC-OTFS I/O relation efficiently.

For channels with small delay and Doppler spreads, the MC-OTFS I/O relation may be acquired efficiently with some accuracy, for example, using a single DD pulse pilot signal. However, for high delay and Doppler spreads, it is challenging to acquire the I/O relation efficiently using a single DD pulse pilot. In comparison, the Zak-OTFS I/O relation in Chapter 2 is matched to the channel action, and, therefore, acquiring the Zak-OTFS I/O relation is the same as estimating the effective channel filter, which is achieved using a single DD pulse pilot signal irrespective of the amount of channel delay/Doppler spread as long as they are less than the respective Zak-OTFS period parameters (i.e., operation is in the crystalline regime).

The two equivalent approaches to the generation of MC pulsones are briefly explained in the following. While the Heisenberg transform-based approach gives the explicit transceiver architecture compatible with an OFDM modulator, the equivalent Zak transform-based approach helps in comparing it with Zak-OTFS modulation, which is known to be predictable and non-fading in the crystalline regime.

Through a series of questions and their answers, Fig. 3.1 illustrates the generation of pulsones using the Heisenberg transform when there is no time or bandwidth limitation. The Heisenberg transform can be logically thought to map a TF function to a TD signal. As discussed in Chapter 2, an infinite time and bandwidth TD pulsones corresponding to a quasi-periodic DD pulse at (τ_0, ν_0) is given by an infinite duration Dirac-delta impulse train multiplied by the tone $e^{j2\pi\nu_0(t-\tau_0)}$. The impulses in the impulse train are located at $t = (\tau_0 + n\tau_p), n \in \mathbb{Z}$. This Zak-OTFS TD pulsones is subsequently denoted by $p_{\tau_0, \nu_0}(t)$.

1. Which TF functions are mapped by Heisenberg transform to TD pulsones?

The TF function which maps to $p_{\tau_0, \nu_0}(t)$ consists of infinitely many Dirac delta impulses on the period lattice Λ_p , i.e., at $(t, f) = (n\tau_p, m\nu_p), n, m \in \mathbb{Z}$, multiplied with the TF function $e^{j2\pi(\nu_0 t - \tau_0 f)}$. Note that $(\nu_0 t - \tau_0 f)$ is the symplectic form $(\nu t - \tau f)$ evaluated at $(\tau, \nu) = (\tau_0, \nu_0)$. As the Heisenberg transform uses a MC generator $g_{\text{tx}}(t)$ for mapping a TF function to a TD signal, the mapping of the above symplectic form TF function is not exactly $p_{\tau_0, \nu_0}(t)$, but is instead the scaled pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0)p_{\tau_0, \nu_0}(t)$, where the constant $G_{\text{tx,dd}}(\tau_0, \nu_0)$ is the Zak transform of the MC generator $g_{\text{tx}}(t)$ evaluated at $(\tau, \nu) = (\tau_0, \nu_0)$ (see Fig. 3.1).

2. What is the DD representation of the TF function which maps to the scaled pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0)p_{\tau_0, \nu_0}(t)$?

The DD domain representation of any TF function is given by its symplectic Fourier transform (SFT), and the TF representation of a DD function is given by its inverse symplectic Fourier transform (ISFT). It turns out that a DD domain periodic pulse at (τ_0, ν_0) , i.e., Dirac-delta impulses at $(\tau, \nu) = (\tau_0 + n\tau_p, \nu_0 + m\nu_p), n, m \in \mathbb{Z}$, is the SFT/DD representation of the TF function which maps to the scaled TD pulsones (see Fig. 3.1).

3. What is the DD domain representation (i.e., Zak transform) of the scaled pulsones?

The DD domain representation of $G_{\text{tx,dd}}(\tau_0, \nu_0)p_{\tau_0, \nu_0}(t)$ is $G_{\text{tx,dd}}(\tau, \nu)$ times the periodic DD pulse at (τ_0, ν_0) . Note that $G_{\text{tx,dd}}(\tau, \nu)$ is the Zak transform of the MC generator $g_{\text{tx}}(t)$.

Based on the above questions and answers, the block diagram for the generation of unlimited time and bandwidth TD pulsones using both approaches is shown in Fig. 3.2. The Heisenberg transform based approach is highlighted in dark gray and the Zak transform based approach is highlighted in light gray.

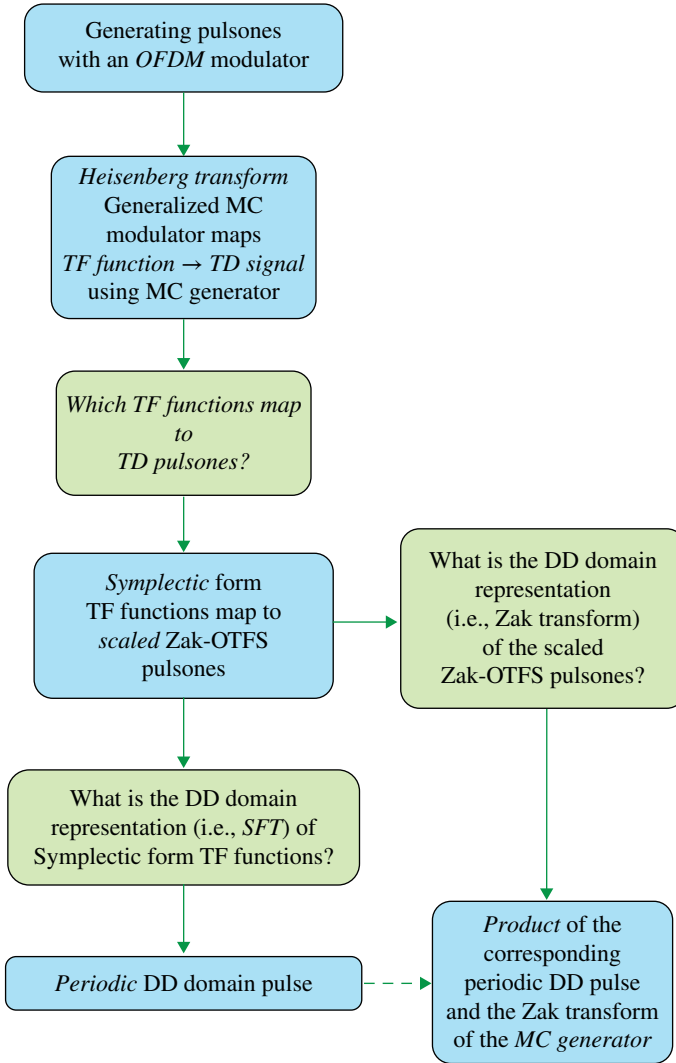


Figure 3.1 Heisenberg transform-based generation of TD pulsones which are *unlimited* in time and bandwidth.

Since time and bandwidth is limited in practice, in the following, again through a series of questions and answers (see Fig. 3.3), the generation of limited time and bandwidth TD pulsones is briefly described using the Heisenberg transform.

1. In Fig. 3.2, the Heisenberg transform maps the symplectic form TF function to a pulsome of unlimited time and bandwidth.

Is it possible to process the symplectic form TF function so as to limit the time duration and bandwidth of the mapped pulsome?

Yes, it is possible. For any arbitrary TF function, TF windowing (i.e., multiplying with a TF window function) limits the time duration and bandwidth of the mapped TD signal (mapped by the Heisenberg transform) to the time and frequency support of the TF window function.

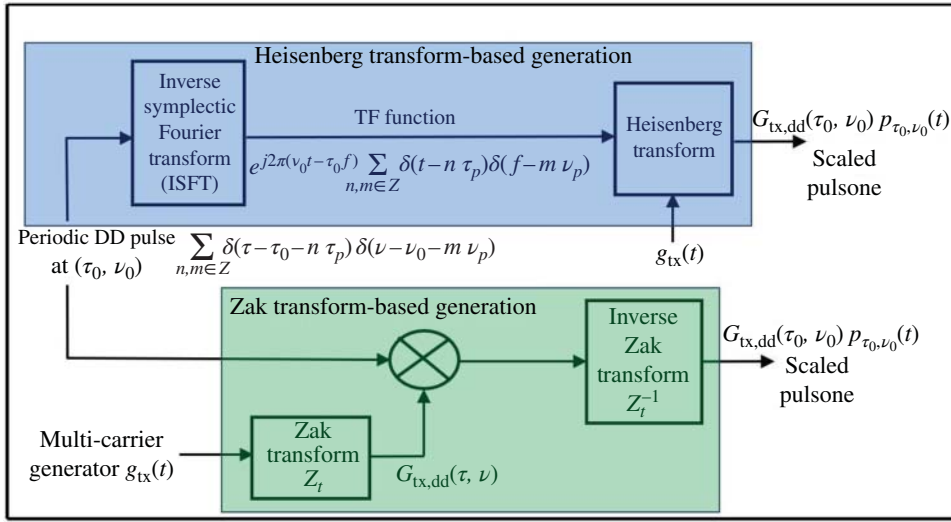


Figure 3.2 Generation of *unlimited* time and bandwidth TD pulsones using Heisenberg transform.

The Heisenberg transform maps a windowed symplectic form TF function to time- and bandwidth-limited pulsones, called *MC pulsones*, which are distinctly different from the Zak-OTFS pulsones in Chapter 2.

2. In OTFS modulation, information is embedded in the DD domain, and therefore, it is natural to shift signal processing to the DD domain. Therefore, the next question is if TF windowing (needed to generate time- and bandwidth-limited pulsones) can be equivalently achieved through signal processing in the DD domain. In Fig. 3.2, the DD representation (i.e., SFT) of the symplectic form TF function $e^{j2\pi(\nu_0 t - \tau_0 f)}$ is the periodic DD pulse at (τ_0, ν_0) .

Can windowing of the TF symplectic form function be achieved through some DD domain processing of its SFT (i.e., the periodic DD pulse at (τ_0, ν_0))? In other words, which DD domain operation on this periodic DD pulse would result in a DD function whose TF representation (i.e., ISFT) is equal to the windowed symplectic form TF function?

The DD domain representation (i.e., SFT) of the product of any two TF functions is equal to the *linear convolution* of their individual DD representations. *Therefore, the DD representation (SFT) of the product of a symplectic form TF function with a TF window $W_{tx}(t, f)$ is equal to the linear convolution of the SFT of the symplectic form TF function (i.e., periodic DD pulse) with $w_{tx}(\tau, \nu)$ (which is the SFT/DD representation of the window $W_{tx}(t, f)$).*

3. In Chapter 2, the generation of Zak-OTFS pulsones is based on the Zak transform. Therefore, in order to compare MC pulsones with Zak-OTFS pulsones, it is important to know the DD domain realization of MC pulsones.

What is the DD domain realization (i.e., Zak transform) of MC pulsones?

The DD domain realization of MC pulsones is the product of the Zak transform of the MC generator with a filtered periodic DD pulse (i.e., periodic DD pulse linearly convolved with the SFT of the TF window).

The block diagram for the generation of time- and bandwidth-limited TD pulsones using the Heisenberg transform is shown in Fig. 3.4.

This chapter discusses the MC-OTFS modulation, where the time- and bandwidth-limited MC pulsones are used to carry information symbols. As shown in Fig. 3.4, there are two equivalent approaches for generating MC pulsones, one based on the Heisenberg transform (filtered periodic

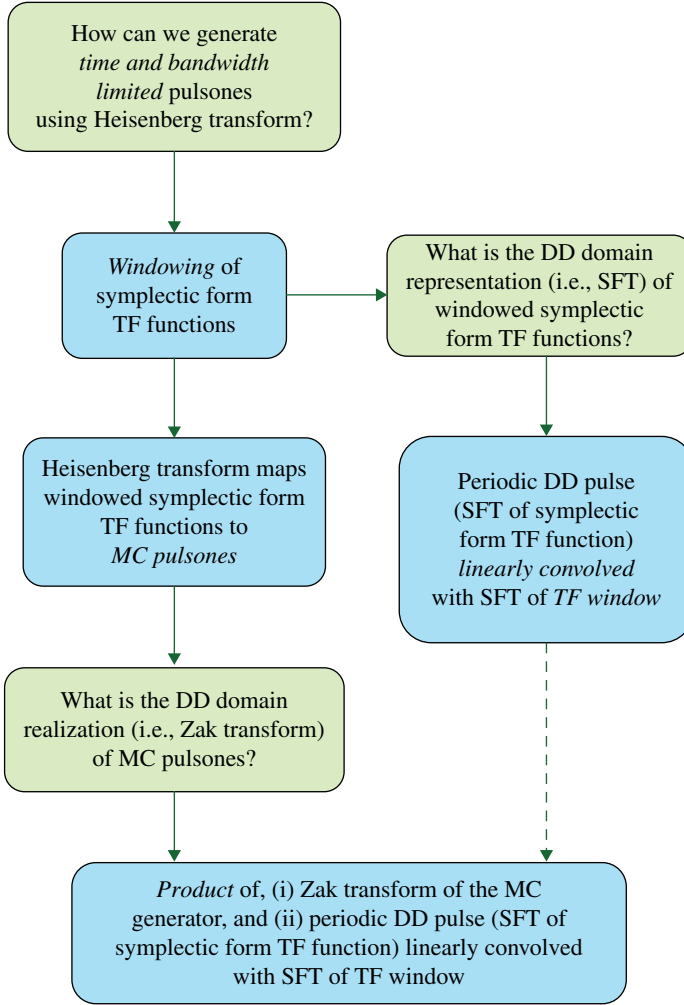


Figure 3.3 Heisenberg transform-based generation of TD pulsones which are *limited* in time and bandwidth.

DD pulse $\rightarrow$ ISFT $\rightarrow$ Heisenberg transform) and the other based on the Zak transform (filtered periodic DD pulse $\rightarrow$ product with the Zak transform of the MC generator $\rightarrow$ Inverse Zak transform). The Heisenberg transform-based approach leads to a two-step MC-OTFS modulator, where the DD domain information symbols are transformed to TF domain symbols in the first step and these TF symbols are then transformed to the MC-OTFS-modulated TD signal in the second step. The Zak transform-based approach leads to a single-step MC-OTFS modulator, where the DD domain information symbols modulate periodic DD domain Dirac-delta impulses located on the information grid Λ_{dd} . This periodic DD function is linearly convolved with the SFT of the TF window followed by multiplication with the Zak transform of the MC generator. The resulting DD domain function is quasi-periodic, and its TD realization (i.e., inverse Zak transform) gives the MC-OTFS-modulated TD signal.

The optimal MC-OTFS receiver consists of projecting the received TD signal onto MC pulsones resulting in DD domain samples which can be jointly processed to decode the transmitted information. The processing of the received TD signal to the DD domain samples can be carried out

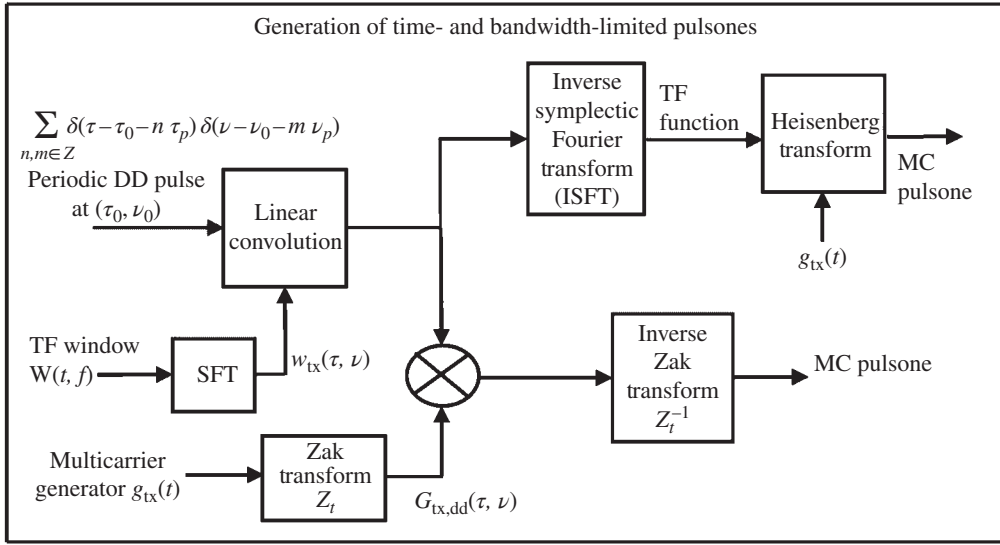


Figure 3.4 Generation of time- and bandwidth-limited TD pulsones using the Heisenberg transform.

using two equivalent but different approaches, namely, (i) the Wigner transform-based approach that converts the received TD signal to TF domain symbols in the first step (the Wigner transform can be thought of as the inverse of the Heisenberg transform), followed by conversion of the received TF symbols to DD domain samples in the second step, and (ii) the Zak transform-based approach, where the Zak transform of the received TD signal is multiplied with the complex conjugate of the Zak transform of the MC generator, followed by linear convolution with a DD domain matched filter, and subsequent sampling on the information grid.

The MC-OTFS I/O relation between the input information symbols and the output DD domain samples is derived for both the Heisenberg and Zak transform-based approaches. Comparison between Zak-OTFS and MC-OTFS I/O relation reveals that the Zak-OTFS I/O relation can be acquired more efficiently (with lesser estimation overhead than for MC-OTFS). Equivalently, for the same estimation overhead (e.g., a single pulson to probe the channel), the Zak-OTFS I/O relation is more predictable specially at higher Doppler spreads and, therefore, has better communication reliability.

This chapter is organized as follows. In Section 3.1, infinite duration and bandwidth TD pulsones (i.e., inverse Zak transform of quasi-periodic Dirac-delta impulses in the DD domain) are shown to form a basis for TD signals. In Section 3.2, these pulsones are generated using the Heisenberg transform. Generation of time- and bandwidth-limited pulsones is then discussed in Section 3.3. These pulsones are called MC pulsones, which are different from the Zak-OTFS pulsones presented in Chapter 2.

Section 3.4 discusses the MC-OTFS modulation, where MC pulsones are information carriers. Two different MC-OTFS modulator architectures based on the Heisenberg transform and the Zak transform are presented. In Section 3.5, two different MC-OTFS demodulator architectures are presented, one based on the Wigner transform and the other based on the Zak transform. In Section 3.6, the MC-OTFS I/O relation is derived for both the Heisenberg/Wigner and Zak transform-based transceivers.

In Section 3.7, the predictability attribute of MC-OTFS is investigated by comparing its bit error rate (BER) performance with that of Zak-OTFS for the imperfect channel state information scenario. A glossary of important signals, filters, and terminology used in this chapter is summarized in Tables 3.1 and 3.2.

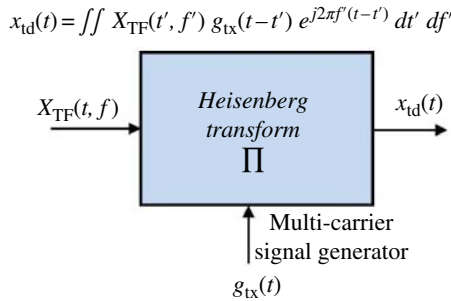
Table 3.1 Glossary-I.

Notation	Description	Reference
$*_{\sigma}$	Twisted convolution	See (2.20)
$\star$	Linear convolution	
$\mathcal{Z}_t$	Zak transform	See (2.23) and Appendix 2.C
$\mathcal{Z}_t^{-1}$	Inverse Zak transform	See Fig. 2.10 and Appendix 2.F
τ_p, ν_p	Delay and Doppler period of Zak-OTFS modulation	See (2.23) and Fig. 2.11
D_0	Fundamental DD period	See (2.18)
B and T	Bandwidth and time duration of modulated TD signals	
$M = B\tau_p$	Delay domain period of discrete DD signals	Section 3.4
$N = T\nu_p$	Doppler domain period of discrete DD signals	Section 3.4
Λ_{dd}	Zak-OTFS information grid	See (2.50)
$h_{\text{phy}}(\tau, \nu)$	DD spreading function of underlying channel	See (2.1)
Quasi-periodic	Attribute of DD domain realization of TD signals	See (2.25) and (2.56) for continuous and discrete DD signals respectively
Periodic DD pulse at (τ_0, ν_0) (continuous)	DD domain Dirac impulses at $(\tau_0 + n\tau_p, \nu_0 + m\nu_p), n, m \in \mathbb{Z}$	$x_p(\tau, \nu; \tau_0, \nu_0)$ in (3.15) See Fig. 3.6
Multicarrier (MC) generator	$g_{\text{tx}}(t)$: MC generator at Tx. $g_{\text{rx}}(t)$: MC generator at Rx.	See (3.8) and Fig. 3.10 See (3.59).
$G_{\text{tx,dd}}(\tau, \nu)$	Zak transform of $g_{\text{tx}}(t)$	See (3.13) and Fig. 3.12
$w_{\text{tx}}(\tau, \nu)$	DD domain transmit pulse shaping filter	See (3.29) and Fig. 3.12
$w_{\text{rx}}(\tau, \nu)$	DD domain receive pulse shaping filter	See (3.67) and Fig. 3.14
$W_{\text{tx}}(t, f)$	TF window at transmitter	See (3.19) and Fig. 3.12
$W_{\text{rx}}(t, f)$	TF window at receiver	See (3.62) and Fig. 3.13
TD pulsone (infinite B and T)	$p_{\tau_0, \nu_0}(t)$	See (3.2)
$\Pi_{X_{\text{TF}}(t,f)}(g_{\text{tx}}(t))$	Heisenberg transform of $g_{\text{tx}}(t)$ corresponding to TF function $X_{\text{TF}}(t, f)$	See (3.8) and Fig. 3.5
$X_{\text{TF}}(t, f; \tau_0, \nu_0)$	TF function Heisenberg transform corresponding to which is $G_{\text{tx,dd}}(\tau_0, \nu_0)p_{\tau_0, \nu_0}(t)$	See (3.12), Fig. 3.8 and Theorem 3.1
SFT, ISFT	Symplectic Fourier Transform (SFT) and inverse SFT	See Appendix 3.D
$x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$	Periodic DD pulse at (τ_0, ν_0) , filtered (twisted convolution) with pulse shaping filter	See (3.30) and Fig. 3.8
$X_{\text{TF}}^{W_{\text{tx}}}(t, f; \tau_0, \nu_0)$	Windowed Symplectic form TF function	See (3.30) and Fig. 3.8
$x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t)$	MC pulsone corresponding to the periodic DD pulse at (τ_0, ν_0) , MC generator $g_{\text{tx}}(t)$, pulse shaping waveform is $w_{\text{tx}}(\tau, \nu)$	See (3.30) and Fig. 3.9

Source: Mohammed et al. [34] IEEE.

Table 3.2 Glossary-II.

Notation	Description	Reference
$x_p[k, l]$	Periodic discrete DD domain information function	See (3.42) and (3.43)
$x_p(\tau, \nu)$	Periodic continuous DD domain information function	See (3.44) and Fig. 3.10
$x_p^{u_{\text{ix}}}(\tau, \nu)$	Filtered DD domain information function	See (3.45) and Fig. 3.10
$X_{\text{TF}}^{W_{\text{ix}}}(t, f)$	TF domain information function ISFT of $x_p^{u_{\text{ix}}}(\tau, \nu)$	See (3.46) and Fig. 3.10
$\Pi_{X_{\text{TF}}^{W_{\text{ix}}}(t, f)}(g_{\text{ix}}(t))$	MC-OTFS-modulated TD transmit signal Heisenberg transform of MC generator $g_{\text{ix}}(t)$ corresponding to TF function $X_{\text{TF}}^{W_{\text{ix}}}(t, f)$	See (3.47)
ISFFT	Inverse Symplectic Finite Fourier Transform	See (3.53) and Fig. 3.11
$X[n, m]$	TF domain information symbols in a two-step MC-OTFS modulator	See (3.53) and Fig. 3.11
$X^{W_{\text{ix}}}[n, m]$	Windowed TF domain information symbols in a two-step MC-OTFS modulator	See (3.54) and Fig. 3.11
$r_{\text{td}}(t)$	Received TD signal	See Figs. 3.13 and 3.14
$y_p[k, l]$	Received discrete DD domain periodic function	See (3.59)
$Y[n, m]$	Received TF domain symbols in a two-step MC-OTFS receiver	See (3.61) and Fig. 3.13
$Y^{W_{\text{ix}}}[n, m]$	Windowed TF domain symbols in a two-step MC-OTFS receiver	See (3.62) and Fig. 3.13
$y_{\text{dd}}(\tau, \nu)$	Received continuous DD domain signal	See (3.68) and Fig. 3.14
$G_{\text{rx,dd}}(\tau, \nu)$	Zak transform of receive MC generator $g_{\text{rx}}(t)$	See Fig. 3.14
$y^G(\tau, \nu)$	Received DD domain signal multiplied with $G_{\text{rx,dd}}^*(\tau, \nu)$	See (3.69) and Fig. 3.14
$H_{\text{TF}}[\cdot, \cdot; \cdot, \cdot]$	Effective TF domain channel for MC-OTFS	See (3.72)
$A_{g_{\text{rx}}g_{\text{ix}}}(\tau, \nu)$	Cross ambiguity function between transmit and receive MC generators	See (3.73)

**Figure 3.5** Heisenberg transform.

3.1 Pulsones as a Basis for TD Signals

From Chapter 2, it is known that the channel interaction of quasi-periodic DD domain pulses is predictable and non-fading in the crystalline regime. The TD realization of a quasi-periodic DD pulse is called a *pulsone* as it is a pulse train modulated by a sinusoid. In this section, we show that pulsones, in fact, form a basis for the space of all TD signals, and the projection of a TD signal $x(t)$ onto this basis gives the DD representation/Zak transform of $x(t)$, i.e., $\mathcal{Z}_t(x(t))$.

The TD realization of a quasi-periodic DD domain pulse at $(\tau, \nu) = (0, 0)$ is given by the TD pulsone (see (2.38))

$$p(t) \triangleq \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - n\tau_p). \quad (3.1)$$

This basic/fundamental TD pulsone can be used to generate any other TD pulsone by modulating it by a sinusoid and shifting it in time. Modulating $p(t)$ by $e^{j2\pi\nu_0 t}$ and then shifting it by τ_0 , we get the TD pulsone

$$\begin{aligned} p_{\tau_0, \nu_0}(t) &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) e^{j2\pi\nu_0(t - \tau_0)} \\ &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) e^{j2\pi n\nu_0 \tau_p}. \end{aligned} \quad (3.2)$$

This TD pulsone is, in fact, the TD realization of a quasi-periodic DD pulse at (τ_0, ν_0) , i.e., the pulse obtained after shifting the DD pulse at $(0, 0)$ by (τ_0, ν_0) (see (2.26)),

$$\mathcal{Z}_t(p_{\tau_0, \nu_0}(t)) = \sum_{n, m \in \mathbb{Z}} e^{j2\pi\nu_0 n\tau_p} \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p). \quad (3.3)$$

The set of all TD pulsone waveforms, i.e.,

$$\mathcal{P} \triangleq \{p_{\tau_0, \nu_0}(t) \mid (\tau_0, \nu_0) \in \mathcal{D}_0\}, \quad (3.4)$$

in fact, forms a basis for the space of all TD signals (see Theorem 1 in [60]). That is, any TD signal $x(t)$ can be expressed as a linear combination of these pulsones as

$$x(t) = \int_0^{\tau_p} \int_0^{\nu_p} c_x(\tau, \nu) p_{\tau, \nu}(t) d\tau d\nu, \quad (3.5)$$

where the coefficient $c_x(\tau, \nu)$ is the projection of $x(t)$ onto $p_{(\tau, \nu)}(t)$, i.e.,

$$c_x(\tau, \nu) = \int_{-\infty}^{\infty} p_{\tau, \nu}^*(t) x(t) dt. \quad (3.6)$$

Substituting the expression of $p_{\tau, \nu}(t)$ from (3.2) on the RHS of the above equation, it follows that $c_x(\tau, \nu)$ is the Zak transform of $x(t)$, i.e.,

$$\begin{aligned} c_x(\tau, \nu) &= \int_{-\infty}^{\infty} x(t) \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau - n\tau_p) e^{-j2\pi n\nu\tau_p} dt \\ &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{-j2\pi n\nu\tau_p} \int_{-\infty}^{\infty} x(t) \delta(t - \tau - n\tau_p) dt \end{aligned}$$

$$\begin{aligned}
&= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{-j2\pi n v \tau_p} \left[\int_{-\infty}^{\infty} x(t) \delta(t - \tau - n\tau_p) dt \right] \\
&= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{-j2\pi n v \tau_p} x(\tau + n\tau_p) = \mathcal{Z}_t(x(t)).
\end{aligned} \tag{3.7}$$

That is, the coefficients in the linear expansion of any TD signal $x(t)$ in the basis of pulsones are nothing but the Zak transform of $x(t)$.

3.2 Generating Pulsones Using the Heisenberg Transform

In this section, we see how pulsones can be generated using the Heisenberg transform. The Heisenberg transform is the basic mathematical transform that can generate most well-known MC-modulated signals, including OFDM.

The Heisenberg transform (denoted by Π) is parameterized by a two-dimensional TF function $X_{\text{TF}}(t, f)$, which we also refer to as the TF information function since it carries information. The Heisenberg transform takes as input a TD signal $g_{\text{tx}}(t)$, called the *MC signal generator*, and outputs the TD signal

$$\begin{aligned}
x_{\text{id}}(t) &= \Pi_{X_{\text{TF}}(t, f)}(g_{\text{tx}}(t)) \\
&= \iint X_{\text{TF}}(t', f') g_{\text{tx}}(t - t') e^{j2\pi f'(t - t')} dt' df'.
\end{aligned} \tag{3.8}$$

Example 3.1 The following example shows how the Heisenberg transform can generate an OFDM-modulated signal with M subcarriers and a subcarrier spacing Δf . Let $x[m]$, $m = 0, 1, \dots, M-1$ denote the M information symbols to be transmitted on the M subcarriers, $x[m]$ being transmitted on the m th subcarrier. For OFDM, the information TF function is given by

$$X_{\text{TF}}(t, f) = \sum_{m=0}^{M-1} x[m] \delta(t) \delta(f - m\Delta f), \tag{3.9}$$

whose Heisenberg transform for the input MC generator

$$g_{\text{tx}}(t) = \begin{cases} 1, & 0 \leq t < \frac{1}{\Delta f} \\ 0, & \text{otherwise} \end{cases} \tag{3.10}$$

is the OFDM-modulated TD signal

$$x_{\text{id}}(t) = \begin{cases} \sum_{m=0}^{M-1} x[m] e^{j2\pi m \Delta f t}, & 0 \leq t < \frac{1}{\Delta f}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.11}$$

Here, $1/\Delta f$ is the OFDM symbol period.

The main question of this section is, how can pulsones be generated using the Heisenberg transform. The following theorem answers this question.

Theorem 3.1 For any given (τ_p, ν_p) , $\tau_p \nu_p = 1$, for the TF function

$$X_{\text{TF}}(t, f; \tau_0, \nu_0) \stackrel{\Delta}{=} e^{j2\pi(\nu_0 t - \tau_0 f)} \sum_{n, m \in \mathbb{Z}} \delta(t - n\tau_p) \delta(f - m\nu_p), \tag{3.12}$$

the corresponding output of the Heisenberg transform is given by the pulsones $p_{\tau_0, \nu_0}(t)$ multiplied by the constant $G_{\text{tx,dd}}(\tau_0, \nu_0)$, which is the Zak transform of the MC generator $g_{\text{tx}}(t)$ evaluated at $(\tau, \nu) = (\tau_0, \nu_0)$, i.e.,

$$\begin{aligned} \Pi_{X_{\text{TF}}(t, f; \tau_0, \nu_0)}(g_{\text{tx}}(t)) &= G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t), \\ G_{\text{tx,dd}}(\tau, \nu) &\stackrel{\Delta}{=} \mathcal{Z}_t \left(g_{\text{tx}}(t) \right). \end{aligned} \quad (3.13)$$

Proof: See Appendix 3.A.

The following corollary gives the expression for the unique MC generator $g_{\text{tx}}(t)$ for which the output $\Pi_{X_{\text{TF}}(t, f; \tau_0, \nu_0)}(g_{\text{tx}}(t))$ is the pulsones $p_{\tau_0, \nu_0}(t)$ for all (τ_0, ν_0) in the fundamental DD period $\mathcal{D}_0$ (see Section 2.3 for definition of $\mathcal{D}_0$). From (3.13), it is clear that this is possible only when $g_{\text{tx}}(t)$ is such that $G_{\text{tx,dd}}(\tau_0, \nu_0) = 1$ for all $(\tau_0, \nu_0) \in \mathcal{D}_0$.

Corollary 3.1 For a given (τ_p, ν_p) , $\tau_p \nu_p = 1$, the unique MC generator $g_{\text{tx}}(t)$ for which $\Pi_{X_{\text{TF}}(t, f; \tau_0, \nu_0)}(g_{\text{tx}}(t)) = p_{\tau_0, \nu_0}(t)$ for all $(\tau_0, \nu_0) \in \mathcal{D}_0$ is the rectangular generator

$$g_{\text{tx}}(t) = \begin{cases} \frac{1}{\sqrt{\tau_p}}, & 0 \leq t < \tau_p, \\ 0, & \text{otherwise.} \end{cases} \quad (3.14)$$

Proof: See Appendix 3.B.

Next, for any $(\tau_0, \nu_0) \in \mathcal{D}_0$, and any MC generator $g_{\text{tx}}(t)$, the following theorem gives the DD domain representation (i.e., Zak transform) of the scaled pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$ generated by the Heisenberg transform corresponding to the TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ (see Theorem 3.1).

Theorem 3.2 For any $(\tau_0, \nu_0) \in \mathcal{D}_0$, the DD domain representation (i.e., Zak transform) of the scaled pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$ is given by the product of a periodic DD pulse at (τ_0, ν_0) and the Zak transform of the MC generator $g_{\text{tx}}(t)$, i.e.,

$$\begin{aligned} \mathcal{Z}_t \left(G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t) \right) &= G_{\text{tx,dd}}(\tau, \nu) x_p(\tau, \nu; \tau_0, \nu_0), \\ x_p(\tau, \nu; \tau_0, \nu_0) &\stackrel{\Delta}{=} \underbrace{\left[\sum_{n, m \in \mathbb{Z}} \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p) \right]}_{\text{Periodic DD domain pulse at } (\tau_0, \nu_0)}. \end{aligned} \quad (3.15)$$

Note that the DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$ on the RHS of (3.15) is periodic and not quasi-periodic.

Proof: See Appendix 3.C.

Theorem 3.2, therefore, suggests an alternate Zak transform-based architecture for the generation of the pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$. First, generate a periodic DD function $\sum_{n, m \in \mathbb{Z}} \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p)$ consisting of infinitely many Dirac-delta at $(\tau_0 + n\tau_p, \nu_0 + m\nu_p)$, $n, m \in \mathbb{Z}$. Clearly, this function is not quasi-periodic in general, and, therefore, its TD realization does not

exist. However, the product of this periodic DD function and the quasi-periodic signal $G_{\text{tx,dd}}(\tau, \nu)$ is quasi-periodic, whose TD realization (i.e., inverse Zak transform) exists and is nothing but $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$. Hence, we simply multiply the periodic DD function with $G_{\text{tx,dd}}(\tau, \nu)$, followed by the inverse Zak transform of the product. Since both the Zak transform-based and the Heisenberg transform-based approaches generate the same pulsone, there must be some relation between $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ and $x_p(\tau, \nu; \tau_0, \nu_0)$.

Interestingly, the TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ which generates the scaled pulsone $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$ (see Theorem 3.1) is related to the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$ through the SFT which converts a TF domain function to a DD domain function (see Appendix 3.D for definition of SFT and its inverse). The following theorem states this fact.

Theorem 3.3 For any given (τ_p, ν_p) , $\tau_p \nu_p = 1$, the SFT of the TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ is the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$, i.e.,

$$\begin{aligned} \text{SFT}(X_{\text{TF}}(t, f; \tau_0, \nu_0)) &= \iint X_{\text{TF}}(t, f; \tau_0, \nu_0) e^{-j2\pi(\nu t - \tau f)} dt df \\ &= x_p(\tau, \nu; \tau_0, \nu_0). \end{aligned} \quad (3.16)$$

Proof: See Appendix 3.E.

Since the SFT of $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ is $x_p(\tau, \nu; \tau_0, \nu_0)$, the inverse operation of SFT (i.e., Inverse SFT (ISFT)) when applied to $x_p(\tau, \nu; \tau_0, \nu_0)$ would give $X_{\text{TF}}(t, f; \tau_0, \nu_0)$. This TF function, as we know, corresponds to the Heisenberg transform which generates the scaled pulsone $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$. Therefore, starting with the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$, there are two ways in which the scaled pulsone could be generated, one using the Zak transform and the other using the Heisenberg transform. Both these are illustrated in Fig. 3.6.

The above equivalence between the Zak transform-based and Heisenberg transform-based generation of pulsones is valid not only for $X_{\text{TF}}(t, f; \tau_0, \nu_0)$, but is, in fact, a more general result which is given by the following theorem, and which is useful later when we discuss the generation of time- and bandwidth-constrained pulsones in Section 3.3.

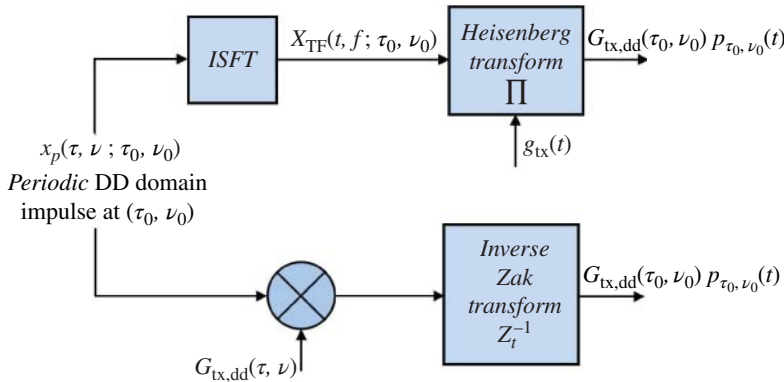


Figure 3.6 Equivalent generation of the TD pulsone using the Heisenberg transform and the inverse Zak transform.

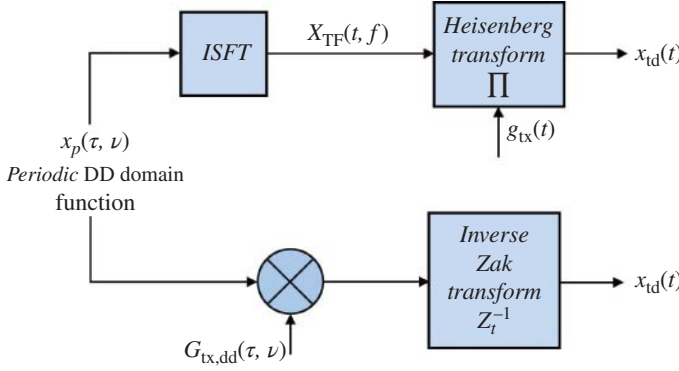


Figure 3.7 Equivalence between the Heisenberg transform- and Zak transform-based generation for arbitrary periodic DD domain functions.

Theorem 3.4 For any given delay and Doppler periods τ_p and ν_p , respectively, $\tau_p \nu_p = 1$, and any arbitrary periodic DD domain function $x_p(\tau, \nu)$, consider the corresponding TD signal $x_{\text{id}}(t)$ generated by the Heisenberg transform corresponding to the TF function $X_{\text{TF}}(t, f) = \text{ISFT}(x_p(\tau, \nu))$, i.e.,

$$x_{\text{id}}(t) = \Pi_{\text{ISFT}(x_p(\tau, \nu))} (g_{\text{tx}}(t)), \quad (3.17)$$

where $g_{\text{tx}}(t)$ is the MC generator (see Fig. 3.7). Then, $x_{\text{id}}(t)$ can also be generated using the Zak transform, i.e.,

$$x_{\text{id}}(t) = \mathcal{Z}_t^{-1} \left(G_{\text{tx,dd}}(\tau, \nu) x_p(\tau, \nu) \right). \quad (3.18)$$

Proof: See Appendix 3.F.

3.3 Generating Time- and Bandwidth-Limited Pulsones

The TD pulsones $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{\tau_0, \nu_0}(t)$ generated by the Heisenberg transform corresponding to the TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ have infinite duration and bandwidth. Practical systems have limited time and bandwidth. The time and bandwidth can be limited simply by multiplying the TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$ with another TF window function, which approximately defines the TD and frequency domain (FD) support of the output of the Heisenberg transform. This is because, in general, the output of the Heisenberg transform corresponding to

$$X_{\text{TF}}^{W_{\text{tx}}}(t, f) \triangleq X_{\text{TF}}(t, f) W_{\text{tx}}(t, f), \quad (3.19)$$

where $W_{\text{tx}}(\cdot, \cdot)$ is the window function, is given by

$$\Pi_{X_{\text{TF}}^{W_{\text{tx}}}(t, f)} (g_{\text{tx}}(t)) = \iint X_{\text{TF}}(t', f') W_{\text{tx}}(t', f') g_{\text{tx}}(t - t') e^{j2\pi f'(t-t')} dt' df', \quad (3.20)$$

where the upper and lower limits of the integration with respect to (t', f') is restricted to the support set of the window $W(t, f)$. Therefore, due to the term $e^{j2\pi f'(t-t')}$ in the integrand, it follows that the frequency components in the output are approximately limited to the support of the window along FD. Similarly, due to the term $g_{\text{tx}}(t - t')$ in the integrand, it follows that the output

lies in a TD interval which is simply the support set of the window along the TD extended by the support of $g_{\text{tx}}(t)$. The following example clarifies this point.

Example 3.2 Consider the TF function

$$X_{\text{TF}}(t, f) = \sum_{n, m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m\nu_p). \quad (3.21)$$

The output of the Heisenberg transform corresponding to this TF function and the rectangular MC generator in (3.14) as input is given by

$$\sum_{n, m \in \mathbb{Z}} X[n, m] g_{\text{tx}}(t - n\tau_p) e^{j2\pi m\nu_p(t - n\tau_p)}, \quad (3.22)$$

which has infinite bandwidth and time duration. Next, for $N, M \in \mathbb{Z}$, consider the window function

$$W_{\text{tx}}(t, f) = \begin{cases} 1, & 0 \leq t < N\tau_p, 0 \leq f < M\nu_p, \\ 0, & \text{otherwise.} \end{cases} \quad (3.23)$$

Since

$$\begin{aligned} X_{\text{TF}}(t, f) W_{\text{tx}}(t, f) &= \left(\sum_{n, m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m\nu_p) \right) W_{\text{tx}}(t, f) \\ &= \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] \delta(t - n\tau_p) \delta(f - m\nu_p), \end{aligned} \quad (3.24)$$

using (3.20), the output of the Heisenberg transform for the windowed TF function $X_{\text{TF}}(t, f) W_{\text{tx}}(t, f)$ is given by

$$\sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] g_{\text{tx}}(t - n\tau_p) e^{j2\pi m\nu_p(t - n\tau_p)}. \quad (3.25)$$

This signal is clearly limited to the time interval $[0, N\tau_p)$ since $g_{\text{tx}}(t)$ is rectangular. In addition, from the $e^{j2\pi m\nu_p(t - n\tau_p)}$ term on the RHS above, it is clear that this signal is approximately limited to $[0, M\nu_p)$ in FD (note that the summation over m on the RHS is limited to $m = 0, 1, \dots, M-1$).

From Theorem 3.3, we know that the ISFT of the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$ is the pulstone generating TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$. It is, therefore, natural to ask as to what is the DD domain function whose ISFT is the windowed TF function $X_{\text{TF}}(t, f; \tau_0, \nu_0) W_{\text{tx}}(t, f)$. An interesting result about SFT is that the SFT of the product of two TF functions $A_{\text{TF}}(t, f)$ and $B_{\text{TF}}(t, f)$ is the linear convolution of their individual SFT,¹ i.e.,

$$\begin{aligned} \text{SFT} \left(A_{\text{TF}}(t, f) B_{\text{TF}}(t, f) \right) &= a(\tau, \nu) \star b(\tau, \nu) \\ &= \iint a(\tau - \tau', \nu - \nu') b(\tau', \nu') d\tau' d\nu' \\ &= \iint b(\tau - \tau', \nu - \nu') a(\tau', \nu') d\tau' d\nu', \end{aligned} \quad (3.26)$$

where $\star$ denotes linear convolution and $a(\tau, \nu)$ and $b(\tau, \nu)$ are the SFT of $A_{\text{TF}}(t, f)$ and $B_{\text{TF}}(t, f)$, respectively. This is equivalent to

$$\text{ISFT}(a(\tau, \nu) \star b(\tau, \nu)) = A_{\text{TF}}(t, f) B_{\text{TF}}(t, f). \quad (3.27)$$

See Theorem 3.15 in Appendix 3.D for a proof of this statement.

¹ Here, we assume that at least one among $A_{\text{TF}}(t, f)$ and $B_{\text{TF}}(t, f)$ is not a discrete TF function like $X_{\text{TF}}(t, f; \tau_0, \nu_0)$.

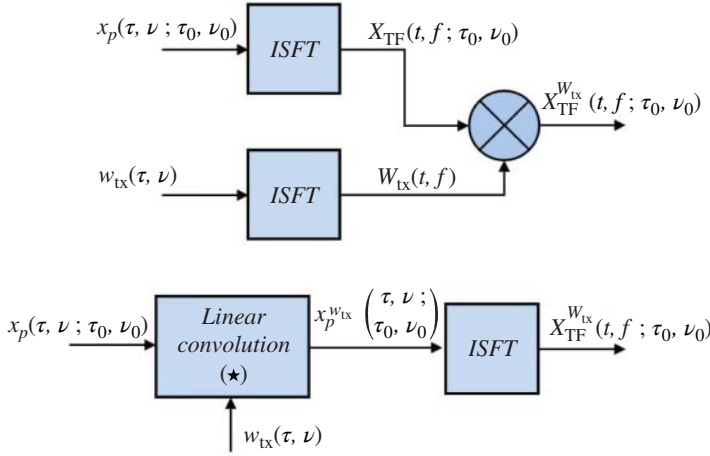


Figure 3.8 TF windowing for limiting the transmit TD signal in time and bandwidth.

Therefore, using (3.27) with $A_{\text{TF}}(t, f) = X_{\text{TF}}(t, f; \tau_0, \nu_0)$ and $B_{\text{TF}}(t, f) = W_{\text{tx}}(t, f)$, it follows that

$$\text{ISFT} \left(x_p(\tau, \nu; \tau_0, \nu_0) \star w_{\text{tx}}(\tau, \nu) \right) = X_{\text{TF}}(t, f; \tau_0, \nu_0) W_{\text{tx}}(t, f), \quad (3.28)$$

where

$$w_{\text{tx}}(\tau, \nu) \triangleq \text{SFT}(W_{\text{tx}}(t, f)) = \iint W_{\text{tx}}(t, f) e^{-j2\pi(\nu t - \tau f)} dt df \quad (3.29)$$

is the SFT of the window $W_{\text{tx}}(t, f)$. In other words, instead of windowing a TF function in the TF domain, alternatively the SFT of the TF function could be linearly convolved with the SFT of the window, followed by the ISFT of the convolution output (see Fig. 3.8).

Now, in Fig. 3.7, substituting the general periodic DD domain function $x_p(\tau, \nu)$ with the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0) \star w_{\text{tx}}(\tau, \nu)$, we get two equivalent methods for generating time- and bandwidth-limited pulsones, viz., one method based on the Heisenberg transform and the other method based on the Zak transform. These two equivalent generation methods are shown in Fig. 3.9. Subsequently, we refer to these pulsones as *MC pulsones* and denote them by $x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t)$, the MC pulsone generated using the periodic DD impulse function at (τ_0, ν_0) with the Heisenberg transform corresponding to the TF function $\text{ISFT} x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ and with MC generator $g_{\text{tx}}(t)$.

Starting with the DD domain periodic impulse function $x_p(\tau, \nu; \tau_0, \nu_0)$ at (τ_0, ν_0) , the Heisenberg transform-based generation of these pulsones is given by

$$\begin{aligned} x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) &\triangleq \Pi_{X_{\text{TF}}^{w_{\text{tx}}}(t, f; \tau_0, \nu_0)}(g_{\text{tx}}(t)), \\ X_{\text{TF}}^{w_{\text{tx}}}(t, f; \tau_0, \nu_0) &\triangleq \text{ISFT} \left(x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) \right), \\ x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) &\triangleq w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu; \tau_0, \nu_0), \end{aligned} \quad (3.30)$$

and the Zak transform-based equivalent pulsone generation is given by

$$x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) = \mathcal{Z}_t^{-1} \left(G_{\text{tx}, \text{dd}}(\tau, \nu) [w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu; \tau_0, \nu_0)] \right). \quad (3.31)$$

The following theorem gives the explicit expression for the time- and bandwidth-limited pulsone generated using the Heisenberg transform (see Fig. 3.9).

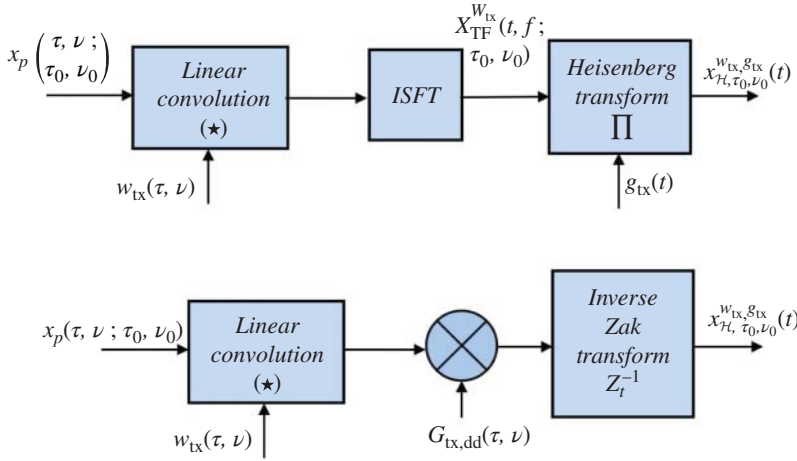


Figure 3.9 Equivalence of the Heisenberg transform- and Zak transform-based generation of time- and bandwidth-limited pulsones.

Theorem 3.5 For a given TF window $W_{\text{tx}}(t, f)$ and a MC generator $g_{\text{tx}}(t)$, the time- and bandwidth-limited MC pulsones $x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t)$ in Fig. 3.9 is given by

$$x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) = \sum_{n \in \mathbb{Z}} e^{j2\pi \nu_0 n \tau_p} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi m \nu_p (t - n\tau_p - \tau_0)} \right]. \quad (3.32)$$

Proof: See Appendix 3.G.

An important property of MC pulsones is given by the following corollary.

Corollary 3.2 For any given $(\tau_0, \nu_0) \in \mathcal{D}_0$, all MC pulsones in the set

$$\left\{ x_{H, \tau, \nu}^{w_{\text{tx}}, g_{\text{tx}}}(t) \middle| (\tau, \nu) = (\tau_0 + n\tau_p, \nu_0 + m\nu_p), n, m \in \mathbb{Z} \right\} \quad (3.33)$$

are identical.

Proof: This follows from the observation that for any $(\tau_0, \nu_0) \in \mathcal{D}_0$ and any $n', m' \in \mathbb{Z}$,

$$\begin{aligned} & x_{H, (\tau_0 + n'\tau_p), (\nu_0 + m'\nu_p)}^{w_{\text{tx}}, g_{\text{tx}}}(t) \\ & \stackrel{(a)}{=} \sum_{n \in \mathbb{Z}} e^{j2\pi (\nu_0 + m'\nu_p) n \tau_p} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi m \nu_p (t - n\tau_p - (\tau_0 + n'\tau_p))} \right] \\ & \stackrel{(b)}{=} \sum_{n \in \mathbb{Z}} e^{j2\pi \nu_0 n \tau_p} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi m \nu_p (t - n\tau_p - \tau_0)} \right] \\ & = x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t), \end{aligned} \quad (3.34)$$

where step (a) follows from the expression of the MC pulsones in (3.32). Step (b) follows from the fact that $\tau_p \nu_p = 1$.

3.3.1 Comparing MC Pulsones with Zak Pulsones

With infinite time duration and bandwidth, the Zak pulson in Chapter 2 corresponding to a DD impulse at $(\tau_0, \nu_0) \in \mathcal{D}_0$ is given by (2.38), i.e.,

$$p_{\tau_0, \nu_0}(t) = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{j2\pi n \nu_0 \tau_p} \delta(t - \tau_0 - n\tau_p). \quad (3.35)$$

With *infinite time and bandwidth*, the MC pulson is $p_{\tau_0, \nu_0}(t)$ times a constant $G_{\text{tx,dd}}(\tau_0, \nu_0)$ (see Fig. 3.6). In other words, *for infinite time and bandwidth, MC pulsones and Zak pulsones are the same. However, for finite time and bandwidth duration, MC and Zak pulsones are different.* This is illustrated in the following through an example.

As explained in Chapter 2, a Zak pulson is generated as follows. A *quasi-periodic* DD domain impulse at (τ_0, ν_0) is *twisted convolved*/filtered with a DD domain pulse shaping filter $w_{\text{tx}}(\tau, \nu)$. The inverse time-Zak transform of this filtered pulse then gives the Zak pulson.

A MC pulson is generated by *linearly* convolving a *periodic* DD domain impulse function at (τ_0, ν_0) with a pulse shaping filter $w_{\text{tx}}(\tau, \nu)$ followed by multiplication with the DD domain representation of the MC generator $g_{\text{tx}}(t)$. The inverse time-Zak transform of this product then gives the MC pulson $x_{\mathcal{H}, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t)$ (see Fig. 3.9).

Clearly, Zak and MC pulsones are not the same as there can be many different MC pulsones for different choice of the MC generator. More importantly, the generation of the *Zak pulson* involves a *quasi-periodic* DD impulse and *twisted* convolution, whereas the generation of a *MC pulson* involves a *periodic* DD impulse and *linear* convolution.

For the TF window $W_{\text{tx}}(t, f)$ in (3.23) and the rectangular MC generator $g_{\text{tx}}(t)$ in (3.14), from the example in (3.21), it is known that the MC pulson $x_{\mathcal{H}, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t)$ is of duration $T = N\tau_p$ and has bandwidth roughly $B = M\nu_p$. For this choice of the TF window and MC generator, from Theorem 3.5, it follows that the corresponding MC pulson is given by

$$x_{\mathcal{H}, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) = \begin{cases} M\sqrt{\nu_p} \left[\frac{1}{M} \sum_{m=0}^{M-1} e^{j2\pi m \nu_p (t - \tau_0)} \right] e^{j2\pi \nu_0 \left\lfloor \frac{t}{\tau_p} \right\rfloor \tau_p}, & 0 \leq t < N\tau_p, \\ 0, & \text{otherwise.} \end{cases} \quad (3.36)$$

Note that $\left[\frac{1}{M} \sum_{m=0}^{M-1} e^{j2\pi m \nu_p (t - \tau_0)} \right]$ is a periodic pulse train having period τ_p where the n th pulse is located around $t = (\tau_0 + n\tau_p)$. The MC pulson consists of only N pulses of this infinite pulse train as its time duration is limited to $T = N\tau_p$, and the adjacent pulses are spaced apart by τ_p seconds.

The term $e^{j2\pi \nu_0 \left\lfloor \frac{t}{\tau_p} \right\rfloor \tau_p}$ corresponds roughly to a Doppler shift of ν_0 . Note that this MC pulson exhibits *discontinuities* at time instances which are integer multiples of τ_p .

The TF window $W_{\text{tx}}(t, f)$ in (3.23) corresponds to the DD domain pulse shaping filter

$$w_{\text{tx}}(\tau, \nu) = \text{SFT} (W_{\text{tx}}(t, f)) = \iint W_{\text{tx}}(t, f) e^{-j2\pi(\nu t - \tau f)} dt df, \quad (3.37)$$

which is given by

$$w_{\text{tx}}(\tau, \nu) = \left[\int_0^{N\tau_p} e^{-j2\pi \nu t} dt \right] \left[\int_0^{M\nu_p} e^{j2\pi \tau f} df \right]. \quad (3.38)$$

With this pulse shaping filter, the Zak pulson generated in Chapter 2 is given by (see (2.41))

$$x_{\mathcal{Z}, \tau_0, \nu_0}^{w_{\text{tx}}}(t) = M\sqrt{\nu_p} \sum_{n=0}^{N-1} e^{j2\pi \nu_0 n \tau_p} e^{j\pi M \nu_p (t - n\tau_p - \tau_0)} \text{sinc} (M \nu_p (t - n\tau_p - \tau_0)). \quad (3.39)$$

Just like the MC pulsone $x_{\mathcal{H}, \tau_0, \nu_0}^{u_{\text{tx}}, g_{\text{tx}}}(t)$, the Zak pulsone $x_{\mathcal{Z}, \tau_0, \nu_0}^{u_{\text{tx}}}(t)$ also consists of a finite train of N pulses located at $t = (\tau_0 + n\tau_p)$, $n = 0, 1, \dots, N-1$. The term $e^{j2\pi\nu_0 n\tau_p}$ corresponds roughly to $e^{j2\pi\nu_0(t-\tau_0)}$ when the pulses are narrow (i.e., when M is large). Note that $e^{j2\pi\nu_0(t-\tau_0)}$ is simply a Doppler shift by ν_0 . However, from the expressions in (3.36) and (3.39), it is clear that the MC and the Zak pulsone are not the same even for the same pulse shaping filter $w_{\text{tx}}(\tau, \nu)$. In addition, the MC pulsone exhibits discontinuities, whereas the Zak pulsone is always continuous.

3.4 Multicarrier OTFS (MC-OTFS) Modulation

This section discusses a MC variant of OTFS (MC-OTFS), where information is carried by MC pulsone. Given a time and bandwidth limitation of $T = N\tau_p$ and $B = M\nu_p$, respectively, in Appendix 3.H, it is argued that the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0) \star w_{\text{tx}}(\tau, \nu)$ (whose ISFT gives the TF function $X_{\text{TF}}^W(t, f; \tau_0, \nu_0)$ used for generating the MC pulsone $x_{\mathcal{H}, \tau_0, \nu_0}^{u_{\text{tx}}, g_{\text{tx}}}(t)$) consists of pulses whose spread along the delay and Doppler domain is approximately $1/B$ and $1/T$, respectively. This suggests that there are approximately BT MC pulsone whose support within the fundamental DD domain period $\mathcal{D}_0$ do not overlap.² The following theorem states an important result that the set of $BT = MN$ MC pulsone (TF window function as in (3.23) and the rectangular MC generator as in (3.14)) corresponding to (τ_0, ν_0) located on the DD domain information grid Λ_{dd} within $\mathcal{D}_0$ form an *orthogonal basis* of waveforms for all TD signals which are approximately time limited to T seconds and bandwidth limited to B Hz.

Theorem 3.6 For a given (τ_p, ν_p) , $\tau_p \nu_p = 1$, and $N = T\nu_p$, $M = B\tau_p$, the set of MN MC pulsone

$$\left\{ \frac{1}{\sqrt{MN}} x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{u_{\text{tx}}, g_{\text{tx}}}(t) \mid k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1 \right\}, \quad (3.40)$$

corresponding to the rectangular TF window function $W_{\text{tx}}(t, f)$ in (3.23) and the rectangular MC generator $g_{\text{tx}}(t)$ as in (3.14), form an orthonormal basis for the set of all TD signals which are approximately time limited to the TD interval $[0, T)$ and bandwidth limited to the FD interval $[0, B)$.

Proof: See Appendix 3.I.

In MC-OTFS, the MN MC pulsone corresponding to periodic DD pulses located at $\left(\frac{k\tau_p}{M}, \frac{l\nu_p}{N}\right)$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, are the MN *carrier waveforms*. Let $x[k, l]$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, denote the MN information symbols to be transmitted in an MC-OTFS frame. The (k, l) th carrier waveform $x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{u_{\text{tx}}, g_{\text{tx}}}(t)$ carries the (k, l) th information symbol $x[k, l]$. The modulated TD MC-OTFS transmit signal is, therefore, given by

$$s_{\text{td}}(t) = \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{u_{\text{tx}}, g_{\text{tx}}}(t). \quad (3.41)$$

The modulated signal $s_{\text{td}}(t)$ can be generated using two different transmitter architectures, as explained in the following. The first architecture is inspired by the generation of MC pulsone

² It suffices to consider MC pulsone $x_{\mathcal{H}, \tau_0, \nu_0}^{u_{\text{tx}}, g_{\text{tx}}}(t)$ corresponding to $(\tau_0, \nu_0) \in \mathcal{D}_0$ since a pulsone corresponding to (τ_0, ν_0) outside $\mathcal{D}_0$ is the same as the pulsone corresponding to $(\tau_0 \bmod \tau_p, \nu_0 \bmod \nu_p)$ which belongs to $\mathcal{D}_0$ (see Corollary 3.2).

using the Heisenberg transform (see top subfigure in Fig. 3.9) and involves two steps, namely, (i) converting the DD domain information symbols to TF domain symbols and (ii) converting the TF domain symbols to the TD transmit signal $s_{\text{td}}(t)$ in (3.41). This architecture is, therefore, also called the two-step MC-OTFS architecture. The second architecture is inspired by the generation of MC pulsones using the Zak transform (see the bottom subfigure in Fig. 3.9).

3.4.1 Two-step MC-OTFS Modulator

In (3.41), the (k, l) th information symbol $x[k, l]$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, M-1$ is carried by the MC pulsone $x_p^{w_{\text{tx}}, g_{\text{tx}}}(\tau, \nu)$ corresponding to the DD domain periodic function $x_p\left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$, which is a periodic impulse function at the DD point $\left(\frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$; see the top subfigure in Fig. 3.9 with $(\tau_0, \nu_0) = \left(\frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$. The transformation from a periodic DD domain impulse function to a TD pulsone is illustrated in Fig. 3.9. Accordingly, for transforming the (k, l) th information symbol $x[k, l]$ to the TD-modulated signal term $x[k, l]x_p^{w_{\text{tx}}, g_{\text{tx}}}(\tau, \nu)$ in (3.41), the (k, l) th periodic DD domain impulse function $x_p\left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$ is filtered with $w_{\text{tx}}(\tau, \nu)$, which is the SFT of the TF window $W_{\text{tx}}(t, f)$. Second, this filtered periodic DD domain impulse function, i.e., $x[k, l]w_{\text{tx}}(\tau, \nu) \star x_p\left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$, is transformed to the TF function $x[k, l]X_{\text{TF}}^{w_{\text{tx}}}\left(t, f; \frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$ using ISFT. Finally, the Heisenberg transform associated with this TF function gives as output the TD-modulated signal $x[k, l]x_p^{w_{\text{tx}}, g_{\text{tx}}}(\tau, \nu)$, which is the (k, l) th term in the summation in (3.41).

Since the linear convolution, ISFT, and Heisenberg transform are all linear operations, the MC-OTFS-modulated signal $s_{\text{td}}(t)$ in (3.41) can be generated using a *two-step* architecture described in the following and illustrated in Fig. 3.10.

First step: In the first step, the information symbols $x[k, l]$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, are transformed to a TF signal $X_{\text{TF}}^{w_{\text{tx}}}(t, f)$. The information symbols are embedded into a periodic

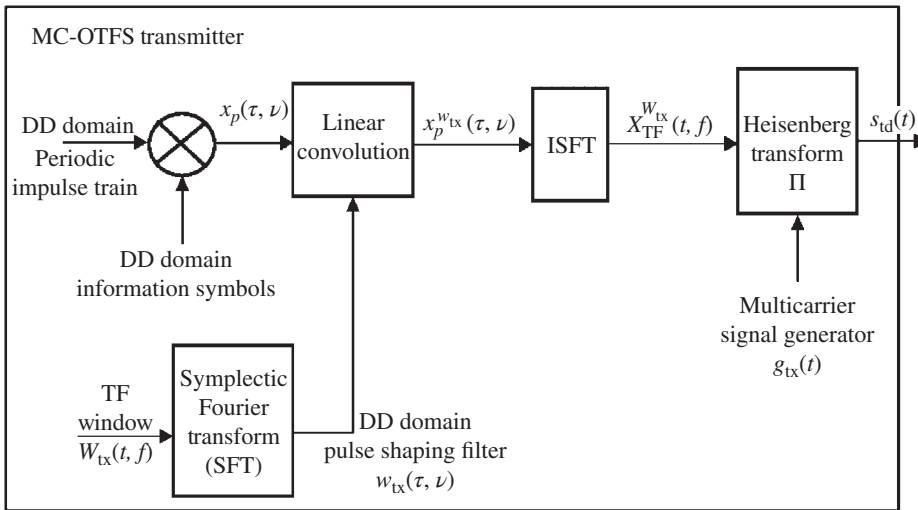


Figure 3.10 Two-step architecture of the MC-OTFS modulator based on the Heisenberg transform.

discrete DD domain function

$$x_p[k, l] \triangleq x[k \bmod M, l \bmod N], \quad k, l \in \mathbb{Z}, \quad (3.42)$$

having periods M and N along the discrete delay and Doppler domains respectively, i.e., for all $n, m \in \mathbb{Z}$,

$$x_p[k + nM, l + mN] = x_p[k, l]. \quad (3.43)$$

This discrete periodic function is then converted to a function $x_p(\tau, \nu)$ in the continuous DD domain by modulating DD domain impulse functions on the information grid Λ_{dd} , i.e.,

$$x_p(\tau, \nu) \triangleq \sum_{k, l \in \mathbb{Z}} x_p[k, l] \delta\left(\tau - \frac{k\tau_p}{M}\right) \delta\left(\nu - \frac{l\nu_p}{N}\right). \quad (3.44)$$

Since $x_p[k, l]$ is periodic with periods M and N , it follows that $x_p(\tau, \nu)$ is periodic with periods τ_p and ν_p along the delay and Doppler domains, respectively. Next, $x_p(\tau, \nu)$ is filtered (linear convolution) with the filter $w_{\text{tx}}(\tau, \nu) = \text{SFT}(W_{\text{tx}}(t, f))$, resulting in the filtered periodic continuous DD domain function

$$\begin{aligned} x_p^{w_{\text{tx}}}(\tau, \nu) &\triangleq w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu), \\ w_{\text{tx}}(\tau, \nu) &= \text{SFT}(W_{\text{tx}}(t, f)). \end{aligned} \quad (3.45)$$

Finally, the ISFT of this filtered periodic function gives the TF function

$$X_{\text{TF}}^{W_{\text{tx}}}(t, f) \triangleq \text{ISFT}\left(x_p^{w_{\text{tx}}}(\tau, \nu)\right). \quad (3.46)$$

Second step: In the second step, the Heisenberg transform corresponding to the TF function $X_{\text{TF}}^{W_{\text{tx}}}(t, f)$ when given the MC generator input $g_{\text{ix}}(t)$ generates the output TD signal, which is given by

$$\Pi_{X_{\text{TF}}^{W_{\text{tx}}}(t, f)}\left(g_{\text{ix}}(t)\right). \quad (3.47)$$

The following theorem states that the output $\Pi_{X_{\text{TF}}^{W_{\text{tx}}}(t, f)}(g_{\text{ix}}(t))$ of the Heisenberg transform in the second step of the two-step MC-OTFS modulator is equal to the MC-OTFS-modulated signal $s_{\text{id}}(t)$ in (3.41).

Theorem 3.7 Consider the two-step MC-OTFS modulator/transmitter architecture shown in Fig. 3.10. For any given TF window function $W_{\text{tx}}(t, f)$, information symbols $x[k, l]$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, and MC generator $g_{\text{ix}}(t)$, the output of the two-step OTFS modulator is equal to the MC-OTFS-modulated signal $s_{\text{id}}(t)$ in (3.41), i.e., for $X_{\text{TF}}^{W_{\text{tx}}}(t, f)$ given by (3.46)

$$\Pi_{X_{\text{TF}}^{W_{\text{tx}}}(t, f)}(g_{\text{ix}}(t)) = \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{tx}}, g_{\text{ix}}}(t). \quad (3.48)$$

Proof: See Appendix 3.J.

The following corollary gives another useful equivalent expression of the MC-OTFS-modulated signal $s_{\text{id}}(t)$ in (3.41).

Corollary 3.3 The MC-OTFS-modulated signal $s_{\text{id}}(t)$ in (3.41) can also be expressed as

$$s_{\text{id}}(t) = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} g_{\text{ix}}(t - n\tau_p) e^{j2\pi m\nu_p(t - n\tau_p)} X^{W_{\text{tx}}}[n, m], \quad (3.49)$$

where the windowed TF symbols are given by

$$X^{W_{\text{tx}}}[n, m] = W_{\text{tx}}(n\tau_p, mv_p) \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] e^{j2\pi \left(\frac{nl}{N} - \frac{mk}{M} \right)}. \quad (3.50)$$

Proof: From the expression of MC pulsones in (3.32), it follows that

$$x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{lv_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t) = \sum_{n \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, mv_p) e^{-j2\pi \frac{mk}{M}} e^{j2\pi mv_p(t - n\tau_p)} \right]. \quad (3.51)$$

Using this expression in (3.41), it follows that

$$\begin{aligned} s_{\text{id}}(t) &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \sum_{n \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, mv_p) e^{-j2\pi \frac{mk}{M}} e^{j2\pi mv_p(t - n\tau_p)} \right] \\ &\stackrel{(a)}{=} \sum_{n, m \in \mathbb{Z}} g_{\text{tx}}(t - n\tau_p) e^{j2\pi mv_p(t - n\tau_p)} \underbrace{W_{\text{tx}}(n\tau_p, mv_p) \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] e^{j2\pi \left(\frac{nl}{N} - \frac{mk}{M} \right)}}_{X^{W_{\text{tx}}}[n, m]}, \end{aligned} \quad (3.52)$$

where step (a) follows by exchanging the order of summation on the RHS.

Due to the time and bandwidth constraints, the TF window $W_{\text{tx}}(t, f)$ is practically zero for $t \notin [0, N\tau_p)$ or $f \notin [0, Mv_p)$. This practical simplification gives the two-step MC-OTFS modulator architecture proposed by Hadani et al. in their IEEE WCNC'2017 paper [28] (see Theorem 3.8).

Theorem 3.8 When the support of the TF window $W_{\text{tx}}(t, f)$ is restricted to $t \in [0, N\tau_p)$ and $f \in [0, Mv_p)$, the two-step modulator can be simplified as follows. In the first step, the information symbols $x[k, l]$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, are converted to MN TF domain symbols $X[n, m]$, $n = 0, 1, \dots, N-1$, $m = 0, 1, \dots, M-1$, through the inverse symplectic finite Fourier transform (ISFFT), i.e.,

$$X[n, m] = \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] e^{j2\pi \left(\frac{ln}{N} - \frac{km}{M} \right)}. \quad (3.53)$$

These TF symbols are then windowed to get the windowed TF symbols

$$X^{W_{\text{tx}}}[n, m] = W_{\text{tx}}(n\tau_p, mv_p) X[n, m]. \quad (3.54)$$

In the second step, these windowed TF symbols then generate the modulated MC-OTFS TD signal as

$$s_{\text{id}}(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X^{W_{\text{tx}}}[n, m] g_{\text{tx}}(t - n\tau_p) e^{j2\pi mv_p(t - n\tau_p)}. \quad (3.55)$$

Proof: The proof follows directly from Corollary 3.3. This is because, on the RHS of (3.49), the summation is restricted to $n = 0, 1, \dots, N-1$, $m = 0, 1, \dots, M-1$. The windowing and the ISFFT follow directly from (3.50).

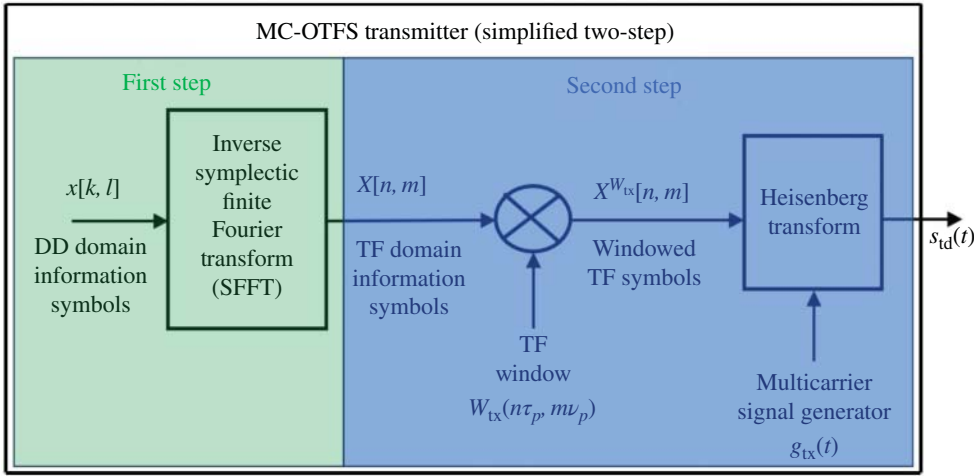


Figure 3.11 Simplified MC-OTFS two-step modulator architecture.

Figure 3.11 shows the simplified MC-OTFS two-step modulator, where the first step is highlighted in light gray and the second step is highlighted in dark gray.

3.4.2 Zak Transform-Based MC-OTFS Modulator

The second architecture for the MC-OTFS modulator is inspired by the generation of MC pulses using the Zak transform (see the bottom subfigure in Fig. 3.9). In Fig. 3.9, the equivalent Zak transform-based generation of a MC pulse consists of a cascade of DD domain linear convolution with $w_{tx}(\tau, \nu)$, followed by multiplication with $G_{tx,dd}(\tau, \nu)$, and subsequently inverse Zak transform ($\mathcal{Z}_t^{-1}$). To be precise, substituting $\tau_0 = k\tau_p/M$ and $\nu_0 = l\nu_p/N$ in (3.31), it follows that

$$x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{tx}g_{tx}}(t) = \mathcal{Z}_t^{-1} \left(G_{tx,dd}(\tau, \nu) \left[w_{tx}(\tau, \nu) \star x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{l\nu_p}{N} \right) \right] \right). \quad (3.56)$$

Since the cascade of operations described above is linear, it follows that the TD MC-OTFS-modulated signal in (3.41) can also be generated using the Zak transform, as shown in Fig. 3.12. The following theorem states this equivalent Zak transform-based MC-OTFS modulator.

Theorem 3.9 The MC-OTFS-modulated TD signal in (3.41) is also given by

$$\begin{aligned} s_{td}(t) &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{tx}g_{tx}}(t) \\ &= \mathcal{Z}_t^{-1} \left(G_{tx,dd}(\tau, \nu) \left[w_{tx}(\tau, \nu) \star x_p(\tau, \nu) \right] \right). \end{aligned} \quad (3.57)$$

Proof: See Appendix 3.K.

The Zak transform-based MC-OTFS modulator architecture follows directly from (3.57) and is illustrated in Fig. 3.12. The steps of this equivalent MC-OTFS modulator are as follows. As shown in

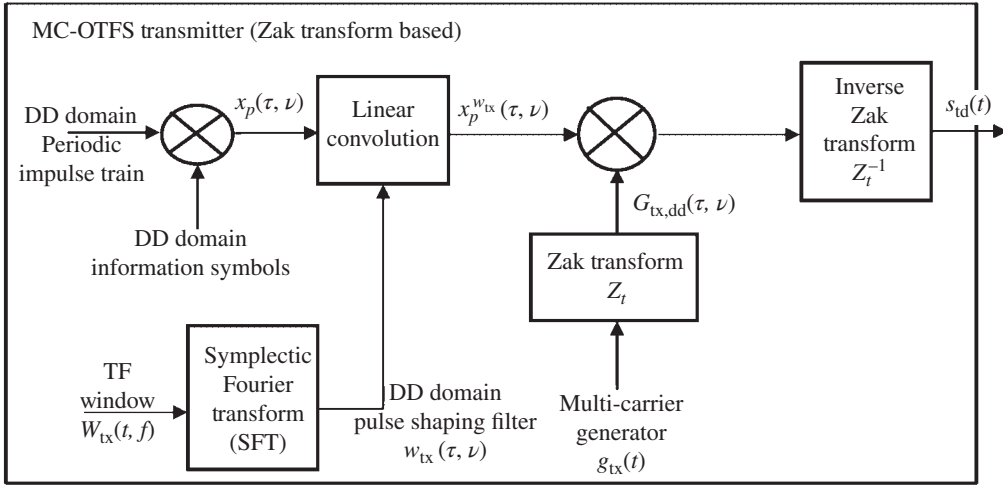


Figure 3.12 MC-OTFS modulator architecture based on the Zak transform.

Fig. 3.12, first, the DD domain periodic function $x_p(\tau, \nu)$ containing information is convolved with the pulse shaping filter $w_{tx}(\tau, \nu)$, resulting in the filtered periodic DD domain function $x_p^{w_{tx}}(\tau, \nu)$. This filtered periodic DD domain function is multiplied with the Zak transform $G_{tx,dd}(\tau, \nu)$ of the MC generator $g_{tx}(t)$. Since the Zak transform of any TD signal is quasi-periodic, and the product of a quasi-periodic DD domain signal and a periodic DD domain function is quasi-periodic, it follows that $G_{tx,dd}(\tau, \nu)x_p^{w_{tx}}(\tau, \nu)$ is *quasi-periodic*. The MC-OTFS-modulated TD signal in (3.41) is then given by

$$s_{td}(t) = \mathcal{Z}_t^{-1} \left(G_{tx,dd}(\tau, \nu) x_p^{w_{tx}}(\tau, \nu) \right). \quad (3.58)$$

3.5 MC-OTFS Receiver

For the rectangular TF window $W_{tx}(t, f)$ in (3.23) and the rectangular MC generator in (3.14), from Theorem 3.6, we know that the MN MC pulsones $x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{tx}, g_{tx}}(t)$, $k = 0, 1, \dots, M-1$, $l = 0, 1, \dots, N-1$, corresponding to periodic DD domain pulses on the information grid Λ_{dd} span the space of TD signals which are approximately limited to the TD interval $[0, N\tau_p)$ and limited to the FD interval $[0, M\nu_p)$. Since the received TD signal $r_{td}(t)$ is also approximately limited to the same TD and FD interval, it suffices to decode information by simply projecting $r_{td}(t)$ onto the MC pulsones.

In general, at the receiver, the signal basis could be a different basis of MC pulsones when compared to that at the transmitter. Therefore, let the TF window and the MC generator for the basis of MC pulsones used at the receiver be $W_{rx}(t, f)$ and $g_{rx}(t)$, respectively. The projection of $r_{td}(t)$ onto the (k, l) th MC pulsone $x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{rx}, g_{rx}}(t)$ is given by

$$\begin{aligned} y_p[k, l] &\triangleq \int r_{td}(t) \left(x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{rx}, g_{rx}}(t) \right)^* dt \\ &= \sum_{n, m \in \mathbb{Z}} W_{rx}^*(n\tau_p, m\nu_p) e^{j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)} \int r_{td}(t) g_{rx}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt, \end{aligned} \quad (3.59)$$

where the second step follows from the pulsed expression in (3.32) corresponding to a periodic DD pulse at $(\tau_0, \nu_0) = (k\tau_p/M, l\nu_p/N)$ on the information grid. In addition, in (3.32), $W_{\text{tx}}(t, f)$ and $g_{\text{tx}}(t)$ are replaced by $W_{\text{rx}}(t, f)$ and $g_{\text{rx}}(t)$, respectively. For any $n, m \in \mathbb{Z}$, the MC pulsed $x_{\text{H}, \frac{(k+nM)\tau_p}{M}, \frac{(l+mN)\nu_p}{N}}^{W_{\text{rx}}, g_{\text{rx}}}(t)$ are all identical (see Corollary 3.2). Therefore, it follows that $y_p[k, l]$ is periodic with periods M and N along the discrete delay and Doppler domains, respectively, i.e.,

$$y_p[k + nM, l + mN] = y_p[k, l], \quad (3.60)$$

for all $n, m \in \mathbb{Z}$. Since $y_p[k, l]$ is periodic, the MN received samples $y[k, l], k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$, are a sufficient statistic for MC-OTFS demodulation.

In the following, we present two different but equivalent receiver architectures for MC-OTFS demodulation. The first architecture is inspired from the two-step MC-OTFS modulator architecture at the transmitter (see Fig. 3.10) and is discussed in Section 3.5.1. The second architecture is inspired from the Zak transform-based MC-OTFS modulator (see Fig. 3.12) and is discussed in Section 3.5.2.

3.5.1 Two-step MC-OTFS Receiver

The integral within the second line on the RHS of (3.59) gives rise to a two-step receiver architecture, where the received TD signal $r_{\text{td}}(t)$ is transformed to TF symbols $Y[n, m], n, m \in \mathbb{Z}$, in the first step, i.e.,

$$Y[n, m] \triangleq \int r_{\text{td}}(t) g_{\text{rx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt. \quad (3.61)$$

This has been referred to as the Wigner transform by Hadani et al. in [28], which can be implemented using an OFDM demodulator. The TF symbols are then windowed with $W_{\text{rx}}(\cdot, \cdot)$ to give $Y^{W_{\text{rx}}}[n, m], n, m \in \mathbb{Z}$, i.e.,

$$Y^{W_{\text{rx}}}[n, m] \triangleq W_{\text{rx}}^*(n\tau_p, m\nu_p) Y[n, m]. \quad (3.62)$$

Using (3.62) in (3.59), we get

$$y_p[k, l] = \sum_{n, m \in \mathbb{Z}} Y^{W_{\text{rx}}}[n, m] e^{j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)}, \quad (3.63)$$

for $k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$. This is the second step of the two-step receiver. In most scenarios, the received TD signal $r_{\text{td}}(t)$ is approximately limited to the TD and FD intervals $[0, N\tau_p)$ and $[0, M\nu_p)$, respectively, and, therefore, the infinite sum on the RHS of (3.63) can be limited to $n = 0, 1, \dots, N-1, m = 0, 1, \dots, M-1$, i.e.,

$$y_p[k, l] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y^{W_{\text{rx}}}[n, m] e^{j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)}. \quad (3.64)$$

This transformation from the MN windowed TF symbols $Y^{W_{\text{rx}}}[n, m], n = 0, 1, \dots, N-1, m = 0, 1, \dots, M-1$, to the MN DD domain symbols $y_p[k, l], k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$, is in fact symplectic finite Fourier transform (SFFT) (i.e., inverse of the ISFFT transform in (3.53)). This two-step receiver architecture is illustrated in Fig. 3.13 and is identical to the receiver structure proposed in [28] by Hadani et al.

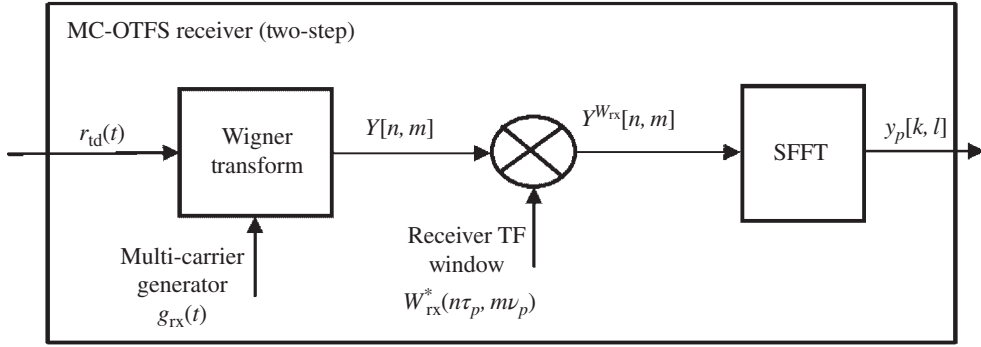


Figure 3.13 Two-step architecture for the MC-OTFS receiver.

3.5.2 Zak Transform-Based MC-OTFS Receiver

Another equivalent receiver is inspired from the Zak transform-based MC-OTFS modulator architecture in Section 3.4.2. From Theorem 2.4, we know that the inner product between any two TD signals is conserved in the DD domain. Therefore, the projection of the received TD signal $r_{\text{id}}(t)$ onto the (k, l) th MC pulsone (i.e., $y_p[k, l]$ in (3.59)) is equivalently given by

$$\begin{aligned}
 y_p[k, l] &= \int r_{\text{id}}(t) \left(x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{rx}} \star g_{\text{rx}}}(t) \right)^* dt \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) \left[\mathcal{Z}_t \left(x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{rx}} \star g_{\text{rx}}}(t) \right) \right]^* d\nu d\tau \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) \left[G_{\text{rx,dd}}(\tau, \nu) \left(w_{\text{rx}}(\tau, \nu) \star x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{l\nu_p}{N} \right) \right) \right]^* d\tau d\nu,
 \end{aligned} \tag{3.65}$$

where

$$G_{\text{rx,dd}}(\tau, \nu) = \mathcal{Z}_t \left(g_{\text{rx}}(t) \right) \tag{3.66}$$

is the Zak transform of the MC generator $g_{\text{rx}}(t)$ at the receiver,

$$w_{\text{rx}}(\tau, \nu) = \text{SFT} \left(W_{\text{rx}}(t, f) \right) \tag{3.67}$$

is the SFT of the TF window $W_{\text{rx}}(t, f)$ at the receiver, and

$$y_{\text{dd}}(\tau, \nu) \triangleq \mathcal{Z}_t \left(r_{\text{id}}(t) \right) \tag{3.68}$$

is the Zak transform of the received TD signal. The third step in (3.65) follows from the expression of the (k, l) th MC pulsone $x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{\text{rx}} \star g_{\text{rx}}}(t)$ in (3.31) with $(\tau_0, \nu_0) = (k\tau_p/M, l\nu_p/N)$, and with $w_{\text{rx}}(\tau, \nu)$ and $G_{\text{rx,dd}}(\tau, \nu)$ replaced by $w_{\text{rx}}(\tau, \nu)$ and $G_{\text{rx,dd}}(\tau, \nu)$, respectively (see also Fig. 3.9 for the Zak transform-based generation of MC pulsones).

The following theorem expresses $y_p[k, l]$ in terms of the Zak transform of the received TD signal $r_{\text{id}}(t)$, and this defines the architecture of the Zak transform-based MC-OTFS demodulator.

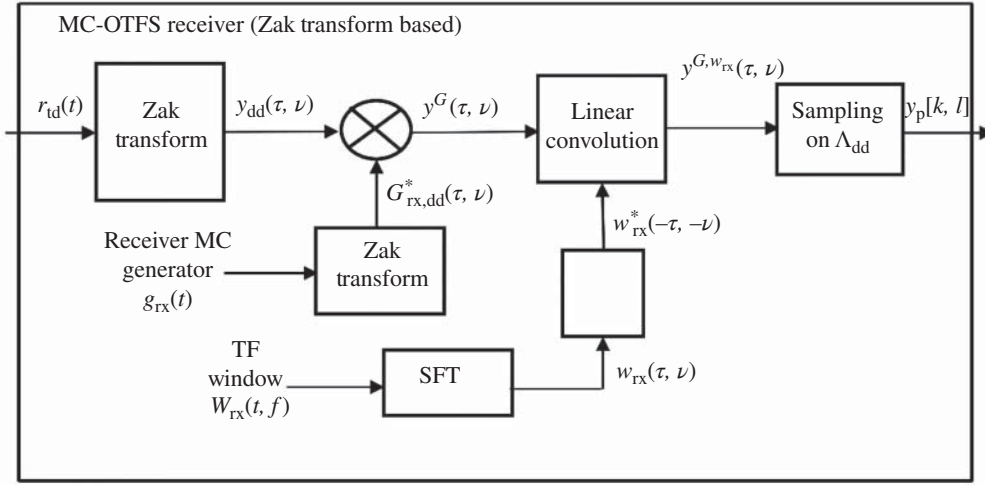


Figure 3.14 Zak transform-based MC-OTFS receiver.

Theorem 3.10 The projection of the received TD signal onto the MN MC pulses $x_{H, \frac{k\tau_p}{M}, \frac{l\nu_p}{N}}^{w_{rx} g_{rx}}(t)$, $k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$, is given by

$$y_p[k, l] = y^{G, w_{rx}} \left(\tau = \frac{k\tau_p}{M}, \nu = \frac{l\nu_p}{N} \right),$$

$$y^{G, w_{rx}}(\tau, \nu) \triangleq y^G(\tau, \nu) \star w_{rx}^*(-\tau, -\nu),$$

$$y^G(\tau, \nu) \triangleq y_{dd}(\tau, \nu) G_{rx, dd}^*(\tau, \nu), \quad (3.69)$$

where $\star$ denotes DD domain linear convolution.

Proof: See Appendix 3.L

Based on Theorem 3.10, the Zak transform-based MC-OTFS receiver architecture is illustrated in Fig. 3.14.

3.6 MC-OTFS I/O Relation

In this section, the explicit relation between the receiver output $y_p[k, l], k, l \in \mathbb{Z}$ and the receiver input $x_p[k, l], k, l \in \mathbb{Z}$ (see (3.42)) is given. Note that both the input $x_p[k, l]$ and the output $y_p[k, l]$ are discrete periodic functions in the DD domain with periods M and N along the delay and Doppler domain, respectively (see (3.43) and (3.60)).

We derive the I/O relation both for the two-step transmitter and receiver, and the Zak transform-based transceiver.

3.6.1 MC-OTFS I/O for a Two-Step Transceiver

The following theorem gives the TF domain I/O relation, i.e., the relation between the received TF symbols $Y[n, m]$ and the transmit windowed TF symbols $X^{W_{tx}}[n, m], n, m \in \mathbb{Z}$ (see (3.50) and Fig. 3.13).

Theorem 3.11 The received TF symbols $Y[n, m]$, $n, m \in \mathbb{Z}$, are given by

$$Y[n, m] = \sum_{n', m' \in \mathbb{Z}} X^{W_{\text{tx}}}[n', m'] H_{TF}[n - n', m - m'; n', m'] + n_{TF}[n, m], \quad (3.70)$$

where

$$n_{TF}[n, m] \triangleq \int n_{\text{id}}(t) g_{\text{rx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt \quad (3.71)$$

and

$$H_{TF}[n - n', m - m'; n', m'] \triangleq \iint [h_{\text{phy}}(\tau, \nu) e^{j2\pi(\nu n' \tau_p - \tau m' \nu_p)} e^{j2\pi\tau(m' - m)\nu_p} A_{g_{\text{rx}}, g_{\text{tx}}}^*(\tau + (n' - n)\tau_p, \nu + (m' - m)\nu_p)] d\tau d\nu. \quad (3.72)$$

Here, $A_{g_{\text{rx}}, g_{\text{tx}}}(\tau, \nu)$ is the cross ambiguity function between $g_{\text{rx}}(t)$ and $g_{\text{tx}}(t)$, which is given by

$$A_{g_{\text{rx}}, g_{\text{tx}}}(\tau, \nu) \triangleq \int g_{\text{rx}}(t) g_{\text{tx}}^*(t - \tau) e^{-j2\pi\nu(t - \tau)} dt. \quad (3.73)$$

Proof: See Appendix 3.M.

From Theorem 3.11, it is clear that, for general doubly spread channels, the transmitted TF symbols interfere with each other. In an ideal channel with no delay/Doppler shift (e.g., $h_{\text{phy}}(\tau, \nu) = \delta(\tau)\delta(\nu)$), the transmit and receive MC generators can be chosen to satisfy the *bi-orthogonality* property which ensures that there is no interference between the TF symbols. The following corollary states this bi-orthogonality condition.

Corollary 3.4 [Bi-orthogonal MC generators] In an ideal channel with no delay/Doppler shift, i.e., $h_{\text{phy}}(\tau, \nu) = \delta(\tau)\delta(\nu)$, there is no interference between TF symbols if the transmit and the receive MC generators satisfy the bi-orthogonality condition, i.e.,

$$A_{g_{\text{rx}}, g_{\text{tx}}}(n\tau_p, m\nu_p) = \int g_{\text{rx}}(t) g_{\text{tx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt = 0, \quad (3.74)$$

for all $(n, m) \neq (0, 0)$, $n, m \in \mathbb{Z}$. For an ideal channel with MC generators satisfying the bi-orthogonality condition, the TF domain I/O relation is, therefore, given by

$$Y[n, m] = X^{W_{\text{tx}}}[n, m] H_{TF}[0, 0; n, m] + n_{TF}[n, m], \quad (3.75)$$

where

$$H_{TF}[0, 0; n, m] = A_{g_{\text{rx}}, g_{\text{tx}}}^*(0, 0) = \int g_{\text{rx}}^*(t) g_{\text{tx}}(t) dt. \quad (3.76)$$

Proof: See Appendix 3.N.

Note that the bi-orthogonality condition in Corollary 3.4 is the same as the condition in Eqn. (4) of the MC-OTFS paper [28] by R. Hadani et al. There exist MC generators which satisfy the bi-orthogonality condition. For example, this condition is satisfied when the transmit and receive MC generators are rectangular, i.e.,

$$g_{\text{tx}}(t) = g_{\text{rx}}(t) = \begin{cases} \frac{1}{\sqrt{\tau_p}}, & 0 \leq t < \tau_p, \\ 0, & \text{otherwise.} \end{cases} \quad (3.77)$$

The next obvious question is whether it is possible to choose the transmit and receive MC generators in such a way so that even in a non-ideal channel, there is no interference between TF domain symbols $X^{W_{\text{tx}}}[n, m]$. The following corollary gives a condition on the MC generators which guarantees non-interfering TF domain symbols. However, this condition cannot be satisfied by practical MC generators, and, therefore, in a doubly spread channel, the TF domain symbols always interfere with each other irrespective of the choice of MC generators.

Corollary 3.5 [Ideal MC generators] Let the support set of the channel DD spreading function $h_{\text{phy}}(\tau, \nu)$ be

$$S_h = \left\{ (\tau, \nu) \mid h_{\text{phy}}(\tau, \nu) \neq 0 \right\}. \quad (3.78)$$

If for all $(n', m') \neq (n, m)$, $n, m, n', m' \in \mathbb{Z}$, the MC generators satisfy the ideal condition

$$A_{g_{\text{rx}}, g_{\text{tx}}}(\tau + (n' - n)\tau_p, \nu + (m' - m)\nu_p) = 0, (\tau, \nu) \in S_h, \quad (3.79)$$

then there is no interference between the TF domain symbols, and the TF domain I/O relation is given by

$$Y[n, m] = X^{W_{\text{tx}}}[n, m] H_{TF}[0, 0; n, m] + n_{TF}[n, m], \quad (3.80)$$

where

$$H_{TF}[0, 0; n, m] = \iint_{(\tau, \nu) \in S_h} h_{\text{phy}}(\tau, \nu) e^{j2\pi(\nu n \tau_p - \tau m \nu_p)} A_{g_{\text{rx}}, g_{\text{tx}}}^*(\tau, \nu) d\tau d\nu. \quad (3.81)$$

As a special case, if MC generators additionally satisfy the condition

$$A_{g_{\text{rx}}, g_{\text{tx}}}(\tau, \nu) = 1, (\tau, \nu) \in S_h, \quad (3.82)$$

then

$$H_{TF}[0, 0; n, m] = \iint h_{\text{phy}}(\tau, \nu) e^{j2\pi(\nu n \tau_p - \tau m \nu_p)} d\tau d\nu, \quad (3.83)$$

which is the ISFT of $h_{\text{phy}}(\tau, \nu)$ evaluated at $(t = n\tau_p, f = m\nu_p)$ and is, in fact, the TF representation of the channel at $t = n\tau_p$ and $f = m\nu_p$.

Proof: See Appendix 3.O.

Note that the ideal condition in (3.79) is the same as that in [28] (see the paragraph before Eqn. (17)), where it is stated for a rectangular support set $S_h = \{(\tau, \nu) \mid |\tau| \leq \tau_{\text{max}}, |\nu| \leq \nu_{\text{max}}\}$. Most practical MC generators (e.g., rectangular) do not satisfy the ideal conditions in (3.79).

The next theorem gives the MC-OTFS I/O relation, i.e., the relation between the discrete DD domain output $y_p[k, l]$ and the input information symbols $x[k, l]$ (see Figs. 3.10 and 3.13).

Theorem 3.12 The I/O relation for the two-step MC-OTFS transceiver is given by

$$y_p[k, l] = \sum_{k'=0}^{M-1N-1} \sum_{l'=0}^{M-1N-1} x[k', l'] h[k', l'; k, l] + n_p[k, l], \quad (3.84)$$

where

$$h[k', l'; k, l] \triangleq \sum_{n, m \in \mathbb{Z}} \sum_{n', m' \in \mathbb{Z}} W_{\text{rx}}^*(n\tau_p, m\nu_p) W_{\text{tx}}(n'\tau_p, m'\nu_p) e^{j2\pi \frac{(n'l' - nl)}{N}} e^{j2\pi \frac{(mk - m'k')}{M}} H_{TF}[n - n', m - m'; n', m']. \quad (3.85)$$

Here, $H_{TF}[n - n', m - m'; n', m']$, $n, m, n', m' \in \mathbb{Z}$, is given by (3.72). The (k, l) th DD domain noise sample in (3.84) is given by

$$n_p[k, l] \stackrel{\Delta}{=} \sum_{n, m \in \mathbb{Z}} e^{j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)} n_{TF}[n, m] W_{rx}^*(n\tau_p, m\nu_p), \quad (3.86)$$

where $n_{TF}[n, m]$ is given by (3.71).

Proof: See Appendix 3.P.

3.6.2 MC-OTFS I/O Relation for Zak Transform-Based Transceiver

Combining the Zak transform-based MC-OTFS transmitter and receiver operations, as illustrated in Figs. 3.12 and 3.14, the continuous DD domain I/O relation between the information carrying DD domain function $x_p(\tau, \nu)$ and the filtered received DD domain function $y^{G, w_{rx}}(\tau, \nu)$ is given by the following theorem.

Theorem 3.13 The received filtered DD domain function $y^{G, w_{rx}}(\tau, \nu)$ is periodic with delay and Doppler domain periods τ_p and ν_p , respectively, i.e., for all $n, m \in \mathbb{Z}$,

$$y^{G, w_{rx}}(\tau + n\tau_p, \nu + m\nu_p) = y^{G, w_{rx}}(\tau, \nu). \quad (3.87)$$

Furthermore, $y^{G, w_{rx}}(\tau, \nu)$ is related to the periodic information function $x_p(\tau, \nu)$ by

$$\begin{aligned} y^{G, w_{rx}}(\tau, \nu) &= w_{rx}^*(-\tau, -\nu) \star \left(G_{rx, dd}^*(\tau, \nu) \left[h_{phy}(\tau, \nu) *_{\sigma} \left(G_{tx, dd}(\tau, \nu) [w_{tx}(\tau, \nu) \star x_p(\tau, \nu)] \right) \right] \right) \\ &\quad + n_p^{G, w_{rx}}(\tau, \nu), \\ n_p^{G, w_{rx}}(\tau, \nu) &\stackrel{\Delta}{=} w_{rx}^*(-\tau, -\nu) \star \left(G_{rx, dd}^*(\tau, \nu) n_{dd}(\tau, \nu) \right). \end{aligned} \quad (3.88)$$

Proof: See Appendix 3.Q.

Remark From (3.88), it is clear that $y^{G, w_{rx}}(\tau, \nu)$ is related to $x_p(\tau, \nu)$ through a mixture of DD domain operations like linear convolution, multiplication, and twisted convolution. Clearly, $y^{G, w_{rx}}(\tau, \nu)$ cannot be expressed as a simple twisted convolution operation of the information function $x_p(\tau, \nu)$ with some effective channel filter. Hence, acquiring the MC-OTFS I/O relation is expected to be more complex and inefficient (i.e., higher pilot overhead) when compared to that for Zak-OTFS, where, in the crystalline regime, the complete I/O relation can be acquired from the channel response to a single DD domain impulse pilot signal (MC-OTFS and Zak-OTFS I/O relations are summarized in Table 3.3). In other words, the Zak-OTFS I/O relation is *more predictable* than the MC-OTFS I/O relation.

Table 3.3 I/O relation for Zak-OTFS and MC-OTFS.

MC-OTFS I/O relation (see (3.88))
$y^{G, w_{rx}}(\tau, \nu) = w_{rx}^*(-\tau, -\nu) \star \left[G_{rx, dd}^*(\tau, \nu) \cdot \left(h_{phy}(\tau, \nu) *_{\sigma} \left\{ G_{tx, dd}(\tau, \nu) \cdot [w_{tx}(\tau, \nu) \star x_p(\tau, \nu)] \right\} \right) \right]$
Zak-OTFS I/O relation (see (2.0.2))
$y_{dd}^{w_{rx}}(\tau, \nu) = w_{rx}(\tau, \nu) *_{\sigma} h_{phy}(\tau, \nu) *_{\sigma} w_{tx}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu) = h_{eff}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu)$

The following corollary to this theorem gives the discrete DD domain I/O relation between $y_p[k, l]$ and $x_p[k, l]$ for the Zak transform-based MC-OTFS transceivers.

Corollary 3.6 The discrete DD domain function $y_p[k, l]$, $k, l \in \mathbb{Z}$, at the output of the MC-OTFS receiver is periodic with periods M and N along the discrete delay and Doppler domains, i.e., for all $n, m \in \mathbb{Z}$,

$$y_p[k + nM, l + mN] = y_p[k, l]. \quad (3.89)$$

Furthermore, the relation between the discrete output $y_p[k, l]$ and the discrete input $x_p[k, l]$, $k, l \in \mathbb{Z}$, is given by

$$\begin{aligned} y_p[k, l] &= \sum_{k', l' \in \mathbb{Z}} x_p[k', l'] h_w[k, l; k', l'] + n_p[k, l], \\ n_p[k, l] &\triangleq n_p^{G, w_{tx}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right), \\ h_w[k, l; k', l'] &\triangleq h_w \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N}; k', l' \right), \\ h_w(\tau, \nu; k', l') &\triangleq w_{rx}^*(-\tau, -\nu) \\ &\quad \star \left(G_{rx, dd}^*(\tau, \nu) \left[h_{phy}(\tau, \nu) *_{\sigma} \left(G_{tx, dd}(\tau, \nu) w_{tx} \left(\tau - k' \frac{\tau_p}{M}, \nu - l' \frac{\nu_p}{N} \right) \right) \right] \right). \end{aligned} \quad (3.90)$$

Proof: See Appendix 3.R.

3.7 Comparing MC-OTFS to Zak-OTFS

It is illuminating to observe how MC-OTFS and Zak-OTFS evolve as we move toward time division multiplexing (TDM) by shrinking the Doppler period ν_p . As we traverse the hyperbola $\tau_p \nu_p = 1$ along the limit $\nu_p \rightarrow 0$, Zak-OTFS converges to TDM, whose I/O relation for delay-only channels is given by linear convolution with an effective channel filter; hence, it is predictable. In contrast, as $\nu_p \rightarrow 0$, MC-OTFS does not converge to TDM. The limit in this case is a non-intuitive modulation whose I/O relation cannot be expressed as a simple linear convolution of an input signal with an effective TD channel filter (see Remark after Theorem 3.13 in Section 3.6.2).

Recall that Fig. 2.27 depicts the relative prediction error (RPE) for Zak-OTFS as a two-dimensional heatmap, for three different points on the period hyperbola $\tau_p \nu_p = 1$. Figure 3.15 depicts the RPE for MC-OTFS with respect to the same two-path channel and the same transmit and receive filters (given by sinc pulses). The MC generator at the transmitter and receiver is taken to be a rectangular TD pulse of duration τ_p as in (3.14). The TF window at the transmitter and receiver is taken to be the rectangular window $W_{tx}(t, f)$ in (3.23). For simulations, a matrix-vector form of the MC-OTFS I/O relation is derived starting from the I/O relation in (3.84). This is along the same lines as the matrix-vector form of I/O relation for Zak-OTFS in Section 2.8.1. Just as in Fig. 2.27, in Fig. 3.15 also, we consider $B = 0.96$ MHz and $T = 1.6$ ms. Just as for Zak-OTFS, for MC-OTFS also, we estimate the I/O relation from the received response to a pilot impulse located at $(M/2, N/2)$.

In Fig. 3.15, the point $\nu_p = 1.25$ kHz, $\tau_p = 800$ μ s represents the delay asymptote outside the crystalline regime, and the point $\nu_p = 240$ kHz, $\tau_p = 4.16$ μ s represents the Doppler asymptote outside the crystalline regime. For these points, the RPE is high for both Zak-OTFS and MC-OTFS. The point $\nu_p = 30$ kHz, $\tau_p = 33.3$ μ s represents the crystalline regime where RPE for MC-OTFS is

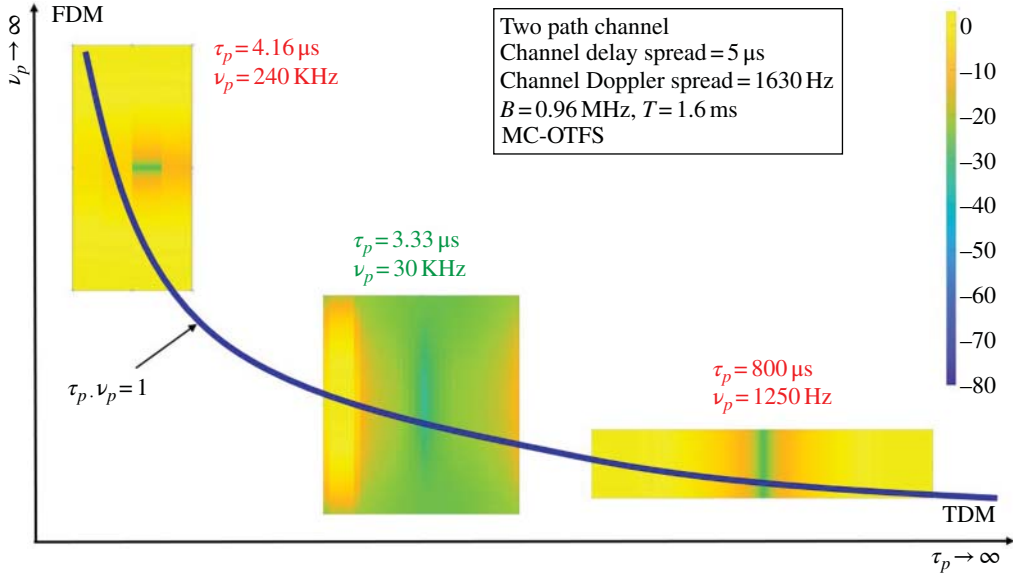


Figure 3.15 Heatmap showing RPE (in dB) for MC-OTFS with sinc pulses as a function of delay (horizontal axis) and Doppler (vertical axis). RPE for MC-OTFS is larger than that for Zak-OTFS. Source: Mohammed et al. [34] IEEE.

larger than that for Zak-OTFS (the heatmap in Fig. 3.15 is a mixture of white and light gray, whereas the heatmap in Fig. 2.27 has a darker gray tone).

Figure 3.16 compares the BER performance of Zak-OTFS and MC-OTFS on the Veh-A channel introduced in Section 2.9 and used in Fig. 2.33. We have changed the channel parameters slightly

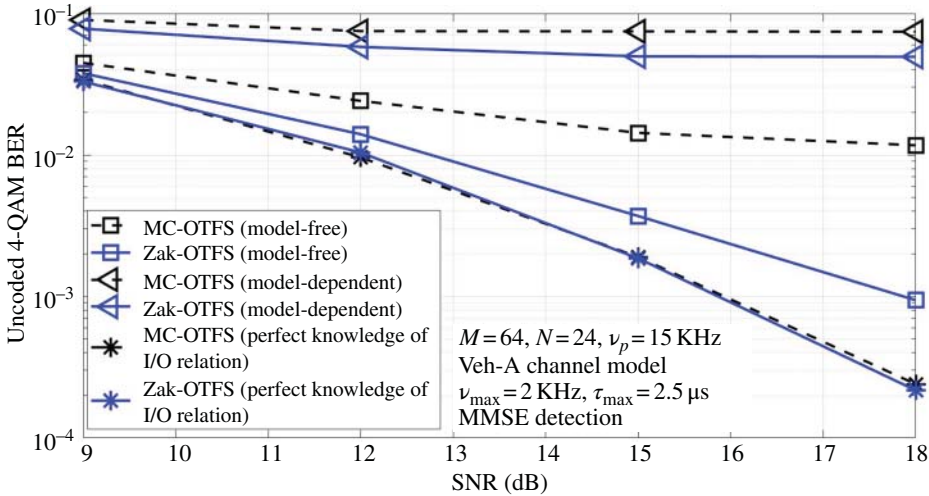


Figure 3.16 Uncoded BER performance for Zak-OTFS and MC-OTFS on a Veh-A channel as a function of increasing SNR. The performance of model-dependent Zak-OTFS and MC-OTFS is poor since the channel bandwidth and duration are not sufficient to estimate channel path gains, delays, and Doppler shifts. The performance of model-free Zak-OTFS is superior to that of model-free MC-OTFS because the Zak-OTFS I/O relation is more predictable. Source: Mohammed et al. [34] IEEE.

by increasing the Doppler shift $\nu_{\max}$ from 815 Hz to 2 kHz. For MC-OTFS, the TF window and the MC signal generator are the same as those used in Fig. 3.15 (see the above discussion). Given perfect knowledge of the I/O relation, the performance of Zak-OTFS and MC-OTFS are quite similar. However, with imperfect knowledge of the I/O relation, the model-free performance of Zak-OTFS is greatly superior to that of MC-OTFS because the Zak-OTFS I/O relation is more predictable than that of MC-OTFS.

Remark Figure 3.16 reveals that with perfect knowledge of the I/O relation, both Zak-OTFS and MC-OTFS exhibit similar BER performance. Acquiring the I/O relation in the model-free approach amounts to estimating the channel response to a pilot at any arbitrary location. In Zak-OTFS, operation in the crystalline regime allows accurate prediction of the channel response to an arbitrary pilot from the response to a single pilot, facilitating efficient acquisition of the complete I/O relation. In contrast, in MC-OTFS, due to the complex nature of its I/O relation, it seems that such a prediction scheme is not possible; hence, accurate acquisition of the complete I/O relation becomes less efficient when compared to that for Zak-OTFS. Since accurate acquisition of the MC-OTFS I/O relation is challenging, several prior studies on MC-OTFS consider the model-dependent approach, where the goal is to estimate the underlying physical channel (e.g., the path gain, delay, and Doppler shifts in case of a discrete multi-path channel). In comparison, for Zak-OTFS, we note that it is the simplicity of its I/O relation which makes the model-free approach practical.

Figure 3.17 compares the model-free performance of various modulation schemes as a function of increasing Doppler spread while operating at a fixed signal-to-noise ratio (SNR) of 16 dB. The simulations use the Veh-A channel introduced in Section 2.9. In particular, the figure illustrates that when $\nu_{\max}$ is less than 1 kHz (Doppler spread less than 2 kHz), the BER performance of Zak-OTFS and MC-OTFS is essentially the same. However, as the Doppler spread increases, the BER performance of MC-OTFS degrades steadily, while that of Zak-OTFS remains almost constant. The reason for that gap is because when the Doppler spread is high, the Zak-OTFS I/O relation is more predictable than that of MC-OTFS.

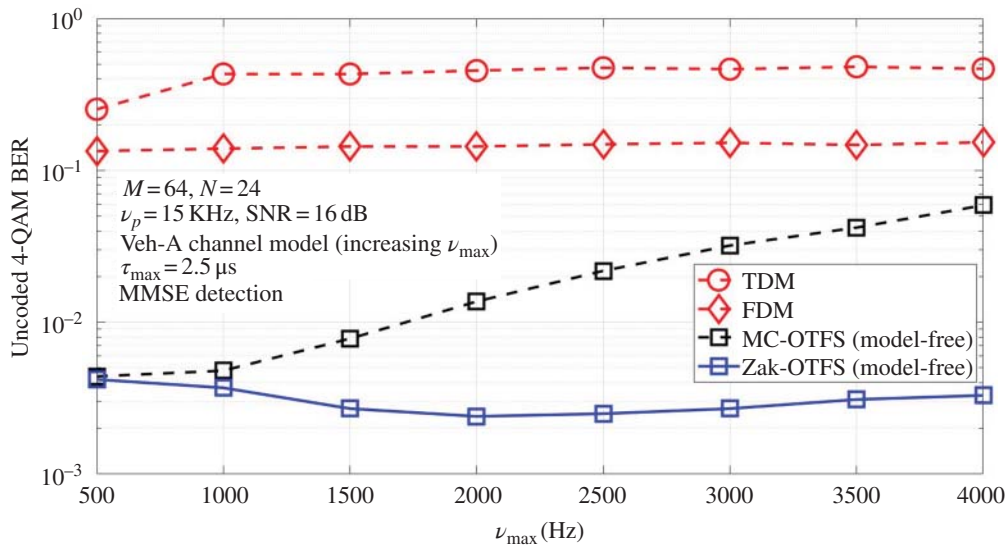


Figure 3.17 Un-coded BER performance of model-free Zak-OTFS and model-free MC-OTFS as a function of $\nu_{\max}$ for the Veh-A channel introduced in Section 2.9 ($\nu_p = 15$ kHz, $M = 64$, $N = 24$). Zak-OTFS is more robust to high Doppler spreads. Source: Mohammed et al. [34] IEEE.

Appendix 3.A Proof of Theorem 3.1

For the TF function in (3.12) and an arbitrary input MC generator $g_{\text{ix}}(t)$, using (3.8), the output of the Heisenberg transform is given by

$$\begin{aligned}
 \Pi_{X_{\text{TF}}(t_f; \tau_0, \nu_0)}(g_{\text{ix}}(t)) &= \iint X_{\text{TF}}(t', f'; \tau_0, \nu_0) g_{\text{ix}}(t - t') e^{j2\pi f'(t-t')} dt' df' \\
 &= \sum_{n, m \in \mathbb{Z}} \iint e^{j2\pi(\nu_0 t' - \tau_0 f')} \delta(t' - n\tau_p) \delta(f' - m\nu_p) g_{\text{ix}}(t - t') e^{j2\pi f'(t-t')} dt' df' \\
 &= \sum_{n, m \in \mathbb{Z}} e^{j2\pi(\nu_0 n\tau_p - \tau_0 m\nu_p)} g_{\text{ix}}(t - n\tau_p) e^{j2\pi m\nu_p(t - n\tau_p)} \\
 &= \left[\sqrt{\tau_p} \sum_{n \in \mathbb{Z}} g_{\text{ix}}(t - n\tau_p) e^{j2\pi \nu_0 n\tau_p} \right] \left[\sqrt{\nu_p} \sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p(t - \tau_0)} \right], \tag{3.A.1}
 \end{aligned}$$

where we have used the fact that $\tau_p \nu_p = 1$. Furthermore,

$$\begin{aligned}
 \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} g_{\text{ix}}(t - n\tau_p) e^{j2\pi \nu_0 n\tau_p} &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} g_{\text{ix}}(t + n\tau_p) e^{-j2\pi \nu_0 n\tau_p} \\
 &= G_{\text{ix,dd}}(t, \nu_0), \tag{3.A.2}
 \end{aligned}$$

where, in the first step, we have simply changed the summation variable from n to $-n$. The second step follows from the comparison of the RHS in the first step with the Zak transform equation in (2.23). Note that $G_{\text{ix,dd}}(\tau, \nu)$ denotes the Zak transform of $g_{\text{ix}}(t)$. In (3.A.1), the factor $\sqrt{\nu_p} \sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p(t - \tau_0)}$ is a periodic TD signal with period τ_p and is simply obtained by delaying the periodic signal $\sqrt{\nu_p} \sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p t}$ by τ_0 seconds. The Fourier transform of $\sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p t}$ consists of a train of Dirac-delta impulses in the FD at integer multiples of the inverse period, i.e., $\nu_p = 1/\tau_p$. Since the inverse Fourier transform of a FD impulse train is a TD impulse train, it follows that

$$\sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p t} = \tau_p \sum_{n \in \mathbb{Z}} \delta(t - n\tau_p), \tag{3.A.3}$$

and, therefore,

$$\sqrt{\nu_p} \sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p(t - \tau_0)} = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p). \tag{3.A.4}$$

Substituting the factors on the RHS of the last line of (3.A.1) with their expressions from (3.A.2) and (3.A.4), it follows that

$$\begin{aligned}
 \Pi_{X_{\text{TF}}(t_f; \tau_0, \nu_0)}(g_{\text{ix}}(t)) &= \sqrt{\tau_p} G_{\text{ix,dd}}(t, \nu_0) \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} G_{\text{ix,dd}}(t, \nu_0) \delta(t - \tau_0 - n\tau_p) \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} G_{\text{ix,dd}}(\tau_0 + n\tau_p, \nu_0) \delta(t - \tau_0 - n\tau_p) \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} G_{\text{ix,dd}}(\tau_0, \nu_0) e^{j2\pi n\nu_0 \tau_p} \delta(t - \tau_0 - n\tau_p), \tag{3.A.5}
 \end{aligned}$$

where the last step follows from the quasi-periodicity of the Zak transform of any TD signal (here, $G_{\text{ix,dd}}(\tau, \nu)$ is quasi-periodic, see (2.25)). Note that the RHS in the equation above is nothing but the pulsone $p_{\tau_0, \nu_0}(t)$ scaled by the constant $G_{\text{ix,dd}}(\tau_0, \nu_0)$.

Appendix 3.B Proof of Corollary 3.14

From (2.23), it follows that the Zak transform of the rectangular MC generator in (3.14) is given by

$$G_{\text{tx,dd}}(\tau, \nu) = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} g_{\text{tx}}(\tau + n\tau_p) e^{-j2\pi n\nu\tau_p}. \quad (3.B.1)$$

Therefore, for any $(\tau_0, \nu_0) \in \mathcal{D}_0$,

$$G_{\text{tx,dd}}(\tau_0, \nu_0) = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} g_{\text{tx}}(\tau_0 + n\tau_p) e^{-j2\pi n\nu_0\tau_p}. \quad (3.B.2)$$

As $(\tau_0, \nu_0) \in \mathcal{D}_0$, $0 \leq \tau_0 < \tau_p$. Furthermore, $g_{\text{tx}}(t)$ is non-zero only for $0 \leq t < \tau_p$, and, therefore, the only non-zero term on the RHS above corresponds to $n = 0$. Hence,

$$G_{\text{tx,dd}}(\tau_0, \nu_0) = \sqrt{\tau_p} g_{\text{tx}}(\tau_0) = 1. \quad (3.B.3)$$

Since $G_{\text{tx,dd}}(\tau_0, \nu_0) = 1$, from (3.13), it follows that $\Pi_{X_{\text{TF}}(t_f; \tau_0, \nu_0)}(g_{\text{tx}}(t)) = p_{\tau_0, \nu_0}(t)$ for all $(\tau_0, \nu_0) \in \mathcal{D}_0$.

Appendix 3.C Proof of Theorem 3.2

Showing (3.15) is equivalent to showing that the inverse Zak transform of $G_{\text{tx,dd}}(\tau, \nu) x_p(\tau, \nu; \tau_0, \nu_0)$ is $G_{\text{tx,dd}}(\tau_0, \nu_0) p_{(\tau_0, \nu_0)}(t)$, i.e.,

$$\mathcal{Z}_t^{-1} \left(G_{\text{tx,dd}}(\tau, \nu) x_p(\tau, \nu; \tau_0, \nu_0) \right) = G_{\text{tx,dd}}(\tau_0, \nu_0) p_{(\tau_0, \nu_0)}(t). \quad (3.C.1)$$

From the expression for the inverse Zak transform in (2.F.1) (see Appendix 2.F.1), it follows that

$$\begin{aligned} & \mathcal{Z}_t^{-1} \left(G_{\text{tx,dd}}(\tau, \nu) x_p(\tau, \nu; \tau_0, \nu_0) \right) \\ &= \sqrt{\tau_p} \int_0^{\nu_p} G_{\text{tx,dd}}(t, \nu) x_p(t, \nu; \tau_0, \nu_0) d\nu \\ &= \sqrt{\tau_p} \int_0^{\nu_p} G_{\text{tx,dd}}(t, \nu) \sum_{n, m \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p) d\nu \\ &\stackrel{(a)}{=} \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) \left[\sum_{m \in \mathbb{Z}} \int_0^{\nu_p} G_{\text{tx,dd}}(\tau_0 + n\tau_p, \nu) \delta(\nu - \nu_0 - m\nu_p) d\nu \right] \\ &\stackrel{(b)}{=} \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) \int_0^{\nu_p} G_{\text{tx,dd}}(\tau_0, \nu) e^{j2\pi n\nu\tau_p} \delta(\nu - \nu_0) d\nu \\ &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \delta(t - \tau_0 - n\tau_p) G_{\text{tx,dd}}(\tau_0, \nu_0) e^{j2\pi n\nu_0\tau_p}, \end{aligned} \quad (3.C.2)$$

where, in step (a), we have used a property of Dirac-delta functions, i.e., $G_{\text{tx,dd}}(t, \nu) \delta(t - \tau_0 - n\tau_p) = G_{\text{tx,dd}}(\tau_0 + n\tau_p, \nu) \delta(t - \tau_0 - n\tau_p)$. In step (b), we have used the fact that since $(\tau_0, \nu_0) \in \mathcal{D}_0$, $0 \leq \nu_0 < \nu_p$, and, therefore, the only non-zero term in the inner summation is for $m = 0$. In this step, we have also used the fact that $G_{\text{tx,dd}}(\tau_0 + n\tau_p, \nu) = e^{j2\pi n\nu\tau_p} G_{\text{tx,dd}}(\tau_0, \nu)$ since $G_{\text{tx,dd}}(\tau, \nu)$ is the Zak transform of $g_{\text{tx}}(t)$ and is therefore a quasi-periodic signal. This completes the proof.

Appendix 3.D SFT and Inverse SFT

SFT gives the DD domain representation of a TF function $X_{\text{TF}}(t, f)$, which is given by

$$x(\tau, \nu) = \text{SFT} \left(X_{\text{TF}}(t, f) \right) \triangleq \iint X_{\text{TF}}(t, f) e^{-j2\pi(\nu t - \tau f)} dt df. \quad (3.D.1)$$

The inverse transform, i.e., ISFT, gives the TF domain representation of a DD domain function, i.e.,

$$\text{ISFT} (x(\tau, \nu)) \triangleq \iint x(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau d\nu. \quad (3.D.2)$$

Since SFT and ISFT are inverse of each other, for any DD domain function $x(\tau, \nu)$,

$$\text{SFT} (\text{ISFT} (x(\tau, \nu))) = x(\tau, \nu). \quad (3.D.3)$$

Similarly, for any TF domain function $X_{\text{TF}}(t, f)$,

$$\text{ISFT} \left(\text{SFT} \left(X_{\text{TF}}(t, f) \right) \right) = X_{\text{TF}}(t, f). \quad (3.D.4)$$

SFT arises from the Fourier series expansion of periodic 2D functions. An important property of SFT is that the SFT of any discrete TF function is a periodic DD domain function, and vice versa.

Theorem 3.14 For any discrete TF function supported on the TF period lattice/grid $\Lambda_{\text{TF}} = \{(t, f) \mid t = n\tau_p, f = m\nu_p, n, m \in \mathbb{Z}\}$, i.e.,

$$X_{\text{TF}}(t, f) = \sum_{n, m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m\nu_p), \quad (3.D.5)$$

$x(\tau, \nu) = \text{SFT} \left(X_{\text{TF}}(t, f) \right)$ is always periodic with periods τ_p and ν_p , respectively, along the delay and Doppler domains, i.e., for all $n, m \in \mathbb{Z}$,

$$x(\tau + n\tau_p, \nu + m\nu_p) = x(\tau, \nu). \quad (3.D.6)$$

Conversely, given a periodic DD domain function $x(\tau, \nu)$ with periods τ_p and ν_p , respectively, along the delay and Doppler domains, its ISFT $X_{\text{TF}}(t, f)$ is a discrete TF function with support only on the TF period lattice/grid Λ_{TF} , i.e., it is given by (3.D.5) with $X[n, m]$ given by

$$X[n, m] \triangleq \int_0^{\nu_p} \int_0^{\tau_p} x(\tau, \nu) e^{j2\pi(n\nu\tau_p - m\tau\nu_p)} d\tau d\nu. \quad (3.D.7)$$

Proof: Using (3.D.1), the SFT of $X_{\text{TF}}(t, f)$ in (3.D.5) is given by

$$\begin{aligned} x(\tau, \nu) &= \text{SFT} \left(X_{\text{TF}}(t, f) \right) \\ &= \iint X_{\text{TF}}(t, f) e^{-j2\pi(\nu t - \tau f)} dt df \\ &= \sum_{n, m \in \mathbb{Z}} X[n, m] \iint \delta(t - n\tau_p) \delta(f - m\nu_p) e^{-j2\pi(\nu t - \tau f)} dt df \\ &= \sum_{n, m \in \mathbb{Z}} X[n, m] e^{-j2\pi(n\nu\tau_p - m\tau\nu_p)}. \end{aligned} \quad (3.D.8)$$

Clearly, $x(\tau, \nu)$ in (3.D.8) is periodic in the DD domain with period τ_p along the delay domain and period ν_p along the Doppler domain. This is because, for any $k, l \in \mathbb{Z}$,

$$\begin{aligned} x(\tau + k\tau_p, \nu + l\nu_p) &= \sum_{n,m \in \mathbb{Z}} X[n, m] e^{-j2\pi(n(\nu + l\nu_p)\tau_p - m(\tau + k\tau_p)\nu_p)} \\ &= \sum_{n,m \in \mathbb{Z}} X[n, m] e^{-j2\pi(n\nu\tau_p - m\tau\nu_p)} = x(\tau, \nu), \end{aligned} \quad (3.D.9)$$

where, in the last step, we have used the fact that $\tau_p \nu_p = 1$.

Conversely, consider an arbitrary periodic DD domain function $x(\tau, \nu)$ satisfying (3.D.6). From (3.D.2), its ISFT is given by

$$X_{\text{TF}}(t, f) = \iint x(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau, d\nu. \quad (3.D.10)$$

Let $x_{\mathcal{D}_0}(\tau, \nu)$ denote $x(\tau, \nu)$ restricted to the fundamental DD domain period $\mathcal{D}_0$ (see (2.18)), i.e.,

$$x_{\mathcal{D}_0}(\tau, \nu) \triangleq \begin{cases} x(\tau, \nu), & (\tau, \nu) \in \mathcal{D}_0, \\ 0, & \text{otherwise.} \end{cases} \quad (3.D.11)$$

Since $x(\tau, \nu)$ is periodic with periods τ_p and ν_p along the delay and Doppler domains, it can be expressed in terms of $x_{\mathcal{D}_0}(\tau, \nu)$ as

$$x(\tau, \nu) = \sum_{n,m \in \mathbb{Z}} x_{\mathcal{D}_0}(\tau - n\tau_p, \nu - m\nu_p). \quad (3.D.12)$$

Substituting the expression of $x(\tau, \nu)$ from (3.D.12) on the RHS of (3.D.10), it follows that

$$X_{\text{TF}}(t, f) = \sum_{n,m \in \mathbb{Z}} \iint x_{\mathcal{D}_0}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi(\nu t - \tau f)} d\tau d\nu. \quad (3.D.13)$$

For any given $(n, m) \in \mathbb{Z}$,

$$\begin{aligned} &\iint x_{\mathcal{D}_0}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi(\nu t - \tau f)} d\tau, d\nu \\ &\stackrel{(a)}{=} \iint x_{\mathcal{D}_0}(\tau, \nu) e^{j2\pi((\nu + m\nu_p)t - (\tau + n\tau_p)f)} d\tau d\nu \\ &\stackrel{(b)}{=} e^{j2\pi m\nu_p t} e^{-j2\pi n\tau_p f} \int_0^{\nu_p} \int_0^{\tau_p} x(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau d\nu, \end{aligned} \quad (3.D.14)$$

where step (a) follows from substituting the integration variables τ and ν with $(\tau + n\tau_p)$ and $(\nu + m\nu_p)$, respectively. Step (b) follows from the fact that the support set of $x_{\mathcal{D}_0}(\tau, \nu)$ is the fundamental DD period $\mathcal{D}_0$, where it is equal to $x(\tau, \nu)$. Using (3.D.14) back in (3.D.13) gives

$$\begin{aligned} X_{\text{TF}}(t, f) &= \sum_{n,m \in \mathbb{Z}} e^{j2\pi m\nu_p t} e^{-j2\pi n\tau_p f} \tilde{X}_{\text{TF}}(t, f), \\ \tilde{X}_{\text{TF}}(t, f) &\triangleq \int_0^{\nu_p} \int_0^{\tau_p} x(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau d\nu. \end{aligned} \quad (3.D.15)$$

Furthermore, from (3.A.3), it follows that $\sum_{m \in \mathbb{Z}} e^{j2\pi m v_p t} = \tau_p \sum_{n \in \mathbb{Z}} \delta(t - n\tau_p)$. Similarly, $\sum_{n \in \mathbb{Z}} e^{-j2\pi n \tau_p f} = v_p \sum_{m \in \mathbb{Z}} \delta(f - m v_p)$. Using these in (3.D.15) gives

$$\begin{aligned} X_{\text{TF}}(t, f) &= \sum_{n, m \in \mathbb{Z}} \delta(t - n\tau_p) \delta(f - m v_p) \tilde{X}_{\text{TF}}(t, f) \\ &= \sum_{n, m \in \mathbb{Z}} \tilde{X}_{\text{TF}}(n\tau_p, m v_p) \delta(t - n\tau_p) \delta(f - m v_p) \\ &= \sum_{n, m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m v_p) \end{aligned} \quad (3.D.16)$$

since $\tilde{X}_{\text{TF}}(n\tau_p, m v_p) = \int_0^{\tau_p} \int_0^{v_p} x(\tau, \nu) e^{j2\pi(n\nu\tau_p - m\tau\nu_p)} d\tau d\nu$, which is nothing but $X[n, m]$ (see (3.D.7)). This completes the proof.

The following useful result shows that the ISFT of the linear convolution of two DD domain functions is the product of their respective ISFT.

Theorem 3.15 The ISFT of the linear convolution of any two DD domain functions is equal to the product of their respective ISFT. Let $a(\tau, \nu)$ and $b(\tau, \nu)$ be the DD domain functions having ISFT $A(t, f) = \text{ISFT}(a(\tau, \nu))$ and $B(t, f) = \text{ISFT}(b(\tau, \nu))$, respectively. Then, the ISFT of the linear convolution $a(\tau, \nu) \star b(\tau, \nu)$ is the product $A(t, f) B(t, f)$.

Proof: First,

$$a(\tau, \nu) \star b(\tau, \nu) = \iint a(\tau - \tau', \nu - \nu') b(\tau', \nu') d\tau' d\nu'. \quad (3.D.17)$$

Next,

$$\begin{aligned} &\text{ISFT}(a(\tau, \nu) \star b(\tau, \nu)) \\ &= \iint \underbrace{\left[\iint a(\tau - \tau', \nu - \nu') b(\tau', \nu') d\tau' d\nu' \right]}_{=a(\tau, \nu) \star b(\tau, \nu)} e^{j2\pi(\nu t - \tau f)} d\tau d\nu \\ &= \iiint a(\tau - \tau', \nu - \nu') b(\tau', \nu') e^{j2\pi((\nu - \nu')t - (\tau - \tau')f)} e^{j2\pi\nu' t} e^{-j2\pi\tau' f} d\tau d\nu d\tau' d\nu' \\ &= \iint b(\tau', \nu') e^{j2\pi\nu' t} e^{-j2\pi\tau' f} \left[\iint a(\tau - \tau', \nu - \nu') e^{j2\pi((\nu - \nu')t - (\tau - \tau')f)} d\tau d\nu \right] d\tau' d\nu'. \end{aligned} \quad (3.D.18)$$

In the above equation, the inner integral within the square brackets evaluates to

$$\begin{aligned} \iint a(\tau - \tau', \nu - \nu') e^{j2\pi((\nu - \nu')t - (\tau - \tau')f)} d\tau d\nu &= \iint a(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau d\nu \\ &= A(t, f), \end{aligned} \quad (3.D.19)$$

where we have simply substituted $(\tau - \tau')$ and $(\nu - \nu')$ by τ and ν , respectively. Using (3.D.19) in (3.D.18) gives

$$\begin{aligned} \text{ISFT}(a(\tau, \nu) \star b(\tau, \nu)) &= A(t, f) \underbrace{\iint b(\tau', \nu') e^{j2\pi\nu' t} e^{-j2\pi\tau' f} d\tau' d\nu'}_{=B(t, f)} \\ &= A(t, f) B(t, f), \end{aligned} \quad (3.D.20)$$

which completes the proof.

Appendix 3.E Proof of Theorem 3.3

Since SFT and ISFT are inverses of each other, it suffices to show that the ISFT of the periodic DD domain function $x_p(\tau, \nu; \tau_0, \nu_0)$ is the TF domain function $X_{\text{TF}}(t, f; \tau_0, \nu_0)$, i.e.,

$$\text{ISFT}(x_p(\tau, \nu; \tau_0, \nu_0)) = X_{\text{TF}}(t, f; \tau_0, \nu_0). \quad (3.E.1)$$

From the expression for ISFT in (3.D.2) (see Appendix 3.D), it follows that

$$\begin{aligned} \text{ISFT}(x_p(\tau, \nu; \tau_0, \nu_0)) &= \iint x_p(\tau, \nu; \tau_0, \nu_0) e^{j2\pi(\nu t - \tau f)} d\tau d\nu \\ &= \sum_{n, m \in \mathbb{Z}} \iint \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m\nu_p) e^{j2\pi(\nu t - \tau f)} d\tau d\nu \\ &= \sum_{n, m \in \mathbb{Z}} e^{j2\pi(\nu_0 t - \tau_0 f)} e^{j2\pi m \nu_p t} e^{-j2\pi n \tau_p f} \\ &= e^{j2\pi(\nu_0 t - \tau_0 f)} \left[\sum_{n \in \mathbb{Z}} e^{-j2\pi n \tau_p f} \right] \left[\sum_{m \in \mathbb{Z}} e^{j2\pi m \nu_p t} \right] \\ &= e^{j2\pi(\nu_0 t - \tau_0 f)} \sum_{n, m \in \mathbb{Z}} \delta(t - n\tau_p) \delta(f - m\nu_p) = X_{\text{TF}}(t, f; \tau_0, \nu_0), \end{aligned} \quad (3.E.2)$$

where the last step follows from the fact that $\sum_{m \in \mathbb{Z}} e^{j2\pi m \nu_p t} = \tau_p \sum_{n \in \mathbb{Z}} \delta(t - n\tau_p)$ (this, in turn, follows from the Fourier series expansion for the TD periodic signal $\sum_{m \in \mathbb{Z}} e^{j2\pi m \nu_p t}$, see (3.A.3)). Similarly, the periodic FD signal $\sum_{n \in \mathbb{Z}} e^{-j2\pi n \tau_p f} = \nu_p \sum_{m \in \mathbb{Z}} \delta(f - m\nu_p)$.

Appendix 3.F Proof of Theorem 3.4

Since $x_p(\tau, \nu)$ is a periodic DD function with delay and Doppler domain periods τ_p and ν_p , respectively, Theorem 3.14 states that the ISFT of $x_p(\tau, \nu)$ has a discrete support on the TF domain period lattice $\Lambda_{\text{TF}} = \{(t, f) \mid t = n\tau_p, f = m\nu_p, n, m \in \mathbb{Z}\}$, i.e.,

$$\begin{aligned} X_{\text{TF}}(t, f) &= \text{ISFT}(x_p(\tau, \nu)) = \sum_{n, m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m\nu_p), \\ X[n, m] &= \int_0^{\nu_p} \int_0^{\tau_p} x_p(\tau, \nu) e^{j2\pi(n\nu\tau_p - m\tau\nu_p)} d\tau d\nu. \end{aligned} \quad (3.F.1)$$

With $X_{\text{TF}}(t, f)$, given by (3.F.1), using (3.8), the TD transmit signal in Fig. 3.7 is given by

$$x_{\text{id}}(t) = \sum_{n, m \in \mathbb{Z}} X[n, m] g_{\text{tx}}(t - n\tau_p) e^{j2\pi m \nu_p (t - n\tau_p)}. \quad (3.F.2)$$

Using (2.23), the Zak transform of $x_{\text{id}}(t)$ is given by

$$\begin{aligned} \mathcal{Z}_t(x_{\text{id}}(t)) &= \sqrt{\tau_p} \sum_{k \in \mathbb{Z}} x_{\text{id}}(\tau + k\tau_p) e^{-j2\pi \nu k \tau_p} \\ &\stackrel{(a)}{=} \sqrt{\tau_p} \sum_{n, m, k \in \mathbb{Z}} X[n, m] g_{\text{tx}}(\tau + k\tau_p - n\tau_p) e^{j2\pi m \nu_p \tau} e^{-j2\pi k \nu \tau_p} \\ &= \sum_{n, m \in \mathbb{Z}} X[n, m] e^{j2\pi m \nu_p \tau} \underbrace{\left(\sqrt{\tau_p} \sum_{k \in \mathbb{Z}} g_{\text{tx}}(\tau - n\tau_p + k\tau_p) e^{-j2\pi k \nu \tau_p} \right)}_{=G_{\text{tx,dd}}(\tau - n\tau_p, \nu)} \end{aligned}$$

$$\begin{aligned}
 & \stackrel{(b)}{=} \sum_{n,m \in \mathbb{Z}} X[n, m] e^{j2\pi m v_p \tau} G_{\text{tx,dd}}(\tau - n\tau_p, \nu) \\
 & \stackrel{(c)}{=} \left[\sum_{n,m \in \mathbb{Z}} X[n, m] e^{j2\pi m v_p \tau} e^{-j2\pi n v \tau_p} \right] G_{\text{tx,dd}}(\tau, \nu),
 \end{aligned} \tag{3.F.3}$$

where step (a) follows from substituting the expression of $x_{\text{id}}(t)$ from (3.F.2) into the first step. Step (b) follows from the fact that $G_{\text{tx,dd}}(\tau, \nu)$ is the Zak transform of $g_{\text{tx}}(t)$ (see (3.13)). Step (c) follows from the quasi-periodicity of $G_{\text{tx,dd}}(\tau, \nu)$ (see (2.25)), i.e., $G_{\text{tx,dd}}(\tau - n\tau_p, \nu) = e^{-j2\pi n v \tau_p} G_{\text{tx,dd}}(\tau, \nu)$.

It suffices to now show that the expression of $\mathcal{Z}_t(x_{\text{id}}(t))$ in (3.F.3) is equal to $x_p(\tau, \nu) G_{\text{tx,dd}}(\tau, \nu)$ (see Fig. 3.7). Since $X_{\text{TF}}(t, f) = \text{ISFT}(x_p(\tau, \nu))$, it is clear that

$$x_p(\tau, \nu) = \text{SFT}(X_{\text{TF}}(t, f)) \tag{3.F.4}$$

since SFT and ISFT are inverses of each other (see Appendix 3.D). Using (3.D.1), the expression for SFT of $X_{\text{TF}}(t, f)$ in (3.F.1) is given by

$$\begin{aligned}
 x_p(\tau, \nu) &= \text{SFT} \left(\sum_{n,m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m v_p) \right) \\
 &= \iint \left(\sum_{n,m \in \mathbb{Z}} X[n, m] \delta(t - n\tau_p) \delta(f - m v_p) \right) e^{-j2\pi(\nu t - \tau f)} dt df \\
 &= \sum_{n,m \in \mathbb{Z}} X[n, m] \iint \delta(t - n\tau_p) \delta(f - m v_p) e^{-j2\pi(\nu t - \tau f)} dt df \\
 &= \sum_{n,m \in \mathbb{Z}} X[n, m] e^{j2\pi m v_p \tau} e^{-j2\pi n v \tau_p}.
 \end{aligned} \tag{3.F.5}$$

This expression is, in fact, equal to the expression within brackets on the RHS of the last step of the derivation of $\mathcal{Z}_t(x_{\text{id}}(t))$ in (3.F.3). Therefore, it follows that

$$\mathcal{Z}_t(x_{\text{id}}(t)) = x_p(\tau, \nu) G_{\text{tx,dd}}(\tau, \nu), \tag{3.F.6}$$

which completes the proof.

Appendix 3.G Proof of Theorem 3.5

First, from the expression of $x_p(\tau, \nu; \tau_0, \nu_0)$ in (3.15), it follows that

$$\begin{aligned}
 x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) &= w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu; \tau_0, \nu_0) \\
 &= w_{\text{tx}}(\tau, \nu) \star \left[\sum_{n,m \in \mathbb{Z}} \delta(\tau - \tau_0 - n\tau_p) \delta(\nu - \nu_0 - m v_p) \right] \\
 &= \sum_{n,m \in \mathbb{Z}} w_{\text{tx}}(\tau - \tau_0 - n\tau_p, \nu - \nu_0 - m v_p).
 \end{aligned} \tag{3.G.1}$$

As shown in Fig. 3.9, the ISFT of $x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ is given by

$$\begin{aligned}
 X_{\text{TF}}^{w_{\text{tx}}}(t, f; \tau_0, \nu_0) &= \text{ISFT}(x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)) \\
 &= \iint x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) e^{j2\pi(\nu t - \tau f)} d\tau d\nu
 \end{aligned}$$

$$\begin{aligned}
& \stackrel{(a)}{=} \sum_{n,m \in \mathbb{Z}} \iint w_{\text{tx}}(\tau - \tau_0 - n\tau_p, \nu - \nu_0 - m\nu_p) e^{j2\pi(\nu t - \tau f)} d\tau d\nu \\
& = \sum_{n,m \in \mathbb{Z}} \left[e^{j2\pi((\nu_0 + m\nu_p)t - (\tau_0 + n\tau_p)f)} \right. \\
& \quad \left. \underbrace{\iint w_{\text{tx}}(\tau - \tau_0 - n\tau_p, \nu - \nu_0 - m\nu_p) e^{j2\pi((\nu - \nu_0 - m\nu_p)t - (\tau - \tau_0 - n\tau_p)f)} d\tau d\nu}_{=W_{\text{tx}}(t,f)} \right] \\
& \stackrel{(b)}{=} W_{\text{tx}}(t,f) e^{j2\pi(\nu_0 t - \tau_0 f)} \left[\sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p t} \right] \left[\sum_{n \in \mathbb{Z}} e^{-j2\pi n\tau_p f} \right], \tag{3.G.2}
\end{aligned}$$

where step (a) follows from substituting the expression of $x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ from (3.G.1). Step (b) follows from the fact that the integral on the RHS of the previous step is the ISFT of $w_{\text{tx}}(\tau, \nu)$, which is the TF window function $W_{\text{tx}}(t, f)$ (see (3.29)). Using $\sum_{m \in \mathbb{Z}} e^{j2\pi m\nu_p t} = \tau_p \sum_{n \in \mathbb{Z}} \delta(t - n\tau_p)$ and $\sum_{n \in \mathbb{Z}} e^{-j2\pi n\tau_p f} = \nu_p \sum_{m \in \mathbb{Z}} \delta(f - m\nu_p)$ in (3.G.2) gives

$$\begin{aligned}
X_{\text{TF}}^{W_{\text{tx}}}(t, f; \tau_0, \nu_0) &= W_{\text{tx}}(t, f) e^{j2\pi(\nu_0 t - \tau_0 f)} \sum_{n,m \in \mathbb{Z}} \delta(t - n\tau_p) \delta(f - m\nu_p) \\
&= \sum_{n,m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi(n\nu_0 \tau_p - m\tau_0 \nu_p)} \delta(t - n\tau_p) \delta(f - m\nu_p). \tag{3.G.3}
\end{aligned}$$

This TF function is similar to that in (3.21) with

$$X[n, m] = W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi(n\nu_0 \tau_p - m\tau_0 \nu_p)}. \tag{3.G.4}$$

Therefore, the TD output of the Heisenberg transform corresponding to $X_{\text{TF}}^{W_{\text{tx}}}(t, f; \tau_0, \nu_0)$ (i.e., the MC pulsone) is given by the expression in (3.22) with $X[n, m]$ given by (3.G.4), i.e.,

$$\begin{aligned}
x_{H, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) &= \sum_{n,m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi(n\nu_0 \tau_p - m\tau_0 \nu_p)} g_{\text{tx}}(t - n\tau_p) e^{j2\pi m\nu_p(t - n\tau_p)} \\
&= \sum_{n \in \mathbb{Z}} e^{j2\pi n\nu_0 \tau_p} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m \in \mathbb{Z}} W_{\text{tx}}(n\tau_p, m\nu_p) e^{j2\pi m\nu_p(t - n\tau_p)} \right], \tag{3.G.5}
\end{aligned}$$

which completes the proof.

Appendix 3.H Impact of Windowing on a Periodic DD Domain Impulse

Here, we study the impact of pulsone time and bandwidth limitation (using TF windowing) on the corresponding periodic DD domain impulse function. From Fig. 3.9, we know that the windowed periodic DD domain impulse function (impulse at (τ_0, ν_0)) is given by

$$\begin{aligned}
x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) &= x_p(\tau, \nu; \tau_0, \nu_0) \star w_{\text{tx}}(\tau, \nu) \\
&\stackrel{(a)}{=} \sum_{n,m \in \mathbb{Z}} w_{\text{tx}}(\tau - n\tau_p - \tau_0, \nu - m\nu_p - \nu_0), \tag{3.H.1}
\end{aligned}$$

where $w_{\text{tx}}(\tau, \nu)$ is the SFT of the TF window $W_{\text{tx}}(t, f)$, and step (a) follows from the expression of $x_p(\tau, \nu; \tau_0, \nu_0)$ in (3.15). From (3.H.1), it follows that $x_p^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0)$ consists of infinitely many

shifted replicas of the pulse $w_{\text{tx}}(\tau, \nu)$. The (n, m) th replica $w_{\text{tx}}(\tau - n\tau_p - \tau_0, \nu - m\nu_p - \nu_0)$ is located at $(\tau_0 + n\tau_p, \nu_0 + m\nu_p)$ and has support over an interval whose widths along the delay and Doppler domain are same as those along the delay and Doppler domain spreads of the pulse $w_{\text{tx}}(\tau, \nu)$. From a previous example starting at (3.21), it is known that for a time and bandwidth limitation of $T = N\tau_p$ and $B = M\nu_p$, respectively, the spread of $W_{\text{tx}}(t, f)$ is roughly T and B along TD and FD, respectively. Since $w_{\text{tx}}(\tau, \nu)$ is the SFT of $W_{\text{tx}}(t, f)$, it follows that the spread of $w_{\text{tx}}(\tau, \nu)$ is approximately $1/T$ along the Doppler domain and $1/B$ along the delay domain. To see this, consider a factorizable $W_{\text{tx}}(t, f) = W_1(t) W_2(f)$, where $W_1(t)$ has a spread of T seconds and $W_2(f)$ has a spread of B Hz. Then, its SFT is given by

$$\begin{aligned} w_{\text{tx}}(\tau, \nu) &= \iint W_{\text{tx}}(t, f) e^{-j2\pi(\nu t - \tau f)} dt df \\ &= \iint W_1(t) W_2(f) e^{-j2\pi(\nu t - \tau f)} dt df \\ &= \underbrace{\left[\int W_1(t) e^{-j2\pi \nu t} dt \right]}_{w_\nu(\nu)} \underbrace{\left[\int W_2(f) e^{j2\pi \tau f} df \right]}_{w_\tau(\tau)}. \end{aligned} \quad (3.H.2)$$

Since $w_\nu(\nu) = \int W_1(t) e^{-j2\pi \nu t} dt$ is essentially the Fourier transform of $W_1(t)$, it is clear that the spread of $w_\nu(\nu)$ along the Doppler domain will be roughly the inverse of the spread of $W_1(t)$ along TD, i.e., $1/T$. Similarly, $w_\tau(\tau) = \int W_2(f) e^{j2\pi \tau f} df$ is the inverse Fourier transform of $W_2(f)$, and, therefore, its spread along the delay domain is roughly the inverse of the spread of $W_2(f)$ along FD, i.e., $1/B$. This result is also valid for a general non-factorizable $W_{\text{tx}}(t, f)$.

Appendix 3.I Proof of Theorem 3.6

Since the TF window $W(t, f)$ is zero outside the TF rectangle $[0, N\tau_p] \times [0, M\nu_p]$ and is one inside it, the pulsone expression in (3.32) is simplified to

$$x_{\mathcal{H}, \tau_0, \nu_0}^{w_{\text{tx}}, g_{\text{tx}}}(t) = \sum_{n=0}^{N-1} e^{j2\pi \nu_0 n \tau_p} g_{\text{tx}}(t - n\tau_p) \left[\sum_{m=0}^{M-1} e^{j2\pi m \nu_p (t - n\tau_p - \tau_0)} \right]. \quad (3.I.1)$$

Using this expression, the inner product between any two pulsones in the basis in (3.40) corresponding to $(\tau_0, \nu_0) = (k_1 \tau_p/M, l_1 \nu_p/N)$ and $(\tau_0, \nu_0) = (k_2 \tau_p/M, l_2 \nu_p/N)$ is given by

$$\begin{aligned} &\frac{1}{MN} \int x_{\mathcal{H}, \frac{k_1 \tau_p}{M}, \frac{l_1 \nu_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t) \left(x_{\mathcal{H}, \frac{k_2 \tau_p}{M}, \frac{l_2 \nu_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t) \right)^* dt \\ &= \frac{1}{MN} \sum_{n_1=0}^{N-1} \sum_{n_2=0}^{N-1} \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} e^{j2\pi \frac{(l_1 n_1 - l_2 n_2)}{N}} e^{j2\pi \frac{m_2 k_2 - m_1 k_1}{M}} \\ &\quad \left[\int g_{\text{tx}}(t - n_1 \tau_p) g_{\text{tx}}^*(t - n_2 \tau_p) e^{j2\pi (m_1 - m_2) \nu_p t} dt \right]. \end{aligned} \quad (3.I.2)$$

From (3.14), it is clear that $g_{\text{tx}}(t - n_1 \tau_p)$ is non-zero only in the TD interval $[n_1 \tau_p, (n_1 + 1)\tau_p]$ and $g_{\text{tx}}(t - n_2 \tau_p)$ is non-zero only in the TD interval $[n_2 \tau_p, (n_2 + 1)\tau_p]$. Since n_1 and n_2 are integers, the product $g_{\text{tx}}(t - n_1 \tau_p) g_{\text{tx}}^*(t - n_2 \tau_p)$ is non-zero only if $n_1 = n_2$. For $n_1 = n_2$, the integral on the RHS

above simplifies to

$$\frac{1}{\tau_p} \int_{n_1 \tau_p}^{(n_1+1)\tau_p} e^{j2\pi(m_1-m_2)v_p t} dt = \delta[m_1 - m_2] \quad (3.I.3)$$

since irrespective of the value of n_1 , this integral is one when $m_1 = m_2$ and is zero otherwise. Therefore, for any $n_1, m_1, n_2, m_2 \in \mathbb{Z}$,

$$\int g_{\text{tx}}(t - n_1 \tau_p) g_{\text{tx}}^*(t - n_2 \tau_p) e^{j2\pi(m_1-m_2)v_p t} dt = \delta[n_1 - n_2] \delta[m_1 - m_2]. \quad (3.I.4)$$

Using this in (3.I.2) gives

$$\begin{aligned} & \frac{1}{MN} \int x_{\mathcal{H}, \frac{k_1 \tau_p}{M}, \frac{l_1 v_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t) \left(x_{\mathcal{H}, \frac{k_2 \tau_p}{M}, \frac{l_2 v_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t) \right)^* dt \\ &= \frac{1}{MN} \sum_{n_1=0}^{N-1} \sum_{n_2=0}^{N-1} \sum_{m_1=0}^{M-1} \sum_{m_2=0}^{M-1} e^{j2\pi \frac{(l_1 n_1 - l_2 n_2)}{N}} e^{j2\pi \frac{m_2 k_2 - m_1 k_1}{M}} \delta[n_1 - n_2] \delta[m_1 - m_2] \\ &= \frac{1}{MN} \sum_{n_1=0}^{N-1} \sum_{m_1=0}^{M-1} e^{j2\pi n_1 \frac{(l_1 - l_2)}{N}} e^{j2\pi m_1 \frac{k_2 - k_1}{M}} \\ &= \left[\frac{1}{N} \sum_{n_1=0}^{N-1} e^{j2\pi n_1 \frac{(l_1 - l_2)}{N}} \right] \left[\frac{1}{M} \sum_{m_1=0}^{M-1} e^{j2\pi m_1 \frac{(k_2 - k_1)}{M}} \right] \\ &= \delta[k_1 - k_2] \delta[l_1 - l_2], \end{aligned} \quad (3.I.5)$$

which completes the proof.

Appendix 3.J Proof of Theorem 3.7

Due to the linearity of linear convolution, SFT, and the Heisenberg transform, it is noted that

$$\begin{aligned} & \Pi_{\text{ISFT}} \left(w_{\text{tx}}(\tau, v) \star \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_p \left(\tau, v; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right) (g_{\text{tx}}(t)) \\ & \stackrel{(a)}{=} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \Pi_{\text{ISFT}} \left(w_{\text{tx}}(\tau, v) \star x_p \left(\tau, v; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right) (g_{\text{tx}}(t)) \\ & \stackrel{(b)}{=} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{lv_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t), \end{aligned} \quad (3.J.1)$$

where step (a) follows from the linearity of the Heisenberg transform with respect to its associated TF function. Step (b) follows from (3.30). Next,

$$\begin{aligned} & \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_p \left(\tau, v; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \\ & \stackrel{(a)}{=} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \left[\sum_{n, m \in \mathbb{Z}} \delta(\tau - k\tau_p/M - n\tau_p) \delta(v - lv_p/N - mv_p) \right] \end{aligned}$$

$$\begin{aligned}
 &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} \left[\sum_{n,m \in \mathbb{Z}} x[k, l] \delta(\tau - (k + nM)\tau_p/M) \delta(\nu - (l + mN)v_p/N) \right] \\
 &\stackrel{(b)}{=} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} \left[\sum_{n,m \in \mathbb{Z}} x_p[k + nM, l + mN] \delta(\tau - (k + nM)\tau_p/M) \delta(\nu - (l + mN)v_p/N) \right] \\
 &\stackrel{(c)}{=} \sum_{k,l \in \mathbb{Z}} x_p[k, l] \delta(\tau - k\tau_p/M) \delta(\nu - lv_p/N) = x_p(\tau, \nu), \tag{3.J.2}
 \end{aligned}$$

where step (a) follows from the expression of $x_p\left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N}\right)$ in (3.15). Step (b) follows from the periodicity of $x_p[k, l]$ in (3.42). Step (c) follows from substitution of summation variables. The last step follows from the definition of $x_p(\tau, \nu)$ in (3.44). Hence,

$$\begin{aligned}
 \text{ISFT} \left(w_{\text{tx}}(\tau, \nu) \star \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right) &\stackrel{(a)}{=} \text{ISFT} (w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu)) \\
 &\stackrel{(b)}{=} X_{\text{TF}}^{W_{\text{tx}}}(t, f), \tag{3.J.3}
 \end{aligned}$$

where step (a) follows from (3.J.2). Step (b) follows from (3.45) and (3.46). From (3.J.3), it follows that

$$\Pi_{\text{ISFT}} \left(w_{\text{tx}}(\tau, \nu) \star \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right) (g_{\text{tx}}(t)) = \Pi_{X_{\text{TF}}^{W_{\text{tx}}}(t, f)} (g_{\text{tx}}(t)). \tag{3.J.4}$$

Using this result in (3.J.1) completes the proof.

Appendix 3.K Proof of Theorem 3.9

Note that

$$\begin{aligned}
 s_{\text{td}}(t) &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{lv_p}{N}}^{w_{\text{tx}} g_{\text{tx}}} (t) \\
 &\stackrel{(a)}{=} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \mathcal{Z}_t^{-1} \left(G_{\text{tx,dd}}(\tau, \nu) \left[w_{\text{tx}}(\tau, \nu) \star x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right] \right) \\
 &\stackrel{(b)}{=} \mathcal{Z}_t^{-1} \left(G_{\text{tx,dd}}(\tau, \nu) \left[w_{\text{tx}}(\tau, \nu) \star \underbrace{\left\{ \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{lv_p}{N} \right) \right\}}_{=x_p(\tau, \nu), \text{ see (3.J.2)}} \right] \right) \\
 &\stackrel{(c)}{=} \mathcal{Z}_t^{-1} (G_{\text{tx,dd}}(\tau, \nu) [w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu)]) \\
 &\stackrel{(d)}{=} \mathcal{Z}_t^{-1} (G_{\text{tx,dd}}(\tau, \nu) x_p^{w_{\text{tx}}}(\tau, \nu)), \tag{3.K.1}
 \end{aligned}$$

where step (a) follows from substituting $x_{\mathcal{H}, \frac{k\tau_p}{M}, \frac{lv_p}{N}}^{w_{\text{tx}} g_{\text{tx}}} (t)$ on the RHS of the previous step with its expression in (3.56). Step (b) follows from the linearity of the cascade of linear convolution of a periodic DD domain function with $w_{\text{tx}}(\tau, \nu)$, followed by multiplication with $G_{\text{tx,dd}}(\tau, \nu)$, and subsequently the inverse Zak transform. Step (c) follows from (3.J.2). Step (d) follows from (3.45).

Appendix 3.L Proof of Theorem 3.10

From (3.65), it follows that

$$\begin{aligned}
 y_p[k, l] &= \int_0^{\tau_p} \int_0^{\nu_p} \underbrace{\left[y_{\text{dd}}(\tau, \nu) G_{\text{rx,dd}}^*(\tau, \nu) \right]}_{=y^G(\tau, \nu)} \left[w_{\text{rx}}(\tau, \nu) \star x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{l\nu_p}{N} \right) \right]^* d\tau d\nu \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} y^G(\tau, \nu) \left[\sum_{n, m \in \mathbb{Z}} w_{\text{rx}}^* \left(\tau - n\tau_p - \frac{k\tau_p}{M}, \nu - m\nu_p - \frac{l\nu_p}{N} \right) \right] d\tau d\nu \\
 &= \sum_{n, m \in \mathbb{Z}} \int_0^{\tau_p} \int_0^{\nu_p} y^G(\tau, \nu) w_{\text{rx}}^* \left(\tau - n\tau_p - \frac{k\tau_p}{M}, \nu - m\nu_p - \frac{l\nu_p}{N} \right) d\tau d\nu, \tag{3.L.1}
 \end{aligned}$$

where the second step follows from the expression of $x_p \left(\tau, \nu; \frac{k\tau_p}{M}, \frac{l\nu_p}{N} \right)$ in (3.15) with $(\tau_0, \nu_0) = (k\tau_p/M, l\nu_p/N)$. Since $y_{\text{dd}}(\tau, \nu)$ and $G_{\text{rx,dd}}(\tau, \nu)$ are DD domain realizations of TD signals, they are both quasi-periodic DD domain functions, i.e., for all $n, m \in \mathbb{Z}$,

$$\begin{aligned}
 y_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) &= e^{j2\pi n\nu\tau_p} y_{\text{dd}}(\tau, \nu) \\
 G_{\text{rx,dd}}(\tau + n\tau_p, \nu + m\nu_p) &= e^{j2\pi n\nu\tau_p} G_{\text{rx,dd}}(\tau, \nu). \tag{3.L.2}
 \end{aligned}$$

Therefore, for all $n, m \in \mathbb{Z}$,

$$\begin{aligned}
 y^G(\tau + n\tau_p, \nu + m\nu_p) &= y_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) G_{\text{rx,dd}}^*(\tau + n\tau_p, \nu + m\nu_p) \\
 &= y_{\text{dd}}(\tau, \nu) e^{j2\pi n\nu\tau_p} G_{\text{rx,dd}}^*(\tau, \nu) e^{-j2\pi n\nu\tau_p} \\
 &= y_{\text{dd}}(\tau, \nu) G_{\text{rx,dd}}^*(\tau, \nu) = y^G(\tau, \nu), \tag{3.L.3}
 \end{aligned}$$

i.e., $y^G(\tau, \nu)$ is a *periodic* DD domain function. Using this fact in (3.L.1), the integral in the (n, m) th term on the RHS is given by

$$\begin{aligned}
 &\int_0^{\tau_p} \int_0^{\nu_p} y^G(\tau, \nu) w_{\text{rx}}^* \left(\tau - n\tau_p - \frac{k\tau_p}{M}, \nu - m\nu_p - \frac{l\nu_p}{N} \right) d\tau d\nu \\
 &= \int_{-n\tau_p}^{-(n-1)\tau_p} \int_{-m\nu_p}^{-(m-1)\nu_p} y^G(\tau + n\tau_p, \nu + m\nu_p) w_{\text{rx}}^* \left(\tau - \frac{k\tau_p}{M}, \nu - \frac{l\nu_p}{N} \right) d\tau d\nu \\
 &= \int_{-n\tau_p}^{-(n-1)\tau_p} \int_{-m\nu_p}^{-(m-1)\nu_p} y^G(\tau, \nu) w_{\text{rx}}^* \left(\tau - \frac{k\tau_p}{M}, \nu - \frac{l\nu_p}{N} \right) d\tau d\nu, \tag{3.L.4}
 \end{aligned}$$

where we have substituted the integration variables τ and ν with $(\tau + n\tau_p)$ and $(\nu + m\nu_p)$, respectively. The last step follows from the periodicity of $y^G(\tau, \nu)$. Using the simplified integral expression from (3.L.4) to (3.L.1), it follows that

$$\begin{aligned}
 y_p[k, l] &= \sum_{n, m \in \mathbb{Z}} \int_{n\tau_p}^{(n+1)\tau_p} \int_{m\nu_p}^{(m+1)\nu_p} y^G(\tau, \nu) w_{\text{rx}}^* \left(\tau - n\tau_p - \frac{k\tau_p}{M}, \nu - m\nu_p - \frac{l\nu_p}{N} \right) d\tau d\nu \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^G(\tau, \nu) w_{\text{rx}}^* \left(\tau - \frac{k\tau_p}{M}, \nu - \frac{l\nu_p}{N} \right) d\tau d\nu. \tag{3.L.5}
 \end{aligned}$$

This equivalent expression for $y[k, l]$ is, in fact, the DD domain filtering of $y^G(\tau, \nu)$ with the receive filter $w_{\text{rx}}^*(-\tau, -\nu)$, followed by DD domain sampling on the information grid Λ_{dd} , i.e.,

$$y_p[k, l] = y^{G, w_{\text{rx}}} \left(\tau = \frac{k\tau_p}{M}, \nu = \frac{l\nu_p}{N} \right),$$

$$y^{G, w_{\text{rx}}}(\tau, \nu) \triangleq y^G(\tau, \nu) \star w_{\text{rx}}^*(-\tau, -\nu). \quad (3.L.6)$$

Appendix 3.M Proof of Theorem 3.11

Substituting the expression of $s_{\text{id}}(t)$ from (3.49) on the RHS of the expression of $r_{\text{id}}(t)$ in (2.90) gives an expression for the received TD signal in terms of the windowed TF domain transmit symbols $X^{W_{\text{tx}}}[n, m]$. Substituting this expression of $r_{\text{id}}(t)$ on the RHS of (3.61) gives the following expression of the received TF symbols $Y[n, m]$ in terms of $X^{W_{\text{tx}}}[n, m]$, i.e.,

$$\begin{aligned} Y[n, m] &= \int r_{\text{id}}(t) g_{\text{rx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt \\ &\stackrel{(a)}{=} \sum_{n', m' \in \mathbb{Z}} X^{W_{\text{tx}}}[n', m'] \iint h_{\text{phy}}(\tau, \nu) g_{\text{rx}}^*(t - n\tau_p) g_{\text{tx}}(t - \tau - n'\tau_p) e^{j2\pi\nu(t - \tau)} \\ &\quad e^{-j2\pi m\nu_p(t - n\tau_p)} e^{j2\pi m'\nu_p(t - n'\tau_p - \tau)} d\tau d\nu \\ &\quad + \underbrace{\int n_{\text{id}}(t) g_{\text{rx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt}_{=n_{\text{TF}}[n, m]}. \end{aligned} \quad (3.M.1)$$

The integral in the (n', m') th term on the RHS above is nothing but $H_{\text{TF}}[n - n', m - m'; n', m']$ given in (3.72). This then completes the proof.

Appendix 3.N Derivation of Bi-orthogonality Condition

For an ideal channel with no delay/Doppler shift, i.e., $h_{\text{phy}}(\tau, \nu) = \delta(\tau)\delta(\nu)$, using (3.72), we get

$$H[n - n', m - m'; n', m'] = A_{g_{\text{rx}}, g_{\text{tx}}}^* ((n' - n)\tau_p, (m' - m)\nu_p). \quad (3.N.1)$$

From the TF domain I/O relation in (3.70), it follows that the TF domain symbols $X^{W_{\text{tx}}}[n, m]$ do not interfere with each other if and only if for all $(n, m) \neq (n', m')$, $n, n', m, m' \in \mathbb{Z}$,

$$H_{\text{TF}}[n - n', m - m'; n', m'] = 0, \quad (3.N.2)$$

which, using (3.N.1), is equivalent to the condition

$$A_{g_{\text{rx}}, g_{\text{tx}}}(n\tau_p, m\nu_p) = \int g_{\text{rx}}(t) g_{\text{tx}}^*(t - n\tau_p) e^{-j2\pi m\nu_p(t - n\tau_p)} dt = 0, \quad (3.N.3)$$

for all $(n, m) \neq (0, 0)$, $n, m \in \mathbb{Z}$. This is referred to as the *bi-orthogonality* property for the MC generators $g_{\text{tx}}(t)$ and $g_{\text{rx}}(t)$. Using (3.N.2) in (3.70) gives the TF domain I/O relation in (3.75) for the ideal channel. The expression for $H_{\text{TF}}[0, 0; n', m']$ in (3.76) follows from substituting $h_{\text{phy}}(\tau, \nu) = \delta(\tau)\delta(\nu)$ in (3.72).

Appendix 3.O Proof of Corollary 3.5

In the integral expression for $H_{\text{TF}}[n - n', m - m'; n', m']$ in (3.72), due to the multiplicative term $h_{\text{phy}}(\tau, \nu)$ in the integrand, the integration variables are clearly limited to the support set of $h_{\text{phy}}(\tau, \nu)$, i.e.,

$$H_{\text{TF}}[n - n', m - m'; n', m'] \stackrel{\Delta}{=} \iint_{(\tau, \nu) \in S_h} [h(\tau, \nu) e^{j2\pi(\nu n' \tau_p - \tau m' \nu_p)} e^{j2\pi\tau(m' - m)\nu_p} A_{g_{\text{tx}}, g_{\text{rx}}}^*(\tau + (n' - n)\tau_p, \nu + (m' - m)\nu_p)] d\tau d\nu. \quad (3.O.1)$$

From this expression, it is clear that for all $(n', m') \neq (n, m)$, $n', m', n, m \in \mathbb{Z}$, $H_{\text{TF}}[n - n', m - m'; n', m'] = 0$ if $A_{g_{\text{tx}}, g_{\text{rx}}}^*(\tau + (n' - n)\tau_p, \nu + (m' - m)\nu_p) = 0$ for all (τ, ν) in the support set of $h_{\text{phy}}(\tau, \nu)$, i.e., if condition (3.79) is satisfied. Therefore, when (3.79) is satisfied, from (3.70), it is clear that the TF domain I/O relation is given by (3.80). The expression for $H_{\text{TF}}[0, 0; n, m]$ in (3.81) follows from (3.O.1) for the special case when $(n, m) = (n', m')$. Additionally, if the cross ambiguity function $A_{g_{\text{tx}}, g_{\text{rx}}}(\tau, \nu)$ between the transmitter and receiver MC generators is one for all (τ, ν) in the support set of $h_{\text{phy}}(\tau, \nu)$, i.e., the condition in (3.82) is satisfied, then from (3.81), it follows that

$$H_{\text{TF}}[0, 0; n, m] = \iint h(\tau, \nu) e^{j2\pi(\nu n \tau_p - \tau m \nu_p)} d\tau d\nu. \quad (3.O.2)$$

The ISFT transform of $h(\tau, \nu)$ gives the TF domain representation of the channel, which is given by [55]

$$H(t, f) \stackrel{\Delta}{=} \iint h(\tau, \nu) e^{j2\pi(\nu t - \tau f)} d\tau d\nu. \quad (3.O.3)$$

Comparing (3.O.2) and (3.O.3), it is clear that

$$H_{\text{TF}}[0, 0; n, m] = H(t = n\tau_p, f = m\nu_p). \quad (3.O.4)$$

In other words, $H_{\text{TF}}[0, 0; n, m]$ is the effective TF domain channel gain at $t = n\tau_p$ and $f = m\nu_p$.

Appendix 3.P Derivation of I/O Relation for the Two-step MC-OTFS Transceiver

Equation (3.70) of Theorem 3.11 expresses $Y[n, m]$ in terms of $X^{W_{\text{ix}}}[n, m]$. Substituting $Y[n, m]$ on the RHS of (3.62) with the expression from (3.70) followed by SFFT in (3.63) results in a relation between $X^{W_{\text{ix}}}[n, m]$ and the DD domain output $y_p[k, l]$. In this relation, substituting $X^{W_{\text{ix}}}[n, m]$ with its expression from (3.50) gives the desired I/O relation in (3.84).

Appendix 3.Q Proof of Theorem 3.13

For the Zak transform-based transceiver, the transmitted TD signal is given by (see (3.58))

$$s_{\text{td}}(t) = \mathcal{Z}_t^{-1} \left(G_{\text{tx}, \text{dd}}(\tau, \nu) x_p^{W_{\text{ix}}}(\tau, \nu) \right). \quad (3.Q.1)$$

Since a doubly spread channel acts on its input through twisted convolution, from Theorem 2.3, it follows that the Zak transform of the received TD signal $r_{\text{td}}(t)$ is given by

$$y_{\text{dd}}(\tau, \nu) \stackrel{(a)}{=} \mathcal{Z}_t \left(\iint h_{\text{phy}}(\tau, \nu) s_{\text{td}}(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu + n_{\text{td}}(t) \right)$$

$$\begin{aligned}
&\stackrel{(b)}{=} h_{\text{phy}}(\tau, \nu) *_{\sigma} \mathcal{Z}_t(s_{\text{td}}(t)) + n_{\text{dd}}(\tau, \nu) \\
&\stackrel{(c)}{=} h_{\text{phy}}(\tau, \nu) *_{\sigma} (G_{\text{tx,dd}}(\tau, \nu) x_p^{w_{\text{rx}}}(\tau, \nu)) + n_{\text{dd}}(\tau, \nu),
\end{aligned} \tag{3.Q.2}$$

where step (a) follows from the expression of $r_{\text{td}}(t)$ in (2.52), step (b) follows from Theorem 2.3, and the last step follows from (3.Q.1). From Fig. 3.14, it follows that the filtered received DD domain function is given by

$$y^{G, w_{\text{rx}}}(\tau, \nu) = w_{\text{rx}}^*(-\tau, -\nu) \star \left(G_{\text{rx,dd}}^*(\tau, \nu) y_{\text{dd}}(\tau, \nu) \right). \tag{3.Q.3}$$

Note that both $G_{\text{rx,dd}}(\tau, \nu)$ and $y_{\text{dd}}(\tau, \nu)$ are quasi-periodic, and, therefore, $G_{\text{rx,dd}}^*(\tau, \nu) y_{\text{dd}}(\tau, \nu)$ is a periodic DD domain function with delay and Doppler domain periods τ_p and ν_p , respectively. Since linear convolution preserves periodicity, $y^{G, w_{\text{rx}}}(\tau, \nu)$ is also periodic, i.e., for all $n, m \in \mathbb{Z}$,

$$y^{G, w_{\text{rx}}}(\tau + n\tau_p, \nu + m\nu_p) = y^{G, w_{\text{rx}}}(\tau, \nu). \tag{3.Q.4}$$

Substituting the expression of $y_{\text{dd}}(\tau, \nu)$ from (3.Q.2) into (3.Q.3) and using the fact that $x_p^{w_{\text{rx}}}(\tau, \nu) = w_{\text{tx}}(\tau, \nu) \star x_p(\tau, \nu)$ (see (3.45)) give (3.88).

Appendix 3.R Proof of Corollary 3.6

Using the expression of $x_p(\tau, \nu)$ from (3.44) on the RHS of (3.88), it follows that

$$y^{G, w_{\text{rx}}}(\tau, \nu) = \sum_{k', l' \in \mathbb{Z}} x_p[k', l'] h_w(\tau, \nu; k', l') + n_p^{G, w_{\text{rx}}}(\tau, \nu), \tag{3.R.1}$$

where for each $k', l' \in \mathbb{Z}$, the DD domain function $h_w(\tau, \nu; k', l')$ is given by (3.90). Next, sampling $y^{G, w_{\text{rx}}}(\tau, \nu)$ on the information grid Λ_{dd} gives the discrete DD domain output function

$$y_p[k, l] = y^{G, w_{\text{rx}}} \left(\tau = k \frac{\tau_p}{M}, \nu = l \frac{\nu_p}{N} \right), \tag{3.R.2}$$

for $k, l \in \mathbb{Z}$. Using (3.R.1) in the above equation then completes the proof. The periodicity of $y_p[k, l]$ follows from the periodicity of $y^{G, w_{\text{rx}}}(\tau, \nu)$ (see (3.87)).

4

Delay-Doppler Diversity in OTFS

A key performance metric that captures the error performance of communication links in the high signal-to-noise ratio (SNR) regime in fading environments is the diversity order. The main idea behind diversity is the availability of multiple copies of the transmitted signal at the receiver. It is less probable that all the signal copies at the receiver are in deep fade simultaneously. Therefore, the receiver can reliably decode the transmitted signal using these received signals. This can be done, for example, by combining the multiple received signals or by picking the signal with the highest SNR. The diversity order characterizes the slope of the probability of error vs. SNR curve in the log–log scale in the high-SNR regime, which is formally defined as

$$D = \lim_{\text{SNR} \rightarrow \infty} \frac{-\log p_e}{\log \text{SNR}}. \quad (4.1)$$

The larger the diversity order, the better the mitigation of fading effects. Diversity opportunities in time, frequency, and space domains are common in communication links that use conventional modulation schemes. Diversity opportunities can be created intentionally by design (e.g., use of multiple antennas that gives spatial diversity and repetition in time/frequency that gives time/frequency diversity) or naturally by multipath environment (e.g., RAKE diversity in frequency-selective fading channels). In orthogonal time frequency space (OTFS), the wireless channel response is represented in the delay-Doppler (DD) domain. The channel taps in this domain correspond to the group of reflectors having a particular delay (depends on the reflectors' relative distance) and Doppler (depends on the reflectors' relative velocity). The DD representation of the channel offers diversity opportunities in the DD domain. This chapter focuses on the achieved DD diversity performance of multicarrier-orthogonal time frequency space (MC-OTFS).

This chapter presents the analysis of the asymptotic diversity performance of MC-OTFS in both single-input single-output (SISO) and multiple-input multiple-output (MIMO) settings. A channel with P DD taps/paths in a SISO system can potentially achieve a maximum diversity order of P . An interesting performance behavior in OTFS is that the diversity slope is high in the low-to-medium SNR regime, whereas in the asymptotic regime where $\text{SNR} \rightarrow \infty$, the diversity slope becomes one. The diversity slope in low-to-medium SNRs and the SNR at which the diversity one slope starts increase for increasing OTFS frame sizes. Hence, for practical values of frame sizes, OTFS can offer diversity orders close to the maximum diversity of P in practical SNRs of interest. Also, a OTFS frame precoded by a diagonal phase matrix can achieve the full diversity order of P . This chapter presents these results through analysis and simulations. It also presents the diversity analysis and results for space–time coded OTFS systems and MIMO OTFS systems with receive antenna selection.

4.1 Diversity in SISO OTFS

The block diagram of the SISO MC-OTFS modulation scheme is shown in Fig. 4.1. The MC-OTFS transmitter multiplexes MN information symbols (e.g., phase shift keying (PSK)/quadrature amplitude modulation (QAM) symbols from a modulation alphabet $\mathbb{A}$), denoted by $x[k, l]$, $k = 0, \dots, N-1$, $l = 0, \dots, M-1$, onto a $N \times M$ DD grid, where N and M denote the number of Doppler and delay bins, respectively. These MN symbols are sent over an OTFS frame of duration NT seconds using a bandwidth of $M\Delta f$, where $\Delta f = 1/T$. At the transmitter, the information symbols in the DD domain are first mapped to the TF domain using ISFFT. The TF symbols are transformed to the time domain using the Heisenberg transform, and the resulting time domain signal is transmitted over the channel. At the receiver, the time domain signal is transformed to the TF domain using the Wigner transform, and the resulting TF domain symbols are mapped to the DD domain using SFFT.

The following section presents the vectorized DD domain system model for the SISO MC-OTFS modulation scheme shown in Fig. 4.1. With this system model, one can carry out performance analysis using established analytical tools and techniques. For example, with this system model, asymptotic diversity orders achieved by MC-OTFS systems can be predicted using pairwise error probability (PEP) analysis. The PEP analysis can also provide bounds on the BER performance of the system.

4.1.1 System Model

OTFS modulation referred to in this chapter is the MC-OTFS modulation discussed in Chapter 3. The delay and Doppler period τ_p and $\nu_p = 1/\tau_p$ in Chapter 3 have been denoted by T and $\Delta f = 1/T$, respectively, in this chapter. As in Chapter 3, here also the total time duration and bandwidth of the MC-OTFS modulated signal are $NT = N\tau_p$ and $M\Delta f = M\nu_p$, respectively. Also, the (k, l) th DD domain information symbol $x[k, l]$ is carried by the MC pulse $x_{\mathcal{H}, \frac{l\tau_p}{M}, \frac{k\nu_p}{N}}^{u_{\text{tx}}, g_{\text{tx}}}(t)$ (see (3.30), (3.31), and (3.41)). As discussed in Chapter 3, there are two transceiver architectures for MC-OTFS modulation, one based on the two-step approach and the other based on the Zak transform approach. In this chapter, the two-step approach has been used for MC-OTFS transceivers (see Sections 3.4.1 and 3.5.1). Specifically, in this chapter, the simplified two-step MC-OTFS transmitter and receiver illustrated through Figs. 3.11 and 3.13, respectively, have been used. Also, in terms of notation, in Chapter 3, k and l denote the discrete delay domain index and Doppler domain index, respectively,

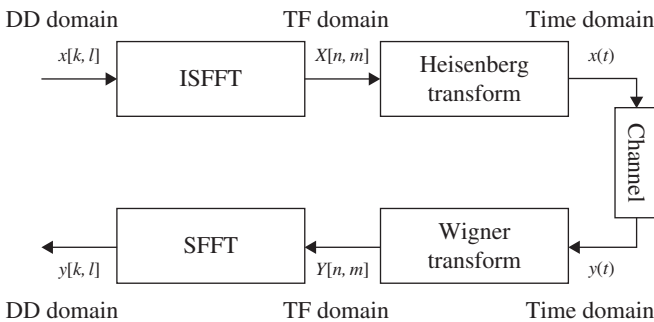


Figure 4.1 MC-OTFS modulation scheme.

whereas here in this chapter (and in Chapters 5–9), it is the opposite, i.e., k denotes the Doppler domain index and l denotes the delay domain index.

The DD domain symbols $x[k, l]$, $k = 0, 1, \dots, N-1$, $l = 0, 1, \dots, M-1$, are mapped to TF domain using ISFFT as

$$X[n, m] = \frac{1}{\sqrt{MN}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] e^{j2\pi \left(\frac{nk}{N} - \frac{ml}{M} \right)}. \quad (4.2)$$

The TF signal in (4.2) is mapped to a TD signal using the Heisenberg transform as

$$x(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] g_{\text{tx}}(t - nT) e^{j2\pi m \Delta f (t - nT)}, \quad (4.3)$$

where $g_{\text{tx}}(t)$ defines the shape of the transmit pulse. The transmitted signal is passed through the channel whose DD domain response is denoted by $h(\tau, \nu)$, where τ and ν are the delay and Doppler variables, respectively. With this representation, the received TD signal at the receiver is given by (see (2.90))

$$y(t) = \int_{\nu} \int_{\tau} h(\tau, \nu) x(t - \tau) e^{j2\pi \nu (t - \tau)} d\tau d\nu + v(t), \quad (4.4)$$

where $v(t)$ is the AWGN. At the receiver, a matched filter computes the cross ambiguity function $A_{g_{\text{rx}}, v}(t, f)$ as

$$A_{g_{\text{rx}}, v}(t, f) = \int g_{\text{rx}}^*(t' - t) y(t) e^{-j2\pi f (t' - t)} dt', \quad (4.5)$$

where $g_{\text{rx}}(t)$ defines the shape of the receive pulse, and the TF domain output samples are obtained by sampling $A_{g_{\text{rx}}, v}(t, f)$ at $t = nT$ and $f = m\Delta f$ as

$$Y[n, m] = A_{g_{\text{rx}}, v}(t, f)|_{t=nT, f=m\Delta f}. \quad (4.6)$$

The operations (4.5) and (4.6) are referred as the Wigner transform. Next, SFFT is applied on the TF domain samples $Y[n, m]$ to obtain the corresponding DD domain samples $y[k, l]$ as

$$y[k, l] = \frac{1}{\sqrt{MN}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y[n, m] e^{-j2\pi \left(\frac{nk}{N} - \frac{ml}{M} \right)}. \quad (4.7)$$

Now, consider that the channel has P paths in the DD domain resulting from P reflectors, where each reflector is associated with a delay and a Doppler, so that the channel response $h(\tau, \nu)$ in the DD domain is given by

$$h(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (4.8)$$

where τ_i , ν_i , and h_i denote the delay, Doppler, and fade coefficient of the i th path, respectively. The path delays and Dopplers are defined as $\tau_i \triangleq \frac{a_i + \alpha_i}{M\Delta f}$ and $\nu_i \triangleq \frac{b_i + \beta_i}{NT}$, $i = 1, \dots, P$, where a_i and b_i are integers, and α_i and β_i are real, $-\frac{1}{2} < \alpha_i, \beta_i \leq \frac{1}{2}$. The fractional parts α_i and β_i can be neglected if M and N are large, and, hence, the delay resolution $\frac{1}{M\Delta f}$ and the Doppler resolution $\frac{1}{NT}$ are sufficiently small to approximate the path delays and Doppler shifts to the nearest integer sampling points. In the diversity analysis of OTFS presented in this chapter, it is assumed that the fractional parts α_i s and β_i s are negligible and taken to be zero. In other words, the delays and Dopplers are assumed to be integer multiples of the delay and Doppler resolutions, respectively. For integer delays and

Dopplers with ideal $g_{\text{rx}}(t)$ and $g_{\text{tx}}(t)$ pulses,¹ the end-to-end DD domain input-output relation can be derived as [38]

$$y[k, l] = \sum_{i=1}^P h'_i x[(k - b_i)_N, (l - a_i)_M] + v[k, l], \quad (4.9)$$

where $h'_i = h_i e^{-j2\pi v_i \tau_i}$ and $v[k, l]$ is the AWGN. Considering uniform scattering profile, h_i s are assumed to be i.i.d. $\mathcal{CN}(0, 1/P)$. The input-output relation in (4.9), $\forall k, l$, can be written in a vectorized form as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}, \quad (4.10)$$

where $\mathbf{x}, \mathbf{y}, \mathbf{v} \in \mathbb{C}^{MN \times 1}$, the $(k + Nl)$ th entry of $\mathbf{x}$, $x_{k+Nl} = x[k, l]$, $k = 0, 1, \dots, N - 1$, $l = 0, 1, \dots, M - 1$. Similarly, $y_{k+Nl} = y[k, l]$ and $v_{k+Nl} = v[k, l]$, $k = 0, 1, \dots, N - 1$, $l = 0, 1, \dots, M - 1$, and $\mathbf{H} \in \mathbb{C}^{MN \times MN}$ is the effective channel matrix, whose j th row ($j = k + Nl$), denoted by $\mathbf{H}[j]$, is given by $\mathbf{H}[j] = [\tilde{h}((k - 0)_N, (l - 0)_M) \ \tilde{h}((k - 1)_N, (l - 0)_M) \ \cdots \ \tilde{h}((k - (N - 1))_N, (l - (M - 1))_M)]$, where $\tilde{h}(k, l)$ is given by

$$\tilde{h}(k, l) = \begin{cases} h'_i & \text{if } k = b_i, l = a_i, i \in \{1, \dots, P\}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.11)$$

It is noted that, due to the modulo operations in (4.9), there are only P non-zero elements in each row and column of the channel matrix $\mathbf{H}$ in (4.10). Also, note that $\mathbf{H}$ has M circulant blocks, each of size $N \times N$, which are cyclically shifted to form a block circulant matrix. This block circulant structure of the $\mathbf{H}$ matrix with circulant blocks is illustrated in the following examples.

Example 4.1 Consider an example of the above system model for the case of $M = 2$, $N = 2$, and $P = 2$ paths, where the Doppler and delay indices for paths P_1 and P_2 are $(b_1, a_1) = (0, 0)$ and $(b_2, a_2) = (1, 1)$, respectively, as shown in Fig. 4.2(a). The output DD equations for $y[k, l]$, $k = 0, 1$ and $l = 0, 1$ are given by

$$\begin{aligned} y[0, 0] &= h'_1 x[0, 0] + h'_2 x[1, 1] + v[0, 0], \\ y[1, 0] &= h'_1 x[1, 0] + h'_2 x[0, 1] + v[1, 0], \\ y[0, 1] &= h'_1 x[0, 1] + h'_2 x[1, 0] + v[0, 1], \\ y[1, 1] &= h'_1 x[1, 1] + h'_2 x[0, 0] + v[1, 1], \end{aligned} \quad (4.12)$$

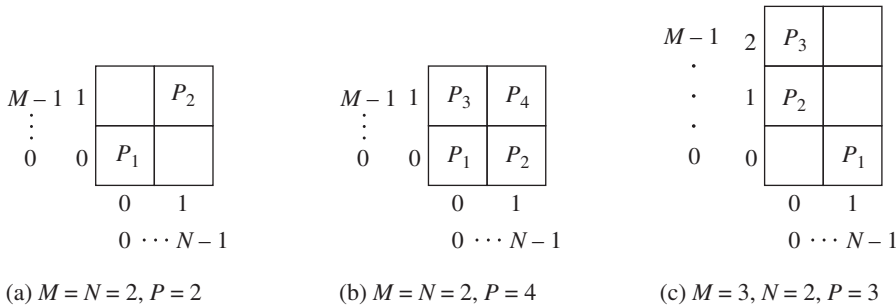


Figure 4.2 Delay-Doppler paths for Examples 1–5.

¹ Ideal pulses have been discussed in detail in (3.79) of Corollary 3.5. See also the discussion in the paragraph just after this corollary but before Theorem 3.12.

which can be written in a vector-matrix form as

$$\underbrace{\begin{bmatrix} y[0,0] \\ y[1,0] \\ y[0,1] \\ y[1,1] \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h'_1 & 0 & 0 & h'_2 \\ 0 & h'_1 & h'_2 & 0 \\ 0 & h'_2 & h'_1 & 0 \\ h'_2 & 0 & 0 & h'_1 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x[0,0] \\ x[1,0] \\ x[0,1] \\ x[1,1] \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} v[0,0] \\ v[1,0] \\ v[0,1] \\ v[1,1] \end{bmatrix}}_{\mathbf{v}}. \quad (4.13)$$

Note that $\mathbf{H}$ in (4.13) is a block circulant matrix with $M (= 2)$ blocks of size $N \times N$ (i.e., 2×2), given by $\begin{bmatrix} h'_1 & 0 \\ 0 & h'_1 \end{bmatrix}$ and $\begin{bmatrix} 0 & h'_2 \\ h'_2 & 0 \end{bmatrix}$.

Example 4.2 Likewise, for the case of $M = N = 2$, and $P = 4$ paths, where the Doppler and delay indices for paths P_1, P_2, P_3 , and P_4 are $(b_1, a_1) = (0, 0)$, $(b_2, a_2) = (1, 0)$, $(b_3, a_3) = (0, 1)$, and $(b_4, a_4) = (1, 1)$, respectively, as shown in Fig. 4.2(b), the vectorized system model in the DD domain is given by

$$\underbrace{\begin{bmatrix} y[0,0] \\ y[1,0] \\ y[0,1] \\ y[1,1] \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h'_1 & h'_2 & h'_3 & h'_4 \\ h'_2 & h'_1 & h'_4 & h'_3 \\ h'_3 & h'_4 & h'_1 & h'_2 \\ h'_4 & h'_3 & h'_2 & h'_1 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x[0,0] \\ x[1,0] \\ x[0,1] \\ x[1,1] \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} v[0,0] \\ v[1,0] \\ v[0,1] \\ v[1,1] \end{bmatrix}}_{\mathbf{v}}. \quad (4.14)$$

Here, the $\mathbf{H}$ matrix in (4.14) is a block circulant matrix with $M (= 2)$ blocks of size $N \times N$ (i.e., 2×2), given by $\begin{bmatrix} h'_1 & h'_2 \\ h'_2 & h'_1 \end{bmatrix}$ and $\begin{bmatrix} h'_3 & h'_4 \\ h'_4 & h'_3 \end{bmatrix}$.

Example 4.3 For the case of $M = 3$, $N = 2$, and $P = 3$ paths, where the Doppler and delay indices for paths P_1, P_2 , and P_3 are $(b_1, a_1) = (1, 0)$, $(b_2, a_2) = (0, 1)$, and $(b_3, a_3) = (0, 2)$, respectively, as shown in Fig. 4.2(c), the vectorized system model in the DD domain is given by

$$\underbrace{\begin{bmatrix} y[0,0] \\ y[1,0] \\ y[0,1] \\ y[1,1] \\ y[0,2] \\ y[1,2] \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 0 & h'_1 & h'_3 & 0 & h'_2 & 0 \\ h'_1 & 0 & 0 & h'_3 & 0 & h'_2 \\ h'_2 & 0 & 0 & h'_1 & h'_3 & 0 \\ 0 & h'_2 & h'_1 & 0 & 0 & h'_3 \\ h'_3 & 0 & h'_2 & 0 & 0 & h'_1 \\ 0 & h'_3 & 0 & h'_2 & h'_1 & 0 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x[0,0] \\ x[1,0] \\ x[0,1] \\ x[1,1] \\ x[0,2] \\ x[1,2] \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} v[0,0] \\ v[1,0] \\ v[0,1] \\ v[1,1] \\ v[0,2] \\ v[1,2] \end{bmatrix}}_{\mathbf{v}}. \quad (4.15)$$

Here, the $\mathbf{H}$ matrix in (4.15) is a block circulant matrix with $M (= 3)$ blocks of size $N \times N$ (i.e., 2×2), given by $\begin{bmatrix} 0 & h'_1 \\ h'_1 & 0 \end{bmatrix}$, $\begin{bmatrix} h'_2 & 0 \\ 0 & h'_2 \end{bmatrix}$, and $\begin{bmatrix} h'_3 & 0 \\ 0 & h'_3 \end{bmatrix}$.

As noted earlier, in the above examples, $\mathbf{H}$ is a block circulant matrix with P unique non-zero entries, and each row and column has only P non-zero elements.

The diversity analysis and results presented in this chapter are for the optimum maximum-likelihood (ML) detector. This is because of the analytical tractability that is possible with ML detection. Other detectors are sub-optimum, and such detectors and their performance are dealt in Chapter 5.

4.1.2 Diversity Analysis

Consider the optimum ML detector for the system model in (4.10). Assuming perfect channel knowledge at the receiver, the optimum ML detection rule is given by

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{A}^{MN}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2. \quad (4.16)$$

A key step in the diversity analysis is writing the system model in (4.10) in a form suitable for analysis. Toward this, the vectorized input–output relation in (4.10) can be rewritten in an alternate form as

$$\mathbf{y}^T = \mathbf{h}'\mathbf{X} + \mathbf{v}^T, \quad (4.17)$$

where $\mathbf{y}^T$ is a $1 \times MN$ received vector, $\mathbf{h}'$ is a $1 \times P$ vector whose i th entry is given by $h'_i = h_i e^{-j2\pi v_i \tau_i}$, $\mathbf{v}^T$ is the $1 \times MN$ noise vector, and $\mathbf{X}$ is a $P \times MN$ matrix whose i th column ($i = k + NL$, $i = 0, 1, \dots, MN - 1$), denoted by $\mathbf{X}[i]$, is given by

$$\mathbf{X}[i] = \begin{bmatrix} \mathcal{X}_{(k-b_1)_N+N(l-a_1)_M} \\ \mathcal{X}_{(k-b_2)_N+N(l-a_2)_M} \\ \vdots \\ \mathcal{X}_{(k-b_P)_N+N(l-a_P)_M} \end{bmatrix}. \quad (4.18)$$

Example 4.4 For the system considered in Example 4.1, the $\mathbf{h}'$ vector and $\mathbf{X}$ matrix in the alternate form of the system model in (4.17) are given by $\mathbf{h}' = [h'_1 \ h'_2]$ and

$$\mathbf{X} = \begin{bmatrix} x_0 & x_1 & x_2 & x_3 \\ x_3 & x_2 & x_1 & x_0 \end{bmatrix}, \quad (4.19)$$

which results in the same set of equations in (4.12), where x_0, x_1, x_2 , and x_3 in (4.19) are the symbols $x[0, 0], x[0, 1], x[1, 0]$, and $x[1, 1]$, respectively, in (4.12).

The representation of $\mathbf{X}$ in the form given in (4.17) allows us to view $\mathbf{X}$ as a $P \times MN$ symbol matrix. For convenience, the elements of $\mathbf{X}$ are normalized so that the average energy per symbol time is one so that the SNR, denoted by γ , is given by $\gamma = 1/N_0$, where N_0 is the noise spectral density. Assuming perfect channel state information (CSI) and ML detection at the receiver, the probability of transmitting the symbol matrix $\mathbf{X}_i$ and deciding in favor of $\mathbf{X}_j$ at the receiver is the PEP between $\mathbf{X}_i$ and $\mathbf{X}_j$, given by [6]

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j | \mathbf{h}', \mathbf{X}_i) = Q \left(\sqrt{\frac{\|\mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)\|^2}{2N_0}} \right). \quad (4.20)$$

The PEP averaged over the channel statistics can be written as

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) = \mathbb{E} \left[Q \left(\sqrt{\frac{\gamma \|\mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)\|^2}{2}} \right) \right], \quad (4.21)$$

which can be simplified by writing $\|\mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)\|^2 = \mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)(\mathbf{X}_i - \mathbf{X}_j)^H \mathbf{h}'^H$. The matrix $(\mathbf{X}_i - \mathbf{X}_j)(\mathbf{X}_i - \mathbf{X}_j)^H$ is a Hermitian matrix that is diagonalizable by unitary transformation. Hence, it can be written as $(\mathbf{X}_i - \mathbf{X}_j)(\mathbf{X}_i - \mathbf{X}_j)^H = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$, where $\mathbf{U}$ is unitary and $\mathbf{\Lambda} = \text{diag}\{\lambda_1^2, \dots, \lambda_P^2\}$, λ_i being the i th singular value of the difference matrix $\mathbf{\Delta}_{ij}$, given by $\mathbf{\Delta}_{ij} = (\mathbf{X}_i - \mathbf{X}_j)$. Using the above and defining $\tilde{\mathbf{h}}^H = \mathbf{U}^H \mathbf{h}'^H$, $\|\mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)\|^2$ can be simplified as

$$\|\mathbf{h}'(\mathbf{X}_i - \mathbf{X}_j)\|^2 = \tilde{\mathbf{h}}\mathbf{\Lambda}\tilde{\mathbf{h}}^H = \sum_{l=1}^r \lambda_l^2 |\tilde{h}_l|^2, \quad (4.22)$$

where r denotes the rank of the difference matrix $\mathbf{\Delta}_{ij}$. Therefore, the average PEP between symbol matrices $\mathbf{X}_i$ and $\mathbf{X}_j$ can be written as

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) = \mathbb{E} \left[Q \left(\sqrt{\frac{\gamma \sum_{l=1}^r \lambda_l^2 |\tilde{h}_l|^2}{2}} \right) \right]. \quad (4.23)$$

Since $\tilde{\mathbf{h}}$ is obtained by multiplying a unitary matrix to $\mathbf{h}'$, it has the same distribution as that of $\mathbf{h}'$. Therefore, $\tilde{h}_l$ s are distributed as $C\mathcal{N}(0, 1/P)$. Using this, the average PEP in (4.23) can be simplified to get the following upper bound on PEP [6]

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) \leq \prod_{l=1}^r \frac{1}{1 + \frac{\gamma \lambda_l^2}{4P}}. \quad (4.24)$$

At high SNRs, (4.24) can be further simplified as

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) \leq \frac{1}{\gamma^r \prod_{l=1}^r \frac{\lambda_l^2}{4P}}. \quad (4.25)$$

The exponent of the SNR term γ in (4.25) is r , which is equal to the rank of the difference matrix $\mathbf{\Delta}_{ij}$. For all i, j , $i \neq j$, the PEP with the minimum value of r dominates the overall BER. Therefore, the achieved diversity order, denoted by $\rho_{\text{siso-otfs}}$, is given by

$$\rho_{\text{siso-otfs}} = \min_{i,j} \text{rank}(\mathbf{\Delta}_{ij}). \quad (4.26)$$

Now, consider a case when $x_i[k, l] = a$ and $x_j[k, l] = a'$, $\forall k = 0, \dots, N-1$ and $l = 0, \dots, M-1$. This corresponds to the case when $\mathbf{X}_i = a \cdot \mathbf{1}_{P \times MN}$ and $\mathbf{X}_j = a' \cdot \mathbf{1}_{P \times MN}$, where $\mathbf{1}_{P \times MN}$ denotes all-ones matrix of size $P \times MN$. Then, $\mathbf{\Delta}_{ij} = (\mathbf{X}_i - \mathbf{X}_j) = (a - a') \cdot \mathbf{1}_{P \times MN}$, whose rank is one, which is the minimum rank of $\mathbf{\Delta}_{ij}$, $\forall i, j$, $i \neq j$. Hence, the asymptotic diversity order of SISO-OTFS with ML detection is one.

The above diversity order result can be verified using bounds on the BER performance of SISO-OTFS. A lower bound on the BER can be computed by summing up the PEPs corresponding to all pairs of data matrices whose difference matrices have rank one as

$$\text{BER} \geq \frac{1}{2^{MN}} \sum_{k=1}^{\kappa} P(\mathbf{X}_i \rightarrow \mathbf{X}_j), \quad (4.27)$$

where κ denotes the number of difference matrices having rank one. When $\mathbf{\Delta}_{ij}$ has rank one, it has only one non-zero singular value (λ_1), which can be computed to be $\sqrt{4PMN}$. With this, the PEP in (8.81) for the pair $(\mathbf{X}_i, \mathbf{X}_j)$ with $\mathbf{\Delta}_{ij}$ having rank one simplifies to

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) = \mathbb{E} \left[Q \left(\sqrt{2\gamma PMN |\tilde{h}_1|^2} \right) \right]. \quad (4.28)$$

Since $\tilde{h}_1 \sim \mathcal{CN}(0, 1/P)$, evaluating the expectation in (4.28) gives [6]

$$P(\mathbf{X}_i \rightarrow \mathbf{X}_j) = \frac{1}{2} \left(1 - \sqrt{\frac{MN}{MN + \gamma^{-1}}} \right). \quad (4.29)$$

Using (4.29) in (4.27), the lower bound becomes

$$\text{BER} \geq \frac{\kappa}{2^{MN}} \frac{1}{2} \left(1 - \sqrt{\frac{MN}{MN + \gamma^{-1}}} \right), \quad (4.30)$$

which, at high SNRs, further simplifies to

$$\text{BER} \geq \frac{\kappa}{2^{MN}} \frac{1}{4\gamma MN}. \quad (4.31)$$

Simulations have shown that the above bound is tight at high SNRs (see Fig. 4.3). The following two observations can be made from (4.31).

- The diversity one result follows from the γ term in the denominator. This is illustrated in Fig. 4.3, which shows the simulated BER in comparison with lower and upper bounds on the BER. The upper bound is based on the union bound.
- The value of the lower bound in (4.31) depends on the ratio $\frac{\kappa}{2^{MN}}$. As the values M and N increase, the 2^{MN} term dominates this ratio. Therefore, increasing M and N reduces the value of the lower bound. This means that the BER can decrease with a higher slope for higher values of M and N before it changes the slope and meets the diversity one lower bound of (4.31). This observation is illustrated in Fig. 4.4 which shows the performance of three systems with BPSK, namely, system-1 with $M = N = 2$, system-2 with $M = 4, N = 2$, and system-3 with $M = N = 4$. The $\frac{\kappa}{2^{MN}}$ values for these systems are $\frac{8}{16}$, $\frac{8}{256}$, and $\frac{8}{65536}$, respectively. Since the $\frac{\kappa}{2^{MN}}$ values are decreasing for increasing MN , (4.31) indicates that the lower bound for system-3 should lie below that of system-2, which, in turn, should lie below that of system-1. This trend is evident from Fig. 4.4. Furthermore, for all the systems, the BER plots show a diversity slope greater than one for low-to-medium SNR values before it meets the diversity one lower bound. An interesting observation, however, is

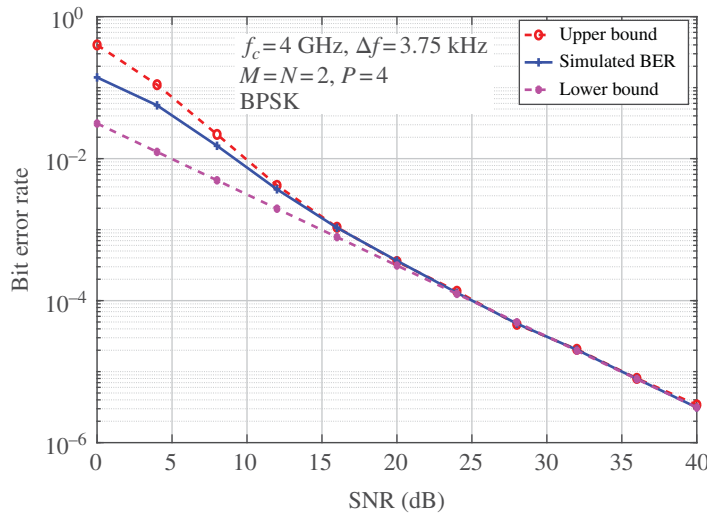


Figure 4.3 BER performance of SISO-OTFS with $M = 2, N = 2, P = 4$, and BPSK. The diversity slope is one at high SNRs.

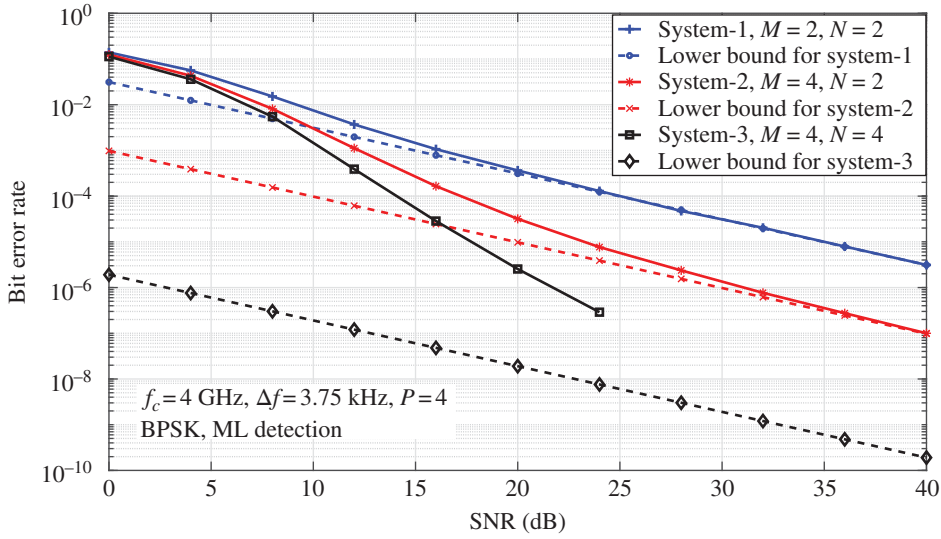


Figure 4.4 BER performance of SISO-OTFS for (i) $M = 2, N = 2$, (ii) $M = 4, N = 2$, (iii) $M = 4, N = 4$, and BPSK.

that the system-3 with higher M and N values achieves a higher diversity slope compared to those of system-1 and system-2 before meeting the lower bound. This is because the lower bound for system-3 lies much below the lower bounds for system-1 and system-2, and the BER curve of system-3 falls with greater slope to meet its lower bound. This shows that, though the asymptotic diversity order is one, increasing the value of MN (i.e., increasing the OTFS frame size) can lead to higher diversity slopes in the finite SNR regime, resulting in improved performance for increased frame sizes.

Note: The above diversity analysis and results show that, although the channel provides the possibility of achieving a maximum DD diversity of P , OTFS does not extract the full diversity available in the DD channel. The full DD diversity of P can be achieved by phase rotation of the OTFS signal vectors before transmission. OTFS with phase rotation and its full diversity result are presented in Section 4.1.3.

4.1.3 Full Diversity with Phase Rotation

This section introduces the OTFS scheme with “phase rotation” which, unlike the basic OTFS scheme, extracts the full diversity offered by the DD channel. From the diversity analysis in the previous section (Section 4.1.2), it is clear that the asymptotic diversity order of OTFS depends on the minimum rank of the difference matrix $\Delta_{ij} = \mathbf{X}_i - \mathbf{X}_j$, over all pairs of symbol matrices $(\mathbf{X}_i, \mathbf{X}_j)$. In order to design a scheme that can extract full diversity, let us take a closer look at the symbol matrix $\mathbf{X}$ whose i th column is given by (4.18). The symbol matrix $\mathbf{X}$ is a $P \times MN$ matrix that has only MN unique entries, which are nothing but the MN symbols of the transmit vector $\mathbf{x}$. The rows of the matrix $\mathbf{X}$ are the permutations of the transmit symbol vector $\mathbf{x}$. When $P = MN$, the matrix $\mathbf{X}$ is a block circulant matrix with circulant blocks, as shown in (4.32), where $x_q^{(l)}$ denotes the q th distinct element of the l th block, $q = 0, \dots, N - 1$ and $l = 0, \dots, M - 1$. Note that $\mathbf{X}$ has M circulant blocks, each of size $N \times N$, which are cyclically shifted to form a block circulant matrix (see (4.32)).

$$\mathbf{X} = \begin{bmatrix} x_0^{(0)} & x_1^{(0)} & \cdots & x_{N-1}^{(0)} & x_0^{(1)} & \cdots & x_{N-1}^{(1)} & \vdots & x_0^{(M-1)} & \cdots & x_{N-1}^{(M-1)} \\ x_{N-1}^{(0)} & x_0^{(0)} & \cdots & x_{N-2}^{(0)} & x_{N-1}^{(1)} & \cdots & x_{N-2}^{(1)} & \vdots & x_{N-1}^{(M-1)} & \cdots & x_{N-2}^{(M-1)} \\ & & \vdots & & & & & \vdots & & & \\ x_1^{(0)} & x_2^{(0)} & \cdots & x_0^{(0)} & x_1^{(1)} & \cdots & x_0^{(1)} & \vdots & x_1^{(M-1)} & \cdots & x_0^{(M-1)} \\ x_0^{(M-1)} & x_1^{(M-1)} & \cdots & x_{N-1}^{(M-1)} & x_0^{(0)} & \cdots & x_{N-1}^{(0)} & \vdots & x_0^{(M-2)} & \cdots & x_{N-1}^{(M-2)} \\ x_{N-1}^{(M-1)} & x_0^{(M-1)} & \cdots & x_{N-2}^{(M-1)} & x_{N-1}^{(0)} & \cdots & x_{N-2}^{(0)} & \vdots & x_{N-1}^{(M-2)} & \cdots & x_{N-2}^{(M-2)} \\ & & \vdots & & & & & \vdots & & & \\ x_1^{(M-1)} & x_2^{(M-1)} & \cdots & x_0^{(M-1)} & x_1^{(0)} & \cdots & x_0^{(0)} & \vdots & x_1^{(M-2)} & \cdots & x_0^{(M-2)} \\ & & \vdots & & & & & \vdots & & & \\ x_0^{(1)} & x_1^{(1)} & \cdots & x_{N-1}^{(1)} & x_0^{(2)} & \cdots & x_{N-1}^{(2)} & \vdots & x_0^{(0)} & \cdots & x_{N-1}^{(0)} \\ x_{N-1}^{(1)} & x_0^{(1)} & \cdots & x_{N-2}^{(1)} & x_{N-1}^{(2)} & \cdots & x_{N-2}^{(2)} & \vdots & x_{N-1}^{(0)} & \cdots & x_{N-2}^{(0)} \\ & & \vdots & & & & & \vdots & & & \\ x_1^{(1)} & x_2^{(1)} & \cdots & x_0^{(1)} & x_1^{(2)} & \cdots & x_0^{(2)} & \vdots & x_1^{(0)} & \cdots & x_0^{(0)} \end{bmatrix}. \quad (4.32)$$

The $MN \times 1$ OTFS transmit vector corresponding to symbol matrix $\mathbf{X}$ in (4.32) is given by

$$\mathbf{x} = [x_0^{(0)}, \dots, x_{N-1}^{(0)}, x_0^{(1)}, \dots, x_{N-1}^{(1)}, \dots, x_0^{(M-1)}, \dots, x_{N-1}^{(M-1)}]^T. \quad (4.33)$$

Example 4.5 For a system with $M = N = 2$ and $P = 4$, the $\mathbf{X}$ matrix is of size 4×4 and is given by

$$\mathbf{X} = \begin{bmatrix} x_0 & x_1 & x_2 & x_3 \\ x_1 & x_0 & x_3 & x_2 \\ x_2 & x_3 & x_0 & x_1 \\ x_3 & x_2 & x_1 & x_0 \end{bmatrix}, \quad (4.34)$$

which can be seen to be a block circulant matrix with $M(= 2)$ blocks of size $N \times N$ (i.e., 2×2), given by $\begin{bmatrix} x_0 & x_1 \\ x_1 & x_0 \end{bmatrix}$ and $\begin{bmatrix} x_2 & x_3 \\ x_3 & x_2 \end{bmatrix}$.

When $P < MN$, the P rows of the matrix $\mathbf{X}$ are a subset of rows from (4.32), and the selected subset depends on the positions of non-zero entries in the DD channel matrix. This structure of $\mathbf{X}$ arises naturally from the pre- and post-processing operations (ISFFT and SFFT), which result in the 2D circular convolution of the transmit vector $\mathbf{x}$ with the channel response in the DD domain.

Example 4.6 Note that for the system in Example 4.1 with $M = N = 2$ and $P = 2$ (i.e., $P < MN$), the $\mathbf{X}$ matrix is given by

$$\mathbf{X} = \begin{bmatrix} x_0 & x_1 & x_2 & x_3 \\ x_3 & x_2 & x_1 & x_0 \end{bmatrix}. \quad (4.35)$$

Note that the rows in the above matrix are a subset of the rows in the $\mathbf{X}$ matrix in (4.34), i.e., the $\mathbf{X}$ matrix for the corresponding $P = MN$ case.

The following theorem shows that multiplying the OTFS transmit vector $\mathbf{x}$ in (4.33) by a diagonal phase rotation matrix $\mathbf{\Phi}$ with distinct transcendental numbers results in full diversity.

Theorem 4.1 Let

$$\mathbf{\Phi} = \text{diag} \left\{ \phi_0^{(0)}, \dots, \phi_{N-1}^{(0)}, \phi_0^{(1)}, \dots, \phi_{N-1}^{(1)}, \dots, \phi_{N-1}^{(M-1)} \right\} \quad (4.36)$$

be the phase rotation matrix and

$$\mathbf{x}' = \mathbf{\Phi} \mathbf{x} = \begin{bmatrix} \phi_0^{(0)} x_0^{(0)} \\ \vdots \\ \phi_{N-1}^{(0)} x_{N-1}^{(0)} \\ \phi_0^{(1)} x_0^{(1)} \\ \vdots \\ \phi_{N-1}^{(1)} x_{N-1}^{(1)} \\ \vdots \\ \phi_{N-1}^{(M-1)} x_{N-1}^{(M-1)} \end{bmatrix} \quad (4.37)$$

be the phase-rotated OTFS transmit vector. OTFS with the above phase rotation achieves the full diversity of P when $\phi_q^{(l)} = e^{j a_q^{(l)}}$, $q = 0, \dots, N-1$, $l = 0, \dots, M-1$, are transcendental numbers with $a_q^{(l)}$ real, distinct, and algebraic.

Proof: Let $\mathbf{x}'_i = \mathbf{\Phi} \mathbf{x}_i$ and $\mathbf{x}'_j = \mathbf{\Phi} \mathbf{x}_j$ be two phase-rotated OTFS transmit vectors. Let $\mathbf{X}'_i$ and $\mathbf{X}'_j$ denote the corresponding phase-rotated symbol matrices.

Case 1: $P = MN$

When $P = MN$, the symbol matrices $\mathbf{X}'_i$ and $\mathbf{X}'_j$ are block circulant with circulant blocks, and hence $\Delta'_{ij} = \mathbf{X}'_i - \mathbf{X}'_j$ also has the same structure, i.e.,

$$\Delta'_{ij} = \begin{bmatrix} \Delta'_{ij}{}^{(0)} & \Delta'_{ij}{}^{(1)} & \dots & \Delta'_{ij}{}^{(M-1)} \\ \Delta'_{ij}{}^{(M-1)} & \Delta'_{ij}{}^{(0)} & \dots & \Delta'_{ij}{}^{(M-2)} \\ \vdots & \vdots & \vdots & \vdots \\ \Delta'_{ij}{}^{(1)} & \Delta'_{ij}{}^{(2)} & \dots & \Delta'_{ij}{}^{(0)} \end{bmatrix}, \quad (4.38)$$

where

$$\Delta'_{ij}{}^{(l)} = \begin{bmatrix} \delta_0^{(l)} \phi_0^{(l)} & \delta_1^{(l)} \phi_1^{(l)} & \dots & \delta_{N-1}^{(l)} \phi_{N-1}^{(l)} \\ \delta_{N-1}^{(l)} \phi_{N-1}^{(l)} & \delta_0^{(l)} \phi_0^{(l)} & \dots & \delta_{N-2}^{(l)} \phi_{N-2}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_1^{(l)} \phi_1^{(l)} & \delta_2^{(l)} \phi_2^{(l)} & \dots & \delta_0^{(l)} \phi_0^{(l)} \end{bmatrix}, \quad (4.39)$$

where $\delta_q^{(l)} = x_{i,q}^{(l)} - x_{j,q}^{(l)}$, with $x_{i,q}^{(l)}$ and $x_{j,q}^{(l)}$ being q th distinct elements in the l th block of $\mathbf{X}_i$ and $\mathbf{X}_j$, respectively. Since Δ'_{ij} is block circulant with circulant blocks, it is diagonalized by $\mathbf{F}_M \otimes \mathbf{F}_N$, where $\mathbf{F}_M$ and $\mathbf{F}_N$ denote the $M \times M$ and $N \times N$ DFT matrices and $\otimes$ denotes the Kronecker product. Therefore, Δ'_{ij} is given by [71]

$$\Delta'_{ij} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \mathbf{D} (\mathbf{F}_M \otimes \mathbf{F}_N), \quad (4.40)$$

where $\mathbf{D}$ is an $MN \times MN$ diagonal matrix whose entries are the eigenvalues of Δ'_{ij} , given by

$$\mathbf{D} = \sum_{l=0}^{M-1} \mathbf{\Omega}_M^l \otimes \mathbf{\Lambda}^{(l)}, \quad (4.41)$$

where $\mathbf{\Omega}_M = \text{diag}\{1, \omega, \omega^2, \dots, \omega^{M-1}\}$ with $\omega = e^{j2\pi/M}$, and $\mathbf{\Lambda}^{(l)}$ is a diagonal matrix whose entries are the eigenvalues of $\Delta'^{(l)}_{ij}$. Let $\lambda_q^{(l)}$ denote the q th eigenvalue of $\Delta'^{(l)}_{ij}$ and $\mu_0, \mu_1, \dots, \mu_{MN-1}$ denote the eigenvalues of Δ'_{ij} . From (4.41), the k th eigenvalue of Δ'_{ij} , that is, μ_k is

$$\mu_k = \sum_{l=0}^{M-1} \lambda_u^{(l)} \omega^{vl}, \quad (4.42)$$

where $u = k - \lfloor \frac{k}{N} \rfloor N$, $v = \lfloor \frac{k}{N} \rfloor$, and $\lambda_u^{(l)}$ is the u th eigenvalue of $\Delta'^{(l)}_{ij}$, given by

$$\lambda_u^{(l)} = \sum_{q=0}^{N-1} \phi_q^{(l)} \delta_q^{(l)} e^{-j2\pi u q / N}. \quad (4.43)$$

Using (4.43) in (4.42), μ_k can be written as

$$\begin{aligned} \mu_k &= \sum_{l=0}^{M-1} \sum_{q=0}^{N-1} \phi_q^{(l)} \delta_q^{(l)} e^{-j2\pi u q / N} \omega^{vl} \\ &= \sum_{l=0}^{M-1} \left\{ \phi_0^{(l)} \delta_0^{(l)} + \phi_1^{(l)} \delta_1^{(l)} e^{-j2\pi u / N} + \dots + \phi_{N-1}^{(l)} \delta_{N-1}^{(l)} e^{-j2\pi u (N-1) / N} \right\} \omega^{vl}, \end{aligned} \quad (4.44)$$

which can be further simplified as

$$\begin{aligned} \mu_k &= \phi_0^{(0)} \delta_0^{(0)} + \phi_1^{(0)} \delta_1^{(0)} e^{-\frac{j2\pi u}{N}} + \dots + \phi_{N-1}^{(0)} \delta_{N-1}^{(0)} e^{-\frac{j2\pi u (N-1)}{N}} \\ &\quad + \phi_0^{(1)} \delta_0^{(1)} \omega^v + \dots + \phi_{N-1}^{(1)} \delta_{N-1}^{(1)} e^{-\frac{j2\pi u (N-1)}{N}} \omega^v + \dots \\ &\quad + \phi_0^{(M-1)} \delta_0^{(M-1)} \omega^{v(M-1)} + \dots + \phi_{N-1}^{(M-1)} \delta_{N-1}^{(M-1)} e^{-\frac{j2\pi u (N-1)}{N}} \omega^{v(M-1)}. \end{aligned} \quad (4.45)$$

Now, invoke Lindemann's theorem [72], which states that if $a_1, a_2, \dots, a_m$ are distinct algebraic numbers and if $c_1, c_2, \dots, c_m$ are algebraic and not all equal to zero, then

$$c_1 e^{a_1} + c_2 e^{a_2} + \dots + c_m e^{a_m} \neq 0. \quad (4.46)$$

It is noted from (4.45) that the terms of the form $\delta_q^{(l)} e^{-\frac{j2\pi u q}{N}} \omega^{vl}$ are all algebraic [72]. Therefore, comparing (4.46) and (4.45), if the terms $\phi_q^{(l)}$ are chosen such that $\phi_q^{(l)} = e^{ja_q^{(l)}}$ are transcendental with $a_q^{(l)}$ real, distinct, and algebraic, then μ_k cannot be zero. Since μ_k s are eigenvalues of Δ'_{ij} , choosing the diagonal phase rotation matrix with entries $\phi_q^{(l)} = e^{ja_q^{(l)}}$ being transcendental with $a_q^{(l)}$ real, distinct, and algebraic ensures that all the eigenvalues of Δ'_{ij} are non-zero, making it full rank (i.e., rank P). Since this is true for Δ'_{ij} for all (i, j) , $i \neq j$, the minimum rank of Δ'_{ij} is equal to MN . Hence, from (4.26), the achieved diversity order of OTFS with the proposed phase rotation is $MN (= P)$.

Case 2: $P < MN$

Next, consider the case when $P < MN$. As mentioned earlier, when $P < MN$, the rows of the transmit symbol matrix $\mathbf{X}$ are the subset of rows from the corresponding $MN \times MN$ matrix in (4.32). If $\mathbf{X}'_i$ and $\mathbf{X}'_j$ are two phase-rotated symbol matrices with $P < MN$, then the rows of $\Delta'_{ij} = \mathbf{X}'_i - \mathbf{X}'_j$ form a subset of the rows of the corresponding $MN \times MN$ matrix in (4.38). Since the matrix in (4.38) is shown to be full rank in Case 1, Δ'_{ij} with $P < MN$ should have a rank equal to P . Therefore, OTFS

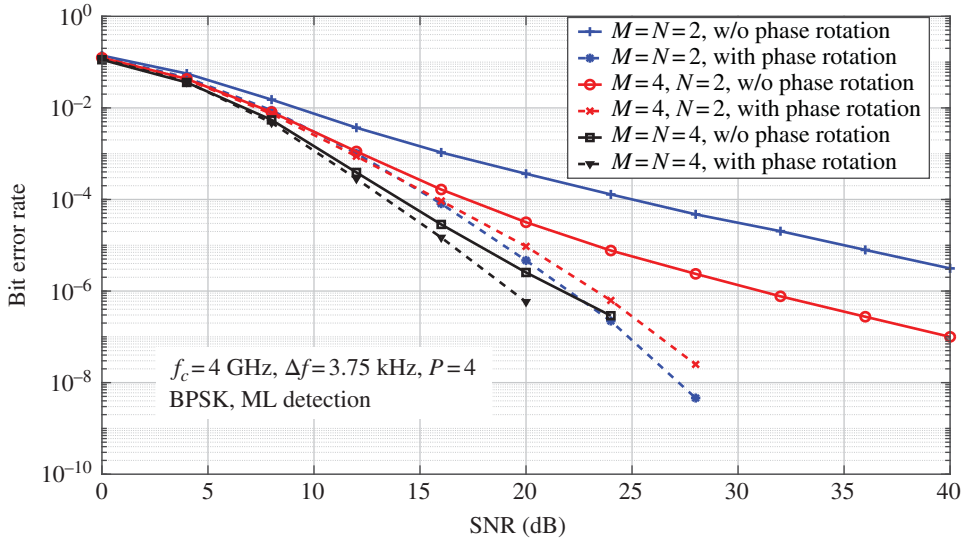


Figure 4.5 BER performance of OTFS without and with phase rotation for (i) $M = N = 2$, (ii) $M = 4, N = 2$, and (iii) $M = N = 4$.

with phase rotation using transcendental numbers of the form $\phi_q^{(l)} = e^{ja_q^{(l)}}$ with $a_q^{(l)}$ real, distinct, and algebraic achieves the full diversity of P in the DD domain.

The simulated BER plots in Fig. 4.5 illustrate the full diversity performance achieved in OTFS with phase rotation. In Fig. 4.5, the BER performance of OTFS without and with phase rotation for the following three systems with BPSK and $P = 4$ are plotted: (i) system-1 with $M = N = 2$, (ii) system-2 with $M = 4, N = 2$, and (iii) system-3 with $M = N = 4$. The phase rotation matrix used is given by $\Phi = \text{diag}\{1, e^{j\frac{1}{MN}}, \dots, e^{j\frac{MN-1}{MN}}\}$. From Fig. 4.5, it is observed that the asymptotic diversity order of all the three systems without phase rotation is one. With phase rotation, these systems exhibit full diversity in the high SNR regime corroborating with the analysis. While full diversity is achieved in all the three systems, the achieved coding gains are slightly different. This is because the phase rotation matrix used in all the systems is the same. The coding gain achieved by a system can be improved by optimizing the phases used in the phase rotation matrix [72].

4.2 Diversity in MIMO-OTFS

Consider a MIMO-OTFS system shown in Fig. 4.6 with n_t transmit and n_r receive antennas. Each antenna transmits an independent OTFS signal vector. The channel gain between the k th transmit antenna and l th receive antenna in the DD domain is given by

$$h_{lk}(\tau, \nu) = \sum_{i=1}^P h_{lk_i} \delta(\tau - \tau_{lk_i}) \delta(\nu - \nu_{lk_i}), \quad (4.47)$$

$k = 1, 2, \dots, n_t, l = 1, 2, \dots, n_r$. Let $\mathbf{H}_{lk}$ denote the $MN \times MN$ equivalent channel matrix between the k th transmit antenna and l th receive antenna. Let $\mathbf{x}_k$ denote the $MN \times 1$ transmit vector from the k th transmit antenna and $\mathbf{y}_l$ denote the $MN \times 1$ received vector at the l th receive antenna. Then,

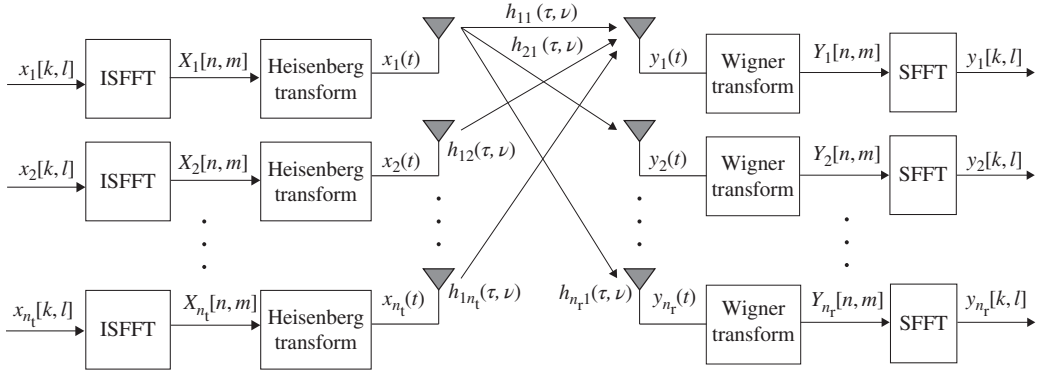


Figure 4.6 MIMO-OTFS modulation scheme.

using the linear vector channel model in (4.10) for SISO-OTFS, the set of equations describing the input-output relation in MIMO-OTFS can be written as

$$\begin{aligned}
 \mathbf{y}_1 &= \mathbf{H}_{11}\mathbf{x}_1 + \mathbf{H}_{12}\mathbf{x}_2 + \cdots + \mathbf{H}_{1n_t}\mathbf{x}_{n_t} + \mathbf{v}_1, \\
 \mathbf{y}_2 &= \mathbf{H}_{21}\mathbf{x}_1 + \mathbf{H}_{22}\mathbf{x}_2 + \cdots + \mathbf{H}_{2n_t}\mathbf{x}_{n_t} + \mathbf{v}_2, \\
 &\vdots \\
 \mathbf{y}_{n_r} &= \mathbf{H}_{n_r,1}\mathbf{x}_1 + \mathbf{H}_{n_r,2}\mathbf{x}_2 + \cdots + \mathbf{H}_{n_r,n_t}\mathbf{x}_{n_t} + \mathbf{v}_{n_r}.
 \end{aligned} \tag{4.48}$$

Defining

$$\mathbf{H}_{\text{mimo}} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} & \cdots & \mathbf{H}_{1n_t} \\ \mathbf{H}_{21} & \mathbf{H}_{22} & \cdots & \mathbf{H}_{2n_t} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}_{n_r,1} & \mathbf{H}_{n_r,2} & \cdots & \mathbf{H}_{n_r,n_t} \end{bmatrix}, \tag{4.49}$$

$\mathbf{x}_{\text{mimo}} = [\mathbf{x}_1^T, \dots, \mathbf{x}_{n_t}^T]^T$, $\mathbf{y}_{\text{mimo}} = [\mathbf{y}_1^T, \dots, \mathbf{y}_{n_r}^T]^T$, and $\mathbf{v}_{\text{mimo}} = [\mathbf{v}_1^T, \dots, \mathbf{v}_{n_r}^T]^T$, (4.48) can be written as

$$\mathbf{y}_{\text{mimo}} = \mathbf{H}_{\text{mimo}}\mathbf{x}_{\text{mimo}} + \mathbf{v}_{\text{mimo}}, \tag{4.50}$$

where $\mathbf{x}_{\text{mimo}} \in \mathbb{C}^{n_t MN \times 1}$, $\mathbf{y}_{\text{mimo}}, \mathbf{v}_{\text{mimo}} \in \mathbb{C}^{n_r MN \times 1}$, and $\mathbf{H}_{\text{mimo}} \in \mathbb{C}^{n_r MN \times n_t MN}$.

4.2.1 MIMO-OTFS Diversity Analysis

In the effective channel matrix $\mathbf{H}_{\text{mimo}}$, each $\mathbf{H}_{lk}$ has only P unique entries, and, hence, $\mathbf{H}_{\text{mimo}}$ has Pn_t unique entries. Furthermore, each row of $\mathbf{H}_{\text{mimo}}$ has only $n_t P$ non-zero elements and each column has only $n_r P$ non-zero elements. Following (4.17), the MIMO-OTFS system model in (4.50) can be written as

$$\underbrace{\begin{bmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \\ \vdots \\ \mathbf{y}_{n_r}^T \end{bmatrix}}_{\tilde{\mathbf{y}}} = \underbrace{\begin{bmatrix} \mathbf{h}'_{11} & \mathbf{h}'_{12} & \cdots & \mathbf{h}'_{1n_t} \\ \mathbf{h}'_{21} & \mathbf{h}'_{22} & \cdots & \mathbf{h}'_{2n_t} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{h}'_{n_r,1} & \mathbf{h}'_{n_r,2} & \cdots & \mathbf{h}'_{n_r,n_t} \end{bmatrix}}_{\tilde{\mathbf{H}}} \underbrace{\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_{n_t} \end{bmatrix}}_{\tilde{\mathbf{x}}} + \underbrace{\begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_{n_r}^T \end{bmatrix}}_{\tilde{\mathbf{v}}} \tag{4.51}$$

or equivalently

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}\tilde{\mathbf{X}} + \tilde{\mathbf{V}}, \quad (4.52)$$

where $\tilde{\mathbf{y}}$ is an $n_r \times MN$ received signal matrix whose l th row is the received vector received in the l th receive antenna, $\tilde{\mathbf{X}}$ is an $n_t P \times MN$ matrix obtained by stacking n_t number of $P \times MN$ sized symbol matrices of the form (4.18), $\tilde{\mathbf{H}} \in \mathbb{C}^{n_r \times n_t P}$ is the channel matrix with $\mathbf{h}'_{lk} \in \mathbb{C}^{1 \times P}$ consisting of P unique non-zero entries of $\mathbf{H}_{lk}$, and $\tilde{\mathbf{V}} \in \mathbb{C}^{n_r \times MN}$ is the noise matrix.

Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be two symbol matrices. Assuming perfect CSI and ML detection at the receiver, the probability of decoding the transmitted symbol matrix $\tilde{\mathbf{X}}_i$ in favor of $\tilde{\mathbf{X}}_j$ is given by

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j | \tilde{\mathbf{H}}) = Q \left(\sqrt{\frac{\|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2}{2N_0}} \right), \quad (4.53)$$

and the average PEP is given by

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) = \mathbb{E} \left[Q \left(\sqrt{\frac{\|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2}{2N_0}} \right) \right]. \quad (4.54)$$

Using the Chernoff bound and the fact that each antenna transmits independent OTFS symbols, an upper bound on the PEP in (4.54) can be obtained as [6]

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \left(\prod_{l=1}^r \frac{1}{1 + \frac{\gamma \lambda_{k,l}^2}{4P}} \right)^{n_r}, \quad (4.55)$$

where γ is the SNR per receive antenna, $\lambda_{k,l}$ is the l th singular value of the difference matrix $\Delta_{k,ij} = (\mathbf{X}_{k,i} - \mathbf{X}_{k,j})$ with $\mathbf{X}_{k,i}$ and $\mathbf{X}_{k,j}$ denoting OTFS symbol matrices transmitted from k th antenna (for some $k \in 1, 2, \dots, n_t$) in $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$, respectively, and r is the rank of $\Delta_{k,ij}$. At high SNR values, (4.55) simplifies to

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \frac{1}{\gamma^{n_r r}} \left(\prod_{l=1}^r \frac{\lambda_{k,l}^2}{4P} \right)^{-n_r}. \quad (4.56)$$

The PEP term with the minimum value of r dominates the overall BER. Therefore, the diversity order achieved by MIMO-OTFS, denoted by $\rho_{\text{mimo-otfs}}$ is given by

$$\rho_{\text{mimo-otfs}} = n_r \cdot \min_{i,j \text{ } i \neq j} \text{rank}(\Delta_{k,ij}). \quad (4.57)$$

Similar to the case of SISO-OTFS, if $\mathbf{X}_{k,i} = a \cdot \mathbf{1}_{P \times MN}$ and $\mathbf{X}_{k,j} = a' \cdot \mathbf{1}_{P \times MN}$, then the difference matrix $\Delta_{k,ij} = (a - a') \cdot \mathbf{1}_{P \times MN}$ has rank one. Hence, the asymptotic diversity order of MIMO-OTFS is n_r .

4.2.2 MIMO-OTFS with Phase Rotation

Consider MIMO-OTFS with phase rotation. The transmit vector in MIMO-OTFS is a concatenation of n_t independent OTFS transmit vectors of size $MN \times 1$, as described by (4.50). The $MN \times 1$ OTFS transmit vector from each antenna is multiplied by the phase rotation matrix Φ in (4.36). The phase-rotated MIMO-OTFS transmit vector is then given by

$$\mathbf{x}'_{\text{mimo}} = (\mathbf{I}_{n_t} \otimes \Phi) \mathbf{x}_{\text{mimo}}, \quad (4.58)$$

where $\mathbf{I}_{n_t}$ denotes the $n_t \times n_t$ identity matrix. Let $\tilde{\mathbf{X}}'$ be the phase-rotated MIMO-OTFS symbol matrix corresponding to $\mathbf{x}'$. From (4.51), $\tilde{\mathbf{X}}'$ is of the form

$$\tilde{\mathbf{X}}' = \begin{bmatrix} \mathbf{X}'_1 \\ \mathbf{X}'_2 \\ \vdots \\ \mathbf{X}'_{n_t} \end{bmatrix}, \quad (4.59)$$

where $\mathbf{X}'_k$ is the phase-rotated OTFS symbol matrix corresponding to the k th transmit antenna. If $\tilde{\mathbf{X}}'_i$ and $\tilde{\mathbf{X}}'_j$ are two phase-rotated MIMO-OTFS symbol matrices corresponding to the transmit vectors $\mathbf{x}'_{i,\text{mimo}}$ and $\mathbf{x}'_{j,\text{mimo}}$, then their difference matrix $\tilde{\Delta}'_{ij}$ is of the form

$$\tilde{\Delta}'_{ij} = \begin{bmatrix} \Delta'_{1,ij} \\ \Delta'_{2,ij} \\ \vdots \\ \Delta'_{n_t,ij} \end{bmatrix}, \quad (4.60)$$

where $\Delta'_{k,ij} = \mathbf{X}'_{k,i} - \mathbf{X}'_{k,j}$ with $\mathbf{X}'_{k,i}$ and $\mathbf{X}'_{k,j}$ being the phase-rotated OTFS symbol matrices corresponding to the k th antenna in $\tilde{\mathbf{X}}'_i$ and $\tilde{\mathbf{X}}'_j$, respectively. From Section 4.1.3, it is known that $\Delta'_{k,ij}$ has rank equal to P for all $k = 1, 2, \dots, n_t$. Using this fact in (4.57), the diversity order achieved by phase-rotated MIMO-OTFS system is equal to Pn_t .

Figure 4.7 shows the BER performance of SISO-OTFS and 2×2 MIMO-OTFS systems, both systems using $M = N = 2$ and BPSK. The number of channel taps is $P = 4$. The carrier frequency and the subcarrier spacing used are 4 GHz and 3.75 kHz, respectively. It is observed that the simulated BER for SISO-OTFS and 2×2 MIMO-OTFS shows diversity orders of one and two, respectively,

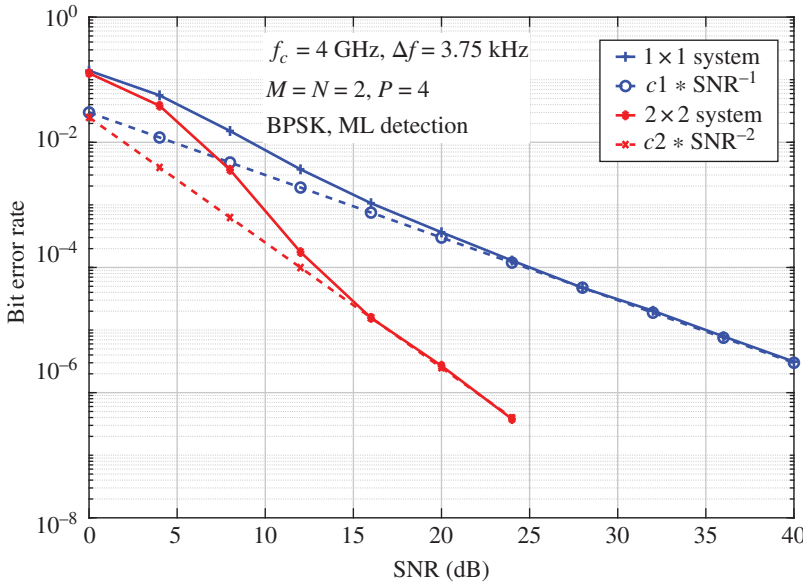


Figure 4.7 BER performance of SISO-OTFS and 2×2 MIMO-OTFS systems.

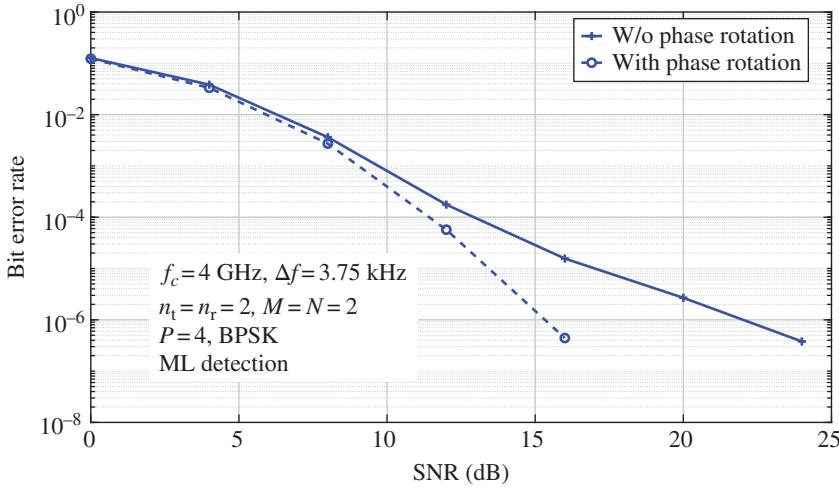


Figure 4.8 BER performance of the 2×2 MIMO-OTFS system without and with phase rotation, $M = N = 2$.

verifying the analytical predicted diversity order of n_r . Figure 4.8 shows the BER performance of 2×2 MIMO-OTFS with and without phase rotation. It is seen that the MIMO-OTFS system with phase rotation achieves the full diversity of Pn_r , whereas the same system without phase rotation achieves a diversity order of only n_r .

4.3 Diversity in Space–Time Coded OTFS

Space–time coding has been a widely researched topic in MIMO wireless communications [73]. Space–time codes provide the opportunity to harness transmit antenna diversity, in addition to receive antenna diversity. In space–time coded systems, coding is done across the spatial domain (across n_t transmit antennas) and time domain (over p channel uses). A space–time code matrix is of size $n_t \times p$. If k symbols are sent using p channel uses, the rate of the code is k/p . A well-known space–time code is the Alamouti code, which uses $n_t = 2$ transmit antennas and $p = 2$ channel uses to send $k = 2$ symbols. The Alamouti space–time code matrix is of size 2×2 and is constructed using the linear combination of two symbols x_1, x_2 and their conjugates as [74]

$$\mathbf{X} = \begin{bmatrix} x_1 & -x_2^* \\ x_2 & x_1^* \end{bmatrix}. \quad (4.61)$$

The rate of the Alamouti code is $k/p = 2/2 = 1$, and the achieved diversity order is $n_t n_r = 2n_r$. That is, the Alamouti code achieves full transmit and receive diversity.

In OTFS systems, diversity opportunities exist in space (transmit and receive antennas) as well as DD domains. In Sections 4.1.3 and 4.2.2, full DD diversity of P was shown to be achieved in SISO-OTFS using phase rotation. Also, MIMO-OTFS with phase rotation was shown to achieve a diversity of $n_r P$, harnessing full DD and receive antenna diversity. It is also possible to harness the transmit antenna diversity in OTFS. This section presents a space–time coded OTFS (STC-OTFS) scheme, which, when used with phase rotation, achieves a diversity of $n_t n_r P$, harnessing full DD diversity as well as full transmit antenna and receive antenna diversity. The idea is to use the structure of the Alamouti code, generalized to matrices [75].

4.3.1 STC-OTFS Scheme

Since OTFS is a 2D modulation, the Alamouti code structure in (4.61) needs to be extended to a matrix form. Toward that, consider the MIMO-OTFS system model in (4.50) and its alternate form in (4.51), i.e., consider the MIMO-OTFS system model given by

$$\underbrace{\begin{bmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \\ \vdots \\ \mathbf{y}_{n_r}^T \end{bmatrix}}_{\tilde{\mathbf{y}}} = \underbrace{\begin{bmatrix} \mathbf{h}'_{11} & \mathbf{h}'_{12} & \cdots & \mathbf{h}'_{1n_t} \\ \mathbf{h}'_{21} & \mathbf{h}'_{22} & \cdots & \mathbf{h}'_{2n_t} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{h}'_{n_r1} & \mathbf{h}'_{n_r2} & \cdots & \mathbf{h}'_{n_rn_t} \end{bmatrix}}_{\tilde{\mathbf{H}}} \underbrace{\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_{n_t} \end{bmatrix}}_{\tilde{\mathbf{X}}} + \underbrace{\begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_{n_r}^T \end{bmatrix}}_{\tilde{\mathbf{v}}}, \quad (4.62)$$

where the vectors and matrices are as defined in (4.51).

4.3.1.1 Encoding and Decoding

Consider OTFS signaling over a MIMO channel with n_t transmit antennas. Assume a quasi-static DD channel over T' frame duration, i.e., NTT' channel uses. An STC-OTFS codeword matrix can be defined as a block matrix $\tilde{\mathbf{X}}$ of size $n_t MN \times T' MN$, in which each block $\tilde{\mathbf{X}}_{kt}$ denotes the $MN \times MN$ OTFS transmit matrix from the k th antenna in the t th frame use. If the space-time code matrix $\tilde{\mathbf{X}}$ contains Q independent OTFS symbol matrices which are transmitted over T' frame uses, then the rate of the code is Q/T' symbols per channel use. Generalizing the Alamouti code to matrices, a rate-1, $n_t = 2$ Alamouti STC-OTFS codeword matrix of size $2MN \times 2MN$ is constructed as

$$\tilde{\mathbf{X}} = \begin{bmatrix} \mathbf{X}_1 & -\mathbf{X}_2^H \\ \mathbf{X}_2 & \mathbf{X}_1^H \end{bmatrix}, \quad (4.63)$$

where $\mathbf{X}_1$ and $\mathbf{X}_2$ are the symbol matrices, as described in (4.17) and (4.18). That is, in the STC-OTFS scheme, the OTFS transmit vectors corresponding to $\mathbf{X}_1$ and $\mathbf{X}_2$ are transmitted from the first and the second antennas, respectively, during the first frame duration. During the second frame duration, the vectors corresponding to $-\mathbf{X}_2^H$ and $\mathbf{X}_1^H$ are transmitted from the first and second antennas, respectively. The block diagram of the STC-OTFS system is shown in Fig. 4.9.

Now, it is important to look at the OTFS transmit vectors that are transmitted during the two-frame duration. The OTFS transmit vector corresponding to the block $\mathbf{X}_k, k \in \{1, 2\}$ in (4.63), denoted by $\mathbf{x}_k$, is nothing but the first row of $\mathbf{X}_k$. The transmitted vector corresponding to $\mathbf{X}_k^H, k \in \{1, 2\}$, is the conjugate of the first column of $\mathbf{X}_k$. The vector corresponding to the first column of

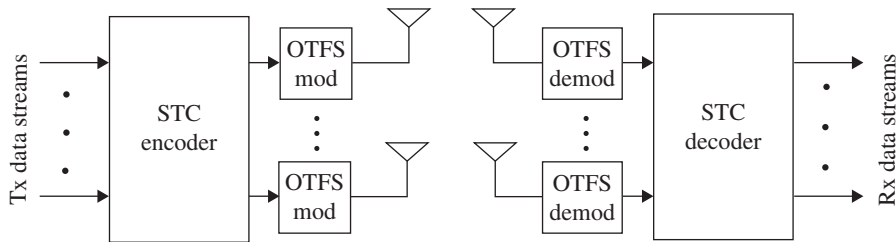


Figure 4.9 STC-OTFS scheme.

$\mathbf{X}_k$ can be obtained from $\mathbf{x}_k$ as $\hat{\mathbf{x}}_k = \mathbf{P}\mathbf{x}_k$, where $\mathbf{P}$ is the $MN \times MN$ permutation matrix given by

$$\mathbf{P} = \mathbf{P}'_M \otimes \mathbf{P}'_N, \quad (4.64)$$

where $\mathbf{P}'_M$ and $\mathbf{P}'_N$ are the left circulant matrices given by

$$\mathbf{P}'_M = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & & & & \\ 0 & 1 & \cdots & 0 & 0 \end{bmatrix}_{M \times M}, \quad \mathbf{P}'_N = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & & & & \\ 0 & 1 & \cdots & 0 & 0 \end{bmatrix}_{N \times N}. \quad (4.65)$$

In general, $\hat{\mathbf{x}} = \mathbf{P}\mathbf{x}$ is of the form

$$\hat{\mathbf{x}} = [x_0 \ x_{N-1} \ \cdots \ x_1 \ x_{(M-1)N} \ x_{MN-1} \ x_{((M-1)N)+1} \ \cdots \ \cdots \ x_N \ x_{2N-1} \ \cdots \ x_{N+1}]^T. \quad (4.66)$$

It is interesting to note that the transmitted OTFS vectors in the second frame duration are not just the conjugated vectors but are conjugated and permuted vectors of those transmitted in the first frame duration. Now, using (4.63) and the alternate representation of the MIMO-OTFS system in (4.62), the input-output equation of STC-OTFS can be written as

$$[\mathbf{y}_1^T \ \mathbf{y}_2^T] = [\mathbf{h}'_1 \ \mathbf{h}'_2] \begin{bmatrix} \mathbf{X}_1 & -\mathbf{X}_2^H \\ \mathbf{X}_2 & \mathbf{X}_1^H \end{bmatrix} + [\mathbf{v}_1^T \ \mathbf{v}_2^T], \quad (4.67)$$

where $\mathbf{y}_1$ and $\mathbf{y}_2$ denote the $MN \times 1$ received vectors at the first and second frame duration, respectively, $\mathbf{h}'_1$ and $\mathbf{h}'_2$ denote the channel from the first and second antennas, respectively, and $\mathbf{v}_1$ and $\mathbf{v}_2$ are the $MN \times 1$ independent noise vectors.

To gain insight into the decoding complexity of this scheme, consider the MIMO-OTFS system model in (4.50). Denoting the channel matrices from the first and second transmit antennas with $\mathbf{H}_1$ and $\mathbf{H}_2$, respectively, the received vectors in the first and second frame duration are

$$\mathbf{y}_1 = \mathbf{H}_1\mathbf{x}_1 + \mathbf{H}_2\mathbf{x}_2 + \mathbf{v}_1, \quad (4.68)$$

$$\mathbf{y}_2 = -\mathbf{H}_1(\hat{\mathbf{x}}_2)^* + \mathbf{H}_2(\hat{\mathbf{x}}_1)^* + \mathbf{v}_2. \quad (4.69)$$

At the receiver, permutation $\mathbf{P}$ and conjugation are applied on $\mathbf{y}_2$. With this,

$$\underbrace{\begin{bmatrix} \mathbf{y}_1 \\ (\hat{\mathbf{y}}_2)^* \end{bmatrix}}_{\triangleq \hat{\mathbf{y}}} = \underbrace{\begin{bmatrix} \mathbf{H}_1 & \mathbf{H}_2 \\ \mathbf{H}_2^H & -\mathbf{H}_1^H \end{bmatrix}}_{\triangleq \tilde{\mathbf{H}}} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{v}_1 \\ (\hat{\mathbf{v}}_2)^* \end{bmatrix}, \quad (4.70)$$

where $\hat{\mathbf{y}}_2 = \mathbf{P}\mathbf{y}_2$ and $\hat{\mathbf{v}}_2 = \mathbf{P}\mathbf{v}_2$. Denoting $\tilde{\mathbf{H}} = \begin{bmatrix} \mathbf{H}_1 & \mathbf{H}_2 \\ \mathbf{H}_2^H & -\mathbf{H}_1^H \end{bmatrix}$ as the effective channel matrix, we observe that the two block columns of $\tilde{\mathbf{H}}$ are orthogonal. Therefore, the decoding problem for $\mathbf{x}_1$ and $\mathbf{x}_2$ can be decomposed into two separate orthogonal problems, thereby keeping the decoding complexity same as that of SISO-OTFS.

4.3.1.2 Diversity Analysis

Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be two distinct STC-OTFS codeword matrices, and let $\tilde{\Delta}_{ij} \triangleq \tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j$ be the difference matrix. The asymptotic diversity order of STC-OTFS with $n_t = 2$ and $n_r = 1$ is

$$\rho_{\text{stc-otfs}} = \min \left\{ \min_{i,j \text{ } i \neq j} \text{rank}(\tilde{\Delta}_{ij}), 2P \right\}. \quad (4.71)$$

Define the difference matrix corresponding to the k th independent OTFS symbol matrix as $\Delta_{k,ij} = \mathbf{X}_{k,i} - \mathbf{X}_{k,j}$. Note that $\text{rank}(\tilde{\Delta}_{ij}) = \text{rank}(\tilde{\Delta}_{ij}^H \tilde{\Delta}_{ij})$. Hence, $\tilde{\Delta}_{ij}^H \tilde{\Delta}_{ij}$ is used for analysis, which can be simplified as

$$\begin{aligned} \tilde{\Delta}_{ij}^H \tilde{\Delta}_{ij} &= \begin{bmatrix} \Delta_{1,ij} & -\Delta_{2,ij}^H \\ \Delta_{2,ij} & \Delta_{1,ij}^H \end{bmatrix}^H \begin{bmatrix} \Delta_{1,ij} & -\Delta_{2,ij}^H \\ \Delta_{2,ij} & \Delta_{1,ij}^H \end{bmatrix} \\ &= \begin{bmatrix} \Delta_{1,ij}^H \Delta_{1,ij} + \Delta_{2,ij}^H \Delta_{2,ij} & \mathbf{0}_{MN \times MN} \\ \mathbf{0}_{MN \times MN} & \Delta_{1,ij}^H \Delta_{1,ij} + \Delta_{2,ij}^H \Delta_{2,ij} \end{bmatrix}. \end{aligned} \quad (4.72)$$

Observe that $\tilde{\Delta}_{ij}^H \tilde{\Delta}_{ij}$ is block diagonal due to the commutative property of the block circulant matrix with circulant blocks [71], and, hence,

$$\text{rank}(\tilde{\Delta}_{ij}^H \tilde{\Delta}_{ij}) = 2 \times \text{rank}(\Delta_{1,ij}^H \Delta_{1,ij} + \Delta_{2,ij}^H \Delta_{2,ij}). \quad (4.73)$$

It has been shown earlier in this chapter (Section 4.1) that the asymptotic diversity order of SISO-OTFS is one. Hence, the minimum rank of $\Delta_{k,ij}$ is one $\forall k$. Now, consider the case when $\Delta_{1,ij} = \mathbf{0}_{MN \times MN}$ and $\Delta_{2,ij} = (a_1 - a_2)\mathbf{1}_{MN \times MN}$, where $a_1, a_2 \in \mathbb{A}$, where $\mathbb{A}$ is the modulation alphabet (e.g., PSK/QAM), in which case the rank of $\Delta_{1,ij}^H \Delta_{1,ij} + \Delta_{2,ij}^H \Delta_{2,ij}$ is one. Therefore, the minimum rank of $\Delta_{1,ij}^H \Delta_{1,ij} + \Delta_{2,ij}^H \Delta_{2,ij}$ is one. Hence, from (4.71), as $P \geq 1$, the asymptotic diversity order of STC-OTFS with $n_t = 2$ and $n_r = 1$ is two. Extension of the system model in (4.67) to STC-OTFS with $n_t = 2$ and $n_r \geq 2$ yields the full spatial diversity order of $2n_r$. Note that for MIMO-OTFS (with no space-time coding) with $n_t = 2$ and $n_r \geq 2$, the achieved diversity order is only n_r . Although STC-OTFS can extract full spatial diversity as discussed above, it fails to extract the DD diversity offered by each link of the MIMO channel. In the following, STC-OTFS is shown to achieve full spatial as well as DD diversity when phase rotation is used.

4.3.1.3 Phase Rotation in STC-OTFS

Let $\Phi = \text{diag}\{\phi_0, \phi_1, \dots, \phi_{MN-1}\}$ be the phase rotation matrix and $\mathbf{x}' = \Phi \mathbf{x}$ be the phase-rotated OTFS transmit vector. As per Theorem 4.1, SISO-OTFS with the above phase rotation achieves the full diversity when $\phi_i = e^{ja_i}$, $i = 0, 1, \dots, MN - 1$, are transcendental numbers with a_i real, distinct, and algebraic. In STC-OTFS, the OTFS transmit vector from each of the transmit antennas during every frame is multiplied by the phase rotation matrix Φ . Let the difference matrix of phase-rotated STC-OTFS codeword matrix be denoted with $\tilde{\Delta}_{ij}^\phi$. Similar to (4.71), the asymptotic diversity of phase-rotated STC-OTFS with $n_t = 2$ and $n_r = 1$ is given by

$$\rho_{\text{stc-otfs}} = \min \left\{ \min_{i,j \text{ } i \neq j} \text{rank}(\tilde{\Delta}_{ij}^\phi), 2P \right\}. \quad (4.74)$$

We denote the difference matrix corresponding to the k th phase-rotated OTFS symbol matrix with $\Delta_{k,ij}^\phi$. Now, as a consequence of Theorem 4.1, $\Delta_{k,ij}^\phi$ has a full rank of MN for all k . As described in Section 4.3.1.2, the minimum rank of $\tilde{\Delta}_{ij}^\phi$ is the same as that of $(\tilde{\Delta}_{ij}^\phi)^H \tilde{\Delta}_{ij}^\phi = 2 \times \text{rank}((\Delta_{1,ij}^\phi)^H \Delta_{1,ij}^\phi + (\Delta_{2,ij}^\phi)^H \Delta_{2,ij}^\phi)$. Now, for any two distinct codewords ($\tilde{\Delta}_{ij}^\phi \neq \mathbf{0}$), either $\Delta_{1,ij}^\phi \neq \mathbf{0}$ or $\Delta_{2,ij}^\phi \neq \mathbf{0}$ or both are non-zero matrices. If $\Delta_{k,ij}^\phi \neq \mathbf{0}$, from Theorem 4.1, it will be full rank. Thus, at least one of $(\Delta_{1,ij}^\phi)^H \Delta_{1,ij}^\phi$ or $(\Delta_{2,ij}^\phi)^H \Delta_{2,ij}^\phi$, or both are positive definite matrices. Therefore, their sum $(\Delta_{1,ij}^\phi)^H \Delta_{1,ij}^\phi + (\Delta_{2,ij}^\phi)^H \Delta_{2,ij}^\phi$ is also positive definite, and, hence, $\tilde{\Delta}_{ij}^\phi$ has the full rank of $2MN$ for any input symbols. Since $P \leq MN$, the asymptotic diversity order of phase-rotated STC-OTFS with $(n_t, n_r) = (2, 1)$ is $2P$. The extension of this scheme to STC-OTFS with $n_t = 2$ and $n_r \geq 2$ yields a full diversity order of $2Pn_r$.

Figure 4.10 shows the BER performance of STC-OTFS with $(n_t, n_r) = (2, 1)$ and $(2, 2)$ for $M = N = 2$, a carrier frequency of 4 GHz, a subcarrier spacing of 3.75 kHz, $P = 2$ with $(\tau_i, \nu_i) = (0, 0)$ and $(\frac{1}{M\Delta f}, \frac{1}{N\Delta T})$, a maximum speed of 506.2 km/h, BPSK, and ML detection. The performance of SISO-OTFS ($n_t = n_r = 1$) with diversity order one is also shown. It is observed that the BER plots for STC-OTFS with $(n_t, n_r) = (2, 1)$ and $(2, 2)$ show diversity orders of two and four, respectively, as predicted by the analysis. Figure 4.11 shows the BER performance of STC-OTFS with phase rotation for $(n_t, n_r) = (2, 1)$, $M = N = 2$, $P = 2, 4$, and phase rotation matrix $\Phi = \text{diag}\{1, e^{j\frac{1}{MN}} \dots, e^{j\frac{MN-1}{MN}}\}$. The BER performance of the system without phase rotation is also plotted. It is observed that

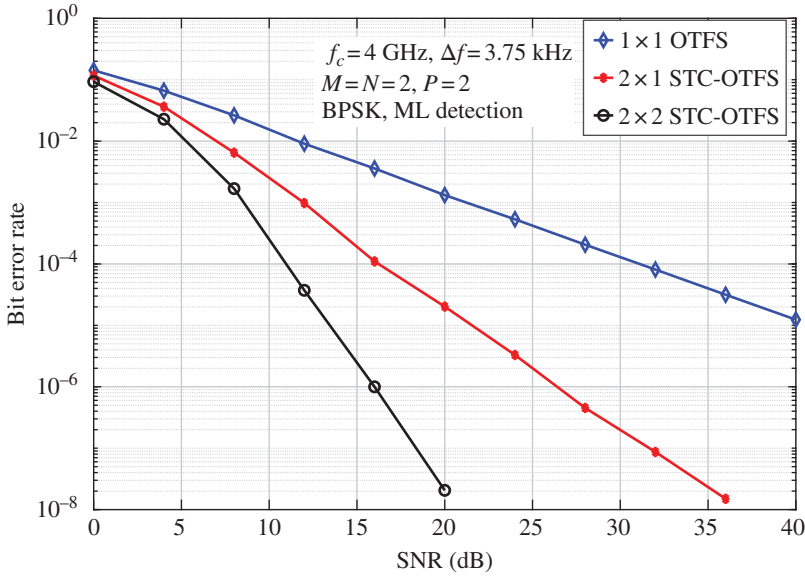


Figure 4.10 BER performance of STC-OTFS with $(n_t, n_r) = (1, 1), (2, 1), (2, 2)$ for $M = N = 2$.

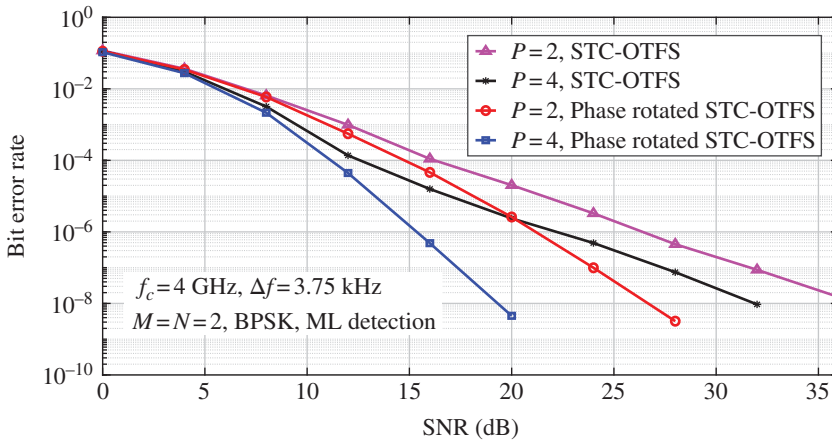


Figure 4.11 BER performance of STC-OTFS with and without phase rotation for $(n_t, n_r) = (2, 1)$, $M = N = 2$, and $P = 2, 4$.

STC-OTFS with phase rotation achieves the predicted diversity order of $2P$, whereas STC-OTFS without phase rotation achieves a diversity of only two. This demonstrates that STC-OTFS with rotation achieves the full spatial and DD diversity of $2n_r P$.

4.4 Diversity in OTFS with Antenna Selection

MIMO wireless systems using multi-antenna techniques such as spatial multiplexing and space-time coding can offer rate and performance benefits. However, increased RF hardware complexity is an important issue in MIMO systems. The increased RF hardware complexity is due to the use of multiple RF chains at the MIMO transmitter and receiver. In traditional MIMO systems, the number of RF chains at the transmitter and receiver is equal to the number of antennas at the transmitter (n_t) and receiver (n_r), respectively. Each RF chain at the transmitter consists of mixers, filters, and amplifiers for up-conversion, and a power amplifier. Likewise, each RF chain at the receiver consists of a low-noise amplifier and mixers, filters, and amplifiers for down-conversion. MIMO systems when used along with antenna selection techniques can allow the use of less number of RF chains than the number of antennas. Antenna selection can be done at the MIMO transmitter or receiver or both. In transmit antenna selection schemes, the number of transmit RF chains ($n_{\text{rf}}^{(t)}$) can be less than the number of transmit antennas, i.e., $n_{\text{rf}}^{(t)} < n_t$, and $n_{\text{rf}}^{(t)}$ antennas out of the n_t available antennas are used for transmission. Likewise, the number of receive RF chains ($n_{\text{rf}}^{(r)}$) can be less than the number of receive antennas, i.e., $n_{\text{rf}}^{(r)} < n_r$, and $n_{\text{rf}}^{(r)}$ antennas out of n_r available antennas are used for reception. Systems with antenna selection can retain the same diversity performance achieved by systems where one RF chain per antenna is used. For example, even though $n_{\text{rf}}^{(r)} < n_r$ receive RF chains are used in a system with receive antenna selection, the full receive diversity order of n_r can still be achieved. Thus, antenna selection offers the benefits of good diversity performance at low RF hardware complexity. In this section, the diversity performance of multi-antenna OTFS systems with receive antenna selection is analyzed.

4.4.1 MIMO-OTFS with RAS

Figure 4.12 shows the block diagram of MIMO-OTFS with receive antenna selection. The transmitter has n_t transmit antennas and the same number of transmit RF chains. The receiver has n_r receive antennas but only n_s receive RF chains, where $n_s \leq n_r$. The receiver selects n_s out of the

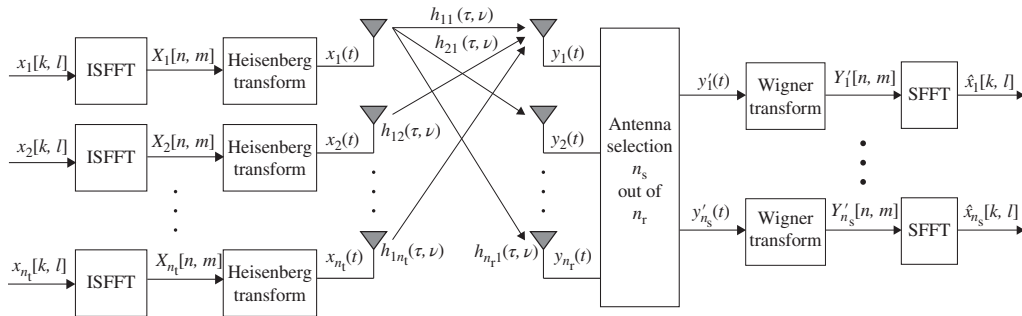


Figure 4.12 MIMO-OTFS with receive antenna selection.

n_r available receive antennas for reception. Demodulation, detection, and decoding operations are carried out on the signals received over these selected antennas. Channel knowledge is often used in selecting the n_s receive antennas. In the analysis, perfect channel knowledge is assumed at the receiver.

Following the MIMO-OTFS system model in (4.50), the input-output relation of the MIMO-OTFS system with n_r receive antennas and n_t transmit antennas without receive antenna selection can be written as

$$\underbrace{\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_{n_r} \end{bmatrix}}_{\triangleq \bar{\mathbf{y}}} = \underbrace{\begin{bmatrix} \mathbf{H}_{11} & \cdots & \mathbf{H}_{1n_t} \\ \vdots & \ddots & \vdots \\ \mathbf{H}_{n_r 1} & \cdots & \mathbf{H}_{n_r n_t} \end{bmatrix}}_{\triangleq \bar{\mathbf{H}}} \underbrace{\begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_{n_t} \end{bmatrix}}_{\triangleq \bar{\mathbf{x}}} + \underbrace{\begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_{n_r} \end{bmatrix}}_{\triangleq \bar{\mathbf{v}}}, \quad (4.75)$$

where $\bar{\mathbf{y}} \in \mathbb{C}^{n_r MN \times 1}$ is the received signal vector, $\bar{\mathbf{H}} \in \mathbb{C}^{n_r MN \times n_t MN}$ is the overall equivalent channel matrix with $\mathbf{H}_{ij}$ being the $MN \times MN$ equivalent channel matrix between the j th transmit and i th receive antennas, $\bar{\mathbf{x}} \in \mathbb{C}^{n_t MN \times 1}$ is the OTFS transmit vector, and $\bar{\mathbf{v}} \in \mathbb{C}^{n_r MN \times 1}$ is the noise vector.

The receive antenna selection is made based on the Frobenius norms of the channel matrices. Let the sum of the Frobenius norms of the channel matrices between the i th receive antenna and all transmit antennas be given by [76]

$$\sum_{j=1}^{n_t} \|\mathbf{H}_{ij}\|^2, \quad i = 1, 2, \dots, n_r. \quad (4.76)$$

The receiver selects those n_s receive antennas for which the corresponding sum of Frobenius norms are the largest among those of all n_r antennas. Observing that each $\mathbf{H}_{ij}$ contains only PMN non-zero elements with P unique elements and using the definition of Frobenius norm, the selection metric in (4.76) can be written as

$$\sum_{k=1}^P \sum_{j=1}^{n_t} |h_{ij}^{(k)}|^2, \quad i = 1, 2, \dots, n_r, \quad (4.77)$$

where $h_{ij}^{(k)}$ are the unique non-zero entries of $\mathbf{H}_{ij}$. Therefore, the input-output relation of the MIMO-OTFS system with antenna selection can be written as

$$\underbrace{\begin{bmatrix} \mathbf{y}'_1 \\ \vdots \\ \mathbf{y}'_{n_s} \end{bmatrix}}_{\triangleq \bar{\mathbf{y}}'} = \underbrace{\begin{bmatrix} \mathbf{H}'_{11} & \cdots & \mathbf{H}'_{1n_t} \\ \vdots & \ddots & \vdots \\ \mathbf{H}'_{n_s 1} & \cdots & \mathbf{H}'_{n_s n_t} \end{bmatrix}}_{\triangleq \bar{\mathbf{H}}'} \underbrace{\begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_{n_t} \end{bmatrix}}_{\triangleq \bar{\mathbf{x}}'} + \underbrace{\begin{bmatrix} \mathbf{v}'_1 \\ \vdots \\ \mathbf{v}'_{n_s} \end{bmatrix}}_{\triangleq \bar{\mathbf{v}}'} \quad (4.78)$$

or equivalently

$$\bar{\mathbf{y}}' = \bar{\mathbf{H}}' \bar{\mathbf{x}} + \bar{\mathbf{v}}', \quad (4.79)$$

where $\bar{\mathbf{y}}' \in \mathbb{C}^{n_s MN \times 1}$, $\bar{\mathbf{H}}' \in \mathbb{C}^{n_s MN \times n_t MN}$ is the equivalent channel matrix with antenna selection, $\bar{\mathbf{x}} \in \mathbb{C}^{n_t MN \times 1}$ is the OTFS transmit vector, and $\bar{\mathbf{v}}' \in \mathbb{C}^{n_s MN \times 1}$ is the noise vector.

An alternate form of MIMO-OTFS with RAS: The input-output relation in (4.79) can be written in an alternate form similar to that in (4.17), by observing that each $\mathbf{H}'_{ij}$ in (4.78) contains only P unique non-zero elements, and, hence, $\bar{\mathbf{H}}'$ in (4.79) contains only $Pn_s n_t$ unique non-zero elements with each row having only Pn_t unique non-zero elements and each column having only $n_s P$ unique

non-zero elements. Therefore, (4.78) can be written as

$$\underbrace{\begin{bmatrix} \mathbf{y}_1^T \\ \vdots \\ \mathbf{y}_{n_s}^T \end{bmatrix}}_{\triangleq \tilde{\mathbf{Y}}} = \underbrace{\begin{bmatrix} \mathbf{h}'_{11} & \cdots & \mathbf{h}'_{1n_t} \\ \vdots & \ddots & \vdots \\ \mathbf{h}'_{n_s 1} & \cdots & \mathbf{h}'_{n_s n_t} \end{bmatrix}}_{\triangleq \tilde{\mathbf{H}}} \underbrace{\begin{bmatrix} \mathbf{X}_1 \\ \vdots \\ \mathbf{X}_{n_t} \end{bmatrix}}_{\triangleq \tilde{\mathbf{X}}} + \underbrace{\begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_{n_s}^T \end{bmatrix}}_{\triangleq \tilde{\mathbf{V}}} \quad (4.80)$$

or equivalently

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{H}}\tilde{\mathbf{X}} + \tilde{\mathbf{V}}, \quad (4.81)$$

where $\tilde{\mathbf{Y}} \in \mathbb{C}^{n_s \times MN}$ with its i th row corresponding to the received signal in the i th selected receive antenna, $\tilde{\mathbf{H}} \in \mathbb{C}^{n_s \times n_t P}$ is the channel matrix with $\mathbf{h}'_{ij} \in \mathbb{C}^{1 \times P}$ containing P unique non-zero entries of $\mathbf{H}'_{ij}$, $\tilde{\mathbf{X}}$ is the $n_t P \times MN$ symbol matrix, and $\tilde{\mathbf{V}} \in \mathbb{C}^{n_s \times MN}$ is the noise matrix.

4.4.2 STC-OTFS with RAS

Figure 4.13 shows the block diagram of the Alamouti code-based STC-OTFS scheme (introduced in Section 4.3.1) with receive antenna selection. Following the development of the STC-OTFS system model without receive antenna selection in 4.3.1, the input-output relation for the Alamouti code-based STC-OTFS scheme with the selection of n_s out of n_t antennas at the receiver can be written in the form

$$\underbrace{\begin{bmatrix} \mathbf{y}'_{11} \\ \vdots \\ \mathbf{y}'_{n_s 1} \\ (\hat{\mathbf{y}}'_{12})^* \\ \vdots \\ (\hat{\mathbf{y}}'_{n_s 2})^* \end{bmatrix}}_{\triangleq \bar{\mathbf{y}}'} = \underbrace{\begin{bmatrix} \mathbf{H}'_{11} & \mathbf{H}'_{12} \\ \vdots & \vdots \\ \mathbf{H}'_{n_s 1} & \mathbf{H}'_{n_s 2} \\ \mathbf{H}'_{12}^H & -\mathbf{H}'_{11}^H \\ \vdots & \vdots \\ \mathbf{H}'_{n_s 2}^H & -\mathbf{H}'_{n_s 1}^H \end{bmatrix}}_{\triangleq \bar{\mathbf{H}}'} \underbrace{\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}}_{\triangleq \bar{\mathbf{x}}} + \underbrace{\begin{bmatrix} \mathbf{v}'_{11} \\ \vdots \\ \mathbf{v}'_{n_s 1} \\ (\hat{\mathbf{v}}'_{12})^* \\ \vdots \\ (\hat{\mathbf{v}}'_{n_s 2})^* \end{bmatrix}}_{\triangleq \bar{\mathbf{v}}'}, \quad (4.82)$$

where $\mathbf{y}'_{ij} \in \mathbb{C}^{MN \times 1}$ is the received signal vector at the i th antenna in the j th time slot with $\hat{\mathbf{y}}'_{ij} = \mathbf{P}\mathbf{y}'_{ij}$, where $\mathbf{P}$ is the $MN \times MN$ permutation matrix given by (4.64), $\mathbf{H}'_{ij} \in \mathbb{C}^{MN \times MN}$ is the equivalent channel matrix between the i th selected receive antenna and j th transmit antenna, and $\mathbf{x}_i \in \mathbb{C}^{MN \times 1}$ is the transmitted OTFS vector. The compact form of (4.82) is given by

$$\bar{\mathbf{y}}' = \bar{\mathbf{H}}'\bar{\mathbf{x}} + \bar{\mathbf{v}}', \quad (4.83)$$

where $\bar{\mathbf{y}}', \bar{\mathbf{v}}' \in \mathbb{C}^{2n_s MN \times 1}$, $\bar{\mathbf{H}}' \in \mathbb{C}^{2n_s MN \times 2MN}$, and $\bar{\mathbf{x}} \in \mathbb{C}^{2MN \times 1}$.

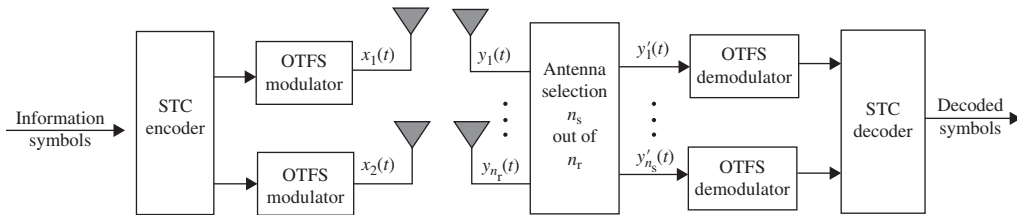


Figure 4.13 STC-OTFS system based on receive antenna selection.

An alternate form of Alamouti STC-OTFS with RAS: The input–output relation in (4.83) can be written in an alternate form, based on (4.17) and (4.80), as

$$\underbrace{\begin{bmatrix} \mathbf{y}'_{11}^T & \mathbf{y}'_{12}^T \\ \vdots & \vdots \\ \mathbf{y}'_{n_s 1}^T & \mathbf{y}'_{n_s 2}^T \end{bmatrix}}_{\triangleq \tilde{\mathbf{Y}}} = \underbrace{\begin{bmatrix} \mathbf{h}'_{11} & \mathbf{h}'_{12} \\ \vdots & \vdots \\ \mathbf{h}'_{n_s 1} & \mathbf{h}'_{n_s 2} \end{bmatrix}}_{\triangleq \tilde{\mathbf{H}}} \underbrace{\begin{bmatrix} \mathbf{X}_1 & -(\hat{\mathbf{X}}_2)^* \\ \mathbf{X}_2 & (\hat{\mathbf{X}}_1)^* \end{bmatrix}}_{\triangleq \tilde{\mathbf{X}}} + \underbrace{\begin{bmatrix} \mathbf{v}'_{11}^T & \mathbf{v}'_{12}^T \\ \vdots & \vdots \\ \mathbf{v}'_{n_s 1}^T & \mathbf{v}'_{n_s 2}^T \end{bmatrix}}_{\triangleq \tilde{\mathbf{V}}}, \quad (4.84)$$

which can be written in the following compact form²:

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{H}}\tilde{\mathbf{X}} + \tilde{\mathbf{V}}, \quad (4.85)$$

where $\tilde{\mathbf{Y}}, \tilde{\mathbf{V}} \in \mathbb{C}^{n_s \times 2MN}$, $\tilde{\mathbf{H}} \in \mathbb{C}^{n_s \times 2P}$, and $\tilde{\mathbf{X}} \in \mathbb{C}^{2P \times 2MN}$. It is observed that $\mathbf{X}_i \neq \hat{\mathbf{X}}_i$ since transmitted OTFS vectors in the second frame are conjugated and permuted vectors of those transmitted in the first frame. In (4.63), $\tilde{\mathbf{X}}$ is defined to be a $2MN \times 2MN$ symbol matrix, but for diversity analysis, $\tilde{\mathbf{X}} \in \mathbb{C}^{2P \times 2MN}$ in (4.84) is convenient.

4.4.3 Rank of Multi-antenna OTFS Systems

To proceed with the diversity analysis, the rank of the considered multi-antenna OTFS systems without and with phase rotation is identified below.

Rank of MIMO-OTFS, SIMO-OTFS: Consider MIMO-OTFS ($n_t \geq 2$) without phase rotation. Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be the two distinct symbol matrices defined in (4.81). The minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is $1 < \min(n_t P, MN)$ [41]. Therefore, MIMO-OTFS without phase rotation is rank deficient. Next, consider MIMO-OTFS with phase rotation. Let $\tilde{\mathbf{x}}'_i = \Phi \tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}'_j = \Phi \tilde{\mathbf{x}}_j$ be the two distinct phase-rotated OTFS transmit vectors in (4.79). Let $\tilde{\mathbf{X}}'_i$ and $\tilde{\mathbf{X}}'_j$ be the corresponding phase-rotated symbol matrices in (4.81). The minimum rank of $(\tilde{\mathbf{X}}'_i - \tilde{\mathbf{X}}'_j)$ is $P < \min(n_t P, MN)$ [41]. Therefore, the MIMO-OTFS system with phase rotation is also rank deficient. SIMO-OTFS can be viewed as a special case of MIMO-OTFS with $n_t = 1$. Therefore, for SIMO-OTFS without phase rotation, the minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is $1 < \min(P, MN)$. Therefore, the SIMO-OTFS system without phase rotation is rank deficient for $P > 1$. For $P = 1$, the dimension of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is $1 \times MN$ and the minimum rank is 1, and so it is full rank. For SIMO-OTFS with phase rotation, the minimum rank of $(\tilde{\mathbf{X}}'_i - \tilde{\mathbf{X}}'_j)$ is $P = \min(P, MN)$. Since $P \leq MN$ and minimum rank is P , it is full rank.

Rank of STC-OTFS: Consider Alamouti STC-OTFS without phase rotation. Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be the two distinct symbol matrices defined in (4.85). The minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is $2 < \min(2P, 2MN)$ [75]. Therefore, for $P > 1$, Alamouti STC-OTFS is rank deficient, and for $P = 1$, it is full rank with rank 2. For Alamouti STC-OTFS with phase rotation, the minimum rank of $(\tilde{\mathbf{X}}'_i - \tilde{\mathbf{X}}'_j)$ is $2P \leq \min(2P, 2MN)$ [75]. Therefore, Alamouti STC-OTFS with phase rotation is full rank with rank $2P$.

4.4.4 Analysis of Multiantenna OTFS with RAS

The diversity performance of multi-antenna OTFS with RAS is analyzed by deriving explicit upper bounds on PEP for full-rank and rank-deficient cases as follows.

² In order to adopt a unified input–output system model in the analysis, the same notation is used in (4.81) and (4.85), where for MIMO-OTFS without space–time coding in (4.81), $\tilde{\mathbf{Y}}, \tilde{\mathbf{V}} \in \mathbb{C}^{n_s \times MN}$, $\tilde{\mathbf{H}} \in \mathbb{C}^{n_s \times n_t P}$, and $\tilde{\mathbf{X}} \in \mathbb{C}^{n_t P \times MN}$, and for space–time coded OTFS in (4.85), $\tilde{\mathbf{Y}}, \tilde{\mathbf{V}} \in \mathbb{C}^{n_s \times 2MN}$, $\tilde{\mathbf{H}} \in \mathbb{C}^{n_s \times 2P}$, and $\tilde{\mathbf{X}} \in \mathbb{C}^{2P \times 2MN}$.

Full rank multi-antenna OTFS with RAS: Consider the case of full rank multi-antenna OTFS with RAS. Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be the two distinct symbol matrices. Assuming perfect DD channel knowledge and ML detection at the receiver, the conditional PEP between the symbol matrices $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$, assuming $\tilde{\mathbf{X}}_i$ to be the transmitted symbol matrix, is given by

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j | \tilde{\mathbf{H}}, \tilde{\mathbf{X}}_i) = Q \left(\sqrt{\frac{\|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2}{2N_0}} \right). \quad (4.86)$$

For convenience, the entries of $\tilde{\mathbf{X}}$ are normalized so that the average energy per symbol time is one and the SNR, denoted by γ , is given by $\gamma = 1/N_0$. Therefore, (4.86) can be written as

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j | \tilde{\mathbf{H}}, \tilde{\mathbf{X}}_i) = Q \left(\sqrt{\frac{\gamma \|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2}{2}} \right). \quad (4.87)$$

Averaging over the distribution of $\tilde{\mathbf{H}}$ and upper bounding using the Chernoff bound, an upper bound on the unconditional PEP can be written as

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \mathbb{E}_{\tilde{\mathbf{H}}} \left[\exp \left(-\frac{\gamma \|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2}{4} \right) \right]. \quad (4.88)$$

The distribution of $\tilde{\mathbf{H}}$ is given by [77, 78]

$$f_{\tilde{\mathbf{H}}}(\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s}) = \binom{n_r}{n_s} \left(\sum_{l=1}^{n_s} \left[1 - e^{-P\|\mathbf{h}'_l\|^2} \sum_{k=0}^{n_t P-1} \frac{P^k \|\mathbf{h}'_l\|^{2k}}{k!} \right]^{n_t - n_s} I_{\tilde{\mathcal{H}}_l}(\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s}) \right) \cdot \frac{P^{n_t n_s}}{\pi^{n_t n_s}} e^{-P(\|\mathbf{h}'_1\|^2 + \dots + \|\mathbf{h}'_{n_s}\|^2)}, \quad (4.89)$$

where $\mathbf{h}'_i$ is the i th row of $\tilde{\mathbf{H}}$, $I_{\tilde{\mathcal{H}}_l}(\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s})$ is the indicator function given by

$$I_{\tilde{\mathcal{H}}_l}(\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s}) = \begin{cases} 1 & \text{if } (\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s}) \in \tilde{\mathcal{H}}_l, \\ 0 & \text{else,} \end{cases} \quad (4.90)$$

and the region $\tilde{\mathcal{H}}_l$ is defined as $\tilde{\mathcal{H}}_l = \{\mathbf{h}'_1, \dots, \mathbf{h}'_{n_s} : \|\mathbf{h}'_l\| < \|\mathbf{h}'_k\|, k = 1, \dots, l-1, l+1, \dots, n_s\}$. The PEP bound can be written as

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \sum_{l=1}^{n_s} \int_{\tilde{\mathcal{H}}_l} e^{-\frac{\gamma}{4} \|\tilde{\mathbf{H}}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2} \binom{n_r}{n_s} \left(1 - e^{-P\|\mathbf{h}'_l\|^2} \sum_{k=0}^{n_t P-1} \frac{P^k \|\mathbf{h}'_l\|^{2k}}{k!} \right)^{n_t - n_s} \cdot \frac{P^{n_s n_t P}}{\pi^{n_s n_t P}} e^{-P(\|\mathbf{h}'_1\|^2 + \dots + \|\mathbf{h}'_{n_s}\|^2)} d\mathbf{h}'_1 \dots d\mathbf{h}'_{n_s}. \quad (4.91)$$

Letting $\sqrt{P}\mathbf{h}'_l = \mathbf{s}_l$, $l = 1, \dots, n_s$, $\mathbf{S}$ to be an $n_s \times n_t P$ matrix whose l th row is $\mathbf{s}_l$, and region $\tilde{\mathcal{H}}_l = \{\mathbf{s}_1, \dots, \mathbf{s}_{n_s} : \|\mathbf{s}_l\| < \|\mathbf{s}_k\|, k = 1, \dots, l-1, l+1, \dots, n_s\}$, (4.91) can be written as

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \sum_{l=1}^{n_s} \int_{\tilde{\mathcal{H}}_l} e^{-\frac{\gamma}{4P} \|\mathbf{S}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2} \binom{n_r}{n_s} \left(1 - e^{-\|\mathbf{s}_l\|^2} \sum_{k=0}^{n_t P-1} \frac{\|\mathbf{s}_l\|^{2k}}{k!} \right)^{n_t - n_s} \cdot \frac{(\sqrt{P})^{n_s n_t P}}{\pi^{n_s n_t P}} e^{-(\|\mathbf{s}_1\|^2 + \dots + \|\mathbf{s}_{n_s}\|^2)} d\mathbf{s}_1 \dots d\mathbf{s}_{n_s}. \quad (4.92)$$

The term $\|\mathbf{S}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2$ in (4.92) can be simplified as

$$\begin{aligned}\|\mathbf{S}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)\|^2 &= \text{Tr}\{\mathbf{S}(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)^H \mathbf{S}^H\} \\ &= \text{Tr}\{\mathbf{S}\mathbf{U}\mathbf{\Lambda}(\mathbf{S}\mathbf{U})^H\} \\ &= \sum_{k=1}^{n_t P} \lambda_k \|\mathbf{c}_k\|^2,\end{aligned}\quad (4.93)$$

where (4.93) uses the eigenvalue decomposition of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)^H$, $\mathbf{U}$ is the unitary matrix whose columns are the eigenvectors of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)^H$, $\mathbf{\Lambda}$ is the diagonal matrix containing its eigenvalues, and $\mathbf{c}_k$ is the k th column of $\mathbf{S}\mathbf{U}$. Let $\mathbf{c}'_l$ be the l th row of $\mathbf{S}\mathbf{U}$ so that $\tilde{\mathcal{H}}_l = \{\mathbf{c}'_1, \dots, \mathbf{c}'_{n_s} : \|\mathbf{c}'_l\| < \|\mathbf{c}'_k\|, k = 1, \dots, l-1, l+1, \dots, n_s\}$. Defining $K \triangleq n_t P$ and $\rho \triangleq (\sqrt{P})^{n_s n_t P}$, and changing variables in (4.92) by substituting $c_{ij} = s_{ij}$ for $i = 1, \dots, n_s$ and $j = 1, \dots, n_t P$, (4.92) can be written as

$$\begin{aligned}P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) &\leq \rho \binom{n_r}{n_s} \sum_{l=1}^{n_s} \int_{\tilde{\mathcal{H}}_l} e^{\frac{-\gamma}{4P}(\lambda_1(|c_{11}|^2 + \dots + |c_{n_s 1}|^2) + \dots + \lambda_K(|c_{1K}|^2 + \dots + |c_{n_s K}|^2))} \\ &\quad \cdot \left(1 - e^{-(|c_{1l}|^2 + \dots + |c_{lK}|^2)} \sum_{k=0}^{K-1} \frac{(|c_{1l}|^2 + \dots + |c_{lK}|^2)^k}{k!}\right)^{n_r - n_s} \\ &\quad \cdot \frac{1}{\pi^{n_s K}} e^{-\sum_{i=1}^{n_s} \sum_{j=1}^K |c_{ij}|^2} dc_{11} \dots dc_{n_s K}.\end{aligned}\quad (4.94)$$

Evaluating the integral in (4.94) over the region is difficult. However, because of the symmetry of pdf, it is possible to evaluate over the whole space which results in an upper bound. Because of the symmetry of the pdf, the integral over $\tilde{\mathcal{H}}_l$ for each l is the same. The l th term in (4.94) can be rewritten using standard integration as

$$\begin{aligned}\mathcal{I}_l &= \rho \binom{n_r}{n_s} \int_0^\infty \dots \int_0^\infty e^{\frac{-\gamma}{4P}(\lambda_1(|c_{11}|^2 + \dots + |c_{n_s 1}|^2) + \dots + \lambda_K(|c_{1K}|^2 + \dots + |c_{n_s K}|^2))} \\ &\quad \cdot \left(1 - e^{-(|c_{1l}|^2 + \dots + |c_{lK}|^2)} \sum_{k=0}^{K-1} \frac{(|c_{1l}|^2 + \dots + |c_{lK}|^2)^k}{k!}\right)^{n_r - n_s} \\ &\quad \cdot \frac{1}{\pi^{n_s K}} e^{-\sum_{i=1}^{n_s} \sum_{j=1}^K |c_{ij}|^2} dc_{11} \dots dc_{n_s K}.\end{aligned}\quad (4.95)$$

Changing the variables $c_{ij} = \sigma_{ij} e^{j\theta_{ij}}$, $i = 1, \dots, n_s$, $j = 1, \dots, K$ (with differential element $dc_{ij} = \sigma_{ij} d\sigma_{ij} d\theta_{ij}$), after evaluating integral with respect to $d\theta_{ij}$ over $[0, 2\pi]$, $\mathcal{I}_l$ can be written as

$$\begin{aligned}\mathcal{I}_l &= 2^{n_s K} \rho \binom{n_r}{n_s} \int_0^\infty \dots \int_0^\infty e^{\frac{-\gamma}{4P}(\lambda_1(\sigma_{11}^2 + \dots + \sigma_{n_s 1}^2) + \dots + \lambda_K(\sigma_{1K}^2 + \dots + \sigma_{n_s K}^2))} \\ &\quad \cdot \left(1 - e^{-(\sigma_{1l}^2 + \dots + \sigma_{lK}^2)} \sum_{k=0}^{K-1} \frac{(\sigma_{1l}^2 + \dots + \sigma_{lK}^2)^k}{k!}\right)^{n_r - n_s} \\ &\quad \cdot e^{-\sum_{i=1}^{n_s} \sum_{j=1}^K \sigma_{ij}^2} \sigma_{11} \dots \sigma_{n_s K} d\sigma_{11} \dots d\sigma_{n_s K}.\end{aligned}\quad (4.96)$$

Substituting $\sigma_{ij}^2 = v_{ij}$, $i = 1, \dots, n_s$, $j = 1, \dots, K$,

$$\mathcal{I}_l = \rho \binom{n_r}{n_s} \int_0^\infty \dots \int_0^\infty e^{\frac{-\gamma}{4P}(\lambda_1(v_{11} + \dots + v_{n_s 1}) + \dots + \lambda_K(v_{1K} + \dots + v_{n_s K}))}$$

$$\begin{aligned}
& \cdot \left(1 - e^{-(v_{l1} + \dots + v_{lK})} \sum_{k=0}^{K-1} \frac{(v_{l1} + \dots + v_{lK})^k}{k!} \right)^{n_r - n_s} \\
& \cdot e^{-\sum_{i=1}^{n_s} \sum_{j=1}^K v_{ij}} dv_{l1} \dots dv_{n_s K}.
\end{aligned} \tag{4.97}$$

Now, (4.97) can be written as

$$\begin{aligned}
\mathcal{I}_l &= \rho \binom{n_r}{n_s} \int_0^\infty \dots \int_0^\infty e^{-\sum_{i=1}^K \sum_{d=1, d \neq l}^{n_s} \lambda_i v_{di}} e^{-\left(\sum_{i=1}^K \sum_{d=1, d \neq l}^{n_s} v_{di} \right)} \prod_{i=1}^K \prod_{d=1, d \neq l}^{n_s} dv_{di} \\
& \cdot \int_0^\infty \dots \int_0^\infty e^{-\sum_{i=1}^K \lambda_i v_{li}} \left(1 - e^{-(v_{l1} + \dots + v_{lK})} \sum_{k=0}^{K-1} \frac{(v_{l1} + \dots + v_{lK})^k}{k!} \right)^{n_r - n_s} \\
& \cdot e^{-(v_{l1} + \dots + v_{lK})} dv_{l1} \dots dv_{lK}.
\end{aligned} \tag{4.98}$$

Let $\mathcal{I}_l^{(1)}$ denote the first set of integrals and $\mathcal{I}_l^{(2)}$ denote the second set of integrals in the above expression. Evaluating $\mathcal{I}_l^{(1)}$ using $\int_0^\infty e^{-\alpha x} dx = \frac{1}{\alpha}$ gives

$$\mathcal{I}_l^{(1)} = \left(\frac{1}{\prod_{i=1}^K \left(1 + \frac{\gamma \lambda_i}{4P} \right)} \right)^{n_s - 1}. \tag{4.99}$$

Let $g(u) = 1 - e^{-u} \sum_{m=0}^{K-1} \frac{u^m}{m!}$ be the incomplete Gamma function satisfying $g(u) \leq \frac{u^K}{K!}$ for $u > 0$. Upper bounding $\mathcal{I}_l^{(2)}$ by $\frac{u^K}{K!}$ with $u = v_{l1} + \dots + v_{lK}$ in $\mathcal{I}_l^{(2)}$ gives

$$\begin{aligned}
\mathcal{I}_l^{(2)} &\leq \frac{1}{(K!)^{n_r - n_s}} \int_0^\infty \dots \int_0^\infty e^{-\sum_{i=1}^K \lambda_i v_{li}} (v_{l1} + \dots + v_{lK})^{K(n_r - n_s)} \\
& \cdot e^{-(v_{l1} + \dots + v_{lK})} dv_{l1} \dots dv_{lK}.
\end{aligned} \tag{4.100}$$

Observe that

$$(v_{l1} + \dots + v_{lK})^{K(n_r - n_s)} = \sum_{i_1=1}^K \dots \sum_{i_{K(n_r - n_s)}=1}^K v_{li_1} \dots v_{li_{K(n_r - n_s)}}, \tag{4.101}$$

where index i_k in v_{li_k} takes values from the set $\zeta = \{1, \dots, K\}$ with $k \in \{1, \dots, K(n_r - n_s)\}$. Let the index j appear m_j times among the subscripts of the term $v_{li_1} \dots v_{li_{K(n_r - n_s)}}$ in (4.101). Then,

$$v_{li_1} \dots v_{li_{K(n_r - n_s)}} = \prod_{j=1}^K (v_{lj})^{m_j} \tag{4.102}$$

such that $\sum_{j=1}^K m_j = K(n_r - n_s)$. Using (4.101) and (4.102) in (4.100) and changing the order of summation and integration give

$$\begin{aligned}
\mathcal{I}_l^{(2)} &\leq \frac{1}{(K!)^{n_r - n_s}} \sum_{i_1=1}^K \dots \sum_{i_{K(n_r - n_s)}=1}^K \\
& \cdot \left(\int_0^\infty \dots \int_0^\infty e^{-\sum_{i=1}^K \left(\frac{\gamma \lambda_i}{4P} + 1 \right) v_{li}} \prod_{i=1}^K (v_{li})^{m_i} dv_{l1} \dots dv_{lK} \right).
\end{aligned} \tag{4.103}$$

Using $\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$, (4.103) can be written as

$$\mathcal{I}_l^{(2)} \leq \frac{1}{(K!)^{n_r-n_s}} \left(\sum_{i_1=1}^K \cdots \sum_{i_{K(n_r-n_s)}=1}^K \frac{m_1! \cdots m_K!}{\left(1 + \frac{\gamma \lambda_1}{4P}\right)^{m_1+1} \cdots \left(1 + \frac{\gamma \lambda_K}{4P}\right)^{m_K+1}} \right). \quad (4.104)$$

Using (4.104) and (4.99) in (4.98), $\mathcal{I}_l$ can be written as

$$\begin{aligned} \mathcal{I}_l \leq & \frac{\rho}{(K!)^{n_r-n_s}} \binom{n_r}{n_s} \left(\frac{1}{\prod_{i=1}^K \left(1 + \frac{\gamma \lambda_i}{4P}\right)} \right)^{n_s-1} \\ & \sum_{i_1=1}^K \cdots \sum_{i_{K(n_r-n_s)}=1}^K \frac{m_1! \cdots m_K!}{\left(1 + \frac{\gamma \lambda_1}{4P}\right)^{m_1+1} \cdots \left(1 + \frac{\gamma \lambda_K}{4P}\right)^{m_K+1}}. \end{aligned} \quad (4.105)$$

The above bound is independent of l . Therefore, substituting (4.105) in (4.94) gives

$$\begin{aligned} P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq & \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r-n_s}} \left(\frac{1}{\prod_{i=1}^K \left(1 + \frac{\gamma \lambda_i}{4P}\right)} \right)^{n_s-1} \\ & \cdot \sum_{i_1=1}^K \cdots \sum_{i_{K(n_r-n_s)}=1}^K \frac{m_1! \cdots m_K!}{\left(1 + \frac{\gamma \lambda_1}{4P}\right)^{m_1+1} \cdots \left(1 + \frac{\gamma \lambda_K}{4P}\right)^{m_K+1}}. \end{aligned} \quad (4.106)$$

In the high SNR regime, with some algebraic manipulations, the above equation can be written as

$$\begin{aligned} P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq & \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r-n_s}} \frac{1}{\left(\prod_{i=1}^K \lambda_i\right)^{n_s}} \\ & \cdot \left(\sum_{i_1=1}^K \cdots \sum_{i_{K(n_r-n_s)}=1}^K \frac{m_1! \cdots m_K!}{\lambda_1^{m_1} \cdots \lambda_K^{m_K}} \right) \left(\frac{\gamma}{4P} \right)^{-\sum_{i=1}^K m_i+1} \\ & \cdot \left(\frac{\gamma}{4P} \right)^{-K(n_s-1)}. \end{aligned} \quad (4.107)$$

Finally, substituting $\sum_{i=1}^K m_i = K(n_r - n_s)$ in (4.107) gives

$$\begin{aligned} P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq & \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r-n_s}} \frac{1}{\left(\prod_{i=1}^K \lambda_i\right)^{n_s}} \\ & \cdot \left(\sum_{i_1=1}^K \cdots \sum_{i_{K(n_r-n_s)}=1}^K \frac{m_1! \cdots m_K!}{\lambda_1^{m_1} \cdots \lambda_K^{m_K}} \right) \left(\frac{\gamma}{4P} \right)^{-Kn_r}. \end{aligned} \quad (4.108)$$

Note that the inequality (4.108) implies that a diversity order of $n_r K (= n_r n_t P)$ is achieved in a full rank multi-antenna OTFS system when n_s antennas are selected at the receiver. The above diversity result can be specialized for the considered multi-antenna OTFS systems which are full rank as follows.

Table 4.1 Summary of diversity order results for multi-antenna OTFS systems with RAS.

OTFS system	# ant. selected	# DD paths	Diversity order	
			Without PR	With PR
SIMO-OTFS, $n_r \geq 1$	$n_s \geq 1$	$P = 1$	n_r	n_r
		$P > 1$	n_s	$n_r P$
MIMO-OTFS, $n_r \geq n_t$	$n_s \geq n_t$	$P \geq 1$	n_s	$n_s P$
STC-OTFS (Alamouti) $n_t = 2, n_r \geq 1$	$n_s \geq 1$	$P = 1$	$2n_r$	$2n_r$
		$P > 1$	$2n_s$	$2n_r P$

- SIMO-OTFS systems without phase rotation for $P = 1$ and with phase rotation for $P > 1$ are full rank. Therefore, in these cases, full spatial and DD diversity of $n_r P$ is achieved when n_s receive antennas are selected.
- STC-OTFS systems with the Alamouti code without phase rotation for $P = 1$ and with phase rotation for $P > 1$ are also full rank. Therefore, in these cases, the full spatial and DD diversity of $2n_r P$ is achieved when n_s received antennas are selected.

The above diversity results have been summarized in Table 4.1.

4.4.5 Rank Deficient Multi-antenna OTFS Systems with RAS

Next, consider the case of rank-deficient multi-antenna OTFS systems with RAS. Let $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$ be two distinct symbol matrices. Let $r < K$ be the minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$. For rank-deficient case, the diversity analysis follows from (4.94) to (4.105), except now $\lambda_1, \dots, \lambda_r > 0, \lambda_{r+1} = \dots = \lambda_K = 0$. Therefore, in the high SNR regime, the average PEP between $\tilde{\mathbf{X}}_i$ and $\tilde{\mathbf{X}}_j$, assuming $\tilde{\mathbf{X}}_i$ to be the transmitted symbol matrix, is given by

$$\begin{aligned}
 P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) &\leq \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r - n_s}} \frac{1}{\left(\prod_{i=1}^r \lambda_i\right)^{n_s}} \\
 &\cdot \left[\sum_{i_1=1}^K \cdots \sum_{i_{K(n_r - n_s)}=1}^K \frac{m_1! \cdots m_K!}{\lambda_1^{m_1} \cdots \lambda_r^{m_r}} \left(\frac{\gamma}{4P}\right)^{-\sum_{i=1}^r m_i} \right] \\
 &\cdot \left(\frac{\gamma}{4P}\right)^{-rn_s}.
 \end{aligned} \tag{4.109}$$

Since $\sum_{i=1}^K m_i = K(n_r - n_s)$, it follows that $0 \leq \sum_{i=1}^r m_i \leq K(n_r - n_s)$. It is observed that the term in the square brackets is a function of $\frac{\gamma}{4P}$ and there exist terms $i_1 \cdots i_{K(n_r - n_s)}$ such that $\sum_{i=1}^r m_i = 0$. Regrouping the terms in (4.109), (4.109) can be written as

$$\begin{aligned}
 P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) &\leq \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r - n_s}} \frac{1}{\left(\prod_{i=1}^r \lambda_i\right)^{n_s}} \\
 &\left(\sum_{j=0}^{K(n_r - n_s)} \psi_j \left(\frac{\gamma}{4P}\right)^{-j} \right) \left(\frac{\gamma}{4P}\right)^{-rn_s},
 \end{aligned} \tag{4.110}$$

where $j = \sum_{i=1}^r m_i$ and ψ_j is the sum of the terms multiplying $\left(\frac{\gamma}{4P}\right)^{-\sum_{i=1}^r m_i}$ with the same exponents.

For sufficiently high SNRs, the term $\left(\frac{\gamma}{4P}\right)^{-j}$ vanishes for $\sum_{i=1}^r m_i > 0$. Thus,

$$P(\tilde{\mathbf{X}}_i \rightarrow \tilde{\mathbf{X}}_j) \leq \frac{\rho \cdot n_r!}{(n_r - n_s)!(n_s - 1)!(K!)^{n_r - n_s}} \frac{1}{\left(\prod_{i=1}^r \lambda_i\right)^{n_s}} \psi_0 \left(\frac{\gamma}{4P}\right)^{-n_s}. \quad (4.111)$$

The above expression shows that a diversity order of $n_s r$ is achieved for a rank-deficient multi-antenna OTFS system when n_s antennas are selected at the receiver. The above diversity result is specialized for the considered multi-antenna OTFS systems which are rank deficient as follows.

- The minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is 1 for SIMO-OTFS ($P > 1$) and MIMO-OTFS ($P \geq 1$) systems without phase rotation. Therefore, these systems achieve a diversity of n_s when n_s antennas are selected at the receiver.
- For MIMO-OTFS ($P \geq 1$) systems with phase rotation, the minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is P , which is rank deficient. Therefore, these systems achieve a diversity of $n_s P$ when n_s receive antennas are selected.
- For STC-OTFS ($P > 1$) systems with the Alamouti code without phase rotation, the minimum rank of $(\tilde{\mathbf{X}}_i - \tilde{\mathbf{X}}_j)$ is 2, which is rank deficient. Therefore, these systems achieve a diversity of $2n_s$ when n_s receive antennas are selected.

The above diversity results are summarized in Table 4.1.

Figure 4.14 shows the BER performance of SIMO-OTFS with RAS for without and with phase rotation, $P = 2$, $M = N = 4$, $n_s = 1$, $n_r = 1, 2$, BPSK, ML detection, 4 GHz carrier frequency, 3.75 kHz subcarrier spacing, and a DD profile of $\{(\tau_1, \nu_1), (\tau_2, \nu_2)\} = \{(0, 0), (\frac{1}{M\Delta f}, \frac{1}{NT})\}$. For $P > 1$, SIMO-OTFS with RAS without phase rotation is rank deficient and the analytical diversity order is n_s . With phase rotation, the system is full ranked and it has a diversity order of $n_r P$. For the

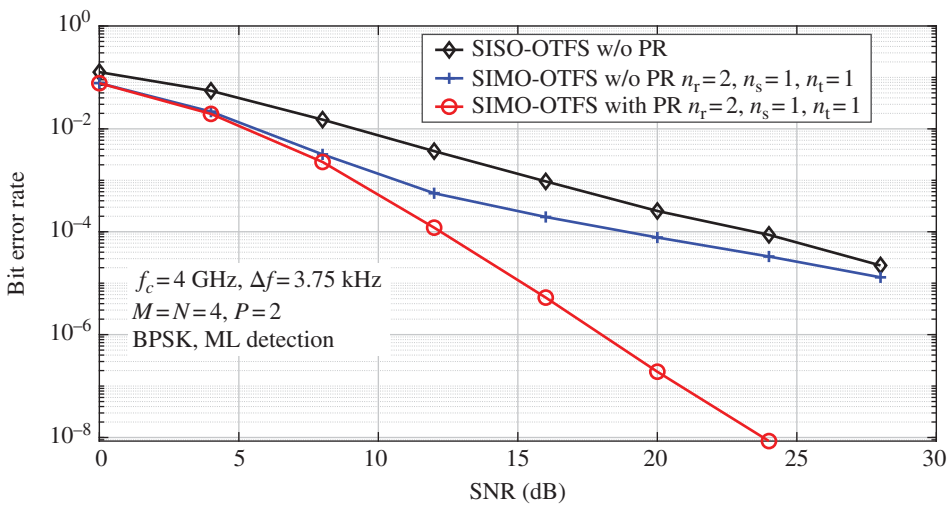


Figure 4.14 BER performance of SIMO-OTFS with RAS for without and with phase rotation, $P = 2$, $M = N = 4$, $n_s = 1$, $n_r = 1, 2$, and BPSK.

considered system, the predicted diversity orders are 1 and 4 for without and with phase rotation, respectively. The slopes in the BER plots in Fig. 4.14 are observed to be in line with these predicted diversity orders.

Figure 4.15 shows the BER performance of MIMO-OTFS with RAS for without and with phase rotation, $P = 2$, $M = 4$, $N = 2$, $n_t = 2$, $n_s = 2$, $n_r = 2, 3$, BPSK, ML detection, and other parameters as in Fig. 4.14. The considered systems are rank deficient, and the predicted diversity orders are n_s and $n_s P$ for without and with phase rotation, respectively. As predicted in Fig. 4.15, MIMO-OTFS

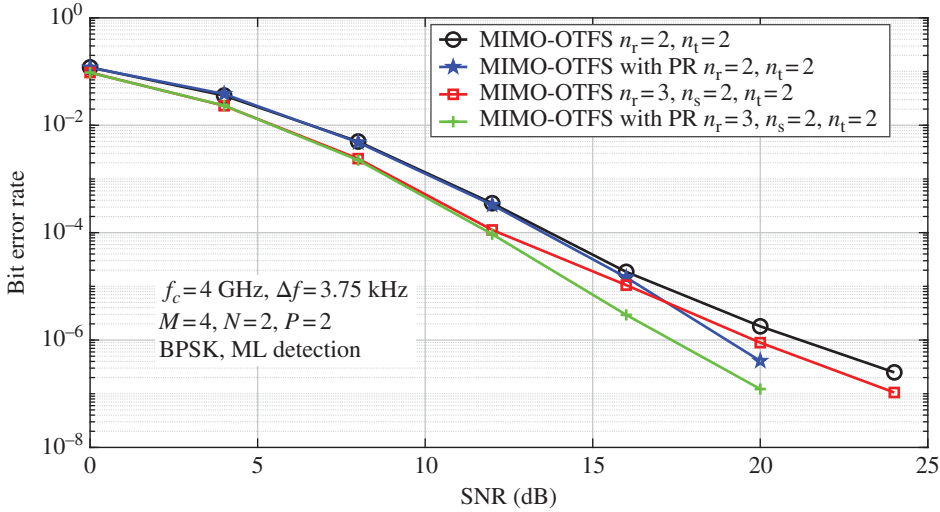


Figure 4.15 BER performance of MIMO-OTFS with RAS for without and with phase rotation, $P = 2$, $M = 4$, $N = 2$, $n_t = 2$, $n_s = 2$, $n_r = 2, 3$, and BPSK.

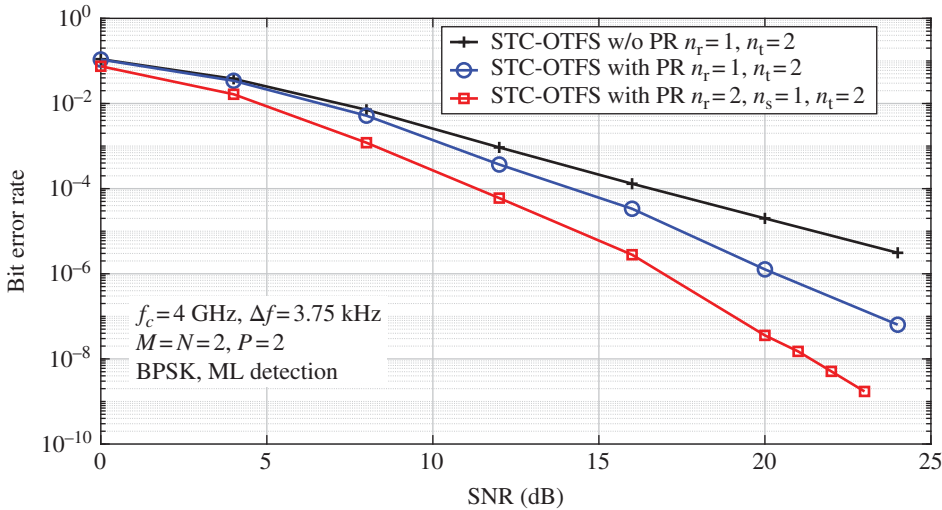


Figure 4.16 BER performance of STC-OTFS with RAS for without and with phase rotation, $P = 2$, $M = N = 2$, $n_t = 2$, $n_s = 1$, $n_r = 1, 2$, and BPSK.

with RAS achieves second order diversity slope without phase rotation and fourth order diversity slope with phase rotation.

Figure 4.16 shows the BER performance of STC-OTFS with RAS for without and with phase rotation, $P = 2$, $M = N = 2$, $n_t = 2$, $n_s = 1$, $n_r = 1, 2$, BPSK, ML detection, and other parameters as in Fig. 4.14. The observed diversity slope is 2 for without phase rotation. This corroborates with the predicted diversity order of $2n_s$ (since the system is rank deficient). With phase rotation, the system is full ranked and the predicted diversity order is $2n_tP$. The diversity slopes observed in this case are in accordance with this analytical prediction.

5

OTFS Signal Detection

Signal detection is a crucial task in any communication receiver. In orthogonal time frequency space (OTFS), signal detection is carried out in the delay-Doppler (DD) domain. In this chapter, we consider signal detection in multicarrier-orthogonal time frequency space (MC-OTFS) systems. The single-input single-output (SISO) MC-OTFS system model in Chapter 4 is considered here. For a MC-OTFS system with M delay bins and N Doppler bins, the input-output relation in the DD domain can be written in a vectorized form as (see (4.10))

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}, \quad (5.1)$$

where $\mathbf{y}$ is the $MN \times 1$ observed vector at the receiver, $\mathbf{x}$ is the $MN \times 1$ transmit symbol vector consisting of MN information symbols in an OTFS frame, $\mathbf{H}$ is the $MN \times MN$ effective DD channel matrix, and $\mathbf{v}$ is the $MN \times 1$ noise vector. Signal detection is the problem of estimating the vector $\mathbf{x}$, given the observed vector $\mathbf{y}$ and the knowledge of the channel matrix $\mathbf{H}$. In this chapter, signal detection algorithms are presented assuming perfect knowledge of the $\mathbf{H}$ matrix at the receiver. In practice, the knowledge of $\mathbf{H}$ is obtained through channel estimation methods using pilot transmission. The topic of DD domain channel estimation for OTFS is discussed in Chapter 6.

This chapter deals with efficient methods for MC-OTFS signal detection in the DD domain. Assuming the noise vector $\mathbf{v}$ to be independent and identically distributed (i.i.d) Gaussian, the optimum maximum-likelihood (ML) decision rule is given by

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{A}^{MN}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2, \quad (5.2)$$

where $\mathbb{A}$ is the modulation alphabet used (e.g., QAM/PSK). The complexity of the optimum detector in (5.2) grows exponentially in MN because of the exhaustive enumeration of all possible vectors in the OTFS signal set. Therefore, low-complexity suboptimum detection methods/algorithms are of interest. In this chapter, OTFS signal detection approaches are classified into three broad categories, namely, (i) low-complexity linear detection, (ii) approximate maximum a posteriori (MAP) detection, and (iii) cross-domain detection, as shown in Fig. 5.1. Popular detection algorithms under these categories are presented in the chapter.

In the linear detection approach, a linear transformation is applied to the received vector $\mathbf{y}$, followed by element-wise mapping to a symbol in the modulation alphabet which is closest in terms of Euclidean distance. An interesting and attractive attribute in MC-OTFS is that the inherent structure in the effective channel matrix $\mathbf{H}$ and the sparse and quasi-banded nature of the OTFS demodulation matrices can be exploited to reduce the complexity of conventional linear detection methods such as minimum mean squared error (MMSE) and zero-forcing (ZF) detection. This chapter emphasizes such low-complexity detection opportunities in OTFS.

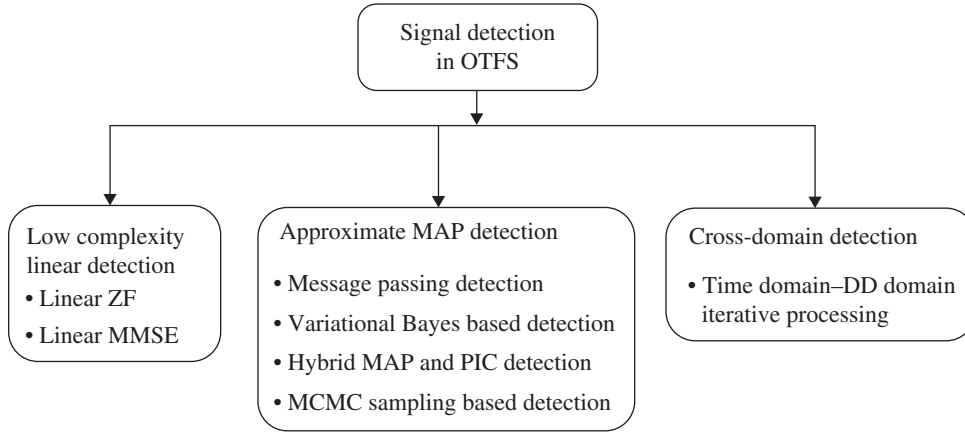


Figure 5.1 OTFS signal detection approaches.

With regard to the approximate MAP detection approach, several algorithms have been reported in the OTFS literature. These algorithms use detection methods based on iterative techniques, message passing, interference cancellation, variational Bayes approach, and Markov chain Monte-Carlo (MCMC) sampling. This chapter presents some of these algorithms, illustrating their performance and complexity attributes.

While the first two approaches perform detection exclusively in the DD domain, the cross-domain approach performs processing in multiple domains and iterates between these domains. This detection approach can bring in certain advantages in OTFS. For example, a cross-domain approach involving processing in the time domain and DD domain can exploit the time domain channel sparsity and the DD domain symbol constellation constraints. This chapter presents an algorithm based on the cross-domain approach.

5.1 Linear Detection

In linear detection methods, a linear transformation is applied to the received vector, and the elements in the resulting vector are mapped to symbols in the modulation alphabet using the minimum Euclidean distance rule. The linear transformation matrix is represented by $\mathbf{G}$, and the mapping function is represented by $f(\cdot)$. The estimate of the transmit vector $\mathbf{x}$ is given by

$$\hat{\mathbf{x}} = f(\mathbf{G}\mathbf{y}). \quad (5.3)$$

The transformation matrix for MMSE detection is given by

$$\mathbf{G}_{\text{mmse}} = (\mathbf{H}^H \mathbf{H} + \sigma^2 \mathbf{I})^{-1} \mathbf{H}^H, \quad (5.4)$$

where $(\cdot)^H$ denotes the Hermitian operation and σ^2 is the noise variance. The transformation matrix for ZF detection is given by

$$\mathbf{G}_{\text{zf}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H. \quad (5.5)$$

These detectors have a computation complexity of $\mathcal{O}(M^3 N^3)$ because of the matrix inversion involved in them. However, the structure of the $\mathbf{H}$ matrix can be exploited so that the transformation $\mathbf{G}$ in (5.4) and (5.5) can be realized using fast Fourier transform (FFT) and inverse fast

Fourier transform (IFFT) operations instead of matrix inverse operation, and this reduces the detection complexity [79–82].

5.1.1 Reduced Complexity MMSE Detection

This section presents two complexity reduction schemes for MMSE detection in MC-OTFS. Both schemes exploit the structure of the $\mathbf{H}$ matrix to achieve complexity reduction.

5.1.1.1 Complexity Reduction Scheme 1

Recall from Chapter 4 that the $MN \times MN$ channel matrix $\mathbf{H}$ in (4.10) has a block circulant structure with M circulant blocks, denoted by $\mathbf{A}_i$, $i = 0, 1, \dots, M-1$, of size $N \times N$. Making use of this structure of the channel matrix, a low-complexity algorithm for implementing MMSE detection can be devised as follows [79]. Let $\mathcal{B}_{M,N}$ denote the class of such block circulant matrices with M circulant blocks of size $N \times N$. Denoting circulant matrices with $\text{circ}()$, any matrix $A \in \mathcal{B}_{M,N}$ will be of the form $\text{circ}(A_0, A_1, \dots, A_{M-1})$, where each block A_i is an $N \times N$ circulant matrix of the form $\text{circ}(a_{i,0}, a_{i,1}, \dots, a_{i,N-1})$ [71]. The matrices in $\mathcal{B}_{M,N}$ have the following exploitable properties.

1. Any matrix $\mathbf{A} \in \mathcal{B}_{M,N}$ can be unitarily diagonalizable as

$$\mathbf{A} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \mathbf{\Lambda} (\mathbf{F}_M \otimes \mathbf{F}_N), \quad (5.6)$$

where $\mathbf{\Lambda} = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_{MN}\}$ such that λ_i is the i th eigenvalue of $\mathbf{A}$, $\mathbf{F}_M$ is the DFT matrix of size M , and $\otimes$ is the Kronecker product operator.

2. The matrix $\mathbf{\Lambda}$ can be written as

$$\mathbf{\Lambda} = \sum_{i=0}^{M-1} \Omega_M^i \otimes \mathbf{\Lambda}_i, \quad (5.7)$$

where $\Omega_M = \text{diag}\{1, \omega, \dots, \omega^{M-1}\}$, $\omega = e^{j2\pi/M}$, and $\mathbf{\Lambda}_i$ is the $N \times N$ diagonal matrix with eigenvalues of $N \times N$ circulant block $\mathbf{A}_i$ on the diagonal.

3. For $\mathbf{A}, \mathbf{B} \in \mathcal{B}_{M,N}$, the matrices $\mathbf{A}^T$, $\mathbf{A}^H$, $\mathbf{AB}(=\mathbf{BA})$, $c_1\mathbf{A} + c_2\mathbf{B}$, $\sum_{r=0}^{R-1} c_r \mathbf{A}_r$ (c_r are all scalars), and $\mathbf{A}^{-1}$ (if exists) are also block circulant and belong to $\mathcal{B}_{M,N}$.

Since the channel matrix $\mathbf{H}$ belongs to $\mathcal{B}_{M,N}$, it can be decomposed as

$$\mathbf{H} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \mathbf{\Lambda} (\mathbf{F}_M \otimes \mathbf{F}_N), \quad (5.8)$$

where $\mathbf{\Lambda} = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_{MN}\}$ such that λ_i is the i th eigenvalue of $\mathbf{H}$. From property 3, it can be seen that if $\mathbf{H} \in \mathcal{B}_{M,N}$, then $\mathbf{G}_{\text{mmse}} \in \mathcal{B}_{M,N}$. Therefore, the eigenvalue decomposition of $\mathbf{G}_{\text{mmse}}$ will be of the form

$$\mathbf{G}_{\text{mmse}} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \mathbf{\Psi} (\mathbf{F}_M \otimes \mathbf{F}_N), \quad (5.9)$$

where $\mathbf{\Psi}$ is a diagonal matrix containing the eigenvalues of $\mathbf{G}_{\text{mmse}}$, given by

$$\mathbf{\Psi} = (\mathbf{\Lambda}^* \mathbf{\Lambda} + \sigma^2 \mathbf{I})^{-1} \mathbf{\Lambda}^*, \quad (5.10)$$

$\Psi_i = \frac{\lambda_i^*}{|\lambda_i|^2 + \sigma^2}$, $i = 1, 2, \dots, MN$. Therefore, the eigenvalues of $\mathbf{G}_{\text{mmse}}$ can be expressed in terms of the eigenvalues of $\mathbf{H}$ as

$$\mathbf{\Psi} = \text{diag} \left\{ \frac{\lambda_1^*}{|\lambda_1|^2 + \sigma^2}, \frac{\lambda_2^*}{|\lambda_2|^2 + \sigma^2}, \dots, \frac{\lambda_{MN}^*}{|\lambda_{MN}|^2 + \sigma^2} \right\}. \quad (5.11)$$

It is clear from (5.11) that the eigenvalues of $\mathbf{G}_{\text{mmse}}$ can be computed from the eigenvalues of $\mathbf{H}$. The computation of eigenvalues of $\mathbf{G}_{\text{mmse}}$ using the eigenvalues of $\mathbf{H}$ is one of the key steps in complexity reduction. Using $\mathbf{G}_{\text{mmse}}$, the transmitted symbols are estimated by computing $\mathbf{G}_{\text{mmse}}\mathbf{y}$ with significantly lower complexity using FFTs, IFFTs, and the properties of $\mathbf{H}$ and $\mathbf{G}_{\text{mmse}}$ as follows.

1. Computation of eigenvalues of each block of $\mathbf{H}$: $\mathbf{H}$ has a block circulant structure with circulant blocks with an eigenvalue decomposition of the form (5.8) which can be written as

$$\mathbf{\Lambda} = \sum_{i=0}^{M-1} \mathbf{\Omega}_M^i \otimes \mathbf{\Lambda}_i = \text{diag} \left\{ \sum_{i=0}^{M-1} \mathbf{\Lambda}_i, \sum_{i=0}^{M-1} e^{j2\pi i/M} \mathbf{\Lambda}_i, \dots, \sum_{i=0}^{M-1} e^{j2\pi(M-1)i/M} \mathbf{\Lambda}_i \right\}, \quad (5.12)$$

where $\mathbf{\Lambda}_i$ denotes the $N \times N$ diagonal matrix containing the eigenvalues of the i th block in $\mathbf{H}$. The computation of the eigenvalues of $\mathbf{H}$ involves the following two steps, namely, computing $\mathbf{\Lambda}_i$ s by computing the eigenvalues of the circulant blocks of $\mathbf{H}$ and computing $\mathbf{\Lambda}$ using them as in (5.12). Since each block in $\mathbf{H}$ has a circulant structure, the eigenvalues of each block are computed by computing the DFT of the first row of each block of $\mathbf{H}$. Therefore, the computation of $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2, \dots, \mathbf{\Lambda}_M$ requires the computation of eigenvalues of M number of $N \times N$ circulant matrices and, hence, has a computational complexity of $\mathcal{O}(MN \log N)$.

2. Computation of the eigenvalues of $\mathbf{H}$: Using $\mathbf{\Lambda}_i$ s, the eigenvalues of $\mathbf{H}$ are computed using (5.12). A closer look at (5.12) reveals that the entries of $\mathbf{\Lambda}$ can be computed with low complexity using IFFT. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M$ denote the M column vectors of dimension $N \times 1$ whose entries contain the diagonal elements of $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2, \dots, \mathbf{\Lambda}_M$, respectively. Let $\mathbf{V} = [\mathbf{v}_1^T \mathbf{v}_2^T \dots \mathbf{v}_M^T]$ denote the $M \times N$ matrix whose rows are the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M$. Now, (5.12) can be computed as

$$\mathbf{\Lambda} = \text{diag}\{(\mathbf{F}_M^H \mathbf{V})^T\}. \quad (5.13)$$

The complexity of obtaining $\mathbf{\Lambda}$ using (5.13) involves the computation of N M -point inverse discrete Fourier transforms (IDFTs). Therefore, the complexity of this step is $\mathcal{O}(MN \log M)$.

3. Computation of eigenvalues of $\mathbf{G}_{\text{mmse}}$: Using the eigenvalues of $\mathbf{H}$, the eigenvalues of $\mathbf{G}_{\text{mmse}}$ are computed using (5.11). This step requires the computation of $\Psi_1, \Psi_2, \dots, \Psi_{MN}$ using $\lambda_1, \lambda_2, \dots, \lambda_{MN}$, respectively. Hence, the order of complexity for computing Ψ is $\mathcal{O}(MN)$.

4. Computation of $\mathbf{G}_{\text{mmse}}\mathbf{y}$: The next step is to compute $\mathbf{G}_{\text{mmse}}\mathbf{y} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \Psi (\mathbf{F}_M \otimes \mathbf{F}_N) \mathbf{y}$. This equation can also be evaluated with low complexity using FFTs and IFFTs. Let $\mathbf{Y}$ denote the $N \times M$ matrix such that $\text{vec}(\mathbf{Y}) = \mathbf{y}$. Then, $(\mathbf{F}_M \otimes \mathbf{F}_N) \mathbf{y}$ can be written as

$$\mathbf{z} = (\mathbf{F}_M \otimes \mathbf{F}_N) \mathbf{y} = \text{vec}(\mathbf{F}_N \mathbf{Y} \mathbf{F}_M^H). \quad (5.14)$$

Observe that $\mathbf{z}$ can be obtained by computing N -point DFT along the columns of $\mathbf{Y}$ and M -point IDFT along the rows of $\mathbf{Y}$. Thus, the complexity of computing $\mathbf{z}$ using FFTs and IFFTs is $\mathcal{O}(MN \log M + MN \log N) = \mathcal{O}(MN \log MN)$. Next, $\Psi(\mathbf{F}_M \otimes \mathbf{F}_N) \mathbf{y}$ can be computed by computing $\Psi \mathbf{z}$. Since Ψ is a diagonal matrix, the computation of $\Psi \mathbf{z}$ has a complexity of $\mathcal{O}(MN)$. Now, let $\mathbf{q} = \Psi \mathbf{z}$ and $\mathbf{Q}$ denote the $N \times M$ matrix such that $\text{vec}(\mathbf{Q}) = \mathbf{q}$. With this, $\mathbf{G}_{\text{mmse}}\mathbf{y} = (\mathbf{F}_M \otimes \mathbf{F}_N)^H \Psi (\mathbf{F}_M \otimes \mathbf{F}_N) \mathbf{y}$ can be computed as

$$\mathbf{G}_{\text{mmse}}\mathbf{y} = \text{vec}(\mathbf{F}_N^H \mathbf{Q} \mathbf{F}_M). \quad (5.15)$$

Again, this step involves the computation of N -point IDFT along the columns of $\mathbf{Q}$ and M -point DFT along the rows of $\mathbf{Q}$. Hence, this step has a complexity of $\mathcal{O}(MN \log MN)$. Therefore, the overall complexity of Step 4 is $\mathcal{O}(2MN \log MN + MN)$.

Summarizing the above, the algorithm to obtain the low-complexity MMSE solution for MC-OTFS is given in **Algorithm 5.1**. Table 5.1 shows the computational complexity associated with each step of the algorithm. The algorithm gives the exact MMSE solution for MC-OTFS at a

Algorithm 5.1 Low-complexity MMSE detection algorithm for MC-OTFS**Input:** $\mathbf{y}$, $\mathbf{H}$.

- Step 1: Compute the eigenvalues of M circulant blocks of $\mathbf{H}$ by computing the DFTs of first row of each circulant block.
- Step 2: Compute $\mathbf{\Lambda}$ using (5.13).
- Step 3: Compute $\mathbf{\Psi} = (\mathbf{\Lambda}^* \mathbf{\Lambda} + \sigma^2 \mathbf{I})^{-1} \mathbf{\Lambda}^*$.
- Step 4: Compute $\mathbf{G}_{\text{mmse}} \mathbf{y}$ by computing $\mathbf{z} = \text{vec}(\mathbf{F}_N \mathbf{Y} \mathbf{F}_M^H)$, $\mathbf{q} = \mathbf{\Psi} \mathbf{z}$, and $\text{vec}(\mathbf{F}_N^H \mathbf{Q} \mathbf{F}_M)$.

Output: $\hat{\mathbf{x}} = f(\mathbf{G}_{\text{mmse}} \mathbf{y})$.**Table 5.1** Order of complexity.

Step #	Order of complexity
1: computation of eigenvalues of circulant blocks of $\mathbf{H}$	$\mathcal{O}(MN \log N)$
2: computation of $\mathbf{\Lambda}$	$\mathcal{O}(MN \log M)$
3: computation of $\mathbf{\Psi}$	$\mathcal{O}(MN)$
4: computation of $\mathbf{G}_{\text{mmse}} \mathbf{y}$	$\mathcal{O}(2MN \log MN + MN)$
Overall complexity (dominated by Step 4)	$\mathcal{O}(MN \log MN)$

much less complexity compared to the $\mathcal{O}(M^3 N^3)$ complexity of conventional MMSE solution using matrix inversion. The above complexity reduction technique has been extended for MIMO MC-OTFS with two transmit/receive antennas in [80] and with generic number of transmit/receive antennas in [81]. The MMSE solution provided by the algorithm can be fed as a low-complexity initial solution to nonlinear detectors, such as local neighborhood search-based algorithms [16], to achieve enhanced bit error performance.

5.1.1.2 Complexity Reduction Scheme 2

This scheme takes advantage of the sparse and quasi-banded nature of the MC-OTFS demodulation matrices to devise another low-complexity algorithm for implementing MMSE detection as follows [82]. Here, a different representation of the system is used. The transmit vector $\mathbf{x}$ in the DD domain is written as a $M \times N$ matrix $\mathbf{X}$ such that $\text{vec}(\mathbf{X}) = \mathbf{x}$. Assume that $\mathbb{E}[x(k, l)x^*(k', l')] = \sigma_x^2 \delta(k - k', l - l')$. Using this representation, vector $\mathbf{s}$ is obtained as [83]

$$\mathbf{s} = \text{vec}(\mathbf{X} \mathbf{F}_N^H). \quad (5.16)$$

The vector $\mathbf{s}$ can also be written in the form $\mathbf{s} = \mathbf{A} \mathbf{x}$, where $\mathbf{A} = \mathbf{F}_N^H \otimes \mathbf{I}_M$ is a unitary matrix. The received vector $\mathbf{r}$ at the receiver is given by

$$\mathbf{r} = \tilde{\mathbf{H}} \mathbf{s} + \mathbf{n} = \tilde{\mathbf{H}} \mathbf{A} \mathbf{x} + \mathbf{n}, \quad (5.17)$$

where

$$\tilde{\mathbf{H}} = \sum_{i=1}^P h_i \mathbf{\Pi}^{a_i} \mathbf{\Delta}^{b_i}, \quad (5.18)$$

$\mathbf{\Pi} = \text{circ}\{[010 \cdots 0]_{MN \times 1}^T\}$ is a circulant delay matrix, $\text{circ}(\mathbf{z})$ denotes a circulant matrix whose first column is $\mathbf{z}$, $\mathbf{\Delta} = \text{diag}\{1, e^{j2\pi \frac{1}{MN}}, \dots, e^{j2\pi \frac{MN-1}{MN}}\}$ is a diagonal Doppler matrix, and $\mathbf{n}$ is a i.i.d Gaussian

noise vector with variance σ_n^2 . The detected symbol vector for this system model in (5.17) using linear MMSE equalization is given by

$$\hat{\mathbf{x}} = (\tilde{\mathbf{H}}\mathbf{A})^H \left[(\tilde{\mathbf{H}}\mathbf{A})(\tilde{\mathbf{H}}\mathbf{A})^H + \frac{\sigma_n^2}{\sigma_x^2} \mathbf{I} \right]^{-1} \mathbf{r}. \quad (5.19)$$

Due to the unitary nature of $\mathbf{A}$, the above equation reduces to

$$\hat{\mathbf{x}} = \mathbf{A}^H \underbrace{\tilde{\mathbf{H}}^H \left[\tilde{\mathbf{H}}\tilde{\mathbf{H}}^H + \frac{\sigma_n^2}{\sigma_x^2} \mathbf{I} \right]^{-1}}_{\mathbf{H}_{\text{eq}}} \mathbf{r}. \quad (5.20)$$

This detected vector $\hat{\mathbf{x}}$ is obtained in two steps. The first step involves the calculation of $\tilde{\mathbf{r}} = \mathbf{H}_{\text{eq}} \mathbf{r}$, and the second step involves the matched filter operation $\mathbf{A}^H \tilde{\mathbf{r}}$ to get $\hat{\mathbf{x}}$. Most complexity is in the first step as it involves an inverse of $\Psi = \left[\tilde{\mathbf{H}}\tilde{\mathbf{H}}^H + \frac{\sigma_n^2}{\sigma_x^2} \mathbf{I} \right]$ to obtain $\mathbf{H}_{\text{eq}}$. This complexity is reduced by using a low-complexity lower-upper (LU) decomposition of Ψ [82]. With LU decomposition of Ψ , $\tilde{\mathbf{r}}$ can be simplified as

$$\tilde{\mathbf{r}} = \tilde{\mathbf{H}}^H \underbrace{\mathbf{U}^{-1} \mathbf{L}^{-1} \mathbf{r}}_{\mathbf{r}_2}. \quad (5.21)$$

The computational complexity can be reduced by exploiting the quasi-banded nature of the lower triangular matrix $\mathbf{L}$ and the upper triangular matrix $\mathbf{U}$. As $\mathbf{L}$ is a quasi-banded lower triangular matrix, $\mathbf{r}_1$ in (5.21) is computed using a low-complexity forward substitution method (see Algorithm 5.2 in [82]). Likewise, since $\mathbf{U}$ is a quasi-banded upper triangular matrix, $\mathbf{r}_2$ in (5.21) is computed using a low-complexity backward substitution method (see Algorithm 5.3 in [82]). Using the definition of $\tilde{\mathbf{H}}$, $\tilde{\mathbf{r}} = \tilde{\mathbf{H}}^H \mathbf{r}_2$ can be written as

$$\tilde{\mathbf{r}} = \sum_{i=1}^P h_i^* \Delta^{-b_i} \underbrace{\mathbf{\Pi}^{-a_i} \mathbf{r}_2}_{\text{circular shift}}. \quad (5.22)$$

To compute $\tilde{\mathbf{r}}$, $\mathbf{r}_2$ is first circularly shifted by delay $-a_i$ and then multiplied by $h_i^* \text{diag}\{\Delta^{-b_i}\}$ by using point-to-point multiplication for each path i . All vectors obtained in above step are finally summed to obtain $\tilde{\mathbf{r}}$. Next, instead of directly computing $\hat{\mathbf{x}} = \mathbf{A}^H \tilde{\mathbf{r}}$, the vector $\tilde{\mathbf{r}}$ is first reshaped into a $M \times N$ -sized $\tilde{\mathbf{R}}$ matrix as

$$\tilde{\mathbf{R}} = \begin{bmatrix} \tilde{\mathbf{r}}(0) & \tilde{\mathbf{r}}(M) & \cdots & \tilde{\mathbf{r}}(MN - N) \\ \tilde{\mathbf{r}}(1) & \tilde{\mathbf{r}}(M + 1) & \cdots & \tilde{\mathbf{r}}(MN - N + 1) \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{\mathbf{r}}(M - 1) & \tilde{\mathbf{r}}(2M - 1) & \cdots & \tilde{\mathbf{r}}(MN - 1) \end{bmatrix}. \quad (5.23)$$

Using the $\tilde{\mathbf{R}}$ matrix, $\hat{\mathbf{x}}$ is obtained using the DFT operation as

$$\hat{\mathbf{x}} = \text{vec}(\tilde{\mathbf{R}}\mathbf{F}_N). \quad (5.24)$$

Figure 5.2 shows the various signal processing steps involved in the algorithm. The main reduction in the complexity is in the computation of $\tilde{\mathbf{r}}$. All the steps involved in obtaining $\mathbf{r}_2$ together have a complexity of $\mathcal{O}(MN)$ and the final computation in (5.24) to obtain $\hat{\mathbf{x}}$ has a complexity of $\mathcal{O}(\frac{MN}{2} \log_2 N)$. On the whole, this detection method has a complexity of just $\mathcal{O}(\frac{MN}{2} \log_2 N + MN)$ compared to the $\mathcal{O}(M^3 N^3)$ complexity of conventional MMSE detection involving matrix inversion.

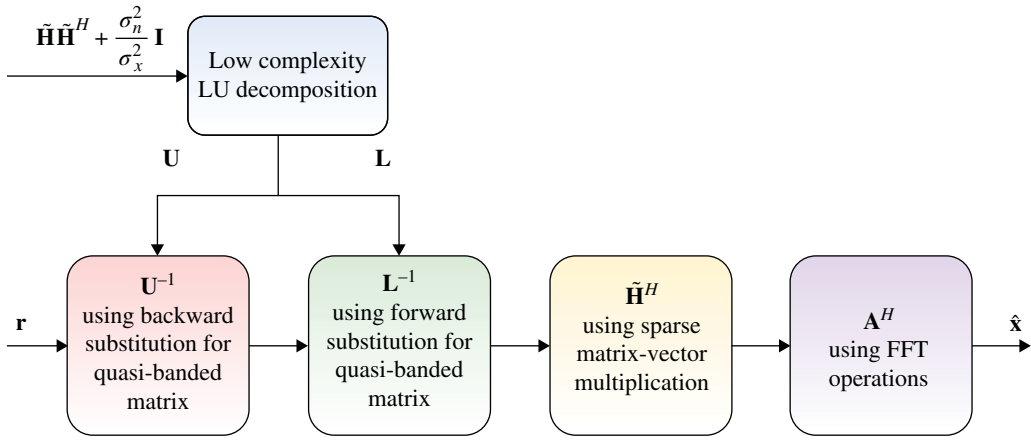


Figure 5.2 Low-complexity linear MMSE detection using LU decomposition.

5.2 Approximate MAP-Based Detection

Linear detection methods, though simple with polynomial complexity, are not optimal. Their performances are far from optimum ML/MAP detectors performance. However, the exponential complexity of optimum ML/MAP detectors makes them impractical for large frame sizes. Nonlinear detection methods perform better than linear methods, and they achieve performance close to optimum performance in polynomial complexity. Often these methods are iterative in nature involving approximations to achieve detection in polynomial complexity. Some of these nonlinear detection methods are presented in this section. From the vectorized representation $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}$ in (5.1), the MAP decision rule for detection of $\mathbf{x}$ is given by

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x} \in \mathbb{A}^{MN}} \Pr(\mathbf{x}|\mathbf{y}, \mathbf{H}). \quad (5.25)$$

In the following, some of the popular low-complexity approximate MAP algorithms for OTFS signal detection are presented.

5.2.1 Message Passing Detection

One of the popular detectors reported in the early OTFS literature is an approximate MAP detector based on low-complexity message passing [38]. The key advantages of this detector are its linear complexity in MN and very good performance. The message passing algorithm involves the computation of approximate a posteriori probabilities of the modulation symbols by passing messages on a sparsely connected factor graph. The transmitted vector $\mathbf{x}$ is represented by MN variable nodes, and the received vector $\mathbf{y}$ is represented by MN observation nodes in the graph. As noted earlier, $\mathbf{H}$ is sparse with only L non-zero elements in any row or column (generally $L \ll MN$ and $L = P$ for non-fractional delays and Dopplers).

Let $\mathcal{I}(r)$ and $\mathcal{J}(c)$ denote the sets of indices corresponding to non-zero elements in the r th row and c th column, respectively, such that $|\mathcal{I}(r)| = |\mathcal{J}(c)| = L$ for all rows and columns. In the graph, each variable node $x[c]$ has connections with L observation nodes $y[r]$, $r \in \mathcal{J}(c)$, and each observation node $y[r]$ has connections with L variable nodes $x[c]$, $c \in \mathcal{I}(r)$, as shown in Fig. 5.3. Since joint MAP

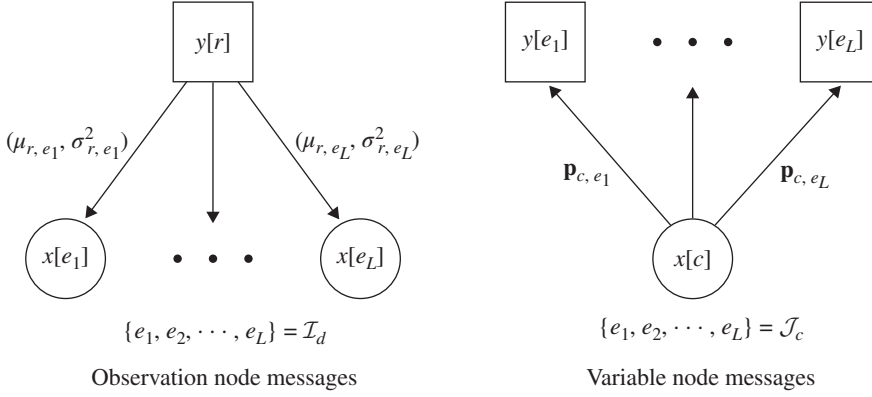


Figure 5.3 Message passing between variable nodes and observation nodes.

detection in (5.25) is exponentially complex, consider the symbol-by-symbol MAP decision

$$\begin{aligned}
 \hat{x}[c] &= \arg \max_{a_j \in \mathbb{A}} \Pr(x[c] = a_j | \mathbf{y}, \mathbf{H}) \\
 &= \arg \max_{a_j \in \mathbb{A}} \frac{1}{|\mathbb{A}|} \Pr(\mathbf{y} | x[c] = a_j, \mathbf{H}) \\
 &\approx \arg \max_{a_j \in \mathbb{A}} \prod_{r \in J_c} \Pr(y[r] | x[c] = a_j, \mathbf{H}),
 \end{aligned} \tag{5.26}$$

where all the symbols $a_j \in \mathbb{A}$ are assumed to be equally likely and the components of $\mathbf{y}$ are assumed to be approximately independent for a given $x[c]$ because of the sparsity of $\mathbf{H}$. The system model in (5.1) can be written in the form

$$y[r] = x[c]H[r, c] + \underbrace{\sum_{e \in I(r), e \neq c} x[e]H[r, e] + v[r]}_{\zeta_{r,c}^{(i)}}, \tag{5.27}$$

where, as in [84], the contribution of a variable $x[c]$ to $y[r]$ is isolated from the contributions from other interfering variables $x[e] \in I(r), e \neq c$, and the interfering terms are approximated as Gaussian noise with an easily computable mean and variance. Specifically, in the i th iteration, the interference plus noise term in (5.27) is approximated as a single Gaussian random variable $\zeta_{r,c}^{(i)}$ with mean $\mu_{r,c}^{(i)}$ and variance $(\sigma_{r,c}^{(i)})^2$. The mean and variance of the interference plus noise term are used as messages from observation nodes to variable nodes. On the other hand, the message passed from a variable node $x[c]$ to observation nodes $y[r], r \in J(c)$, is the probability mass function (pmf) of the alphabet $\mathbf{p}_{c,r} = \{p_{c,r}(a_j) | a_j \in \mathbb{A}\}$. Figure 5.3 shows the connections and the messages passed between the observation and variable nodes.

• **Message from observation node $y[r]$ to variable node $x[c]$**

The observation nodes pass the mean and variance information to the variable nodes. The mean and variance of $\zeta_{r,c}^{(i)}$ are computed as

$$\mu_{r,c}^{(i)} = \sum_{e \in I(r), e \neq c} \sum_{j=1}^{|\mathbb{A}|} p_{e,r}^{(i-1)}(a_j) a_j H[r, e] \tag{5.28}$$

and

$$(\sigma_{r,c}^{(i)})^2 = \sum_{e \in \mathcal{I}(r), e \neq c} \left(\sum_{j=1}^{|\mathbb{A}|} p_{e,r}^{(i-1)}(a_j) |a_j|^2 |H[r, e]|^2 - \left| \sum_{j=1}^{|\mathbb{A}|} p_{e,r}^{(i-1)}(a_j) a_j H[r, e] \right|^2 \right) + \sigma^2, \quad (5.29)$$

and σ^2 is the variance of $v[r]$.

- **Message from variable node $x[c]$ to observation node $y[r]$**

The variable nodes compute the pmf of the alphabet and pass it to the observation nodes. The pmf vector $\mathbf{p}_{c,r}^{(i)}$ is updated as

$$p_{(c,r)}^{(i)}(a_j) = \delta \tilde{p}_{(c,r)}^{(i)}(a_j) + (1 - \delta) p_{(c,r)}^{(i-1)}(a_j), \quad (5.30)$$

where $\delta \in (0, 1]$ is the damping factor, the choice of which can influence the performance and convergence speed [85], and

$$\begin{aligned} \tilde{p}_{(c,r)}^{(i)}(a_j) &= \prod_{e \in \mathcal{J}(c), e \neq r} \Pr(y[e] | x[c] = a_j, \mathbf{H}) \\ &= \prod_{e \in \mathcal{J}(c), e \neq r} \frac{\xi_{(i)}(e, c, j)}{\sum_{t=1}^{|\mathbb{A}|} \xi_{(i)}(e, c, t)} \end{aligned} \quad (5.31)$$

and

$$\xi_{(i)}(e, c, t) = \exp \left(\frac{-|y[e] - \mu_{e,c}^{(i)} - H_{e,c} a_t|^2}{(\sigma_{e,c}^{(i)})^2} \right). \quad (5.32)$$

- **Computation of the convergence indicator**

The convergence indicator $\eta^{(i)}$ for some small $\epsilon > 0$ is computed as

$$\eta^{(i)} = \frac{1}{MN} \sum_{c=1}^{MN} \mathbb{I} \left(\max_{a_j \in \mathbb{A}} p_c^{(i)}(a_j) \geq 1 - \epsilon \right), \quad (5.33)$$

where $\mathbb{I}(\cdot)$ is an indicator function which gives a value of 1 if the expression in the argument is true, and 0 otherwise, and

$$p_c^{(i)}(a_j) = \prod_{e \in \mathcal{J}(c)} \frac{\xi_{(i)}(e, c, j)}{\sum_{t=1}^{|\mathbb{A}|} \xi_{(i)}(e, c, t)}. \quad (5.34)$$

- **Decision update**

If $\eta^{(i)} < \eta^{(i-1)}$, then the decision of the transmitted symbol is updated as

$$\hat{x}[c] = \arg \max_{a_j \in \mathbb{A}} p_c^{(i)}(a_j), \quad c = 1, 2, \dots, MN. \quad (5.35)$$

That is, the decision on the transmitted symbols is updated only when the current iteration can provide better estimates than the previous iteration.

- **Stopping criteria**

The algorithm stops when at least one of the following conditions is satisfied:

1. $\eta^{(i)} = 1$.
2. $\eta^{(i)} < \eta^{(i')} - \epsilon$ for some small ϵ , where $\eta^{(i')} = \max_{k < i} \eta^{(k)}$.
3. The maximum limit set for the number of iterations, n_{iter} , is reached.

The first condition occurs in the best case, where all the symbols have converged. The second condition is useful to stop the algorithm if the current iteration provides a worse decision compared to

Algorithm 5.2 Message passing algorithm for OTFS signal detection**Inputs:** $\mathbf{y}$, $\mathbf{H}$, ϵ , ϵ , n_{iter} **Initialize:** pmf vector $\mathbf{p}_{c,r}^{(0)} = \frac{1}{|\mathbb{A}|}$, $c = 1, \dots, MN$, $r \in \mathcal{J}(c)$, $i = 1$ **repeat**

- Observation nodes $\mathbf{y}[r]$ compute the means $\mu_{r,c}^{(i)}$ and variances $(\sigma_{r,c}^{(i)})^2$ of Gaussian random variables $\zeta_{r,c}^{(i)}$ using $\mathbf{p}_{c,r}^{(i-1)}$ and pass them to variable nodes $\mathbf{x}[c]$, $c \in \mathcal{I}(r)$
- Variable nodes $\mathbf{x}[c]$ update $\mathbf{p}_{c,r}^{(i)}$ using $\mu_{r,c}^{(i)}$, $(\sigma_{r,c}^{(i)})^2$, and $\mathbf{p}_{c,r}^{(i-1)}$ and pass them to observation nodes $\mathbf{y}[r]$, $r \in \mathcal{J}_c$
- Compute convergence indicator $\eta^{(i)}$
- Update the decision of the transmitted symbols $\hat{\mathbf{x}}[c]$, $c = 1, \dots, MN$, if needed
- $i = i + 1$

until *Stopping criteria***Output:** The decision on the transmitted symbols $\hat{\mathbf{x}}[c]$

the one in previous iterations. The message passing detection algorithm is listed in **Algorithm 5.2**. This algorithm has been shown to achieve near-optimal performance for large frame sizes at a low complexity of order $\mathcal{O}(MN)$. It has been widely used in the literature for the performance evaluation of MC-OTFS. Several variants of this algorithm have also been proposed. For example, a low-complexity variant that exploits channel hardening through match filtering operation on $\mathbf{y}$ and message passing on the resulting system model is presented in [46]. Another variant in [47] uses an alternate approach, where the a posteriori probability of each transmitted symbol is assumed to be Gaussian distributed and message passing is carried out the resulting system model. These variants are also shown to achieve good bit error performance at low detection complexity.

5.2.2 Variational Bayes-Based Detection

This section presents an iterative algorithm that approximates the optimal MAP detection by approximating the joint a posteriori distribution by the product of local marginals. The approximation of the a posteriori probability $p(\mathbf{x}|\mathbf{y})$ is obtained by using Kullback–Leibler (KL) divergence $D(q||p)$, and the corresponding evidence lower bound (ELBO) is maximized iteratively using the variational Bayes (VB) approach. The marginal distribution for each symbol is obtained which is used for symbol-by-symbol MAP detection. A key advantage of this approach compared to the message passing algorithm presented in Section 5.2.1 is that by choosing the distribution family for the VB method appropriately, the ELBO maximization problem is strictly convex such that the globally optimal solution can be obtained efficiently, leading to a convergence guaranteed receiver. Another advantage is that the VB approach requires a lower complexity and achieves better performance compared to that of the message passing (MP) algorithm in Section 5.2.1.

To proceed with the development of the algorithm, the input–output relation in the DD domain is written in the form [43]

$$y[k, l] = \sum_{k'=0}^{N-1} \sum_{l'=0}^{M-1} x[k', l'] h_{k,l}[k', l'] + v[k, l], \quad (5.36)$$

where $x[k, l] \in \mathbb{A}$ are the data symbols,

$$h_{k,l}[k', l'] = \sum_{i=1}^P h_i v[k - k' - b_i, l - l' - a_i] e^{-j2\pi \frac{b_i a_i}{MN}}, \quad (5.37)$$

and

$$v[k, l] = \frac{1}{MN} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} e^{-j2\pi nT \frac{k}{NT}} e^{-j2\pi m\Delta f \frac{l}{M\Delta f}}, \quad (5.38)$$

where b_i and a_i denote the Doppler index and delay index associated with the i th path, respectively, i.e., $v_i = \frac{b_i}{NT}$ and $\tau_i = \frac{a_i}{M\Delta f}$. The transmitted symbols and equivalent channels are stacked into vectors, yielding the following compact form of (5.36):

$$y[k, l] = \mathbf{h}_{k,l}^T \mathbf{x} + v[k, l], \quad (5.39)$$

where $\mathbf{x}$ and $\mathbf{h}_{k,l}$ denote the data symbol vector and equivalent channel vector, respectively, whose $(k'M + l')$ th entries are $x[k', l']$ and $h_{k,l}[k', l']$, respectively.

The idea behind the VB approach is finding a distribution $q(\mathbf{x})$ from a tractable distribution family $\mathcal{Q}$ as an optimized approximation of the a posteriori distribution $p(\mathbf{x}|\mathbf{y})$. The approximation $q(\mathbf{x})$ can be obtained by minimizing the Kullback–Leibler divergence $D(q||p)$ [86], i.e.,

$$\begin{aligned} q^*(\mathbf{x}) &= \arg \min_{q \in \mathcal{Q}} D(q||p) \\ &= \arg \max_{q \in \mathcal{Q}} \underbrace{\mathbb{E}_q[-\ln q(\mathbf{x}) + \ln p(\mathbf{x}|\mathbf{y})]}_{\mathcal{L}}, \end{aligned} \quad (5.40)$$

where the expectation is over $\mathbf{x}$ according to the probability density function $q(\mathbf{x})$. $\mathcal{L}$ in (5.40) is referred to as the ELBO, which characterizes the difference of the distribution of latent variables and the distribution of respective observed variables [87]. The family $\mathcal{Q}$ manages the complexity of the optimization problem in (5.40). Consider a family $\mathcal{Q}$ that all variables in $q(\mathbf{x})$ are mutually independent, satisfying

$$q(\mathbf{x}) = \prod_{k,l} q_{k,l}(x[k, l]), \quad (5.41)$$

also known as mean field approximation. Consequently, each latent variable $x[j, k]$ is characterized by its own variational factor $q_{k,l}(x[k, l])$. With the mean field approximation, $q(\mathbf{x})$ can be determined iteratively by maximizing the ELBO. Since the noise samples and data symbols are independent, $p(\mathbf{x}|\mathbf{y})$ can be written as

$$p(\mathbf{x}|\mathbf{y}) = \prod_{k,l} p(x[k, l])p(y[k, l]|\mathbf{x}). \quad (5.42)$$

Since symplectic finite Fourier transform (SFFT) is an orthogonal transformation that does not change the statistics of noise samples, leading to a Gaussian-likelihood function $p(y[k, l]|\mathbf{x})$, $p(\mathbf{x}|\mathbf{y})$ can be rewritten in a pairwise form as

$$p(\mathbf{x}|\mathbf{y}) \propto \prod_{k,l} \zeta_{k,l}(x[k, l]) \prod_{k',l'} \psi_{k,l}(x[k, l], x[k', l']), \quad (5.43)$$

where

$$\zeta_{k,l}(x[k, l]) = p(x[k, l]) \exp \left(-\frac{\rho_{k,l}|x[k, l]|^2 + \epsilon_{k,l}x[k, l]}{\sigma^2} \right), \quad (5.44)$$

$$\psi_{k,l}(x[k, l], x[k', l']) = \exp \left(-\frac{\delta_{k,l,k',l'}x[k, l]x[k', l']}{\sigma^2} \right), \quad (5.45)$$

where $\rho_{k,l} = \sum_{k',l'} |h_{k',l'}[k,l]|^2$, $\epsilon_{k,l} = 2 \sum_{k',l'} \mathcal{R}\{h_{k',l'}[k,l]y[k',l']\}$, and $\delta_{k,l,k',l'} = 2\mathcal{R}\{h_{k,l}[k,l]h_{k',l'}^*[k',l']\}$. Substituting (5.43) and $q(\mathbf{x})$ into $\mathcal{L}$ yields

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q \left[\sum_{k,l} \ln(\psi_{k,l}(x[k,l], x[k',l'])) - \sum_{k,l} \ln \left(\frac{q_{k,l}(x[k,l])}{\zeta_{k,l}(x[k,l])} \right) \right] \\ &= \mathbb{E}_q \left[-\frac{\sum_{k,l} \delta_{k,l,k',l'} x[k,l]x[k',l']}{\sigma^2} - \sum_{k,l} \ln \left(\frac{q_{k,l}(x[k,l])}{\zeta_{k,l}(x[k,l])} \right) \right].\end{aligned}\quad (5.46)$$

A stationary point of $\mathcal{L}$ is given by the solution of $\partial \mathcal{L} / \partial q(\mathbf{x}) = 0$, which requires the partial derivatives of $\mathcal{L}$ with respect to all local functions $q_{k,l}(x[k,l])$, $\forall k, l$ being zero. Toward this, a simple iterative algorithm to update each local function is devised as follows. As an example to state the algorithm, consider the latent variable $x[k,l]$ in the n th iteration. Given the obtained local functions in the $(n-1)$ th iteration $q_{k',l'}^{(n-1)}(x[k',l'])$, $\forall \{k',l'\} \neq \{k,l\}$, setting the partial derivative $\partial \mathcal{L} / \partial q_{k,l}(x[k,l])$ to zero leads to

$$\begin{aligned}\mathbb{E}_{q \setminus k,l} \left[-\frac{1}{\sigma^2} \sum_{k',l'} \delta_{k,l,k',l'} x[k,l]x[k',l'] \right] + \ln(\zeta_{k,l}(x[k,l])) \\ - \ln(q_{k,l}^{(n)}(x[k,l])) + C = 0,\end{aligned}\quad (5.47)$$

where $q \setminus \{k,l\} = \prod_{\{k',l'\} \neq \{k,l\}} q_{k',l'}^{(n-1)}(x[k',l'])$ and C is a constant. Solving (5.47) for $q_{k,l}(x[k,l])$ results in the local distribution, expressed as

$$\begin{aligned}q_{k,l}^{(n)}(x[k,l]) &\propto \zeta_{k,l}(x[k,l]) \exp \left(\mathbb{E}_{q \setminus k,l} \left[-\frac{1}{\sigma^2} \sum_{k',l'} \delta_{k,l,k',l'} x[k,l]x[k',l'] \right] \right) \\ &\propto p(x[k,l]) \exp \left(-\frac{\rho_{k,l}|x[k,l]|^2 - m_{k,l}x[k,l]}{\sigma^2} \right),\end{aligned}\quad (5.48)$$

where $m_{k,l} = \epsilon_{k,l} - \sum_{\{k',l'\} \neq \{k,l\}} \delta_{k,l,k',l'} \mathbb{E}_{q_{k',l'}^{(n-1)}}[x[k',l']]$. From (5.48), it is observed that the variance of $x[k,l]$ is underestimated and only the noise variance is considered. This is due to the fact that the mean field approximation assumes that all variables are independent even though they are not. A possible solution to fix this underestimation is to adopt the linear response methods to correct the variance, as proposed in [88]. In a similar way, the above procedure is repeated to approximate a posteriori distributions for all the data symbols iteratively, resulting in the approximate marginals $q_{k,l}^*(x[k,l])$, $\forall k, l$. The sophisticated MAP detection in (5.25) is simplified to a problem of finding the maximum of marginal distribution $q_{k,l}^*(x[k,l])$, i.e.,

$$\hat{x}[k,l] = \arg \max_{x[k,l] \in \mathbb{A}} q_{k,l}^*(x[k,l]).\quad (5.49)$$

The complexity of this detection algorithm is reduced to the order of $|\mathbb{A}|NM$, which is significantly lower than that of the conventional MAP detection. The algorithm also achieves a better detection performance compared to that of the MP algorithm presented in Section 5.2.1.

5.2.3 Hybrid MAP and PIC Detection

This section first considers another low-complexity message passing algorithm for symbol-wise MAP detection, inspired by the sum-product algorithm [89]. Subsequently, it considers a reduced

complexity variant of this algorithm, where complexity reduction is achieved based on a hybrid MAP and parallel interference cancellation (PIC) approach [90]. The symbol-wise MAP decision rule is given by

$$\hat{x}[k, l] = \arg \max_{x[k, l] \in \mathbb{A}} \Pr(x[k, l] | \mathbf{Y}), \quad (5.50)$$

where $\mathbf{Y}$ is the received signal matrix such that $\mathbf{y} = \text{vec}(\mathbf{Y})$. In order to factorize $\Pr(x[k, l] | \mathbf{Y})$, define the following sets:

$$\mathbb{H}^{(i)} \triangleq \{h_j | 1 \leq j \leq P, j \neq i\},$$

$$\mathbb{Y}_{k,l} \triangleq \{y[(k + b_j)_N, (l + a_j)_M] | 1 \leq j \leq P\},$$

$$\mathbb{X}_{k,l}^{(i)} \triangleq \{y[(k + b_i - b_j)_N, (l + a_i - a_j)_M] | 1 \leq j \leq P, j \neq i\},$$

where P is the number of channel paths and i is the path index. The set $\mathbb{Y}_{k,l}$ contains the P received symbols that are associated with the DD domain transmitted symbol $x[k, l]$, while the set $\mathbb{X}_{k,l}^{(i)}$ contains $P - 1$ DD domain transmitted symbols that are related to the received symbol $\mathbb{Y}_{k,l}[i]$. Following the standard sum-product algorithm, $\Pr(x[k, l] | \mathbf{Y})$ can be approximated as [90]

$$\begin{aligned} \Pr\{x[k, l] | \mathbf{Y}\} &\approx \prod_{i=1}^P \sum_{\mathbb{X}_{k,l}^{(i)}} \Pr\left\{\mathbb{Y}_{k,l}[i] | \mathbb{X}_{k,l}^{(i)}, x[k, l]\right\} \\ &\Pr\left\{\mathbb{X}_{k,l}^{(i)} | \mathbf{Y}_{\neq \mathbb{Y}_{k,l}[i]}\right\} \Pr\{x[k, l]\}, \end{aligned} \quad (5.51)$$

where $\mathbf{Y}_{\neq \mathbb{Y}_{k,l}[i]}$ denotes the set of $\mathbf{Y}$ excluding the element $\mathbb{Y}_{k,l}[i]$. The probability factorization in (5.51) can be fully characterized by a probabilistic graphical model, as shown in Fig. 5.4. The observation node $\mathbb{Y}_{k,l}[i]$ receives the messages passed from the variable nodes $x[k, l]$ and $\mathbb{X}_{k,l}^{(i)}[j]$, for $1 \leq j \leq P, j \neq i$, where the a priori probability $\Pr\{x[k, l]\}$ is passed from $x[k, l]$ and the

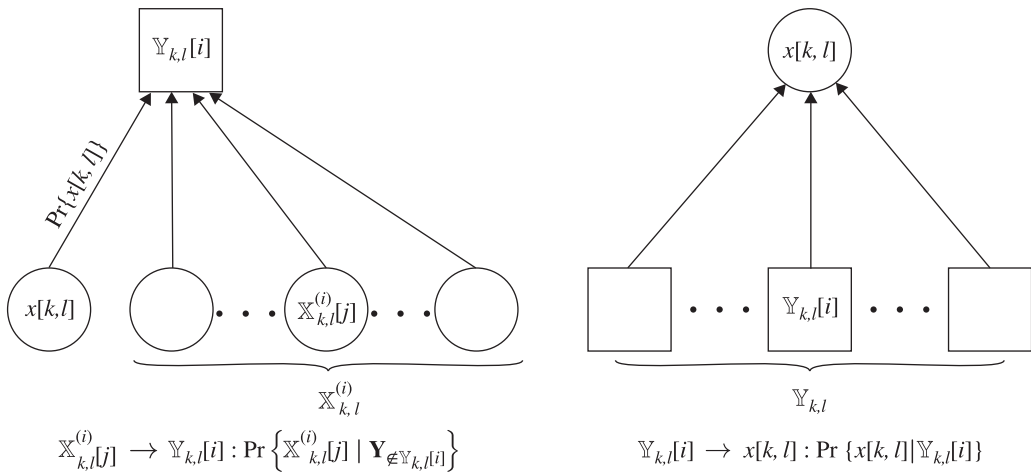


Figure 5.4 Graphical model for the near-optimal symbol-wise MAP detection algorithm. Message passing from the variable nodes to observation node, and vice versa.

probability $\Pr \{ \mathbb{X}_{k,l}^{(i)}[j] | \mathbf{Y}_{\notin \mathbb{Y}_{k,l}[i]} \}$ is passed from $\mathbb{X}_{k,l}^{(i)}[j]$. On the other hand, the messages of probability $\Pr \{ x[k, l] | \mathbb{Y}_{k,l}[i] \}$ are passed from $\mathbb{Y}_{k,l}[i]$, $1 \leq i \leq P$, to the variable node $x[k, l]$. In particular,

$$\Pr \{ \mathbb{Y}_{k,l}[i] | \mathbb{X}_{k,l}^{(i)}, x[k, l] \} = \frac{1}{\sqrt{\pi N_0}} \exp \left(- \left| \mathbb{Y}_{k,l}[i] - \sum_{j=1}^P \mathbb{H}^{(i)}[j] \mathbb{X}_{k,l}^{(i)}[j] - h_i x[k, l] \right|^2 / N_0 \right) \quad (5.52)$$

and

$$\Pr \{ x[k, l] | \mathbf{Y}_{\notin \mathbb{Y}_{k,l}[i]} \} \propto \prod_{j=1, j \neq i}^P \Pr \{ x[k, l] | \mathbb{Y}_{k,l}[j] \}. \quad (5.53)$$

According to the framework of the sum-product algorithm, the approximation in (5.51) becomes exact when the model does not contain any cycles. It can be seen that the enumeration and computation for all possible combinations of $\mathbb{X}_{k,l}^{(i)}$ result in exponential complexity in the number of paths P , which becomes prohibitive when the number of paths becomes large. A complexity-reduction approach applied on the above algorithm that can strike a good balance between the detection complexity and performance is described below.

It can be observed from (5.51) that the detection complexity mainly arises from the enumeration of all possible combinations of $\mathbb{X}_{k,l}^{(i)}$. To reduce the detection complexity, partition the set $\mathbb{X}_{k,l}^{(i)}$ into two subsets and enumerate only the combinations of one subset. This partitioning is done based on the path gains of the channel. Let L be the size of the subset with good path gains. All possible combinations in this set with good path gains are enumerated, and MAP detection is used. On the other subset, the PIC method is used. Toward this, the path gains in $\mathbb{H}^{(i)}$ are arranged in decreasing order such that $|h_m|^2 > |h_n|^2$ for $m < n$. The two subsets of $\mathbb{X}_{k,l}^{(i)}$ are defined as

$$\tilde{\mathbb{X}}_{k,l}^{(i)} \triangleq \left\{ \mathbb{X}_{k,l}^{(i)}[j] | 1 \leq j \leq L \right\}, \quad (5.54)$$

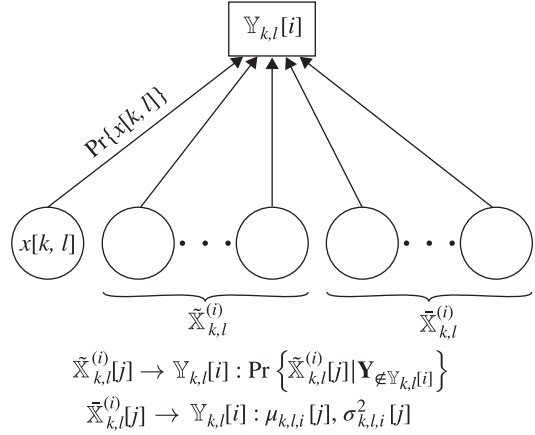
$$\bar{\mathbb{X}}_{k,l}^{(i)} \triangleq \left\{ \mathbb{X}_{k,l}^{(i)}[j] | L + 1 \leq j \leq P - 1 \right\}. \quad (5.55)$$

MAP detection is done for the subset $\tilde{\mathbb{X}}_{k,l}^{(i)}$, and PIC detection is done for the subset $\bar{\mathbb{X}}_{k,l}^{(i)}$. Specifically, the elements of $\bar{\mathbb{X}}_{k,l}^{(i)}$ are assumed to be Gaussian variables, i.e., $\bar{\mathbb{X}}_{k,l}^{(i)}[j]$ has a mean $\mu_{k,l,i}[j]$ and variance $\sigma_{k,l,i}^2[j]$. The values of $\mu_{k,l,i}[j]$ and $\sigma_{k,l,i}^2[j]$ can be derived from the a posteriori probabilities from the previous iteration. Thus, (5.52) can be modified as

$$\Pr \left\{ \mathbb{Y}_{k,l}[i] | \tilde{\mathbb{X}}_{k,l}^{(i)}, \bar{\mathbb{X}}_{k,l}^{(i)}, x[k, l] \right\} = \frac{1}{\sqrt{\pi(N_0 + \sigma^2)}} \exp \left(- \left| \mathbb{Y}_{k,l}[i] - h_i x[k, l] - \sum_{j=1}^L \mathbb{H}^{(i)}[j] \tilde{\mathbb{X}}_{k,l}^{(i)}[j] - \sum_{j=1}^{P-L-1} \mathbb{H}^{(i)}[j+L] \mathbb{E} \left\{ \bar{\mathbb{X}}_{k,l}^{(i)}[j] \right\} \right|^2 / (N_0 + \sigma^2) \right), \quad (5.56)$$

where $\sigma^2 = \sum_{j=1}^{P-L-1} |\mathbb{H}^{(i)}[j+L]|^2 \sigma_{k,l,i}^2[j]$. In terms of message passing, changing (5.52) to (5.56) does not alter the message passing from observation nodes to variable nodes, as shown in Fig. 5.4. On the other hand, for each variable node $\bar{\mathbb{X}}_{k,l}^{(i)}[j]$ that contributes less interference to the detected symbol $x[k, l]$, the corresponding mean $\mu_{k,l,i}[j]$ and variance $\sigma_{k,l,i}^2[j]$ are passed to the observation node $\mathbb{Y}_{k,l}[i]$, as shown in Fig. 5.5. The hybrid MAP and PIC algorithm described above is listed in **Algorithm 5.3**. It can be seen that $L = 0$ in this hybrid algorithm corresponds to the symbol-wise MAP algorithm

Figure 5.5 Graphical model for the hybrid MAP and PIC detection algorithm. Message passing from the variable nodes to observation node.



Algorithm 5.3 Hybrid MAP and PIC detection algorithm

Input: $\mathbf{Y}$, $\mathbb{A}$, M , N , P , $\Pr\{x[k, l]\}$, (h_i, a_i, b_i) , $i = 1, \dots, P$, n_{iter}

```

for  $n = 1; n \leq n_{\text{iter}}$  do
  for  $i = 1; i \leq P$  do
    for  $k = 0; k \leq N - 1$  do
      for  $l = 0; l \leq M - 1$  do
        Calculate the mean and variance of each element in  $\tilde{\mathbf{X}}_{k,l}^{(i)}$ 
        Enumerate all combinations of  $\tilde{\mathbf{X}}_{k,l}^{(i)}$ 
        For each possible combination of  $\tilde{\mathbf{X}}_{k,l}^{(i)}$ , compute (5.56) and
         $\Pr \left\{ \tilde{\mathbf{X}}_{k,l}^{(i)}[j] | \mathbf{Y}_{k,l}[i] \right\}$  based on (5.53)
        Compute  $\Pr \{x[k, l] | \mathbf{Y}\}$  using (5.51)
        Make hard decision of  $x[k, l]$  based (5.50)
      end
    end
  end
end

```

described earlier. Performance results have shown that choosing $L = P/2$ gives good error performance [90]. A trade-off can be established between BER performance and computation complexity by choosing a suitable value for L .

5.2.4 MCMC Sampling-Based Detection

MCMC sampling techniques are computational techniques commonly used for calculating complex integrals by expressing the integral as an expectation of some probability distribution and then estimating this expectation from the generated samples of that distribution. In this technique, a new sample value is generated randomly from the most recent sample. The transition probabilities between samples are only a function of the previous sample, and, hence, the name MCMC. MCMC techniques have been used for signal detection in multiuser and MIMO systems [16, 91–94]. Gibbs sampling is a well-known MCMC technique. Approximate solutions to the MAP decision rule in (5.25) can be obtained efficiently using Gibbs sampling-based MCMC techniques [36]. The joint

probability distribution of interest for detection is

$$\begin{aligned} \Pr(\mathbf{x}|\mathbf{y}, \mathbf{H}) &= \Pr(x_1, x_2, \dots, x_{MN}|\mathbf{y}, \mathbf{H}) \\ &\propto \exp\left(-\frac{\|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2}{\sigma^2}\right). \end{aligned} \quad (5.57)$$

The algorithm starts by initializing a random initial vector $\mathbf{x}^{(t=0)}$, where t denotes the iteration number. The MN coordinates of the $\mathbf{x}$ vector are updated in each iteration based on the coordinates of the previous iteration by sampling from the following distributions:

$$\begin{aligned} x_1^{(t+1)} &\sim \Pr(x_1|x_2^{(t)}, x_3^{(t)}, \dots, x_{MN}^{(t)}, \mathbf{y}, \mathbf{H}), \\ x_2^{(t+1)} &\sim \Pr(x_2|x_1^{(t+1)}, x_3^{(t)}, \dots, x_{MN}^{(t)}, \mathbf{y}, \mathbf{H}), \\ x_3^{(t+1)} &\sim \Pr(x_3|x_1^{(t+1)}, x_2^{(t+1)}, x_4^{(t)}, \dots, x_{MN}^{(t)}, \mathbf{y}, \mathbf{H}), \\ &\vdots \\ x_{MN}^{(t+1)} &\sim \Pr(x_{MN}|x_1^{(t+1)}, x_2^{(t+1)}, \dots, x_{MN-1}^{(t+1)}, \mathbf{y}, \mathbf{H}). \end{aligned} \quad (5.58)$$

After updating over a certain number of iterations, the distribution obtained approximately converges to the distribution in (5.57). For a received vector, the symbol vector which has the minimum ML cost $f_{\text{ML}}(\mathbf{x}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2$ in all the iterations is chosen as the detected symbol vector. This basic version of the algorithm, through simple, suffers from the problem of *stalling*, which refers to a condition of the algorithm getting trapped in some poor local solution for a long time (i.e., for many iterations). Stalling degrades the BER performance at high SNRs [91]. In order to alleviate the stalling problem and to reduce the expected number of iterations for convergence, a temperature parameter c can be introduced in the sampling distribution as

$$\Pr(x_1, x_2, \dots, x_{MN}|\mathbf{y}, \mathbf{H}) \propto \exp\left(-\frac{\|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2}{c^2\sigma^2}\right). \quad (5.59)$$

Though the use of temperature parameter c in the distribution improves performance and convergence speed, the choice of this parameter depends on the operating SNR [93]. Therefore, a sampling distribution which works for a wide range of SNRs is of interest. To this end, a possible approach is to sample from a mixture distribution that is obtained by a weighted mixing of the following two distributions: (i) the regular distribution in (5.57), which corresponds to $c = 1$ in (5.59), and (ii) a random uniform distribution, which corresponds to $c = \infty$ in (5.59). That is, the mixture distribution for sampling is given by

$$\Pr(x_1, x_2, \dots, x_{MN}|\mathbf{y}, \mathbf{H}) \propto (1 - q)\psi(c_1) + q\psi(c_2), \quad (5.60)$$

where $\psi(c) = \exp\left(-\frac{\|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2}{c^2\sigma^2}\right)$ and q is the mixing ratio. While different values for (c_1, c_2) can be chosen, the combination of $(c_1 = 1, c_2 = \infty)$, i.e., a mixture of regular distribution and uniform distribution, with a mixing ratio of $q = \frac{1}{MN}$ has been found to alleviate the stalling problem and achieve good BER performance at low complexity [16, 36]. The algorithm for the mixed-Gibbs sampling-based detection described above is listed in **Algorithm 5.4**

5.3 Cross-Domain Detection

Detection methods described in Sections 5.1 and 5.2 processed the received signal in the DD domain for detection. Most of these methods take advantage of the DD domain channel sparsity and structure to reduce detection complexity. The channel sparsity is more pronounced when the delay and

Algorithm 5.4 Mixed-Gibbs sampling-based detection algorithm**Input:** $\mathbf{y}, \mathbf{H}, \mathbf{x}^{(t=0)} \in \mathbb{A}^{MN}, n_{\text{iter}}$ **Initialize:** $t = 0, \mathbf{z} = \mathbf{x}^{(t=0)}, \theta = f_{\text{ML}}(\mathbf{x}^{(t=0)}), \mathbb{S} = \{1, 2, \dots, MN\}$ **while** ($t < n_{\text{iter}}$) **do** **for** $k = 1$ to MN **do** choose an index i randomly from $\mathbb{S}$ **if** ($k \neq i$) **then**

$$x_k^{(t+1)} \sim p\left(x_k | x_1^{(t+1)}, \dots, x_{k-1}^{(t+1)}, x_{k+1}^{(t)}, \dots, x_{MN}^{(t)}, \mathbf{y}, \mathbf{H}\right)$$

else generate pmf $p(x_k^{(t+1)} = a) \sim U[0, 1], \forall a \in \mathbb{A}, \text{ s.t. } \sum_{a \in \mathbb{A}} p(x_k^{(t+1)} = a) = 1$ sample $x_k^{(t+1)}$ from this pmf **end if** **end for** **if** ($f_{\text{ML}}(\mathbf{x}^{(t+1)}) \leq \theta$) **then** $\mathbf{z} = \mathbf{x}^{(t+1)}$ and $\theta = f_{\text{ML}}(\mathbf{x}^{(t+1)})$ **end if** $t = t + 1$ **end while****Output:** $\mathbf{z}$, the solution vector

Doppler resolutions are high. However, when an OTFS frame duration is not sufficiently long, the resultant DD domain effective channel can be dense due to the insufficient resolution of the Doppler frequency, i.e., fractional Doppler. In such a case, conventional detection methods may experience a high detection complexity since the channel response leaks into adjacent bins and channel sparsity no longer holds. This motivates to consider OTFS detection based on different domains.

In cross-domain detection [95], basic estimation/detection techniques are applied in both the time domain and DD domain, and extrinsic information from the two domains are iteratively updated. In doing so, the time domain channel sparsity and the DD domain symbol constellation constraints are exploited to achieve good performance at low complexity. In this approach, the extrinsic information is passed between the time domain and DD domain via the corresponding unitary transformations. The unitary transformation between the time and DD domain allows the detection/estimation errors in one domain to be principally orthogonal to the errors in the other domain. This can suppress the correlation between the input and output errors for each domain detection/estimation and improve the error performance. An example pair of estimation/detection methods in each of the domains can be linear MMSE (LMMSE) estimation in the time domain and symbol-by-symbol detection in the DD domain. Low complexity is achieved through an efficient implementation based on the discrete Fourier transform (DFT) kernel for cross-domain message passing between the time and DD domains. The cross-domain detection algorithm based on the above observations is presented below, considering integer delays and fractional Dopplers.

Let $\mathbf{X} \in \mathbb{A}^{M \times N}$ and $\mathbf{Y} \in \mathbb{C}^{M \times N}$ denote the DD domain transmitted symbol matrix and received symbol matrix, respectively. Furthermore, let $\mathbf{X}_{\text{TF}} \in \mathbb{C}^{M \times N}$ and $\mathbf{Y}_{\text{TF}} \in \mathbb{C}^{M \times N}$ denote the TF domain transmitted symbol matrix and received symbol matrix, respectively. Likewise, the time domain transmitted symbol matrix and received symbol matrix, respectively, are denoted by $\mathbf{Z} \in \mathbb{C}^{M \times N}$ and

$\mathbf{R} \in \mathbb{C}^{M \times N}$. Two normalized DFT matrices $\mathbf{F}_M$ and $\mathbf{F}_N$ of sizes $M \times M$ and $N \times N$, respectively, can be used to characterize the ISFFT in (4.2) [83]. Accordingly, $\mathbf{X}_{\text{TF}}$ can be written as

$$\mathbf{X}_{\text{TF}} = \mathbf{F}_M \mathbf{X} \mathbf{F}_N^H. \quad (5.61)$$

For rectangular pulse at the transmitter, the time domain transmitted symbol matrix can be obtained as [83]

$$\mathbf{Z} = \mathbf{I}_M \mathbf{F}_M^H \mathbf{X}_{\text{TF}} = \mathbf{X} \mathbf{F}_N^H. \quad (5.62)$$

The corresponding vector form of the transmitted symbols in the DD domain, TF domain, and time domain are defined as

$$\mathbf{x} \stackrel{\Delta}{=} \text{vec}(\mathbf{X}), \quad (5.63)$$

$$\mathbf{x}_{\text{TF}} \stackrel{\Delta}{=} \text{vec}(\mathbf{X}_{\text{TF}}) = (\mathbf{F}_N^H \otimes \mathbf{F}_M) \mathbf{x}, \quad (5.64)$$

$$\mathbf{z} \stackrel{\Delta}{=} \text{vec}(\mathbf{Z}) = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{x}, \quad (5.65)$$

respectively. Corresponding to the time domain transmitted symbol vector $\mathbf{z}$, the time domain effective channel $\mathbf{H}_T^{\text{eff}}$ with a reduced CP frame format is given by [83]

$$\mathbf{H}_T^{\text{eff}} = \sum_{i=1}^P h_i \mathbf{\Pi}^{a_i} \mathbf{\Delta}^{b_i + \beta_i}, \quad (5.66)$$

where $\mathbf{\Pi}$ is the permutation matrix (forward cyclic shift), i.e.,

$$\mathbf{\Pi} = \begin{bmatrix} 0 & \cdots & 0 & 1 \\ 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix}_{MN \times MN},$$

and $\mathbf{\Delta} = \text{diag}\{\alpha^0, \omega^1, \dots, \omega^{MN-1}\}$ is a diagonal matrix with $\omega \stackrel{\Delta}{=} e^{\frac{j2\pi}{MN}}$. The received time domain symbol vector $\mathbf{r}$ is given by

$$\mathbf{r} = \mathbf{H}_T^{\text{eff}} \mathbf{z} + \mathbf{v}, \quad (5.67)$$

which can be rearranged into the 2D time domain received symbol matrix $\mathbf{R}$. Assuming a rectangular pulse at the receiver, the corresponding TF domain and DD domain received symbol matrices can be obtained as

$$\mathbf{Y}_{\text{TF}} = \mathbf{F}_M \mathbf{I}_M \mathbf{R} = \mathbf{F}_M \mathbf{R} \quad (5.68)$$

and

$$\mathbf{Y} = \mathbf{F}_M^H \mathbf{Y}_{\text{TF}} \mathbf{F}_N = \mathbf{R} \mathbf{F}_N. \quad (5.69)$$

The corresponding vector form of the received symbols in the TF domain and DD domain is given by

$$\mathbf{y}_{\text{TF}} \stackrel{\Delta}{=} \text{vec}(\mathbf{Y}_{\text{TF}}) = (\mathbf{I}_N \otimes \mathbf{F}_M) \mathbf{r} \quad (5.70)$$

and

$$\mathbf{y} \stackrel{\Delta}{=} \text{vec}(\mathbf{Y}) = (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{r}. \quad (5.71)$$

Let $\mathbf{H}_{\text{TF}}^{\text{eff}}$ and $\mathbf{H}_{\text{DD}}^{\text{eff}}$ denote the effective channel matrices in the TF domain and DD domain, respectively. From (5.64), (5.66), and (5.70), $\mathbf{H}_{\text{TF}}^{\text{eff}}$ can be written as

$$\mathbf{H}_{\text{TF}}^{\text{eff}} = \sum_{i=1}^P h_i (\mathbf{I}_N \otimes \mathbf{F}_M) \mathbf{\Pi}^{a_i} \mathbf{\Delta}^{b_i + \beta_i} (\mathbf{I}_N \otimes \mathbf{F}_M^H). \quad (5.72)$$

Similarly, from (5.63), (5.66), and (5.71), $\mathbf{H}_{\text{DD}}^{\text{eff}}$ can be written as

$$\mathbf{H}_{\text{DD}}^{\text{eff}} = \sum_{i=1}^P h_i (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{\Pi}^{a_i} \mathbf{\Delta}^{b_i + \beta_i} (\mathbf{F}_N^H \otimes \mathbf{I}_M). \quad (5.73)$$

In the presence of fractional Dopplers, the DD domain effective channel matrix $\mathbf{H}_{\text{DD}}^{\text{eff}}$ in (5.73) can be dense due to power leakage, and, therefore, detection based on $\mathbf{H}_{\text{DD}}^{\text{eff}}$ may require a high complexity. The time domain effective channel matrix $\mathbf{H}_{\text{T}}^{\text{eff}}$ in (5.66) remains to be sparse and banded even in the presence of fractional Dopplers. Specifically, there are only at most P non-zero entries in each row and column of $\mathbf{H}_{\text{T}}^{\text{eff}}$. The above observations on $\mathbf{H}_{\text{DD}}^{\text{eff}}$ and $\mathbf{H}_{\text{T}}^{\text{eff}}$ matrices are shown pictorially in Fig. 5.6. This provides a motivation to devise the cross-domain detection based on the effective time domain channel $\mathbf{H}_{\text{T}}^{\text{eff}}$ in (5.67), instead of $\mathbf{H}_{\text{TF}}^{\text{eff}}$ and $\mathbf{H}_{\text{DD}}^{\text{eff}}$, in the presence of fractional Doppler. The detection algorithm is presented below.

Let $z[k]$ denote the k th element in $\mathbf{z}$, $0 \leq k \leq N - 1$. Assume that the entries in $\mathbf{x}$ independently take values from a normalized constellation $\mathbb{A}$ with equal probabilities so that $\mathbb{E}[\mathbf{x}\mathbf{x}^H] = \mathbf{I}_{MN}$. Then, the entries in the time domain OTFS signal vector $\mathbf{z}$ are also independent of each other, i.e.,

$$\mathbb{E}[\mathbf{z}\mathbf{z}^H] = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbb{E}[\mathbf{x}\mathbf{x}^H] (\mathbf{F}_N \otimes \mathbf{I}_M) = \mathbf{I}_{MN}. \quad (5.74)$$

Recalling (5.62), the time domain OTFS signal is a superposition of N DD domain OTFS symbols with specific phase rotations due to the spreading effect of IFFT. Therefore, in practical OTFS systems, where N is sufficiently large, the time domain OTFS signal behaves like a Gaussian variable according to the central limit theorem. Therefore, the entries in $\mathbf{z}$ are assumed to be i.i.d. Gaussian variables with unit variance according to (5.74). The cross-domain detection structure is shown in

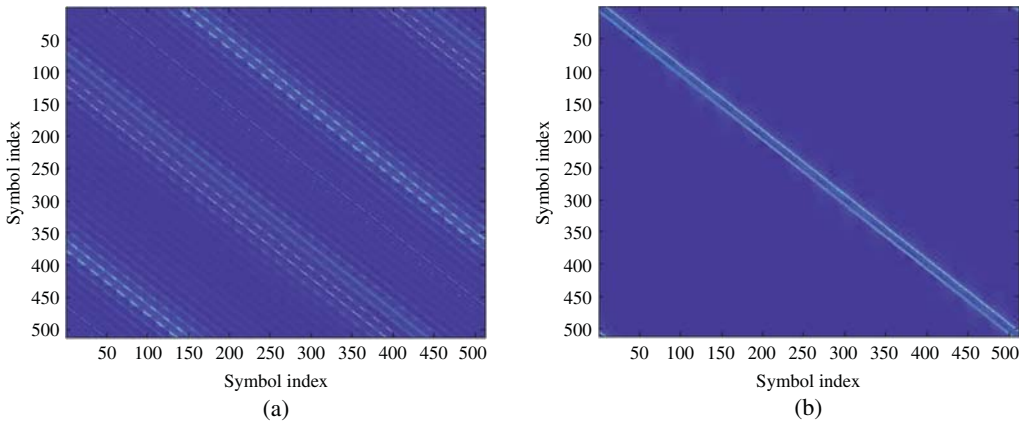


Figure 5.6 Magnitude plots of the elements of $\mathbf{H}_{\text{DD}}^{\text{eff}}$ and $\mathbf{H}_{\text{T}}^{\text{eff}}$ matrices for $P = 5$. The channel coefficient, delay spread, and Doppler spread values for the P paths are $[0.28 + 0.45i, -0.36 + 0.10i, -0.34 - 0.23i, -0.27 - 0.43i, -0.93 - 0.54i]$, $[0, 1, 14, 15, 11]$, and $[7.33, 6.92, -3.07, 4.42, -9.35]$. (a) DD domain effective channel matrix $\mathbf{H}_{\text{DD}}^{\text{eff}}$. (b) Time domain effective channel matrix $\mathbf{H}_{\text{T}}^{\text{eff}}$. Source: Li et al. [95] IEEE.

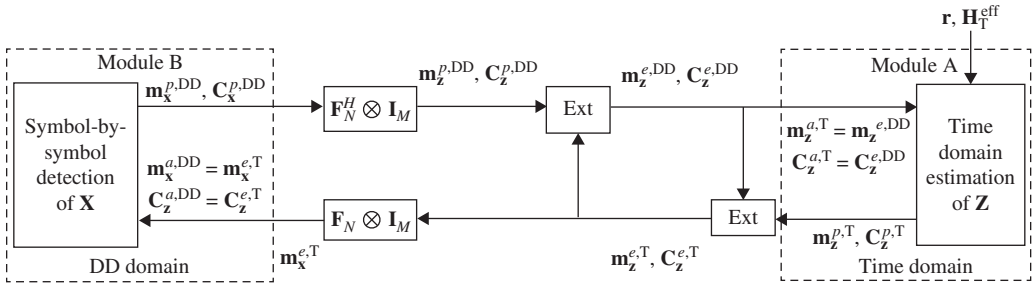


Figure 5.7 Cross-domain detection structure.

Fig. 5.7. It consists of two individual modules, one corresponding to the time domain (module A) and another corresponding to the DD domain (module B). Module A aims to estimate the time domain OTFS signal $\mathbf{z}$ and passes the estimates to module B for the detection of the DD domain OTFS symbol vector $\mathbf{x}$. Module B carries out a simple symbol-by-symbol detection for the DD domain transmitted symbol vector $\mathbf{x}$ according to the estimates of the time domain OTFS signal $\mathbf{z}$. In Fig. 5.7, “Ext” denotes the calculation of extrinsic information, while “ $\mathbf{F}_N^H \otimes \mathbf{I}_M$ ” and “ $\mathbf{F}_N \otimes \mathbf{I}_M$ ” denote the corresponding unitary transformation from the time domain to the DD domain, and vice versa, respectively. Notation-wise, $(\mathbf{m}_z^{a,T}, \mathbf{C}_z^{a,T})$, $(\mathbf{m}_z^{p,T}, \mathbf{C}_z^{p,T})$, and $(\mathbf{m}_z^{e,T}, \mathbf{C}_z^{e,T})$ denote the a priori, a posteriori and extrinsic information (denoted by a , p , and e , respectively, in the superscript) of the means (denoted by $\mathbf{m}$) and covariance matrices (denoted by $\mathbf{C}$) of the time domain (denoted by T in the superscript) OTFS signal vector $\mathbf{z}$. Likewise, $(\mathbf{m}_x^{a,DD}, \mathbf{C}_x^{a,DD})$, $(\mathbf{m}_x^{p,DD}, \mathbf{C}_x^{p,DD})$, and $(\mathbf{m}_x^{e,DD}, \mathbf{C}_x^{e,DD})$ denote the a priori, a posteriori and extrinsic information of the means and covariance matrices for the DD domain (denoted by DD in the superscript) symbol vector $\mathbf{x}$.

The algorithm is based on the idea of iteratively updating the extrinsic information based on a unitary transformation, where the multiplication of a diagonal covariance matrix of the interested variables and the underlying unitary matrix is usually performed for each iteration. However, the diagonal covariance matrix usually becomes a non-diagonal matrix after the multiplication, even if the corresponding variables are independently distributed. This is because the diagonal entries of the covariance matrix may not be of the same value due to the channel conditions, such as distortion and noise. A common solution for this issue is to simply discard the non-diagonal entries (treated as zeros), according to the fact that unitary transformation does not affect independency between variables.

Module A: An LMMSE estimator is used to estimate the TD OTFS signal $\mathbf{z}$ in module A using the TD received symbol vector $\mathbf{r}$ and the TD effective channel $\mathbf{H}_T^{\text{eff}}$ with the aid of a priori information $\mathbf{m}_z^{a,T}$ and $\mathbf{C}_z^{a,T}$ fed back from module B. Due to the i.i.d. assumption, the $\mathbf{C}_z^{a,T}$ diagonal matrix is initialized as $\mathbf{I}_{MN}$ for the first iteration. The LMMSE estimation matrix $\mathbf{W}_{\text{MMSE}}$ is given by

$$\mathbf{W}_{\text{MMSE}} = \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H \left(\mathbf{H}_T^{\text{eff}} \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H + N_0 \mathbf{I}_{MN} \right)^{-1}. \quad (5.75)$$

The a posteriori estimation output $\mathbf{m}_z^{p,T}$ of $\mathbf{z}$ is given by

$$\begin{aligned} \mathbf{m}_z^{p,T} &= \mathbf{m}_z^{a,T} + \mathbf{W}_{\text{MMSE}} (\mathbf{r} - \mathbf{H}_T^{\text{eff}} \mathbf{m}_z^{a,T}) \\ &= \mathbf{m}_z^{a,T} + \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H \left(\mathbf{H}_T^{\text{eff}} \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H + N_0 \mathbf{I}_{MN} \right)^{-1} \\ &\quad \times (\mathbf{r} - \mathbf{H}_T^{\text{eff}} \mathbf{m}_z^{a,T}), \end{aligned} \quad (5.76)$$

Algorithm 5.5 LMMSE estimation for time domain OTFS signal $\mathbf{z}$ **Input:** $\mathbf{r}$, $\mathbf{H}_T^{\text{eff}} \mathbf{m}_z^{a,T}$, $\mathbf{C}_z^{a,T}$ **Steps:**

1. Compute the LMMSE estimator matrix $\mathbf{W}_{\text{MMSE}}$ using (5.75)
2. Calculate the estimation output $\mathbf{m}_z^{p,T}$ using (5.76).
3. Calculate the MSE matrix $\mathbf{C}_z^{p,T}$ using (5.77).

return $\mathbf{m}_z^{p,T}$, $\mathbf{C}_z^{p,T}$

and the a posteriori covariance matrix $\mathbf{C}_z^{p,T}$ associated with $\mathbf{z}$ is given by

$$\mathbf{C}_z^{p,T} = \mathbf{C}_z^{a,T} - \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H \times \left(\mathbf{H}_T^{\text{eff}} \mathbf{C}_z^{a,T} (\mathbf{H}_T^{\text{eff}})^H + N_0 \mathbf{I}_{MN} \right)^{-1} \mathbf{H}_T^{\text{eff}} \mathbf{C}_z^{a,T}. \quad (5.77)$$

The diagonal entries of $\mathbf{C}_z^{p,T}$ are the a posteriori MSEs of the estimates of $\mathbf{z}$ after the LMMSE estimation, and the non-diagonal entries can be discarded (treated as zeros) because only the diagonal entries are of interest according to the i.i.d. assumption. The details of the LMMSE estimation in module A are summarized in **Algorithm 5.5**.

Cross-domain messaging from the TD to DD domain: Let $\mathbf{m}_z^{e,T}$ and $\mathbf{C}_z^{e,T}$ denote the extrinsic mean and covariance matrix from the LMMSE estimation, defined by [96]

$$\mathbf{C}_z^{e,T} = \left(\left(\mathbf{C}_z^{p,T} \right)^{-1} - \left(\mathbf{C}_z^{a,T} \right)^{-1} \right)^{-1}, \quad (5.78)$$

$$\mathbf{m}_z^{e,T} = \mathbf{C}_z^{e,T} \left(\left(\mathbf{C}_z^{p,T} \right)^{-1} \mathbf{m}_z^{p,T} - \left(\mathbf{C}_z^{a,T} \right)^{-1} \mathbf{m}_z^{a,T} \right). \quad (5.79)$$

The DD domain a priori mean of $\mathbf{x}$, i.e., $\mathbf{m}_x^{a,DD}$, and DD domain a priori covariance matrix of $\mathbf{z}$, i.e., $\mathbf{C}_z^{a,DD}$, are needed for the DD domain detection in module B. The time domain extrinsic mean of $\mathbf{x}$, obtained based on the time domain extrinsic mean of $\mathbf{z}$, is forwarded to module B as the a priori information, i.e.,

$$\mathbf{m}_x^{a,DD} = \mathbf{m}_x^{e,T} = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{m}_z^{e,T}. \quad (5.80)$$

The extrinsic covariance matrix $\mathbf{C}_z^{e,T}$ is directly passed to module B as a priori information, i.e., $\mathbf{C}_z^{a,T} = \mathbf{C}_z^{e,T}$. With $\mathbf{m}_x^{a,DD}$ and $\mathbf{C}_z^{a,T}$, the DD domain detection in module B is performed.

Module B: Considering the relation between the TD OTFS signal $\mathbf{z}$ and the DD domain OTFS symbols $\mathbf{x}$ in (5.65), the DD domain detection can be performed on the system equation given by

$$\mathbf{m}_z^{e,T} = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{x} + \hat{\mathbf{v}}, \quad (5.81)$$

where the k th element of $\hat{\mathbf{v}}$ is the white Gaussian noise with zero mean and variance $C_z^{a,DD}[k, k]$. The detection output $\hat{\mathbf{x}}$ as per the ML detection rule should satisfy

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x}} \Pr (\mathbf{m}_z^{e,T} | \mathbf{x}), \quad (5.82)$$

where

$$\Pr (\mathbf{m}_z^{e,T} | \mathbf{x}) \propto \exp \left(-(\mathbf{m}_z^{e,T} - (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{x})^H \times (\mathbf{C}_z^{a,DD})^{-1} (\mathbf{m}_z^{e,T} - (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{x}) \right). \quad (5.83)$$

The optimal maximum-likelihood sequence estimation (MLSE) detection complexity based on (5.83) is exponential in the number of non-zero entries in each row of the matrix $\mathbf{F}_N^H \otimes \mathbf{I}_M$. As the DFT matrix $\mathbf{F}_N$ is a dense matrix, the detection complexity of (5.83) can be high. Therefore, a

different probability factorization that is equivalent to (5.83) but only requires a linear detection complexity is considered by taking advantage of the unitary property of the matrix $\mathbf{F}_N^H \otimes \mathbf{I}_M$. Toward that, the ISFFT kernel $\mathbf{F}_{\text{ISFFT}} \triangleq \mathbf{F}_N^H \otimes \mathbf{I}_M$ is separated into row vectors, i.e.,

$$\mathbf{F}_{\text{ISFFT}} = \left[(\mathbf{F}_{\text{ISFFT}} [0])^T, (\mathbf{F}_{\text{ISFFT}} [1])^T, \dots, (\mathbf{F}_{\text{ISFFT}} [MN-1])^T \right]^T. \quad (5.84)$$

Noting that $\mathbf{C}_z^{a,DD}$ is a diagonal matrix and $\mathbf{m}_x^{a,DD} = (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{m}_z^{e,T}$, (5.83) can be equivalently expressed as

$$\begin{aligned} \Pr(\mathbf{m}_z^{e,T} | \mathbf{x}) &\propto \exp \left(- \sum_{k=0}^{MN-1} \frac{1}{C_z^{a,DD}[k, k]} |\mathbf{F}_{\text{ISFFT}}[k] \mathbf{m}_x^{a,DD} - \mathbf{F}_{\text{ISFFT}}[k] \mathbf{x}|^2 \right) \\ &\propto \exp \left(\sum_{k=0}^{MN-1} \frac{1}{C_z^{a,DD}[k, k]} (-|\mathbf{F}_{\text{ISFFT}}[k] \mathbf{m}_x^{a,DD}|^2 + 2\text{Re}\{x^*[k] m_x^{a,DD}[k]\} - |x[k]|^2) \right) \\ &\propto \prod_{k=0}^{MN-1} \Pr(\mathbf{m}_z^{e,T} | x[k]), \end{aligned} \quad (5.85)$$

where

$$\begin{aligned} \Pr(\mathbf{m}_z^{e,T} | x[k]) &= \Pr(m_x^{a,DD}[k] | x[k]) \\ &\propto \exp \left(\frac{1}{C_z^{a,DD}[k, k]} (2\text{Re}\{x^*[k] m_x^{a,DD}[k]\} - |x[k]|^2) \right). \end{aligned} \quad (5.86)$$

It can be observed from (5.86) that the optimal MLSE detection for DD domain symbols can be carried out in a simple symbol-by-symbol form, and the corresponding inputs for detection are the DD domain symbol estimates $\mathbf{m}_x^{a,DD}$ and the covariance matrix $\mathbf{c}_z^{a,DD}$ of the time domain OTFS signal $\mathbf{z}$. With the i.i.d. assumption of $x[k]$, the a posteriori probability $\Pr(x[k] | \mathbf{m}_z^{e,T})$ is essentially the same as that of $\Pr(\mathbf{m}_z^{e,T} | x[k])$, i.e., $\Pr(x[k] | \mathbf{m}_z^{e,T}) \propto \Pr(\mathbf{m}_z^{e,T} | x[k])$. Therefore, the a posteriori mean $m_x^{p,DD}[k]$ of $x[k]$ can be obtained as

$$\begin{aligned} m_x^{p,DD}[k] &= \mathbb{E}[x[k] | \mathbf{m}_z^{e,T}] \\ &= \sum_{i=1}^{|\mathbb{A}|} \Pr(x[k] = \mathbb{A}[i] | \mathbf{m}_z^{e,T}) \mathbb{A}[i], \end{aligned} \quad (5.87)$$

where $\mathbb{A}[i]$ is the i th symbol in the modulation alphabet $\mathbb{A}$. Also, the k th diagonal element of $\mathbf{C}_x^{p,DD}$ is the a posteriori variance of $x[k]$, given by

$$\begin{aligned} C_x^{p,DD}[k, k] &= \mathbb{E} \left[|x[k] - \mathbb{E}[x[k] | \mathbf{m}_z^{e,T}]|^2 \right] \\ &= \sum_{i=1}^{|\mathbb{A}|} \Pr(x[k] = \mathbb{A}[i] | \mathbf{m}_z^{e,T}) |\mathbb{A}[i]|^2 - |m_x^{p,DD}[k]|^2. \end{aligned} \quad (5.88)$$

Based on the a priori and a posteriori information, the extrinsic information of the DD domain detection is ready to be computed. However, the DD domain detection with (5.86) is a component-wise operation, and, therefore, in principle, the DD domain detection cannot provide any extrinsic information. Hence, the a posteriori probability of the DD domain symbols $\mathbf{x}$ is first converted to the a posteriori probability of time domain signal $\mathbf{z}$, and then the extrinsic information is computed. Specifically, the a posteriori mean $\mathbf{m}_x^{p,DD}$ and covariance matrix $\mathbf{C}_x^{p,DD}$ are served as the outputs of module B. The details of how the extrinsic information is computed based

Algorithm 5.6 Symbol-by-symbol detection for DD domain symbols $\mathbf{x}$

Input: $\mathbf{m}_x^{a,DD}$, $\mathbf{C}_x^{a,T}$, $\mathbb{A}$
for k from 0 to $MN - 1$ **do**
 Calculate the a posteriori probability of $x[k]$ using (5.86)
 Make hard decision on $x[k]$, denoted as $\hat{x}[k]$
 Compute the a posteriori mean $m_x^{p,DD}[k]$ of $x[k]$ using (5.87).
 Compute the a posteriori variance $C_x^{p,DD}[k, k]$ of $x[k]$ using (5.88)
end for
Compute $\mathbf{m}_x^{p,DD}$ of x based on $m_x^{p,DD}[k], \leq k \leq MN - 1$
Compute $\mathbf{C}_x^{p,DD}$ of x based on $C_x^{p,DD}[k, k], \leq k \leq MN - 1$
return $\mathbf{m}_x^{p,DD}$, $\mathbf{C}_x^{p,DD}$, $\hat{\mathbf{x}}$

on the outputs of module B are discussed next. Summarizing the above discussion, **Algorithm 5.6** shows the symbol-by-symbol detection for DD domain symbols.

Cross-domain messaging from the DD domain to TD: The details of message passing from the DD domain to time domain are presented here. According to (5.65), $\mathbf{m}_x^{p,DD}$ and $\mathbf{C}_x^{p,DD}$ are converted to the a posteriori mean $\mathbf{m}_z^{p,DD}$ and covariance matrix $\mathbf{C}_z^{p,DD}$ of the time domain OTFS signal $\mathbf{z}$ as

$$\mathbf{m}_z^{p,DD} = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{m}_x^{p,DD}, \quad (5.89)$$

$$\mathbf{C}_z^{p,DD} = (\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{C}_x^{p,DD} (\mathbf{F}_N \otimes \mathbf{I}_M). \quad (5.90)$$

It is noted that $\mathbf{C}_z^{p,DD}$ can be a dense matrix if the diagonal entries of $\mathbf{C}_x^{p,DD}$ are not of the same value. Here, the non-diagonal entries are discarded (i.e., treated as zeros) according to the i.i.d. assumption [97]. Similar to (5.78) and (5.79), the extrinsic information of $\mathbf{z}$ can be computed in terms of the mean and covariance matrix as

$$\mathbf{C}_z^{e,DD} = \left(\left(\mathbf{C}_z^{p,DD} \right)^{-1} - \left(\mathbf{C}_z^{a,DD} \right)^{-1} \right)^{-1}, \quad (5.91)$$

$$\mathbf{m}_z^{e,DD} = \mathbf{C}_z^{e,DD} \left(\left(\mathbf{C}_z^{p,DD} \right)^{-1} \mathbf{m}_z^{p,DD} - \left(\mathbf{C}_z^{a,DD} \right)^{-1} \mathbf{m}_z^{e,DD} \right). \quad (5.92)$$

Summarizing the above discussion, the complete cross-domain detection algorithm is listed in **Algorithm 5.7**. The complexity of the algorithm is $\mathcal{O}(M^3N^3 + 2MN \log N + MN)$ per iteration. In terms of performance, it has been shown that this cross-domain detection algorithm achieves significantly improved performance compared to the conventional detection algorithms in the presence of fractional Dopplers [95].

Algorithm 5.7 Complete cross-domain detection algorithm

Input: $\mathbf{r}$, $\mathbf{H}_T^{\text{eff}}$, $\mathbb{A}$, n_{iter}
Initialization: Set $m_z^{a,T}[k] = 0$ for $\leq k \leq MN - 1$ and $\mathbf{C}_z^{a,T} = \mathbf{I}_{MN}$
for n from 1 to n_{iter} **do**
 Perform LMMSE estimation for time domain signal $\mathbf{z}$ using **Algorithm 5.5**
 Compute $\mathbf{C}_z^{a,DD}$ based on (5.78) and $\mathbf{m}_x^{a,DD}$ based on (5.79)
 Perform symbol-by-symbol detection for DD domain $\mathbf{x}$ using **Algorithm 5.6**
 Compute $\mathbf{m}_z^{p,DD}$ and $\mathbf{C}_z^{p,DD}$ based on (5.89) and (5.90)
 Compute $\mathbf{C}_z^{a,T}$ and $\mathbf{m}_z^{a,T}$ based on (5.91) and (5.92)
end for
return $\hat{\mathbf{x}}$

6

Delay-Doppler Channel Estimation

Joint detection of all the information symbols in an OTFS frame using robust detection algorithms, as described in the Chapter 5, is necessary to ensure the robustness of OTFS modulation toward channel-induced delays and Dopplers. The signal detection operation requires the knowledge of the channel state information (CSI) at the receiver. Channel estimation at the receiver is crucial for this purpose as the detection performance depends on the accuracy of the channel estimates. In OTFS, channel estimation is carried out in the DD domain, and the channel estimation problem amounts to obtaining the estimates of the delays, Dopplers, and fade coefficients of all the resolvable paths in the DD domain. That is, if (τ_i, ν_i, h_i) denotes the three-tuple corresponding to the delay, Doppler, and fade coefficient of the i th path and P denotes the number of paths, then the task of channel estimation is to obtain the estimates of the three-tuple for all the P paths, $(\hat{\tau}_i, \hat{\nu}_i, \hat{h}_i)$, $i = 1, 2, \dots, P$. These estimates, in turn, are used to construct the corresponding estimated channel matrix $\hat{\mathbf{H}}$ for use in the detection algorithms. Although the DD domain representation of the channel response is sparse, the number of paths to be estimated can still be large since the bandwidth and time duration of an OTFS frame is typically large. This chapter deals with approaches and algorithms for DD domain channel estimation in MC-OTFS. Recall that the values of τ_i s and ν_i s can be real, with an integer part and a fractional part. That is, $\tau_i = \frac{a_i + \alpha_i}{M\Delta f}$ and $\nu_i = \frac{b_i + \beta_i}{NT}$, $i = 1, \dots, P$, where a_i and b_i are integers, and α_i and β_i are real, $-\frac{1}{2} < \alpha_i, \beta_i \leq \frac{1}{2}$. If M and N are large such that the delay resolution $\frac{1}{M\Delta f}$ and the Doppler resolution $\frac{1}{NT}$ are sufficiently small, the path delays and Dopplers can be approximated to the nearest integer values. In such scenarios, the task of estimating τ_i s and ν_i s gets limited to finding the indices of the delay and Doppler bins in the DD grid in which the different paths occur (i.e., a_i s and b_i s). This task is often termed as integer DD estimation or on-grid DD estimation in the literature. On the other hand, in situations where the fractional parts of delays and Dopplers cannot be ignored, they tend to be a source of inter-path interference (IPI) in the DD domain which can degrade performance. Therefore, the estimation of the fractional parts of τ_i s and ν_i s, in addition to the integer parts, in such cases becomes necessary for effective equalization/signal detection. In the literature, this task is termed as fractional DD estimation or off-grid DD estimation. The integer and fractional DD estimation are also termed as coarse and fine estimation, respectively. In this chapter, algorithms for integer and fractional DD estimation are introduced. The MC-OTFS system model in Chapters 4 and 5 is considered here.

In order to estimate the three-tuple $(\hat{\tau}_i, \hat{\nu}_i, \hat{h}_i)$ s and obtain an estimate of the channel matrix $\hat{\mathbf{H}}$, OTFS signal vectors $\mathbf{x}$ consisting of known pilot symbols in the DD domain are sent, and channel estimation is carried out in the DD domain. Alternately, known sequences in the time domain, e.g., pseudo-noise (PN) sequences, can be sent as pilot sequences, and the estimation problem can be

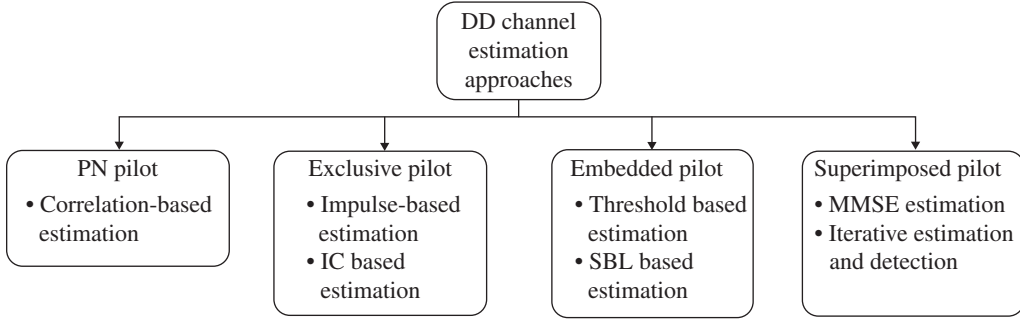


Figure 6.1 DD domain pilot-based channel estimation approaches in OTFS.

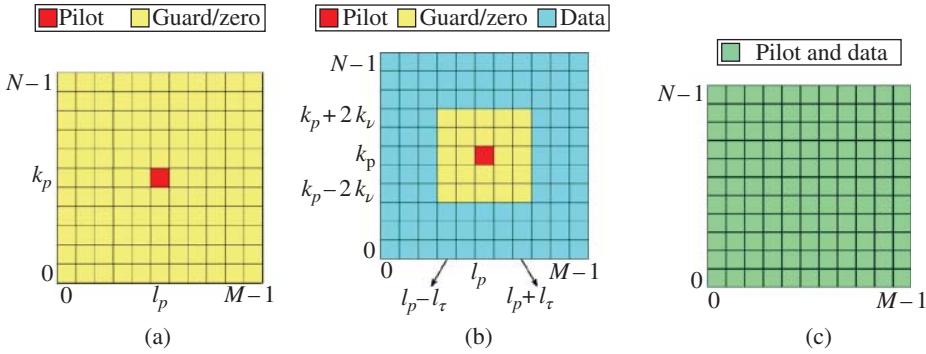


Figure 6.2 Pilot and data placement approaches for DD channel estimation: (a) exclusive pilot, (b) embedded pilot, and (c) superimposed pilots.

solved as a time-frequency shift problem [98]. For the purpose of exposition in this chapter, the DD domain pilot-based channel estimation approaches are classified into four broad categories based on how the data and pilot symbols are multiplexed in an OTFS frame. Figure 6.1 shows this classification, consisting of (i) a PN pilot approach, (ii) an exclusive pilot approach, (iii) an embedded pilot approach, and (iv) a superimposed pilot (SP) approach. In the PN pilot approach, PN sequences are sent as pilot sequences and correlation-based estimation is carried out at the receiver. In the exclusive pilot approach, OTFS frames consisting of only pilot symbols are sent for the purpose of channel estimation. The estimated channel matrix obtained during a pilot frame is used for detection in subsequent data frames. The embedded pilot approach involves embedding both pilot and data symbols in a frame with guard bins isolating the two from each other. In the superimposed pilot approach, pilot symbols are superimposed on data symbols without any guard bins. Figure 6.2 shows the above three approaches of pilot and data symbol placement in an OTFS frame. Algorithms for DD channel estimation for MC-OTFS are described in the following Sections 6.1–6.4.

6.1 PN Pilot-Based Estimation

A PN sequence in the time domain can be used as the pilot sequence for the estimation of (τ_i, ν_i, h_i) s as follows [36]. Consider the delay-Doppler channel impulse response

$$h(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i). \quad (6.1)$$

The coupling between the input signal $x(t)$ and the channel can be written as

$$\begin{aligned} y(t) &= \sum_{i=1}^P h_i x(t - \tau_i) e^{j2\pi v_i(t - \tau_i)} + v(t) \\ &= \sum_{i=1}^P h'_i e^{j2\pi v_i t} x(t - \tau_i) + v(t), \end{aligned} \quad (6.2)$$

where $h'_i = e^{-j2\pi v_i \tau_i}$, $\tau_i \in \mathbb{R}_+$, and $v_i \in \mathbb{R}$ need to be estimated. The continuous time channel model in (6.2) is converted to a discrete channel model. The discrete channel model simplifies the problem of channel estimation since waveforms become sequences. The discretization process is described as follows.

Let $\mathcal{H}$ denote the vector space of complex-valued functions on the set of finite integers $\mathbb{Z}_{N_p} = \{0, 1, \dots, N_p - 1\}$ equipped with addition and multiplication modulo N_p . The inner product in $\mathcal{H}$ is defined as

$$\langle g_1, g_2 \rangle = \sum_{n \in \mathbb{Z}_{N_p}} g_1[n] g_2^*[n], \quad g_1, g_2 \in \mathcal{H}. \quad (6.3)$$

Start with a sequence $S \in \mathcal{H}$ and transmit the following analog signal of bandwidth W :

$$S_A(t) = \sum_{n=0}^{M-1} S[n \bmod N_p] \text{sinc}(Wt - n), \quad (6.4)$$

where $M \geq N_p$. Let $M = N_p + K$, where $K = \lceil W \max(\tau_i) \rceil$. Transmit the analog signal $S_A(t)$ from time $t = 0$ to $t = M/W$. The received signal $R_A(t)$ is related to $S_A(t)$ through (6.2). $R_A(t)$ is sampled at an interval $T_s = 1/W$ from time K/W , and the sequence $R[n] = R_A(\frac{K+n}{W})$ for $n = 0, 1, \dots, N_p - 1$ is obtained. The following proposition in [98] gives the discrete channel model.

Proposition 6.1 Let $\tau_i \in \frac{1}{W}\mathbb{Z}_+$ and $v_i \in \frac{W}{N_p}\mathbb{Z}$ for $i = 1, 2, \dots, P$. Then, $R[n]$ given above satisfies

$$R[n] = \sum_{i=1}^P \alpha_i e(\omega_i n) S[K + n - \delta_i] + v[n], \quad n \in \mathbb{Z}_{N_p}, \quad (6.5)$$

where $e(t) = e^{\frac{j2\pi}{N_p} t}$, $\alpha_i = h'_i e^{(j2\pi v_i K/W)}$, $\delta_i = \tau_i W$, and $\omega_i = N_p v_i / W$.

Now, let $S, R \in \mathcal{H}$ where $R[n]$ is given by

$$R[n] = \sum_{i=1}^P \alpha_i e(\omega_i n) S[n - \delta_i] + v[n], \quad n \in \mathbb{Z}_{N_p}, \quad (6.6)$$

where $\alpha_i \in \mathbb{C}$, $\delta_i, \omega_i \in \mathbb{Z}_{N_p}$, and $v[n] \in \mathcal{H}$. Once α_i , δ_i , and ω_i are estimated, h'_i , τ_i , and v_i for $i = 1, 2, \dots, P$ can be computed using Proposition 6.1. δ_i , ω_i , and α_i are estimated using a PN pilot-based scheme described as follows.

6.1.1 Solving for $(\delta_i, \omega_i, \alpha_i)$

Consider the following simpler variant of (6.6):

$$R[n] = e(\omega_0 n) S[n - \delta_0] + v[n], \quad (6.7)$$

where $n \in \mathbb{Z}_{N_p}$ and $(\delta_0, \omega_0) \in \mathbb{Z}_{N_p} \times \mathbb{Z}_{N_p}$. The problem of estimating (δ_0, ω_0) is called the time-frequency shift problem, which is solved as follows. Define the matched filter matrix of R and S as

$$\mathcal{M}(R, S)[\delta, \omega] = \langle R[n], e(\omega n) S[n - \delta] \rangle, \quad (\delta, \omega) \in \mathbb{Z}_{N_p} \times \mathbb{Z}_{N_p}. \quad (6.8)$$

Suppose R and S satisfy (6.7), and $S \in \mathcal{H}$ is a PN sequence of norm one and length N_p . Then, the following expression for $\mathcal{M}(R, S)[\delta, \omega]$ can be obtained by the law of iterated logarithm, with probability going to one, as N_p goes to infinity [98]:

$$\begin{aligned} \mathcal{M}(R, S)[\delta, \omega] &= 1 + \epsilon'_{N_p} \quad \text{if } (\delta, \omega) = (\delta_0, \omega_0) \\ &= \epsilon_{N_p} \quad \text{if } (\delta, \omega) \neq (\delta_0, \omega_0), \end{aligned} \quad (6.9)$$

where $|\epsilon'_{N_p}| \leq \frac{1}{\sqrt{N_p}}$ and $|\epsilon_{N_p}| \leq \frac{(C+1)}{\sqrt{N_p}}$ for some constant $C > 0$. Hence, the solution to the time-frequency shift problem is to compute $\mathcal{M}(R, S)$ and choose (δ_0, ω_0) for which $\mathcal{M}(R, S)[\delta_0, \omega_0] \approx 1$. Once (δ_i, ω_i) s are obtained as above, the identity (6.9) along with the bi-linearity of the inner product gives

$$\alpha_i \approx \mathcal{M}(R, S)[\delta_i, \omega_i], \quad i = 1, 2, \dots, P. \quad (6.10)$$

Example 6.1 For illustration, the plots of the auto-correlation function (ACF) of the PN sequence and the magnitudes of the entries of the matched filter matrix are shown in Fig. 6.3. Figure 6.3(a) shows the ACF of the PN sequence with $N_p = 127$, and Figs. 6.3(b) and (c) show the magnitudes of the entries of the matched filter matrix for the time-frequency shift problem with $\alpha_0 = 1$ at

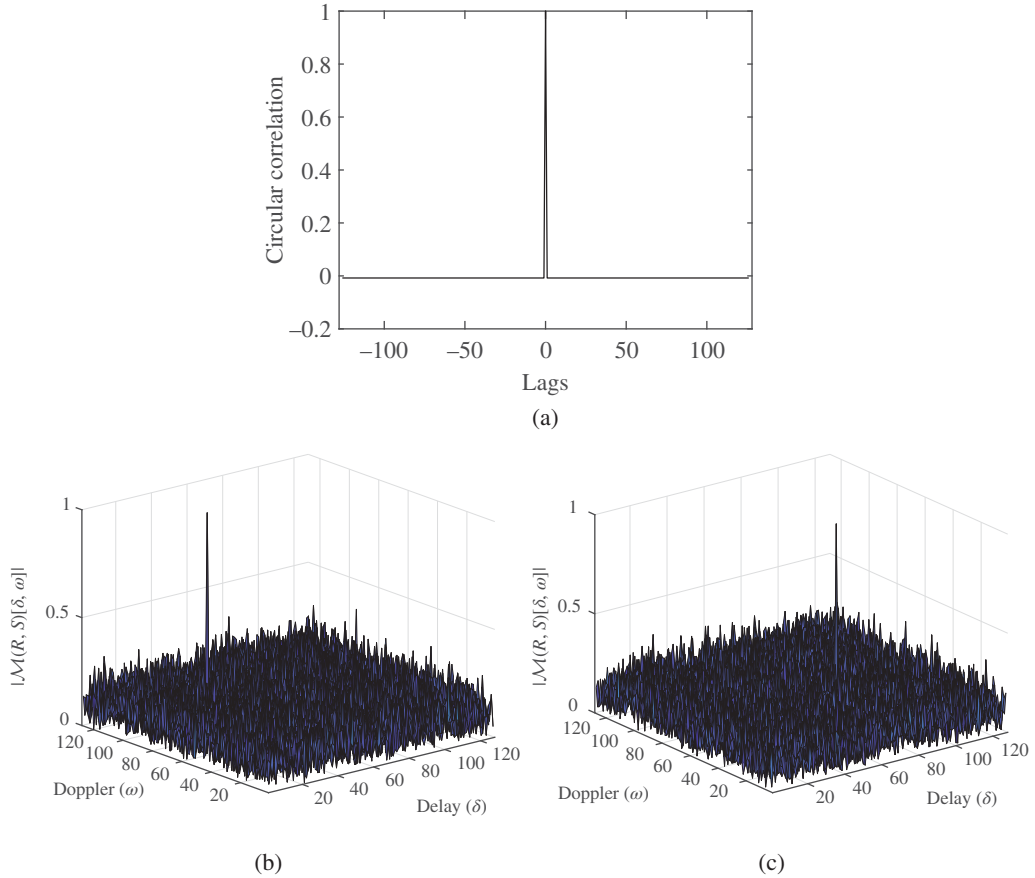


Figure 6.3 (a) Auto-correlation function of the PN sequence with $N_p = 127$. (b), (c) Magnitude of the entries of the matched filter matrix at SNR = 0 dB and $N_p = 127$ for the time-frequency shift problem with $(\delta_0, \omega_0) = (40, 90)$ and $(\delta_0, \omega_0) = (80, 60)$, respectively.

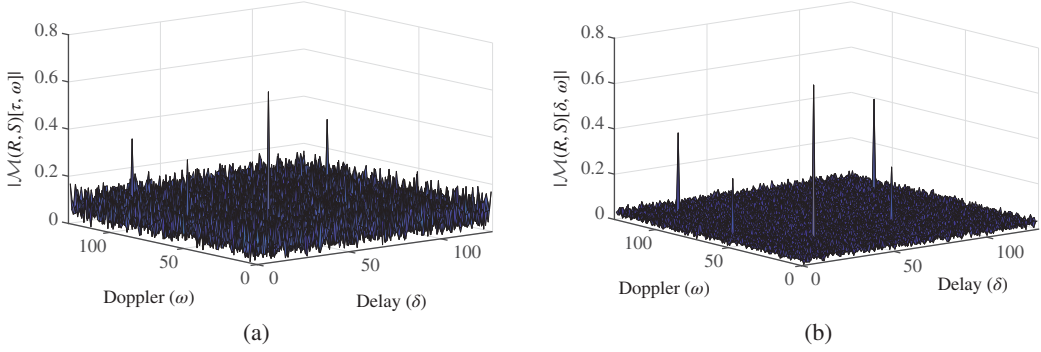


Figure 6.4 Magnitude of the entries of the matched filter matrix at SNR = 20 dB for a channel with $P = 5$, $(\delta_i, \omega_i) \in \{(10, 60), (20, 110), (30, 30), (80, 40), (110, 90)\}$ for (a) $N_p = 127$ and (b) $N_p = 1023$.

SNR = 0 dB for $(\delta_0, \omega_0) = (40, 90)$ and $(\delta_0, \omega_0) = (80, 60)$, respectively. From these figures, it is seen that the magnitude of the matched filter matrix is close to one when $(\delta, \omega) = (\delta_0, \omega_0)$ and is close to zero when $(\delta, \omega) \neq (\delta_0, \omega_0)$, as indicated by (6.9).

Example 6.2 Figure 6.4 shows the magnitude plots for a channel with $P = 5$ at SNR = 20 dB and $(\delta_i, \omega_i) \in \{(10, 60), (20, 110), (30, 30), (80, 40), (110, 90)\}$. Two values of N_p are considered. Figure 6.4(a) considers $N_p = 127$, and Fig. 6.4(b) considers $N_p = 1023$. It can be seen that the magnitudes for $(\delta, \omega) \neq (\delta_i, \omega_i)$ for $N_p = 1023$ are low compared to those for $N_p = 127$. That is, the estimate can be more accurate for larger values of N_p , which can be seen from (6.9) and the fact that $|\epsilon_{N_p}|, |\epsilon'_{N_p}| \propto 1/\sqrt{N_p}$.

6.1.2 Performance Results

Figures 6.5 and 6.6 show the performance achieved by the PN-based channel estimation. The worst case delay and Doppler spreads are assumed to be known so that the parameter K can be

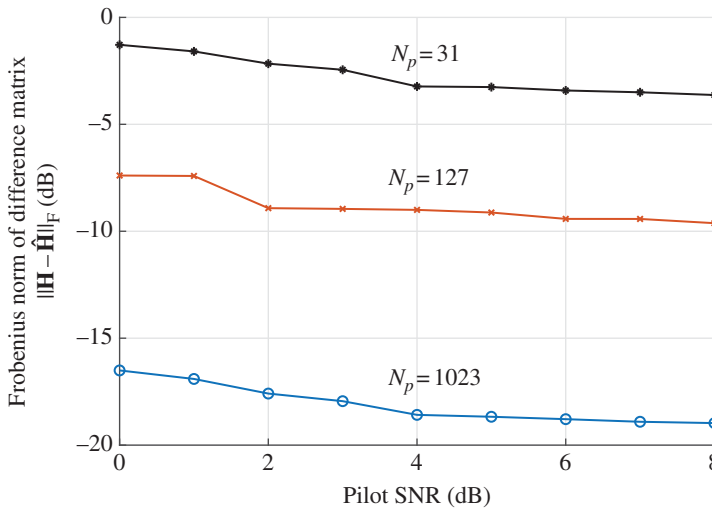


Figure 6.5 Frobenius norm of the difference between the effective channel matrix and the estimated effective channel matrix as a function of pilot SNR for various N_p values.

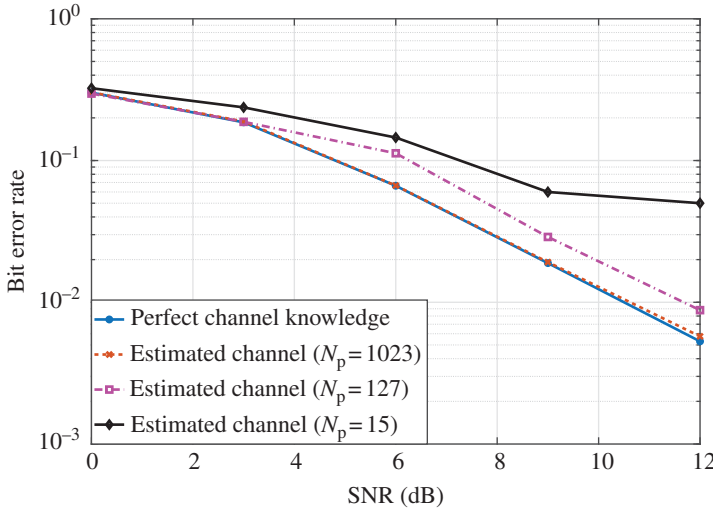


Figure 6.6 BER performance with PN-based channel estimation and MCMC detection.

chosen a priori in the process of continuous time to discrete time conversion. A frame size of $(M, N) = (32, 32)$, a subcarrier spacing of $\Delta f = 15$ kHz, a carrier frequency of 4 GHz, a channel model with $P = 5$, integer delays and Dopplers, and BPSK modulation are considered. The delay-Doppler profile considered is given by $\{(\tau_i \mu\text{s}, \nu_i \text{Hz})\} = \{(2.1, 0), (4.2, 470), (6.3, 940), (8.4, 1410), (10.4, 1880)\}$. Figure 6.5 shows the estimation error given by the Frobenius norm of the difference between the effective channel matrix and the estimated effective channel matrix as a function of pilot SNR for $N_p = 31, 127, 1023$. As expected, the estimation performance is found to improve as pilot SNR and N_p are increased. A similar trend is observed in the corresponding BER performance, as shown in Fig. 6.6, where the BER performance with increasing N_p approaches close to the performance with perfect channel knowledge.

6.2 Exclusive Pilot Approach

In the exclusive pilot approach, separate frames are used for sending pilot symbols and data symbols. Pilot frames carry only pilot symbol(s), and data frames carry only data symbols. One pilot frame per spatial coherence interval of the DD channel is sent. The channel estimated during the pilot frame is used for the detection of symbols in the data frames in that coherence interval.

6.2.1 Impulse-Based Channel Estimation

In this method, an impulse in the DD domain is sent as the pilot symbol in a pilot frame [39]. The impulse is located in the (k_p, l_p) th DD bin in the frame, and the remaining bins are filled with zeros, i.e.,

$$\begin{aligned} x_p[k, l] &= 1 \quad \text{if } (k, l) = (k_p, l_p) \\ &= 0 \quad \forall (k, l) \neq (k_p, l_p). \end{aligned} \quad (6.11)$$

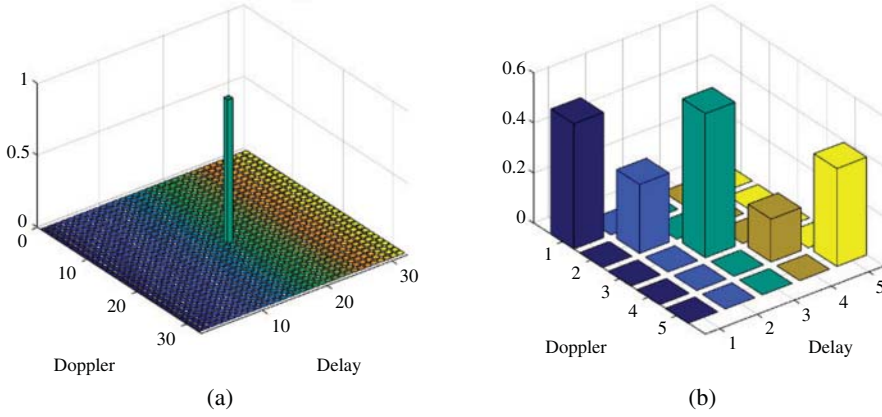


Figure 6.7 Impulse-based pilot scheme in SISO-OTFS. (a) Pilot frame and (b) channel response.

The impulse at $(k, l) = (k_p, l_p)$ is spread by the channel to the extent of the support of the channel in the DD domain. Considering the delays and Dopplers to be integers, the DD domain output at the receiver when the channel is excited by the pilot frame in (6.11) is given by (see (4.9))

$$y_p[k, l] = \sum_{i=1}^P h'_i x_p[(k - b_i)_N, (l - a_i)_M] + v[k, l], \quad (6.12)$$

where $h'_i = h_i e^{-j2\pi v_i \tau_i}$. From (6.11) and (6.12), it can be seen that the output will consist of peaks in the DD bins that are offset relative to the (x_p, l_p) th bin by (b_i, a_i) for the i th path (as illustrated Fig. 6.7). The strong peaks which are above a certain threshold are picked up as valid paths, and the indices of their DD bins normalized by the respective resolutions are declared as the estimates of τ_i s and v_i s. Also, the complex coefficients in these bins serve as estimates of h'_i s. With sufficiently large pilot SNRs, this scheme provides good estimation performance. A key advantage of this scheme is its simplicity. Also, where fractional DDs need to be estimated, the integer estimates obtained from this scheme can serve as the coarse input around which fine estimation can be carried out. The main drawbacks, however, are the large overhead involved and the need for high pilot SNRs. This method may not be effective if the channel estimation obtained during a pilot frame becomes outdated in the subsequent data frames because of small spatial coherence times. In this case, pilot frames need to be sent more often, which further compromises the throughput. Also, pilot concentration to a single/few bins can lead to high PAPR in pilot frame transmissions. Despite these drawbacks, the simplicity in using a pilot-only frame for channel estimation can serve the purpose of motivating and bringing out key features and insights of various channel estimation algorithms [99–101].

Another advantage of the impulse-based pilot approach is its suitability for simple channel estimation in MIMO-OTFS [39]. Each transmit–receive antenna pair in a MIMO-OTFS system sees a different channel having a finite support in the DD domain determined by the delay and Doppler spread of the channel. This fact can be used to estimate the channel for all the transmit–receive antenna pairs using a single pilot frame. If pilot impulses are sent from multiple transmit antennas with sufficient spacing in the DD domain, they will be received without overlap. Hence, the channel responses corresponding to different transmit–receive antenna pairs can be estimated using a single pilot frame. This is illustrated in Fig. 6.8 for a 2×1 MIMO-OTFS system with a frame size of $(M, N) = (32, 32)$ at an SNR of 4 dB. The first antenna transmits its pilot impulse at location $(k_1, l_1) = (0, 0)$, and the second antenna transmits its pilot impulse at location $(k_2, l_2) = (16, 16)$ in the same pilot frame. The impulse responses are observed to be non-overlapping at the receiver

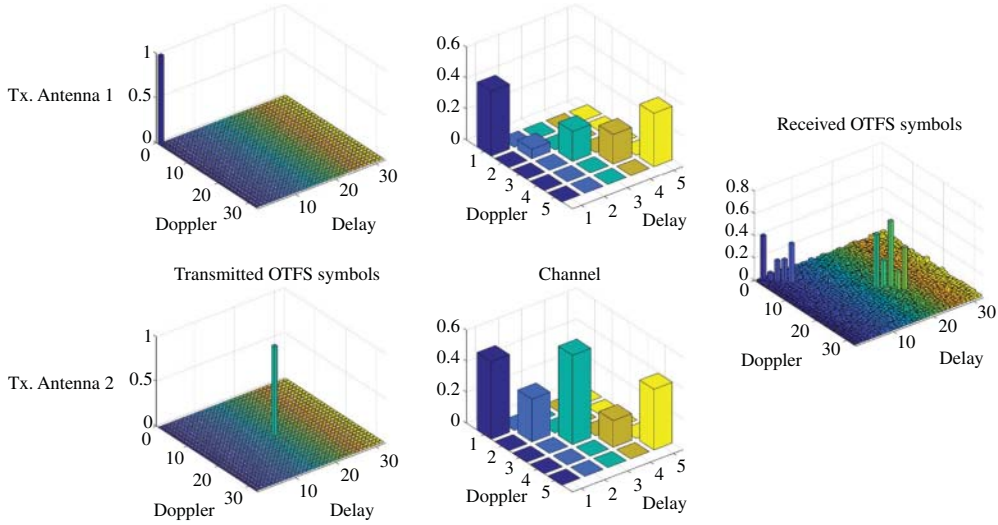


Figure 6.8 Impulse-based pilot scheme in MIMO-OTFS.

which can be estimated using the same received pilot frame. The estimated channel parameters for different transmit–receive antenna pairs can then be used to construct the estimated effective MIMO channel matrix $\hat{\mathbf{H}}_{\text{MIMO}}$ [39]. As a measure of estimation error, Fig. 6.9 shows the plot of the Frobenius norm of the difference between the effective channel matrix $\mathbf{H}_{\text{MIMO}}$ and the estimated effective channel matrix $\hat{\mathbf{H}}_{\text{MIMO}}$ as a function of pilot SNR for a 2×2 MIMO-OTFS system with the following system parameters: carrier frequency = 4 GHz, subcarrier spacing = 15 kHz, frame size $(M, N) = (32, 32)$, BPSK modulation, maximum speed = 507.6 kmph, and delay-Doppler profile $\{(\tau_i \mu\text{s}, \nu_i \text{Hz}), i = 1, \dots, 5\} = \{(2.08, 0), (4.164, 470), (6.246, 940), (8.328, 1410), (10.41, 1880)\}$. As expected, the Frobenius norm of the difference matrix decreases with pilot SNR. Figure 6.10 shows the corresponding BER performance of the system with perfect channel knowledge and estimated channel knowledge, where it is seen that the BER performance with estimated channel knowledge is quite close to that with perfect channel knowledge.

6.2.1.1 Integer DD and Fractional DD

As mentioned before, integer DD refers to the scenario where the channel path delays and Dopplers are integer multiples of the delay resolution and the Doppler resolution, respectively. On the other hand, fractional DD refers to a general scenario where the delays are not integer multiples of the delay resolution and the path Dopplers are not integer multiples of the Doppler resolution. It is noted that simple estimation schemes such as the impulse-based channel estimation scheme described above can work well for integer DDs. This is because the channel response is well localized for integer DDs. This can be seen in the channel response shown in Fig. 6.11(a), where the paths corresponding to different reflectors do not interfere with each other significantly. On the other hand, with fractional DDs, the channel response is not well localized as shown in Fig. 6.11(b), resulting in IPI. The extent of IPI is dependent on how close or far the channel paths are in the DD grid (e.g., paths 2 and 3 in Fig. 6.11(b) exhibit higher IPI compared to other paths). Because of this high IPI, the estimation performance is compromised. Therefore, more sophisticated estimation schemes are needed to achieve better performance. In Section 6.2.2, one such scheme which exploits techniques such as interference cancellation to achieve improved performance with fractional DD channel estimation under the exclusive pilot framework is presented.

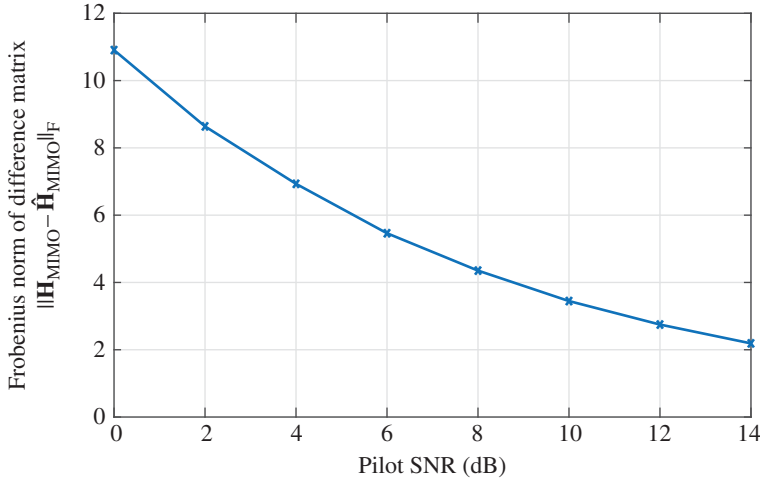


Figure 6.9 Frobenius norm of the difference between the effect channel matrix H_{MIMO} and the estimated effective channel matrix $\hat{H}_{\text{MIMO}}$ as a function of pilot SNR in a 2×2 MIMO-OTFS system.

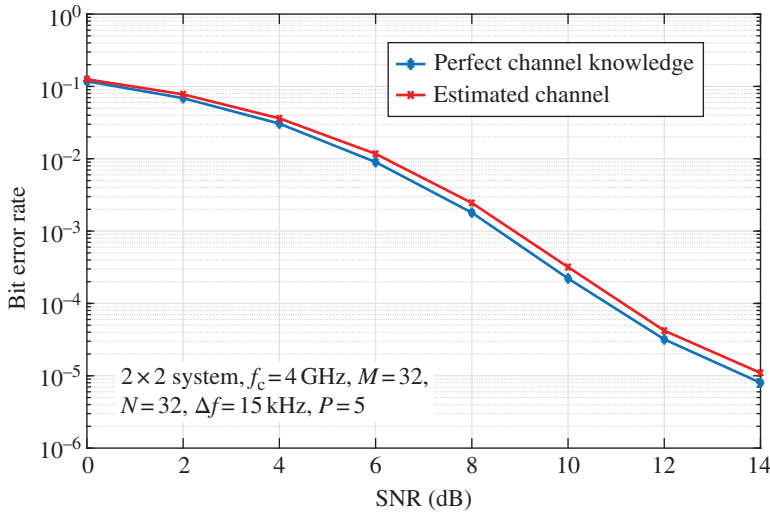


Figure 6.10 BER performance of the MIMO-OTFS system using message passing detection with the estimated channel matrix in a 2×2 MIMO-OTFS system.

6.2.2 DDIPIC-Based Estimation

This subsection presents an algorithm for channel estimation in the presence of fractional DDs considering rectangular pulses and an exclusive pilot frame. The algorithm is based on IPI cancellation. Recall that fractional DDs are sources of IPI in the DD channel response, as observed in Fig. 6.11(b). Motivated by this observation, in the estimation algorithm in this section, channel estimates are refined by sequentially cancelling the DD domain IPI. The algorithm, therefore,

is called DD domain IPI cancellation (DDIPIC) algorithm [101]. Consider the following form of input-output relation in the DD domain [100]:

$$\mathbf{y}_{\text{DD}} = \sum_{p=1}^P \alpha_p \mathbf{A}_p(\tau_p, \nu_p) \mathbf{x}_{\text{DD}} + \mathbf{w}, \quad (6.13)$$

where $(\alpha_p, \tau_p, \nu_p)$ are the channel fade coefficient, delay, and Doppler of the p th path, $p = 1, 2, \dots, P$, $\mathbf{x}_{\text{DD}}$ is the $MN \times 1$ DD domain pilot frame vector, $\mathbf{y}_{\text{DD}} \in \mathbb{C}^{MN \times 1}$ is the corresponding DD domain received vector, $\mathbf{w} \in \mathbb{C}^{MN \times 1}$ is the noise vector whose entries are distributed as i.i.d. $\mathcal{CN}(0, \sigma^2)$, $\mathbf{A}_p$ is an $MN \times MN$ matrix whose entries are given by $\mathbf{A}_p[q', q] = e^{-j2\pi\tau_p\nu_p a a'}$, where

$$a = \frac{1}{N} \sum_{n=0}^{N-1} e^{-j2\pi n \left(\frac{k' - k}{N} - \frac{\nu_p}{\Delta f} \right)}, \quad (6.14)$$

$$a' = \frac{1}{M} \sum_{m=0}^{M-1} e^{j2\pi \frac{m}{M} \left(l' - l - M \frac{\tau_p}{T} \right)} r_{\tau_p, \nu_p, k, l'}(m), \quad (6.15)$$

and

$$\begin{aligned} r_{\tau_p, \nu_p, k, l'}(m) = & \sum_{s=-m}^{M-1-m} e^{j2\pi \frac{s l'}{M}} \left[\left(1 - \frac{\tau_p}{T} \right) e^{j\pi \left(1 + \frac{\tau_p}{T} \right) \left(\frac{\nu_p}{\Delta f} - s \right)} \text{sinc} \left(\left(1 - \frac{\tau_p}{T} \right) \left(\frac{\nu_p}{\Delta f} - s \right) \right) \right. \\ & \left. + e^{-j2\pi \frac{k}{N}} \left(\frac{\tau_p}{T} \right) e^{j\pi \frac{\tau_p}{T} \left(\frac{\nu_p}{\Delta f} - s \right)} \text{sinc} \left(\left(\frac{\tau_p}{T} \right) \left(\frac{\nu_p}{\Delta f} - s \right) \right) \right]. \end{aligned} \quad (6.16)$$

An exclusive pilot frame is transmitted for the purpose of obtaining estimates of the DD channel parameters, $(\hat{\alpha}_p, \hat{\tau}_p, \hat{\nu}_p)$, $p = 1, 2, \dots, P$, at the receiver. These estimated parameters are then used to construct the effective channel matrix $\hat{\mathbf{H}} = \sum_{p=1}^P \hat{\alpha}_p \mathbf{A}_p \in \mathbb{C}^{MN \times MN}$ (see (6.13)), which is used for the detection of data symbols. Equation (6.13) can be written in an alternate form as

$$\begin{aligned} \mathbf{y} &= \sum_{p=1}^P \mathbf{b}_p(\tau_p, \nu_p) \alpha_p + \mathbf{w} \\ &= \mathbf{B}(\boldsymbol{\tau}, \boldsymbol{\nu}) \boldsymbol{\alpha} + \mathbf{w}, \end{aligned} \quad (6.17)$$

where $\mathbf{b}_p(\tau_p, \nu_p) = \mathbf{A}_p \mathbf{x}_{\text{DD}} \in \mathbb{C}^{MN \times 1}$, $\mathbf{B}(\boldsymbol{\tau}, \boldsymbol{\nu}) = [\mathbf{b}_1(\tau_1, \nu_1) \cdots \mathbf{b}_P(\tau_P, \nu_P)] \in \mathbb{C}^{MN \times P}$, $\boldsymbol{\alpha} = [\alpha_1 \cdots \alpha_P]^T \in \mathbb{C}^{P \times 1}$, and $\mathbf{w} \sim \mathcal{CN}(0, \sigma^2) \in \mathbb{C}^{MN \times 1}$. The maximum-likelihood estimate of the $(\boldsymbol{\alpha}, \boldsymbol{\tau}, \boldsymbol{\nu})$ -tuple can

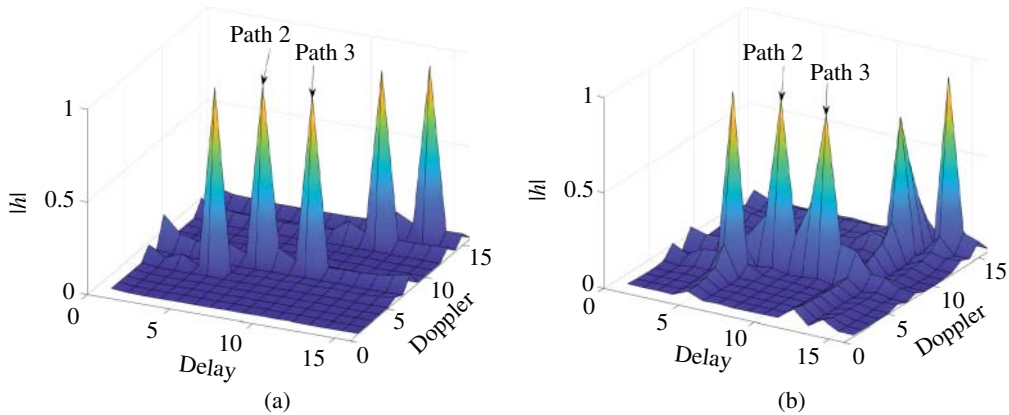


Figure 6.11 Received DD domain pilot signal for integer and fractional DDs. (a): Integer DD and (b) fractional DD.

be evaluated as

$$(\hat{\alpha}, \hat{\tau}, \hat{\nu}) = \underset{\alpha, \tau, \nu}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{B}(\tau, \nu)\alpha\|^2, \quad (6.18)$$

which is an optimization problem with three unknowns. In order to reduce the complexity, τ and ν are estimated first and α is estimated subsequently. Toward this, note that for a given (τ, ν) , the maximum-likelihood estimate of α is given by

$$\hat{\alpha} = [\mathbf{B}^H(\tau, \nu)\mathbf{B}(\tau, \nu)]^{-1} \mathbf{B}^H(\tau, \nu)\mathbf{y}. \quad (6.19)$$

Representing (6.18) as $(\mathbf{y} - \mathbf{B}(\tau, \nu)\alpha)^H (\mathbf{y} - \mathbf{B}(\tau, \nu)\alpha)$, simplifying, and substituting in (6.19) yield the estimate of (τ, ν) as

$$[\hat{\tau}, \hat{\nu}] = \underset{\tau, \nu}{\operatorname{argmax}} [\Phi(\mathbf{B})], \quad (6.20)$$

where $\Phi(\mathbf{B}) = \mathbf{y}^H \mathbf{B}(\tau, \nu) (\mathbf{B}^H(\tau, \nu) \mathbf{B}(\tau, \nu))^{-1} \mathbf{B}^H(\tau, \nu) \mathbf{y}$. Using the estimates $\hat{\tau}$ and $\hat{\nu}$, the estimate of α is obtained as

$$\hat{\alpha} = [\mathbf{B}^H(\hat{\tau}, \hat{\nu})\mathbf{B}(\hat{\tau}, \hat{\nu})]^{-1} \mathbf{B}^H(\hat{\tau}, \hat{\nu})\mathbf{y}. \quad (6.21)$$

Equation (6.20) is solved by the DDIPIC algorithm. The estimation of delay and Doppler values is carried out on a path by path basis. Furthermore, this estimation is carried out in two phases, namely, (i) coarse estimation phase, where the search space is over integer multiples of $(\frac{1}{M\Delta f}, \frac{1}{NT})$ and (ii) fine estimation phase, where the search space is over fractional values. The algorithm runs for a maximum of $P_{\max}$ iterations. In each iteration, one path is estimated so that a maximum of $P_{\max}$ paths are estimated. $P_{\max}$ is chosen to be more than the number of paths P in the channel. The algorithm begins by initializing $\mathbf{B}(\tau, \nu) = [\mathbf{b}_1(\tau_1, \nu_1) \ \mathbf{b}_2(\tau_2, \nu_2) \ \cdots \ \mathbf{b}_{P_{\max}}(\tau_{P_{\max}}, \nu_{P_{\max}})] = \mathbf{0}_{MN \times P_{\max}}$.

6.2.2.1 Coarse Estimation

Coarse estimation is carried out over the search space defined by $\tau \in \left\{ \frac{0}{M\Delta f}, \frac{1}{M\Delta f}, \dots, \frac{L}{M\Delta f} \right\}$, $\nu \in \left\{ -\frac{K}{NT}, \dots, \frac{0}{NT}, \dots, \frac{K}{NT} \right\}$, where $L = \lceil \tau_{\max} M\Delta f \rceil$, $K = \lceil \nu_{\max} NT \rceil$, and $\tau_{\max}, \nu_{\max}$ are the maximum delay and Doppler, respectively. For estimating the first path, $\mathbf{b}_1(\tau_1, \nu_1)$ is evaluated with (τ_1, ν_1) taking all possible combinations of (τ, ν) in the search space. To obtain the optimal coarse estimate (τ'_1, ν'_1) , the cost function $\Phi(\mathbf{B})$ in (6.20) is maximized.

6.2.2.2 Fine Estimation

Fine estimation is carried out in a search space around the optimal coarse value obtained in the previous step. The fine estimation of the parameters is carried out iteratively, where the search space is narrowed down as the iterations progress. The search space along the delay and Doppler axes is divided into $(2\lfloor \frac{m_\tau}{2} \rfloor + 1)$ and $(2\lfloor \frac{n_\nu}{2} \rfloor + 1)$ equally spaced bins, respectively. This iterative procedure is presented in Algorithm 6.1, where $\Gamma = \{0, \dots, \lfloor \frac{m_\tau}{2} \rfloor\}$ for $\tau'_1 = 0$ and $\Gamma = \{-\lfloor \frac{m_\tau}{2} \rfloor, \dots, 0, \dots, \lfloor \frac{m_\tau}{2} \rfloor\}$ for $\tau'_1 > 0$, $\Lambda = \{-\lfloor \frac{n_\nu}{2} \rfloor, \dots, 0, \dots, \lfloor \frac{n_\nu}{2} \rfloor\}$, and the operator $\otimes$ denotes the Cartesian product of two sets. At the end of **Algorithm 6.1**, fine estimates for the first path $\hat{\tau}_1$ and $\hat{\nu}_1$ are obtained.

For the estimation of parameters of the t th path ($t > 1$), columns 1, 2, $\dots$, $t - 1$ in $\mathbf{B}$ are filled using the already obtained $(\hat{\tau}_p, \hat{\nu}_p)$ s, $p = 1, 2, \dots, t - 1$. The coarse estimates for τ_t and ν_t are obtained by maximizing the cost function in (6.20) over different values of (τ, ν) in the t th column of $\mathbf{B}$, as described in the coarse estimation stage. This is followed by the fine estimation stage using **Algorithm 6.1**, which gives the fine estimates $(\hat{\tau}_t, \hat{\nu}_t)$.

Algorithm 6.1 Fine estimation of channel parameters

Input: Coarse estimates (τ'_p, ν'_p) , refinements (m_τ, n_ν) , convergence indicators $(\epsilon_\tau, \epsilon_\nu)$, and max_iter

Initialize: $s = 1$, $\hat{\tau}^{(1)} = \tau'_p$, and $\hat{\nu}^{(1)} = \nu'_p$

repeat

- search width in delay, $w_\tau^{(s)} = \frac{1}{M\Delta f m_\tau^{s-1}}$
- search width in Doppler, $w_\nu^{(s)} = \frac{1}{NT n_\nu^{s-1}}$
- $\mathbf{F}^{(s)} = \{w_\tau^{(s)}\Gamma + \hat{\tau}^{(s)}\} \otimes \{w_\nu^{(s)}\Lambda + \hat{\nu}^{(s)}\}$, search space for fractional DD
- $\hat{\tau}^{(s+1)}, \hat{\nu}^{(s+1)} = \underset{(\tau, \nu) \in \mathbf{F}^{(s)}}{\operatorname{argmax}} \Phi(\mathbf{B})$
- $s = s + 1$

until $s = \text{max_iter}$ or $(|\hat{\tau}^{(s+1)} - \hat{\tau}^{(s)}| < \epsilon_\tau \text{ and } |\hat{\nu}^{(s+1)} - \hat{\nu}^{(s)}| < \epsilon_\nu)$

Output: $\hat{\tau}_p = \hat{\tau}^{(s+1)}$ and $\hat{\nu}_p = \hat{\nu}^{(s+1)}$

6.2.2.3 Stopping Criterion

At the end of fine estimation for t th path, $t > 1$, the matrix $\mathbf{B}(\hat{\tau}, \hat{\nu})$ is obtained, using which $\hat{\mathbf{a}}$ is obtained using (6.21). A residue vector $\mathcal{E}^{(t)}$ is obtained as $\mathcal{E}^{(t)} = \mathbf{y} - \mathbf{B}(\hat{\tau}, \hat{\nu})\hat{\mathbf{a}}$. If $\|\mathcal{E}^{(t)} - \mathcal{E}^{(t-1)}\|^2 > \epsilon$ and $t < P_{\max}$, then the algorithm makes $t = t + 1$ and continues with the estimation of the next path. The algorithm stops if $\|\mathcal{E}^{(t)} - \mathcal{E}^{(t-1)}\|^2 < \epsilon$ or $t = P_{\max}$.

6.2.2.4 Refinement of Parameters

After the estimation of the t th path, $1 < t < P_{\max}$, if the stopping criterion is not met, the previously obtained estimates are refined before estimating the parameters of the $(t + 1)$ th path. This refinement proceeds as follows. For the z th path, with $1 \leq z \leq t$, the estimates $(\hat{\tau}_i, \hat{\nu}_i)$ s with $i = 1, 2, \dots, z - 1, z + 1, \dots, t$ are used to obtain the matrix $\mathbf{B}$ (i.e., fill all the columns in $\mathbf{B}$ except the z th column). Then, the refined estimates of the z th path are evaluated again, by optimizing the cost function in (6.20) for the z th column of $\mathbf{B}$. This is carried out first over the coarse search space and then over the fine search space. Following this, the estimate of parameters of the $(t + 1)$ th path is obtained using the refined estimates of all the paths till t . This is illustrated with the following example. When $t = 3$, three paths are estimated, and the stopping criterion is not met, then

- the first path is refined using the second and third paths (both unrefined),
- the second path is refined using the refined first path and the third path (unrefined), and
- the third path is refined using the refined first and second paths.

The estimate of the parameters of the fourth path is obtained using the refined first, second, and third path estimates. It is noted that an estimated path maybe refined multiple times depending on how many paths are estimated before the convergence of the DDIPIC algorithm.

6.2.2.5 Choice of ϵ

A high value of ϵ can lead to paths (with small amplitudes) being missed, while a small value of ϵ can lead to false paths or multi-checks, wherein an already estimated path is picked for estimation again. Simulations have shown that the chances of multi-checks and misses are reduced when ϵ is chosen to be about 1% of the noise energy in the frame. Consequently, a suggested value for ϵ is $0.01 MN\sigma^2$ [101].

6.2.2.6 Complexity

Table 6.1 shows the computation complexity of the estimator (6.20) in terms of the number of complex additions and multiplications. The sum of the last column $C = 2P_{\max}MN + 2P_{\max}^2MN + P_{\max}^2 + \mathcal{O}(P_{\max}^3) - 1$ is the total number of operations required to compute (6.20). Table 6.2 shows

Table 6.1 Computational complexity for cost function.

Operation	Complex multiplications	Complex additions	Total complexity
$\Xi = \mathbf{y}^H \mathbf{B}(\boldsymbol{\tau}, \boldsymbol{\nu})$	$P_{\max} MN$	$P_{\max} (MN - 1)$	$2P_{\max} MN - P_{\max}$
$\Upsilon = \mathbf{B}^H(\boldsymbol{\tau}, \boldsymbol{\nu}) \mathbf{B}(\boldsymbol{\tau}, \boldsymbol{\nu})$	$P_{\max}^2 MN$	$P_{\max}^2 (MN - 1)$	$2P_{\max}^2 MN - P_{\max}^2$
Υ^{-1}	—	—	$\mathcal{O}(P_{\max}^3)$
$\Xi \Upsilon^{-1}$	$P_{\max}^2$	$P_{\max} (P_{\max} - 1)$	$2P_{\max}^2 - P_{\max}$
$\Xi \Upsilon^{-1} \Xi^H$	$P_{\max}$	$P_{\max} - 1$	$2P_{\max} - 1$

Table 6.2 Complexity in different stages of estimation.

Estimation stage	Complexity (value $\times C$)
Coarse estimation	$C' = (L + 1)(2K + 1)P_{\max}$
Fine estimation	$C'' = \left(2 \lfloor \frac{m_\tau}{2} \rfloor + 1\right) \left(2 \lfloor \frac{n_\nu}{2} \rfloor + 1\right) P_{\max}$
Refinement	$\left(\frac{(P_{\max} - 1)P_{\max}}{2} - 1\right) (C' + C'')$
Channel coefficient	1

the maximum number of times the cost function in (6.20) is evaluated in each stage of estimation. The sum of the last column multiplied with C denotes the maximum complexity of the proposed algorithm.

6.2.2.7 Performance

Consider a system with $M = 64$, $N = 32$, $\Delta f = 30$ kHz, and $f_c = 5.1$ GHz, a channel with $P = 5$ paths with a line-of-sight (LOS) path, and a Rice factor of 15 dB [100]. The delays of the first and second paths are taken to be 0.667 μ s and 0.867 μ s, respectively, and the delays of other paths are uniformly distributed in (0.867 μ s, 7 μ s]. Doppler frequencies for all the paths are generated from Jake's Doppler spectrum using $\nu_p = \nu_{\max} \cos(\theta_p)$, where θ_p is uniformly distributed in $(0, 2\pi]$ and $\nu_{\max}$ is 1700 Hz. The fixed absolute squared value of the channel gain of the LOS path is taken according to the Rice factor, and the exponential power delay profile is used for the other paths as in [100]. Furthermore, $m_\tau = n_\nu = 10$, $\epsilon_\tau = 10^{-10}$, $\epsilon_\nu = 10^{-2}$, $P_{\max} = 15$, and $\epsilon = 0.01 MN \sigma^2$.

Figure 6.12 shows the MSE performance with estimation refinement using the DDIPIC algorithm, as a function of pilot SNR. The MSE of the channel coefficient, delay, and Doppler values are presented in Figs. 6.12(a)–(c), respectively. The MSE performance without estimation refinement is also presented for comparison. It is seen that the MSE performance achieved without refinement floors, whereas the MSE performance improves significantly with refinement. The estimated channel parameters are used to obtain the estimated effective channel matrix, whose normalized mean square error (NMSE) is evaluated as $\mathbb{E} \left[\frac{\|\mathbf{H} - \hat{\mathbf{H}}\|_F^2}{\|\mathbf{H}\|_F^2} \right]$. Figure 6.13(a) shows the NMSE performance without and with estimation refinement. It is seen that while the NMSE performance improves with increasing values of pilot SNRs in both cases of with and without estimation refinement, the performance achieved with estimation refinement is better. For example, an NMSE value of -30 dB is obtained at around 10 dB of pilot SNR with estimation refinement, compared to at around 15 dB of pilot SNR without estimation refinement. Figure 6.13(b) shows the BER vs. SNR

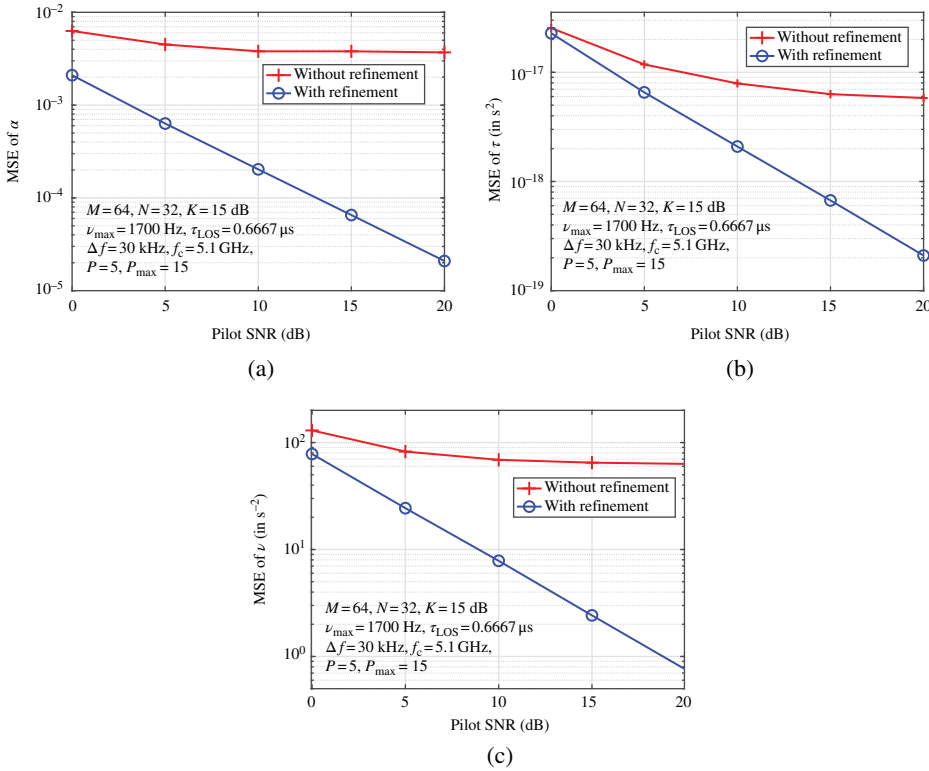


Figure 6.12 MSE of the estimated parameters as a function of pilot SNR. (a) Channel coefficient, (b) delay, and (c) Doppler.

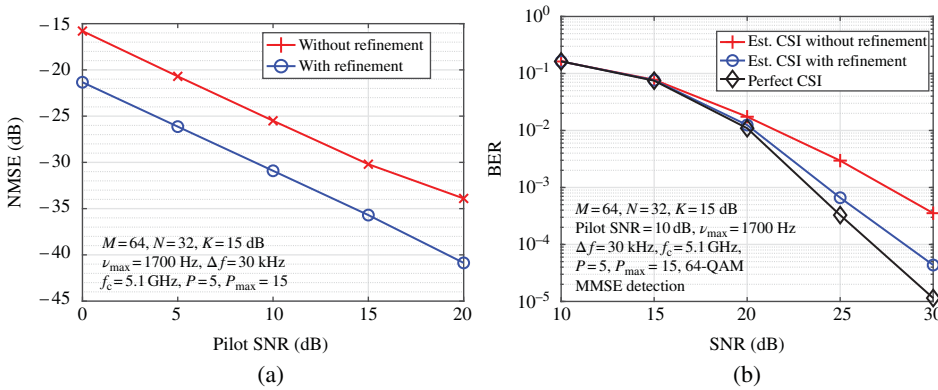


Figure 6.13 NMSE and BER performance with estimated channel parameters. (a) NMSE performance and (b) BER performance.

performance with estimation refinement using the DDIPIC algorithm for a pilot SNR of 10 dB. The BER performance with perfect CSI is also added as a benchmark, along with the performance without estimation refinement. MMSE detection is used, and the data symbols are drawn from the 64-QAM modulation alphabet. It is observed that the scheme with estimation refinement achieves better performance compared to that without estimation refinement. For example, a BER

of 10^{-3} is achieved at 24 dB SNR with estimation refinement, while the same BER is achieved at about 28 dB SNR without refinement. Furthermore, the performance with estimation refinement is observed to close to that with perfect CSI.

6.3 Embedded Pilot Approach

The drawbacks of a large overhead and aging of channel estimates in the exclusive pilot approach can be alleviated by the embedded pilot approach. In the embedded pilot approach, both pilot and data symbols are embedded in each frame. The channel is estimated every frame, and the estimated channel is used for the detection of data symbols in the same frame. An embedded frame can be constructed in such a way that pilot and data symbols are separated by guard symbols in between so that pilot and data symbols do not interfere with each other significantly. In such cases, simple threshold-based estimation schemes (as described for exclusive pilot frames in Section 6.2) can work well. Alternately, an embedded frame can consist of pilot and data symbols without any guard symbols in between. In such cases, mutual interference between pilot and data symbols can degrade performance, and, hence, more sophisticated estimation schemes will be needed. In Sections 6.3.1 and 6.3.2, channel estimation for both the above embedded frame constructions are considered. A threshold-based channel estimation for the first construction [102] and a sparse Bayesian learning (SBL)-based channel estimation for the second construction [103] are presented.

6.3.1 Embedded Pilot with Guard Symbols

In embedded frames with guard symbols, each frame is constructed by suitably arranging a pilot symbol, guard symbols, and data symbols (see Fig. 6.2(b)). Threshold-based channel estimation for such embedded frames is described below.

6.3.1.1 Threshold-Based Estimation

This estimation scheme is described for SISO-OTFS with integer delays and Dopplers, although it can be extended to MIMO and multiuser uplink/downlink with fractional Dopplers [102]. Considering integer delays and Dopplers, let l_τ and k_v denote the delay and Doppler taps corresponding to the largest $\tau_{\max}$ and $\nu_{\max}$, respectively. The input-output relation between $y[k, l]$ and $x[k, l]$ can be written as [104]

$$y[k, l] = \sum_{k'=-k_v}^{k_v} \sum_{l'=0}^{l_\tau} b[k', l'] \hat{h}[k', l'] x[[k - k']_N, [l - l']_M] + v[k, l], \quad (6.22)$$

where $\hat{h}[k', l'] = h[k', l'] e^{-j2\pi \frac{k'}{NT} \frac{l'}{M\Delta f}}$, $b[k', l'] \in \{0, 1\}$ is the path indicator, i.e., $b[k', l'] = 1$ indicates that there is a path with Doppler tap k' and delay tap l' with the corresponding path magnitude $\hat{h}[k', l']$ and $b[k', l'] = 0$ indicates that there is no path with Doppler tap k' and delay tap l' and corresponding $\hat{h}[k', l'] = 0$, $v[k, l] \sim \mathcal{CN}(0, \sigma^2)$ is the additive noise, and $[\cdot]_N$ and $[\cdot]_M$ denote modulo- N and modulo- M operations, respectively. Note that

$$\sum_{k'=-k_v}^{k_v} \sum_{l'=0}^{l_\tau} b[k', l'] = P, \quad (6.23)$$

where P is the total number of paths, and that each path circularly shifts the transmitted symbols by the delay and Doppler taps.

Let x_p denote the pilot symbol with a pilot SNR given by $\text{SNR}_p = \frac{|x_p|^2}{\sigma^2}$, x_d denote the data symbols with data SNR of $\text{SNR}_d = \frac{\mathbb{E}(|x_d|^2)}{\sigma^2}$, and 0 denote the guard symbols. The number of pilot, guard, and data symbols in a frame are 1, N_g , and $MN - N_g - 1$, respectively. These symbols are arranged in the frame as follows. The pilot is placed in the DD grid location $[k_p, l_p]$ such that $0 \leq k_p \leq N - 1$ and $0 \leq l_p \leq M - 1$ such that $0 \leq l_p - l_\tau \leq l_p \leq l_p + l_\tau \leq M - 1$ and $0 \leq k_p - 2k_v \leq k_p \leq k_p + 2k_v \leq N - 1$, where l_τ and k_v denote the taps corresponding to the maximum delay and Doppler values. The symbol placement in a frame is given by (see Fig. 6.2(b))

$$x[k, l] = \begin{cases} x_p & k = k_p, l = l_p, \\ 0 & k_p - 2k_v \leq k \leq k_p + 2k_v, \\ & l_p - l_\tau \leq l \leq l_p + l_\tau, \\ x_d[k, l] & \text{otherwise.} \end{cases} \quad (6.24)$$

In this arrangement, the number of guard symbols is $N_g = (2l_\tau + 1)(4k_v + 1) - 1$ and the overhead for pilot and guard symbols is $\frac{N_g + 1}{MN}$. This overhead can be small for typical channel models and frame sizes. For example, for long-term evolution (LTE) channels with $l_\tau = 20$, $k_v = 4$, the overhead is about 4% of the frame for $M = 256$ and $N = 64$.

At the receiver, the received symbols $y[k, l]$, $k_p - k_v \leq k \leq k_p + k_v$, $l_p \leq l \leq l_p + l_\tau$, are used for channel estimation and the remaining symbols are used for data detection. Due to the symbol arrangement in (6.24), using (6.22), the received symbols for channel estimation can be expressed as

$$y[k, l] = b[k - k_p, l - l_p] \hat{h}[k - k_p, l - l_p] x_p + v[k, l] \quad (6.25)$$

for $k \in [k_p - k_v, k_p + k_v]$, $l \in [l_p, l_p + l_\tau]$. It can be seen that if there is a path with Doppler tap $k - k_p$ and delay tap $l - l_p$, i.e., $b[k - k_p, l - l_p] = 1$, then $y[k, l] = \hat{h}[k - k_p, l - l_p] x_p + v[k, l]$. Otherwise, $y[k, l] = v[k, l]$.

Now, the channel estimation algorithm is as follows. For $k \in [k_p - k_v, k_p + k_v]$, $l \in [l_p, l_p + l_\tau]$, if the magnitude $|y[k, l]| \geq \mathcal{T}$, where $\mathcal{T}$ is some positive detection threshold, then the estimates are $b[k - k_p, l - l_p] = 1$ and $\hat{h}[k - k_p, l - l_p] = y[k, l] / x_p$. Otherwise, $b[k - k_p, l - l_p] = \hat{h}[k - k_p, l - l_p] = 0$. This scheme relies on the fact that if a path exists, the received symbol is the scaled pilot signal with additive white Gaussian noise (see (6.25)). Otherwise, it is only noise. By varying the threshold $\mathcal{T}$, the miss detection and false alarm probabilities of path detection can be altered. As a result, the error performance of data detection is affected by $\mathcal{T}$. The estimated information is then used for data detection, where the received symbols $y[k, l]$ for data detection are

$$y[k, l] = \sum_{k' = -k_v}^{k_v} \sum_{l' = 0}^{l_\tau} b[k', l'] \hat{h}[k', l'] x_d[[k - k']_N, [l - l']_M] + v[k, l] \quad (6.26)$$

for $k \notin [k_p - k_v, k_p + k_v]$ or $l \notin [l_p, l_p + l_\tau]$. Note that there is a total of $MN - (2k_v + 1)(l_\tau + 1)$ received symbols to detect a smaller number of $MN - (2l_\tau + 1)(4k_v + 1)$ data symbols.

In the above, ideal transmit and receive pulses, $g_{tx}(t)$ and $g_{rx}(t)$, have been assumed. The above embedded channel estimation scheme can, however, be employed with commonly used pulses, such as the rectangular pulse. With rectangular pulses, the input-output symbol relationship can be rewritten as ([38], Eq. (24))

$$y[k, l] = \sum_{k' = -k_v}^{k_v} \sum_{l' = 0}^{l_\tau} b[k', l'] \hat{h}[k', l'] \beta[k, l] x[[k - k']_N, [l - l']_M] + v[k, l], \quad (6.27)$$

where

$$\beta[k, l] = \begin{cases} e^{j2\pi \left(\frac{l-l'}{M} \right) \frac{k'}{N}} & l' \leq l < M \\ \frac{N-1}{N} e^{j2\pi \left(\frac{l-l'}{M} \right) \frac{k'}{N}} e^{-j2\pi \left(\frac{(k-k')N}{N} \right)} & 0 \leq l < l'. \end{cases} \quad (6.28)$$

Hence, the above threshold-based estimation scheme can be straightforwardly employed by introducing a known phase $\beta[k, l]$ in the detection process. The thresholds for the rectangular waveforms remain the same as the ideal waveforms since the channel differs only by a phase.

6.3.2 Embedded Pilot Without Guard Symbols

In embedded frames without guard symbols, each frame consists of only pilot and data symbols and no guard symbols. The system model, embedded frame structure, and SBL algorithm for channel estimation are presented below.

6.3.2.1 System Model

Considering integer delays, fractional Dopplers, and rectangular pulses, the DD domain input-output relation can be written as [38, 102]

$$y[k, l] \approx \sum_{i=1}^P \sum_{q=-Q}^Q \beta_i(k, l, q) \chi_i(q) h_i x \left[[k - k_i + q]_N, [l - l_i]_M \right] + w[k, l], \quad (6.29)$$

where h_i is the fade coefficient of the i th path,

$$\beta_i[k, l, q] = \begin{cases} e^{j2\pi \frac{(l-l_i)(k_i+k_l)}{MN}}, & l_i \leq l < M \\ \frac{N-1}{N} e^{j2\pi \frac{(l-l_i)(k_i+k_l)}{MN}} e^{-j2\pi \frac{[k-k_i+q]_N}{N}}, & 0 \leq l < l_i, \end{cases} \quad (6.30)$$

and

$$\chi_i(q) = \frac{e^{-j2\pi(-q-k_i)} - 1}{Ne^{-j\frac{2\pi}{N}(-q-k_i)} - N}, \quad (6.31)$$

where $\tau_i = \frac{l_i}{M\Delta f}$ is the integer delay and $\nu_i = \frac{k_i+k_l}{NT}$ is the fractional Doppler, respectively, of the i th path, and $\kappa_i \in (-0.5, 0.5]$ is the fractional part of the Doppler of the i th path. The effect of the fractional Doppler is approximated by selecting an appropriate Q , and $Q = 5$ gives a good approximation [102]. From (6.29), it can be seen that each received signal $y[k, l]$ is affected by $P(2Q + 1)$ transmitted signals around it. Also, for a given $y[k, l]$, the positions of the transmitted signals that influence $y[k, l]$ are in the delay and Doppler ranges of $[l - l_\tau, l]$ and $[k - k_\nu - Q, k + k_\nu + Q]$, respectively. It can be seen from (6.29) that if channel estimation for K received signals has to be realized, then corresponding $L = (2k_\nu + 2Q + 1)(l_\tau + 1)$ pilot symbols $x_p[k, l]$ need to be inserted according to the position of each received signal $y[k, l]$. Furthermore, it can be seen from (6.30) that the rectangular pulse will affect the phase of the pilot symbol. Therefore, the pilot signal is multiplied by the conjugate of $\beta_i[k, l, q]$ to perform phase compensation. Now, the matrix form of (6.29) can be written as [103]

$$\mathbf{Y} = (\mathbf{X}_p \odot \mathbf{B}) \mathbf{h} + \mathbf{W} = \mathbf{\Phi} \mathbf{h} + \mathbf{W}, \quad (6.32)$$

where $\mathbf{Y} \in \mathbb{C}^{K \times 1}$, $\mathbf{X}_p \in \mathbb{C}^{K \times L}$, $\mathbf{h} \in \mathbb{C}^{L \times 1}$, and $\mathbf{W} \in \mathbb{C}^{K \times 1}$ are the received signal, transmitted signal, channel, and noise in the DD domain, respectively, $\odot$ is the Hadamard product, and

$\mathbf{B} = [\text{conj}(\beta_1), \dots, \text{conj}(\beta_K)]^T$ is a phase compensation matrix, where $\beta_i \in \mathbb{C}^{1 \times L}$ is the vector form of $\beta[k, l, q]$, and $\text{conj}(\cdot)$ denotes conjugation operation. Bayesian inference can be used to reconstruct $\mathbf{h}$, exploiting the sparsity of the channel. Once $\mathbf{h}$ is reconstructed, the delay taps l_i s, Doppler taps v_i s, and fade coefficients h_i s, $i = 1, \dots, P(2Q + 1)$, of the channel are determined by the positions and values of the $P(2Q + 1)$ non-zero entries in the reconstructed $\mathbf{h}$.

6.3.2.2 Embedded Frame Structure

Figure 6.14(a) shows the embedded frame structure considered. The placement of pilot symbols $x_p[k, l]$ s and data symbols $x_d[k, l]$ s in a frame is expressed as

$$x[k, l] = \begin{cases} x_p[k, l], & k_p - 2k_v - Q \leq k \leq k_p + 2k_v + Q, \\ & l_p - l_r \leq l \leq l_p + l_r, \\ x_d[k, l], & \text{otherwise,} \end{cases} \quad (6.33)$$

where $k_p = \frac{N}{2}$, $l_p = \frac{M}{2}$, and l_r and k_v denote the indices of the taps corresponding to $\tau_{\max}$ and $\nu_{\max}$, respectively. The number of pilot symbols in a frame is $N_p = (4k_v + 2Q + 1)(l_r + l_r + 1)$, and the number of channel estimated symbols is $K = (2k_v + 1)(l_r + 1)$. According to (6.29), the receiver selects $y[k, l]$, $k \in [k_p - k_v, k_p + k_v]$, $l \in [l_p, l_p + l_r]$ for channel estimation, as shown in Fig. 6.14(b).

6.3.2.3 SBL-Based Estimation

The SBL-based estimation approach exploits the channel sparsity in the DD domain, where the sparse channel parameter update is performed according to Bayesian inference.

Bayesian Inference The noise in (6.32) is modeled as Gaussian white noise with mean zero and precision (reciprocal of variance) α_0 , i.e.,

$$p(\mathbf{W} | \alpha_0) = \mathcal{N}(\mathbf{W} | 0, \alpha_0^{-1} \mathbf{I}). \quad (6.34)$$

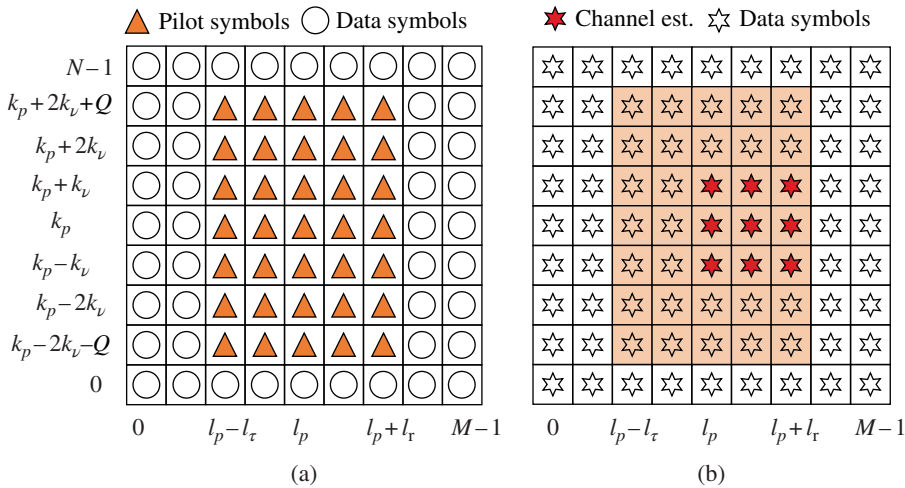


Figure 6.14 Embedded pilot frame without guard symbols. (a) Transmitted symbols pattern, and (b) received symbols pattern. Source: Zhao et al. [103] IEEE.

Assume that α_0 follows Gamma distribution controlled by hyperparameters a and b , i.e.,

$$p(\alpha_0; a, b) = \Gamma(\alpha_0 | a, b). \quad (6.35)$$

The likelihood function of the received signal can be expressed as

$$p(\mathbf{Y} | \Phi, \alpha_0) = \mathcal{N}(\mathbf{Y} | \Phi \mathbf{h}, \alpha_0^{-1} \mathbf{I}). \quad (6.36)$$

From the perspective of hierarchical prior and conjugate prior, the sparse channel in the DD domain is modeled as

$$p(\mathbf{h} | \alpha) = \mathcal{N}(\mathbf{h} | 0, \Lambda), \quad (6.37)$$

$$p(\alpha; \lambda) = \prod_{i=1}^L \Gamma\left(\alpha_i | 1, \frac{\lambda}{2}\right), \quad (6.38)$$

where $\Lambda = \text{diag}(\alpha) = \text{diag}([\alpha_1, \dots, \alpha_L]^T)$ is the covariance matrix and α_i is the variance of h_i . All h_i s share the common Laplace sparse prior governed by λ as [105]

$$p(h_i | \lambda) = \text{Laplace}(0, 1/\sqrt{\lambda}), \quad \text{for } i = 1, \dots, L. \quad (6.39)$$

Combining the stages of Bayesian hierarchical modeling, the joint pdf of the model can be expressed as

$$p(\mathbf{Y}, \mathbf{h}, \alpha, \alpha_0) = p(\mathbf{Y} | \mathbf{h}, \alpha_0) p(\mathbf{h} | \alpha) p(\alpha) p(\alpha_0). \quad (6.40)$$

The posterior probability of $\mathbf{h}$ can be derived as Gaussian [106], given by

$$p(\mathbf{h} | \mathbf{Y}, \alpha, \alpha_0) = \mathcal{N}(\mathbf{h} | \mu, \Sigma), \quad (6.41)$$

with $\mu = \alpha_0 \Sigma \Phi^T \mathbf{Y}$ and $\Sigma = (\alpha_0 \Phi^T \Phi + \Lambda^{-1})^{-1}$. To calculate μ and Σ , the values of the hyperparameters α and α_0 need to be obtained. Since the analytical calculation of $p(\alpha, \alpha_0 | \mathbf{Y})$ is difficult, the expectation maximization (EM) algorithm is used to solve it. The expression for obtaining (α, α_0) through EM is given by [107]

$$(\alpha^{\text{new}}, \alpha_0^{\text{new}}) = \underset{\alpha, \alpha_0}{\text{argmax}} \mathbb{E} \{\ln p(\mathbf{Y}, \mathbf{h}, \alpha, \alpha_0)\}. \quad (6.42)$$

α^{new} and α_0^{new} are updated as [108]

$$\alpha_i^{\text{new}} = \frac{\sqrt{1 + 4\lambda (\Sigma_{ii} + \mu_i^2)} - 1}{2\lambda}, \quad \text{for } i = 1, \dots, L, \quad (6.43)$$

$$\alpha_0^{\text{new}} = \frac{2a - 2 + K}{2b + \mathbb{E} \{\|\mathbf{Y} - \Phi \mathbf{h}\|_2^2\}}, \quad (6.44)$$

with $\mathbb{E} \{\|\mathbf{Y} - \Phi \mathbf{h}\|_2^2\} = \alpha_0^{-1} \sum_{i=1}^L (1 - \alpha_i^{-1} \Sigma_{ii}) + \|\mathbf{Y} - \Phi \mu\|_2^2$. The updates are continued until a maximum number of iterations, max_iter , is reached or the normalized error, $\frac{\|\alpha^{\text{new}} - \alpha\|_2^2}{\|\alpha\|_2^2}$, goes below a certain threshold, ϵ . Finally, the first $P(2Q + 1)$ largest amplitude elements in μ are taken to $\mathbf{h}$, which is output as the reconstructed $\mathbf{h}$. The iterative procedure of the SBL-based channel estimation algorithm is given in **Algorithm 6.2**. As mentioned earlier, the delay taps l_i , Doppler taps k_i , and fade coefficient h_i , $i \in [1, P(2Q + 1)]$ of the channel are determined by the positions and values of the $P(2Q + 1)$ non-zero entries in $\mathbf{h}$. The computation complexity of the algorithm is mainly due to the matrix inversion operation involved in computing Σ . Using the Woodbury matrix identity, the dimension of the inverse matrix can be reduced from L to K , resulting in a complexity of order $\mathcal{O}(JKL^2)$, where J is the number of iterations. The SBL algorithm has been shown to be convergent in many cases [106].

Algorithm 6.2 SBL-based channel estimation algorithm**Input:** $\mathbf{Y}, \Phi, P, L, K, Q, \max_iter, \epsilon$ **Initialize:** $\alpha_0, \alpha, a, b, j = 0$ **repeat**

- Calculate μ and Σ using current α_0 and α
- Update α and α_0 according to (6.43) and (6.44)
- $j = j + 1$

until $j = \max_iter$ or $\frac{\|\alpha^{new} - \alpha\|_2^2}{\|\alpha\|_2^2} < \epsilon$

- Select the first $P(2Q + 1)$ largest amplitude elements of μ to $\mathbf{h}$

Output: $\mathbf{h}$

Performance Figure 6.15(a) shows the NMSE performance of the SBL-based channel estimation algorithm for different pilot SNRs in comparison with the performance achieved by the threshold-based estimation in [102] and the orthogonal matching pursuit (OMP)-based estimation [109], which is another sparse recovery algorithm. The pilot overhead for the three schemes is $\eta_{sbl} = \eta_{omp} = \frac{(4k_v + 2Q + 1)(l_r + l_t + 1)}{MN}$ and $\eta_{threshold} = \frac{(4k_v + 4Q + 1)(l_r + l_t + 1)}{MN}$. The NMSE is defined as $10 \log_{10} \left(\frac{\mathbb{E}(\|\mathbf{h} - \hat{\mathbf{h}}\|_2^2)}{\mathbb{E}(\|\mathbf{h}\|_2^2)} \right)$, where $\mathbf{h}$ is the true CSI and $\hat{\mathbf{h}}$ is the estimated CSI. The pilot SNR is defined as $\text{SNR}_p = 10 \log_{10}(\mathbb{E}(|x_p|^2)/\sigma^2)$. The following system parameters are considered: carrier frequency = 4 GHz, $\Delta f = 15$ kHz, $M = 512$, $N = 128$, $P = 9$, maximum delay tap $l_r = 20$, and Doppler tap $k_v = 16$ with 500 km/h speed. The channel model uses an extended vehicle model [110], where each delay tap generates its own Doppler through Jakes' formula. From Fig. 6.15(a), it can be seen that the SBL-based estimation achieves better NMSE performance compared to the other two schemes. Also, increasing Q improves the performance at the cost of additional overhead. Figure 6.15(b) shows the BER performance using the message passing detection in Section 5.2.1. The better NMSE performance translates to better BER performance in favor of the SBL-based estimation.

Another algorithm for channel estimation in the presence of fractional DDs with rectangular pulses and the embedded pilot frame with guard symbols is presented in [100]. This algorithm exploits the fine DD domain resolution of the OTFS-modulated signal to decouple the joint

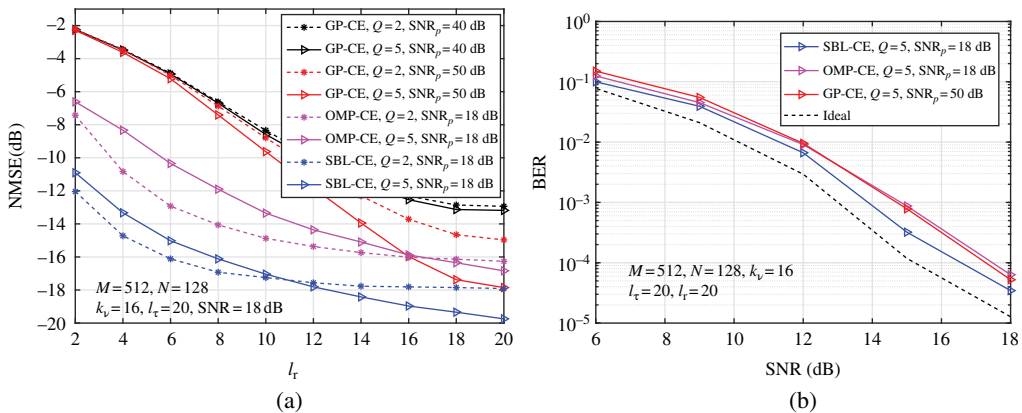


Figure 6.15 NMSE and BER performance of SBL-based channel estimation. (a) NMSE performance and (b) BER performance. Source: Zhao et al. [103] IEEE.

estimation of the channel parameters (fade coefficient, delay, and Doppler) of all paths into separate estimation of the channel parameters for each path, thereby achieving a lower complexity. This algorithm has also been shown to achieve better performance compared to the SBL-based channel estimation. The SBL-based estimation presented in this subsection is for SISO-OTFS. In [111], this SBL-based approach has been extended to channel estimation for MIMO-OTFS under a block SBL framework. In contrast to the traditional block SBL method, the algorithm in [111] iteratively updates the size of non-sparse blocks to obtain a better channel estimation accuracy.

6.4 Superimposed Pilot Approach

The loss in throughput due to pilot and guard symbols in exclusive and embedded pilot approaches can be alleviated by superimposing pilot symbols on to information symbols. All DD bins in a frame carry information symbols without any guard symbols. Low-power pilot symbols can be superimposed on all the data symbols in the frame, as shown in Fig. 6.2(c) [112]. Alternately, pilot symbols can be superimposed on only one data symbol [113] or only few data symbols in the frame [114, 115]. The idea of using superimposed pilot signals for channel estimation in non-OTFS systems, e.g., SISO, MIMO, and OFDM systems, has been reported in the literature [116–118]. This section considers the superimposed pilot approach in the context of DD channel estimation in OTFS systems. In this approach, the pilot and data symbols interfere with each other making the channel estimation and data detection tasks more involved. Judicious allocation of power among data and pilot symbols in a frame and signal processing based on the iterative channel estimation and data detection can help. This section presents two schemes for channel estimation using superimposed pilots. One is a non-iterative scheme, termed as the SP-NI scheme, and another is an iterative scheme termed as the SP-I scheme. The SP-NI scheme treats the data symbols in the received frame as noise and obtains the MMSE estimate of the DD channel, which is then used to recreate and cancel the pilot component in the received signal. The resulting received signal is used for data detection using a detection algorithm. The SP-I scheme iterates between channel estimation and data detection to obtain better channel estimation and data detection performance. It takes the data estimate from the SP-NI scheme as an initial data estimate and uses it along with the superimposed pilot to perform data-aided channel estimation.

6.4.1 System Model

The pilot symbol $x_p[l, k]$ is superimposed on the data symbol $x_d[l, k]$ in the DD domain, $k = 0, \dots, N - 1, l = 0, \dots, M - 1$, as

$$x[l, k] = x_d[l, k] + x_p[l, k], \quad (6.45)$$

where $\mathbb{E}\{|x_d[l, k]|^2\} = \sigma_d^2$ and $\mathbb{E}\{|x_p[l, k]|^2\} = \sigma_p^2$ with the constraint $\mathbb{E}\{|x[l, k]|^2\} = \sigma_d^2 + \sigma_p^2 = 1$. Let $\mathbf{x}_d \in \mathbb{C}^{NM \times 1}$ denote the data vector consisting of data symbols, $x_d[l, k]$ s. Likewise, let $\mathbf{x}_p \in \mathbb{C}^{NM \times 1}$ denote the pilot vector consisting of pilot symbols, $x_p[l, k]$ s. The DD domain transmit vector $\mathbf{x} \in \mathbb{C}^{NM \times 1}$ is then given by

$$\mathbf{x} = \mathbf{x}_d + \mathbf{x}_p. \quad (6.46)$$

Assuming integer delays and Dopplers, the received signal vector can be written as [112]

$$\mathbf{y} = \mathbf{H}_{\text{eff}} \mathbf{x} + \tilde{\mathbf{w}}, \quad (6.47)$$

where the effective channel matrix $\mathbf{H}_{\text{eff}} \in \mathbb{C}^{MN \times MN}$ is given by

$$\mathbf{H}_{\text{eff}} = (\mathbf{F}_N \otimes \mathbf{G}_{\text{rx}}) \mathbf{H} (\mathbf{F}_N^H \otimes \mathbf{G}_{\text{tx}}), \quad (6.48)$$

$\tilde{\mathbf{w}} = (\mathbf{F}_N \otimes \mathbf{G}_{\text{tx}}) \mathbf{w}$, and $\mathbf{w} \in \mathbb{C}^{MN \times 1}$ is the noise vector with i.i.d. complex Gaussian entries with zero mean and variance σ_w^2 . In the above, $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ is the normalized DFT matrix with $\mathbf{F}_N(p, q) = (1/\sqrt{N}) \exp(-j2\pi pq/N)$, and $\mathbf{G}_{\text{tx}}, \mathbf{G}_{\text{rx}} \in \mathbb{C}^{M \times M}$ are diagonal matrices obtained by uniformly sampling the transmit pulse $g_{\text{tx}}(t)$ and receive pulse $g_{\text{rx}}(t)$, respectively, at time instants $t = \frac{mT}{M}$, $m = 0, 1, \dots, M-1$. The channel matrix $\mathbf{H} \in \mathbb{C}^{MN \times MN}$ in (6.48) is given by [83]

$$\mathbf{H} = \sum_{i=1}^P h_i \mathbf{\Pi}^i \mathbf{\Delta}^{k_i}, \quad (6.49)$$

where $\mathbf{\Pi}$ is an $MN \times MN$ permutation (forward cyclic shift) matrix, $\mathbf{\Delta}$ is a diagonal matrix given by $\text{diag}\{z^0, z^1, \dots, z^{MN-1}\}$ with $z = \exp(2\pi j/MN)$, $j = 0, 1, \dots, MN-1$, and k_i and l_i are integers denoting the indices of the Doppler and delay taps, respectively, of the i th path. For rectangular transmit and receive pulses of duration T at the transmitter and receiver, $\mathbf{G}_{\text{rx}} = \mathbf{G}_{\text{tx}} = \mathbf{I}_M$. With the above, the $\mathbf{H}_{\text{eff}}$ matrix in (6.48) can be written as

$$\begin{aligned} \mathbf{H}_{\text{eff}} &= (\mathbf{F}_N \otimes \mathbf{I}_M) \left(\sum_{i=1}^P h_i \mathbf{\Pi}^i \mathbf{\Delta}^{k_i} \right) (\mathbf{F}_N^H \otimes \mathbf{I}_M) \\ &= \mathbf{B}_{\text{rx}} \left(\sum_{i=1}^P h_i \mathbf{\Theta}_i \right) \mathbf{B}_{\text{tx}}, \end{aligned} \quad (6.50)$$

where $\mathbf{B}_{\text{tx}} = (\mathbf{F}_N^H \otimes \mathbf{I}_M)$, $\mathbf{B}_{\text{rx}} = (\mathbf{F}_N \otimes \mathbf{I}_M)$, and $\mathbf{\Theta}_i = \mathbf{\Pi}^i \mathbf{\Delta}^{k_i}$ are the matrices of size $MN \times MN$. Now, substituting $\mathbf{H}_{\text{eff}}$ from (6.50) in (6.47), the received vector $\mathbf{y}$ can be written as

$$\mathbf{y} = \left(\sum_{i=1}^P h_i \mathbf{\Gamma}_i \right) (\mathbf{x}_p + \mathbf{x}_d) + \tilde{\mathbf{w}}, \quad (6.51)$$

where the matrix $\mathbf{\Gamma}_i = \mathbf{B}_{\text{rx}} \mathbf{\Theta}_i \mathbf{B}_{\text{tx}} \in \mathbb{C}^{MN \times MN}$. The above equation can be written in the form

$$\mathbf{y} = \mathbf{\Omega}_p \mathbf{h} + \mathbf{\Omega}_d \mathbf{h} + \tilde{\mathbf{w}}, \quad (6.52)$$

where the $\mathbf{\Omega}_p, \mathbf{\Omega}_d \in \mathbb{C}^{MN \times P}$ matrices are concatenated matrices corresponding to the pilot vector $\mathbf{x}_p$ and data vector $\mathbf{x}_d$, given by

$$\mathbf{\Omega}_p = [\mathbf{\Gamma}_1 \mathbf{x}_p \quad \mathbf{\Gamma}_2 \mathbf{x}_p \quad \cdots \quad \mathbf{\Gamma}_P \mathbf{x}_p], \quad (6.53)$$

$$\mathbf{\Omega}_d = [\mathbf{\Gamma}_1 \mathbf{x}_d \quad \mathbf{\Gamma}_2 \mathbf{x}_d \quad \cdots \quad \mathbf{\Gamma}_Q \mathbf{x}_d]. \quad (6.54)$$

The DD domain fade coefficient vector $\mathbf{h} = [h_1, h_2, \dots, h_P]^T \in \mathbb{C}^{P \times 1}$ has a zero mean and covariance matrix

$$\mathbf{C}_h = \mathbb{E}\{\mathbf{h}\mathbf{h}^H\} = \text{diag}\{\sigma_{h_1}^2, \dots, \sigma_{h_P}^2\}, \quad (6.55)$$

which is assumed to be known at the receiver. The matrix $\mathbf{\Omega}_d$ has $\mathbb{E}\{\mathbf{\Omega}_d\} = \mathbf{0}_{MN \times P}$ and $\mathbb{E}\{\mathbf{\Omega}_d \mathbf{\Omega}_d^H\} = \sigma_d^2 P \mathbf{I}_{MN}$. The noise vector $\tilde{\mathbf{w}}$ has

$$\mathbb{E}\{\tilde{\mathbf{w}}\} = \mathbf{0}_{MN \times 1}, \quad \mathbf{C}_{\tilde{\mathbf{w}}} = \mathbb{E}\{\tilde{\mathbf{w}}\tilde{\mathbf{w}}^H\} = \sigma_w^2 \mathbf{I}_{MN}. \quad (6.56)$$

With the above system model in place, the SP-NI and SP-I schemes are described below.

6.4.2 SP-NI Scheme

Consider the received signal vector in (6.52), given by

$$\begin{aligned} \mathbf{y} &= \mathbf{\Omega}_p \mathbf{h} + \underbrace{\mathbf{\Omega}_d \mathbf{h}}_{\triangleq \tilde{\mathbf{w}}_d} + \tilde{\mathbf{w}} \\ &= \mathbf{\Omega}_p \mathbf{h} + \tilde{\mathbf{w}}_d, \end{aligned} \quad (6.57)$$

where $\tilde{\mathbf{w}}_d$ is the interference-plus-noise vector. The mean vector and covariance matrix of $\tilde{\mathbf{w}}_d$ are given by [112]

$$\boldsymbol{\mu}_{\tilde{\mathbf{w}}_d} = \mathbb{E} \{ \tilde{\mathbf{w}}_d \} = \mathbf{0}_{MN \times 1} \quad (6.58)$$

and

$$\mathbf{C}_{\tilde{\mathbf{w}}_d} = \mathbb{E} \{ \tilde{\mathbf{w}}_d \tilde{\mathbf{w}}_d^H \} = \left(\left(\sum_{i=1}^P \sigma_{h_i}^2 \right) \sigma_d^2 + \sigma_w^2 \right) \mathbf{I}_{MN}. \quad (6.59)$$

The MMSE estimate of the channel vector $\mathbf{h}$ is given by

$$\hat{\mathbf{h}}_{\text{NI}} = \left(\mathbf{\Omega}_p^H \mathbf{C}_{\tilde{\mathbf{w}}_d}^{-1} \mathbf{\Omega}_p + \mathbf{C}_{\mathbf{h}}^{-1} \right)^{-1} \mathbf{\Omega}_p^H \mathbf{C}_{\tilde{\mathbf{w}}_d}^{-1} \mathbf{y}, \quad (6.60)$$

where $\mathbf{C}_{\mathbf{h}}$ is the covariance matrix of $\mathbf{h}$, given by

$$\mathbf{C}_{\mathbf{h}} = \mathbb{E} \{ \mathbf{h} \mathbf{h}^H \} = \text{diag} \left\{ \sigma_{h_1}^2, \dots, \sigma_{h_P}^2 \right\}. \quad (6.61)$$

The covariance matrix of the error vector $(\mathbf{h} - \hat{\mathbf{h}}_{\text{NI}})$, denoted by $\boldsymbol{\Sigma}_{\text{NI}}$, can be obtained as [119]

$$\boldsymbol{\Sigma}_{\text{NI}} = \left(\mathbf{\Omega}_p^H \mathbf{C}_{\tilde{\mathbf{w}}_d}^{-1} \mathbf{\Omega}_p + \mathbf{C}_{\mathbf{h}}^{-1} \right)^{-1}, \quad (6.62)$$

and the MSE of the channel estimate is given by

$$\mathbb{E} \left\{ \|\mathbf{h} - \hat{\mathbf{h}}_{\text{NI}}\|^2 \right\} = \text{Tr} \{ \boldsymbol{\Sigma}_{\text{NI}} \}. \quad (6.63)$$

The signal used for detecting data is obtained by subtracting the pilot signal from the received signal in (6.47) as

$$\begin{aligned} \mathbf{y}_d &= \mathbf{y} - \hat{\mathbf{H}}_{\text{eff-NI}} \mathbf{x}_p \\ &= \mathbf{H}_{\text{eff}} \mathbf{x}_d + \tilde{\mathbf{w}}_e. \end{aligned} \quad (6.64)$$

The vector $\tilde{\mathbf{w}}_e = \mathbf{e}_p + \tilde{\mathbf{w}}$ consists of channel estimation error and noise, with the error vector $\mathbf{e}_p = (\mathbf{H}_{\text{eff}} - \hat{\mathbf{H}}_{\text{eff-NI}}) \mathbf{x}_p$. The estimate of the effective channel matrix $\mathbf{H}_{\text{eff}}$, defined in (6.50), is obtained by substituting the expression of channel estimate $\hat{\mathbf{h}}_{\text{NI}}$ and the indices of the delays and Dopplers of various paths in (6.50) as

$$\hat{\mathbf{H}}_{\text{eff-NI}} = \mathbf{B}_{\text{rx}} \left(\sum_{i=1}^P \hat{h}_{\text{NI},i} \boldsymbol{\Theta}_i \right) \mathbf{B}_{\text{tx}}. \quad (6.65)$$

Now, using the $\mathbf{y}_d$ vector in (6.64), the estimated effective channel matrix $\hat{\mathbf{H}}_{\text{eff-NI}}$, and the MSE of the channel estimate in (6.63), the data symbols in the frame can be estimated using a detection algorithm (e.g., message passing detection algorithm). Let $\hat{\mathbf{x}}_{d,\text{NI}}$ denote the data estimate of the SP-NI scheme. It is noted that the computation of $\hat{\mathbf{H}}_{\text{eff-NI}}$ in (6.65) requires the knowledge of the number of paths and their delay and Doppler indices. These can be estimated using other simple estimation schemes such as impulse-based estimation.

6.4.3 SP-I Scheme

The SP-I scheme takes the data estimate from the SP-NI scheme as an initial data estimate, i.e., $\hat{\mathbf{x}}_d^{(0)} = \hat{\mathbf{x}}_{d,\text{NI}}$, and performs data-aided channel estimation. Let $\hat{\mathbf{\Omega}}_d^{(0)}$ be the initial estimate of the data matrix $\mathbf{\Omega}_d$. It is obtained from the SP-NI scheme by substituting the data estimate $\mathbf{x}_d = \hat{\mathbf{x}}_d^{(0)} = \hat{\mathbf{x}}_{d,\text{NI}}$ in (6.54) as

$$\hat{\mathbf{\Omega}}_d^{(0)} = \begin{bmatrix} \mathbf{\Gamma}_1 \hat{\mathbf{x}}_d^{(0)} & \mathbf{\Gamma}_2 \hat{\mathbf{x}}_d^{(0)} & \cdots & \mathbf{\Gamma}_Q \hat{\mathbf{x}}_d^{(0)} \end{bmatrix}. \quad (6.66)$$

The received signal vector in (6.52) can be expressed as

$$\begin{aligned} \mathbf{y} &= \left(\mathbf{\Omega}_p + \hat{\mathbf{\Omega}}_d^{(0)} \right) \mathbf{h} + \left(\mathbf{\Omega}_d - \hat{\mathbf{\Omega}}_d^{(0)} \right) \mathbf{h} + \tilde{\mathbf{w}} \\ &= \mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(0)} \mathbf{h} + \xi_{\tilde{\mathbf{w}}}^{(0)}, \end{aligned} \quad (6.67)$$

where the data-aided pilot matrix $\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(0)}$ and the noise-plus-interference vector $\xi_{\tilde{\mathbf{w}}}^{(0)}$ are defined as

$$\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(0)} \triangleq \mathbf{\Omega}_p + \hat{\mathbf{\Omega}}_d^{(0)}, \quad \xi_{\tilde{\mathbf{w}}}^{(0)} \triangleq \mathbf{\Xi}_{\mathbf{x}_d}^{(0)} \mathbf{h} + \tilde{\mathbf{w}}, \quad (6.68)$$

where $\mathbf{\Xi}_{\mathbf{x}_d}^{(0)}$ is the estimation error matrix given by $\mathbf{\Xi}_{\mathbf{x}_d}^{(0)} \triangleq \mathbf{\Omega}_d - \hat{\mathbf{\Omega}}_d^{(0)}$. Using the above initial data estimate, the data-aided estimation of the channel vector $\mathbf{h}$ in (6.67) in the n th iteration is performed as

$$\hat{\mathbf{h}}^{(n)} = \left(\left(\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)} \right)^H \left(\mathbf{C}_{\xi_{\tilde{\mathbf{w}}}}^{(n-1)} \right)^{-1} \mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)} + \mathbf{C}_{\mathbf{h}}^{-1} \right)^{-1} \left(\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)} \right)^H \left(\mathbf{C}_{\xi_{\tilde{\mathbf{w}}}}^{(n-1)} \right)^{-1} \mathbf{y}. \quad (6.69)$$

The covariance matrix $\mathbf{C}_{\xi_{\tilde{\mathbf{w}}}}^{(n)}$ of the vector $\xi_{\tilde{\mathbf{w}}}^{(n)}$ in the n th iteration can be computed as

$$\begin{aligned} \mathbf{C}_{\xi_{\tilde{\mathbf{w}}}}^{(n)} &= \mathbb{E} \left\{ \xi_{\tilde{\mathbf{w}}}^{(n)} \left(\xi_{\tilde{\mathbf{w}}}^{(n)} \right)^H \right\} \\ &= 2 \left(\sum_{i=1}^Q \sigma_{h_i}^2 \right) \sigma_d^2 \mathbf{I}_{MN} + \sigma_w^2 \mathbf{I}_{MN}. \end{aligned} \quad (6.70)$$

The MSE of the data-aided channel estimator is

$$B_{\mathbf{h}}^{(n)} = \text{Tr} \left(\mathbf{\Sigma}_{\mathbf{h}}^{(n)} \right), \quad (6.71)$$

where $\mathbf{\Sigma}_{\mathbf{h}}^{(n)} = \left(\left(\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)} \right)^H \left(\mathbf{C}_{\xi_{\tilde{\mathbf{w}}}}^{(n-1)} \right)^{-1} \mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)} + \mathbf{C}_{\mathbf{h}}^{-1} \right)^{-1}$ is the covariance matrix of the error vector $(\mathbf{h} - \hat{\mathbf{h}}^{(n)})$. Let $\hat{\mathbf{H}}_{\text{eff}}^{(n)}$ be the estimate of the effective channel matrix corresponding to the data-aided channel estimate $\hat{\mathbf{h}}^{(n)}$ in the n th iteration. It is obtained by substituting $\hat{\mathbf{h}}^{(n)}$ in (6.50) as

$$\hat{\mathbf{H}}_{\text{eff}}^{(n)} = \mathbf{B}_{\text{rx}} \left(\sum_{i=1}^P \hat{h}_i^{(n)} \mathbf{\Theta}_i \right) \mathbf{B}_{\text{tx}}. \quad (6.72)$$

The receive signal $\tilde{\mathbf{y}}_d^{(n)} \in \mathbb{C}^{MN \times 1}$ used for detecting data in the n th iteration is obtained from the received vector in (6.47) as

$$\begin{aligned} \tilde{\mathbf{y}}_d^{(n)} &= \mathbf{y} - \hat{\mathbf{H}}_{\text{eff}}^{(n)} \mathbf{x}_p \\ &= \mathbf{H}_{\text{eff}} \mathbf{x}_d + \tilde{\mathbf{a}}_p^{(n)} + \tilde{\mathbf{w}} \\ &= \mathbf{H}_{\text{eff}} \mathbf{x}_d + \tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)}, \end{aligned} \quad (6.73)$$

Algorithm 6.3 Iterative channel estimation/data detection in the SP-I algorithm**Input:** Observation vector $\mathbf{y}$, pilot matrix $\mathbf{\Omega}_p$, $\mathbb{E}\{\mathbf{h}\}$, $\mathbf{C}_h = \mathbb{E}\{\mathbf{h}\mathbf{h}^H\}$, initial estimate $\hat{\mathbf{x}}_d^{(0)} = \hat{\mathbf{x}}_{d,\text{NI}}$ **Output:** Channel estimate $\hat{\mathbf{h}}$, data estimate $\hat{\mathbf{x}}_d$ **Initialize:** counter $n = 1$ **repeat**

- Compute $\hat{\mathbf{\Omega}}_d^{(n-1)}$ as given in (6.66). Compute $\mathbf{\Omega}_{\mathbf{x}_p \hat{\mathbf{x}}_d}^{(n-1)}$ and $\mathbf{C}_{\xi \tilde{\mathbf{w}}}^{(n-1)}$ as in (6.68) and (6.70), respectively.
- Compute $\hat{\mathbf{h}}^{(n)}$ and $\hat{\mathbf{H}}_{\text{eff}}^{(n)}$ as in (6.69) and (6.72), respectively.
- Feed the estimate $\hat{\mathbf{H}}_{\text{eff}}^{(n)}$ and $\tilde{\mathbf{y}}_d^{(n)}$ to detection algorithm to calculate the data estimate $\hat{\mathbf{x}}_d^{(n)}$

 $n = n + 1$ **until** Stopping criteria**Output:** $\hat{\mathbf{x}}_d$

where $\tilde{\mathbf{e}}_p^{(n)} = (\mathbf{H}_{\text{eff}} - \hat{\mathbf{H}}_{\text{eff}}^{(n)})\mathbf{x}_p$ and $\tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)} = \tilde{\mathbf{e}}_p^{(n)} + \tilde{\mathbf{w}}$. The first- and second-order statistics of $\tilde{\mathbf{e}}_p^{(n)}$ can be obtained as $\mathbb{E}\{\tilde{\mathbf{e}}_p^{(n)}(j)\} = 0$ and $\text{Var}[\tilde{\mathbf{e}}_p^{(n)}(j)] = \sigma_p^2 B_h^{(n)}$, respectively. Also, since $\tilde{\mathbf{e}}_p^{(n)}$ and $\tilde{\mathbf{w}}$ are statistically independent random vectors, the statistics of $\tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)}$ can be obtained as $\mathbb{E}\{\tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)}(j)\} = 0$ and $\text{Var}[\tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)}(j)] = \sigma_p^2 B_h^{(n)} + \sigma_w^2$, respectively. Now, using $\tilde{\mathbf{y}}_d^{(n)}$ in (6.73), $\hat{\mathbf{H}}_{\text{eff}}^{(n)}$, and the statistics of $\tilde{\mathbf{e}}_p^{(n)}$ and $\tilde{\mathbf{w}}_{\tilde{\mathbf{e}}}^{(n)}$, the data estimate in the n th iteration $\hat{\mathbf{x}}_d^{(n)}$ can be obtained using the detection algorithm. This detected data in the n th iteration is fed to the next iteration of channel estimation. This iterative process of channel estimation and data detection is continued till a maximum number of iterations or stopping criteria is reached. The detected data in the last iteration are output as the final detected data $\hat{\mathbf{x}}_d$, as given in **Algorithm 6.3**.

6.4.4 Performance

Figure 6.16(a) shows the MSE performance of channel estimation using SP-NI and SP-I schemes. The following system parameters are considered: carrier frequency = 4 GHz, subcarrier spacing = 15 kHz, $M = 64$, $N = 12$, number of paths $P = 6$, and BPSK modulation. The delay of each path is obtained from the Vehicular A (Veh-A) channel model [70], where the delay coefficient is rounded of the nearest integer. The Doppler spread of each path is obtained using Jakes' Doppler

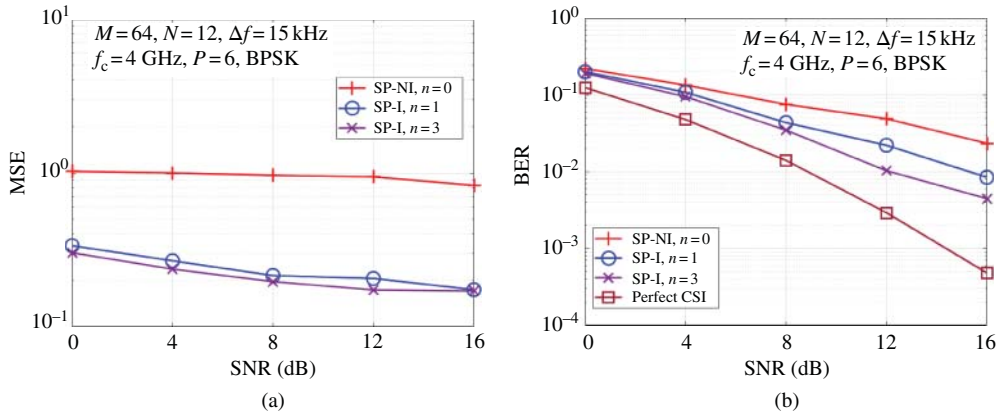


Figure 6.16 MSE and BER performance of SP-NI and SP-I schemes. (a) NMSE performance and (b) BER performance.

spectrum given by $v_i = v_{\max} \cos(\theta_i)$ for $i = 1, 2, \dots, P$, where θ_i is uniformly distributed in $-\pi$ to π , $v_{\max} = 815$ Hz, and the Doppler coefficient of each path is rounded off to the nearest integer. MSE plots for number of iterations $n = 0, 1, 3$ are shown, where $n = 0$ corresponds to the SP-NI scheme. As expected, the iterative scheme is found to achieve better MSE performance compared to the non-iterative scheme. Also, the first iteration ($n = 1$) gives a large improvement in performance compared to the non-iterative scheme, and increasing n further gives diminishing returns in terms of performance improvement. Figure 6.16(b) shows the corresponding BER performance with the message passing algorithm for detection. The BER performance is found to improve and get closer to the perfect CSI performance as n is increased.

7

Multuser OTFS

This chapter considers OTFS-based multuser communication, where multiple-user terminals (UTs) share the same physical resource. Sharing of physical resource can be orthogonal or non-orthogonal. In orthogonal sharing of resource, the resource elements (REs) allocated to a UT are not allocated to any other UT. In non-orthogonal sharing of resource, a RE could be allocated to more than one UT. In this chapter, state-of-the-art methods for orthogonal/non-orthogonal multuser communication on the uplink and downlink based on OTFS modulation are described in detail and their performance and signal processing complexity are compared. Section 7.1 introduces the notion of RE (the smallest unit of physical resource) in the context of OTFS modulation. OTFS-based state-of-the-art multiple access (MA) methods in the uplink are introduced in Section 7.2. Two types of uplink MA methods are described: (i) orthogonal multiple access (OMA), where no RE is allocated to more than one UT (Section 7.2.1) and (ii) non-orthogonal MA, where a RE could be allocated to more than one UT (Section 7.2.2).

Uplink OMA methods are characterized by simple receiver implementation, since at most one UT transmits information on an RE. A simple OMA method is to allocate non-overlapping portions of the delay-Doppler (DD) domain resource to different UTs, with DD domain guard bands (GBs) separating the portions allocated to different UTs. Although this simple strategy can reduce multuser interference (MUI), GB overhead reduces the effective spectral efficiency (SE). Non-GB-based OMA methods avoid this SE loss by time-frequency (TF) domain or DD domain interleaving of REs allocated to different UTs. Non-GB-based OMA methods, therefore, achieve higher SE when compared to GB-based methods. The GB-based OMA method is presented in Section 7.2.1.1, and the non-GB OMA methods based on the DD domain and TF domain interleaving are presented in Sections 7.2.1.2 and 7.2.1.3, respectively. A comparison of the sum SE performance achieved by these OMA methods is given in Section 7.2.1.4.

Non-orthogonal MA methods have the potential to achieve higher SE when compared to OMA methods, as the same RE is allocated to more than one UT. However, for the receiver to be able to separate the information streams of different UTs communicating on the same RE, extra signal dimensions are required. A well-known way to increase the received signal dimension is to use multiple antennas at the base station (BS) receiver. Multuser OTFS where the BS is provided with multiple receive antennas is introduced in Section 7.2.2.1. It is observed that the DD domain input-output relation between the received DD domain symbols at the BS and the symbols transmitted by the UTs is similar to that of a point-to-point single-user OTFS system with single antenna each at the transmitter and receiver. Therefore, state-of-the-art single-user OTFS detection algorithms (e.g., based on message passing) can be used for multuser OTFS detection as well.

Although MIMO-OTFS-based non-orthogonal MA methods achieve higher SE than OMA methods, the receiver complexity is higher due to larger signal dimensionality. At the same time, a large number of antennas at a massive MIMO BS allows for angular domain differentiation between signals arriving from different UTs and also improves energy efficiency. In Section 7.2.2.2, an OTFS-based massive MIMO system, where low-complexity multiuser detection can be performed using maximum ratio combining (MRC) in the angle-delay-Doppler domain, is presented.

In the downlink, orthogonal communication can be achieved by using similar GB/interleaving-based strategies of OMA on the uplink. Although downlink orthogonal communication requires simple transmitter implementation at the BS, each UT needs to perform joint detection of all DD domain information symbols intended for it. When the number of information symbols intended for a UT is large, joint DD domain detection may be challenging for UTs which often have low signal processing capability. Also, higher SE can be achieved through non-orthogonal downlink communication with multiple antennas at the BS and downlink precoding in the DD domain, using either linear or non-linear precoding (refer to Sections 7.3.1.1 and 7.3.1.2). Downlink precoding also has the advantage of reducing the burden of joint DD domain detection at the UT. Low-complexity non-orthogonal downlink precoding for massive MIMO-OTFS is presented in Section 7.3.2.

Accurate knowledge of the effective DD domain channel is required at the BS for both multiuser detection in the uplink and precoding in the downlink. The DD channel estimation methods for single-user OTFS in a point-to-point setting (presented in Chapter 6) are applicable to multiuser OTFS as well. State-of-the-art multiuser DD channel estimation methods popularly known in the literature are presented in Section 7.4.

In communication receivers, symbol timing and carrier frequency synchronization is needed to ensure reliable communication. A unique advantage of OTFS-based communication systems is that frequency synchronization is not required as channel-induced Doppler shifts are part of the effective DD domain channel response. However, OTFS frame/timing synchronization may be required in scenarios where the difference between the propagation delays of two UTs is significant in comparison to the OTFS frame duration. In such scenarios, the absence of frame synchronization could lead to substantial interference between unsynchronized OTFS frames received at the BS from different UTs. OTFS frame synchronization requires the BS to estimate the propagation delay between a UT and the BS, which is then fed back to the UT for it to advance its uplink transmission timing. Traditionally, the estimation of propagation delay is based on TF domain waveforms, for which the estimation accuracy deteriorates significantly in high-mobility scenarios. In Section 7.5.2, a DD domain-based waveform is discussed for the estimation of propagation delay in high-mobility scenarios. In high-mobility scenarios, the estimation accuracy of this DD domain-based waveform is significantly better than that of traditional TF-based waveforms.

7.1 Resource Elements

In multiuser communication, physical resource is shared between multiple UTs. The smallest possible unit of physical resource that can be allocated to a UT is referred to as a RE. For example, in OFDM-based communication systems where information symbols are embedded in the TF domain, a RE is the combination of a subcarrier along the frequency domain (FD) and an OFDM symbol along the time domain. In OTFS modulation, the information symbols are embedded in the DD domain. The notion of RE in the context of OTFS modulation is defined in this section.

Consider the two-dimensional interval $[0, T) \times [0, \Delta f)$ in the DD domain, which is T seconds wide along the delay domain and $\Delta f = 1/T$ Hz wide along the Doppler domain. The delay domain

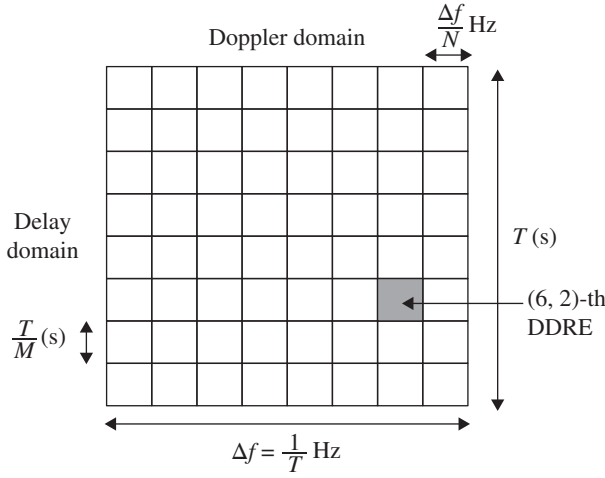


Figure 7.1 Two-dimensional DD domain interval $[0, T) \times [0, \Delta f)$ with $MN = 64$ DDREs, $M = N = 8$.

interval $[0, T]$ is divided into M divisions each T/M seconds wide, and the Doppler domain interval $[0, \Delta f]$ is divided into N divisions each $\Delta f/N$ Hz wide. The combination of a division along the delay domain and a division along the Doppler domain is referred to as a DD domain resource element (DDRE). A DDRE is the smallest unit of physical resource that can be used to communicate one information symbol. The l th delay domain division, $l = 0, 1, \dots, M - 1$, corresponds to the delay domain interval $\left[\frac{lT}{M}, \frac{(l+1)T}{M}\right]$, and the k th Doppler domain division, $k = 0, 1, \dots, N - 1$, corresponds to the Doppler domain interval $\left[\frac{k\Delta f}{N}, \frac{(k+1)\Delta f}{N}\right]$. The (k, l) th DDRE is a combination of the k th Doppler domain division and the l th delay domain division. An illustration of the DD domain interval and DDREs is presented in Fig. 7.1 for $M = N = 8$. There are $MN = 64$ DDREs, each shown by a small rectangle of length T/M sec along the delay domain and $\Delta f/N$ Hz along the Doppler domain.

Remark OTFS modulation referred to in this chapter is the multicarrier OTFS (MC-OTFS) modulation discussed in Chapter 3. The delay and Doppler period τ_p and $\nu_p = 1/\tau_p$ in Chapter 3 have been denoted by T and $\Delta f = 1/T$, respectively, in this chapter. As in Chapter 3, here also the total time duration and bandwidth of the signal received/transmitted at/from the BS are $NT = N\tau_p$ and $M\Delta f = M\nu_p$, respectively. Also, the information symbol transmitted on the (k, l) th DDRE is carried by the MC pulsone $x_{H, \frac{lT}{M}, \frac{k\Delta f}{N}}^{w_{tx}, g_{tx}}(t)$ (see (3.30), (3.31), and (3.41)). As discussed in Chapter 3, there are two transceiver architectures for MC-OTFS modulation, one based on the two-step approach and the other based on the Zak transform approach. In this chapter, the two-step approach has been used for MC-OTFS transceivers (see Sections 3.4.1 and 3.5.1). Specifically, in this chapter, the simplified two-step MC-OTFS transmitter and receiver illustrated through Figs. 3.11 and 3.13, respectively, have been used. Also, in terms of notation, in Chapter 3, k and l denote the discrete delay and Doppler domain index respectively, whereas here in this chapter, it is the opposite, i.e., k denotes the Doppler domain index and l denotes the delay domain index.

In OTFS modulation, the DD domain information signal is transformed to the TF domain and then to the time-domain (TD) transmit signal [28]. Therefore, TF resource can also be shared between UTs. The TD transmit signal is band limited to $M\Delta f$ Hz and is limited to a time duration of NT seconds. In other words, the total available TF resource is restricted to the frequency

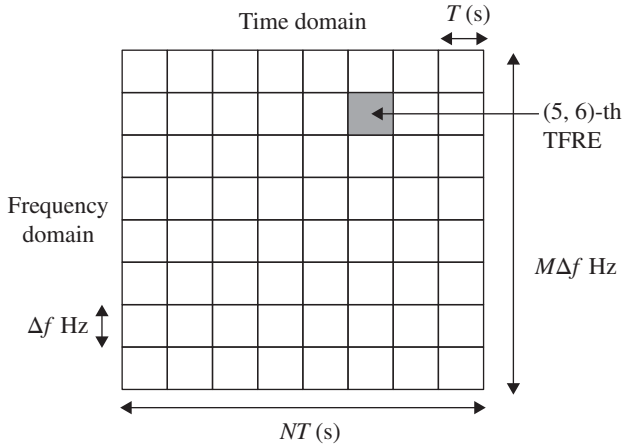


Figure 7.2 TF domain grid, $M = N = 8$.

domain interval $[0, M\Delta f]$ Hz and the TD interval $[0, NT]$ seconds [28–30]. An OFDM modulator with symbol length T and subcarrier spacing Δf can be used to convert the TF domain signal to the TD transmit signal. The FD interval is divided into M subcarriers each Δf Hz wide, and the TD interval is divided into N divisions each T seconds wide. Therefore, the smallest unit of TF resource, i.e., a time-frequency resource element (TFRE), is a combination of a subcarrier along the FD and a TD division of duration T seconds. To be precise, the (n, m) th TFRE is limited to the interval $[nT, (n+1)T]$ along TD and the interval $[m\Delta f, (m+1)\Delta f]$ along FD, $n = 0, 1, \dots, N-1$, $m = 0, 1, \dots, M-1$. Figure 7.2 shows an illustration of the TF domain grid for $M = N = 8$.

7.2 OTFS Uplink

Consider uplink multiuser communication where multiple UTs transmit independent information to the BS using OTFS modulation, as shown in Fig. 7.3. Let there be Q single-antenna UTs. The TD

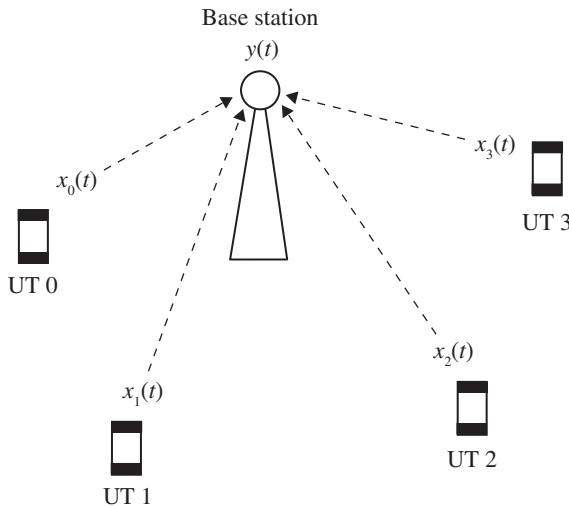


Figure 7.3 OTFS-based uplink communication with four UTs.

signal transmitted by the q th UT is denoted by $x_q(t)$, $q = 0, 1, \dots, Q - 1$. Consider scenarios where the wireless channel between a UT and the BS is characterized by multiple channel paths. Each path introduces a time delay and a Doppler shift. A time delay τ and a Doppler shift ν on a transmit signal $x(t)$ results in a received signal $x(t - \tau)e^{j2\pi\nu(t - \tau)}$ [55]. Let there be L_q channel paths between the q th UT and the BS, with the i th path introducing a time delay $\tau_{q,i}$ and a Doppler shift $\nu_{q,i}$. The complex gain of the i th channel path is denoted by $h_{q,i}$ which models fading and path loss. The TD signal received at the BS is then given by (see (2.2) and (2.3))

$$y(t) = \sum_{q=0}^{Q-1} \sum_{i=1}^{L_q} h_{q,i} x_q(t - \tau_{q,i}) e^{j2\pi\nu_{q,i}(t - \tau_{q,i})} + w(t), \quad (7.1)$$

where $w(t)$ is the AWGN at the BS having power spectral density (PSD) N_0 W/Hz. Also, let $\tau_{\max}$ and $\nu_{\max}$ denote the maximum possible delay and Doppler shift of any channel path, i.e.,

$$|\tau_{q,i}| \leq \tau_{\max}, \quad |\nu_{q,i}| \leq \nu_{\max}, \quad \forall q = 0, \dots, Q - 1, \quad i = 1, \dots, L_q. \quad (7.2)$$

7.2.1 Orthogonal Multiple Access (OMA)

In traditional OMA in OFDM-based systems (referred to as OFDMA), a RE is allocated to only one UT at a time. In OTFS-based systems, information can be restricted to a sub-region of the DD domain or the TF domain or both. Therefore, in OTFS-based OMA methods, we can have orthogonal access either in the DD domain or in the TF domain, or in both DD and TF domains. OMA allows for simple receiver processing as each RE can carry at most one information symbol. Also, OMA methods can be suitably designed to minimize MUI. The primary disadvantage of OMA is that resource utilization is limited to a single information symbol on every RE, which limits the achievable SE. State-of-the-art OMA methods/strategies for OTFS-based communication systems and their SE performances are described in Sections 7.2.1.1–7.2.1.3.

7.2.1.1 Guard Band (GB)-Based OMA

In GB-based OMA methods, orthogonal access is made in the DD domain and not in the TF domain. Each UT is allocated a set of DDREs which is not allocated to any other UT. Due to multipath delay and Doppler shift, the information transmitted by a UT on a DDRE spreads to DDREs which have been allocated to other UTs. This causes MUI which degrades performance. Therefore, a simple strategy is to separate the DDREs allocated to different UTs by a region of “guard” DDREs which do not carry any information and are not allocated to any UT [120].

A simple illustration of GB-based OMA, referred to as delay domain GB-OMA, is shown in Fig. 7.4 for $M = N = 8$ and $Q = 4$ UTs, where the DDREs allocated to a particular UT span the entire Doppler domain but are restricted to a single RE along the delay domain. The DDREs allocated to a UT are shaded, and the guard DDREs are unshaded. For example, the first UT (i.e., UT0) is allocated the DDREs in the set $\{(k, l) | l = 0, k = 0, 1, \dots, N - 1\}$. No other UT is allocated any DDRE which lies adjacent to a DDRE allocated to UT0, and therefore, all DDREs in the set $\{(k, l) | l = 1, k = 0, 1, \dots, N - 1\}$ are guard DDREs. Similarly, all DDREs in the set $\{(k, l) | l = M - 1, k = 0, 1, \dots, N - 1\}$ are used as guard DDREs to reduce interference between UT0 and UT3. The width of each GB in terms of the number of delay domain divisions is referred to as the GB size G (in Fig. 7.4, $G = 1$). Another type of GB-based OMA is Doppler domain GB-OMA, where the GBs are used to separate the DDREs along the Doppler domain, as shown through the example in Fig. 7.5 for $M = N = 8$ and $Q = 4$.

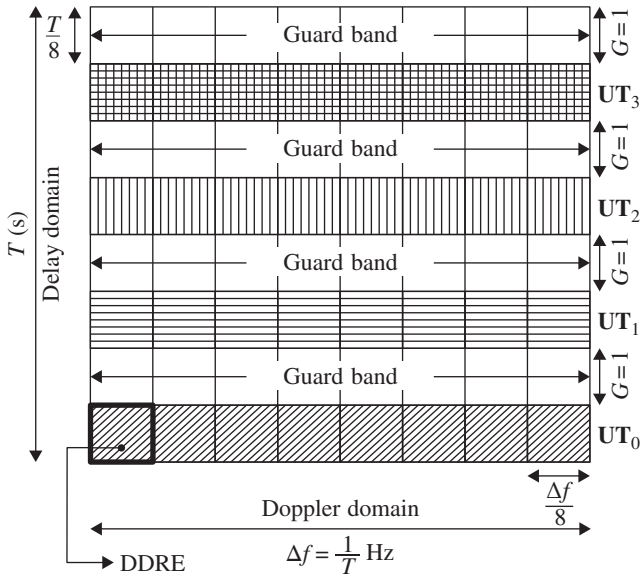


Figure 7.4 Delay domain GB-OMA. $M = N = 8$, $Q = 4$ UTs, and $G = 1$.

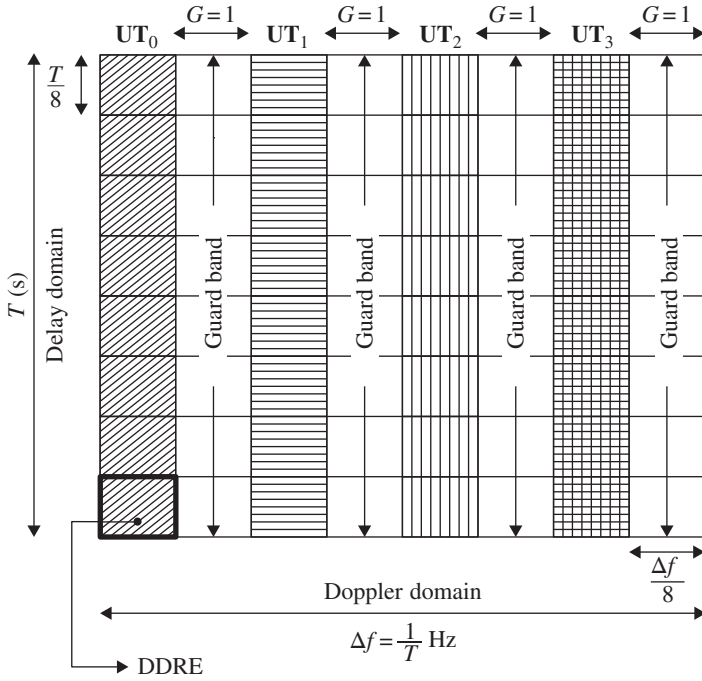


Figure 7.5 Doppler domain GB-OMA. $M = N = 8$, $Q = 4$ UTs, and $G = 1$.

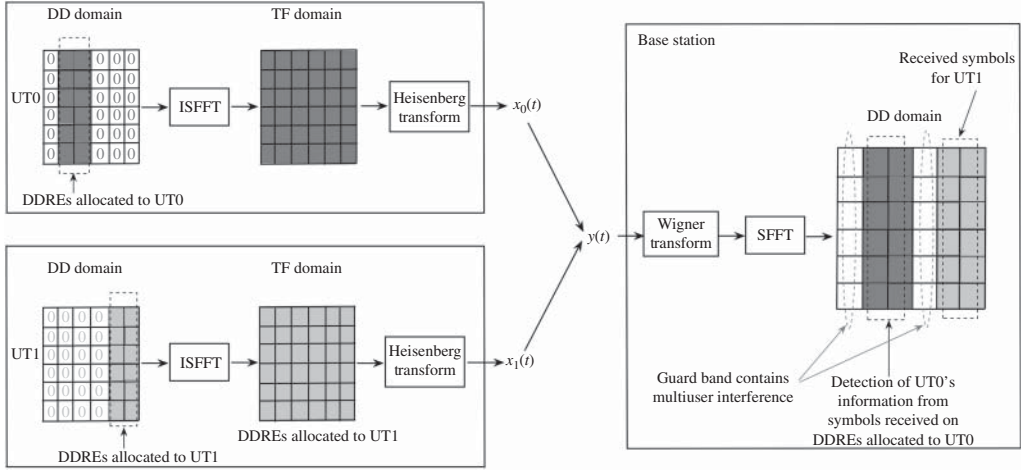


Figure 7.6 OTFS-based uplink communication with Doppler domain GB-OMA for $M = N = 6$, $Q = 2$.

In GB-OMA, each UT transmits its DD domain information symbols only on the DDREs allocated to it and transmits zero on the DDREs which are not allocated to it. See Fig. 7.6 for an example of Doppler domain GB-OMA with $M = N = 6$ and $Q = 2$ UTs. The signal processing at each UT is the same as that in an OTFS modulation-based transmitter. As shown in the figure, at each UT, the DD domain signal (with zeroes transmitted on DDREs not allocated to that UT) is transformed to the TF domain using the ISFFT. The resulting TF domain symbols are converted to the TD transmit signal using the Heisenberg transform. Note that even though the two UTs are allocated non-overlapping sets of DDREs in the DD domain, their TF domain symbols occupy the entire TF grid. At the BS receiver, the received TD signal is first transformed to the TF domain (using the Wigner transform) and then to the DD domain (using SFFT), just as in an OTFS receiver of a point-to-point system. The DD domain signal at the BS contains MUI in the GB DDREs. Hence, the GB DDREs are not used for detection of information symbols transmitted by the UTs. For a given UT, the DD domain symbols received in the DDREs allocated to that UT are used to detect the information symbols transmitted by that UT. Since delay and Doppler spreads cause MUI, the GB size should be sufficiently large so as to reduce the amount of MUI that leaks into the DDREs allocated to other UTs. For example, in Fig. 7.6, the GB size is $G = 1$ along the Doppler domain and the amount of interference leakage is expected to be small if the maximum Doppler shift is less than $G\Delta f/N = \frac{\Delta f}{6}$ Hz. Therefore, in high delay and Doppler scenarios (e.g., high mobility), the GB size has to be larger when compared to that required when the delay and Doppler spread is relatively small. Since GB DDREs do not carry information, the effective SE of GB-OMA methods reduces significantly in high delay and Doppler scenarios.

Remark For both the delay domain and Doppler domain GB-OMA methods, a larger GB size G would imply less MUI. However, this would also result in smaller sum SE as fewer information carrying DDREs would be allocated to each UT. Therefore, there is a trade-off between reducing MUI and the number of allocated DDREs. Also, in high Doppler spread scenarios, it is expected that delay domain GB-based MA would achieve higher sum SE than Doppler domain GB-based MA. This is because, in delay domain GB-based MA, the DDREs allocated to different UTs are separated along the delay domain, whereas channel-induced Doppler spread can cause MUI only along the Doppler domain.

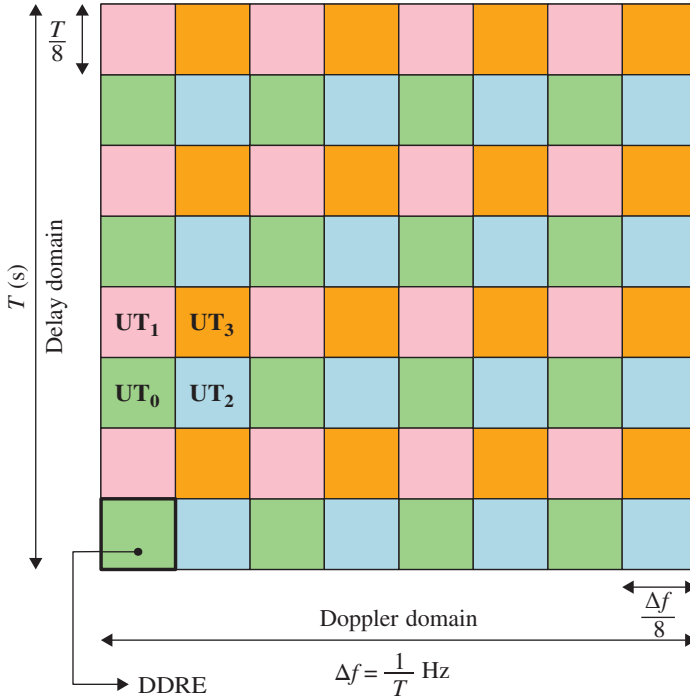


Figure 7.7 DDRE allocation in IDD-OMA. $M = N = 8$, $Q = 4$ UTs, and $g_1 = g_2 = 2$.

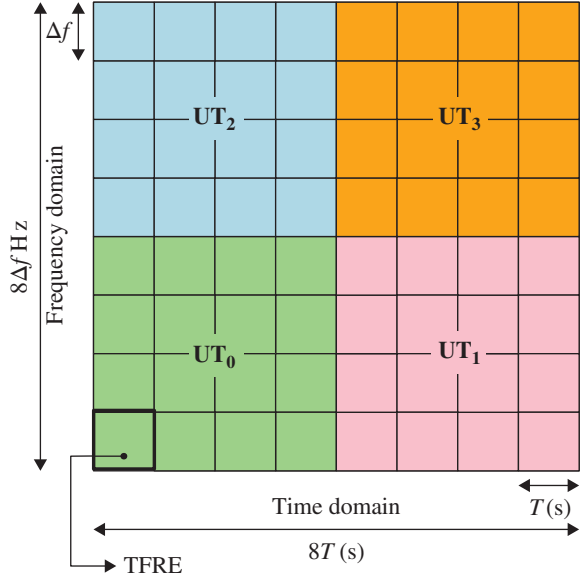
7.2.1.2 Interleaved Delay-Doppler OMA

In interleaved delay-Doppler OMA (IDD-OMA), the DDREs allocated to each UT are interleaved in the DD domain, i.e., they are spaced apart by g_1 DDREs along the delay domain and by g_2 DDREs along the Doppler domain [121]. No GB-DDREs are used. An example interleaved allocation of DDREs in IDD-OMA is shown in Fig. 7.7 for $M = N = 8$, $Q = 4$ UTs, and $g_1 = g_2 = 2$. Let g_1 and g_2 divide M and N , respectively. Due to interleaving, each UT is allocated $\frac{MN}{g_1 g_2}$ DDREs out of the total MN DDREs. Therefore, the number of UTs $Q = g_1 g_2$. Due to the regular spacing of allocated DDREs, the TF domain symbols of each UT can be restricted to a subset of the TF grid. The allocation of non-overlapping TFREs to the TF domain symbols of different UTs (i.e., OMA in the TF domain) results in better resilience to MUI when compared to GB-based OMA. For the interleaved DDRE allocation in Fig. 7.7, the corresponding TFRE allocation is shown in Fig. 7.8. Note that there are no GBs used in the DD or TF domain.

At the BS, the BS receiver separates the TF symbols received from different UTs. There is minimal MUI as the UTs are allocated non-overlapping TFREs. The received TF signal of each UT is then separately converted to the DD domain, followed by separate detection for each UT. An example of IDD-OMA-based uplink OTFS is illustrated in Fig. 7.9 for $M = N = 8$, $Q = 4$ UTs, and $g_1 = g_2 = 2$. Each UT transmits zero symbols in the DDREs which are not allocated to it. The q th UT ($0 \leq q < g_1 g_2$) is allocated DDREs in the set S_q , given by

$$S_q \triangleq \left\{ (k, l) \left| k = \left\lfloor \frac{q}{g_1} \right\rfloor + g_2 u, l = (q)_{g_1} + g_1 v, 0 \leq u < \frac{N}{g_2}, 0 \leq v < \frac{M}{g_1} \right. \right\}, \quad (7.3)$$

Figure 7.8 TFRE allocation in IDD-OMA corresponding to the DDRE allocation in Fig. 7.7. $M = N = 8$, $Q = 4$ UTs, and $g_1 = g_2 = 2$.



where, for any real number a , $\lfloor a \rfloor$ denotes the greatest integer less than or equal to a . Also, for any two integers q and g_1 , $(q)_{g_1}$ denotes the unique smallest non-negative integer which is congruent to q modulo g_1 . For all $(k, l) \in S_q$, let $x_q[k, l]$ denote the DD domain information symbol transmitted by the q th UT on the (k, l) th DDRE. Note that $x_q[k, l] = 0$ for $(k, l) \notin S_q$. At the q th UT, these DD domain information symbols are transformed to the TF domain through ISFFT (see Fig. 7.9). The corresponding TF symbols are given by (see (3.53))

$$\begin{aligned}
 X_q[n, m] &= \frac{1}{MN} \sum_{(k, l) \in S_q} x_q[k, l] e^{-j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)}, \\
 &\stackrel{(a)}{=} \lambda(m, n) \sum_{u=0}^{\frac{N}{g_2}-1} \sum_{v=0}^{\frac{M}{g_1}-1} \tilde{x}_q[u, v] e^{-j2\pi \left(\frac{mv}{M/g_1} - \frac{nu}{N/g_2} \right)}, \quad (7.4)
 \end{aligned}$$

where $n = 0, 1, \dots, (N/g_2) - 1$, $m = 0, 1, \dots, (M/g_1) - 1$, $\lambda(m, n) \triangleq e^{-j2\pi \left(\frac{m(q)_{g_1}}{M} - \frac{n(q)_{g_1}}{N} \right)} / (MN)$, and step (a) follows from the fact that $x_q[k, l]$ is zero for all $(k, l) \notin S_q$. The information symbols of the q th UT $\tilde{x}_q[u, v] \triangleq x_q[k = \lfloor q/g_1 \rfloor + g_1 u, l = (q)_{g_1} + g_1 v]$ are transmitted on DDREs which are spaced apart by g_1 DDREs along the delay domain and by g_2 DDREs along the Doppler domain. Due to this, from (7.4), it follows that for any integers (t_1, t_2) , $X_q[n + t_1(N/g_2), m + t_2(M/g_1)] = e^{-j2\pi \left(\frac{t_2(q)_{g_1}}{g_1} - \frac{t_1 \lfloor q/g_1 \rfloor}{g_2} \right)} X_q[n, m]$, i.e., the TF-modulated symbols are invariant to translations by integer multiples of N/g_2 and M/g_1 in the time and frequency domain, respectively (invariance up to a fixed complex exponential factor). This implies that the TF symbols for each UT can be confined to a (NT/g_2) seconds $\times$ $(M\Delta f/g_1)$ Hz region of the TF domain. The confinement of the TF symbols to a (NT/g_2) seconds $\times$ $(M\Delta f/g_1)$ Hz region of the TF domain is achieved through the following conversion of TF domain symbols to the TD transmit signal. At the q th UT, the TF symbols $X_q[n, m]$, $n = 0, 1, \dots, N/g_2 - 1$, $m = 0, 1, \dots, M/g_1 - 1$ are converted to the transmit TD signal $x_q(t)$ using the Heisenberg transform (see Fig. 7.9). The TD signal transmitted from the q th UT is

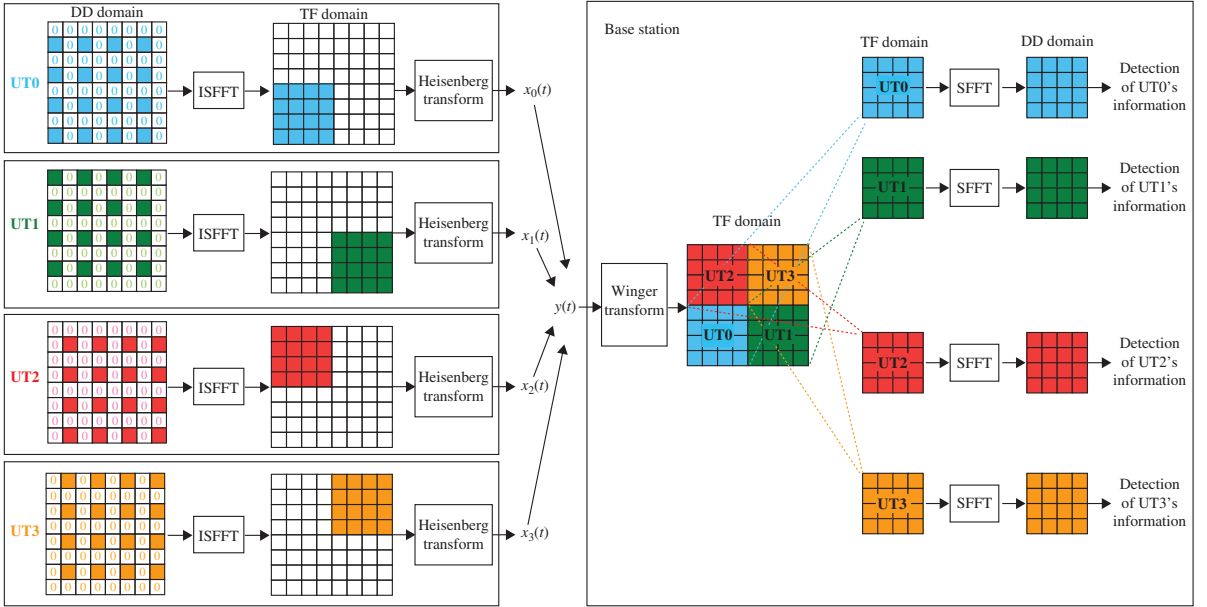


Figure 7.9 OTFS-based uplink communication with IDD-OMA. $M = N = 8$, $Q = 4$ UTs, and $g_1 = g_2 = 2$.

therefore given by (see (3.55) and Fig. 3.11, with the TF window $W_{\text{tx}}(\cdot, \cdot)$ being constant within the TF region allocated to the q th UT and zero outside this region)

$$x_q(t) = \sum_{m=0}^{M/g_1-1} \sum_{n=0}^{N/g_2-1} \left[X_q[n, m] g_{\text{tx}}(t - (n + (N/g_2)(q)_{g_2})T) e^{j2\pi(m + (M/g_1)\lfloor q/g_2 \rfloor)\Delta f(t - (n + (N/g_2)(q)_{g_2})T)} \right], \quad (7.5)$$

where $g_{\text{tx}}(\cdot)$ is the transmit pulse (referred to as the MC generator in Chapter 3) which satisfies

$$\int g_{\text{tx}}^*(t - nT) g_{\text{tx}}(t) e^{-j2\pi m \Delta f(t - nT)} dt = \delta[m] \delta[n]. \quad (7.6)$$

Here, $\delta[m]$, $m \in \mathbb{Z}$, is one when $m = 0$ and is zero otherwise. The conversion in (7.5) guarantees that the TF symbols of the q th UT are confined to the region $\left[(NT/g_2)(q)_{g_2}, (NT/g_2)((q)_{g_2} + 1) \right]$ seconds $\times$ $\left[(M/g_1)\lfloor q/g_2 \rfloor \Delta f, (M/g_1)(\lfloor q/g_2 \rfloor + 1)\Delta f \right]$ Hz of the TF domain (see Fig. 7.8). Also, for two distinct UTs, their corresponding TF regions are non-overlapping.

The signal received at the BS is given by (7.1) with the signal transmitted by each UT given by (7.5). The BS generates $MN/(g_1 g_2)$ TF symbols for each UT using the Wigner transform (see (3.61) and Fig. 3.13). As illustrated in Fig. 7.9, the TF symbols for each UT at the BS are separable as the TFREs allocated to the UTs are non-overlapping. As the TFREs allocated to the q' th UT, $q' = 0, 1, \dots, Q - 1$, belong to the region $\left[(NT/g_2)(q')_{g_2}, (NT/g_2)((q')_{g_2} + 1) \right]$ seconds $\times$ $\left[(M/g_1)\lfloor q'/g_2 \rfloor \Delta f, (M/g_1)(\lfloor q'/g_2 \rfloor + 1)\Delta f \right]$ Hz of the TF domain, the $MN/(g_1 g_2)$ TF symbols for the q' th UT are given by [121] (see also (3.62) with the TF window $W_{\text{rx}}(\cdot, \cdot)$ being constant within the TF region allocated to the q' th UT and zero outside this region)

$$Y_{q'}[\tilde{n}, \tilde{m}] = \int \left\{ g_{\text{rx}}^*(t - (\tilde{n} + (N/g_2)(q')_{g_2})T) y(t) e^{-j2\pi(\tilde{m} + (M/g_1)\lfloor q'/g_2 \rfloor)\Delta f(t - (\tilde{n} + (N/g_2)(q')_{g_2})T)} \right\} dt, \quad (7.7)$$

where $\tilde{n} = 0, 1, \dots, N/g_2 - 1$, $\tilde{m} = 0, 1, \dots, M/g_1 - 1$. Here, $g_{\text{rx}}(\cdot)$ is the receive pulse. For any $(m, n) \neq (0, 0)$ and any (τ, ν) such that $|\tau - nT| \leq \tau_{\text{max}}$ and $|\nu - m\Delta f| \leq \nu_{\text{max}}$, the transmit and receive pulses satisfy¹

$$\int g_{\text{rx}}^*(t - \tau) g_{\text{tx}}(t) e^{-j2\pi \nu(t - \tau)} dt = 0. \quad (7.8)$$

Furthermore,

$$\int g_{\text{rx}}^*(t - \tau) g_{\text{tx}}(t) e^{-j2\pi \nu(t - \tau)} dt = 1, \quad |\tau| \leq \tau_{\text{max}}, \quad |\nu| \leq \nu_{\text{max}}. \quad (7.9)$$

Like the transmit pulse, the receive pulse $g_{\text{rx}}(\cdot)$ also satisfies

$$\int g_{\text{rx}}^*(t - nT) g_{\text{rx}}(t) e^{-j2\pi m \Delta f(t - nT)} dt = \delta[m] \delta[n]. \quad (7.10)$$

The following development of the input–output relation for IDD-OMA shows that IDD-OMA is resilient to MUI.

Input–output relation for IDD-OMA: First consider the TF domain input–output relation between the TF domain symbols transmitted by that UT and the TF domain symbols received at the BS.

¹ This is an ideal condition not satisfied by practical pulses/MC generators. The performance with ideal pulses gives performance bounds and fundamental insights. The performance of these OMA methods with practical rectangular pulses has been studied in [122], and the results have been included here in Section 7.2.1.5. The ideal condition in (7.8) has been discussed in detail in (3.79) of Corollary 3.5. See also the discussion in the paragraph just after this corollary but before Theorem 3.12.

Using (7.5) for $x_q(t)$ in (7.1) gives the expression for $y(t)$, which when used in (7.7) gives the expression for the received TF domain symbols of each UT. The expression for the received TF domain symbols of the q' th UT (in the set of TFREs allocated to the q' th UT) is given by

$$Y_{q'}[\tilde{n}, \tilde{m}] = \sum_{n=0}^{N/g_2-1} \sum_{m=0}^{M/g_1-1} \sum_{q=0}^{Q-1} \sum_{i=1}^{L_q} [X_q[n, m] h_{q,i} e^{-j2\pi v_{q,i} \tau_{q,i}} e^{-j2\pi \tau_{q,i} \Delta f (m + (M/g_1) \lfloor q/g_2 \rfloor)} F_i(\tilde{n}, \tilde{m}, n, m, q, q')] + W_{q'}[\tilde{n}, \tilde{m}], \quad (7.11)$$

where

$$F_i(\tilde{n}, \tilde{m}, n, m, q, q') \stackrel{\Delta}{=} \int \left[g_{\text{rx}}^*(t - (\tilde{n} + (N/g_2)(q')_{g_2})T) g_{\text{tx}}(t - \tau_{q,i} - nT - (N/g_2)(q)_{g_2}T) e^{j2\pi t(v_{q,i} + \Delta f(m - \tilde{m} + (M/g_1)(\lfloor q/g_2 \rfloor - \lfloor q'/g_2 \rfloor)))} \right] dt, \quad (7.12)$$

$\tilde{n} = 0, 1, \dots, N/g_2 - 1$, $\tilde{m} = 0, 1, \dots, M/g_1 - 1$. $(\tau_{q,i}, v_{q,i})$ satisfies (7.2), and for $q \neq q'$, $((q)_{g_2}, \lfloor q/g_2 \rfloor) \neq ((q')_{g_2}, \lfloor q'/g_2 \rfloor)$. Therefore, from the property of the transmit and receive pulses in (7.8), $F_i(\tilde{n}, \tilde{m}, n, m, q, q')$ can be written as

$$F_i(\tilde{n}, \tilde{m}, n, m, q, q') = e^{j2\pi v_{q',i} T (\tilde{n} + (N/g_2)(q')_{g_2})} \delta[m - \tilde{m}] \delta[n - \tilde{n}] \delta[q - q']. \quad (7.13)$$

Using (7.13) in (7.11) gives the TF domain input–output relation as

$$Y_{q'}[\tilde{n}, \tilde{m}] = X_{q'}[\tilde{n}, \tilde{m}] \tilde{H}_{q'}[\tilde{n}, \tilde{m}] + W_{q'}[\tilde{n}, \tilde{m}], \quad (7.14)$$

where

$$\tilde{H}_{q'}[\tilde{n}, \tilde{m}] \stackrel{\Delta}{=} \sum_{i=1}^{L_{q'}} \left(h_{q',i} e^{-j2\pi v_{q',i} \tau_{q',i}} e^{j2\pi v_{q',i} (\tilde{n} + (N/g_2)(q')_{g_2})T} e^{-j2\pi \tau_{q',i} \Delta f (\tilde{m} + (M/g_1) \lfloor q'/g_2 \rfloor)} \right) \quad (7.15)$$

and

$$W_{q'}[\tilde{n}, \tilde{m}] \stackrel{\Delta}{=} \int \left\{ g_{\text{rx}}^*(t - (\tilde{n} + (N/g_2)(q')_{g_2})T) w(t) e^{-j2\pi (\tilde{m} + (M/g_1) \lfloor q'/g_2 \rfloor) \Delta f (t - (\tilde{n} + (N/g_2)(q')_{g_2})T)} \right\} dt. \quad (7.16)$$

From the above, it is clear that $Y_{q'}[\tilde{n}, \tilde{m}]$ depends only on the TF symbols transmitted by the q' th UT, and, therefore, there is no MUI in the TF domain.

Now consider the DD domain input–output relation for each UT. For the q th UT, the BS transforms the received TF domain symbols $Y_q[\tilde{n}, \tilde{m}]$ to the DD domain using SFFT, i.e., for all $0 \leq \tilde{u} < N/g_2$ and $0 \leq \tilde{v} < M/g_1$, the received DD domain symbols $\hat{x}_q[\tilde{u}, \tilde{v}]$ are given by (see (3.64))

$$\hat{x}_q[\tilde{u}, \tilde{v}] \stackrel{\Delta}{=} \sum_{\tilde{n}=0}^{N/g_2-1} \sum_{\tilde{m}=0}^{M/g_1-1} Y_q[\tilde{n}, \tilde{m}] e^{j2\pi \left(\frac{\tilde{m}\tilde{v}}{M/g_1} - \frac{\tilde{n}\tilde{u}}{N/g_2} \right)}. \quad (7.17)$$

Substituting this expression for $X_q[n, m]$ from (7.4) into (7.14) gives the expression for $Y_q[\tilde{n}, \tilde{m}]$ in terms of the DD domain information symbols $\tilde{x}_q[., .]$. Substituting the expression of $Y_q[\tilde{n}, \tilde{m}]$ on the RHS of (7.17) gives the DD domain input–output relation between the DD domain information symbols transmitted by a UT and the DD domain symbols received at the BS as

$$\hat{x}_q[\tilde{u}, \tilde{v}] = \sum_{u=0}^{N/g_2-1} \sum_{v=0}^{M/g_1-1} \tilde{x}_q[u, v] \tilde{h}_q \left[(\tilde{u} - u)_{N/g_2}, (\tilde{v} - v)_{M/g_1} \right] + w_q[\tilde{u}, \tilde{v}], \quad (7.18)$$

where

$$\begin{aligned}
\tilde{h}_q[k, l] &\triangleq \sum_{i=1}^{L_q} \left[h_{q,i} e^{-j2\pi \left(\nu_{q,i} \tau_{q,i} + \frac{\tau_{q,i}}{T} \frac{M}{g_1} \left\lfloor \frac{q}{g_2} \right\rfloor - \frac{\nu_{q,i}}{\Delta f} \frac{N}{g_2} (q)_{g_2} \right)} D_{q,i}[l] E_{q,i}[k] \right], \\
D_{q,i}[l] &\triangleq \frac{1}{M} \sum_{m=0}^{M/g_1-1} e^{-j2\pi m \left(\frac{(q)_{g_1}}{M} - \frac{l}{M/g_1} + \frac{\tau_{q,i}}{T} \right)}, \\
E_{q,i}[k] &\triangleq \frac{1}{N} \sum_{n=0}^{N/g_2-1} e^{j2\pi n \left(\frac{(q)_{g_1}}{N} - \frac{k}{N/g_2} + \frac{\nu_{q,i}}{\Delta f} \right)}, \\
w_q[\tilde{u}, \tilde{v}] &\triangleq \sum_{\tilde{n}=0}^{N/g_2-1} \sum_{\tilde{m}=0}^{M/g_1-1} W_q[\tilde{n}, \tilde{m}] e^{j2\pi \left(\frac{\tilde{m}\tilde{v}}{M/g_1} - \frac{\tilde{n}\tilde{u}}{N/g_2} \right)}, \tag{7.19}
\end{aligned}$$

where $W_q[\tilde{n}, \tilde{m}]$ is given by (7.14). Since AWGN at the BS has PSD N_0 (see (7.1)), using (7.10) in (7.14), it follows that $W_q[\tilde{n}, \tilde{m}] \sim \text{i.i.d. } \mathcal{CN}(0, N_0)$, $\tilde{n} = 0, 1, \dots, (N/g_2) - 1$, $\tilde{m} = 0, 1, \dots, (M/g_1) - 1$. From (7.18), it then follows that $w_q[\tilde{u}, \tilde{v}] \sim \text{i.i.d. } \mathcal{CN}(0, MNN_0/Q)$. The 2D convolution in (7.18) can be written in a matrix-vector form as

$$\hat{\mathbf{x}}_q = \hat{\mathbf{H}}_q \tilde{\mathbf{x}}_q + \tilde{\mathbf{w}}_q, \tag{7.20}$$

where

$$\begin{aligned}
\tilde{\mathbf{x}}_q &\triangleq \left[\tilde{x}_q[0, 0], \dots, \tilde{x}_q \left[\frac{N}{g_2} - 1, 0 \right], \dots, \tilde{x}_q \left[0, \frac{M}{g_1} - 1 \right], \dots, \tilde{x}_q \left[\frac{N}{g_2} - 1, \frac{M}{g_1} - 1 \right] \right]^T, \\
\hat{\mathbf{x}}_q &\triangleq \left[\hat{x}_q[0, 0], \dots, \hat{x}_q \left[\frac{N}{g_2} - 1, 0 \right], \dots, \hat{x}_q \left[0, \frac{M}{g_1} - 1 \right], \dots, \hat{x}_q \left[\frac{N}{g_2} - 1, \frac{M}{g_1} - 1 \right] \right]^T, \\
\tilde{\mathbf{w}}_q &\triangleq \left[w_q[0, 0], \dots, w_q \left[\frac{N}{g_2} - 1, 0 \right], \dots, w_q \left[0, \frac{M}{g_1} - 1 \right], \dots, w_q \left[\frac{N}{g_2} - 1, \frac{M}{g_1} - 1 \right] \right]^T.
\end{aligned}$$

The matrix $\hat{\mathbf{H}}_q \in \mathbb{C}^{\frac{MN}{g_1 g_2} \times \frac{MN}{g_1 g_2}}$ is derived from the 2D convolution in (7.18). To be precise, from (7.18), it follows that the element of $\hat{\mathbf{H}}_q$ in its $(\tilde{v}(N/g_2) + \tilde{u} + 1)$ th row and $(v(N/g_2) + u + 1)$ th column is given by $\tilde{h}_q \left[(\tilde{u} - u)_{N/g_2}, (\tilde{v} - v)_{M/g_1} \right]$, $\tilde{u}, u = 0, 1, \dots, (N/g_2) - 1$ and $\tilde{v}, v = 0, 1, \dots, (M/g_1) - 1$. With the DD domain information symbols $\tilde{x}_q[u, v]$ distributed as i.i.d. $\mathcal{CN}(0, E_s)$, from (7.20), it follows that the achievable sum SE (bits/s/Hz) for the IDD-OMA method is given by [6, 123]

$$R \triangleq \frac{1}{MN} \sum_{q=0}^{Q-1} \log_{g_2} \left| \mathbf{I} + \frac{E_s Q}{MNN_0} \hat{\mathbf{H}}_q \hat{\mathbf{H}}_q^H \right|, \tag{7.21}$$

where for any square matrix $\mathbf{A}$, $|\mathbf{A}|$ denotes its determinant. With i.i.d. $\mathcal{CN}(0, E_s)$ symbols $\tilde{x}_q[u, v]$, from (7.5) and (7.6), it follows that the average total transmitted energy from each UT is $\mathbb{E} \left[\int |x_q(t)|^2 dt \right] = \frac{E_s}{Q^2}$. Since the transmission time is NT seconds, the average transmitted power for each UT is $\frac{E_s}{NTQ^2}$. The total noise power in the communication bandwidth is $M\Delta f N_0$. Therefore, the ratio of the average power transmitted by each UT to the AWGN power at the BS is $\rho \triangleq \frac{E_s}{MNN_0 Q^2}$.

7.2.1.3 Interleaved Time-Frequency OMA

Interleaved time-frequency OMA (ITF-OMA) is another OMA method which is also resilient to MUI and does not use DD domain GBs [124]. The main difference when compared to IDD-OMA is that, in ITF-OMA, interleaving is performed in the TF domain.

In ITF-OMA, each UT transmits MN/Q DD domain information symbols. Let $Q = r_1 r_2$, where r_1 and r_2 are positive integers. At each UT, the MN/Q information symbols are quasi-periodically repeated $Q = r_1 r_2$ times in the DD domain so that there are MN DD domain symbols after this repetition (a quasi-periodic repetition by r_1 times along the delay domain and by r_2 times along the Doppler domain). These quasi-periodically repeated MN symbols are then transmitted on the MN DDREs. Therefore, each UT is allocated the entire DD domain. The manner in which DD domain quasi-periodic repetition is carried out is different for each UT and is such that after SFFT, the corresponding TF symbols of each UT are interleaved in the TF domain. That is, for each UT, only MN/Q out of the MN TF symbols are non-zero. These non-zero TF symbols occupy TFREs which are regularly spaced apart by r_1 TFREs along the frequency domain and by r_2 TFREs along the time domain. Also, different UTs occupy non-overlapping TFREs due to which the TF symbols of each UT are received at the BS with minimal MUI. Therefore, just like IDD-OMA, ITF-OMA also has the potential to achieve interference-free multiuser communication without the need for GBs.

An illustration of the ITF-OMA method for $M = N = 8$, $Q = 4$ UTs, and $r_1 = r_2 = 2$ is provided in Fig. 7.10. Let $u_q[k', l']$, $k' = 0, 1, \dots, (N/r_2) - 1$, $l' = 0, 1, \dots, (M/r_1) - 1$ denote the $MN/Q = MN/(r_1 r_2)$ DD domain information symbols transmitted by the q th UT, $q = 0, 1, \dots, Q - 1$. Each information symbol is repeated $Q = r_1 r_2$ times, i.e., r_1 times along the delay domain and r_2 times along the Doppler domain. That is, for a given (k', l') , the DD domain information symbol $u_q[k', l']$ is transmitted on all the $r_1 r_2$ DDREs in the set

$$\mathcal{R}_{q,k',l'} \triangleq \{ (k, l) \mid k = k' + d(N/r_2), l = l' + c(M/r_1), \\ d = 0, 1, \dots, (r_2 - 1), c = 0, 1, \dots, (r_1 - 1) \}. \quad (7.22)$$

To be precise, the symbol transmitted by the q th UT on any DDRE in this set is

$$x_q[k = k' + d(N/r_2), l = l' + c(M/r_1)] = u_q[k', l'] \underbrace{e^{j2\pi \left(c \frac{(q)r_1}{r_1} - d \frac{\lfloor q/r_1 \rfloor}{r_2} \right)}}_{\text{Quasi-periodic repetition factor}}. \quad (7.23)$$

Note that the same information symbol $u_q[k', l']$ is transmitted on all the $r_1 r_2$ DDREs in the set $\mathcal{R}_{q,k',l'}$, but with different complex exponential term $e^{j2\pi \left(c \frac{(q)r_1}{r_1} - d \frac{\lfloor q/r_1 \rfloor}{r_2} \right)}$. Note the dependence of this complex exponential term (i.e., the quasi-periodic repetition factor) depends on the UT index q . Using ISFFT, the MN symbols $x_q[k, l]$ s are then transformed to the TF symbols $X_q[n, m]$, $n = 0, 1, \dots, (N - 1)$, $m = 0, 1, \dots, M - 1$, given by

$$X_q[n, m] = \frac{1}{MN} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x_q[k, l] e^{-j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)} \\ \stackrel{(a)}{=} \frac{1}{MN} \sum_{k'=0}^{\frac{N}{r_2}-1} \sum_{l'=0}^{\frac{M}{r_1}-1} u_q[k', l'] T_q[k', l', n, m], \quad (7.24)$$

where

$$T_q[k', l', n, m] \triangleq \sum_{d=0}^{r_2-1} \sum_{c=0}^{r_1-1} e^{j2\pi \left(c \frac{(q)r_1}{r_1} - d \frac{\lfloor q/r_1 \rfloor}{r_2} \right)} e^{-j2\pi \left(\frac{m(l'+c(M/r_1))}{M} - \frac{n(k'+d(N/r_2))}{N} \right)}$$

and step (a) follows from (7.23). Further simplification of $T_q[k', l', n, m]$ gives

$$T_q[k', l', n, m] = e^{-j2\pi \left(\frac{ml'}{M} - \frac{nk'}{N} \right)} \sum_{d=0}^{r_2-1} e^{j2\pi \frac{d}{r_2} (n - \lfloor q/r_1 \rfloor)} \sum_{c=0}^{r_1-1} e^{-j2\pi \frac{c}{r_1} (m - (q)r_1)}. \quad (7.25)$$

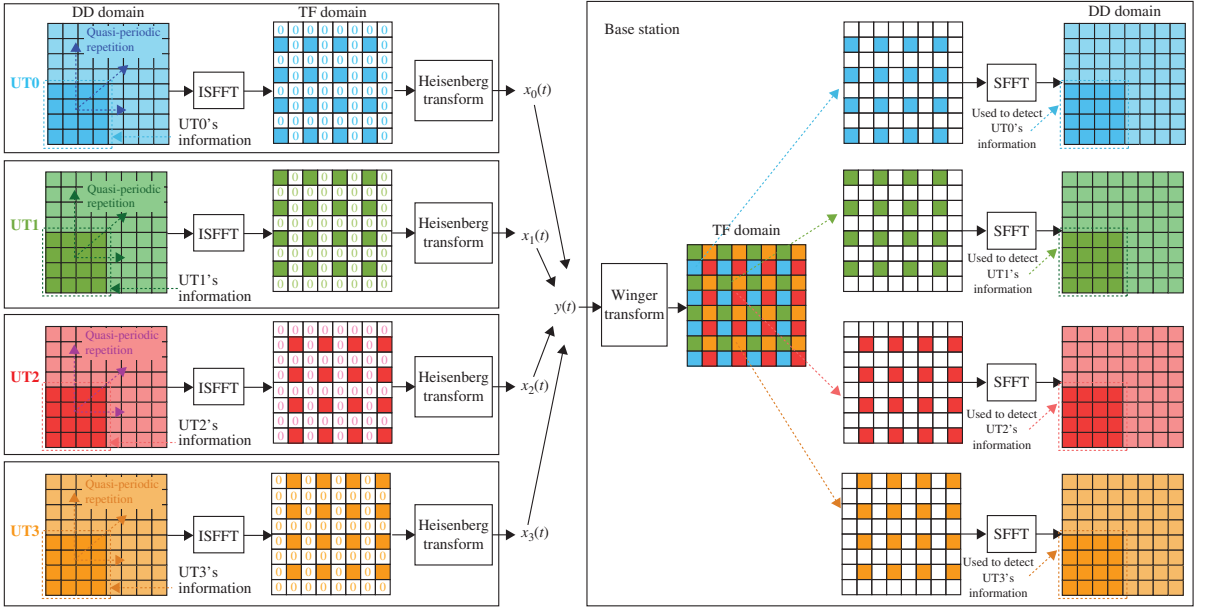


Figure 7.10 OTFS-based uplink communication with ITF-OMA. $M = N = 8$, $Q = 4$ UTs, $r_1 = r_2 = 2$.

Here, it is clear that $\sum_{d=0}^{r_2-1} e^{j2\pi \frac{d}{r_2} (n - \lfloor q/r_1 \rfloor)} = r_2 \delta \left[(n - \lfloor q/r_1 \rfloor)_{r_2} \right]$, i.e., it is zero for all n such that n is not congruent to $\lfloor q/r_1 \rfloor$ modulo r_2 (that is, all n such that $(n - \lfloor q/r_1 \rfloor)$ is not divisible by r_2). Similarly, $\sum_{c=0}^{r_1-1} e^{-j2\pi \frac{c}{r_1} (m - (q)_{r_1})} = r_1 \delta \left[(m - (q)_{r_1})_{r_1} \right]$. Using these facts in (7.25) gives a simplified expression for $T_q[k', l', n, m]$, using which in step (a) of (7.24) gives

$$X_q[n, m] = \begin{cases} 0, & \text{for } (n - \lfloor q/r_1 \rfloor)_{r_2} \neq 0 \\ & \text{or } (m - (q)_{r_1})_{r_1} \neq 0 \\ \frac{1}{\left(\frac{M}{r_1}\right)\left(\frac{N}{r_2}\right)} \sum_{k'=0}^{\frac{N}{r_2}-1} \sum_{l'=0}^{\frac{M}{r_1}-1} u_q[k', l'] e^{-j2\pi \left(\frac{ml'}{M} - \frac{nk'}{N}\right)}, & \text{for } (n - \lfloor q/r_1 \rfloor)_{r_2} = 0 \\ & \text{and } (m - (q)_{r_1})_{r_1} = 0. \end{cases} \quad (7.26)$$

From (7.26), it is clear that for the q th UT, only those TF symbols $X[n, m]$, $n = 0, 1, \dots, N-1$, $m = 0, 1, \dots, M-1$, are non-zero for which $(n - \lfloor q/r_1 \rfloor)$ is an integer multiple of r_2 and $(m - (q)_{r_1})$ is an integer multiple of r_1 . Therefore, the q th UT is only allocated those TFREs where $X_q[n, m]$ is not zero, i.e., the TFREs in the set

$$\mathcal{T}_q \triangleq \left\{ (n, m) \mid n = \lfloor q/r_1 \rfloor + d'r_2, m = (q)_{r_1} + c'r_1, d' = 0, 1, \dots, \frac{N}{r_2} - 1, c' = 0, 1, \dots, \frac{M}{r_1} - 1 \right\}. \quad (7.27)$$

Since $q = 0, 1, \dots, Q-1$ and $Q = r_1 r_2$, from (7.27), it follows that for any $q_2 \neq q_1$ and $q_1, q_2 \in \{0, 1, \dots, Q-1\}$, $\mathcal{T}_{q_1}$ and $\mathcal{T}_{q_2}$ are disjoint sets. In other words, the set of TFREs allocated to any two UTs are non-overlapping. An illustration of this fact is shown in Fig. 7.10 for $M = N = 8$, $Q = 4$ UTs, and $r_1 = r_2 = 2$. These TF symbols are then converted to the TD transmit signal using the Heisenberg transform. The TD signal transmitted by the q th UT is, therefore, given by

$$x_q(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X_q[n, m] g_{\text{tx}}(t - nT) e^{j2\pi m \Delta f (t - nT)}, \quad (7.28)$$

where $g_{\text{tx}}(\cdot)$ is the transmit pulse which satisfies (7.6). The TD signal $y(t)$ received at the BS is given by (7.1). At the BS, the received TD signal is converted to the TF domain using the Wigner transform. The received TF symbols are therefore given by

$$Y[n, m] = \int y(t) g_{\text{rx}}^*(t - nT) e^{-j2\pi m \Delta f (t - nT)} dt, \quad (7.29)$$

where the receive pulse $g_{\text{rx}}(\cdot)$ satisfies (7.10). The transmit and receive pulses also satisfy (7.8) and (7.9). Since non-overlapping TFREs are allocated to the UTs, their respective TF symbols are separable. Furthermore, since the information of each UT is restricted to TFREs allocated to that UT, the DD domain information symbols transmitted by each UT can be separately detected with minimal MUI. The received TF domain symbols for the q th UT are given by

$$Y_q[n, m] = \begin{cases} Y[n, m], & (n, m) \in \mathcal{T}_q \\ 0, & (n, m) \notin \mathcal{T}_q. \end{cases} \quad (7.30)$$

At the BS, the received TF domain symbols for each UT are then separately converted to the DD domain using SFFT (see Fig. 7.10). The received DD domain symbols for the q th UT are given by

$$\hat{x}_q[k, l] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y_q[n, m] e^{j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)}$$

$$k = 0, 1, \dots, \frac{N}{r_2} - 1, \quad l = 0, 1, \dots, \frac{M}{r_1} - 1, \quad (7.31)$$

where $Y_q[n, m]$ is given by (7.30). Substituting (7.26) in (7.28) gives an expression for $x_q(t)$ in terms of the DD domain information symbols $u_q[\cdot, \cdot]$. Substituting $x_q(t)$ in (7.1) gives $y(t)$. Using $y(t)$ in (7.29) gives $Y[n, m]$. Using $Y[n, m]$ in (7.30) gives an expression for $Y_q[\cdot, \cdot]$. Using $Y_q[\cdot, \cdot]$ in (7.31) results in the final input-output relation, i.e., an expression for $\hat{x}_q[k, l]$ in terms of $u_q[\cdot, \cdot]$, the DD domain information symbols of the q th UT. This input-output relation can be expressed in the matrix-vector form (as done for IDD-OMA). Using this matrix-vector representation of the input-output relation, the expression for an achievable sum SE can be derived.

7.2.1.4 Performance of OTFS-Based OMA Methods with Ideal Pulses

In this section, the sum spectral efficiency of GB-OMA, IDD-OMA, and ITF-OMA are compared when the pulses $g_{tx}(t)$ and $g_{rx}(t)$ at the transmitter and receiver are ideal, i.e., they satisfy the condition in (7.8). The GB-OMA method aims to reduce MUI through the use of GBs in the DD domain. Since GBs do not carry information, this reduces the effective sum SE. The other methods, i.e., IDD-OMA and ITF-OMA, use interleaving of allocated REs in the DD and TF domains, respectively, to achieve MUI resilient communication without the need for GBs. Therefore, the IDD-OMA and ITF-OMA methods are expected to be more spectral efficient when compared to the GB-OMA method.

The achievable sum SE of GB-OMA, IDD-OMA, and ITF-OMA methods are compared in this section. Consider an OTFS-based uplink communication system, where $\Delta f = 1/T = 15$ kHz, $M = 24$, and $N = 24$, and the channel model is the same for all the UTs. 3GPP standardized Extended Typical Urban (ETU 300) channel model [125] is considered. In this model, there are nine channel paths and the channel path delay profile $\{\tau_{q,i}\}_{i=1}^{L_q=9}$ is $[0, 50, 120, 200, 230, 500, 1600, \dots]$ ns. The corresponding power profile $\{\mathbb{E}[|h_{q,i}|^2]\}_{i=1}^{L_q=9}$ is $[-1, -1, -1, 0, 0, 0, -3, -5, -7]$ dB. This power profile is normalized such that $\sum_{i=1}^9 \mathbb{E}[|h_{q,i}|^2] = 1$. The channel path gains $|h_{q,i}|$, $q = 0, 1, \dots, (Q-1)$, $i = 1, 2, \dots, L_q$, are modeled as independent zero mean Rayleigh faded random variables. The Doppler shift for the i th channel path of the q th UT is taken to be $\nu_{q,i} = \nu_{\max} \cos(\theta_{q,i})$, where $\nu_{\max}$ is the maximum Doppler shift and $\theta_{q,i}$ are independent and uniformly distributed in $[0, 2\pi)$.

Figure 7.11 shows the average sum SE of the OMA methods versus the received SNR. The sum SE is averaged over the channel path gains and the path Doppler shifts. The received SNR, ρ , is given by the ratio of the signal power received at the BS from a particular UT to the AWGN power at the BS in the total communication bandwidth $M\Delta f$. In this figure, and in the subsequent figures, the GB size G of the GB-OMA methods is chosen so as to maximize the sum SE. In Fig. 7.11, the number of UTs is $Q = 6$. For the IDD-OMA method $g_1 = 3$, $g_2 = 2$, and for the ITF-OMA method $r_1 = 3$, $r_2 = 2$.

From Fig. 7.11, it is observed that the IDD-OMA and ITF-OMA methods indeed achieve a significantly higher sum SE compared to that achieved by the GB-OMA methods. This is primarily due to the fact that IDD-OMA and ITF-OMA achieve MUI resilient communication without the use of GBs, whereas the SE of the GB-OMA method degrades due to the overhead of GBs and also due to MUI. It is also observed that at high SNRs, the sum SE achieved by the GB-OMA methods

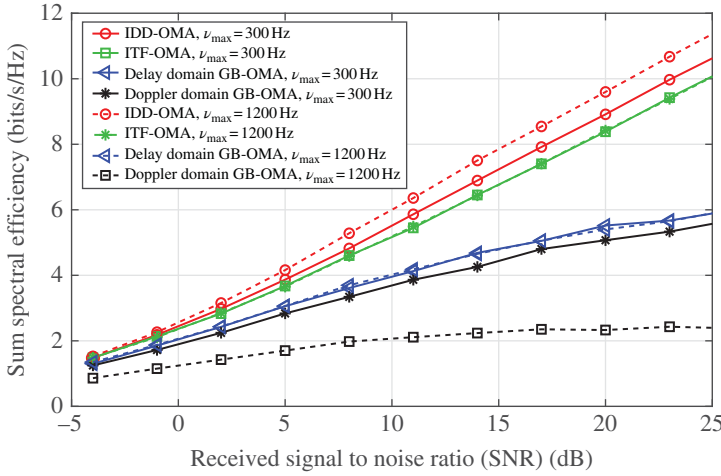


Figure 7.11 Sum spectral efficiency vs. received SNR for different OMA methods. $M = N = 24$, $Q = 6$ UTs, $g_1 = r_1 = 3$, and $g_2 = r_2 = 2$.

increases slowly with increasing SNR (when compared to the other two methods). This is because, in GB-OMA, even with GBs there is always residual MUI which increases with increasing SNR, which dominates the overall noise and interference at high SNRs. It is also observed that the sum SE performance of all methods (except the Doppler domain GB-OMA method) is almost the same even when the maximum Doppler shift $\nu_{\max}$ is increased from 300 to 1200 Hz. However, the sum SE of the Doppler domain GB-OMA method degrades with increasing $\nu_{\max}$ since DDREs allocated to different UTs are separated by GBs in the Doppler domain which are ineffective when the Doppler shift is large.

Figure 7.12 confirms the above observation of the degradation of the sum SE of Doppler domain GB-OMA with increasing $\nu_{\max}$. Figure 7.12 shows the sum SE performance of the OMA methods as a function of increasing $\nu_{\max}$ for $M = N = 24$, received SNR $\rho = 25$ dB, and the number of

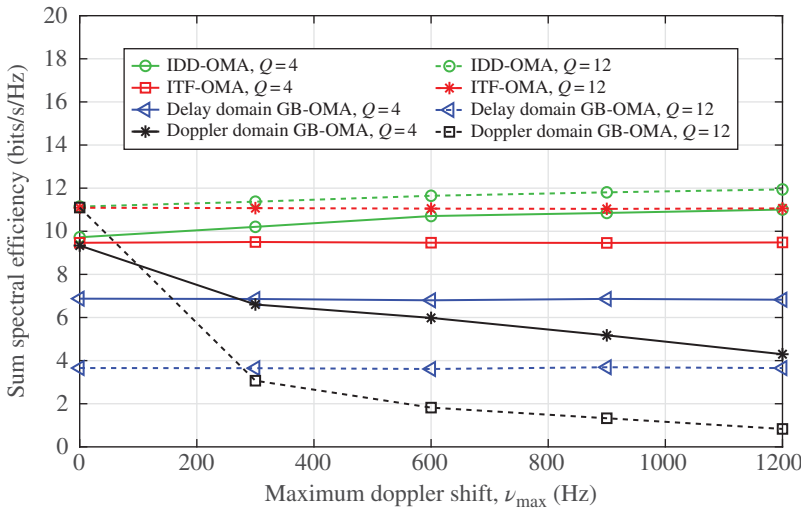


Figure 7.12 Sum spectral efficiency vs. maximum Doppler shift, $\nu_{\max}$. $M = N = 24$, received SNR $\rho = 25$ dB.

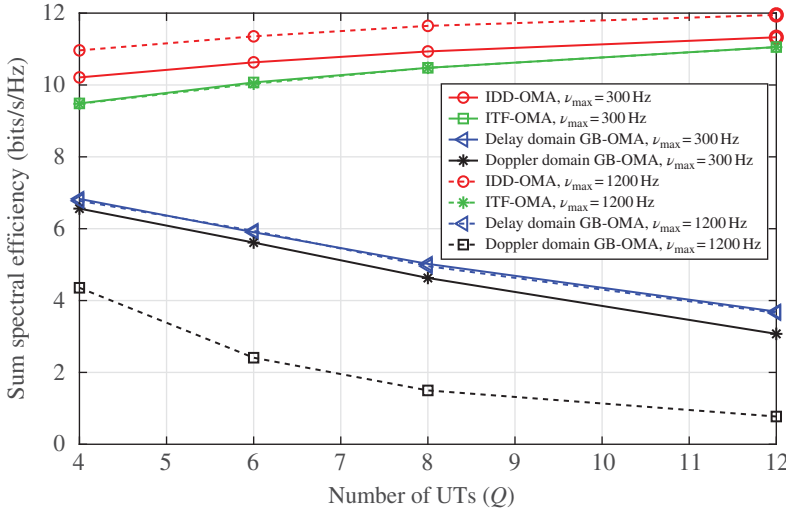


Figure 7.13 Sum spectral efficiency vs. number of UTs, Q . $M = N = 24$, received SNR $\rho = 25$ dB.

UTs is $Q = 4, 12$. For $Q = 12$, $g_1 = r_1 = 4$ and $g_2 = r_2 = 3$, and for $Q = 4$, $g_1 = r_1 = g_2 = r_2 = 2$. It is observed that indeed the sum SE of the Doppler domain GB-OMA method decreases with increasing $\nu_{\max}$. The sum SE performance of the other methods is almost the same irrespective of $\nu_{\max}$. This also highlights the robustness of OTFS-based communication systems to multipath Doppler shifts. Figure 7.13 shows the sum SE performance of the OMA methods as a function of increasing number of UTs for a fixed $\nu_{\max} = 300, 1200$ Hz, fixed received SNR $\rho = 25$ dB, and $M = N = 24$. For $Q = 4$ UTs, $g_1 = r_1 = g_2 = r_2 = 2$, and for $Q = 6$ UTs, $g_1 = r_1 = 3$ and $g_2 = r_2 = 2$. Also, for $Q = 8$ UTs, $g_1 = r_1 = 4$ and $g_2 = r_2 = 2$, and for $Q = 12$ UTs, $g_1 = r_1 = 4$ and $g_2 = r_2 = 3$. From this figure, it is observed that the sum SE performance of the GB-OMA methods decreases with increasing number of UTs, whereas those of the other two methods (IDD-OMA and ITF-OMA) do not decrease. This is primarily due to the fact that with increasing number of UTs, the GB size needs to be reduced for allocation of DDREs to UTs, which results in an increase in MUI. Therefore, it can be concluded that the simple GB-OMA methods have poor SE performance due to the use of GBs, whereas the interleaving-based IDD-OMA and ITF-OMA methods achieve significantly higher SE as they enable MUI resilient multiuser communication without the use of GBs.

7.2.1.5 Performance of OTFS-Based OMA Methods with Practical Rectangular Pulses

With ideal pulses, both ITF-OMA and IDD-OMA achieve almost MUI-free communication without the need for GBs, and, therefore, they achieve superior sum SE performance when compared to GB-OMA. However, with practical pulses, there will be MUI in the TF domain due to the channel delay and Doppler spread, which is expected to degrade the sum SE performance of all methods when compared to their performance with ideal pulses.

In [122], the sum SE achieved by these OMA methods has been studied for practical rectangular pulses. It is observed that although ITF-OMA and IDD-OMA have nearly the same performance with ideal pulses, with practical rectangular pulses, the degradation in the sum SE performance of ITF-OMA is much more significant than the degradation for IDD-OMA. The primary reason is that the TF domain resource allocation for ITF-OMA allocates neighboring TFREs to different UTs due to which there is high MUI in the TF domain specially when the channel delay/Doppler spread is high. At the same time, the TF allocation in IDD-OMA is contiguous due to which only a small

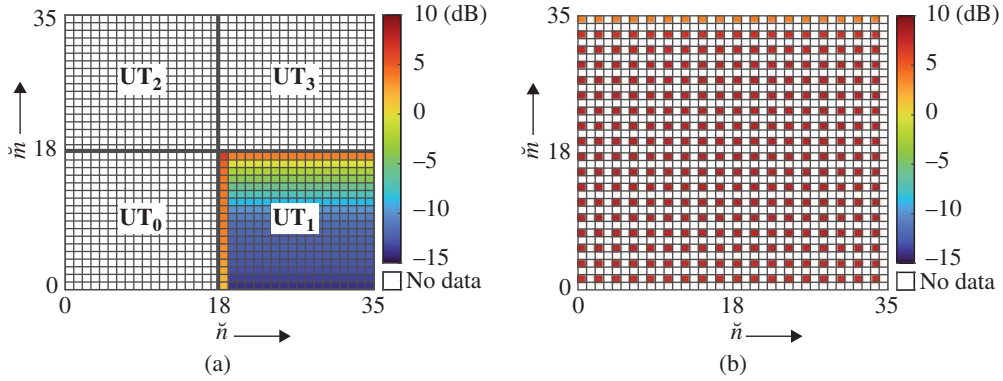


Figure 7.14 Color plot for the ratio of variance of MUI to the variance of useful term in TFREs allocated to UT1 (IDD-OMA (a) and ITF-OMA (b)). $M = N = 36$, $Q = 4$ UTs, $g_1 = g_2 = r_1 = r_2 = 2$, and $v_{\max} = 5000$ Hz.

fraction of TFREs allocated to a particular UT are adjacent to TFREs of another UT (see the contiguous TFRE allocation for IDD-OMA in Fig. 7.9 and the interleaved TFRE allocation for ITF-OMA in Fig. 7.10).

This fact is also evident from Fig. 7.14, where for each TFRE allocated to a particular UT (UT1), the ratio of the variance of MUI received from other UTs on that TFRE to the variance of the useful intended signal on that TFRE is plotted. For the sake of illustration, a single channel path is considered with delay $5 \mu\text{s}$ and maximum Doppler shift $v_{\max} = 5000$ Hz, $\Delta f = 15$ kHz, $M = N = 36$, and $Q = 4$ UTs. The channel path gain is $\mathcal{CN}(0, 1)$ distributed, and the Doppler shift is modeled as $v = v_{\max} \cos(\theta)$ where θ is uniformly distributed in $[0, 2\pi)$. The plotted ratio is averaged over a large number of random channel realizations. From the figure, it is clear that the MUI is significantly higher in ITF-OMA than that in IDD-OMA.

Next, the sum SE performance of GB-OMA, ITF-OMA, and IDD-OMA are compared. In the corresponding figures, the legends “IDDMA” and “ITFMA” refer to IDD-OMA and ITF-OMA, respectively, and “GB” refers to “GB-OMA.” The derivation of the I/O relation and the expression for the achievable sum SE for these OMA methods are provided in [122], and, therefore, they are not included here. In the simulations reported in this section, $\Delta f = 15$ kHz and $M = N = 36$. The 3GPP-standardized ETU300 channel model is considered (same as in Section 7.2.1.4).

In Fig. 7.15, sum SE is plotted as a function of increasing SNR for $Q = 6$ UTs and $v_{\max} = 300$ Hz. For IDD-OMA, $g_1 = 3$ and $g_2 = 2$, and for ITF-OMA $r_1 = 3$ and $r_2 = 2$. For the GB-based methods, the GB size G is chosen such that it maximizes sum SE. With rectangular pulses, it is observed that for a desired per-UT SE of more than 0.5 bits/s/Hz, the SNR required by GB-based methods is greater than that required by ITF-OMA, which, in turn, is greater than that required by IDD-OMA. This is because, the GB-based methods use GBs to reduce MUI. These GBs do not carry information and are an overhead.

In Fig. 7.15, the sum SE achieved with ideal pulses is also plotted. Both ITF-OMA and IDD-OMA require extra SNR to achieve a desired sum SE when compared to the SNR required with ideal pulses. This is due to MUI when rectangular pulses are used.

For GB-OMA methods, the extra SNR required with rectangular pulses is smaller when compared to that required for ITF-OMA and IDD-OMA. This is because, in GB-OMA, even with ideal pulses, there is MUI due to the Doppler and delay spread of the wireless channel. The additional

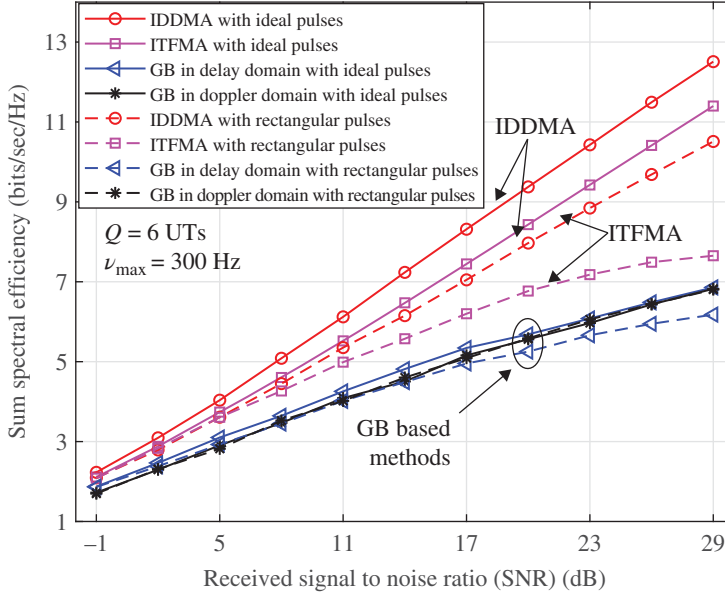


Figure 7.15 Sum SE vs SNR. $M = 36, N = 36, Q = 6, g_1 = r_1 = 3, g_2 = r_2 = 2$, and $\nu_{\max} = 300$ Hz.

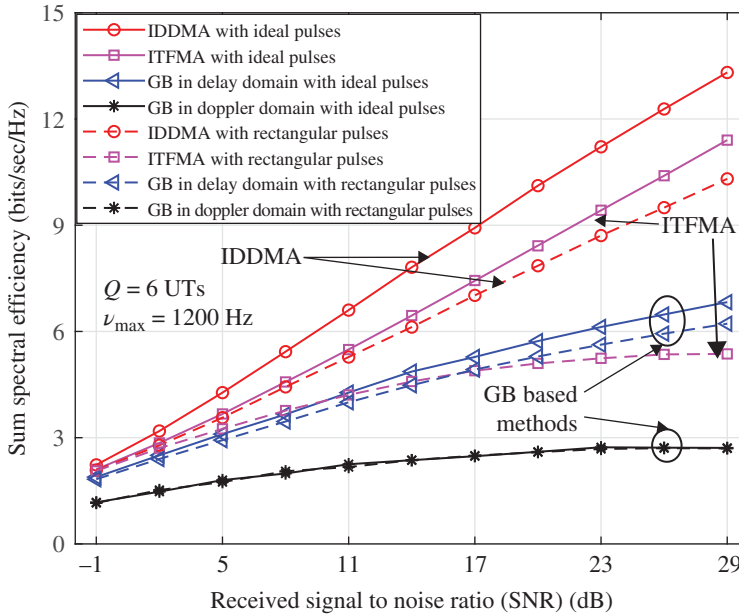


Figure 7.16 Sum SE vs SNR. $M = 36, N = 36, Q = 6, g_1 = r_1 = 3, g_2 = r_2 = 2$, and $\nu_{\max} = 1200$ Hz.

MUI due to the use of rectangular pulses appears to be not as significant as the MUI due to multi-path delay and Doppler spread.

For the same setting as in Fig. 7.15, the sum SE versus SNR is plotted again in Fig. 7.16, but with $\nu_{\max} = 1200$ Hz. The observations made for $\nu_{\max} = 1200$ Hz are similar to those made for $\nu_{\max} = 300$ Hz in Fig. 7.15. There are however two new observations. First, with rectangular pulses, at

high SNR, the sum SE of delay domain GB-OMA is better than that of ITF-OMA. This is because, in ITF-OMA, the TF domain MUI increases significantly when $\nu_{\max}$ is increased from 300 to 1200 Hz. Interestingly, even when $\nu_{\max}$ is increased from 300 to 1200 Hz, the sum SE of IDD-OMA (with rectangular pulses) remains almost the same. This is because, due to the contiguous TFRE allocation in IDD-OMA, it experiences significantly lesser MUI when compared to ITF-OMA where the TFRE allocation is interleaved. Second, the sum SE of Doppler domain GB-OMA decreases significantly when $\nu_{\max}$ is increased from 300 to 1200 Hz. This is because of a significant increase in Doppler domain MUI (see the remark at the end of Section 7.2.1.1).

In Fig. 7.17, the sum SE achieved by the IDD-OMA and the ITF-OMA methods is plotted as a function of increasing SNR for $\nu_{\max} = 0, 300, 600, 1200, 5000$ Hz. The number of UTs is $Q = 6$, i.e., $g_1 = r_1 = 3, g_2 = r_2 = 2$. With rectangular pulses, for a fixed SNR, the sum SE of ITF-OMA *decreases monotonically* when $\nu_{\max}$ is increased from 0 to 5000 Hz. Also, at high SNR, the sum SE of ITF-OMA appears to saturate for $\nu_{\max} \neq 0$. In comparison, for a fixed SNR, the reduction in the sum SE of IDD-OMA with increasing $\nu_{\max}$ is much smaller. This is due to the contiguous TFRE allocation in IDD-OMA which reduces the impact of MUI when compared to the interleaved allocation in ITF-OMA (see Fig. 7.14).

In Fig. 7.18, with rectangular pulses, the sum SE of the OMA methods is plotted as a function of increasing number of UTs Q for a fixed received SNR of 15 dB and fixed $\nu_{\max} = 1200$ Hz. The sum SE of IDD-OMA and ITF-OMA appears to converge to a limiting value for large Q . This is expected since the increase in Q reduces the number of information symbols transmitted by each UT which reduces ISI between the DD/TF domain symbols of the same UT. At the same time, a large Q implies more MUI.

In Fig. 7.18, with ideal pulses, the sum SE of both ITF-OMA and IDD-OMA increases with increasing Q . This is because, although there is no MUI/ISI in the TF domain (as the pulses are ideal), there is still DD domain ISI between the information symbols transmitted by the same UT. This is due to the fact that, even with ideal pulses, the received DD signal for a UT is given by a 2D convolution between the effective DD domain channel for that UT and the DD domain information

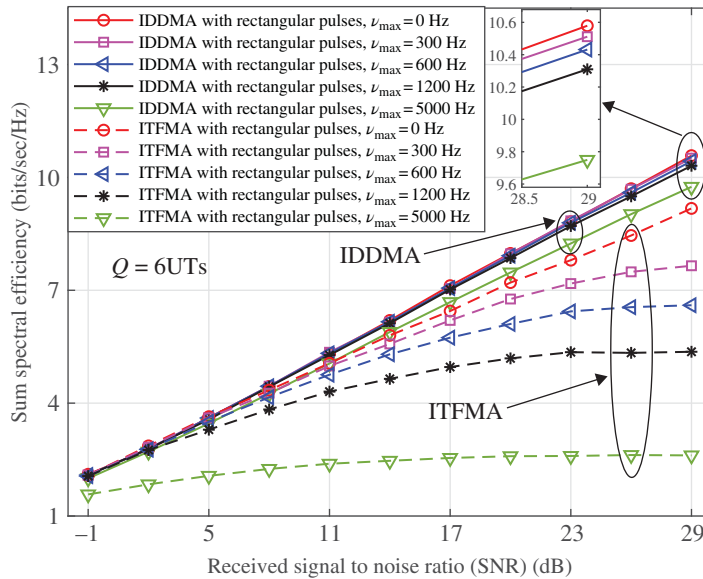


Figure 7.17 Sum SE vs. SNR. $M = 36, N = 36, Q = 6, g_1 = r_1 = 3$, and $g_2 = r_2 = 2$.

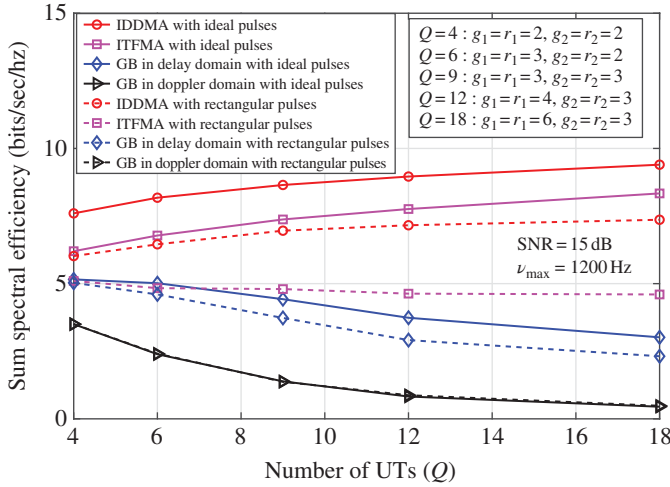


Figure 7.18 Sum SE vs. number of UTs, Q . $M = 36, N = 36$, and $\nu_{\max} = 1200$ Hz.

signal transmitted by that UT. With an increase in the number of UTs, the number of DD domain information symbols transmitted by a single UT decreases since the total number of DDREs is fixed (i.e., MN). The decrease in the number of symbols transmitted by a UT results in a decrease in DD domain ISI, which, in turn, increases the sum SE.

Figure 7.19 studies the impact of the increase in path delay spread on the sum SE of IDD-OMA and ITF-OMA as a function of increasing SNR for fixed $\nu_{\max} = 600$ Hz. Sum SE is evaluated for the path delays in the ETU300 channel model and also for twice the path delays in ETU300. It is observed that at high SNRs, the sum SE of IDD-OMA degrades negligibly even when the delays are doubled, whereas there is a significant reduction in the sum SE of ITF-OMA.

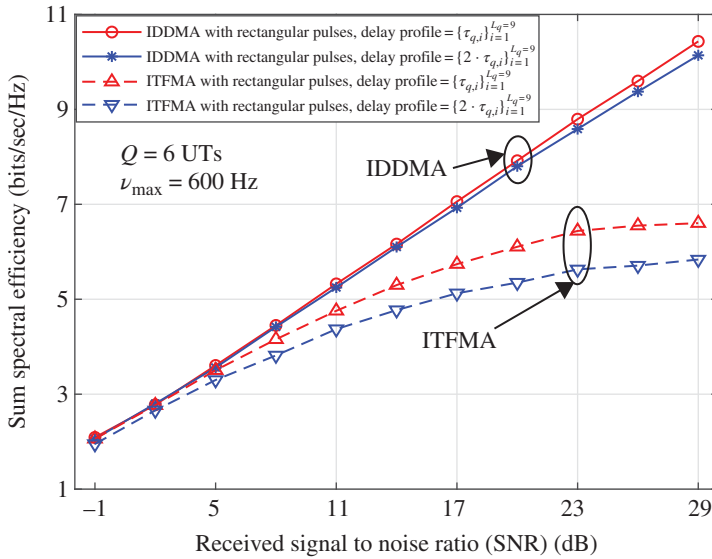


Figure 7.19 Sum SE vs. SNR. $M = N = 36, g_1 = r_1 = 3, g_2 = r_2 = 2$, and $\nu_{\max} = 600$ Hz.

7.2.2 Non-orthogonal Multiple Access

In non-orthogonal multiple access, the set of REs allocated to different UTs overlap in both TF and DD domains. This allows for achieving higher spectral efficiency but at the cost of the increase in signal processing complexity when compared to OMA. Higher signal processing complexity is due to the need for mitigating MUI between UTs sharing the same RE. However, each DDRE or TFRE offers only a single dimension, and, therefore, the dimensionality needs to be increased in order to further increase spectral efficiency. A well-known technique to increase signal dimensionality is to use multiple antennas for transmission and reception [6]. OTFS-based MIMO systems, therefore, have the potential to achieve very high spectral efficiency even in high-mobility scenarios. In addition, MIMO systems are known to increase communication reliability as well. Therefore, MIMO-OTFS systems can improve the communication reliability in high-mobility scenarios, where channel-induced Doppler shift is known to degrade the communication reliability of OFDM-based systems [39].

7.2.2.1 Multiuser MIMO-OTFS Uplink

Consider an OTFS-based multiuser MIMO uplink system in which the UTs and the BS have multiple antennas, where $N_{t,q}$ is the number of antennas at the q th UT and N_r is the number of antennas at the BS. Let L_q denote the number of channel paths between the q th UT and the BS. Let $\phi_{t,q,i}$ and $\theta_{t,q,i}$ denote the azimuthal and zenith angles of departure (AoD), respectively, from the q th UT along the i th path, $i = 1, 2, \dots, L_q$. Similarly, let $\phi_{r,q,i}$ and $\theta_{r,q,i}$ denote the azimuthal and the zenith angles of arrival (AoA), respectively, at the BS along the i th path from the q th UT. An example of multipath propagation between the antenna array at the q th UT and the BS antenna array is depicted in Fig. 7.20.

Let $x_{q,s}(t)$ denote the TD signal transmitted from the s th antenna of the q th UT, $s = 1, 2, \dots, N_{t,q}$. For any general angles of departure (ϕ_D, θ_D) , let $\mathbf{a}_t(\phi_D, \theta_D) \in \mathbb{C}^{N_{t,q} \times 1}$ denote the array response vector of the transmit antenna array at the q th UT. Similarly, for any general angles of arrival (ϕ_A, θ_A) , let $\mathbf{a}_r(\phi_A, \theta_A) \in \mathbb{C}^{N_r \times 1}$ denote the array response vector of the receive antenna array at the BS. These array response vectors are given by

$$\begin{aligned} \mathbf{a}_t(\phi_D, \theta_D) &\triangleq \left(a_{t,1}(\phi_D, \theta_D), a_{t,2}(\phi_D, \theta_D), \dots, a_{t,N_{t,q}}(\phi_D, \theta_D) \right)^T, \\ \mathbf{a}_r(\phi_A, \theta_A) &\triangleq \left(a_{r,1}(\phi_A, \theta_A), a_{r,2}(\phi_A, \theta_A), \dots, a_{r,N_r}(\phi_A, \theta_A) \right)^T. \end{aligned} \quad (7.32)$$

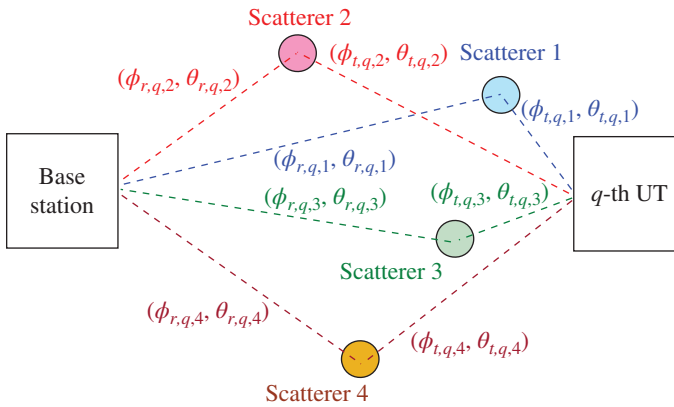


Figure 7.20 Multipath propagation between a multi-antenna UT and the BS antenna array.

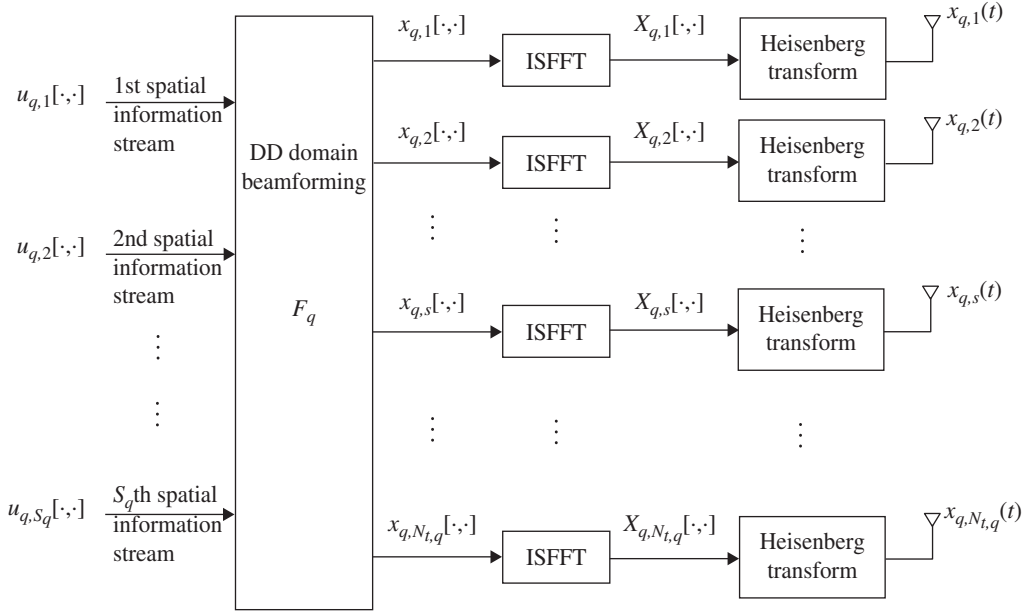


Figure 7.21 Transmit signal processing at the q th UT.

The TD noise-free signal received at the p th BS antenna from the s th antenna of the q th UT along the i th path is $h_{q,i}a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i})a_{t,s}^*(\phi_{t,q,i}, \theta_{t,q,i})x_{q,s}(t - \tau_{q,i})e^{j2\pi\nu_{q,i}(t - \tau_{q,i})}$ [55]. Here, $h_{q,i}$ models the complex channel gain of the i th path from the q th UT to the BS. The TD signal received at the p th BS antenna is then given by

$$y_p(t) = \sum_{q=0}^{Q-1} \sum_{i=1}^{L_q} \sum_{s=1}^{N_{t,q}} [h_{q,i}a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i})a_{t,s}^*(\phi_{t,q,i}, \theta_{t,q,i})x_{q,s}(t - \tau_{q,i})e^{j2\pi\nu_{q,i}(t - \tau_{q,i})}] + w_p(t), \quad (7.33)$$

where $w_p(t)$ is the AWGN at the p th BS antenna. The transmit signal processing at the q th UT is shown in Fig. 7.21.

At the q th UT, let there be $S_q < N_{t,q}$ different spatial streams of information to be transmitted. For the v th spatial stream at the q th UT, the information symbol to be transmitted on the (k, l) th DDRE is denoted by $u_{q,v}[k, l]$, $k = 0, 1, \dots, N - 1$, $l = 0, 1, \dots, M - 1$, $v = 1, 2, \dots, S_q$. The DD domain information symbols of all spatial streams are jointly transformed to create $N_{t,q}$ streams of DD domain transmit symbols (one stream for each transmit antenna at the q th UT). The DD domain symbols for the s th transmit stream are given by

$$x_{q,s}[k, l] = \sum_{v=1}^{S_q} f_{q,s,v} u_{q,v}[k, l], \quad (7.34)$$

where $f_{q,s,v}$, $s = 1, 2, \dots, N_{t,q}$ are the beamforming coefficients for the v th spatial stream. Each stream of DD domain transmit symbols is separately converted to the TF domain using ISFFT. For the s th transmit antenna at the q th UT, the TF symbols are given by

$$X_{q,s}[n, m] = \frac{1}{MN} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x_{q,s}[k, l] e^{-j2\pi\left(\frac{ml}{M} - \frac{nk}{N}\right)}. \quad (7.35)$$

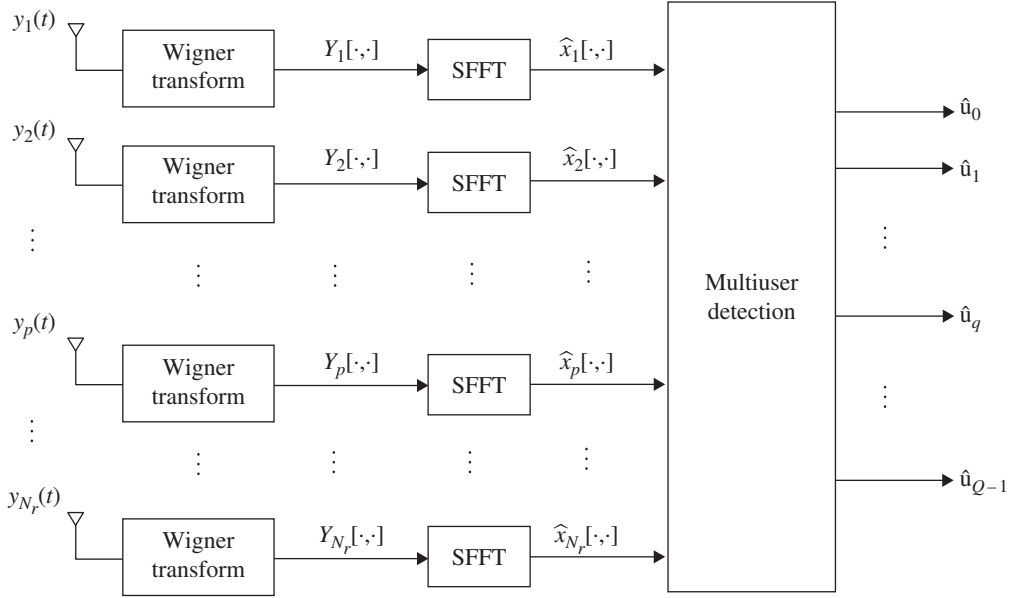


Figure 7.22 Receive signal processing at the BS in the uplink.

These TF symbols are then converted to the TD transmit signal using the Heisenberg transform. The TD signal transmitted by the q th UT on the s th antenna is then given by

$$\mathbf{x}_{q,s}(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X_{q,s}[n, m] g_{\text{tx}}(t - nT) e^{j2\pi m \Delta f (t - nT)}. \quad (7.36)$$

The signal processing at the BS receiver is shown in Fig. 7.22. At the p th BS antenna, the received TD signal $y_p(t)$ is converted to the TF symbols using the Wigner transform as

$$Y_p[n, m] = \int y_p(t) g_{\text{rx}}^*(t - nT) e^{-j2\pi m \Delta f (t - nT)} dt. \quad (7.37)$$

These TF symbols are converted to the DD domain using SFFT, i.e., the received DD domain symbols at the p th BS antenna are given by

$$\hat{\mathbf{x}}_p[k, l] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y_p[n, m] e^{j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)}. \quad (7.38)$$

Let $\hat{\mathbf{x}}_p \in \mathbb{C}^{MN \times 1}$ be the vector of all MN DD domain symbols received at the p th BS antenna having $\hat{\mathbf{x}}_p[k, l]$ as its $(kM + l + 1)$ th element. Similarly, for the q th UT, let $\mathbf{x}_{q,s} \in \mathbb{C}^{MN \times 1}$ denote the vector of DD domain transmit symbols having $x_{q,s}[k, l]$ as its $(kM + l + 1)$ th element. From (7.33), (7.35), (7.36), (7.37), and (7.38), it follows that

$$\hat{\mathbf{x}}_p = \sum_{q=0}^{Q-1} \sum_{s=1}^{N_{t,q}} \sum_{i=1}^{L_q} \mathbf{H}_{p,q,i,s} \mathbf{x}_{q,s} + \tilde{\mathbf{w}}_p, \quad (7.39)$$

where $\mathbf{H}_{p,q,i,s} \in \mathbb{C}^{MN \times MN}$ is the effective DD domain channel matrix between the p th BS antenna and the s th antenna of the q th UT along the i th path. Also, $\tilde{\mathbf{w}}_p$ is the vector of DD domain AWGN samples at the p th BS antenna. Assuming that the AWGN $w_p(t)$ in (7.33) has PSD N_0 W/Hz, from

(7.37) and (7.38), it follows that each element of $\tilde{\mathbf{w}}_p$ is i.i.d. $\mathcal{CN}(0, MNN_0)$ distributed. Furthermore, $\mathbf{H}_{p,q,i,s}$ is given by

$$\mathbf{H}_{p,q,i,s} = a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i}) a_{t,s}^*(\phi_{t,q,i}, \theta_{t,q,i}) \mathbf{G}_{q,i}, \quad (7.40)$$

where $\mathbf{G}_{q,i} \in \mathbb{C}^{MN \times MN}$ depends only on the complex channel gain $h_{q,i}$, the delay $\tau_{q,i}$, and Doppler shift $\nu_{q,i}$ along the i th path from the q th UT to the BS. Note that $\mathbf{G}_{q,i}$ does not depend on the indices s and p of the transmit antenna and receive antenna, respectively. In fact, for a single UT scenario with both the BS and UT having one antenna each, $\mathbf{G}_{q,i}$ is the effective $MN \times MN$ DD domain channel matrix along the i th path and is sparse (i.e., few of its elements have significant values). The element of $\mathbf{G}_{q,i}$ in its $(k'M + l' + 1)$ th row and $(kM + l + 1)$ th column is given by

$$\begin{aligned} & \mathbf{G}_{q,i}[k'M + l' + 1, kM + l + 1] \\ &= \frac{h_{q,i}}{MN} \sum_{m=0}^{M-1} \sum_{m'=0}^{M-1} \sum_{n=0}^{N-1} \sum_{n'=0}^{N-1} \left[e^{-j2\pi m' \Delta f \tau_{q,i}} e^{j2\pi \left(\frac{m'l' - m'l}{M} \right)} e^{-j2\pi \left(\frac{nk' - n'l}{N} \right)} \right. \\ & \quad \left. \int g_{\text{rx}}^*(t - nT) g_{\text{tx}}(t - n'T - \tau_{q,i}) e^{j2\pi((m' - m)\Delta f + \nu_{q,i})t} dt \right]. \end{aligned} \quad (7.41)$$

For transmit and receive pulses satisfying the conditions in (7.6), (7.8), (7.9), and (7.10), from (7.41), it follows that

$$\mathbf{G}_{q,i}[k'M + l' + 1, kM + l + 1] = h_{q,i} \left[\frac{1}{M} \sum_{m=0}^{M-1} e^{j2\pi m \frac{(l' - l - M\Delta f \tau_{q,i})}{M}} \right] \left[\frac{1}{N} \sum_{n=0}^{N-1} e^{-j2\pi n \frac{(k' - k - N\tau_{q,i})}{N}} \right]. \quad (7.42)$$

From this expression, it is clear that $\mathbf{G}_{q,i}[k'M + l' + 1, kM + l + 1]$ has significant values only when $l' \approx l + M\Delta f \tau_{q,i}$ and $k' \approx k + N\tau_{q,i}$. In other words, $\mathbf{G}_{q,i}$ is a sparse matrix. Using (7.40) in (7.39) gives

$$\hat{\mathbf{x}}_p = \sum_{q=0}^{Q-1} \sum_{s=1}^{N_{t,q}} \mathbf{H}_{p,q,s} \mathbf{x}_{q,s} + \tilde{\mathbf{w}}_p, \quad (7.43)$$

where

$$\mathbf{H}_{p,q,s} \triangleq \sum_{i=1}^{L_q} a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i}) a_{t,s}^*(\phi_{t,q,i}, \theta_{t,q,i}) \mathbf{G}_{q,i} \quad (7.44)$$

is the effective DD domain channel between the s th antenna of the q th UT and the p th antenna of the BS. Next, let the received DD domain vectors at all the N_r BS antennas be stacked into a single vector $\hat{\mathbf{x}} \in \mathbb{C}^{MNN_r \times 1}$. Similarly, let all the DD domain transmit vectors $\mathbf{x}_{q,s}$, $s = 1, 2, \dots, N_{t,q}$ be stacked up into a single vector $\mathbf{x}_q \in \mathbb{C}^{MNN_{t,q} \times 1}$. That is,

$$\begin{aligned} \hat{\mathbf{x}} &\triangleq \left[\hat{\mathbf{x}}_1^T \hat{\mathbf{x}}_2^T \cdots \hat{\mathbf{x}}_{N_r}^T \right]^T, \\ \mathbf{x}_q &\triangleq \left[\mathbf{x}_{q,1}^T \mathbf{x}_{q,2}^T \cdots \mathbf{x}_{q,N_{t,q}}^T \right]^T. \end{aligned} \quad (7.45)$$

Using (7.45) in (7.43) gives

$$\hat{\mathbf{x}} = \sum_{q=0}^{Q-1} \mathbf{H}_q \mathbf{x}_q + \tilde{\mathbf{w}}, \quad (7.46)$$

where $\tilde{\mathbf{w}} = [\tilde{\mathbf{w}}_1^T \tilde{\mathbf{w}}_2^T \cdots \tilde{\mathbf{w}}_{N_r}^T]^T$ and

$$\mathbf{H}_q \triangleq \begin{bmatrix} \mathbf{H}_{1,q,1} & \mathbf{H}_{1,q,2} & \cdots & \mathbf{H}_{1,q,N_{t,q}} \\ \mathbf{H}_{2,q,1} & \mathbf{H}_{2,q,2} & \cdots & \mathbf{H}_{2,q,N_{t,q}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{H}_{N_r,q,1} & \mathbf{H}_{N_r,q,2} & \cdots & \mathbf{H}_{N_r,q,N_{t,q}} \end{bmatrix} \in \mathbb{C}^{MNN_r \times MNN_{t,q}}. \quad (7.47)$$

Next, let $\mathbf{u}_{q,v} \in \mathbb{C}^{MN \times 1}$ denote the vector of MN DD domain information symbols of the v th spatial stream of the q th UT having $u_{q,v}[k, l]$ at its $(kM + l + 1)$ th element. Also, let the vector of all MNS_q DD domain information symbols transmitted by the q th UT be given by

$$\mathbf{u}_q \triangleq \left[\mathbf{u}_{q,1}^T \mathbf{u}_{q,2}^T \cdots \mathbf{u}_{q,S_q}^T \right]^T \in \mathbb{C}^{MNS_q \times 1}. \quad (7.48)$$

Then, using (7.34) along with (7.45) and (7.48) gives

$$\mathbf{x}_q = \mathbf{F}_q \mathbf{u}_q, \quad (7.49)$$

where

$$\mathbf{F}_q \triangleq \begin{bmatrix} f_{q,1,1} \mathbf{I} & f_{q,1,2} \mathbf{I} & \cdots & f_{q,1,S_q} \mathbf{I} \\ f_{q,2,1} \mathbf{I} & f_{q,2,2} \mathbf{I} & \cdots & f_{q,2,S_q} \mathbf{I} \\ \vdots & \vdots & \ddots & \vdots \\ f_{q,N_{t,q},1} \mathbf{I} & f_{q,N_{t,q},2} \mathbf{I} & \cdots & f_{q,N_{t,q},S_q} \mathbf{I} \end{bmatrix} \in \mathbb{C}^{MNN_{t,q} \times MNS_q}. \quad (7.50)$$

In (7.50), $\mathbf{I}$ denotes the $MN \times MN$ identity matrix. Using (7.49) in (7.46), the input–output relation between the DD information symbols transmitted by all UTs and the DD domain symbols received at the BS is given by

$$\hat{\mathbf{x}} = \sum_{q=0}^{Q-1} \tilde{\mathbf{H}}_q \mathbf{u}_q + \tilde{\mathbf{w}}, \quad (7.51)$$

where

$$\tilde{\mathbf{H}}_q \triangleq \mathbf{H}_q \mathbf{F}_q \in \mathbb{C}^{MNN_r \times MNS_q}. \quad (7.52)$$

Clearly, $\text{rank}(\tilde{\mathbf{H}}_q) \leq MN \min(N_r, N_{t,q}, S_q) = MN \min(N_r, S_q)$ since $S_q \leq N_{t,q}$. The input–output relation in (7.51) can also be written as

$$\hat{\mathbf{x}} = \tilde{\mathbf{H}} \mathbf{u} + \tilde{\mathbf{w}}, \quad (7.53)$$

where

$$\mathbf{u} \triangleq [\mathbf{u}_0^T \mathbf{u}_1^T \cdots \mathbf{u}_{Q-1}^T]^T \in \mathbb{C}^{(MN \sum_{q=0}^{Q-1} S_q) \times 1}, \quad (7.54)$$

$$\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{H}}_0 \tilde{\mathbf{H}}_1 \cdots \tilde{\mathbf{H}}_{Q-1}] \in \mathbb{C}^{MNN_r \times (MN \sum_{q=0}^{Q-1} S_q)}. \quad (7.55)$$

Here, $\mathbf{u}$ is the vector of all DD domain information symbols transmitted by all UTs and $\tilde{\mathbf{H}}$ is the effective DD domain channel matrix of the uplink multiuser system. It is clear that $\text{rank}(\tilde{\mathbf{H}}) \leq \min(MNN_r, (MN \sum_{q=0}^{Q-1} S_q))$. With sufficient scattering and spatial distribution, (i) for all UTs, the number of distinct paths between a UT and BS is generally more than the number of spatial information streams of that UT, i.e., $L_q \geq S_q$, and (ii) no path of a UT has

the same angle of arrival and departure as that of any path of any other UT. In such scenarios, $\text{rank}(\tilde{\mathbf{H}}) = \min \left(MNN_r, \left(MN \sum_{q=0}^{Q-1} S_q \right) \right)$, and, therefore, with a large antenna array at the BS, i.e., $N_r > \sum_{q=0}^{Q-1} S_q$, the effective multiuser channel matrix has the rank equal to the total number of spatial information streams transmitted by all UTs. In such a scenario, reliable multiuser OTFS detection is possible, i.e., all spatial information streams can be separated from each other.

From (7.43), (7.47), and (7.49), it follows that the effective channel matrix $\tilde{\mathbf{H}}_q = \mathbf{H}_q \mathbf{F}_q$ for the q th UT (see (7.51)) can be written as

$$\tilde{\mathbf{H}}_q = \sum_{i=1}^{L_q} (\mathbf{B}_{q,i} \otimes \mathbf{G}_{q,i}), \quad (7.56)$$

where $\mathbf{B}_{q,i} \in \mathbb{C}^{N_r \times S_q}$ is such that its (p, v) th element is given by

$$\mathbf{B}_{q,i}(p, v) = a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i}) \sum_{s=1}^{N_{t,q}} f_{q,s,v} a_{t,s}^*(\phi_{t,q,i}, \theta_{t,q,i}). \quad (7.57)$$

Since the DD domain matrices $\mathbf{G}_{q,i}$, $i = 1, 2, \dots, L_q$ are sparse, from (7.56), it follows that the effective channel matrix $\tilde{\mathbf{H}}_q$ is also sparse. Therefore, low-complexity OTFS signal detection techniques for single-user systems can also be used for multiuser OTFS detection.

In the MIMO-OTFS-based non-orthogonal multiple access system discussed here, each RE carries $\sum_{q=0}^{Q-1} S_q$ information symbols, and, therefore, it is clear that the sum SE of MIMO-OTFS-based non-orthogonal multiple access is greater than that of the OTFS OMA methods discussed in Section 7.2.1, where each RE carries only one information symbol. Also, the communication reliability improves with increasing number of antennas at the BS. This is shown below by illustrating the simulated BER performance of the linear minimum mean squared error (LMMSE) equalization-based multiuser detector as a function of received SNR for a scenario with $Q = 2$ single-antenna UTs (i.e., $S_q = 1$, $q = 0, 1$) communicating with a multi-antenna BS on all the MN DDREs. Therefore, compared to OMA, the effective SE is almost doubled. A uniform rectangular array (URA) at the BS with adjacent antennas separated by d in both the horizontal and the vertical directions is considered. The URA is N_h -antenna wide in the horizontal direction and N_v -antenna wide in the vertical direction so that $N_r = N_h N_v$. Received SNR is defined as the ratio of the signal power received at each BS antenna to the AWGN power at each receive antenna. Let the mean-square value of the i.i.d. DD domain information symbol transmitted by each UT (i.e., $u_{q,1}[k, l]$) be denoted by E . Then, from (7.35) and (7.36), it follows that the average transmitted power, i.e., $\frac{1}{NT} \int |x_{q,1}(t)|^2 dt$ is $\frac{E}{NT}$. The communication bandwidth is $M\Delta f$, and the AWGN PSD is N_0 . Since the channel path gains are normalized, the SNR of the signal received at each BS antenna is $\frac{E/(NT)}{M\Delta f N_0} = \frac{E}{MNN_0}$. For the LMMSE detector, the vector of detected DD domain information symbols at the output of the multiuser detector in Fig. 7.22 is given by

$$\hat{\mathbf{u}} = \mathbf{A}_{\text{mmse}}^H \hat{\mathbf{x}}, \quad (7.58)$$

where

$$\mathbf{A}_{\text{mmse}} \triangleq \tilde{\mathbf{H}}^H \left(\tilde{\mathbf{H}}^H \tilde{\mathbf{H}} + \frac{MNN_0}{E} \mathbf{I} \right)^{-1} \quad (7.59)$$

is the linear MMSE equalizer matrix. Also, $\hat{\mathbf{x}}$ is the vector of received DD domain symbols at the N_r BS antennas and $\tilde{\mathbf{H}}$ is the effective DD domain multiuser channel matrix (see (7.53)).

The OTFS modulation parameters considered are $\Delta f = 15$ kHz, $M = 12$, and $N = 14$, which corresponds to a communication bandwidth of $M\Delta f = 180$ kHz and an OTFS frame duration of

$NT = 0.933$ ms. A single 120° sector of a circular cell of radius 1 km is considered with the BS at the center and two UTs distributed uniformly in the sector except within a distance of 100 m around the BS. There are L scatterers around each UT, distributed uniformly on a ring of radius 200 m with the UT at its center. The path loss exponent is 3.8, and the circularly symmetric complex Gaussian-distributed channel path gains model the effect of path loss and fast fading. For each UT, the sum of the mean square values of its channel path gains is normalized to unity. The height of the BS is 35 m and that of the UTs is 1.5 m. The carrier frequency is 4.8 GHz. For each channel path, the angles of arrival are calculated based on the location of the scatterers and the BS, while the delay of each path depends on its total path length (i.e., sum of the distance between the UT and the scatterer and the distance between the scatterer and the BS). The Doppler shift of each path depends on the angle between the direction of travel of the UT and the direction in which the scatterer is located with respect to that UT. The array response at the p th BS antenna to a signal arriving from the direction (ϕ, θ) is given by [126]

$$a_{r,p}(\theta, \phi) = e^{j2\pi \frac{d}{\lambda} \left(\text{mod}(p-1, N_h) \sin(\phi) \cos(\theta) + \lfloor \frac{p-1}{N_h} \rfloor \sin(\theta) \right)}, \quad p = 1, 2, \dots, N_r, \quad (7.60)$$

where λ is the carrier wavelength and $d = \lambda/2$ is the inter-antenna spacing in both directions.

Figure 7.23 shows the BER performance of the LMMSE detector as a function of received SNR for $N_r = 4$ BS antennas and $L = 8$ channel paths. The DD domain information symbols belong to 4-QAM constellation. The following two scenarios are considered: (i) no-mobility scenario, where the maximum path Doppler shift is $\nu_{\max} = 0$, and (ii) high-mobility scenario with $\nu_{\max} = 1600$ Hz (corresponding to a UT speed of 360 km/hr). The figure also shows the BER performance of a CP-OFDM-based uplink multiuser MIMO system where there are $M = 12$ subcarriers and the subcarrier spacing is $\Delta f = 15$ kHz, i.e., the total communication bandwidth is $M\Delta f = 180$ kHz (same as that of the OTFS-based MIMO uplink). Separate LMMSE multiuser detection is performed for each subcarrier. The information symbols transmitted on the subcarriers belong to 4-QAM constellation. From Fig. 7.23, it is observed that the BER performance of MIMO-OTFS uplink is better compared to that of MIMO-OFDM uplink for both $\nu_{\max} = 0$ and 1600 Hz. The performance difference for $\nu_{\max} = 0$ Hz is less compared to that for $\nu_{\max} = 1600$ Hz. This is because when $\nu_{\max} = 1600$ Hz, the maximum Doppler shift is more than 10% of the subcarrier spacing due to which there is considerable ICI in the OFDM system which severely degrades its BER performance.

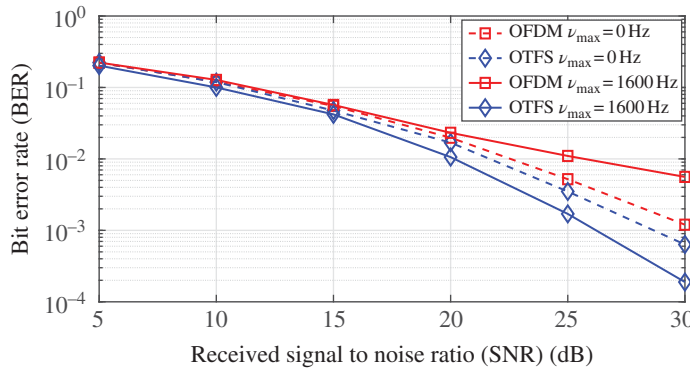


Figure 7.23 BER performance of MIMO-OTFS- and MIMO-OFDM-based uplink with LMMSE multiuser detection. $Q = 2$, $N_r = 4$, and $L = 8$.

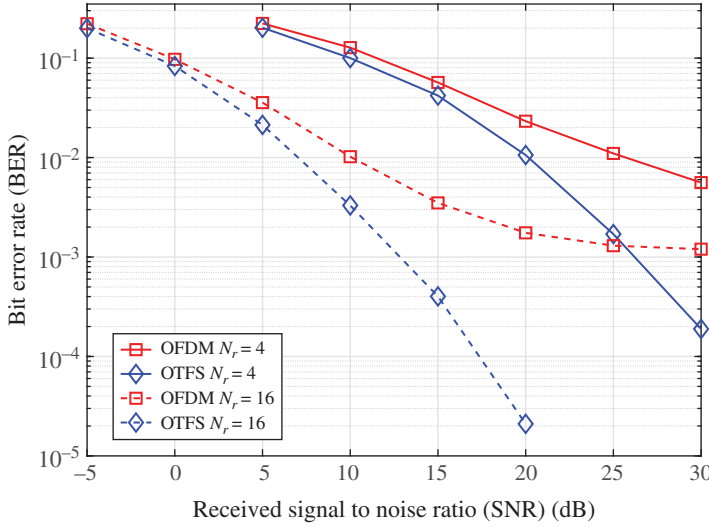


Figure 7.24 BER performance of MIMO-OTFS and MIMO-OFDM uplink with LMMSE multiuser detection. $Q = 2$, $N_r = 4, 16$, $L = 8$, and $v_{\max} = 1600$ Hz.

On the other hand, the BER performance of the OTFS-based system does not degrade even when $v_{\max} = 1600$ Hz. This shows the robustness of OTFS-based uplink multiuser MIMO system toward the channel Doppler spread when compared to OFDM-based systems.

For the same system setting in Fig. 7.23, a comparison of the BER performance of OTFS- and OFDM-based uplink MIMO systems showing the effect of increasing the number of BS antennas is provided in Fig. 7.24 for $N_r = 4, 16$ and $v_{\max} = 1600$ Hz. From Fig. 7.24, it is observed that, even under the considered high-mobility scenario with $v_{\max} = 1600$ Hz, the BER performance of the OTFS-based system improves significantly when the number of BS antennas is increased from $N_r = 4$ to $N_r = 16$. On the other hand, the BER performance of the OFDM-based system exhibits error floor at high SNRs, primarily due to ICI.

Although communication reliability and overall system SE can be improved by using more number of antennas at the BS, the larger size of the effective multiuser channel matrix results in high detection complexity for linear detectors like LMMSE. This is because the computation of $\mathbf{A}_{\text{lmmse}}$ requires matrix inversion which can have high complexity when M and N are large. One solution is to use approximations to the matrix inverse, e.g., those based on the Neumann approximation of matrix inverse [127, 128]. In [82], the structure of the LMMSE equalizer matrix is exploited to propose a low-complexity LU factorization-based LMMSE detector.

7.2.2.2 Massive MIMO-OTFS

Massive MIMO systems with a large antenna array at the BS are known to achieve near-optimal performance even with simple linear processing-based multiuser detection (e.g., MRC, zero-forcing) [16, 17]. Due to the fine angular domain resolution of large antenna arrays and finite number of channel paths between a UT and the BS, the channel representation is generally sparse in the angle domain [6, 129]. Therefore, signals arriving at the BS from UTs at distinct angular locations can be easily separated in the angle domain. A path division multiple access (PDMA)-based OTFS multiuser massive MIMO uplink system is proposed in [130, 131], where UTs having different angular

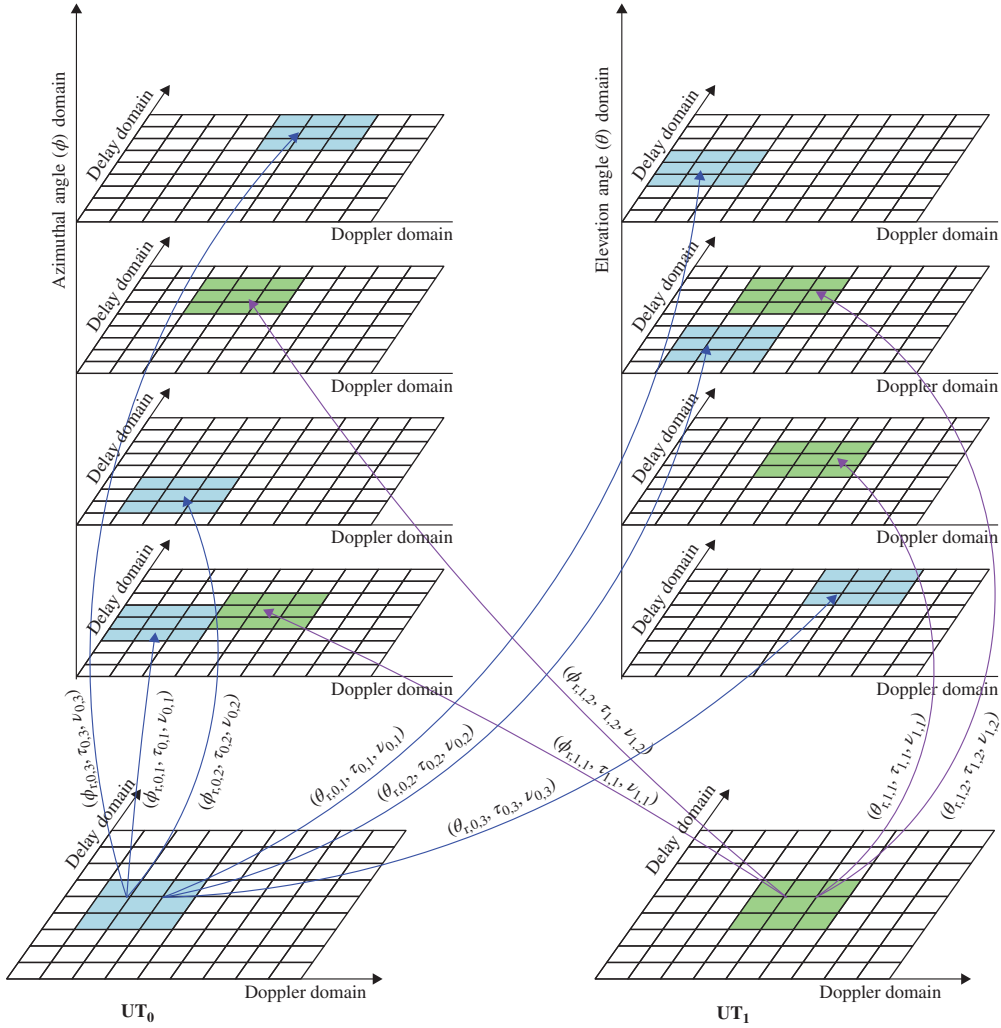


Figure 7.25 PDMA principle.

locations (i.e., having paths with a different angle of arrival at BS) can be scheduled on the same DDRE, as their information is separable in the angle-delay-Doppler domain using simple MRC. A simple illustration of the PDMA principle is shown in Fig. 7.25 for a BS having a rectangular array. Due to the rectangular array, the BS can distinguish received signals along two different angular dimensions, i.e., azimuthal and elevation. As shown in the figure, two users UT_0 and UT_1 are allocated nine DDREs each, out of which three DDREs are allocated to both, i.e., both the UTs transmit information symbols on these three DDREs. There are three channel paths between UT_0 and the BS and two channel paths between UT_1 and the BS. However, due to the different angles of arrivals of the paths corresponding to these two UTs, the received DD domain signals from these two UTs do not overlap in the azimuthal angle-delay-Doppler domain as well as in the elevation angle-delay-Doppler domain. However, this is feasible only when the BS has a large antenna array so that there is sufficient resolution in the angle domains. Figure 7.26 shows the PDMA multiuser detection scheme based on simple MRC on the received signal in the angle-delay-Doppler domain.

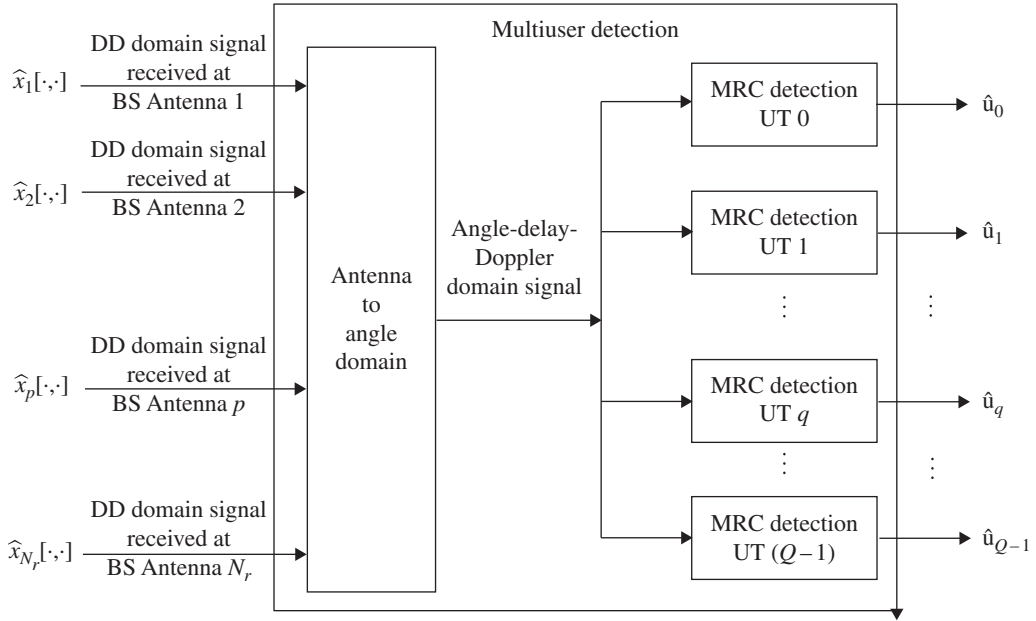


Figure 7.26 PDMA multiuser detection in the angle-delay-Doppler domain.

In [132], an OTFS massive MIMO-based non-orthogonal multiple access method is proposed where even UTs overlapping in the angle domain are also allocated common DDREs. At the BS receiver, a combination of successive interference cancellation and MRC is used for multiuser detection.

7.3 OTFS Downlink

In multiuser downlink systems, a BS sends information to multiple UTs sharing a common physical resource. An illustration of a multiuser downlink communication system is shown in Fig. 7.27. Let $x(t)$ denote the TD signal transmitted by the BS. The TD signal received at the q th UT is then given by

$$y_q(t) = \sum_{i=1}^{L_q} h_{q,i} x(t - \tau_{q,i}) e^{j2\pi v_{q,i}(t - \tau_{q,i})} + w_q(t), \quad (7.61)$$

where $w_q(t)$ is the AWGN at the q th UT having a PSD of N_0 W/Hz.

The BS multiplexes DD domain information for all UTs either by sending information to the UTs on non-overlapping REs, i.e., orthogonal downlink communication (ODC), or by precoding the information to UTs on common REs in such a way that there is little MUI when the signal is received at the respective UTs (i.e., downlink precoding). Clearly, ODC is similar to OMA in the uplink, and, therefore, the OMA methods (i.e., GB-OMA, IDD-OMA, and ITF-OMA) discussed in Section 7.2.1 are applicable to ODC. For example, in GB-based ODC (GB-ODC), the BS sends information to UTs on non-overlapping DDREs separated by GBs. Similarly, the BS could send information to a UT on REs which are regularly spaced apart either in the DD domain or in the TF domain, i.e., DD domain interleaving or TF domain interleaving, respectively, as in IDD-OMA

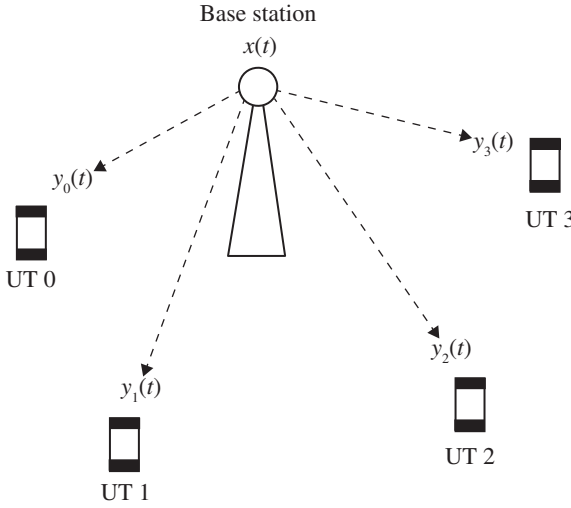


Figure 7.27 Multiuser downlink system.

and ITF-OMA. The advantage of ODC is the simplified receiver processing which is important for downlink systems, where UTs have limited signal processing capability. The disadvantage of ODC is low SE as each RE can carry information for only one UT. With downlink precoding, on the other hand, each RE can carry information of more than one UT which increases the overall SE. The BS uses channel state information of the UTs to precode their information in such a way that there is minimal MUI at the UTs.

7.3.1 Downlink Precoding

The block diagram of a general DD domain-based OTFS MIMO precoder on the downlink is shown in Fig. 7.28. Let $x_p[k, l]$, $k = 0, 1, \dots, N - 1$, $l = 0, 1, \dots, M - 1$ denote the (k, l) th DD domain symbol to be transmitted on the (k, l) th DDRE from the p th BS antenna, $p = 1, 2, \dots, N_r$. The DD domain symbols intended to be transmitted from a particular antenna are converted to the TF domain using ISFFT and then to the TD transmit signal using the Heisenberg transform (see Fig. 7.28). The TD signal transmitted from the p th BS antenna is denoted by $x_p(t)$, $p = 1, 2, \dots, N_r$. Assume each UT has a single antenna. At each UT, the received TD signal is converted to the TF domain using the Wigner transform and then these are transformed to the DD domain using SFFT. Let $y_q[k, l]$ denote the DD domain symbol received at the q th UT in the (k, l) th DDRE. Also, let $\mathbf{y}_q \in \mathbb{C}^{MN \times 1}$ denote the vector of DD domain symbols received at the q th UT, having $y_q[k, l]$ as its $(kM + l + 1)$ th element, $k = 0, 1, \dots, N - 1$, $l = 0, 1, \dots, M - 1$. Furthermore, $\mathbf{x}_p \in \mathbb{C}^{MN \times 1}$ denotes the vector of DD domain symbols transmitted from the p th BS antenna having $x_p[k, l]$ as its $(kM + l + 1)$ th element. Similar to the derivation in Section 7.2.2.1, the vector of received DD domain symbols at the q th UT can be derived in the form

$$\mathbf{y}_q = \mathbf{H}_q^T \mathbf{x} + \mathbf{w}_q, \quad (7.62)$$

where $\mathbf{x} \triangleq [\mathbf{x}_1^T \mathbf{x}_2^T \dots \mathbf{x}_{N_r}^T]^T \in \mathbb{C}^{MNN_r \times 1}$ and $\mathbf{w}_q \in \mathbb{C}^{MN \times 1}$ is the vector of AWGN samples at the q th UT with i.i.d. $\mathcal{CN}(0, MNN_0)$ elements. In (7.62), $\mathbf{H}_q \in \mathbb{C}^{MNN_r \times MN}$ is the effective DD domain channel between the BS and the q th UT and is given by (7.47). Let $\mathbf{y} \triangleq [\mathbf{y}_0^T \mathbf{y}_1^T \dots \mathbf{y}_{Q-1}^T]^T \in \mathbb{C}^{MNQ \times 1}$

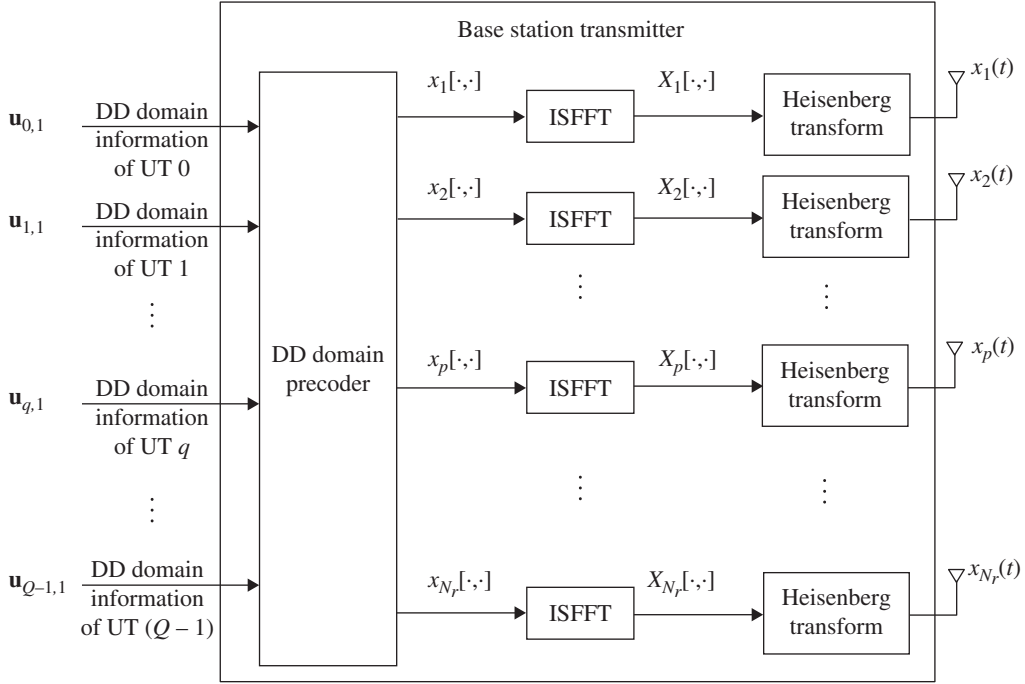


Figure 7.28 DD domain-based OTFS MIMO precoding at the BS.

denote the vector of DD domain symbols received at all the Q UTs. Then, $\mathbf{y}$ can be written in the form

$$\mathbf{y} = \mathbf{H}^T \mathbf{x} + \mathbf{w}, \quad (7.63)$$

where $\mathbf{H} \triangleq [\mathbf{H}_0 \mathbf{H}_1 \cdots \mathbf{H}_{Q-1}] \in \mathbb{C}^{MNN_r \times MNQ}$ and $\mathbf{w} \triangleq [\mathbf{w}_0^T \mathbf{w}_1^T \cdots \mathbf{w}_{Q-1}^T]^T$. The multiuser input-output relation in (7.63) is similar to that in OFDM-based multiuser downlink systems, and, therefore, traditional multiuser precoding techniques can be used. Two types of precoders, namely, linear and non-linear precoders, are common. Linear precoders have lower precoding complexity but are sub-optimal. On the other hand, non-linear precoding can achieve better performance compared to linear precoders but have higher complexity. Although the input-output relation of OTFS-based multiuser downlink systems is similar to that of OFDM-based systems, there is one important difference. In OFDM-based systems, separate downlink precoding is performed for each RE (i.e., each subcarrier). However, in OTFS-based systems, the DD domain information symbol transmitted on a DDRE spreads along both the delay and Doppler domains due to which joint precoding of all DDREs is required to ensure robustness against channel-induced Doppler shift. This explains why for a given number of antennas at the BS and the UT, the dimensions of the effective multiuser channel matrix is MN times larger in OTFS-based systems when compared to that in OFDM-based systems.

7.3.1.1 Linear Precoding

Linear precoding in the downlink is considered in this section. Denote the vector of DD domain information symbols intended for the q th single-antenna UT by $\mathbf{u}_{q,1} \in \mathbb{C}^{MN \times 1}$, $q = 0, 1, \dots, Q - 1$.

In downlink precoding, the DD domain transmit vector $\mathbf{x}$ is chosen in such a way that $\mathbf{y}_q \approx \alpha_q \mathbf{u}_{q,1}$, i.e., the received vector of DD domain symbols at the q th UT is approximately equal to a scalar multiple α_q of the vector of DD domain information symbols intended for that UT. For linear downlink precoders (e.g., linear MMSE, zero-forcing, and maximum ratio transmission (MRT) precoders), the DD domain transmit vector is given by

$$\mathbf{x} = \mathbf{A}\mathbf{u}, \quad (7.64)$$

where $\mathbf{u} = [\mathbf{u}_{0,1}^T \mathbf{u}_{1,1}^T \cdots \mathbf{u}_{Q-1,1}^T]^T \in \mathbb{C}^{MNQ \times 1}$ is the vector of MNQ DD domain information symbols intended for all Q UTs and $\mathbf{A} \in \mathbb{C}^{MNN_r \times MNQ}$ is the multiuser downlink precoding matrix. Consider the DD domain information symbols to be i.i.d. zero mean having unit variance. From the OTFS modulation equations (i.e., ISFFT followed by the Heisenberg transform at each BS antenna), it follows that for a given DD domain transmit vector $\mathbf{x} \in \mathbb{C}^{MNN_r \times 1}$, the total transmitted energy is given by

$$\sum_{p=1}^{N_r} \int |x_p(t)|^2 dt = \frac{\|\mathbf{x}\|^2}{MN} = \frac{\|\mathbf{A}\mathbf{u}\|^2}{MN}. \quad (7.65)$$

Since the OTFS frame duration is NT , the average total power transmitted by the BS in the downlink is $\mathbb{E} [\|\mathbf{A}\mathbf{u}\|^2] / (MN^2T)$, where the expectation is with respect to $\mathbf{u}$. At a given UT receiver, the noise power is $M\Delta f N_0$ since the communication bandwidth is $M\Delta f$ and N_0 is AWGN PSD. Therefore, the ratio of the average total power transmitted by the BS to the AWGN power at the UT, i.e., transmit signal to noise ratio (TSNR), is given by

$$\text{TSNR} \triangleq \frac{\mathbb{E}_{\mathbf{u}} [\|\mathbf{A}\mathbf{u}\|^2]}{M^2 N^2 N_0}. \quad (7.66)$$

Substituting (7.64) in (7.63) gives

$$\mathbf{y} = \mathbf{H}^T \mathbf{A}\mathbf{u} + \mathbf{w}. \quad (7.67)$$

If $\mathbf{A}$ is chosen such that $\mathbf{H}^T \mathbf{A}$ is approximately diagonal, i.e., $\mathbf{H}^T \mathbf{A} \approx \mathbf{D}$, where $\mathbf{D}$ is diagonal, then $\mathbf{y} \approx \mathbf{D}\mathbf{u} + \mathbf{w}$ and there is almost no MUI. However, for such a precoding matrix to exist, the rank of $\mathbf{H}$ must be MNQ for which the number of BS antennas needs to be greater than the number of UTs, i.e., $N_r \geq Q$.² Examples of linear multiuser precoders include (i) zero-forcing (ZF) precoder, where the precoding matrix is given by $\mathbf{A} = \gamma \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1}$ and the scalar γ is chosen to achieve a desired TSNR in (7.66), (ii) linear MMSE (LMMSE) precoder, where the precoding matrix is given by $\mathbf{A} = \gamma \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^* + MN N_0 \mathbf{J})^{-1}$, where $\mathbf{J}$ is a diagonal matrix, and (iii) MRT precoder, where $\mathbf{A} = \gamma \mathbf{H}^*$.

Zero-forcing Precoder For a desired TSNR ρ_{tx} , the power normalization constant which satisfies (7.66) is given by

$$\gamma_{\text{zf}} \triangleq \sqrt{\frac{\rho_{\text{tx}} M^2 N^2 N_0}{\text{Tr}((\mathbf{H}^T \mathbf{H}^*)^{-1})}}. \quad (7.68)$$

² For the general case where each UT has multiple antennas and multiple spatial streams are communicated to each UT, the number of BS antennas must be at least equal to the total number of spatial streams of all UTs.

From (7.67), it follows that, for the ZF detector, the vector of DD domain symbols received at all UTs is given by

$$\begin{aligned}
 \mathbf{y} &= \mathbf{H}^T \mathbf{A} \mathbf{u} + \mathbf{w} \\
 &= \mathbf{H}^T \underbrace{\gamma_{\text{zf}} \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1}}_{=\mathbf{A}} \mathbf{u} + \mathbf{w} \\
 &= \gamma_{\text{zf}} \mathbf{u} + \mathbf{w},
 \end{aligned} \tag{7.69}$$

which indicates that each DD domain information symbol is received without any interference. From (7.62), the elements of $\mathbf{w}$ are i.i.d. $\mathcal{CN}(0, MNN_0)$. Therefore, from (7.69) and (7.68), it follows that the effective received DD domain SNR at each UT is given by

$$\rho_{\text{dl,zf}} \triangleq \frac{\gamma_{\text{zf}}^2}{MNN_0} = \frac{\rho_{\text{tx}}}{\text{Tr}((\mathbf{H}^T \mathbf{H}^*)^{-1})}. \tag{7.70}$$

In scenarios where the matrix $(\mathbf{H}^T \mathbf{H}^*)$ is ill conditioned, $\text{Tr}((\mathbf{H}^T \mathbf{H}^*)^{-1})$ can be large, resulting in a reduction in the DD domain SNR at each UT.

Linear MMSE Precoder In ill-conditioned scenarios, the performance of ZF precoders can be significantly improved by regularizing it, i.e., instead of completely nullifying interference, a controlled amount of interference is allowed by choosing a precoding matrix which is better conditioned than the ZF precoding matrix. One way of improving conditioning is to modify the ZF precoding matrix by adding a diagonal matrix (with non-zero diagonal elements) to the matrix $(\mathbf{H}^T \mathbf{H}^*)$ before inverting it. The precoding matrix thus obtained is of the general form $\mathbf{A} = \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^* + \mathbf{J})^{-1}$, where $\mathbf{J}$ is diagonal. This diagonal matrix $\mathbf{J}$ could be chosen appropriately so as to optimize performance. When the channel statistics for all UTs is same, it is likely that $\mathbf{J}$ is a scaled identity matrix. For example, in [133], it was shown that for OFDM-based multiuser MIMO systems with i.i.d. Rayleigh fading, $\mathbf{J} = Q/\rho_{\text{tx}} \mathbf{I}$ maximizes the received signal-to-interference-plus-noise ratio (SINR) at each single-antenna UT in the downlink (when the number of UTs is sufficiently large). For more details on MMSE-based MIMO precoders in OFDM-based systems, refer to [134].

Performance of ZF and LMMSE Precoders Figure 7.29 shows the uncoded 4-QAM BER performance of the ZF and LMMSE precoders for 4-QAM as a function of increasing transmit SNR in a downlink MIMO-OTFS system where a BS with $N_r = 16$ antennas (array with $N_h = N_v = 4$) serves $Q = 2$ UTs. The channel model and the OTFS modulation parameters are the same as those considered for the simulation of the uplink multiuser MIMO-OTFS system in Fig. 7.23. It is observed from Fig. 7.29 that, for both ZF and LMMSE precoding, the OTFS-based system performs better than the OFDM-based system. While their performance difference is small for the no-mobility case ($v_{\text{max}} = 0$ Hz), it is quite significant for the high-mobility case ($v_{\text{max}} = 1600$ Hz, corresponding to a UT speed of 360 km/h at 4.8 GHz carrier frequency). For example, the OFDM-based system performance for $v_{\text{max}} = 1600$ Hz degrades compared to no-mobility performance and exhibits an error floor. This is because of the increased ICI in OFDM-based systems due to mobility-induced Doppler spread. However, the corresponding performance of the OTFS-based system is significantly better. The BER performance of the OTFS-based system is observed to be almost invariant to the increase in v_{max} from 0 to 1600 Hz. Also, the LMMSE precoder performs better than the ZF precoder in both OTFS- and OFDM-based systems. For example, to achieve a BER of 1×10^{-3} with $v_{\text{max}} = 1600$ Hz, the transmit SNR required by the LMMSE precoder is about 1.5 dB less compared to that required by the ZF precoder.

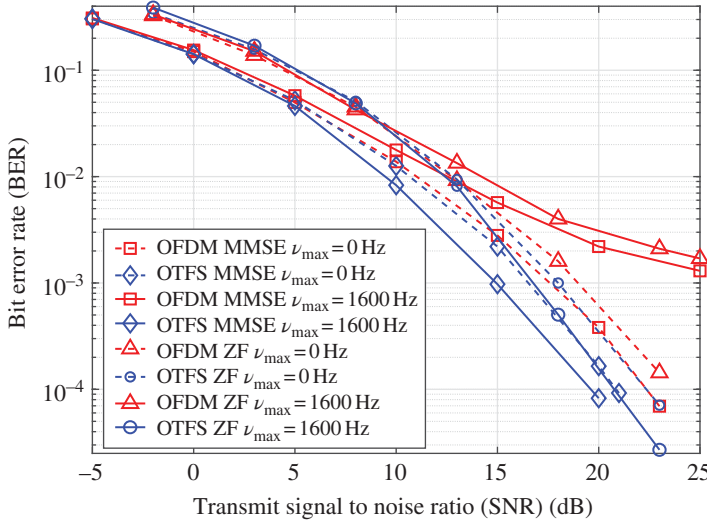


Figure 7.29 BER performance of ZF and LMMSE multiuser precoders in multiuser MIMO-OTFS downlink. $Q = 2, N_r = 16, L = 8$.

The LMMSE and ZF precoders require matrix inversion. Therefore, they have high complexity, especially when M and N are large. MRT-based linear precoding; on the other hand, does not require matrix inversion, but generally has relatively poor performance due to high MUI. MRT precoding is appropriate in certain scenarios. For example, for OFDM-based downlink massive MIMO systems, it is known that under favorable propagation conditions, interference is dominated by AWGN, and, in such cases, MRT can achieve better energy efficiency than the ZF precoder when the desired spectral efficiency is not large [135, 136]. An MRT approach for OTFS-based massive MIMO downlink is of interest, which is considered in Section 7.3.2.

7.3.1.2 Nonlinear Precoding

Nonlinear precoding techniques (e.g., vector perturbation [137]) developed for OFDM-based multiuser MIMO systems are also applicable to OTFS-based multiuser systems. In vector perturbation (VP)-based OTFS ZF precoder, the DD domain transmit vector is given by $\mathbf{x} = \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1} \tilde{\mathbf{u}}$, where $\tilde{\mathbf{u}}$ is obtained from the DD domain information symbol vector $\mathbf{u}$ in such a way that $\|\mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1} \tilde{\mathbf{u}}\|$ is smaller than $\|\mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1} \mathbf{u}\|$. This effectively reduces the required transmit power from the BS. The DD domain signal vector received at the UTs is $\mathbf{y} = \mathbf{H}^T \mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1} \tilde{\mathbf{u}} + \mathbf{w}$, i.e., $\mathbf{y} = \tilde{\mathbf{u}} + \mathbf{w}$. Since $\tilde{\mathbf{u}} + \mathbf{w}$ is received instead of the intended $\mathbf{u} + \mathbf{w}$, $\tilde{\mathbf{u}}$ must be such that $\mathbf{u}$ can be recovered from $\tilde{\mathbf{u}}$. The idea of Tomlinson–Harashima precoding (THP) can be used here [138, 139]. As an example, let $\mathbf{u}$ belong to a regular square QAM alphabet, i.e., the real and imaginary part of each QAM symbol belongs to the set $\{(2k-1)|k = -(K-1), -(K-2), \dots, -1, 0, 1, \dots, K\}$, where the number of symbols in the QAM alphabet is $4K^2$. Then, $\tilde{\mathbf{u}}$ can be chosen as $\tilde{\mathbf{u}} = \mathbf{u} + 4K\mathbf{v}$, where each element of $\mathbf{v} \in \mathbb{C}^{MN \times 1}$ has integer real and imaginary parts. This way, each element in $\mathbf{u}$ maps to an infinite set of values for the corresponding element in $\tilde{\mathbf{u}}$. For example, let $u_i = (2i_1 - 1) + j(2i_2 - 1)$ be the i th element of $\mathbf{u}$. Then, the corresponding element in $\tilde{\mathbf{u}}$, i.e., $\tilde{u}_i$ belongs to an infinite set $\mathcal{S}_{(i_1, i_2)}$, given by

$$\mathcal{S}_{(i_1, i_2)} \triangleq \{((2i_1 - 1) + 4Ka_i) + j((2i_2 - 1) + 4Kb_i) \mid a_i, b_i \in \mathbb{Z}\}. \quad (7.71)$$

It can be seen that for any two distinct pairs of integers (i_1, i_2) and $(i_3, i_4) \neq (i_1, i_2)$, their corresponding sets $S_{(i_1, i_2)}$ and $S_{(i_3, i_4)}$ are disjoint. Clearly, in the absence of AWGN at the UTs, each element of $\mathbf{u}$ can be recovered from the corresponding element of the received $\tilde{\mathbf{u}}$ through the element-wise operation $\mathbf{u} = \text{mod}(\tilde{\mathbf{u}}, 4K)$, where for any complex number z and positive integer K , $\text{mod}(z, 4K)$ denotes a complex number whose real and imaginary parts are unique numbers which lie in the interval $[-2K, 2K)$ and which are congruent (modulo $4K$) to the real and imaginary part of z , respectively. Also $j \triangleq \sqrt{-1}$. In the presence of AWGN, the DD domain information symbols can be recovered from $\mathbf{y}$ by firstly mapping the elements of $\mathbf{y}$ to the interval $[-2K, 2K)$, i.e., $\mathbf{y}' \triangleq \text{mod}(\mathbf{y}, 4K)$, followed by element-wise rounding to the nearest QAM alphabet. As an example with 64-QAM modulation (i.e., $K = 4$), let $y_i = -4.7 + j21.8$ (the i th element of $\mathbf{y}$). From the above discussion, it follows that $y'_i = \text{mod}(-4.7, 16) + j \text{mod}(21.8, 16) = -4.7 + j5.8$. The recovered i th DD domain information symbol is obtained by rounding y'_i to the nearest 64-QAM symbol which is $(-5 + j5)$.

In a ZF-THP precoder, for a given $\mathbf{u}$, $\tilde{\mathbf{u}}$ is chosen in such a way that $\mathbf{u}$ is recoverable from $\tilde{\mathbf{u}}$ and the effective required transmit power from the BS is minimized. For example, for the $4K^2$ -QAM information alphabet,

$$\tilde{\mathbf{u}} = \mathbf{u} + 4K\mathbf{v}^*, \quad (7.72)$$

where

$$\mathbf{v}^* \triangleq \arg \min_{\mathbf{v} \in \mathbb{Z}[j]^{MN_Q \times 1}} \|\mathbf{H}^* (\mathbf{H}^T \mathbf{H}^*)^{-1} (\mathbf{u} + 4K\mathbf{v})\|, \quad (7.73)$$

where $\mathbb{Z}[j]$ is the set of Gauss integers (i.e., complex numbers whose real and imaginary parts are both integers). THP-based nonlinear multiuser precoder is known to achieve superior performance than linear precoders (ZF, MMSE, and MRT). The optimization in (7.72) is essentially a closest point search on a lattice [140]. Techniques for solving (7.72) include those based on sphere decoding [141] and lattice-aided reduction [142]. These techniques are, however, not computationally feasible for OTFS-based systems, where the signal dimensionality is significantly larger than that in OFDM-based systems.

7.3.2 Massive MIMO-Based OTFS Downlink

Under favorable conditions, OFDM-based massive MIMO systems are known to achieve near-optimal downlink performance with low-complexity MRT-based linear precoding [135]. In [143], a MRT-based low-complexity multi-antenna massive MIMO precoding technique has been proposed for OTFS-based systems. It is shown that for a scenario where no two UTs share a common channel path (i.e., same zenith and azimuthal angles of arrival), the total interference and noise are dominated by AWGN when the number of BS antennas is sufficiently large. In other words, the effects of MUI and ISI between different DD domain symbols of the same UT become vanishingly small in comparison to AWGN when the number of BS antennas is large. Since ISI is small, single-symbol DD domain detection is possible at the UTs. That is, each UT does not need to perform joint demodulation of all DD domain information symbols which may be computationally challenging given the limited signal processing capability at a UT [143]. The MRT precoder-based massive MIMO-OTFS downlink system is described in further detail in Section 7.3.2.1.

7.3.2.1 MRT-Based Massive MIMO-OTFS Precoding

For the MRT precoder, the precoding matrix is given by $\mathbf{A} = \gamma_{\text{MRT}} \mathbf{H}^*$. From (7.66), it follows that the TSNR is given by

$$\text{TSNR} = \frac{\mathbb{E}[\|\mathbf{A}\mathbf{u}\|^2]}{M^2 N^2 N_0} = \gamma_{\text{MRT}}^2 \frac{\text{Tr}(\mathbf{H}^T \mathbf{H}^*)}{M^2 N^2 N_0}. \quad (7.74)$$

For a desired TSNR, ρ_{tx} , the required power normalization constant is, therefore, given by

$$\gamma_{\text{MRT}} = \sqrt{\frac{\rho_{\text{tx}} M^2 N^2 N_0}{\text{Tr}(\mathbf{H}^T \mathbf{H}^*)}}. \quad (7.75)$$

From (7.62) and (7.64), it follows that the vector of received DD domain symbols at the q th UT is given by

$$\begin{aligned} \mathbf{y}_q &= \mathbf{H}_q^T \mathbf{A} \mathbf{u} + \mathbf{w}_q \\ &= \gamma_{\text{MRT}} \mathbf{H}_q^T \mathbf{H}^* \mathbf{u} + \mathbf{w}_q \\ &= \gamma_{\text{MRT}} \sum_{i=0}^{Q-1} \mathbf{H}_q^T \mathbf{H}_i^* \mathbf{u}_{i,1} + \mathbf{w}_q \\ &= \gamma_{\text{MRT}} \mathbf{H}_q^T \mathbf{H}_q^* \mathbf{u}_{q,1} + \gamma_{\text{MRT}} \sum_{i=0, i \neq q}^{Q-1} \mathbf{H}_q^T \mathbf{H}_i^* \mathbf{u}_{i,1} + \mathbf{w}_q. \end{aligned} \quad (7.76)$$

For single-symbol detection of the DD domain information symbol $u_{q,1}[k, l]$ transmitted on the (k, l) th DDRE, the q th UT does not perform joint DD domain demodulation. Instead, $u_{q,1}[k, l]$ is simply decoded based on the symbol received at the q th UT on the (k, l) th DDRE, i.e., $y_q[k, l]$. Let $\mathbf{h}_{q,k,l} \in \mathbb{C}^{MN_r \times 1}$ be the $(kM + l + 1)$ th column of $\mathbf{H}_q$. Then, using (7.76),

$$\begin{aligned} \frac{y_q[k, l]}{\gamma_{\text{MRT}} N_r} &= \frac{\|\mathbf{h}_{q,k,l}\|^2}{N_r} u_{q,1}[k, l] + \overbrace{\sum_{(k', l') \neq (k, l)} \frac{\mathbf{h}_{q,k,l}^T \mathbf{h}_{q,k',l'}^*}{N_r} u_{q,1}[k', l']}^{\text{ISI}} \\ &\quad + \underbrace{\sum_{i=0, i \neq q}^{Q-1} \sum_{(k', l')} \frac{\mathbf{h}_{q,k,l}^T \mathbf{h}_{i,k',l'}^*}{N_r} u_{q,i}[k', l']}_{\text{MUI}} + \underbrace{\frac{w_q[k, l]}{N_r \gamma_{\text{MRT}}}}_{\text{AWGN}}. \end{aligned} \quad (7.77)$$

In [143], for a BS with a URA, it is shown that, in (7.77), under favorable conditions (i.e., when no two UTs share a channel path), for a constant $\rho_{\text{tx}} N_r$ (i.e., ρ_{tx} proportional to $1/N_r$), the MUI and ISI terms in (7.77) become vanishingly small compared to AWGN as the number of BS antennas $N_r \rightarrow \infty$ (i.e., $N_h \rightarrow \infty$, $N_v \rightarrow \infty$ and fixed inter-antenna spacing). For sufficiently large (N_h, N_v) (i.e., large N_r), a good approximation for the SINR for single-symbol detection at the q th UT is given by

$$\text{SINR}_q \approx \rho_{\text{tx}} N_r \frac{\sum_{i=1}^{L_q} |h_{q,i}|^2}{\sum_{q'=0}^{Q-1} \sum_{i=1}^{L_{q'}} |h_{q',i}|^2}. \quad (7.78)$$

The derivation of (7.78) is presented below. Note that in (7.78), SINR_q depends on ρ_{tx} and N_r only through their product $\rho_{\text{tx}} N_r$. Therefore, with increasing N_r , for a fixed desired SINR_q , the required ρ_{tx} decreases as $1/N_r$. Also, in (7.78), $\frac{\sum_{i=1}^{L_q} |h_{q,i}|^2}{\sum_{q'=0}^{Q-1} \sum_{i=1}^{L_{q'}} |h_{q',i}|^2}$ is simply the ratio of the total channel path gain for the q th UT to the sum of the channel path gains of all Q UTs. For example, in a scenario where the total channel path gain for each UT is the same, this ratio is $1/Q$. In other words, the effective received SINR at each UT is proportional to its total channel path gain.

Derivation of SINR for Single-symbol Detection Since each UT has a single antenna, from (7.43) and (7.47),

$$\mathbf{h}_{q,k,l} = \left[\mathbf{h}_{q,1,k,l}^T \mathbf{h}_{q,2,k,l}^T \cdots \mathbf{h}_{q,p,k,l}^T \cdots \mathbf{h}_{q,N_r,k,l}^T \right]^T, \quad (7.79)$$

where $\mathbf{h}_{q,p,k,l} \in \mathbb{C}^{MN \times 1}$ is the $(kM + l + 1)$ th column of $\mathbf{H}_{p,q,1}$ (see (7.47)). From the expression for $\mathbf{H}_{p,q,s}$ in (7.43), it follows that

$$\begin{aligned} \mathbf{h}_{q,p,k,l} &= \sum_{i=1}^{L_q} a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i}) a_{t,1}^*(\phi_{t,q,i}, \theta_{t,q,i}) \mathbf{g}_{q,i,k,l} \\ &= \sum_{i=1}^{L_q} a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i}) \mathbf{g}_{q,i,k,l}, \end{aligned} \quad (7.80)$$

where $\mathbf{g}_{q,i,k,l} \in \mathbb{C}^{MN \times 1}$ denotes the $(kM + l + 1)$ th column of $\mathbf{G}_{q,i}$ and the last step follows from the fact that each UT has a single antenna. The elements of $\mathbf{G}_{q,i}$ are given by (7.42). The expression for $\mathbf{h}_{q,k,l}^T \mathbf{h}_{q,k',l'}^*$ (see the ISI term in (7.77)) is then given by

$$\begin{aligned} \mathbf{h}_{q,k,l}^T \mathbf{h}_{q,k',l'}^* &\stackrel{(a)}{=} \sum_{p=1}^{N_r} \mathbf{h}_{q,p,k,l}^T \mathbf{h}_{q,p,k',l'}^* \\ &\stackrel{(b)}{=} \sum_{p=1}^{N_r} \sum_{i=1}^{L_q} |a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i})|^2 \mathbf{g}_{q,i,k,l}^T \mathbf{g}_{q,i,k',l'}^* \\ &\quad + \sum_{p=1}^{N_r} \sum_{i_1=1}^{L_q} \sum_{i_2=1, i_2 \neq i_1}^{L_q} \mathbf{g}_{q,i_1,k,l}^T \mathbf{g}_{q,i_2,k',l'}^* a_{r,p}(\phi_{r,q,i_1}) a_{r,p}^*(\phi_{r,q,i_2}), \end{aligned} \quad (7.81)$$

where step (a) and step (b) follow from (7.79) and (7.80), respectively, and

$$\mathbf{g}_{q,i_1,k,l}^T \mathbf{g}_{q,i_2,k',l'}^* \stackrel{(a)}{=} \frac{h_{q,i_1} h_{q,i_2}^*}{MN} \left[\sum_{m=0}^{M-1} e^{j2\pi m \frac{(l' - l + M\Delta f(\tau_{q,i_2} - \tau_{q,i_1}))}{M}} \right] \left[\sum_{n=0}^{N-1} e^{-j2\pi n \frac{(k' - k + NT(\nu_{q,i_2} - \nu_{q,i_1}))}{N}} \right], \quad (7.82)$$

where (a) follows from the fact that $\mathbf{g}_{q,i,k,l}$ is the $(kM + l + 1)$ th column of $\mathbf{G}_{q,i}$ and the expression for the elements of $\mathbf{G}_{q,i}$ in (7.42). From (7.82), it is clear that, for $i_1 = i_2 = i$,

$$\mathbf{g}_{q,i,k,l}^T \mathbf{g}_{q,i,k',l'}^* = |h_{q,i}|^2 \delta[l' - l] \delta[k' - k]. \quad (7.83)$$

Also, from (7.60), we have

$$\begin{aligned} \frac{1}{N_r} \sum_{p=1}^{N_r} a_{r,p}(\phi_{r,q,i_1}, \theta_{r,q,i_1}) a_{r,p}^*(\phi_{r,q,i_2}, \theta_{r,q,i_2}) &= \frac{1}{N_r} \sum_{p=1}^{N_r} e^{j2\pi \frac{d}{\lambda} \bmod(p-1, N_h) c_{q,i_1,i_2}} e^{j2\pi \frac{d}{\lambda} \left\lfloor \frac{p-1}{N_h} \right\rfloor c'_{q,i_1,i_2}}, \\ c_{q,i_1,i_2} &\triangleq \sin(\phi_{r,q,i_1}) \cos(\theta_{r,q,i_1}) - \sin(\phi_{r,q,i_2}) \cos(\theta_{r,q,i_2}), \\ c'_{q,i_1,i_2} &\triangleq \sin(\theta_{r,q,i_1}) - \sin(\theta_{r,q,i_2}). \end{aligned} \quad (7.84)$$

Further simplification gives

$$\begin{aligned} \frac{1}{N_r} \sum_{p=1}^{N_r} a_{r,p}(\phi_{r,q,i_1}, \theta_{r,q,i_1}) a_{r,p}^*(\phi_{r,q,i_2}, \theta_{r,q,i_2}) \\ = \frac{\text{sinc}\left(\frac{d}{\lambda} N_h c_{q,i_1,i_2}\right)}{\text{sinc}\left(\frac{d}{\lambda} c_{q,i_1,i_2}\right)} \frac{\text{sinc}\left(\frac{d}{\lambda} N_v c'_{q,i_1,i_2}\right)}{\text{sinc}\left(\frac{d}{\lambda} c'_{q,i_1,i_2}\right)} e^{j\pi \frac{d}{\lambda} (N_h - 1) c_{q,i_1,i_2}} e^{-j\pi \frac{d}{\lambda} (N_v - 1) c'_{q,i_1,i_2}}, \end{aligned} \quad (7.85)$$

Under favorable conditions, both angles of arrivals for any two paths between the q th UT and the BS are not the same, and, therefore, for $i_1 \neq i_2$, both c_{q,i_1,i_2} and c'_{q,i_1,i_2} cannot be zero. Hence,

$$\lim_{N_h \rightarrow \infty, N_v \rightarrow \infty} \frac{\text{sinc}\left(\frac{d}{\lambda} N_h c_{q,i_1,i_2}\right)}{\text{sinc}\left(\frac{d}{\lambda} c_{q,i_1,i_2}\right)} \frac{\text{sinc}\left(\frac{d}{\lambda} N_v c'_{q,i_1,i_2}\right)}{\text{sinc}\left(\frac{d}{\lambda} c'_{q,i_1,i_2}\right)} = 0. \quad (7.86)$$

Using this in (7.81) gives

$$\begin{aligned} \lim_{N_h \rightarrow \infty, N_v \rightarrow \infty} \frac{\mathbf{h}_{q,k,l}^T \mathbf{h}_{q,k',l'}^*}{N_r} &= \sum_{p=1}^{N_r} \sum_{i=1}^{L_q} |a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i})|^2 \mathbf{g}_{q,i,k,l}^T \mathbf{g}_{q,i,k',l'}^* \\ &\stackrel{(a)}{=} \sum_{i=1}^{L_q} \mathbf{g}_{q,i,k,l}^T \mathbf{g}_{q,i,k',l'}^*, \end{aligned} \quad (7.87)$$

where step (a) follows from the fact that $|a_{r,p}(\phi_{r,q,i}, \theta_{r,q,i})|^2 = 1$ (see (7.60)). Furthermore, from (7.83), it follows that, for $(k, l) \neq (k', l')$,

$$\mathbf{g}_{q,i,k,l}^T \mathbf{g}_{q,i,k',l'}^* = 0, \quad (k, l) \neq (k', l'). \quad (7.88)$$

Using (7.88) in (7.87) gives

$$\lim_{N_h \rightarrow \infty, N_v \rightarrow \infty} \frac{\mathbf{h}_{q,k,l}^T \mathbf{h}_{q,k',l'}^*}{N_r} = 0, \quad (k, l) \neq (k', l'). \quad (7.89)$$

From (7.83) and (7.87), with $(k, l) = (k', l')$, it also follows that

$$\lim_{N_h \rightarrow \infty, N_v \rightarrow \infty} \frac{\|\mathbf{h}_{q,k,l}\|^2}{N_r} = \sum_{i=1}^{L_q} |h_{q,i}|^2. \quad (7.90)$$

From (7.89), it is clear that the ISI term in (7.77) vanishes as $(N_h, N_v) \rightarrow \infty$. At the same time from (7.90), it is clear that the desired signal term in (7.77), i.e., $\frac{\|\mathbf{h}_{q,k,l}\|^2}{N_r} u_{q,1}[k, l]$, is approximately $\left(\sum_{i=1}^{L_q} |h_{q,i}|^2\right) u_{q,1}[k, l]$ for a large number of BS antennas. Furthermore, the variance of the effective AWGN term in (7.77) is given by

$$\begin{aligned} \mathbb{E} \left[\frac{|w_q[k, l]|^2}{N_r^2 \gamma_{\text{MRT}}} \right] &\stackrel{(a)}{=} \frac{M N N_0 \text{Tr}(\mathbf{H}^T \mathbf{H}^*)}{N_r^2 \rho_{\text{tx}} M^2 N^2 N_0} \\ &= \frac{\sum_{q=0}^{Q-1} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} \|\mathbf{h}_{q,k,l}\|^2}{N_r^2 M N \rho_{\text{tx}}}, \end{aligned} \quad (7.91)$$

where step (a) follows from the expression of γ_{MRT} in (7.75) and the fact that $w_q[k, l] \sim \mathcal{CN}(0, M N N_0)$. Using (7.90) in (7.91), it follows that with $(N_h, N_v) \rightarrow \infty$ and $\rho_{\text{tx}} = \frac{E_{\text{tx}}}{N_r}$ (i.e., decreasing TSNR with increasing N_r),

$$\lim_{N_h \rightarrow \infty, N_v \rightarrow \infty} \mathbb{E} \left[\frac{|w_q[k, l]|^2}{N_r^2 \gamma_{\text{MRT}}} \right] = \frac{\sum_{q'=0}^{Q-1} \sum_{i=1}^{L_{q'}} |h_{q',i}|^2}{E_{\text{tx}}}. \quad (7.92)$$

Similarly, it can also be shown that the MUI term in (7.77) vanishes as $(N_h, N_v) \rightarrow \infty$. Therefore, from (7.89), (7.90), and (7.91), it follows that for sufficiently large (N_h, N_v) , i.e., large N_r , a good approximation for the SINR for single-symbol detection at the q th UT is given by (7.78).

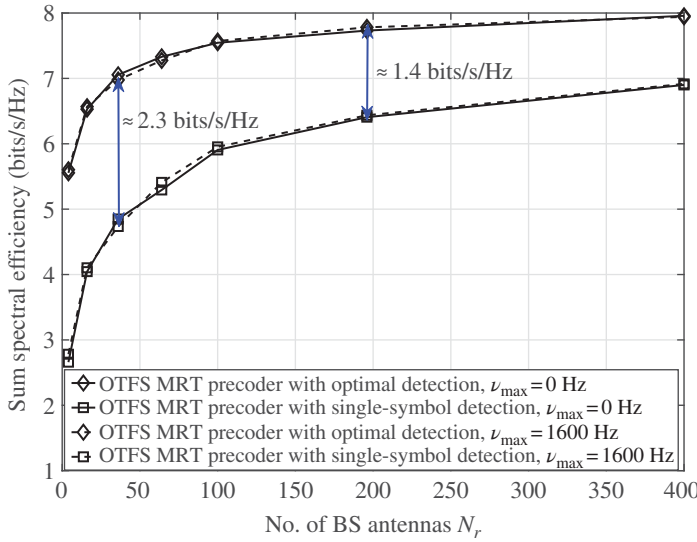


Figure 7.30 Sum spectral efficiency (bits/s/Hz) of the OTFS MRT precoder as a function of the number of BS antennas, N_r .

Numerical studies in [143] reveal that, although the OTFS MRT precoder has low complexity and MUI and ISI vanish for large N_r , the convergence of the variance of the MUI and ISI terms to zero (when compared to the variance of the desired signal term in (7.77)) may be slow. In Fig. 7.30, the variation of the sum spectral efficiency achieved by the OTFS MRT precoder is shown with increasing number of BS antennas and a fixed $Q = 4$ UTs. OTFS modulation parameters are $\Delta f = 15$ kHz and $M = 330, N = 4$. A rural macro non-NLOS (RMa-NLOS) channel model has been considered (for more details on the channel model and system/simulation parameters, refer to Section-VI in [143]). In Fig. 7.30, the sum spectral efficiency for optimal joint demodulation of all MN DD domain symbols at each UT is also plotted. With increasing N_r , fixed inter-antenna spacing $d = \lambda/2$, and fixed $\rho_{\text{tx}} N_r = 0.1$, it is observed that the sum spectral efficiency with single-symbol detection increases and appears to approach a limit. Also, the sum spectral efficiency gap between single-symbol detection and optimal detection reduces with increasing N_r . However, even with $N_r = 400$ BS antennas, there is a significant gap of about 1 bits/s/Hz. Next, in Fig. 7.31, the sum spectral efficiency of the OTFS MRT precoder is compared with that of the OFDM MRT precoder as a function of increasing $\nu_{\max}$ for a fixed $N_r = 196$ (i.e., $N_h = N_v = 14$) and fixed $Q = 4$ UTs. The subcarrier spacing for OFDM is 15 kHz, and independent information symbols are transmitted on each subcarrier. Also, MRT precoding at the BS and detection at the UT are performed separately for each subcarrier. It is observed that, with increasing Doppler shift, the sum spectral efficiency of the OFDM MRT precoder decreases, whereas that of the OTFS MRT precoder is almost the same. For high Doppler shifts, the sum spectral efficiency of the OTFS MRT precoder with single-symbol detection is greater than that achieved by the OFDM MRT precoder. This is primarily because of the increased ICI due to high Doppler shifts in OFDM which degrades performance. On the other hand, OTFS modulation is inherently robust to Doppler shifts, and, therefore, its performance does not degrade even when Doppler shifts are high.

One reason for the slow convergence of the OTFS MRT precoder is the presence of MUI and ISI because each UT is allocated the entire physical resource. A possible way to control MUI is to allocate physical resource to UTs in such a way that we achieve better spectral efficiency than that of

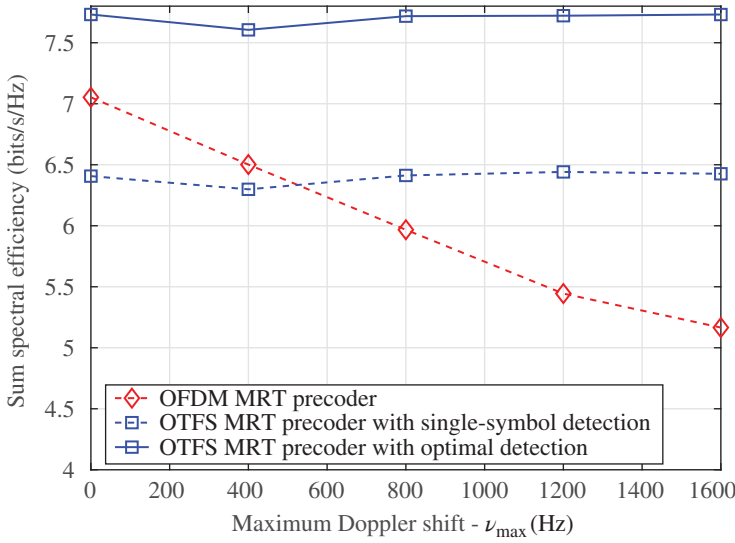


Figure 7.31 Sum spectral efficiency (bits/s/Hz) of OTFS/OFDM MRT precoder as a function of $\nu_{\max}$. $N_r = 196$ and $Q = 4$ UTs.

ODC, but at the same time, MUI is also sufficiently small. One such OTFS multiuser massive MIMO precoder has been proposed in [131], where UL/DL communication with UTs is performed in the angle-delay-Doppler domain. The main idea in [131] is as follows. The high angular resolution of the massive MIMO array at the BS is exploited to distinguish between channel paths from different UTs by viewing the DD domain signal also along the angular domains, i.e., angle-delay-Doppler domain. It is possible that even though two UTs are allocated the same DDREs, information transmitted to these two UTs on common DDREs is received in two different sets of non-overlapping DDREs at these UT receivers. This allows for almost interference-free detection at the two UTs even though their information was transmitted on common DDREs. In [131], uplink-downlink channel reciprocity has been exploited to estimate the effective angle-delay-Doppler domain channel of each UT in the uplink. The estimated channel is then used to allocate DDREs to the UTs in such a way that MUI is small.

7.4 Multiuser Channel Estimation

Multiple access techniques discussed in Section 7.2 rely on the availability of timely and accurate knowledge of the uplink channel at the BS. Similarly, downlink access (e.g., downlink precoding) discussed in Section 7.3 also relies on the availability of the downlink channel at the BS. A distinct advantage of OTFS when compared to OFDM is that the effective DD domain channel representation in OTFS-based systems depends on the delays and Doppler shifts of the channel paths, which usually change slowly when compared to the corresponding time-frequency domain representation of the same channel. Therefore, the effective channel in OTFS-based systems changes slowly when compared to that in OFDM-based systems. This results in significantly reduced channel estimation overhead in OTFS-based systems when compared to OFDM-based systems [28–30]. Several channel estimation techniques have been proposed for OTFS-based systems (see Chapter 6). In most of these methods, channel estimation is performed in the DD

domain [32], for which a subset of DDREs is reserved and no information is transmitted on these DDREs. The response of the effective DD domain channel to the transmitted pilot signal (in the reserved subset of DDREs) is used to estimate the DD domain channel at the receiver. The remaining DDREs carry information symbols [102].

In impulse-based channel estimation methods, a pilot signal is transmitted on a single DDRE and the coefficients of the effective DD domain channel are estimated directly from the received DD domain signal in the reserved subset of DDREs [32, 39, 102]. As the DD domain channel response is generally sparse (i.e., the number of non-zero channel coefficients is much smaller than MN), several works have formulated OTFS channel estimation as a sparse recovery problem which is solved using orthogonal matching pursuit (OMP) [144–146]. In other works, sparse Bayesian learning (SBL) has been used to acquire accurate channel estimates [103, 147, 148]. A central theme for most OTFS channel estimation methods is the need for dedicated resource where no data/information is transmitted. The fraction of resource dedicated for channel estimation increases with increasing delay and/or Doppler spread. Also, in multiuser scenarios, the channel estimation resources for different UTs (i.e., reserved subset of DDREs) must not overlap. To reduce the channel estimation overhead, recently, learning-based techniques have been proposed (e.g., one-shot learning through reservoir computing [149]).

7.5 Synchronization

In traditional communication systems (e.g., OFDM), timing and frequency synchronization is needed to achieve reliable and high throughput communication [150, 151]. Frequency acquisition refers to the receiver acquiring the frequency/phase of the carrier signal, while timing synchronization refers to acquiring symbol timing at the receiver [8, 152].

7.5.1 Frequency Synchronization

Frequency synchronization is needed in OFDM-based systems since residual frequency synchronization errors would lead to ICI which would degrade performance [153]. Frequency synchronization methods generally require pilots/reference signals to be transmitted periodically, which increases overhead and reduces the effective throughput. In high-mobility scenarios, these pilots need to be transmitted more frequently. The impact of ICI can be reduced by increasing subcarrier spacing. However, this results in reduced OFDM symbol duration which increases the effective cyclic prefix overhead and reduces the effective spectral efficiency. This is because the duration of the cyclic prefix is independent of the subcarrier spacing and depends only on the maximum possible channel delay spread. On the other hand, OTFS-based systems do not require separate frequency synchronization as mobility-induced Doppler shift is a part of the effective DD domain channel. This is a distinct advantage of OTFS-based systems when compared to OFDM systems. However, timing synchronization may be required in OTFS-based systems. The need for timing synchronization in OTFS-based systems and the appropriateness of DD domain waveforms for better timing synchronization are highlighted in Section 7.5.2.

7.5.2 Timing Synchronization

Timing synchronization is required in wireless communication systems in order to avoid interference between two consecutive symbol transmissions in time. For example, in the absence of uplink

timing synchronization in OFDM systems, a previous OFDM symbol transmitted by some UT could interfere with the current OFDM symbol of a UT, leading to MUI.

7.5.2.1 Uplink Timing Synchronization in OFDM-Based Systems

In OFDM-based systems, ISI between consecutive symbol transmissions is avoided by introducing a guard time duration between two adjacent/consecutive OFDM symbols in the time domain. However, since the UTs are located at different distances from the BS, this guard time must be at least equal to the maximum round-trip propagation delay between the BS and any UT in the cell. Round-trip propagation delay is considered instead of one-way delay since a UT acquires BS timing only through downlink signals transmitted by the BS. Since round-trip propagation delay can be a significant fraction of the OFDM symbol duration, this guard time overhead can reduce the effective throughput/spectral efficiency. In order to reduce the guard time duration, OFDM-based systems estimate the round-trip propagation delay between the BS and the UT. This estimated round-trip propagation delay is then fed back to the UT, for it to advance its own uplink transmission timing. This guarantees that the OFDM symbols transmitted by different UTs would arrive at the BS within a time window of duration equal to the maximum channel delay spread. Through this “timing advancement” method, the required duration of the guard time is reduced to the maximum channel delay spread which is usually much smaller than the maximum round-trip propagation delay between the BS and a UT [154]. In CP-OFDM, a copy of a portion of the OFDM symbol (cyclic prefix) is transmitted in this guard time interval prior to the transmission of the OFDM symbol.

In CP-OFDM-based 4G/5G systems, the BS estimates the round-trip propagation delay between itself and a UT, through a random access (RA) procedure. In the RA procedure [151], a UT which wants access to physical resource (for communicating information) needs to request the BS by randomly choosing and transmitting a RA preamble waveform from a list of known waveforms. From the received RA waveform, the BS estimates the round-trip propagation delay, also known as timing advance (TA), between that UT and the BS and then sends the TA estimate back to the UT for timing advancement. In 4G/5G systems, the Zadoff–Chu (ZC) sequence [155] is used to generate the RA preamble waveform. In no-mobility scenarios, ZC sequences are known to possess good auto-correlation properties. However, in high-mobility scenarios, the channel-induced Doppler shift degrades the auto-correlation properties of ZC sequences which results in inaccurate TA acquisition at the BS [156]. This increases interference between consecutive symbol transmissions.

7.5.2.2 Uplink Timing Synchronization in OTFS-Based Systems

One OTFS frame duration (i.e., NT seconds) is usually designed to be longer than the coherence time interval, which, in many scenarios can be larger than the round-trip propagation delay between the BS and a UT. In such scenarios, timing advancement is not required, as a guard time equal to the maximum round-trip propagation delay is significantly smaller than the OTFS frame duration, and is, therefore, an insignificant overhead. However, timing advancement may be required in scenarios where the round-trip propagation delay is significant compared to the OTFS frame duration (e.g., air-to-ground communication). There are other scenarios where timing advancement may still be required irrespective of whether or not the round-trip propagation delay is a significant fraction of the OTFS frame duration. For example, for reducing MUI in uplink delay-domain GB-based OTFS OMA systems (see Section 7.2.1.1), the size of the delay domain GB (separating DDREs allocated to distinct UTs) is usually chosen to be greater than the maximum possible propagation delay between the BS and a UT. Since the maximum possible propagation

delay is generally much larger than the channel delay spread, the required GB size is larger when compared to a scenario where timing advancement is used to reduce the effective maximum propagation delay.

For example, consider two UTs, one which is near the BS and another which is far away (at cell edge). Let the channel between the BS and the near UT consists of two paths having round-trip propagation delays 1 and 2.5 μs , respectively. Let the channel between the BS and the faraway UT consists of three paths having round-trip propagation delays 3, 4, and 5.3 μs , respectively. The maximum round-trip propagation delay is, therefore, 5.3 μs . Hence, in the absence of timing advancement, the GB size needs to be at least 5.3 μs . Let the BS estimate the round-trip propagation delay of the near UT to be 1 μs and that of the faraway UT to be 3 μs for timing advancement. The BS sends a timing advance command to the near UT to advance its uplink transmission timing by 1 μs and a separate timing advance command to the faraway UT to advance its timing by 3 μs . After timing advancement, the effective path delays for the near UT is 0 and 1.5 μs , respectively, while that for the faraway UT is 0, 1, and 2.3 μs , respectively. Therefore, after timing advancement, the effective maximum round-trip propagation delay is reduced to 2.3 μs , which results in a smaller GB size compared to the GB size requirement of 5.3 μs when timing advancement is not performed. A smaller GB size implies a larger effective spectral efficiency since the GB DDREs do not carry information.

The RA procedure for OFDM-based 4G/5G systems can also be used to achieve timing advancement in OTFS-based systems. However, ZC sequence-based RA preamble waveform in 4G/5G systems cannot be used for the estimation of round-trip propagation delay in high-mobility scenarios. This is because the auto-correlation properties of ZC sequences degrade in high-mobility scenarios [156]. In [157], a DD domain-based RA preamble waveform has been proposed for TA estimation in high-mobility scenarios. It is shown that, in high-mobility scenarios, the TA estimation accuracy with the proposed OTFS-based RA waveform is superior to that of ZC sequence-based TA estimation in 4G/5G systems. The key to the superior performance of DD domain-based TA estimation is that the amount of delay domain shift induced by a channel path is almost invariant of the Doppler shift of that path.

Figure 7.32 shows an illustration of DD domain-based RA preambles, as proposed in [157]. The main idea is that each RA preamble is a collection of DDREs which occupy the entire delay domain of width T , but only a contiguous portion of the Doppler domain of width $N_1 < N$ DDREs. Therefore, there are $R = \lfloor N/N_1 \rfloor$ RA preambles. The q th RA preamble occupies all DDREs whose Doppler domain index is between qN_1 and $qN_1 + (N_1 - 1)$ (both inclusive). For example, in Fig. 7.32 where $N_1 = 5$, the $q = 0$ th RA preamble occupies all DDREs whose Doppler domain index is between 0 and 4 (there are totally $M \times N_1 = 90$ such DDREs). In each RA preamble, energy is transmitted on only a single DDRE (see the shaded DDRE in each RA preamble). Let (k_q, l_q) denote the DD domain location of the shaded DDRE for the q th RA preamble. A UT which needs access to physical resource selects one of these R RA preambles at random and transmits it. For example, if a UT selects the q th RA preamble, then the DD domain signal transmitted by it is given by

$$x_q[k, l] = \begin{cases} \sqrt{MNE}, & k = k_q, l = l_q \\ 0, & (k, l) \neq (k_q, l_q). \end{cases} \quad (7.93)$$

Using ISFFT, this DD domain signal is transformed to a TF domain signal. This TF domain signal is then converted to the transmit TD signal $x_q(t)$ using the Heisenberg transform. The energy of $x_q(t)$ is $\int |x_q(t)|^2 dt = E$. At the BS receiver, the received TD signal is converted to the TF domain using

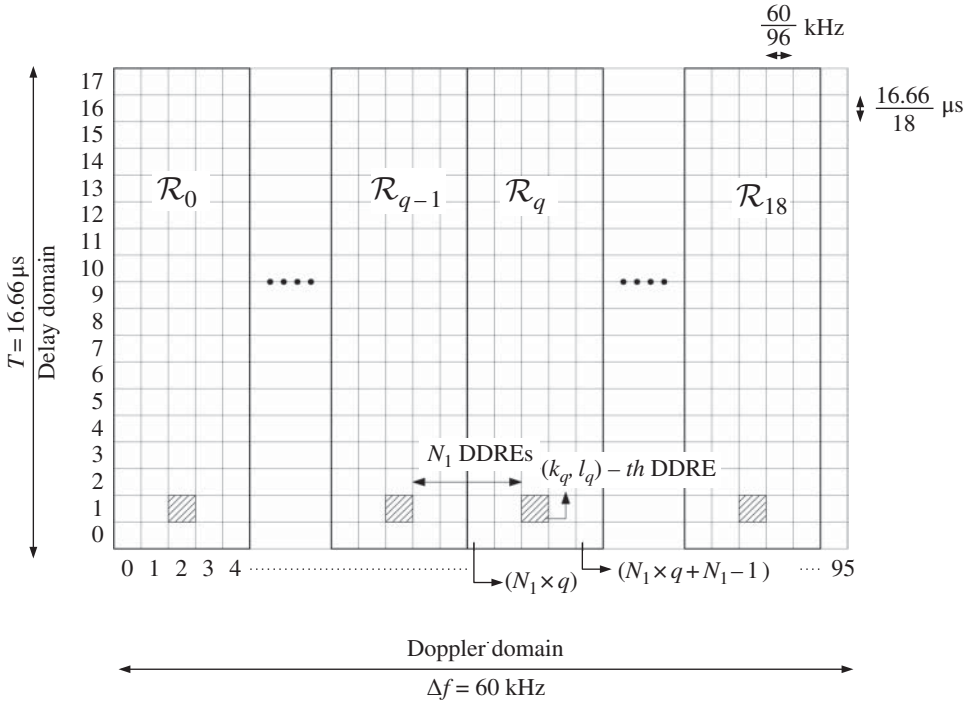


Figure 7.32 DD Domain RA preambles.

the Wigner transform, followed by SFFT which gives the received DD domain signal. For each RA preamble, the BS finds the largest energy received in any of the DDREs occupied by that RA preamble. If this maximum received energy is greater than a pre-determined threshold μ , then that RA preamble is said to be detected at the BS. For a detected RA preamble, the delay domain location of the highest energy DDRE is used to determine the round-trip propagation delay between the BS and the UT which transmitted that RA preamble.

For example, in Fig. 7.32, let the $q = 0$ th RA preamble be detected at the BS and let the location of the highest energy DDRE be $(k'', l'') = (3, 8)$. For the $q = 0$ th RA preamble, energy is transmitted only on the $(k_0, l_0) = (2, 1)$ DDRE, and, therefore, the channel delay is most likely equal to the shift in the delay domain location of the DDRE on which energy was transmitted. Hence, in this example, a good estimate of the propagation delay between the UT which transmitted the RA preamble and the BS is $(l'' - l_0)T/M = (8 - 1)T/M = 7/(M\Delta f)$ seconds, where $T/M = 1/(M\Delta f)$ is the width of each DDRE along the delay domain (also referred to as the delay domain resolution). Since $M\Delta f$ is the bandwidth of the RA preamble waveform and the delay-domain resolution is equal to the inverse bandwidth (i.e., $1/(M\Delta f)$), it is clear that the accuracy of the delay estimate cannot be better than half of the inverse bandwidth.

The design of the DD domain RA preamble (i.e., choice of M, N, T , and N_1) depends primarily on (i) the maximum round-trip propagation delay (denoted by τ_{rndTrip}), (ii) the maximum possible Doppler shift (v_{max}), and (iii) the total time (T_c) and bandwidth (B_c) resource allocated for the DD domain RA preambles. The OTFS modulation parameters M, N , and T for the DD domain RA preamble must be chosen in such a way that $M\Delta f \leq B_c$ and $NT \leq T_c$. Since delay is measured in integer multiples of the delay domain resolution, the maximum measurable propagation delay is

$(M - 1)T/M$. Also, since delay estimation accuracy is limited by the bandwidth of the RA preamble waveform, (M, T) is chosen such that $M\Delta f$ equals its largest possible value, i.e., B_c . Therefore, (M, T) is chosen in such a way that [157]

$$\frac{(M - 1)T}{M} \geq \tau_{\text{rndTrip}}, \quad \frac{M}{T} = B_c. \quad (7.94)$$

It is also shown in [157] that the peak-to-average power ratio (PAPR) of the TD transmit RA preamble is M , and, therefore, M should be sufficiently small. The smallest value of integer M which satisfies (7.94) and the corresponding choice of T are given by

$$M = 1 + \lceil \tau_{\text{rndTrip}} B_c \rceil, \quad T = \frac{1 + \lceil \tau_{\text{rndTrip}} B_c \rceil}{B_c}. \quad (7.95)$$

Regarding the choice of N , note that $NT \leq T_c$. Furthermore, if two or more UTs randomly select and transmit the same RA preamble, then it is likely that delay estimation accuracy would degrade for UTs which are far away from the BS and whose RA preamble is received at the BS with less energy. This event where two or more UTs transmit the same RA preamble is referred to as a collision event. To reduce the probability of occurrence collision events, the number of different RA preambles (i.e., $R = \lfloor N/N_1 \rfloor$) should be large. Hence, N should be the largest integer which satisfies $NT = N \left(\frac{1 + \lceil \tau_{\text{rndTrip}} B_c \rceil}{B_c} \right) \leq T_c$, i.e.,

$$N = \left\lfloor \frac{T_c}{T} \right\rfloor = \left\lfloor \frac{B_c T_c}{1 + \lceil \tau_{\text{rndTrip}} B_c \rceil} \right\rfloor. \quad (7.96)$$

As an example, the RA preamble design in Fig. 7.32 is for $B_c = 1.08$ MHz, $T_c = 1.6$ ms, and $\tau_{\text{rndTrip}} = 15 \mu\text{s}$. For these values, (7.95) and (7.96) give $M = 18$, $T = 16.66 \mu\text{s}$, and $N = 96$. Due to the Doppler shift, energy transmitted on the shaded DDRE of an RA preamble could be received in a DDRE which is occupied by/belongs to another RA preamble, leading to missed detection of the transmitted RA preamble and false detection of another RA preamble at the BS.

Consider another scenario, where two UTs (one near to the BS and another far away) transmit RA preambles which are adjacent to each other in the DD domain. At the BS receiver, the maximum energy DDRE received from the near UT could be shifted along the Doppler domain and could interfere strongly with the adjacent RA preamble transmitted by a faraway UT (which is received with much less energy). To avoid interference in such near-far situations, the RA preamble design parameter N_1 is chosen in such a way that it is greater than the sum of the largest possible Doppler domain shift of both the UTs. Since the largest possible Doppler shift is v_{max} , the largest possible Doppler domain shift for the energy received from some RA preamble (in number of DDREs along the Doppler domain) is $\left\lceil \frac{v_{\text{max}}}{\Delta f/N} \right\rceil = \lceil v_{\text{max}} NT \rceil$. Therefore, N_1 must be chosen to satisfy $N_1 \geq 2\lceil v_{\text{max}} NT \rceil$. However, due to fractional Doppler (i.e., channel path Doppler shift not being an integer multiple of the Doppler domain resolution $\Delta f/N$), in [157], it is recommended that N_1 satisfies

$$2\lceil v_{\text{max}} NT \rceil + 3 \leq N_1 \leq N. \quad (7.97)$$

To control the amount of interference leakage due to a received RA preamble from a near UT to an adjacent RA preamble received from a faraway UT, the received TF symbols at the BS can be windowed in time, i.e., the effective received TF domain symbols at the BS are given by $Y'[n, m] = Y[n, m] W[n]$, $n = 0, 1, \dots, N - 1$, $m = 0, 1, \dots, M - 1$, where $Y[n, m]$ is the output of the Wigner transform [157]. Windowing in time helps in achieving a trade-off between the Doppler domain width (in the number of DDREs along the Doppler domain) of the main lobe of the received RA

preamble energy and the maximum side lobe energy level. A larger main lobe width reduces the accuracy of the delay estimate, while a higher side lobe energy level results in more interference between adjacent RA preambles.

The performance of DD domain-based TA acquisition can be studied in terms of timing error probability (TEP) [157]. A timing error event is said to occur if, when an RA preamble is transmitted, either the energy received on all DDREs occupied by that RA preamble is below the threshold μ or the received energy on some DDRE is greater than threshold, but the estimated propagation delay does not lie in the interval between the minimum and maximum path delay between the BS and the UT which transmitted that RA preamble. On the other hand, a false alarm event is said to occur when an RA preamble is not transmitted, but the energy received on some DDRE occupied by that RA preamble is greater than the threshold μ . The probability of false alarm is denoted by P_{fa} . The energy threshold μ is chosen based on the desired P_{fa} and the AWGN power at the BS [157].

Simulations in [157] show that the TEP of DD domain-based RA preamble is robust to channel-induced Doppler shifts when compared to the ZC sequence-based RA preamble used in OFDM-based 4G-LTE systems. In Fig. 7.33, the TEP of DD domain-based RA preamble and that of 4G-LTE RA preamble are plotted as a function of transmit SNR for different values of the maximum Doppler shift $\nu_{\max}$. The transmit SNR is the ratio of the transmit power of an RA preamble to the AWGN power at the BS receiver. A single cell of radius $r_c = 1500$ m is considered with the BS at its center and no UT located within a distance of $r_a = 100$ m from the BS. The power delay profile is as per the ETU model [158] with nine channel paths between a UT and the BS. For a UT, the path delays are given by $\tau_q = [0, 50, 120, 200, 230, 500, 1600, 2300, 5000]$ ns (i.e., a delay spread of $5 \mu\text{s}$), where τ_q is the delay of the shortest path between that UT and the BS. The path loss exponent is 3, and the channel path gains are normalized in such a way that the sum of the absolute mean squared values of the channel path gains between a cell edge UT and the BS is unity, i.e., for an RA preamble transmitted by a cell-edge UT, the transmit SNR is the same as the received SNR at the BS. Rayleigh fading is considered, i.e., each complex channel path gain is modeled as a zero mean complex Gaussian random variable which is statistically independent of the other channel path gains. The Doppler shift of each channel path is $\nu_{\max} \cos(\theta)$, where θ is uniformly distributed in the interval $[0, 2\pi)$ and θ s corresponding to different paths are statistically independent of each other.

In Fig. 7.33, $B_c = 1.08$ MHz, $T_c = 1.6$ ms, and $\tau_{\text{rndTrip}} = 15 \mu\text{s}$ (time taken for an electromagnetic wave to travel 2×1500 m, plus the delay spread of $5 \mu\text{s}$). Hence, the OTFS modulation parameters are $M = 18$, $N = 96$, and $\Delta f = 60$ kHz. For the OTFS-based RA preamble, a Hamming window $W[\cdot]$ is used at the BS receiver. The number of simultaneous RA requests is Poisson distributed with mean $\lambda_c T_a \pi(r_c^2 - r_a^2)$, where $\lambda_c = 1$ call/s/sq. Km is the spatial density of the RA call rate, and $T_a = 10$ ms is the time duration between two consecutive RA preamble slots. In Fig. 7.33, the desired P_{fa} is fixed to 1×10^{-2} . From the figure, it is clear that in the absence of Doppler shift, i.e., $\nu_{\max} = 0$ Hz, the TEP performance of both OFDM (LTE) and OTFS-based RA preambles are close. However, with increasing $\nu_{\max}$, the TEP performance of OFDM (LTE)-based RA preamble deteriorates significantly whereas that of the OTFS-based RA preamble does not change as much.

Since $N_1 > 2 \lceil \nu_{\max} NT \rceil > 2\nu_{\max} NT$ (see (7.97)), it follows that the number of RA preambles $R = \left\lfloor \frac{N}{N_1} \right\rfloor < \frac{1}{2\nu_{\max} T}$. From (7.95), it is clear that $T > \tau_{\text{rndTrip}}$ and therefore $R < \frac{1}{2\nu_{\max} \tau_{\text{rndTrip}}}$. This shows that irrespective of the time duration and bandwidth of the RA preamble waveform, the maximum number of distinct RA preambles is upper bounded by $\frac{1}{2\nu_{\max} \tau_{\text{rndTrip}}}$ which determines a lower bound on TEP (due to probability of collisions). Therefore, in scenarios where the spatial RA call density

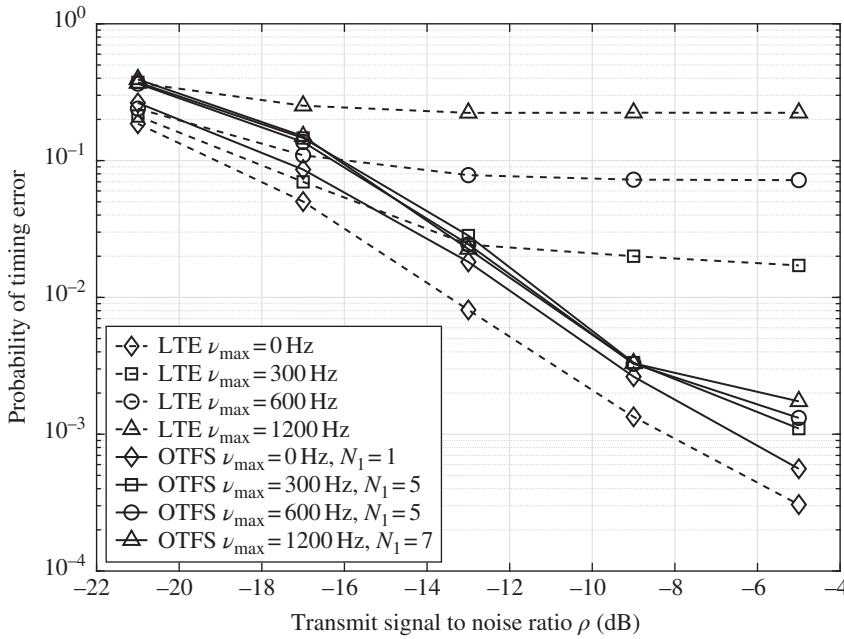


Figure 7.33 Timing error probability vs. transmit SNR.

λ_c is high and the maximum Doppler shift ($\nu_{\max}$) or the round-trip propagation delay (τ_{rndTrip}) is very high (e.g., extreme mobility or large cell size), the maximum possible number of RA preambles decreases as $2\nu_{\max} \tau_{\text{rndTrip}}$ increases. In such scenarios, the TEP performance of the DD domain RA preamble proposed in [157] may degrade due to increased probability of collisions. One solution to this problem is to have more frequent RA preamble slots so that the mean number of simultaneous RA call requests reduce. Another possibility is to allow each RA preamble to occupy all DDREs along the Doppler domain but only a contiguous portion of DDREs along the delay domain (of width at least τ_{rndTrip} seconds). With this approach, the number of RA preambles will be at most $\left\lfloor \frac{T}{\tau_{\text{rndTrip}}} \right\rfloor \leq \left\lfloor \frac{T_c}{\tau_{\text{rndTrip}}} \right\rfloor$. This upper bound on the number of distinct RA preambles is independent of the maximum Doppler shift. For scenarios where τ_{rndTrip} is very high, a sufficiently large T_c can be chosen.

7.5.2.3 Downlink Timing Synchronization in OTFS-Based Systems

In OFDM-based systems, the BS sends periodic synchronization signals in the downlink for synchronization (e.g., primary synchronization signal [PSS] and secondary synchronization signal [SSS] in OFDM-based 4G/5G systems) [151]. These signals also carry cell identity information. However, mobility-induced Doppler shift can lead to incorrect cell identity and timing acquisition. It was seen that DD domain-based uplink timing synchronization achieves lower TEP (i.e., better timing acquisition) compared to time domain/frequency domain-based synchronization in traditional OFDM-based systems. Therefore, downlink synchronization based on DD domain signals is also expected to be robust to mobility-induced Doppler shift. However, compared to uplink, an additional constraint in the design of DD domain downlink synchronization preambles/signals is that the number of distinct synchronization signals should be large due to the requirement of a

large number of cell identities (e.g., 3GPP 5G NR [159]). Therefore, the DD domain-based uplink RA preamble waveform design in [157] may not be directly applicable for downlink synchronization as the maximum number of allowed RA preambles is upper bounded by $\frac{1}{2 v_{\max} \tau_{\text{rndTrip}}}$ irrespective of the time duration and bandwidth of the waveforms. In [160], a DD domain-based downlink synchronization waveform has been proposed which is robust to mobility-induced Doppler shift and also supports a large number of cell identities.

8

Practical Considerations in OTFS Systems

This chapter deals with some of the key considerations in the implementation of orthogonal time frequency space (OTFS)-based communication systems. These considerations include practical pulse shaping in OTFS waveforms [83], peak-to-average-power-ratio (PAPR) of OTFS-modulated signals [161–168], impact of transceiver impairments on such as oscillator phase noise and IQ imbalance on the performance of OTFS modulation [169, 170], and Zak OTFS implementation using the discrete Zak transform (DZT) [64, 65]. Early works on OTFS have assumed ideal pulses which are not practically feasible, although they provide performance bounds and fundamental insights.¹ Moving beyond the assumption of ideal pulses, practical rectangular pulse shapes in orthogonal frequency division multiplexing (OFDM)-based OTFS have been considered in later works [171, 172]. OTFS signaling using arbitrary practical pulse shaping waveforms that are applied to the time domain signal has also been considered [83], where input–output relations for OTFS with arbitrary pulse shaping waveforms have been derived. These input–output relations have aided the development of receiver algorithms for OTFS with practical pulse shapes and their performance evaluation. Other than rectangular pulses, some of the practical pulse shaping waveforms of interest include sinc, raised cosine (RC), Gaussian, and prolate spheroidal waveforms. The choice of different pulse shapes offers different degrees of tradeoff between out-of-band emission in the spectral domain and bit error performance. The time duration and bandwidth of OTFS signals are generally large in order to guarantee channel sparsity in the DD domain, which allows joint detection of information symbols at low complexities. Due to the large time duration and bandwidth, the OTFS-modulated signal in the time domain is a sum of a large number of carrier waveforms which increases the PAPR. The maximum PAPR of an OTFS-modulated signal depends only on the time duration of the OTFS frame and increases with increasing time duration. When the time duration is not too large, the PAPR of an OTFS-modulated signal can be smaller than that of multi-carrier modulation like OFDM and generalized FDM (GFDM). A high PAPR necessitates the use of power amplifiers with good linearity and/or high backoffs in order to reduce output signal distortion due to non-linearity. This leads to reduced power efficiency. To overcome this, PAPR reduction techniques are often employed. Clipping and companding are examples of well-known techniques to reduce PAPR. However, PAPR reduction schemes often result in some loss in bit error performance. This chapter discusses the PAPR characteristics of OTFS-modulated signals and points to some of the PAPR reduction schemes investigated in the literature.

¹ See (3.79) of Corollary 3.5. See also the discussion in the paragraph just after this corollary but before Theorem 3.12.

Practical transceivers are characterized by several impairments, including oscillator phase noise, sampling jitter, IQ imbalance (IQI) in transmit and receive chains, carrier frequency offset (CFO), sampling frequency offset (SFO), and device non-linearities [173–175]. This chapter discusses the impact of some of the key transceiver impairments on the performance of OTFS modulation. In transceivers, oscillators are used to generate RF carrier frequencies for the purpose of RF up conversion and down conversion. Complementary metal-oxide semiconductor (CMOS) technology is a naturally preferred technology for digital very large-scale integration (VLSI) implementations, but not for RF devices. However, CMOS technology is being adopted for RF system-on-chip (SOC) solutions (e.g., in smartphones), in order to take advantage of its cost, size, low-power benefits. In the process, the RF device specifications are compromised. For example, carrier oscillators implemented using CMOS technology often possess high-phase noise, which is detrimental to performance. Likewise, another impairment arises when the orthogonality between the I and Q arms in the transmit/receive chain is compromised when there are gain and phase imbalances between the two arms due to imperfections in implementation. Therefore, modeling, estimation, and compensation of such impairments are important in transceiver implementations. This chapter discusses the effect of oscillator phase noise and transmit/receive IQI on the performance of OTFS. This chapter also introduces a DZT-based implementation of OTFS, which has implementation simplicity and performance advantages.

8.1 PAPR of MC-OTFS Signals

PAPR characteristics of any modulation waveform is important as it influences the linearity requirement of the power amplifier (PA) in transceivers. A higher PAPR of the modulated signal demands a larger backoff in the PA, leading to reduced power efficiency. Hence, the PAPR characterization of modulation waveforms is of interest. This section presents the PAPR characterization of the MC-OTFS signal and techniques to reduce its PAPR [161–163].

MC-OTFS modulation has been discussed in Chapter 3. The delay and Doppler period τ_p and $\nu_p = 1/\tau_p$ in Chapter 3 have been denoted by T and $\Delta f = 1/T$, respectively, in this section. As in Chapter 3, here also the total time duration and bandwidth of the MC-OTFS-modulated signal is $NT = N\tau_p$ and $M\Delta f = M\nu_p$, respectively. Also, the (k, l) th DD domain information symbol $x[k, l]$ is carried by the MC pulsone $x_{\mathcal{H}, \frac{l_p}{M}, \frac{k_p}{N}}^{w_{\text{tx}}, g_{\text{tx}}}(t)$ (see (3.30), (3.31) and (3.41)). As discussed in Chapter 3, there are two transceiver architectures for MC-OTFS modulation, one based on the two-step approach, and the other based on the Zak transform approach. In this section, the two-step approach has been used for MC-OTFS transceivers (see Sections 3.4.1 and 3.5.1). Specifically, in this section, the simplified two-step MC-OTFS transmitter and receiver illustrated through Figs. 3.11 and 3.13, respectively, have been used. Also, in terms of notation, in Chapter 3, k and l denote the discrete delay and Doppler domain index, respectively, whereas here in this section, it is the opposite, i.e., k denotes the Doppler domain index and l denotes the delay domain index.

At the MC-OTFS transmitter, the information symbols are treated as points in the DD grid and are mapped to the TF plane through ISFFT. The TF signal so obtained is passed through a multicarrier modulation system. The TF signal is transformed to a time domain signal for transmission using the Heisenberg transform. The output of the Heisenberg transform is transmitted over the doubly selective channel. At the receiver, the received time domain signal is transformed to the TF domain using the Wigner transform, which is mapped back to the DD domain using symplectic finite Fourier transform (SFFT). The time domain signal at the transmitter is amplified using a power

amplifier before it is transmitted through the wireless channel. The expression for the discrete-time samples of the MC-OTFS signal transmitted through the channel is obtained as follows.

8.1.1 MC-OTFS Transmit Signal

The information symbols denoted by $x[k, l] \in \mathbb{A}$, $k = 0, 1, \dots, N-1$, $l = 0, 1, \dots, M-1$, where $\mathbb{A}$ is the modulation alphabet (e.g., QAM/PSK), are treated as points in the DD grid. The transmitter maps these symbols in the DD domain to the TF domain using ISFFT as

$$X[n, m] = \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] e^{-j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)}. \quad (8.1)$$

The TF signal obtained as the output of ISFFT operation is converted to the time domain using the Heisenberg transform for transmission as

$$s(t) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] g_{\text{tx}}(t - nT) e^{j2\pi m \Delta f (t - nT)}, \quad (8.2)$$

where $g_{\text{tx}}(t)$ is the transmit periodic prototype pulse shape of duration NT as in [176]. The discrete time representation of (8.2) can be obtained by sampling (8.2) with a sampling rate $F_s = \frac{1}{T_s} = B$ (Nyquist sampling) as

$$s(uT_s) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X[n, m] g_{\text{tx}}(uT_s - nT) e^{j2\pi m \Delta f (uT_s - nT)}, \quad (8.3)$$

where $u = 0, 1, \dots, MN-1$. In the analysis that follow, Nyquist sampling (oversampling ratio = 1) is considered. In the simulation results presented later, oversampling ratio more than one is considered. It is convenient to express the samples $s(u)$, $u = 0, 1, \dots, MN-1$, as an $N \times M$ matrix with entries denoted by $s(q, r)$ such that $u = r + qM$, where $r = 0, 1, \dots, M-1$ and $q = 0, 1, \dots, N-1$. Substituting $u = r + qM$ and (8.1) in (8.3) gives

$$s(r + qM) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] e^{-j2\pi \left(\frac{ml}{M} - \frac{nk}{N} \right)} g_{\text{tx}}([r + qM - nM]_{MN}) e^{j\frac{2\pi}{M} m(r + qM)}. \quad (8.4)$$

Here, mod- MN operation, denoted by $[\cdot]_{MN}$, confines the frame transmission within the duration NT [176]. Furthermore, (8.4) can be simplified as

$$s(r + qM) = \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] \sum_{m=0}^{M-1} e^{\frac{j2\pi m}{M} (-l+r)} \sum_{n=0}^{N-1} g_{\text{tx}}([r + qM - nM]_{MN}) e^{\frac{j2\pi nk}{N}}. \quad (8.5)$$

Now, defining $\mathcal{G} \triangleq \sum_{m=0}^{M-1} e^{\frac{j2\pi m}{M} (-l+r)}$,

$$\mathcal{G} = \begin{cases} 0 & \text{if } r \neq l \\ M & \text{if } r = l. \end{cases} \quad (8.6)$$

Substituting (8.6) in (8.5) gives

$$s(r + qM) = M \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} x[k, r] e^{\frac{j2\pi nk}{N}} g_{\text{tx}}([r + qM - nM]_{MN}). \quad (8.7)$$

It is observed that $\sum_{k=0}^{N-1} x[k, r] e^{\frac{j2\pi nk}{N}}$ is the n th ($n = 0, 1, \dots, N-1$) N -point DFT of $x[k, r]$, $k = 0, 1, \dots, N-1$, for a given r . Denoting $\tilde{x}_r[n] = \sum_{k=0}^{N-1} x[k, r] e^{\frac{j2\pi nk}{N}}$, (8.7) can be written as

$$s(r + qM) = M \sum_{n=0}^{N-1} \tilde{x}_r[n] g_{\text{tx}}([r + qM - nM]_{MN}). \quad (8.8)$$

8.1.2 Upper Bound on PAPR

Consider the samples of the transmitted MC-OTFS signal in (8.8). The PAPR of the discrete-time samples of one frame of the MC-OTFS transmit signal is defined as

$$\text{PAPR} = \frac{\max_{r,q} \{|s(r + qM)|^2\}}{P_{\text{avg}}}, \quad (8.9)$$

where $P_{\text{avg}} = \frac{1}{MN} \sum_{r=0}^{M-1} \sum_{q=0}^{N-1} \mathbb{E}\{|s(r + qM)|^2\}$. From (8.8), the numerator in (8.9) can be written as

$$\max_{r,q} |s(r + qM)|^2 = M^2 \max_{r,q} \left| \sum_{n=0}^{N-1} \tilde{x}_r[n] g_{\text{tx}}([r + qM - nM]_{MN}) \right|^2. \quad (8.10)$$

Using the Cauchy–Schwartz inequality, (8.10) can be bounded as

$$\max_{r,q} |s(r + qM)|^2 \leq M^2 \max_{r,q} \sum_{n=0}^{N-1} |\tilde{x}_r[n]|^2 \cdot \max_{r,q} \sum_{n=0}^{N-1} |g_{\text{tx}}([r + qM - nM]_{MN})|^2. \quad (8.11)$$

Recalling that $\tilde{x}_r[n] = \sum_{k=0}^{N-1} x[k, r] e^{\frac{j2\pi nk}{N}}$ for a given r and using Parseval's theorem gives

$$\max_r \sum_{n=0}^{N-1} |\tilde{x}_r[n]|^2 \leq N^2 \max_{k,l} |x[k, l]|^2. \quad (8.12)$$

Substituting (8.12) in (8.11) gives

$$\max_{r,q} |s(r + qM)|^2 \leq M^2 N^2 \max_{k,l} |x[k, l]|^2 \cdot \max_{r,q} \sum_{n=0}^{N-1} |g_{\text{tx}}([r + qM - nM]_{MN})|^2. \quad (8.13)$$

Defining $B_1 \triangleq \max_{r,q} \sum_{n=0}^{N-1} |g_{\text{tx}}([r + qM - nM]_{MN})|^2$, (8.13) can be written as

$$\max_{r,q} |s(r + qM)|^2 \leq M^2 N^2 \max_{k,l} |x[k, l]|^2 B_1. \quad (8.14)$$

Now, the denominator in (8.9) can be written as

$$P_{\text{avg}} = \frac{M^2}{MN} \sum_{r=0}^{M-1} \sum_{q=0}^{N-1} \mathbb{E} \left\{ \left| \sum_{n=0}^{N-1} \tilde{x}_r[n] g_{\text{tx}}([r + qM - nM]_{MN}) \right|^2 \right\}. \quad (8.15)$$

It is noted that for a given value of r , $\tilde{x}_r[n]$ s are the IDFT values of $x[k, r]$, $k = 0, 1, \dots, N - 1$. Since $x[k, l]$ s are information symbols from a modulation alphabet $\mathbb{A}$, they can be treated i.i.d. with zero mean and variance $\sigma_a^2 = \mathbb{E}\{|x[k, l]|^2\}$. Also, due to the Nyquist sampling of the transmit signal, the samples $\tilde{x}_r[n]$ s are mutually uncorrelated, and hence, (8.15) can be simplified as

$$P_{\text{avg}} = \frac{M^2}{MN} \sum_{r=0}^{M-1} \sum_{q=0}^{N-1} \sum_{n=0}^{N-1} \mathbb{E}\{|\tilde{x}_r[n]|^2\} |g_{\text{tx}}([r + qM - nM]_{MN})|^2. \quad (8.16)$$

It is observed that $\mathbb{E}\{|\tilde{x}_r[n]|^2\} = N\sigma_a^2$, $\forall r, n$. With this, (8.16) can be further simplified as

$$P_{\text{avg}} = \frac{M^2 N \sigma_a^2}{MN} \sum_{n=0}^{N-1} \sum_{r=0}^{M-1} \sum_{q=0}^{N-1} |g_{\text{tx}}([r + qM - nM]_{MN})|^2. \quad (8.17)$$

Defining $B_2 \triangleq \frac{1}{MN} \sum_{n=0}^{N-1} \sum_{r=0}^{M-1} \sum_{q=0}^{N-1} |g_{\text{tx}}([r + qM - nM]_{MN})|^2$, the PAPR of the MC-OTFS transmit signal can be upper bounded as

$$\begin{aligned} \text{PAPR} &\leq \frac{M^2 N^2 \max_{k,l} |x[k, l]|^2 B_1}{M^2 N \sigma_a^2 B_2} \\ &= \frac{N \max_{k,l} |x[k, l]|^2 B_1}{\sigma_a^2 B_2}. \end{aligned} \quad (8.18)$$

It is noted that (8.18) is a bound on the PAPR of transmit signal in MC-OTFS with any pulse shape $g_{\text{tx}}(t)$. Now, consider MC-OTFS with a rectangular pulse shape. In this case, $B_1/B_2 = 1$. Hence, the maximum PAPR will meet the bound in (8.18), i.e.,

$$\text{PAPR}_{\max} = \frac{N \max_{c \in \mathbb{A}} |c|^2}{\sigma_a^2}. \quad (8.19)$$

It is noted that the upper bound on the PAPR of the MC-OTFS transmit signal grows linearly with N and not with M . This is an interesting feature of MC-OTFS compared to other multicarrier modulations where the PAPR depends on the number of subcarriers (M) and grows linearly with the number of subcarriers used. In practice, the number of delay bins/subcarriers (M) is chosen such that $\Delta f = B/M$ satisfies $\nu_{\max} < \Delta f < 1/\tau_{\max}$, where $\tau_{\max}$ and $\nu_{\max}$ denote the maximum delay and Doppler spreads of the channel, respectively. The value of N decides the Doppler resolution, and it is generally chosen such that the maximum latency constraint of the application is satisfied (larger N results in larger decoding delay). Hence, in practice, N is less than M .

8.1.3 CCDF of PAPR

While the above analysis characterizes the maximum PAPR, the characterization of the complementary cumulative distribution function (CCDF) of the PAPR is important. For the ease of analysis, consider MC-OTFS with a rectangular pulse. From (8.8), it can be seen that the time domain samples of the MC-OTFS transmit signal with a rectangular pulse are nothing but the N -point IDFT values of the symbols in the DD domain. Hence, if N is large, then by the central limit theorem, the transmitted samples can be approximated to have complex Gaussian distribution with zero mean [7]. Therefore, the instantaneous envelope $|s[u]|$ is Rayleigh distributed, and hence, the instantaneous-to-average power ratio ($\text{IAPR} = |s[u]|^2 / \mathbb{E}\{|s|^2\}$) of each time domain sample follows exponential distribution. Therefore, the probability that the IAPR does not exceed a threshold γ_0 is given by

$$P(\text{IAPR} \leq \gamma_0) \approx (1 - e^{-\gamma_0}). \quad (8.20)$$

Assuming the transmitted samples to be mutually uncorrelated, which is true for Nyquist sampling (oversampling ratio = 1), the probability that the PAPR of the transmit MC-OTFS signal in a frame does not exceed γ_0 is given by

$$P(\text{PAPR} \leq \gamma_0) \approx \prod_{i=0}^{NM-1} (1 - e^{-\gamma_0}) = (1 - e^{-\gamma_0})^{MN}. \quad (8.21)$$

It is noted that the assumption that the transmitted samples are uncorrelated does not hold when the transmit signal is oversampled [177]. Now, the CCDF of the PAPR is given by

$$P(\text{PAPR} > \gamma_0) \approx 1 - (1 - e^{-\gamma_0})^{MN}. \quad (8.22)$$

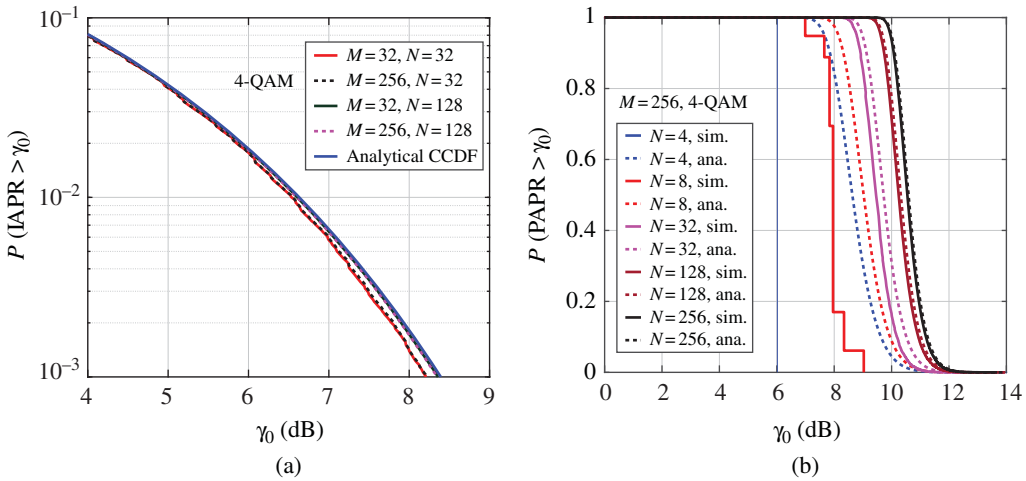


Figure 8.1 Analytical and simulated CCDF of IAPR and PAPR of MC-OTFS. (a) CCDF of IAPR in OTFS and (b) CCDF of PAPR in OTFS.

It is noted that the PAPR in OTFS is bounded by $\text{PAPR}_{\max}$ derived in the above, i.e., $P(\text{PAPR} > \gamma_0) = 0$ for $\gamma_0 > \text{PAPR}_{\max}$. However, the CCDF in (8.22) does not capture this. Hence, (8.22) is an approximation to the actual CCDF. The actual CCDF of PAPR converges to the CCDF in (8.22) as $N \rightarrow \infty$.

8.1.4 Results

Figure 8.1(a) shows the simulated CCDF of IAPR of the transmitted MC-OTFS signal with a rectangular pulse and Nyquist sampling (oversampling ratio = 1) using 4-QAM for $M = 32$ and 256 and $N = 32$ and 128. The analytical CCDF of IAPR using (8.20) is also plotted. From Fig. 8.1(a), it is observed that the simulated CCDF gets closer to the analytical CCDF for higher values of N , indicating that the transmitted samples attain Gaussian distribution as N grows. It is noted that the CCDF of IAPR does not depend on M . Figure 8.1(b) shows the CCDF of PAPR for MC-OTFS with a rectangular pulse for systems with $M = 256$ and $N = 4, 8, 32, 128$, and 256. All the systems use 4-QAM modulation and Nyquist sampling. The simulated CCDF of PAPR along with the analytical CCDF in (8.22) for all the systems is plotted. It is seen that the simulated CCDF matches very closely with the analytical CCDF in (8.22) for higher values of N , indicating that the analytical CCDF is more accurate in the large N regime. It is also observed that the systems with a smaller value of N have lower PAPR and the PAPR increases with the increase in the value of N . Hence, it is evident that the maximum PAPR in MC-OTFS grows with N , as inferred from the upper bound.

8.1.4.1 Effect of Increasing M and N on the CCDF of PAPR

Figure 8.2 shows the CCDF of the PAPR of MC-OTFS with $M = 32, 64$, and 128 for $N = 4$ and 8. All the systems use 4-QAM and Nyquist sampling. As observed earlier in the analysis, $\text{PAPR}_{\max}$ depends only on N and grows linearly with N . That is, the maximum PAPR for $N = 4$ and 8 are $10 \log 4 = 6.02$ dB and $10 \log 8 = 9.03$ dB, respectively. From Fig. 8.2, it can be observed that the CCDF is zero for $\gamma_0 > 6.02$ dB for $N = 4$ and $\gamma_0 > 9.03$ dB for $N = 8$. Also, with the increase in the value of M , the probability to have large peaks increases, and hence, the CCDF of PAPR increases, as indicated by (8.22). Therefore, although increasing M increases the CCDF, the maximum PAPR depends only on N and $P(\text{PAPR} > \gamma_0) = 0$ for $\gamma_0 > \text{PAPR}_{\max}$.

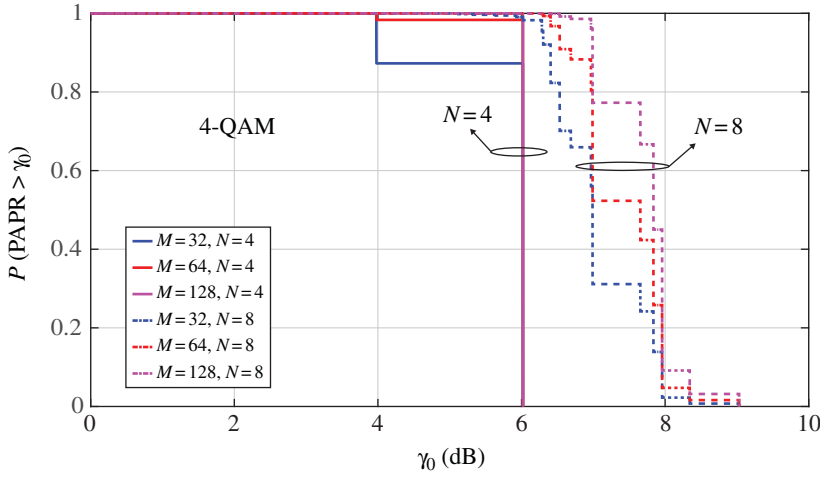


Figure 8.2 Effect of M and N on the CCDF of PAPR in MC-OTFS.

8.1.4.2 Effect of Pulse Shaping on PAPR

Figure 8.3 shows the effect of pulse shape on the PAPR of MC-OTFS. It shows a comparison of the CCDF of PAPR with $M = 256$, $N = 8$ with (i) a rectangular pulse defined as $\text{rect}(t/T)$, (ii) an RC pulse defined as $\frac{\text{sinc}(t/T) \cos(\pi a t/T)}{(1-4a^2(t/T)^2)}$ with $a = 0.5$, and (iii) a Gaussian pulse defined as $e^{-a\pi(t/T)^2}$ with $a = 1$, for $-NT/2 \leq t < NT/2$. All the pulses are sampled at a rate $1/T_s$ to obtain MN discrete time samples which are normalized to have unit energy. All the systems use 16-QAM and are four times oversampled using an RC filter of length $6T$ with a roll-off factor of 0.5. From Fig. 8.3, it is observed that the PAPR of MC-OTFS with pulse shaping can increase as indicated by the upper bound in (8.18). It is seen that MC-OTFS with the rectangular pulse has a better PAPR compared to that of the RC pulse, which, in turn, has a better PAPR compared to the Gaussian pulse.

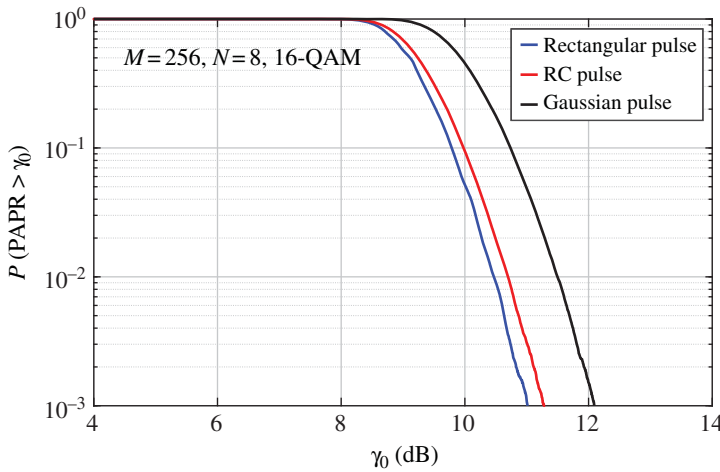


Figure 8.3 CCDF of the PAPR of MC-OTFS for different pulse shapes.

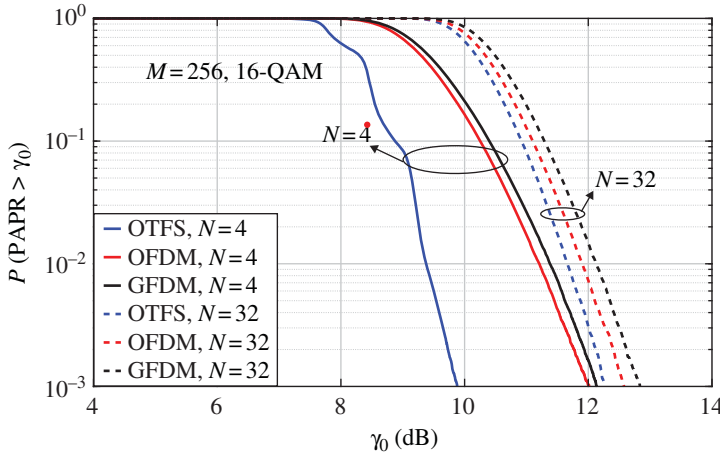


Figure 8.4 Comparison of the CCDF of the PAPR of MC-OTFS with those of OFDM and GFDM with 16-QAM.

8.1.4.3 Comparison with GFDM and OFDM

GFDM is a block-based TF multicarrier transmission scheme that offers flexible pulse shaping of individual subcarriers [178]. The transmit waveform of GFDM follows (8.3) in which $X[n, m]$ s are the information symbols from the modulation alphabet $\mathbb{A}$. OFDM can be viewed as a special case of GFDM where the pulse shape is rectangular and $N = 1$. In Fig. 8.4, a comparison is made between the CCDF of the PAPR of MC-OTFS with those of GFDM and OFDM systems with $M = 256, N = 4$ and $M = 256, N = 32$. All the systems use 16-QAM, and an oversampling ratio of four is used. For all the considered systems, oversampling is done using an RC filter of length $6T$ and a roll-off factor of 0.5. It is noted that, for the comparison of M -subcarrier OFDM with MC-OTFS and GFDM, which have MN symbols in a frame, the CCDF of the concatenation of N OFDM symbols is considered. From Fig. 8.4, it is observed that, for the same number of subcarriers, OFDM has a slightly better PAPR compared to GFDM, as was also shown in [179] and [180]. It is also observed that MC-OTFS with a rectangular pulse can have better PAPR compared to OFDM and GFDM (using RC pulse shaping with roll-off factor 0.5) systems. For example, MC-OTFS with $N = 4$ has a PAPR which is approximately 2.2 dB lower than that of OFDM and approximately 2.4 dB lower than that of GFDM at a probability of 10^{-3} . However, increasing N to 32 shifts the CCDF of MC-OTFS closer to those of GFDM and OFDM, and the gain in PAPR compared to that of OFDM reduces to 0.3 dB at a probability of 10^{-3} . Therefore, as seen in the analysis, the maximum PAPR in MC-OTFS increases with N , and hence, MC-OTFS can have better PAPR compared to OFDM and GFDM when $N < M$. This demonstrates the good PAPR characteristics of the MC-OTFS waveform. The PAPR of MC-OTFS can be reduced further using precoding of information symbols prior to OTFS modulation (e.g., DFT-precoded OTFS), iterative clipping and filtering, and deep neural network-based autoencoders [163–167].

8.2 Impact of Phase Noise on OTFS

In communication transceivers, carrier signals used for up/down conversion are generated using oscillators whose outputs are contaminated by random phase noise. The variance of oscillator phase noise increases with increasing carrier frequency. The demand for high data rates and the

availability of large amount of spectrum in mmWave frequency bands have motivated the design of high data rate wireless systems that operate at mmWave frequencies [181, 182]. However, there are several challenges associated with mmWave communication. For example, the use of mmWave carrier frequencies results in high Doppler spreads even in low/medium mobility scenarios, making mmWave systems more vulnerable to mobility. Another challenge is the high oscillator phase noise in mmWave frequencies, which is detrimental to performance.

In the following, the OTFS system model is discussed in both the absence and presence of phase noise. MC-OTFS modulation discussed in Chapter 3 is considered here. A discrete-time formulation of OTFS modulation in [171] where OFDM is used as the inner TF signal modulation block is considered.

8.2.1 System Model Without Phase Noise

The transmitter maps the information symbols in the DD domain, $x_{\text{DD}}[k, l] \in \mathbb{A}$, $k = 0, \dots, M-1$, $l = 0, \dots, N-1$, to the TF domain through ISFFT operation as

$$X_{\text{TF}}[m, n] = \frac{1}{\sqrt{MN}} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x_{\text{DD}}[k, l] e^{-j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)}. \quad (8.23)$$

Let $\mathbf{X}_{\text{DD}} \in \mathbb{A}^{M \times N}$ denote the matrix with information symbols $x_{\text{DD}}[k, l]$ in the DD domain and $\mathbf{X}_{\text{TF}}$ denote the $M \times N$ matrix with entries $X_{\text{TF}}[m, n]$, $m = 0, \dots, M-1$, $n = 0, \dots, N-1$ in the TF domain. A closer look at (8.23) reveals that ISFFT of $\mathbf{X}_{\text{DD}}$ is equivalent to computing M -point DFT along the columns of $\mathbf{X}_{\text{DD}}$ and N -point IDFT along the rows of $\mathbf{X}_{\text{DD}}$. Denoting the M -point DFT matrix by $\mathbf{F}_M$ and N -point IDFT matrix by $\mathbf{F}_N^H$, (8.23) can be written as [171]

$$\mathbf{X}_{\text{TF}} = \mathbf{F}_M \mathbf{X}_{\text{DD}} \mathbf{F}_N^H. \quad (8.24)$$

The matrix operation in (8.24) can be vectorized as

$$\mathbf{x}_{\text{TF}} = (\mathbf{F}_N^H \otimes \mathbf{F}_M) \mathbf{x}_{\text{DD}}, \quad (8.25)$$

where $\mathbf{x}_{\text{TF}} = \text{vec}(\mathbf{X}_{\text{TF}})$, $\mathbf{x}_{\text{DD}} = \text{vec}(\mathbf{X}_{\text{DD}})$, and $\otimes$ denotes the Kronecker product operation. The signal vector $\mathbf{x}_{\text{TF}} \in \mathbb{C}^{MN \times 1}$ is then partitioned into N blocks, each of length M , denoted by $\mathbf{x}_{\text{TF}}^{(n)}$, $n = 0, \dots, N-1$. Each block $\mathbf{x}_{\text{TF}}^{(n)}$ of length M is then fed to an OFDM modulator with M subcarriers. The OFDM modulator multiplies each block with an M -point IDFT matrix $\mathbf{F}_M^H$ and adds cyclic prefix (CP) to each block. If L_{cp} denotes the length of the CP used, then the length of each block after cyclic prefixing will be $L = M + L_{\text{cp}}$. Let $\mathbf{I}_{\text{cp}} = [\mathbf{A}_{\text{cp}}^T \mathbf{I}_M]^T$ denote the $L \times M$ CP insertion matrix, where $\mathbf{A}_{\text{cp}}$ is the matrix with last L_{cp} rows of the identity matrix $\mathbf{I}_M$. These operations are performed on each of the N blocks as

$$\tilde{\mathbf{x}}_{\text{TF}} = (\mathbf{I}_N \otimes \mathbf{I}_{\text{cp}}) (\mathbf{I}_N \otimes \mathbf{F}_M^H) \mathbf{x}_{\text{TF}}. \quad (8.26)$$

The vector $\tilde{\mathbf{x}}_{\text{TF}}$ is transmitted through the linear time-varying (LTV) channel, and the received signal vector at the receiver is given by

$$\tilde{\mathbf{y}}_{\text{TF}} = \mathbf{H}_{\text{TV}} \tilde{\mathbf{x}}_{\text{TF}} + \mathbf{w}, \quad (8.27)$$

where $\mathbf{H}_{\text{TV}} \in \mathbb{C}^{NL \times NL}$ denotes the channel matrix and $\mathbf{w} \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ is the noise vector. Assuming that the channel has P paths and denoting the channel coefficient, delay, and Doppler associated with the i th path by h_i , τ_i , and ν_i , respectively, the entries of $\mathbf{H}_{\text{TV}}$ are given by [36, 39]

$$h_{\text{TV}}(\tau, n) = \sum_{i=1}^P h_i e^{j \frac{2\pi \nu_i n}{M \Delta f}} \delta(\tau - \tau_i). \quad (8.28)$$

At the receiver, the received vector is partitioned into N blocks of length L , and the CP is removed from each block. Let $\mathbf{R}_{\text{cp}} = [\mathbf{0}_{M \times L_{\text{cp}}} \mathbf{I}_M]$ denote the $M \times L$ matrix that removes the CP from each OFDM symbol. This is followed by an M -point DFT on each block. CP removal and DFT operations on each of the N blocks can be vectorized as

$$\mathbf{y}_{\text{TF}} = (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \tilde{\mathbf{y}}_{\text{TF}}. \quad (8.29)$$

Now, let $\mathbf{Y}_{\text{TF}}$ denote the $M \times N$ matrix with entries $Y_{\text{TF}}[m, n]$, $m = 0, \dots, M-1$, $n = 0, \dots, N-1$ in the TF domain, such that $\mathbf{y}_{\text{TF}} = \text{vec}(\mathbf{Y}_{\text{TF}})$. An SFFT operation is performed on $\mathbf{Y}_{\text{TF}}$ that maps the points $Y_{\text{TF}}[m, n]$ in the TF domain to points in the DD domain, denoted by $y_{\text{DD}}[k, l]$, given by

$$y_{\text{DD}}[k, l] = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y_{\text{TF}}[m, n] e^{j2\pi \left(\frac{mk}{M} - \frac{nl}{N} \right)}. \quad (8.30)$$

The SFFT of $\mathbf{Y}_{\text{TF}}$ is equivalent to computing N -point DFT along the rows of $\mathbf{Y}_{\text{TF}}$ and M -point IDFT along the columns of $\mathbf{Y}_{\text{TF}}$. Hence, (8.30) can be written as

$$\mathbf{Y}_{\text{DD}} = \mathbf{F}_M^H \mathbf{Y}_{\text{TF}} \mathbf{F}_N. \quad (8.31)$$

With $\mathbf{y}_{\text{DD}} = \text{vec}(\mathbf{Y}_{\text{DD}})$, (8.31) can be vectorized as

$$\mathbf{y}_{\text{DD}} = (\mathbf{F}_N \otimes \mathbf{F}_M^H) \mathbf{y}_{\text{TF}}. \quad (8.32)$$

Using (8.25) to (8.29) in (8.32), the end-to-end system model is given by

$$\begin{aligned} \mathbf{y}_{\text{DD}} &= (\mathbf{F}_N \otimes \mathbf{F}_M^H) (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{H}_{\text{TV}} (\mathbf{I}_N \otimes \mathbf{I}_{\text{cp}}) \\ &\quad \cdot (\mathbf{I}_N \otimes \mathbf{F}_M^H) (\mathbf{F}_N^H \otimes \mathbf{F}_M) \mathbf{x}_{\text{DD}} + \hat{\mathbf{w}}. \end{aligned} \quad (8.33)$$

It is noted that $\hat{\mathbf{w}} = (\mathbf{F}_N \otimes \mathbf{F}_M^H) (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{w}$ has the same statistics as $\mathbf{w}$. From (8.33), the effective channel matrix in the DD domain is given by

$$\mathbf{H}_{\text{DD}} = (\mathbf{F}_N \otimes \mathbf{F}_M^H) (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{H}_{\text{TV}} (\mathbf{I}_N \otimes \mathbf{I}_{\text{cp}}) \cdot (\mathbf{I}_N \otimes \mathbf{F}_M^H) (\mathbf{F}_N^H \otimes \mathbf{F}_M). \quad (8.34)$$

If $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are square matrices, then $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC} \otimes \mathbf{BD})$. Using this property, (8.34) can be simplified as

$$\mathbf{H}_{\text{DD}} = (\mathbf{F}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{H}_{\text{TV}} (\mathbf{F}_N^H \otimes \mathbf{I}_{\text{cp}}). \quad (8.35)$$

The input–output relation in (8.33) can then be expressed as

$$\mathbf{y}_{\text{DD}} = \mathbf{H}_{\text{DD}} \mathbf{x}_{\text{DD}} + \hat{\mathbf{w}}. \quad (8.36)$$

8.2.2 System Model with Phase Noise

The phase noise model of the carrier oscillator in the transmitter/receiver can be incorporated in the end-to-end system model in (8.33) as follows. The transmitter oscillator output with carrier frequency f_c and phase noise process $\theta_{\text{tx}}(t)$ can be modeled as [175]

$$s_{\text{tx}}(t) = e^{j2\pi f_c t + \theta_{\text{tx}}(t)}, \quad (8.37)$$

whose discrete-time baseband model can be written as

$$s_{\text{tx}}(n) = e^{j\theta_{\text{tx}}(n)}. \quad (8.38)$$

Similarly, the discrete-time baseband model for the receiver oscillator output with phase noise is given by $s_{\text{rx}}(n) = e^{j\theta_{\text{rx}}(n)}$. The received signal in (8.33) affected by transmitter and receiver oscillator phase noise can then be written in the form

$$\tilde{\mathbf{y}}_{\text{DD}} = (\mathbf{F}_N \otimes \mathbf{F}_M^H) (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{\Theta}_{\text{rx}} \mathbf{H}_{\text{TV}} \mathbf{\Theta}_{\text{tx}} (\mathbf{I}_N \otimes \mathbf{I}_{\text{cp}}) \cdot (\mathbf{I}_N \otimes \mathbf{F}_M^H) (\mathbf{F}_N^H \otimes \mathbf{F}_M) \mathbf{x}_{\text{DD}} + \tilde{w}, \quad (8.39)$$

where $\mathbf{\Theta}_{\text{tx}}$ is the phase noise matrix at the transmitter, given by

$$\mathbf{\Theta}_{\text{tx}} = \text{diag} \{s_{\text{tx}}(0), s_{\text{tx}}(1), \dots, s_{\text{tx}}(NL - 1)\}. \quad (8.40)$$

Similarly, $\mathbf{\Theta}_{\text{rx}}$ is the phase noise matrix at the receiver, given by

$$\mathbf{\Theta}_{\text{rx}} = \text{diag} \{s_{\text{rx}}(0), s_{\text{rx}}(1), \dots, s_{\text{rx}}(NL - 1)\}. \quad (8.41)$$

Thus, from (8.35) and (8.39), the effective channel matrix taking into account the transmitter and receiver phase noise matrices $\mathbf{\Theta}_{\text{tx}}$ and $\mathbf{\Theta}_{\text{rx}}$ can be written in the form

$$\begin{aligned} \tilde{\mathbf{H}}_{\text{DD}} &= (\mathbf{F}_N \otimes \mathbf{F}_M^H) (\mathbf{I}_N \otimes \mathbf{F}_M) (\mathbf{I}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{\Theta}_{\text{rx}} \mathbf{H}_{\text{TV}} \mathbf{\Theta}_{\text{tx}} \\ &\quad (\mathbf{I}_N \otimes \mathbf{I}_{\text{cp}}) (\mathbf{I}_N \otimes \mathbf{F}_M^H) (\mathbf{F}_N^H \otimes \mathbf{F}_M) \\ &= (\mathbf{F}_N \otimes \mathbf{R}_{\text{cp}}) \mathbf{\Theta}_{\text{rx}} \mathbf{H}_{\text{TV}} \mathbf{\Theta}_{\text{tx}} (\mathbf{F}_N^H \otimes \mathbf{I}_{\text{cp}}). \end{aligned} \quad (8.42)$$

Consequently, as in (8.36), the received signal with transmitter and receiver oscillator phase noise is given by

$$\tilde{\mathbf{y}}_{\text{DD}} = \tilde{\mathbf{H}}_{\text{DD}} \mathbf{x}_{\text{DD}} + \tilde{\mathbf{w}}. \quad (8.43)$$

Signal detection algorithms such as the message passing (MP) algorithm [38] can be applied on the system model in (8.43) to recover the transmitted symbols.

8.2.3 Performance in the Presence of Phase Noise

The bit error performance of OTFS in the presence of phase noise is presented here. A carrier frequency of $f_c = 28$ GHz with oscillator phase noise characteristics in 3GPP standards [183, 184] is considered. The MP algorithm is used for signal detection. The other system parameters considered in the performance evaluation are listed in Table 8.1. For the same system settings, the performance of OFDM is also presented for comparison. The oscillator phase noise at the transmitter

Table 8.1 System parameters for performance evaluation with oscillator phase noise.

Parameter	Value
Carrier frequency	28 GHz
Bandwidth	10 MHz
Subcarrier spacing, Δf	78.125 kHz
Frame size (M, N)	(128, 64)
Modulation	BPSK
Number of paths, P	5
Delay profile (μs)	0.3, 1, 1.7, 2.4, 3.1
Doppler profile (Hz)	0, -400, 400, -1220, 1220

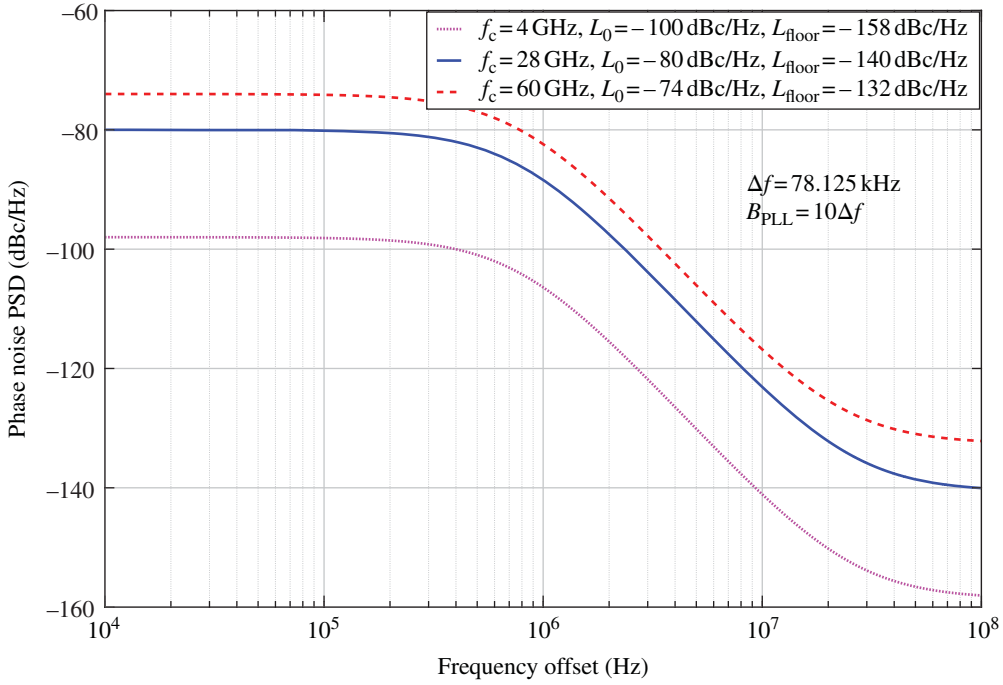


Figure 8.5 PSD of oscillator phase noise for various carrier frequencies (4, 28, and 60 GHz).

and receiver is modeled as in [175]. The power spectral density (PSD) of the phase noise is given by [175]

$$L(f_m) = \frac{B_{\text{PLL}}^2 L_0}{B_{\text{PLL}}^2 + f_m^2} + L_{\text{floor}}, \quad (8.44)$$

where B_{PLL} is the phase locked loop (PLL) -3 dB bandwidth, f_m is the frequency offset from the carrier frequency (f_c), L_0 is the in-band phase noise level in rad^2/Hz , and L_{floor} is the noise floor. It is noted that the oscillator phase noise increases with increased carrier frequencies [184]. This is illustrated in Fig. 8.5, where the phase noise PSD for carrier frequencies of 4, 28, and 60 GHz are plotted. These PSDs are plotted using (8.44), and the parameters obtained from practical oscillators reported in [184] for a base carrier frequency ($f_{c,\text{base}}$) of 30 GHz and shifting its PSD by $20 \log(f_c/f_{c,\text{base}})$ dBc/Hz. The variance of the phase noise samples generated corresponding to the PSDs in Fig. 8.5 as a function of B_{PLL} (parameterized by n , where $B_{\text{PLL}} = n\Delta f$ and Δf is the subcarrier spacing) is shown in Fig. 8.6. It can be seen that the phase noise variance increases as the PLL bandwidth relative to the subcarrier spacing increases, and this results in increased inter-carrier interference.

A channel model with $P = 5$ paths (taps) and exponential power delay profile is considered. The propagation in mmWave frequencies is dominated by a line-of-sight component due to which the first path is considered to follow a Rician fading distribution and the subsequent paths are considered to follow the Rayleigh distribution, as in the tapped delay line-D (TDL-D) model in [183]. The Doppler and delay profiles considered are given in Table 8.1. As can be seen, a maximum Doppler shift of 1.22 kHz (which corresponds to a maximum speed of 47 km/h at 28 GHz carrier frequency) is considered. A Rice factor of 13 dB, BPSK modulation, and a damping factor of 0.5 for the MP algorithm are used.

Figure 8.7 shows a BER performance comparison between OTFS and OFDM for the system settings described above with a B_{PLL} value of $10\Delta f$ (which corresponds to a high phase noise scenario).

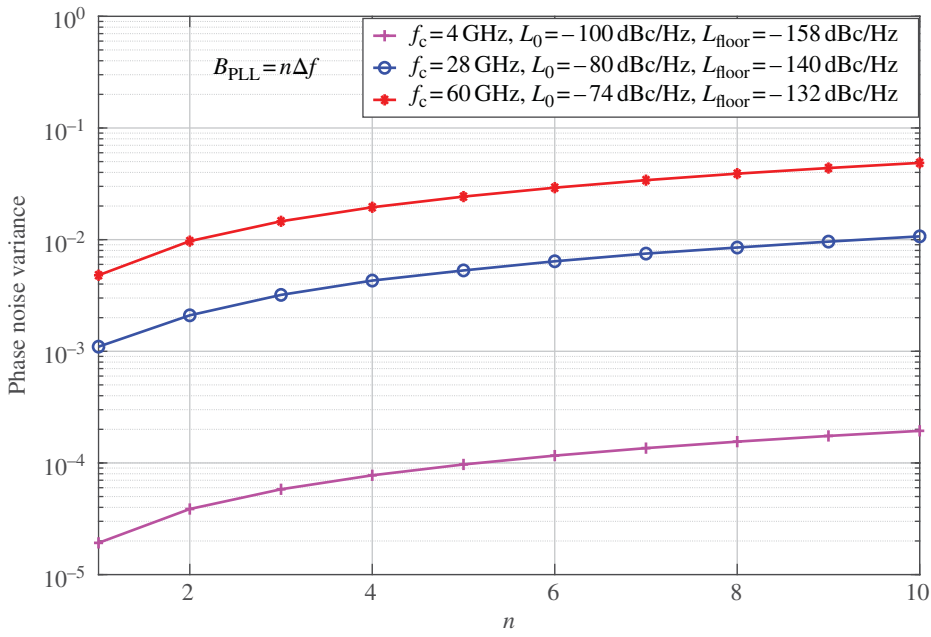


Figure 8.6 Variance of the oscillator phase noise as a function of n where $B_{\text{PLL}} = n\Delta f$.

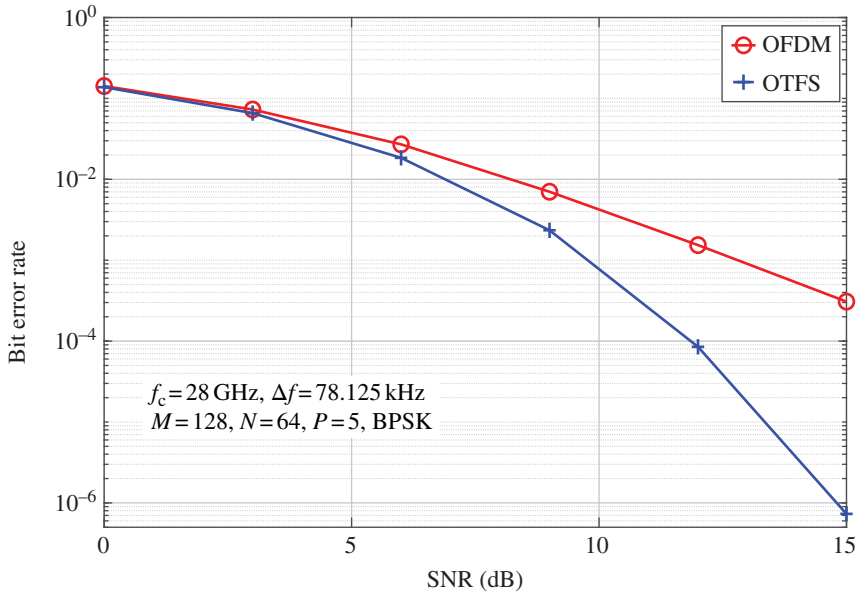


Figure 8.7 BER performance comparison between OTFS and OFDM systems with phase noise and Doppler shifts.

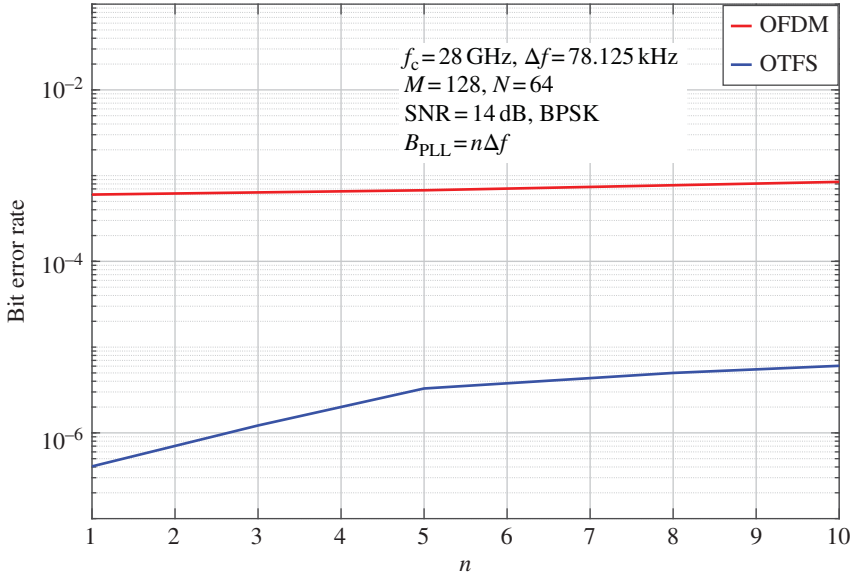


Figure 8.8 BER performance comparison between OTFS and OFDM as a function of $B_{\text{PLL}} = n\Delta f$ with zero Doppler.

From this figure, it is observed that the performance of OTFS is superior compared to that of OFDM in the presence of phase noise and Doppler shifts. For example, at a BER of 10^{-3} , the performance of OTFS is better than that of OFDM by about 3 dB. Next, Figure 8.8 shows the effect of phase noise alone (i.e., zero Doppler) on the performance of OTFS and OFDM. This figure shows the BER as a function of $B_{\text{PLL}} = n\Delta f$ at an SNR of 14 dB. A two-order better BER performance is observed in favor of OTFS compared to OFDM (10^{-5} BER in OTFS vs. 10^{-3} BER in OFDM) for $B_{\text{PLL}} = 10\Delta f$. It can also be observed that while OTFS achieves better than 10^{-5} BER for B_{PLL} values from Δf up to $10\Delta f$, OFDM is unable to achieve even 10^{-4} BER at a B_{PLL} of just Δf . It is seen that there is a three-order better BER performance in favor of OTFS for $B_{\text{PLL}} = \Delta f$ (i.e., $n = 1$). These results illustrate that OTFS is more resilient to oscillator phase noise in the mmWave band compared to OFDM.

8.3 Impact of IQ Imbalance on OTFS

Transceiver designs using direct conversion architecture in the RF front-end are becoming popular as they provide a highly integrated architecture at low cost [173]. In direct conversion receivers, the received RF signal is down converted directly to baseband without any intermediate frequency (IF) stage in between. This is unlike in heterodyne architecture receivers which have one or more IF stages in the RF to baseband conversion. Since there is no IF stage in the direct conversion architecture, it is also called as zero-IF architecture. Similarly, in a direct conversion transmitter, the baseband to RF conversion is carried out in a single stage without any intermediate stage. A key issue with the direct conversion approach is the need to deal with RF impairments [174, 175]. IQ imbalance (IQI) is one such impairment that compromises the orthogonality between the in-phase (I) and quadrature phase (Q) arms in a transceiver chain. It is a challenge to achieve perfect match/balance between the I and Q arms, more so at mmWave frequencies. The IQI in transceivers is modeled using two parameters, namely, gain imbalance (ΔG) and phase imbalance

($\Delta\phi$) between the I and Q arms. Large gain and phase imbalances between I and Q arms can lead to severe performance degradation [185–187]. This section presents an assessment of the effect of IQI on the performance of OTFS. Toward this, the IQI model at the transmitter/receiver and the OTFS system model with IQI are presented next, followed by BER results in the presence of IQI.

8.3.1 OTFS System Model with IQI

In the following, the system model is discussed for both transmitter and receiver IQI. MC-OTFS modulation discussed in Chapter 3 is considered here.

8.3.1.1 System Model with Transmitter IQI

A transmit RF chain which is impaired by IQI is modeled, as shown in Fig. 8.9, where ΔG_T and $\Delta\phi_T$ denote the gain and phase imbalances, respectively, between the I and Q arms of the transmit chain. In the ideal case of $\Delta G_T = \Delta\phi_T = 0$, the I and Q arms are in perfect quadrature. However, in the non-ideal case where ΔG_T and $\Delta\phi_T$ are not zero, the I and Q arms are not in perfect quadrature. A consequence of this is that a complex symbol fed at the input gets distorted in amplitude and phase at the output. This can be seen in the figure where an ideal 4-QAM alphabet at the input gets skewed at the output. Alternately viewed, IQI generates an image signal at the output corresponding to the complex signal at the input. That is, if x is the complex symbol at the input, then the output y is a weighted sum of the input symbol x and its conjugate (image), i.e.,

$$y = \alpha_T x + \beta_T x^*, \quad (8.45)$$

where α_T and β_T are complex scalars which are dependent on the imbalances ΔG_T and $\Delta\phi_T$ and can be derived as [174, 175]

$$\alpha_T = \frac{1}{2} \left[\cos\left(\frac{\Delta\phi_T}{2}\right) + j \frac{\Delta G_T}{2} \sin\left(\frac{\Delta\phi_T}{2}\right) \right] \quad (8.46)$$

and

$$\beta_T = \frac{1}{2} \left[-\frac{\Delta G_T}{2} \cos\left(\frac{\Delta\phi_T}{2}\right) - j \sin\left(\frac{\Delta\phi_T}{2}\right) \right]. \quad (8.47)$$

Based on the above, if $\mathbf{x}$ is the OTFS signal vector at the input to the transmit chain, then the corresponding vector at the output of the transmit chain is given by

$$\mathbf{x}_{IQ} = \alpha_T \mathbf{x} + \beta_T \mathbf{x}^*. \quad (8.48)$$

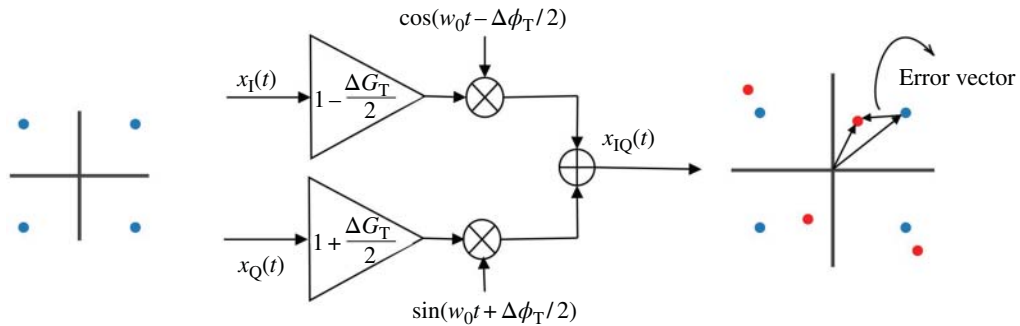


Figure 8.9 IQ imbalance at the transmitter.

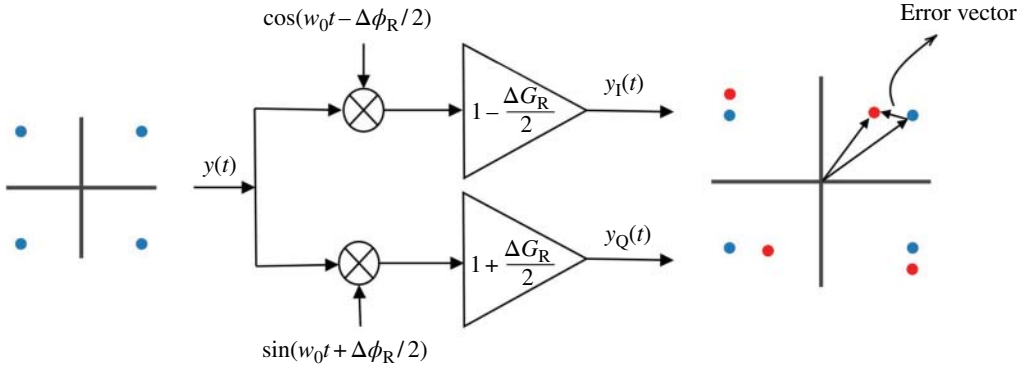


Figure 8.10 IQ imbalance at the receiver.

It is noted that, while the intent is to transmit the vector $\mathbf{x}$, the actual transmitted vector becomes $\mathbf{x}_{\text{IQ}}$ because of IQI. Now, assuming an ideal RF chain at the receiver (without receiver IQI), the received vector can be written as

$$\mathbf{y} = \mathbf{H}\mathbf{x}_{\text{IQ}} + \mathbf{v}. \quad (8.49)$$

8.3.1.2 System Model with Receiver IQI

Similar to the transmitter IQI model, the IQI in the receiver RF chain can be modeled, as shown in Fig. 8.10, where ΔG_R and $\Delta\phi_R$ denote the gain and phase imbalances, respectively, between the I and Q arms in the receive chain.

As in the case of transmitter with IQI, here also, the output of the receive chain will have an image signal generated due to receiver IQI. That is, the received vector in the presence of receiver IQI can be written as

$$\mathbf{y}_{\text{IQ}} = \alpha_R \mathbf{y} + \beta_R \mathbf{y}^*, \quad (8.50)$$

where

$$\alpha_R = \frac{1}{2} \left[\cos\left(\frac{\Delta\phi_R}{2}\right) + j \frac{\Delta G_R}{2} \sin\left(\frac{\Delta\phi_R}{2}\right) \right], \quad (8.51)$$

$$\beta_R = \frac{1}{2} \left[-\frac{\Delta G_R}{2} \cos\left(\frac{\Delta\phi_R}{2}\right) - j \sin\left(\frac{\Delta\phi_R}{2}\right) \right], \quad (8.52)$$

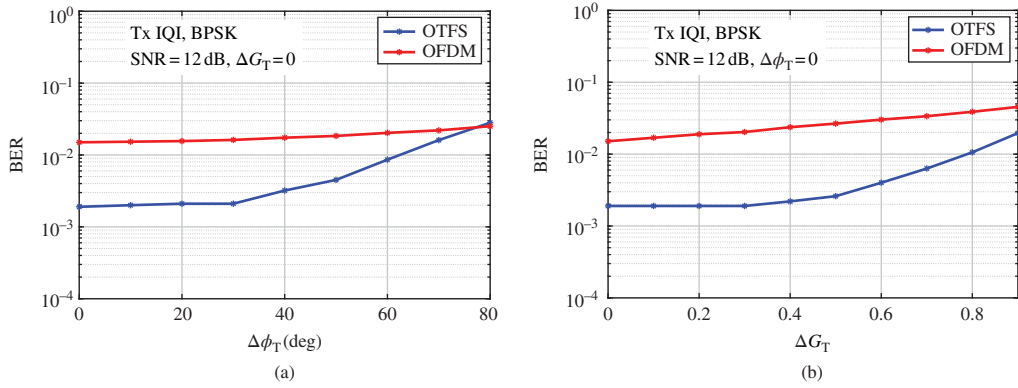
$\mathbf{y}$ is the received signal in the absence of receiver IQI, and $\mathbf{y}^*$ is the image signal caused due to receiver IQI. Signal detection algorithms can be applied on the above system model to recover the information symbols. Because of the presence of the image signal as interference, the BER performance gets degraded.

8.3.2 Sensitivity of OTFS and OFDM to IQI

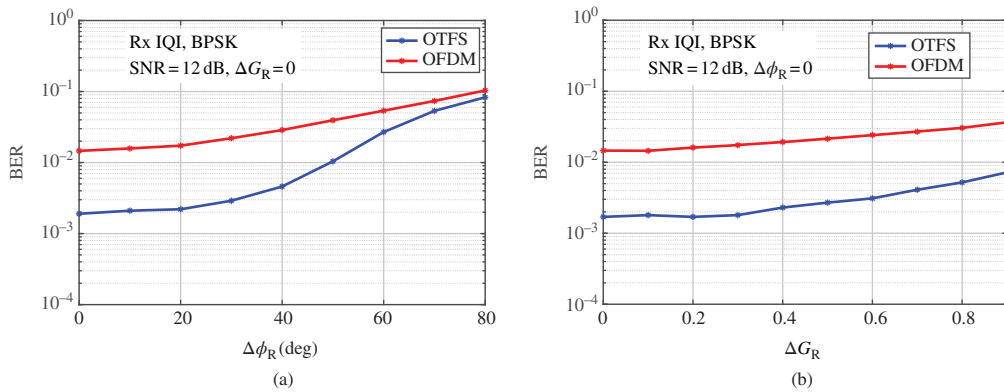
The effect of transmitter (Tx) IQI (assuming ideal receiver) and receiver (Rx) IQI (assuming ideal transmitter) on MC-OTFS is presented here. The system parameters considered for the performance evaluation are listed in Table 8.2. Figure 8.11 shows the BER performance comparison between OTFS and OFDM as a function of IQI parameters ΔG_T and $\Delta\phi_T$ at the transmitter. The gain imbalance ΔG_T is varied by considering $\Delta\phi_T = 0$. Similarly, $\Delta\phi_T$ is varied by fixing the value of ΔG_T at 0. The SNR considered is 12 dB. It can be seen that the BER performance of OTFS is better

Table 8.2 System parameters used for performance evaluation with Tx/Rx IQI.

Parameter	Value
Frame size (M, N)	(4, 4)
Carrier frequency (GHz)	4
Subcarrier spacing (kHz)	3.75
No. of DD channel paths (P)	4
Delay-Doppler profile (τ_i, ν_i)	$\left(0, \frac{1}{NT}\right), \left(\frac{1}{M\Delta f}, \frac{1}{NT}\right),$ $\left(\frac{2}{M\Delta f}, \frac{1}{NT}\right), \left(\frac{3}{M\Delta f}, \frac{1}{NT}\right)$
Power delay profile	Uniform

**Figure 8.11** Sensitivity of OTFS and OFDM to transmitter IQI. (a) $\Delta\phi_T$ sensitivity and (b) ΔG_T sensitivity.

compared to OFDM in the presence of Tx IQI. In the case of OTFS, the BER is in the order of 10^{-3} , while it is in the order of 10^{-2} in OFDM. OTFS is more resilient to IQI when $\Delta G_T \in [0, 0.3]$ and $\Delta\phi_T \in [0^\circ, 20^\circ]$. Though the BER of OTFS increases for higher values of gain and phase imbalances, it is still better than the BER of OFDM. A similar performance behavior is observed in Fig. 8.12 which shows the effect of receiver IQI. In practice, techniques to estimate and compensate the

**Figure 8.12** Sensitivity of OTFS and OFDM to receiver IQI. (a) $\Delta\phi_R$ sensitivity and (b) ΔG_R sensitivity.

transmitter and receiver IQI can alleviate the performance degradation. In Chapter 9, a deep neural network-based IQI estimation/compensation scheme and its performance in OTFS are presented.

8.4 Discrete Zak Approach to OTFS

In this section, a DZT approach toward the practical realization of Zak-based OTFS transceivers is considered. As seen in Chapter 2, the Zak transform framework [56] provides a natural means to directly convert the information symbols multiplexed in the DD domain to the time domain for transmission, and vice versa at the receiver. The Zak transform framework for OTFS in continuous time has been derived from first principles in [60], where it is shown that the spectral efficiency performance of the Zak approach remains invariant to user velocity, whereas it degrades in OFDM as the velocity increases. A DZT-based [188] implementation of OTFS, which has the benefit of implementation simplicity, has been introduced in [64]. This section introduces this DZT-based OTFS system (referred to as DZT-OTFS), its vectorized DD domain input–output relation, and its performance in terms of diversity and BER [65]. In Appendix 8.A, digital implementations for Zak-OTFS receiver and transmitter are derived based on DZT and inverse DZT (IDZT). In Appendix 8.B, DZT, IDZT, and discrete time and FD pulsones are discussed.²

8.4.1 DZT-OTFS System Model

Figure 8.13 shows the block diagram of a DZT-OTFS transceiver [64, 65]. At the transmitter, KL information symbols from a modulation alphabet $\mathbb{A}$ are multiplexed on the DD grid. That is, KL information symbols, denoted by $Z_x[k, l]$ s, are multiplexed over a $K \times L$ DD grid given by $\left\{ \left(k\Delta\nu = \frac{k}{KL T_s}, l\Delta\tau = lT_s \right), k = 0, \dots, K-1, l = 0, \dots, L-1 \right\}$, where $T_s = 1/B$ is the symbol duration, B is the bandwidth available for communication, and $\Delta\tau$ and $\Delta\nu$ are the delay and Doppler resolutions, respectively.

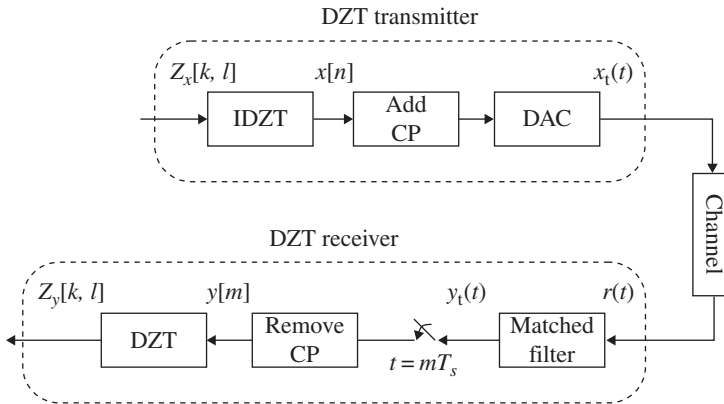


Figure 8.13 Block diagram of the DZT-OTFS transceiver.

² The notation used in Appendix 8.A and Appendix 8.B is the same as that in Chapter 2, which is different from the notation used in this section. In this section, k and l denote the Doppler and delay domain indices, respectively, whereas in Chapter 2, it is the opposite. Also, in Chapter 2, the delay and Doppler domain periods are denoted by M and N , respectively, whereas in this section, they are denoted by L and K , respectively.

The DD domain symbols, $Z_x[k, l]$ s, are converted to the time domain using IDZT as [64, 188]

$$x[n] = \frac{1}{\sqrt{K}} \sum_{k=0}^{K-1} Z_x[k, (n)_L] e^{j2\pi \frac{[n/L]}{K} k}, \quad (8.53)$$

where $(\cdot)_L$ denotes modulo- L operation and $[\cdot]$ denotes the floor function. To mitigate the inter-frame interference, a cyclic prefix (CP) of length $N_{cp} \geq \lceil \tau_{\max}/T_s \rceil$ is added in the time domain, where $\tau_{\max}$ is the maximum delay spread of the channel and $\lceil \cdot \rceil$ denotes the ceiling function. The time domain sequence $x[n]$ is then mounted on the time-shifted transmit pulse $g_{tx}(t)$, $0 \leq t \leq T_s$, resulting in a continuous time signal $x_t(t)$ given by

$$x_t(t) = \sum_{n=0}^{KL+N_{cp}-1} x[(n - N_{cp})_{KL}] g_{tx}(t - nT_s), \quad (8.54)$$

where $0 \leq t \leq (KL + N_{cp})T_s$. The transmit signal $x_t(t)$ passes through a time-varying channel whose DD domain response is given by

$$h(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (8.55)$$

where P denotes the number of resolvable paths in the DD domain, h_i is the channel coefficient of the i th path, and $\tau_i = (\alpha_i + a_i)T_s$ is the delay associated with the i th path, where α_i is the integer part and $a_i \in [-0.5, 0.5]$ is the fractional part of the i th path's delay. Similarly, $\nu_i = \frac{(\beta_i + b_i)}{KLT_s}$ is the Doppler associated with the i th path, where β_i is the integer part and $b_i \in [-0.5, 0.5]$ is the fractional part of the i th path's Doppler. The received time domain signal is given by

$$r(t) = \int_{\nu} \int_{\tau} h(\tau, \nu) x_t(t - \tau) e^{j2\pi \nu(t - \tau)} d\tau d\nu + w(t), \quad (8.56)$$

where $w(t)$ denotes the additive noise. Substituting (8.54) and (8.55) in (8.56),

$$r(t) = \sum_{i=1}^P h_i \sum_{n=0}^{KL+N_{cp}-1} x[(n - N_{cp})_{KL}] g_{tx}(t - \tau_i - nT_s) e^{j2\pi \nu_i t} + w(t). \quad (8.57)$$

At the receiver, the received signal after matched filtering with receive pulse $g_{rx}(t)$ can be written as

$$y_t(t) = \int_{\tau} r(\tau) g_{rx}^*(t - \tau) d\tau. \quad (8.58)$$

Let $g_{tx}(t) = g_{rx}(t) = p(t)$. Substituting (8.57) in (8.58),

$$y_t(t) = \sum_{i=1}^P h_i \sum_{n=0}^{KL+N_{cp}-1} x[(n - N_{cp})_{KL}] \int_{\tau} p(\tau - \tau_i - nT_s) p^*(t - \tau) e^{j2\pi \nu_i \tau} d\tau + v(t), \quad (8.59)$$

where $v(t)$ is the match filtered noise. Assuming the pulse bandwidth to be much larger than the maximum Doppler, the integral in (8.59) can be approximated as [64]

$$\int_{\tau} p(\tau - \tau_i - nT_s) p^*(t - \tau) e^{j2\pi \nu_i \tau} d\tau \approx e^{j2\pi \nu_i (\tau_i + nT_s)} g(t - \tau_i - nT_s), \quad (8.60)$$

where $g(t) \triangleq \int_{\tau} p(\tau) p^*(t - \tau) d\tau$. For example, for a RC pulse with a roll-off factor γ , $g(t)$ is given by

$$g(t) = \frac{\sin(\pi t/T_s) \cos(\gamma \pi t/T_s)}{(\pi t/T_s)(1 - (2\gamma t/T_s)^2)}. \quad (8.61)$$

The output of the matched filter is sampled at $t = mT_s$, $m = 0, 1, 2, \dots$. The first N_{cp} samples corresponding to the CP are discarded. The resulting discrete time-domain signal is given by

$$y[m] = \sum_{i=1}^P h_i \sum_{n=0}^{KL-1} x[n] g[(m-n)T_s - \tau_i] e^{j2\pi v_i(\tau + nT_s)} + v[m]. \quad (8.62)$$

Substituting for τ_i and v_i in (8.62),

$$y[m] = \sum_{i=1}^P h_i e^{j2\pi \tau_i v_i} \sum_{n=0}^{KL-1} x[n] e_i[n] g_i[m-n] + v[m], \quad (8.63)$$

where $g_i[n] = g[n - l_i]$ and $e_i[n] = e^{j2\pi \frac{k_i}{KL} n}$, $n = 0, \dots, KL - 1$, $l_i = \alpha_i + a_i$, and $k_i = \beta_i + b_i$. Here, g_i and e_i are the discrete sequences that capture the effect of delay and Doppler of the i th path in the channel, respectively. Now, letting $h'_i = h_i e^{j2\pi \tau_i v_i}$ and

$$y_i[m] = \sum_{n=0}^{KL-1} x[n] e_i[n] g_i[m-n], \quad (8.64)$$

(8.63) can be written as

$$y[m] = \sum_{i=1}^P h'_i y_i[m] + v[m]. \quad (8.65)$$

8.4.2 Vectorized DD Domain I/O Relation

Here, the time-domain input–output relation in (8.65) is first written in a vectorized form, and it is then transformed to a DD domain input–output relation in a vectorized form using matrix decomposition. The relation between $y_i[m]$, $x[m]$, $e_i[m]$, and $g_i[m]$ can be expressed in a vector form as

$$\mathbf{y}_i = (\mathbf{x} \odot \mathbf{e}_i) \circledast \mathbf{g}_i, \quad (8.66)$$

where $\mathbf{y}_i, \mathbf{e}_i, \mathbf{g}_i \in \mathbb{C}^{1 \times KL}$ are obtained by stacking the elements $y_i[m], e_i[m]$, and $g_i[m]$, respectively, $\odot$ denotes the Hadamard product, and $\circledast$ denotes the circular convolution operation. The input–output relation in (8.65) can then be written in a vectorized form as

$$\mathbf{y} = [1 \ 1 \ \dots \ 1] \text{diag}\{h'_1, \dots, h'_P\} \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_P \end{bmatrix} + \mathbf{v}. \quad (8.67)$$

The discrete time-domain signal $\mathbf{y}$ at the receiver is converted to the DD domain using DZT as

$$Z_y[k, l] = \frac{1}{\sqrt{K}} \sum_{n=0}^{K-1} y[l + nL] e^{-j2\pi \frac{k}{K} n}. \quad (8.68)$$

The equivalent DD domain representations of (8.64) and (8.65) are given by

$$Z_{y_i}[k, l] = \sum_{m=0}^{L-1} \sum_{n=0}^{K-1} Z_x[n, m] Z_{e_i}[k-n, m] Z_{g_i}[k, l-m], \quad (8.69)$$

and

$$Z_y[k, l] = \sum_{i=1}^P h'_i Z_{y_i}[k, l] + Z_v[k, l], \quad (8.70)$$

respectively, where $Z_{y_i}[:, \cdot]$, $Z_{e_i}[:, \cdot]$, $Z_{g_i}[:, \cdot]$, and $Z_v[:, \cdot]$ denote the elements of the DZT of the sequences $\mathbf{y}_i$, $\mathbf{e}_i$, $\mathbf{g}_i$, and $\mathbf{v}$, respectively.

Now, for vectorizing (8.70), let $\mathbf{B}_u$ be a $K \times K$ matrix whose j th row is

$$\mathbf{B}_u(j-1, :) = (Z_{e_i}(:, u-1))^T \mathbf{P}_K^{j-1}, \quad (8.71)$$

where $u = 1, \dots, L$, $j = 1, \dots, K$, and $\mathbf{P}_K$ is a $K \times K$ basic circulant permutation matrix (BPCM) [189]. Furthermore, define a block diagonal matrix $\mathbf{E}_i$ with matrices $\{\mathbf{B}_u\}_{u=1}^L$ along the diagonal, i.e.,

$$\mathbf{E}_i = \text{diag}\{\mathbf{B}_1, \dots, \mathbf{B}_L\}. \quad (8.72)$$

Likewise, define a $K \times KL$ block matrix $\mathbf{A} = [\text{diag}\{Z_{g_i}[:, 0]\}, \dots, \text{diag}\{Z_{g_i}[:, L-1]\}]$. Let $\mathbf{Q}_u$ be an $KL \times KL$ matrix, given by

$$\mathbf{Q}_u = \mathbf{P}_L^{u-1} \otimes \mathbf{I}_K, \quad (8.73)$$

where $\mathbf{P}_L$ is an $L \times L$ permutation matrix and $\otimes$ operator denotes the Kronecker product. Also, define a $KL \times KL$ matrix $\mathbf{G}_i$ as

$$\mathbf{G}_i = \begin{bmatrix} \mathbf{A}\mathbf{Q}_1 \\ \vdots \\ \mathbf{A}\mathbf{Q}_L \end{bmatrix}. \quad (8.74)$$

Using $\mathbf{E}_i$ and $\mathbf{G}_i$, the effective channel matrix $\mathbf{H}$ in the DD domain can be written as

$$\mathbf{H} = \sum_{i=1}^P h'_i \mathbf{E}_i \mathbf{G}_i. \quad (8.75)$$

Remark Since $\mathbf{G}_i$ s capture the effect of delays (α_i s) and $\mathbf{E}_i$ s capture the effect of Dopplers (β_i s), the channel representation in (8.75) is in a form that decouples the effect of channel gains (h_i s), delays (α_i s), and Dopplers (β_i s). This decoupled representation can be exploited to devise low-complexity DD domain channel estimation algorithms for DZT-OTFS systems [190].

Now, vectorizing (8.70) using the above, the end-to-end DD domain input-output relation is obtained as

$$\mathbf{y}_{\text{DD}} = \mathbf{x}_{\text{DD}} \mathbf{H} + \mathbf{v}_{\text{DD}}, \quad (8.76)$$

where $\mathbf{y}_{\text{DD}}, \mathbf{x}_{\text{DD}}, \mathbf{v}_{\text{DD}} \in \mathbb{C}^{1 \times KL}$, such that $\mathbf{y}_{\text{DD}}(k + Kl) = Z_y[k, l]$, $\mathbf{x}_{\text{DD}}(k + Kl) = Z_x[k, l]$, and $\mathbf{v}_{\text{DD}}(k + Kl) = Z_v[k, l]$, for $k = 0, 1, \dots, K-1$ and $l = 0, 1, \dots, L-1$, and $\mathbf{H} \in \mathbb{C}^{KL \times KL}$ is the end-to-end effective channel matrix in the DD domain given by (8.75). The matrix-vector representation of DZT-OTFS in (8.76) is used for the diversity analysis and BER performance evaluation in Sections 8.4.3 and 8.4.4.

8.4.3 Diversity Analysis of DZT-OTFS

For the purpose of diversity analysis, the input-output relation in (8.76) can be written in an alternate form as

$$\mathbf{y}_{\text{DD}} = \mathbf{h}\mathbf{X} + \mathbf{v}_{\text{DD}}, \quad (8.77)$$

where $\mathbf{X}$ is a $P \times KL$ matrix defined as

$$\mathbf{X} = (\mathbf{I}_P \otimes \mathbf{x}_{\text{DD}}) \begin{bmatrix} \mathbf{G}_1 \mathbf{E}_1 \\ \vdots \\ \mathbf{G}_P \mathbf{E}_P \end{bmatrix}, \quad (8.78)$$

and $\mathbf{h} = [h'_1, \dots, h'_P]$ is a $1 \times P$ vector. Let $\mathbf{x}_{\text{DD}}^i$ and $\mathbf{x}_{\text{DD}}^j$ be two distinct DD symbol vectors, as defined in (8.76), and let the corresponding symbol matrices be $\mathbf{X}_i$ and $\mathbf{X}_j$, respectively. Let ρ denote the normalized SNR, given by $\rho = 1/N_0$, where N_0 is the noise power. Assuming maximum-likelihood (ML) detection and perfect channel knowledge at the receiver, the conditional pairwise error probability (PEP) between $\mathbf{X}_i$ and $\mathbf{X}_j$ is given by

$$\begin{aligned} P(\mathbf{x}_{\text{DD}}^i \rightarrow \mathbf{x}_{\text{DD}}^j | \mathbf{h}) &= P(\|\mathbf{y} - \mathbf{h}\mathbf{X}_j\|^2 < \|\mathbf{y} - \mathbf{h}\mathbf{X}_i\|^2) \\ &= Q\left(\sqrt{\frac{\rho \|\mathbf{h}(\mathbf{X}_i - \mathbf{X}_j)\|^2}{2}}\right). \end{aligned} \quad (8.79)$$

Using the Chernoff bound and averaging over the statistics of $\mathbf{h}$, the average PEP can be bounded as

$$P(\mathbf{x}_{\text{DD}}^i \rightarrow \mathbf{x}_{\text{DD}}^j) \leq \mathbb{E}_{\mathbf{h}} \left\{ \exp \left(-\frac{\rho \|\mathbf{h}(\mathbf{X}_i - \mathbf{X}_j)\|^2}{4} \right) \right\}, \quad (8.80)$$

where $\mathbb{E}_{\mathbf{h}}(\cdot)$ denotes the expectation operation. Assuming the elements of $\mathbf{h}$ are independent and identically distributed complex Gaussian random variables with zero mean and variance $1/P$, (8.80) can be simplified as [6]

$$P(\mathbf{x}_{\text{DD}}^i \rightarrow \mathbf{x}_{\text{DD}}^j) \leq \prod_{l=1}^r \frac{1}{1 + \frac{\rho \lambda_{lij}}{4P}}, \quad (8.81)$$

where r is the rank of the difference matrix $(\mathbf{X}_i - \mathbf{X}_j)$, and λ_{lij} is the eigenvalue of $(\mathbf{X}_i - \mathbf{X}_j)(\mathbf{X}_i - \mathbf{X}_j)^H$. Using $1 + \frac{\rho \lambda_{lij}}{4P} \approx \frac{\rho \lambda_{lij}}{4P}$ at high SNRs, (8.81) can be written as

$$P(\mathbf{x}_{\text{DD}}^i \rightarrow \mathbf{x}_{\text{DD}}^j) \leq \frac{1}{\rho^r \prod_{l=1}^r \frac{\lambda_{lij}}{4P}}. \quad (8.82)$$

Using the above average PEP expression, an upper bound on the bit error probability, P_e , can be obtained as

$$\begin{aligned} P_e &\leq \frac{1}{QKL \log_2 |\mathbb{A}|} \sum_{i=1}^Q \sum_{j=1, j \neq i}^Q d(\mathbf{x}_{\text{DD}}^i, \mathbf{x}_{\text{DD}}^j) P(\mathbf{x}_{\text{DD}}^i \rightarrow \mathbf{x}_{\text{DD}}^j) \\ &= \frac{1}{QKL \log_2 |\mathbb{A}|} \sum_{i=1}^Q \sum_{j=1, j \neq i}^Q \frac{d(\mathbf{x}_{\text{DD}}^i, \mathbf{x}_{\text{DD}}^j) \rho^{-r}}{\prod_{l=1}^r \frac{\lambda_{lij}}{4P}}, \end{aligned} \quad (8.83)$$

where $d(\mathbf{x}_{\text{DD}}^i, \mathbf{x}_{\text{DD}}^j)$ is the Hamming distance between $\mathbf{x}_{\text{DD}}^i$ and $\mathbf{x}_{\text{DD}}^j$, $Q = |\mathbb{A}^{KL}|$, and $|\cdot|$ denotes the cardinality of a set. Also, a lower bound on P_e is obtained by summing the PEPs corresponding to the pairs $\mathbf{X}_i$ and $\mathbf{X}_j$ such that the rank of $(\mathbf{X}_i - \mathbf{X}_j)$ is least amongst all the pairs. From (8.83), it is observed that the exponent of the SNR is r , which is the rank of the symbol difference matrix $(\mathbf{X}_i - \mathbf{X}_j)$, and, for all i, j , $i \neq j$, the minimum rank dominates the overall BER. Therefore, the diversity order (DO) is given by

$$\text{DO} = \min_{i,j, i \neq j} \text{rank}(\mathbf{X}_i - \mathbf{X}_j). \quad (8.84)$$

Table 8.3 Rank profile of the difference matrices.

System parameters				Number of difference matrices with rank				Minimum rank
				1	2	3	4	
System-1 $K = 2,$ $L = 2,$ $P = 2$	DD	$\tau = [0, 0] \Delta \tau$	w/o PR	32	208	—	—	1
	Profile-A	$\nu = [0, 1] \Delta \nu$						
			with PR	0	240	—	—	2
	DD	$\tau = [0, 1] \Delta \tau$	w/o PR	0	240	—	—	2
	Profile-B	$\nu = [0, 1] \Delta \nu$	with PR	0	240	—	—	2
System-2 $K = 2,$ $L = 4,$ $P = 4$	DD	$\tau = [0, 0, 1, 1] \Delta \tau$	w/o PR	0	136	2072	63072	2
	Profile-C	$\nu = [0, 1, 0, 1] \Delta \nu$						
			with PR	0	0	0	65280	4
	DD	$\tau = [0.2, 1.4, 2.3, 3.6] \Delta \tau$	w/o PR	0	0	0	65280	4
	Profile-D	$\nu = [-0.3, 1.4, 0.8, -0.2] \Delta \nu$	with PR	0	0	0	65280	4

It has been observed through simulations that the rank profile of the difference matrices can be such that r is less than P for several $(\mathbf{X}_i, \mathbf{X}_j)$ pairs [65]. This means the full DD diversity of P is not always achieved. To alleviate this, phase rotation (PR) of the transmit vector $\mathbf{x}_{\text{DD}}$ can be carried out before transmission. That is, instead of transmitting the $\mathbf{x}_{\text{DD}}$ vector as such, a phase-rotated vector $\tilde{\mathbf{x}}_{\text{DD}} = \Theta \mathbf{x}_{\text{DD}}$ is transmitted, where Θ is a PR matrix given by $\Theta = \text{diag}\{e^{j\frac{q}{KL}}\}$, $q = 0, \dots, KL - 1$. Simulations have shown that with this PR scheme, the full diversity of P is achieved. The above observations are illustrated in Table 8.3, which shows the rank profile of difference matrices in two systems, namely, System-1 with $K = L = P = 2$ and System-2 with $K = 2$ and $L = P = 4$. The rank profiles for these two systems with and without PR are shown. It can be seen that, without PR, 32 out of 240 rank matrices in System-1 with DD Profile-A have rank 1 (which is less than $P = 2$), resulting in a diversity order of 1. Likewise, without PR, many difference matrices in System-2 with DD Profile-C have ranks 2 and 3 (which are less than $P = 4$). On the other hand, with PR, all the ranks are P in both the systems for all the DD Profiles-A to D, achieving the full diversity order of P .

8.4.4 Performance Results

Here, the simulated BER results that validate the analytical BER bounds and diversity orders obtained in the previous subsection are presented. The simulated BER performance of DZT-OTFS for different system settings in comparison with those of MC-OTFS and OFDM are also presented.

8.4.4.1 Diversity Performance

Figure 8.14 shows the simulated BER performance of a DZT-OTFS system with $K = L = 2$ and $P = 2$ and 4. A carrier frequency (f_c) of 4 GHz and ν_p of 3.75 kHz (hence, $\tau_p = \frac{1}{\nu_p} = 0.267$ ms) are considered. The information symbols are chosen from the BPSK modulation alphabet. Perfect channel knowledge and ML detection are assumed at the receiver. Simulated BER curves without PR for System-1 with DD Profile-A and DD Profile-C in Table 8.3 are plotted. The corresponding upper bound and lower bound on BER are also plotted for comparison. The following observations

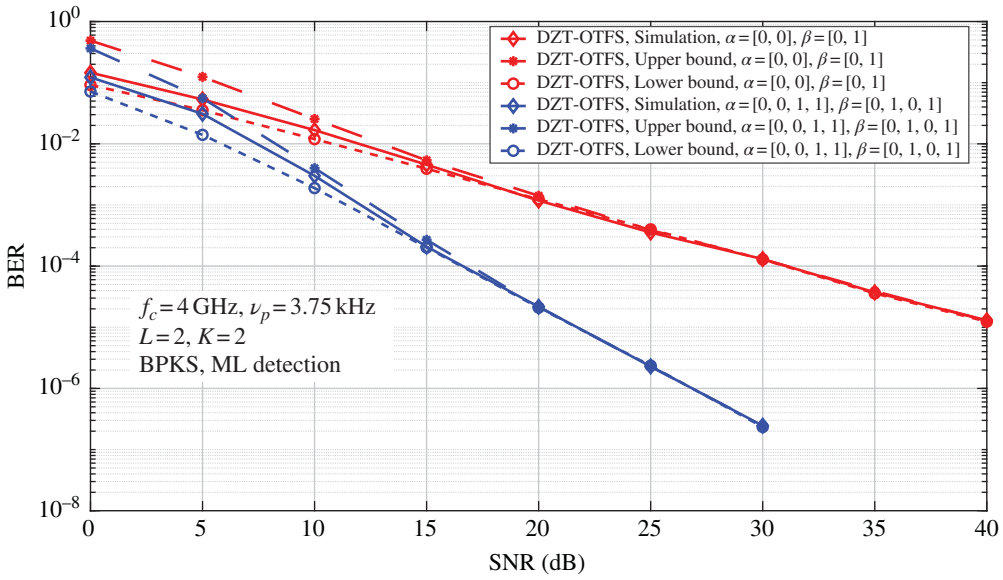


Figure 8.14 Upper and lower bounds on the BER and simulated BER performance of DZT-OTFS.

can be made from Fig. 8.14. The upper and lower bounds on BER and the simulated BER curves almost merge at high SNRs validating the analytical BER results. Also, the diversity slopes in the high SNR regime are 1 and 2, respectively, for System-1 with DD Profile-A and System-1 with DD Profile-C. These are in agreement with the diversity orders predicted according to minimum ranks shown in Table 8.3.

Figure 8.15 shows the simulated BER curves without PR for (i) System-1 with DD Profile-A, (ii) System-2 with DD Profile-C, and (iii) System-2 with DD Profile-D. The corresponding BER

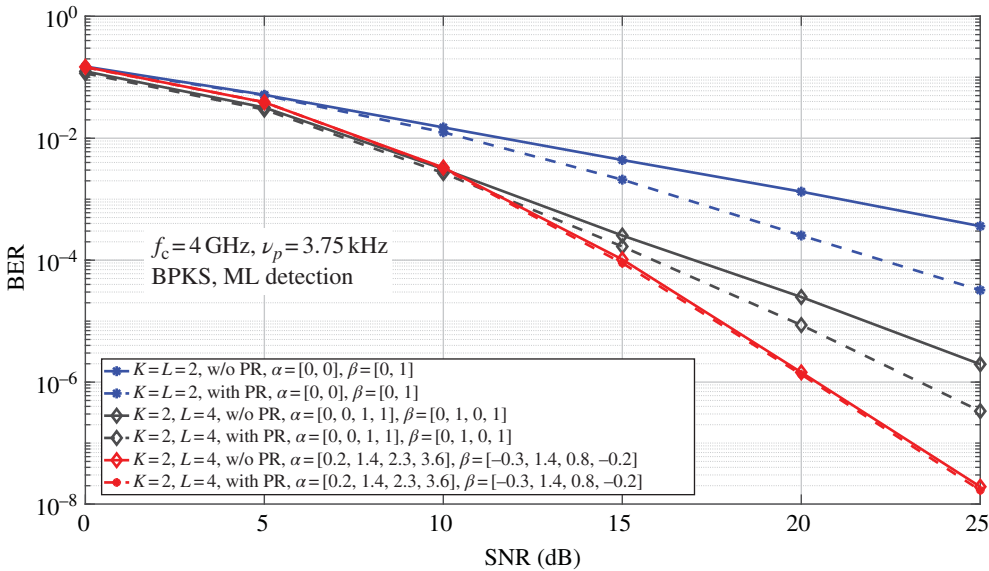


Figure 8.15 BER performance of DZT-OTFS with and without PR.

curves with PR are also plotted. It is observed that the diversity slopes of the BER curves are in accordance with the minimum ranks shown in Table 8.3. It is also observed that the full diversity of P is achieved with PR.

8.4.4.2 BER Performance

The BER performance of DZT-OTFS for different system settings is presented here. The number of delay bins L and the number of Doppler bins K are chosen to meet the system specifications and channel constraints as follows. A fundamental rectangle in the DD domain of width τ_p and height ν_p is chosen such that $\tau_p \nu_p = 1$, and for a given available bandwidth (B) and delay requirement (T), K and L are chosen such that $L = \left\lceil \frac{B}{\nu_p} \right\rceil$ and $K = \left\lceil \frac{T}{\tau_p} \right\rceil$. The following parameters are considered in the simulations: $B = 60$ kHz, $T = 4.27$ ms, and $T_s = \frac{1}{B} = 16.67$ μ s. Fixing $\nu_p = 3.75$ kHz gives $\tau_p = \frac{1}{\nu_p} = 0.267$ ms, $L = \left\lceil \frac{B}{\nu_p} \right\rceil = 16$, and $K = \left\lceil \frac{T}{\tau_p} \right\rceil = 16$. Fractional delay-Dopplers are considered and $g(t)$ is taken to be a RC pulse with roll-off factor γ . The channel is considered to have $P = 4$ paths with uniform power delay profile. For a given maximum Doppler $\nu_{\max}$ and maximum delay $\tau_{\max}$, the i th path's α_i is a random integer uniformly sampled from $\{0, 1, \dots, \alpha_{\max}\}$, $a_i \in \text{Unif}[-0.5, 0.5]$, where $\alpha_{\max} = \text{round}\left(\frac{\tau_{\max}}{T_s}\right)$, $\text{Unif}[\cdot, \cdot]$ denotes the uniform distribution, and $\nu_i = \nu_{\max} \cos(\theta)$, where $\theta \in \text{Unif}[-\pi, \pi]$. The performance of MC-OTFS and OFDM is also shown for comparison. BPSK modulation and minimum mean square error (MMSE) detection are used in all the systems. Perfect channel knowledge at the receiver is assumed.

Figure 8.16 shows the BER performance of DZT-OTFS, MC-OTFS, and OFDM as a function of SNR. The maximum delay and Doppler spreads are taken to be $\tau_{\max} = 8T_s = 133.33$ μ s and $\nu_{\max} = 937$ Hz, and the roll-off factor of the RC pulse γ is taken to be 0. From Fig. 8.16, it is observed that both DZT-OTFS and MC-OTFS achieve significantly better performance compared to OFDM. Also, DZT-OTFS is seen to achieve better performance compared to MC-OTFS, illustrating the inherent strength of the DZT-OTFS waveform compared to the MC-OTFS waveform. For example, at a BER of 10^{-3} , DZT-OTFS has an SNR gain of about 1 and 10 dB compared to MC-OTFS and OFDM,

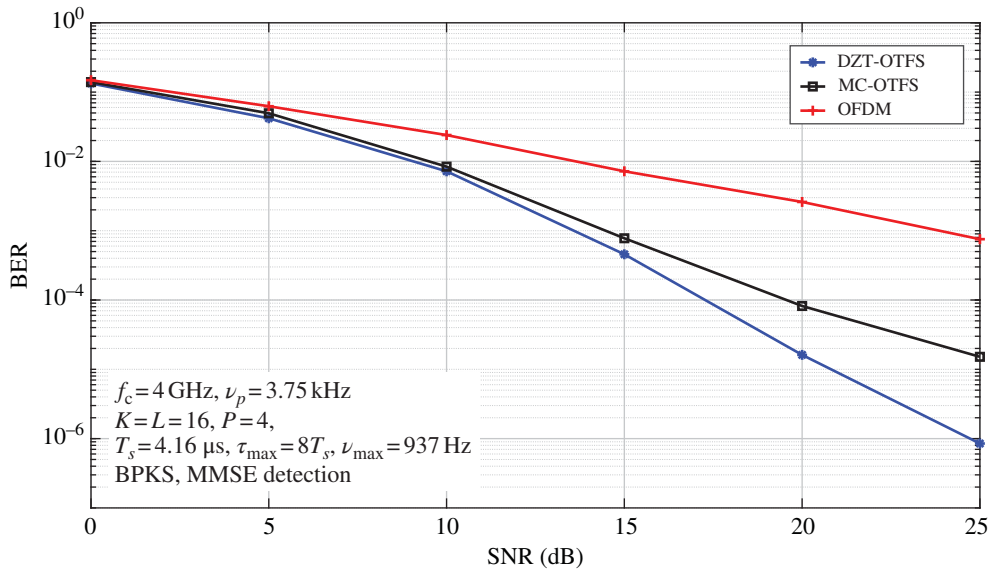


Figure 8.16 BER performance of DZT-OTFS, MC-OTFS, and OFDM as a function of SNR.

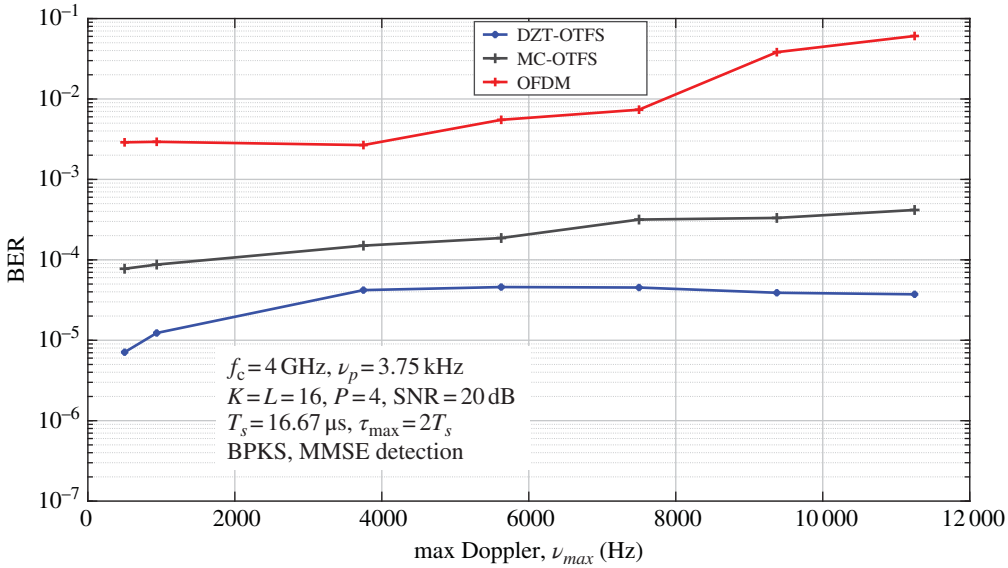


Figure 8.17 BER performance of DZT-OTFS, MC-OTFS, and OFDM as a function of maximum Doppler, $\nu_{\max}$.

respectively. Furthermore, the performance gap between DZT-OTFS and MC-OTFS increases in favor of DZT-OTFS as SNR increases.

Figure 8.17 shows the BER as a function of $\nu_{\max}$ at an SNR of 20 dB and $\gamma = 0$. The simulations are done for a fixed $\tau_{\max} = 2T_s = 33.34$ μ s and $\nu_{\max}$ varied in the range 500 Hz to 11.25 kHz. It is observed that, compared to OFDM, both DZT-OTFS and MC-OTFS are significantly more resilient to the increase in the maximum Doppler. In addition, DZT-OTFS is found to achieve better performance compared to MC-OTFS over a wide range of $\nu_{\max}$. This illustrates that the DZT-OTFS waveform is more resilient to high Doppler spreads compared to the MC-OTFS waveform.

The choice of roll-off factor (γ) of the RC pulse plays a crucial role in the performance of DZT-OTFS. This is due to the DD domain spread of the RC pulse. As γ increases from 0 to 1, the spread of the RC pulse in the transform domain increases considerably. This spread of the DD domain symbols into the adjacent bins leads to a degraded BER performance. The simulation results in Figs. 8.18(a) and (b) illustrate this point, where the BER is shown as a function of $\nu_{\max}$ and $\tau_{\max}$, respectively, for $\gamma = 0.1, 0.5, 0.9$ at an SNR of 15 dB. The performance is observed to degrade as γ is increased from 0.1 to 0.9 because of the increased spread of the RC pulse.

8.4.4.3 Complexity

A complexity comparison between DZT-OTFS and MC-OTFS in transforming a time domain sequence to the DD domain and vice versa is presented here. For DZT-OTFS, from (8.68), it is observed that for a given (k, l) , $Z_y[k, l]$ can be calculated for all $k = 0, 1, \dots, K - 1$ using the K -point DFT, whose complexity is $O(K \log_2 K)$. Therefore, for $k = 0, 1, \dots, K - 1$ and $l = 0, 1, \dots, L - 1$, the overall complexity for DZT-OTFS is $O(KL \log_2 K)$. For MC-OTFS, transforming a signal from DD to TD via the TF domain involves SFFT, whose complexity is of the order $O(KL \log_2 KL)$, which is higher than the complexity of DZT-OTFS.

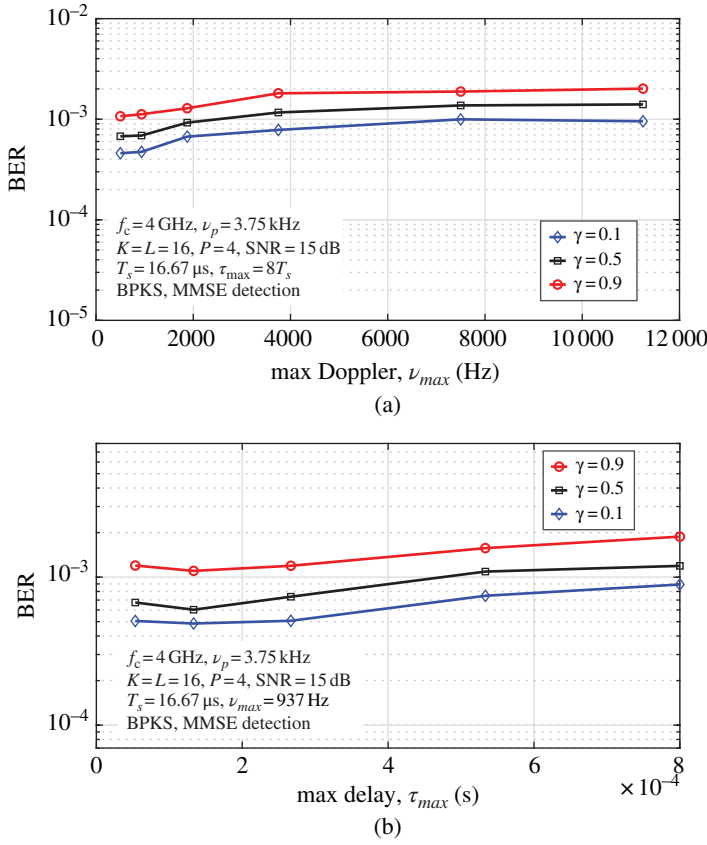


Figure 8.18 Effect of the RC pulse roll-off factor, γ , on the BER performance for varying $\nu_{\max}$ and $\tau_{\max}$. (a) BER vs. $\nu_{\max}$ and (b) BER vs. $\tau_{\max}$.

Appendix 8.A Digital Implementation of Zak-OTFS Modulation

Modern communication systems are based on digital signal processing due to its low cost, low power consumption, and high programmability. This is why contemporary communication systems are realized with digital transceivers, where continuous time signals are sampled at discrete time instances, followed by discrete-time signal processing. In the following, digital implementation of the Zak-OTFS receiver and transmitter is derived in Appendix 8.A.1 and Appendix 8.A.2, respectively. These implementations are based on the discrete counterpart of the continuous Zak transform, suitably referred to as the DZT.

8.A.1 Digital Implementation of the Zak-OTFS Receiver

The Zak-OTFS receiver implementation in Chapter 2 is based on continuous time and continuous DD domain signal processing, which is not amenable to digital implementation (see Table 2.4 and Section 2.6.3). Figure 8.A.1 shows the continuous time and continuous DD domain-based transceiver implementation of Zak-OTFS modulation. Clearly, implementing the continuous DD domain Zak transform of the received TD signal $r_{\text{id}}(t)$ at the receiver (i.e., $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(r_{\text{id}}(t))$)

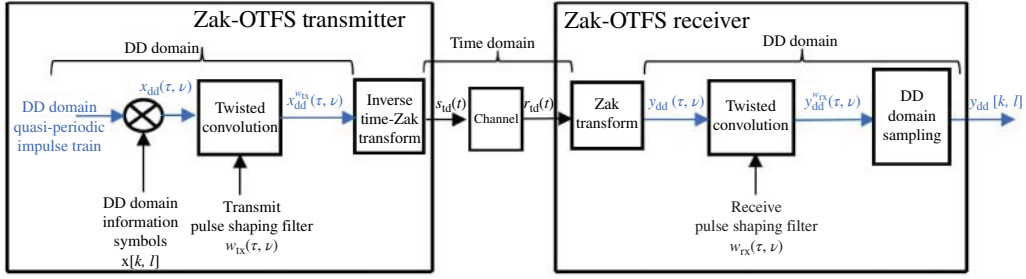


Figure 8.A.1 Zak-OTFS modulation based on the continuous Zak transform.

and the matched filtering operation $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu)$ in the continuous DD domain would require complex analog signal processing.

In the following, it is shown that the receiver output, i.e., $y_{\text{dd}}[k, l] = y_{\text{dd}}^{w_{\text{rx}}}(\tau = \frac{k\tau_p}{M}, \nu = \frac{l\nu_p}{N})$, can be computed through suitable processing of the discrete time samples of the received continuous TD signal $r_{\text{id}}(t)$. Recall from Chapter 2 that $M = B\tau_p$ and $N = T\nu_p$, where B and T are the bandwidth and time, respectively, of the Zak-OTFS-modulated TD signal $s_{\text{id}}(t)$. Let $y^{w_{\text{rx}}}(t)$ denote the TD realization of $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$, i.e.,

$$y^{w_{\text{rx}}}(t) \triangleq \mathcal{Z}_t^{-1}(y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)). \quad (8.A.1)$$

Then, from the definition of the Zak transform in (2.23), it follows that

$$y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} y^{w_{\text{rx}}}(\tau + n\tau_p) e^{-j2\pi n\nu\tau_p}. \quad (8.A.2)$$

Let

$$y^{w_{\text{rx}}}[n] \triangleq y^{w_{\text{rx}}}\left(t = \frac{n\tau_p}{M} = \frac{n}{B}\right), \quad n \in \mathbb{Z}, \quad (8.A.3)$$

denote the discrete time samples of $y^{w_{\text{rx}}}(t)$ at integer multiples of the inverse bandwidth $1/B$. Then,

$$\begin{aligned} y_{\text{dd}}[k, l] &= y_{\text{dd}}^{w_{\text{rx}}}\left(\tau = \frac{k\tau_p}{M}, \nu = \frac{l\nu_p}{N}\right) \\ &\stackrel{(a)}{=} \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} y^{w_{\text{rx}}}\left((k + nM)\frac{\tau_p}{M}\right) e^{-j2\pi \frac{nl}{N}} \\ &\stackrel{(b)}{=} \sqrt{\tau_p} \sum_{p \in \mathbb{Z}} \sum_{q=0}^{N-1} y^{w_{\text{rx}}}[k + (pN + q)M] e^{-j2\pi \frac{(pN+q)l}{N}} \\ &= \sqrt{\tau_p} \sum_{p \in \mathbb{Z}} \sum_{q=0}^{N-1} y^{w_{\text{rx}}}[k + qM + pNM] e^{-j2\pi \frac{ql}{N}} \\ &= \sqrt{\tau_p} \sum_{q=0}^{N-1} \left[\sum_{p \in \mathbb{Z}} y^{w_{\text{rx}}}[k + qM + pNM] \right] e^{-j2\pi \frac{ql}{N}} \\ &= \sqrt{\tau_p} \sum_{q=0}^{N-1} y_p^{w_{\text{rx}}}[k + qM] e^{-j2\pi \frac{ql}{N}}, \end{aligned} \quad (8.A.4)$$

where the discrete time signal

$$y_p^{w_{\text{rx}}}[k] \triangleq \sum_{p \in \mathbb{Z}} y^{w_{\text{rx}}} [k + pMN] \quad (8.A.5)$$

is the periodization of $y^{w_{\text{rx}}}$ with period MN , i.e., $y_p^{w_{\text{rx}}}$ is a periodic discrete time signal with period MN . Also, in (8.A.4), step (a) follows from the expression of $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)$ in (8.A.2). In step (b), we substitute the summation variable n with two variables, $q = n \bmod N$ and $p = \left\lfloor \frac{n}{N} \right\rfloor$, i.e., $n = (pN + q)$.

This transformation from the discrete-time periodic signal $y_p^{w_{\text{rx}}}$ to the discrete DD domain signal $y_{\text{dd}}[k, l]$ is known as the DZT. DZT and its properties are discussed in Appendix 8.B. In Theorem 8.1 in Appendix 8.B, it is shown that the DZT of any periodic signal is a quasi-periodic discrete DD domain signal. Therefore, $y_{\text{dd}}[k, l]$ is quasi-periodic and it suffices to compute it only for $k = 0, 1, \dots, M-1, l = 0, 1, \dots, N-1$.

Next, it is shown how discrete time samples $y^{w_{\text{rx}}}[n]$ can be computed from the received continuous TD signal $r_{\text{td}}(t)$. From Theorem 2.3 in Appendix 2.C, it is known that for a DD domain function $f(\tau, \nu)$ and a quasi-periodic DD domain signal $a_{\text{dd}}(\tau, \nu)$ having TD realization $a(t) = \mathcal{Z}_t^{-1}(a_{\text{dd}}(\tau, \nu))$, the DD domain realization of the TD signal $b(t) = \iint f(\tau, \nu) a(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu$ is $f(\tau, \nu) *_{\sigma} a_{\text{dd}}(\tau, \nu)$. In other words,

$$\mathcal{Z}_t^{-1}(f(\tau, \nu) *_{\sigma} a_{\text{dd}}(\tau, \nu)) = \iint f(\tau, \nu) a(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu. \quad (8.A.6)$$

Using this result with $f(\tau, \nu) = w_{\text{rx}}(\tau, \nu)$, $a(\tau, \nu) = y_{\text{dd}}(\tau, \nu)$, it follows that

$$\begin{aligned} y^{w_{\text{rx}}}(t) &= \mathcal{Z}_t^{-1}(y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu)) \\ &\stackrel{(a)}{=} \mathcal{Z}_t^{-1}(w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu)) \\ &\stackrel{(b)}{=} \iint w_{\text{rx}}(\tau, \nu) r_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu, \end{aligned} \quad (8.A.7)$$

where step (a) follows from the fact that $y_{\text{dd}}^{w_{\text{rx}}}(\tau, \nu) = w_{\text{rx}}(\tau, \nu) *_{\sigma} y_{\text{dd}}(\tau, \nu)$. Step (b) follows from the fact that $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(r_{\text{td}}(t))$ (see Table 2.4). For factorizable receive pulse shaping filter $w_{\text{rx}}(\tau, \nu) = w_{\text{rx},1}(\tau) w_{\text{rx},2}(\nu)$,

$$\begin{aligned} y^{w_{\text{rx}}}(t) &= \int w_{\text{rx},1}(\tau) r_{\text{td}}(t - \tau) \left[\int w_{\text{rx},2}(\nu) e^{j2\pi\nu(t-\tau)} d\nu \right] d\tau \\ &= \int w_{\text{rx},1}(\tau) r_{\text{td}}^w(t - \tau) d\tau \\ &= w_{\text{rx},1}(t) \star r_{\text{td}}^w(t), \end{aligned} \quad (8.A.8)$$

where $\star$ denotes the linear convolution operation and

$$r_{\text{td}}^w(t) \triangleq r_{\text{td}}(t) W_{\text{rx},2}(t), \quad W_{\text{rx},2}(t) \triangleq \int w_{\text{rx},2}(\nu) e^{j2\pi\nu t} d\nu. \quad (8.A.9)$$

In other words, $y^{w_{\text{rx}}}(t)$ is derived from the received TD signal $r_{\text{td}}(t)$ by firstly time limiting it to the TD support of the inverse Fourier transform of the Doppler domain pulse $w_{\text{rx},2}(\cdot)$, resulting in the time-limited TD signal $r_{\text{td}}^w(t)$, followed by band-limiting $r_{\text{td}}^w(t)$ by linearly convolving it with the delay domain pulse $w_{\text{rx},1}(\cdot)$, resulting in $y^{w_{\text{rx}}}(t)$. Both these operations of limiting the received continuous TD signal in time and bandwidth can be achieved by discrete time processing of samples of $r_{\text{td}}(t)$, resulting in discrete time samples $y^{w_{\text{rx}}}[n]$ of $y^{w_{\text{rx}}}(t)$. An illustration of the DZT-based Zak-OTFS receiver is presented in Fig. 8.A.2.

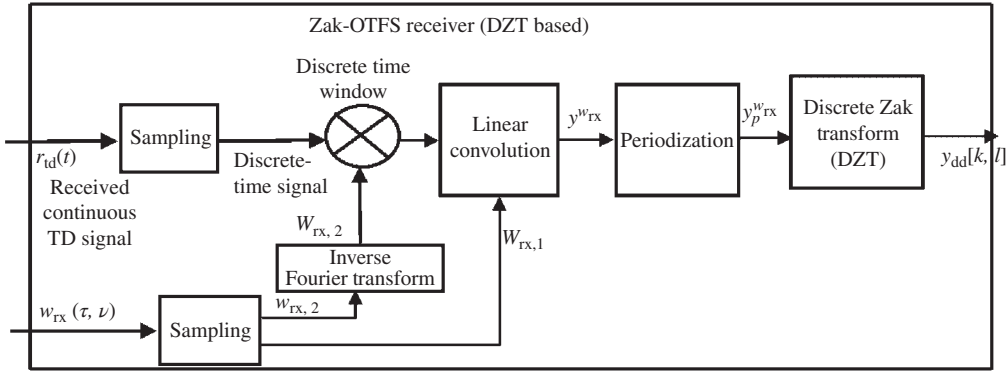


Figure 8.A.2 DZT-based Zak-OTFS receiver.

8.A.2 Digital Implementation of the Zak-OTFS Transmitter

From Fig. 8.A.1, the transmit TD signal is given by $s_{td}(t) = \mathcal{Z}_t^{-1} (x_{dd}^{w_{tx}}(\tau, \nu))$. Also, $x_{dd}^{w_{tx}}(\tau, \nu) = w_{tx}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu)$. Using the result in (8.A.6) with $f(\tau, \nu) = w_{tx}(\tau, \nu)$, $a(\tau, \nu) = x_{dd}(\tau, \nu)$, it follows that

$$\begin{aligned} s_{td}(t) &= \mathcal{Z}_t^{-1} (x_{dd}^{w_{tx}}(\tau, \nu)) = \mathcal{Z}_t^{-1} (w_{tx}(\tau, \nu) *_{\sigma} x_{dd}(\tau, \nu)) \\ &= \iint w_{tx}(\tau, \nu) x(t - \tau) e^{j2\pi\nu(t-\tau)} d\tau d\nu, \end{aligned} \quad (8.A.10)$$

where $x(t)$ is the TD realization of the DD domain information signal $x_{dd}(\tau, \nu)$, i.e.,

$$x(t) = \mathcal{Z}_t^{-1} (x_{dd}(\tau, \nu)). \quad (8.A.11)$$

From Table 2.4, it follows that

$$x_{dd}(\tau, \nu) = \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \underbrace{\sum_{n,m \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \delta\left(\tau - \frac{(k+nM)}{B}\right) \delta\left(\nu - \frac{(l+mN)}{T}\right)}_{=x_{dd}(\tau, \nu; \tau_0=k\tau_p/M, \nu_0=l\nu_p/N)}, \quad (8.A.12)$$

where the expression of $x_{dd}(\tau, \nu; \tau_0, \nu_0)$ follows from (2.26). Recall that $x_{dd}(\tau, \nu; \tau_0 = k\tau_p/M, \nu_0 = l\nu_p/N)$ is the information carrying quasi-periodic DD domain pulse at $(\tau_0 = k\tau_p/M, \nu_0 = l\nu_p/N)$, whose TD realization is given by (2.38), i.e.,

$$\mathcal{Z}_t^{-1} (x_{dd}(\tau, \nu; \tau_0, \nu_0)) = \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} e^{j2\pi n \nu_0 \tau_p} \delta(t - \tau_0 - n\tau_p). \quad (8.A.13)$$

Using this in (8.A.11) gives

$$\begin{aligned} x(t) &= \mathcal{Z}_t^{-1} \left(\sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \sum_{n,m \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \delta\left(\tau - \frac{(k+nM)}{B}\right) \delta\left(\nu - \frac{(l+mN)}{T}\right) \right) \\ &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \mathcal{Z}_t^{-1} \left(\sum_{n,m \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \delta\left(\tau - \frac{(k+nM)}{B}\right) \delta\left(\nu - \frac{(l+mN)}{T}\right) \right) \\ &= \sqrt{\tau_p} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \sum_{n \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \delta\left(t - \frac{(k+nM)}{B}\right), \end{aligned} \quad (8.A.14)$$

where the last step follows from (8.A.13). Using this expression of $x(t)$ on the RHS of (8.A.10) gives

$$\begin{aligned}
 s_{\text{td}}(t) &= \sqrt{\tau_p} \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} x[k, l] \sum_{n \in \mathbb{Z}} e^{j2\pi \frac{nl}{N}} \iint w_{\text{tx}}(\tau, \nu) \delta \left(t - \tau - \frac{(k + nM)}{B} \right) e^{j2\pi \nu(t-\tau)} d\tau d\nu \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \sum_{k=0}^{M-1} \left[\sum_{l=0}^{N-1} x[k, l] e^{j2\pi \frac{nl}{N}} \right] \iint w_{\text{tx},1}(\tau) w_{\text{tx},2}(\nu) \delta \left(t - \tau - \frac{(k + nM)}{B} \right) e^{j2\pi \nu(t-\tau)} d\tau d\nu \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \sum_{k=0}^{M-1} \left[\sum_{l=0}^{N-1} x[k, l] e^{j2\pi \frac{nl}{N}} \right] \int w_{\text{tx},1}(\tau) \delta \left(t - \tau - \frac{(k + nM)}{B} \right) \underbrace{\left[\int w_{\text{tx},2}(\nu) e^{j2\pi \nu(t-\tau)} d\nu \right]}_{\triangleq W_{\text{tx},2}(t-\tau)} d\tau,
 \end{aligned} \tag{8.A.15}$$

where we consider the factorizable transmit pulse shaping filter $w_{\text{tx}}(\tau, \nu) = w_{\text{tx},1}(\tau) w_{\text{tx},2}(\nu)$. Note that

$$W_{\text{tx},2}(t) \triangleq \int w_{\text{tx},2}(\nu) e^{j2\pi \nu t} d\nu \tag{8.A.16}$$

is the inverse Fourier transform of the Doppler domain pulse $w_{\text{tx},2}(\nu)$.

Furthermore, consider a quasi-periodic discrete DD domain signal $x_{\text{dd}}[k, l]$, which is the quasi-periodic extension of the information symbols $x[k, l]$, i.e.,

$$x_{\text{dd}}[k, l] = x[k \bmod M, l \bmod N] e^{j2\pi \frac{\lfloor \frac{k}{M} \rfloor l}{N}}, \quad k, l \in \mathbb{Z}. \tag{8.A.17}$$

Next, for any quasi-periodic discrete DD domain signal, its discrete time realization is given by the inverse discrete Zak transform (IDZT),

$$\begin{aligned}
 s &\triangleq \text{IDZT} \left(x_{\text{dd}} \right), \quad \text{i.e.,} \\
 s[q] &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_{\text{dd}}[q, l], \quad q \in \mathbb{Z},
 \end{aligned} \tag{8.A.18}$$

is the discrete time realization of $x_{\text{dd}}[k, l]$ (see Appendix 8.B on DZT, IDZT, and their properties). Therefore, from (8.A.17) and (8.A.18), it follows that

$$s[q] = \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x[q \bmod M, l] e^{j2\pi \frac{\lfloor \frac{q}{M} \rfloor l}{N}}, \quad q \in \mathbb{Z}. \tag{8.A.19}$$

Therefore, for any $n \in \mathbb{Z}$ and $k = 0, 1, \dots, M-1$,

$$s[k + nM] = \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x[k, l] e^{j2\pi \frac{nl}{N}}. \tag{8.A.20}$$

Using (8.A.20) in (8.A.15) gives

$$\begin{aligned}
 s_{\text{td}}(t) &= \sqrt{N\tau_p} \sum_{n \in \mathbb{Z}} \sum_{k=0}^{M-1} s[k + nM] \int w_{\text{tx},1}(\tau) \delta \left(t - \tau - \frac{(k + nM)}{B} \right) W_{\text{tx},2}(t - \tau) d\tau \\
 &= \sqrt{N\tau_p} \sum_{n \in \mathbb{Z}} \sum_{k=0}^{M-1} s[k + nM] W_{\text{tx},2}((k + nM)/B) w_{\text{tx},1} \left(t - \frac{(k + nM)}{B} \right) \\
 &= \sqrt{\tau_p} \sum_{n \in \mathbb{Z}} \sum_{k=0}^{M-1} s_w[k + nM] w_{\text{tx},1} \left(t - \frac{(k + nM)}{B} \right)
 \end{aligned}$$

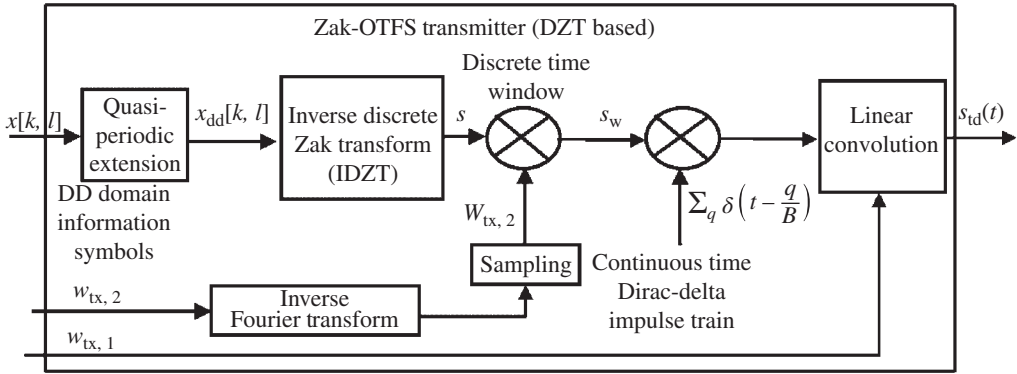


Figure 8.A.3 DZT-based Zak-OTFS transmitter.

$$\begin{aligned}
 &= \sqrt{\tau_p} \sum_{q \in \mathbb{Z}} s_w[q] w_{tx,1}(t - q/B) \\
 &= \sqrt{\tau_p} \left[\sum_{q \in \mathbb{Z}} s_w[q] \delta(t - q/B) \right] \star w_{tx,1}(t),
 \end{aligned} \tag{8.A.21}$$

where

$$s_w[q] \triangleq \sqrt{N} s[q] W_{tx,2}(q/B), \tag{8.A.22}$$

and, in the second last step in (8.A.21), we have substituted the summation over k and n with a single summation over $q = (k + nM)$ since the RHS in the previous step depends on k and M only through $(k + nM)$. Note that $W_{tx,2}(\cdot)$ limits the duration of the discrete time signal $s[q]$, i.e., it behaves as a time windowing waveform.

The DZT-based digital implementation of the Zak-OTFS transmitter is shown in Fig. 8.A.3.

Appendix 8.B Discrete Zak Transform

The DZT and its inverse are the discrete counterparts of the continuous Zak transform and its inverse. DZT gives the discrete DD domain realization of a discrete time signal, and IDZT gives the discrete time realization of a quasi-periodic discrete DD domain signal.

Definition DZT is parameterized by discrete delay and Doppler periods M and N , respectively. Given a periodic discrete time signal with period MN , i.e., $x[q] = x[q + MN]$ for all $q \in \mathbb{Z}$, its discrete DD realization $x_{dd}[k, l]$ is given by

$$\begin{aligned}
 x_{dd} &= \text{DZT}(x) \quad , \quad \text{i.e.,} \\
 x_{dd}[k, l] &= \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} x[k + qM] e^{-j2\pi \frac{ql}{N}}, \quad k, l \in \mathbb{Z}.
 \end{aligned} \tag{8.B.1}$$

The discrete DD realization of any discrete time signal is quasi-periodic, i.e., it satisfies (2.L.2) (see Appendix 2.L).

Theorem 8.1 The DZT $x_{dd}[k, l]$ of a periodic discrete time signal $x[q]$ having period MN is a quasi-periodic discrete DD domain signal, i.e., for any $n, m \in \mathbb{Z}$,

$$x_{dd}[k + nM, l + mN] = x_{dd}[k, l] e^{j2\pi \frac{nl}{N}}. \tag{8.B.2}$$

Proof: From (8.B.1), it follows that

$$\begin{aligned}
 x_{\text{dd}}[k + nM, l + mN] &= \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} x[k + nM + qM] e^{-j2\pi \frac{q(l+mN)}{N}} \\
 &\stackrel{(a)}{=} \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{i=n}^{n+N-1} x[k + iM] e^{-j2\pi \frac{il}{N}} = \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{iM=nM}^{M(n+N-1)} x[k + iM] e^{-j2\pi \frac{iMl}{MN}} \\
 &\stackrel{(b)}{=} \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{p=nM, p \bmod M=0}^{M(n+N-1)} x[k + p] e^{-j2\pi \frac{pl}{MN}} \\
 &= \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{p=nM, p \bmod M=0}^{M(n+N-1)} x \left[k + MN \left\lfloor \frac{p}{MN} \right\rfloor + p \bmod MN \right] \\
 &\quad \times e^{-j2\pi \frac{(MN \lfloor \frac{p}{MN} \rfloor + p \bmod MN)l}{MN}} \\
 &\stackrel{(c)}{=} \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{p=nM, p \bmod M=0}^{M(n+N-1)} x[k + p \bmod MN] e^{-j2\pi \frac{(p \bmod MN)l}{MN}} \\
 &\stackrel{(d)}{=} \frac{1}{\sqrt{N}} e^{j2\pi \frac{nl}{N}} \sum_{q'=0}^{(N-1)} x[k + Mq'] e^{-j2\pi \frac{q'l}{N}} \\
 &= e^{j2\pi \frac{nl}{N}} x_{\text{dd}}[k, l], \tag{8.B.3}
 \end{aligned}$$

where step (a) follows from substituting the summation variable q with $i = (q + n)$. Step (b) follows from substituting iM with p . Step (c) follows from the fact that the discrete time signal $x[q]$ is periodic with period MN . Step (d) follows from the fact that the RHS in step (c) depends on p only through $p \bmod MN$. Since p is also an integer multiple of M , $p \bmod MN = nM \bmod MN$ on the RHS can be substituted with Mq' , where $q' = 0, \dots, N-1$.

Definition IDZT is the inverse of DZT. Given a quasi-periodic DD domain signal $x_{\text{dd}}[k, l]$, $k, l \in \mathbb{Z}$, IDZT gives its discrete time realization $s[q]$, which is given by

$$\begin{aligned}
 s &\stackrel{\Delta}{=} \text{IDZT} \left(x_{\text{dd}} \right), \text{ i.e.,} \\
 s[q] &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_{\text{dd}}[q, l], \quad q \in \mathbb{Z}. \tag{8.B.4}
 \end{aligned}$$

Note that $s[q]$ is periodic with period MN since

$$\begin{aligned}
 s[q + MN] &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_{\text{dd}}[q + MN, l] \\
 &\stackrel{(a)}{=} \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_{\text{dd}}[q, l] \\
 &= s[q], \tag{8.B.5}
 \end{aligned}$$

where step (a) follows from the fact that a quasi-periodic DD domain signal is periodic along both delay and Doppler domains with period MN , i.e., $x_{\text{dd}}[k + nMN, l + mMN] = x_{\text{dd}}[k, l]$ for all $n, m \in \mathbb{Z}$.

Theorem 8.2 DZT and IDZT are inverses of each other. Consider a periodic discrete time signal $s[q], q \in \mathbb{Z}$, having period MN . The DZT of $s[q]$ is the quasi-periodic signal $x_{\text{dd}}[k, l]$ given by

$$x_{\text{dd}}[k, l] = \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} s[q + qM] e^{-j2\pi \frac{ql}{N}}, \quad k, l \in \mathbb{Z}. \quad (8.B.6)$$

Then, the IDZT of $x_{\text{dd}}[k, l]$ is $s[q]$, i.e.,

$$\text{IDZT}(\text{DZT}(s)) = s. \quad (8.B.7)$$

Proof: Let $g[q], q \in \mathbb{Z}$, be the IDZT of $x_{\text{dd}}[k, l]$. Then,

$$\begin{aligned} g[q] &\stackrel{(a)}{=} \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} x_{\text{dd}}[q, l] \\ &\stackrel{(b)}{=} \frac{1}{N} \sum_{l=0}^{N-1} \sum_{k=0}^{N-1} s[q + kM] e^{-j2\pi \frac{kl}{N}} \\ &= \sum_{k=0}^{N-1} s[q + kM] \left(\frac{1}{N} \sum_{l=0}^{N-1} e^{-j2\pi \frac{kl}{N}} \right) \\ &= \sum_{k=0}^{N-1} s[q + kM] \delta[k \bmod N] \\ &= s[q], \end{aligned} \quad (8.B.8)$$

where step (a) follows from (8.B.4) and step (b) follows from using the expression of $x_{\text{dd}}[k, l]$ from the RHS in (8.B.6).

8.B.1 Discrete Delay and Doppler Shifts

Consider a periodic discrete time signal $x[k], k \in \mathbb{Z}$, having period MN and whose DD domain realization is a quasi-periodic DD domain signal $x_{\text{dd}}[k, l]$.

A discrete delay of k_0 samples and discrete Doppler shift of l_0 ($k_0, l_0 \in \mathbb{Z}$) to the discrete time periodic signal $x[k]$ results in another periodic signal $x[k - k_0] e^{j2\pi \frac{l_0(k-k_0)}{MN}}, k \in \mathbb{Z}$. The effect of a delay and Doppler shift on a signal's DD representation is given by the following theorem.

Theorem 8.3 For any discrete time periodic signal $x[k]$ having DD realization (i.e., DZT) $x_{\text{dd}}[k, l]$, the DD domain realization of the delay and Doppler shifted signal $x[k - k_0] e^{j2\pi \frac{l_0(k-k_0)}{MN}}$ is given by

$$\begin{aligned} \text{DZT} \left(x[k - k_0] e^{j2\pi \frac{l_0(k-k_0)}{MN}} \right) &= (\delta[k - k_0] \delta[l - l_0]) *_\sigma x_{\text{dd}}[k, l] \\ &= e^{j2\pi \frac{l_0(k-k_0)}{MN}} x_{\text{dd}}[k - k_0, l - l_0], \end{aligned} \quad (8.B.9)$$

where $*_\sigma$ denotes the twisted convolution in the discrete DD domain.

Proof: The DZT of $x[k - k_0] e^{j2\pi \frac{l_0(k-k_0)}{MN}}$ at the discrete DD point (k, l) is given by

$$\begin{aligned} \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} x[k - k_0 + qM] e^{j2\pi \frac{l_0(k+qM-k_0)}{MN}} e^{-j2\pi q \frac{l}{N}} &= e^{j2\pi \frac{l_0(k-k_0)}{MN}} \underbrace{\frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} x[k - k_0 + qM] e^{-j2\pi q \frac{(l-l_0)}{N}}}_{=x_{\text{dd}}[k-k_0, l-l_0]} \\ &= e^{j2\pi \frac{l_0(k-k_0)}{MN}} x_{\text{dd}}[k - k_0, l - l_0]. \end{aligned} \quad (8.B.10)$$

Furthermore, from the definition of discrete twisted convolution in (2.65), it follows that

$$\begin{aligned} (\delta[k - k_0] \delta[l - l_0]) *_{\sigma} x_{\text{dd}}[k, l] &= \sum_{k', l' \in \mathbb{Z}} \delta[k' - k_0] \delta[l' - l_0] x_{\text{dd}}[k - k', l - l'] e^{j2\pi \frac{l'(k-k')}{MN}} \\ &= e^{j2\pi \frac{l_0(k-k_0)}{MN}} x_{\text{dd}}[k - k_0, l - l_0]. \end{aligned} \quad (8.B.11)$$

In Chapter 2, the noise-free discrete DD domain output is given by $y_{\text{dd}}[k, l] = h_{\text{eff}}[k, l] *_{\sigma} x_{\text{dd}}[k, l]$, where $x_{\text{dd}}[k, l]$ is the DD domain input information signal and $h_{\text{eff}}[k, l]$ is the effective discrete DD domain channel filter. Let the non-zero taps of $h_{\text{eff}}[k, l]$ have support set $\mathcal{S}_{\text{ch}}$, i.e.,

$$h_{\text{eff}}[k, l] = \sum_{(k_i, l_i) \in \mathcal{S}_{\text{ch}}} h_{\text{eff}}[k_i, l_i] \delta[k - k_i] \delta[l - l_i]. \quad (8.B.12)$$

Therefore, the discrete DD domain output is

$$\begin{aligned} y_{\text{dd}}[k, l] &= h_{\text{eff}}[k, l] *_{\sigma} x_{\text{dd}}[k, l] \\ &= \sum_{(k_i, l_i) \in \mathcal{S}_{\text{ch}}} h_{\text{eff}}[k_i, l_i] \underbrace{((\delta[k - k_i] \delta[l - l_i]) *_{\sigma} x_{\text{dd}}[k, l])}_{x_{\text{dd}}[k, l] \text{ DD shifted by } (k_i, l_i), \text{ see Theorem 8.3}}. \end{aligned} \quad (8.B.13)$$

Therefore, the channel output is a weighted superposition of the input signal delay/Doppler shifted by the channel filter taps.

8.B.2 Discrete Pulsones

In Chapter 2, it was shown that continuous TD/FD pulsones are the Zak-OTFS carrier waveforms. Similarly, for the discrete time Zak-OTFS-modulated signal $s[q]$ (see (8.A.19)), in the following, it is shown that the carrier waveforms are discrete time pulsones which are the DD domain realization of quasi-periodic discrete DD domain pulses. The following theorem gives the expression for the discrete-time realization of discrete DD domain quasi-periodic pulses.

Theorem 8.4 The discrete time realization of the quasi-periodic discrete DD pulse at (k_0, l_0) , $k_0, l_0 \in \mathbb{Z}$, is given by

$$\phi[k; k_0, l_0] \triangleq \frac{1}{\sqrt{N}} \sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] e^{j2\pi \frac{nl_0}{N}}. \quad (8.B.14)$$

Proof: The discrete DD domain quasi-periodic pulse at the origin (i.e., $(0, 0)$) is given by

$$\phi_{\text{dd}}[k, l; 0, 0] = \sum_{n, m \in \mathbb{Z}} \delta[k - nM] \delta[l - mN]. \quad (8.B.15)$$

The quasi-periodic pulse at any other location (k_0, l_0) is obtained by shifting the pulse at the origin along delay by k_0 and along Doppler by l_0 . From Theorem 8.3, it then follows that the quasi-periodic DD pulse at (k_0, l_0) is given by (see also (2.68))

$$\begin{aligned}\phi_{\text{dd}}[k, l; k_0, l_0] &= (\delta[k - k_0] \delta[l - l_0]) *_\sigma \phi_{\text{dd}}[k, l; 0, 0] \\ &= \sum_{n, m \in \mathbb{Z}} \delta[k - k_0 - nM] \delta[l - l_0 - mN] e^{j2\pi \frac{l_0(k-k_0)}{MN}} \\ &= \sum_{n, m \in \mathbb{Z}} \delta[k - k_0 - nM] \delta[l - l_0 - mN] e^{j2\pi \frac{nl_0}{N}}.\end{aligned}\quad (8.B.16)$$

The discrete time realization of this DD pulse is given by its IDZT, i.e.,

$$\begin{aligned}\phi[k; k_0, l_0] &\triangleq \text{IDZT}(\phi_{\text{dd}}[k, l; k_0, l_0]) \\ &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} \phi_{\text{dd}}[k, l; k_0, l_0] \\ &= \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} \delta[k - k_0 - nM] \delta[l - l_0 - mN] e^{j2\pi \frac{l_0(k-k_0)}{MN}} \\ &= \frac{1}{\sqrt{N}} \sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] \underbrace{\left(\sum_{l=0}^{N-1} \sum_{m \in \mathbb{Z}} \delta[l - l_0 - mN] \right)}_{=1} e^{j2\pi \frac{l_0(k-k_0)}{MN}} \\ &= \frac{1}{\sqrt{N}} \underbrace{\left[\sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] \right]}_{\text{pulse train}} \underbrace{\left[e^{j2\pi \frac{l_0(k-k_0)}{MN}} \right]}_{\text{tone of freq. } \frac{l_0}{N}} \\ &= \frac{1}{\sqrt{N}} \sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] e^{j2\pi \frac{nl_0}{N}}.\end{aligned}\quad (8.B.17)$$

From the above equation, it is clear that the discrete time realization of a quasi-periodic discrete DD pulse at (k_0, l_0) is simply a Dirac-delta pulse train (with the n th pulse at discrete time $(k_0 + nM)$) modulated by a discrete tone of frequency l_0/N .

The following theorem shows that the set of MN discrete time pulsones $\phi[k; k_0, l_0]$, $k_0 = 0, 1, \dots, M-1$, $l_0 = 0, 1, \dots, N-1$ are the discrete time carrier waveforms.

Theorem 8.5 Any discrete time periodic signal $x[k]$ having period MN can be expressed as a linear combination of the discrete time pulsones, where the weight for the (k_0, l_0) th pulsone is $x_{\text{dd}}[k_0, l_0]$ (the DZT of $x[k]$ evaluated at (k_0, l_0)), i.e.,

$$x[k] = \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \phi[k; k_0, l_0]. \quad (8.B.18)$$

A special case of this equation is with $x_{\text{dd}}[k, l]$ as the DD domain quasi-periodic information signal in the Zak-OTFS digital transmitter architecture shown in Fig. 8.A.3. For this special case, $x[k]$ in (8.B.18) is indeed the discrete time Zak-OTFS-modulated signal $s[\cdot]$ (output of IDZT in Fig. 8.A.3, see (8.A.18)).

Proof: The RHS in (8.B.18) is given by

$$\begin{aligned}
 & \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \phi[k; k_0, l_0] \\
 & \stackrel{(a)}{=} \frac{1}{N} \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} \sum_{q=0}^{N-1} \sum_{n \in \mathbb{Z}} x[k_0 + qM] \delta[k - k_0 - nM] e^{j2\pi \frac{(n-q)l_0}{N}} \\
 & \stackrel{(a)}{=} \sum_{k_0=0}^{M-1} \sum_{q=0}^{N-1} \sum_{n \in \mathbb{Z}} x[k_0 + qM] \delta[k - k_0 - nM] \underbrace{\left[\frac{1}{N} \sum_{l_0=0}^{N-1} e^{j2\pi \frac{(n-q)l_0}{N}} \right]}_{=\delta[(n-q) \bmod N]} \\
 & \stackrel{(b)}{=} \sum_{k_0=0}^{M-1} \sum_{q=0}^{N-1} \sum_{m \in \mathbb{Z}} x[k_0 + qM] \delta[k - k_0 - (mN + q)M], \tag{8.B.19}
 \end{aligned}$$

where step (a) follows from the expressions of $x_{\text{dd}}[k, l]$ and $\phi[k; k_0, l_0]$ in (8.B.1) and (8.B.14), respectively. Step (b) follows from the fact that non-zero contribution to the sum is only for those terms for which n is congruent to q modulo N , i.e., n is of the form $(mN + q)$ for integer m . On the RHS, the Dirac delta's are located at $k = (k_0 + qM + mMN)$, $q = 0, 1, \dots, N-1$, $m \in \mathbb{Z}$. Therefore, for a given k , the summations over k_0, m , and q give a non-zero term only for $k_0 = k \bmod M$, $q = \left\lfloor \frac{(k-k_0)}{M} \right\rfloor \bmod N$, and $m = \frac{(k-k_0-qM)}{MN}$. Therefore,

$$\begin{aligned}
 \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \phi[k; k_0, l_0] &= x \left[k_0 + M \left(\left\lfloor \frac{(k-k_0)}{M} \right\rfloor \bmod N \right) \right] \\
 &= x \left[k_0 + M \left(\frac{(k-k_0)}{M} - \left\lfloor \frac{(k-k_0)}{MN} \right\rfloor N \right) \right] \\
 &= x \left[k - MN \left\lfloor \frac{(k-k_0)}{MN} \right\rfloor \right] \\
 &= x[k], \tag{8.B.20}
 \end{aligned}$$

where the last step follows from the fact that $x[k]$ has period MN .

Next, with $x_{\text{dd}}[k, l]$ as the quasi-periodic Zak-OTFS information signal, it follows that the linear combination on the RHS of (8.B.18) gives

$$\begin{aligned}
 & \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \phi[k; k_0, l_0] \\
 & \stackrel{(a)}{=} \frac{1}{\sqrt{N}} \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] e^{j2\pi \frac{n l_0}{N}} \\
 & = \frac{1}{\sqrt{N}} \sum_{l_0=0}^{N-1} \sum_{k_0=0}^{M-1} \underbrace{\sum_{n \in \mathbb{Z}} x_{\text{dd}}[k_0, l_0] e^{j2\pi \frac{n l_0}{N}}}_{=x_{\text{dd}}[k_0+nM, l_0]} \delta[k - k_0 - nM] \\
 & \stackrel{(b)}{=} \frac{1}{\sqrt{N}} \sum_{l_0=0}^{N-1} \sum_{k_0=0}^{M-1} \sum_{n \in \mathbb{Z}} x_{\text{dd}}[k_0 + nM, l_0] \delta[k - k_0 - nM]
 \end{aligned}$$

$$\begin{aligned}
& \stackrel{(c)}{=} \frac{1}{\sqrt{N}} \sum_{l_0=0}^{N-1} \underbrace{\sum_{q \in \mathbb{Z}} x_{\text{dd}}[q, l_0] \delta[k - q]}_{=x_{\text{dd}}[k, l_0]} \\
& = \frac{1}{\sqrt{N}} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k, l_0] \\
& = s[k],
\end{aligned} \tag{8.B.21}$$

where step (a) follows from substituting $\phi[k; k_0, l_0]$ on the RHS with its expression in (8.B.14). Step (b) follows from the fact that $x_{\text{dd}}[k, l]$ is quasi-periodic (see (2.L.2) in Appendix 2.L), and, therefore, $x_{\text{dd}}[k_0 + nM, l_0] = e^{j2\pi \frac{n l_0}{N}} x_{\text{dd}}[k_0, l_0]$. In step (b), the RHS depends on the summation variables k_0 and n only through $(k_0 + nM)$. Therefore, in step (c), these two summation variables are substituted with a single summation variable $q = (k_0 + nM)$. Since $k_0 = 0, 1, \dots, M-1$ and $n \in \mathbb{Z}$, it follows that $q \in \mathbb{Z}$. Step (d) follows from the expression of the discrete time Zak-OTFS-modulated signal $s[\cdot]$ in (8.A.18).

The inner product between any two periodic discrete time signals, $a[k]$ and $b[k]$, both having period MN , is given by

$$\langle a, b \rangle = \sum_{k=0}^{MN-1} a[k] b^*[k], \tag{8.B.22}$$

where $*$ denotes the complex conjugation. Two discrete time periodic signals are orthogonal if and only if their inner product is zero. The following theorem shows that any two distinct discrete time pulsones $\phi[k; k_1, l_1]$ and $\phi[k; k_2, l_2]$, $(k_1, l_1) \neq (k_2, l_2)$, $k_1, k_2 \in \{0, 1, \dots, M-1\}$, $l_1, l_2 \in \{0, 1, \dots, N-1\}$, are orthogonal. In other words, the MN discrete time pulsones form an orthogonal basis for discrete time signals having period MN .

Theorem 8.6 The MN discrete time pulsones $\phi[k; k_0, l_0]$, $k_0 = 0, 1, \dots, M-1$, $l_0 = 0, 1, \dots, N-1$, form an orthonormal basis for discrete time periodic signals having period MN .

Proof: From the expression of the discrete time pulsones in (8.B.14), it follows that the (k_1, l_1) th discrete time pulsones $\phi[k; k_1, l_1]$ has Dirac-delta impulses at the discrete time instances $(k_1 + nM)$, $n \in \mathbb{Z}$, whereas the (k_2, l_2) th discrete time pulsones has Dirac-delta impulses at the discrete time instances $(k_2 + nM)$, $n \in \mathbb{Z}$. When $k_1 \neq k_2$, the time location of the Dirac-delta impulses for the two pulsones do not overlap, and, therefore, their inner product is zero.

Next, if the delay domain index of two pulsones is the same, i.e., $k_1 = k_2$, but their Doppler domain index is distinct, i.e., $l_1 \neq l_2$, then the inner product between them is given by

$$\begin{aligned}
& \sum_{k=0}^{MN-1} \phi[k; k_1, l_1] \phi^*[k; k_1, l_2] \\
& \stackrel{(a)}{=} \frac{1}{N} \sum_{k=0}^{MN-1} \sum_{n_1 \in \mathbb{Z}} \sum_{n_2 \in \mathbb{Z}} \delta[k - k_1 - n_1 M] e^{j2\pi \frac{n_1 l_1}{N}} \delta[k - k_1 - n_2 M] e^{-j2\pi \frac{n_2 l_2}{N}} \\
& \stackrel{(b)}{=} \frac{1}{N} \sum_{k=0}^{MN-1} \sum_{n_1 \in \mathbb{Z}} \delta[k - k_1 - n_1 M] e^{j2\pi n_1 \frac{(l_1 - l_2)}{N}}
\end{aligned}$$

$$\begin{aligned}
& \stackrel{(c)}{=} \frac{1}{N} \sum_{n_1=0}^{N-1} e^{j2\pi n_1 \frac{(l_1-l_2)}{N}} \\
& = \delta[l_1 - l_2],
\end{aligned} \tag{8.B.23}$$

where step (a) follows from the expression of the discrete time pulsones in (8.B.14). Step (b) follows from the fact that on the RHS in step (a), only those terms are non-zero for which $n_1 = n_2$ since otherwise $\delta[k - k_1 - n_1 M] \delta[k - k_1 - n_2 M] = 0$. On the RHS of step (b), the Dirac-delta impulses are located at discrete time instances $(k_1 + n_1 M)$. Since the summation variable $k = 0, \dots, MN - 1$ and $k_1 \in \{0, 1, \dots, M - 1\}$, in step (c), the summation over n_1 is restricted to $n_1 = 0, 1, \dots, N - 1$. Therefore, from (8.B.23) and the paragraphs preceding it, it is clear that any two pulsones whose delay or Doppler index differ are orthogonal.

With $l_1 = l_2$ in (8.B.23), it also follows that the inner product of a pulsone with itself is unity. Therefore,

$$\begin{aligned}
\langle \phi[k; k_1, l_1], \phi[k; k_2, l_2] \rangle &= \sum_{k=0}^{MN-1} \phi[k; k_1, l_1] \phi^*[k; k_2, l_2] \\
&= \delta[k_1 - k_2] \delta[l_1 - l_2].
\end{aligned} \tag{8.B.24}$$

Furthermore, from Theorem 8.5, we know that any discrete time periodic signal can be expressed as a linear combination of the MN pulsones. Therefore, the MN discrete time pulsones $\phi[k; k_0, l_0]$, $k_0 = 0, \dots, M - 1$, $l_0 = 0, \dots, N - 1$, form an orthonormal basis for discrete time periodic signals having period MN .

The following corollary to Theorem 8.6 shows that the DZT of any periodic discrete time signal $x[k]$ computed at (k_0, l_0) , i.e., $x_{\text{dd}}[k_0, l_0]$, is equal to the inner product between $x[k]$ and the (k_0, l_0) th discrete time pulsone $\phi[k; k_0, l_0]$.

Corollary 8.1 For any discrete time periodic signal $x[k]$ having period MN , the inner product between the (k_0, l_0) th discrete time pulsone $\phi[k; k_0, l_0]$ and $x[k]$ is the DZT of $x[k]$ evaluated at (k_0, l_0) , i.e.,

$$\langle x[k], \phi[k; k_0, l_0] \rangle = x_{\text{dd}}[k_0, l_0]. \tag{8.B.25}$$

Proof: From (8.B.18), it follows that

$$\begin{aligned}
\langle x[k], \phi[k; k_0, l_0] \rangle &= \sum_{k_1=0}^{M-1} \sum_{l_1=0}^{N-1} x_{\text{dd}}[k_1, l_1] \langle \phi[k; k_1, l_1], \phi[k; k_0, l_0] \rangle \\
&= x_{\text{dd}}[k_0, l_0] \langle \phi[k; k_0, l_0], \phi[k; k_0, l_0] \rangle,
\end{aligned} \tag{8.B.26}$$

where the last step follows from the fact that distinct discrete time pulsones are orthogonal to each other (see Theorem 8.6). The proof now follows from the fact that the inner product of a discrete time pulsone with itself is 1.

This leads to an important result that the inner product between any two discrete time signals is preserved in the discrete DD domain as well. Specifically, the energy of a discrete time periodic signal is preserved in the discrete DD domain.

Corollary 8.2 The inner product between any two discrete time periodic signals $x[k]$ and $y[k]$ both having period MN is preserved in the discrete DD domain. That is, if $x_{\text{dd}}[k, l]$ and $y_{\text{dd}}[k, l]$ denote their respective DZT (i.e., discrete DD domain realizations), then

$$\langle x[k], y[k] \rangle = \langle x_{\text{dd}}[k, l], y_{\text{dd}}[k, l] \rangle. \quad (8.B.27)$$

Proof: The inner product in the discrete time domain is

$$\begin{aligned} \langle x[k], y[k] \rangle &= \left\langle \left[\sum_{k_1=0}^{M-1} \sum_{l_1=0}^{N-1} x_{\text{dd}}[k_1, l_1] \phi[k; k_1, l_1] \right], \left[\sum_{k_2=0}^{M-1} \sum_{l_2=0}^{N-1} y_{\text{dd}}[k_2, l_2] \phi[k; k_2, l_2] \right] \right\rangle \\ &= \sum_{k_1=0}^{M-1} \sum_{l_1=0}^{N-1} \sum_{k_2=0}^{M-1} \sum_{l_2=0}^{N-1} \langle \phi[k; k_1, l_1], \phi[k; k_2, l_2] \rangle x_{\text{dd}}[k_1, l_1] y_{\text{dd}}^*[k_2, l_2], \end{aligned} \quad (8.B.28)$$

where the first step follows from the expression of $x[k]$ and $y[k]$ as a linear combination of the discrete time pulsones given in Theorem 8.5. Since the discrete pulsones are orthonormal to each (see Theorem 8.6), it follows that

$$\begin{aligned} \langle x[k], y[k] \rangle &= \sum_{k_1=0}^{M-1} \sum_{l_1=0}^{N-1} \sum_{k_2=0}^{M-1} \sum_{l_2=0}^{N-1} \delta[k_1 - k_2] \delta[l_1 - l_2] x_{\text{dd}}[k_1, l_1] y_{\text{dd}}^*[k_2, l_2] \\ &= \sum_{k_1=0}^{M-1} \sum_{l_1=0}^{N-1} x_{\text{dd}}[k_1, l_1] y_{\text{dd}}^*[k_1, l_1] \\ &= \langle x_{\text{dd}}[k, l], y_{\text{dd}}[k, l] \rangle. \end{aligned} \quad (8.B.29)$$

A special case of Corollary 8.2 is when $x[k] = y[k]$. Then, the inner product of $x[k]$ with itself is its energy which is the same as the energy of its DD domain realization, i.e.,

$$\begin{aligned} \sum_{k=0}^{MN-1} |x[k]|^2 &= \langle x[k], x[k] \rangle \stackrel{(a)}{=} \langle x_{\text{dd}}[k, l], x_{\text{dd}}[k, l] \rangle \\ &= \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} |x_{\text{dd}}[k, l]|^2, \end{aligned} \quad (8.B.30)$$

which is the energy of the DD domain realization of $x[k]$. Here, step (a) follows from Corollary 8.2.

The discrete time pulsones are periodic signals having period MN , and, therefore, their discrete frequency domain representation is given by their discrete Fourier transform (DFT). The following theorem shows that the discrete frequency domain representation of a discrete time pulson is a discrete frequency domain pulse train modulated by a sinusoid (i.e., a discrete frequency domain pulson).

Theorem 8.7 The DFT of the discrete time pulson corresponding to the DD pulse at (k_0, l_0) is given by

$$\begin{aligned} \phi_{\text{fd}}[f; k_0, l_0] &\triangleq \text{DFT}(\phi[k; k_0, l_0]) \\ &= \frac{1}{\sqrt{M}} \underbrace{e^{-j2\pi \frac{fk_0}{MN}}}_{\text{tone in FD with freq. } \frac{k_0}{M}} \underbrace{\sum_{m \in \mathbb{Z}} \delta[f - l_0 - mN]}_{\text{discrete FD pulse train}}, \quad f \in \mathbb{Z}. \end{aligned} \quad (8.B.31)$$

In other words, the discrete FD realization of a discrete quasi-periodic DD pulse at (k_0, l_0) is a discrete FD pulse train modulated by a FD tone, i.e., a discrete FD pulsone. The pulses are located at $f = (l_0 + mN)$, $m \in \mathbb{Z}$. Furthermore, for any (k_0, l_0) , the corresponding discrete FD pulsone is also periodic with period MN , i.e.,

$$\phi_{\text{fd}}[f + MN; k_0, l_0] = \phi_{\text{fd}}[f; k_0, l_0], \quad \forall f \in \mathbb{Z}. \quad (8.B.32)$$

Also, the set of MN discrete FD pulsones, $\phi_{\text{fd}}[f; k_0, l_0]$, $k_0 = 0, 1, \dots, M-1$, $l_0 = 0, 1, \dots, N-1$, form an orthonormal set. That is, for all $k_1, k_2 = 0, 1, \dots, M-1$, $l_1, l_2 = 0, 1, \dots, N-1$,

$$\begin{aligned} \langle \phi_{\text{fd}}[f; k_1, l_1], \phi_{\text{fd}}[f; k_2, l_2] \rangle &= \sum_{f=0}^{MN-1} \phi_{\text{fd}}[f; k_1, l_1] \phi_{\text{fd}}^*[f; k_2, l_2] \\ &= \delta[k_1 - k_2] \delta[l_1 - l_2]. \end{aligned} \quad (8.B.33)$$

Proof: The DFT of the periodic discrete time pulsone $\phi[k; k_0, l_0]$ having period MN is given by

$$\phi_{\text{fd}}[f; k_0, l_0] = \frac{1}{\sqrt{MN}} \sum_{k=0}^{MN-1} \phi[k; k_0, l_0] e^{-j2\pi \frac{f k}{MN}}, \quad f \in \mathbb{Z}. \quad (8.B.34)$$

Substituting the expression of $\phi[k; k_0, l_0]$ from (8.B.14) on the RHS above gives

$$\phi_{\text{fd}}[f; k_0, l_0] = \frac{1}{\sqrt{MN}} \frac{1}{\sqrt{N}} \sum_{k=0}^{MN-1} \sum_{n \in \mathbb{Z}} \delta[k - k_0 - nM] e^{j2\pi \frac{n l_0}{N}} e^{-j2\pi \frac{f k}{MN}}. \quad (8.B.35)$$

Clearly, due to the Dirac-delta on the RHS at $k = (k_0 + nM)$, the summation variable k is restricted to $k = (k_0 \bmod M) + qM$, $q = 0, 1, \dots, N-1$, and for each such k , the corresponding summation variable n is restricted to $n = (k - k_0)/M = (k_0 \bmod M - k_0 + qM)/M$. This gives

$$\begin{aligned} \phi_{\text{fd}}[f; k_0, l_0] &= \frac{1}{\sqrt{MN}} \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} e^{j2\pi \frac{(k_0 \bmod M - k_0 + qM) l_0}{MN}} e^{-j2\pi \frac{f (k_0 \bmod M + qM)}{MN}} \\ &= \frac{1}{\sqrt{M}} e^{-j2\pi \frac{f k_0}{MN}} \left(\frac{1}{N} \sum_{q=0}^{N-1} e^{-j2\pi q \frac{(f - l_0)}{N}} \right) \\ &= \frac{1}{\sqrt{M}} \underbrace{e^{-j2\pi \frac{f k_0}{MN}}}_{\text{tone in FD with freq. } \frac{k_0}{M}} \underbrace{\sum_{m \in \mathbb{Z}} \delta[f - l_0 - mN]}_{\text{discrete FD pulse train}}, \end{aligned} \quad (8.B.36)$$

where the second step follows from the fact that $k_0 \in \{0, 1, \dots, M-1\}$, and, therefore, $k_0 = k_0 \bmod M$.

In the above expression, the pulse train is periodic with period MN and the term $e^{-j2\pi \frac{f k_0}{MN}}$ is also periodic in f with period MN . Therefore, a discrete FD pulsone is also periodic in FD with period MN .

Since the inner product of discrete time periodic signals is preserved in the discrete FD, it follows that the inner product between two discrete time pulsones $\phi[k; k_1, l_1]$ and $\phi[k; k_2, l_2]$ is equal to the inner product between their respective discrete FD representations, i.e., $\phi_{\text{fd}}[f; k_1, l_1]$ and $\phi_{\text{fd}}[f; k_2, l_2]$. Since the discrete time pulsones form an orthonormal set of discrete time periodic signals, it immediately follows that the discrete FD pulsones also form an orthonormal set of periodic discrete FD signals.

For a given quasi-periodic discrete DD domain signal, IDZT gives its discrete time realization whose DFT gives its discrete FD realization. Therefore, there must be a transform (inverse discrete frequency Zak transform – IDFZT) which directly gives the discrete FD realization of a quasi-periodic discrete DD domain signal. The following theorem gives this transform.

Theorem 8.8 For a given quasi-periodic discrete DD signal $x_{\text{dd}}[k, l]$, its discrete FD realization is given by

$$\begin{aligned} x_{\text{fd}}[f] &= \text{IDFZT} (x_{\text{dd}}[k, l]) \\ &\stackrel{\Delta}{=} \text{DFT} (\text{IDZT} (x_{\text{dd}}[k, l])) \\ &= \frac{1}{\sqrt{M}} \sum_{k_0=0}^{M-1} x_{\text{dd}}[k_0, f] e^{-j2\pi \frac{fk_0}{MN}}. \end{aligned} \quad (8.B.37)$$

Proof: From Theorem 8.5, it is known that any discrete time periodic signal $x[k], k \in \mathbb{Z}$, having period MN can be expressed as a linear combination of the MN discrete time pulsones. Taking DFT of both sides of (8.B.18) gives

$$\begin{aligned} x_{\text{fd}}[f] &= \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \text{DFT} (\phi[k; k_0, l_0]) \\ &\stackrel{(a)}{=} \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} x_{\text{dd}}[k_0, l_0] \phi_{\text{fd}}[f; k_0, l_0], \end{aligned} \quad (8.B.38)$$

where step (a) follows from (8.B.31). Substituting the expression for $\phi_{\text{fd}}[f; k_0, l_0]$ from (8.B.31) on the RHS of step (a) gives

$$\begin{aligned} x_{\text{fd}}[f] &= \frac{1}{\sqrt{M}} \sum_{k_0=0}^{M-1} \sum_{l_0=0}^{N-1} \sum_{m \in \mathbb{Z}} x_{\text{dd}}[k_0, l_0] e^{-j2\pi \frac{fk_0}{MN}} \delta[f - l_0 - mN] \\ &\stackrel{(a)}{=} \frac{1}{\sqrt{M}} \sum_{k_0=0}^{M-1} e^{-j2\pi \frac{fk_0}{MN}} x_{\text{dd}}[k_0, f \bmod N] \\ &\stackrel{(b)}{=} \frac{1}{\sqrt{M}} \sum_{k_0=0}^{M-1} e^{-j2\pi \frac{fk_0}{MN}} x_{\text{dd}}[k_0, f], \end{aligned} \quad (8.B.39)$$

where step (a) follows from the fact that with regard to the summation variable l_0 , the only non-zero contribution on the RHS is for $l_0 = f \bmod N$. Step (b) follows from the fact that $x_{\text{dd}}[k, l]$ being a quasi-periodic discrete DD signal is periodic along the discrete Doppler axis with period N .

9

Deep Learning for OTFS Transceiver Design

Deep neural networks (DNNs) have found applications in a wide range of fields. They are being increasingly used for inference tasks like regression and classification. With the advent of libraries available for training, it has become easier than ever to train and deploy application-specific neural networks (NNs). This is aided by the availability of hardware tailor made for training NNs. The time required to train NNs, even complex ones having a large number of parameters, has reduced drastically. The networks also have the advantage that, once trained, they are computationally efficient when compared to the conventional optimal algorithms.

In the field of communications, research on the use of deep learning (DL) to solve a wide range of problems in the physical layer is being pursued actively [191–209]. Some of the physical layer tasks for which the deep learning approach is promising include the design of channel codes and algorithm design for channel decoding, signal detection, channel estimation, beamforming, and precoding. These tasks can be posed as classification and regression problems in a deep learning framework and implemented using suitably architected DNNs. In addition to classification and regression problems, the deep learning approach finds application in prediction tasks like channel prediction and tracking.

This chapter focuses on introducing and describing deep learning techniques that find application in the design of efficient orthogonal time frequency space (OTFS) transceivers. It presents a short primer on the learning framework and DNNs and introduces popular NN architectures such as fully connected neural networks (FCNNs), recurrent neural networks (RNNs), and convolutional neural networks (CNNs). It provides insights on which architecture will be appropriate for which communication task. A key requirement in OTFS transceivers design is signal processing in the delay-Doppler (DD) domain. The chapter presents DNN-based solutions to various OTFS transceiver tasks such as signal detection in the DD domain, channel estimation and tracking in the DD domain, and IQ imbalance (IQI) estimation and compensation. For this, it draws illustrative examples from recent literature. It provides key highlights on the performance and complexity aspects of these solutions.

Another point of focus in the chapter is to showcase scenarios where learning-based approaches can outperform conventional optimum approaches. For example, such performance advantages of learning approaches can be predominantly seen when there are deviations from the standard channel/noise models assumed in deriving conventional optimum solutions. Such deviations can happen in practice, e.g., noise can be colored/correlated and non-Gaussian (deviating from the standard i.i.d. Gaussian noise model assumed in the derivation of optimum detection). Learning-based techniques will be a natural fit in such scenarios because of their ability to learn the underlying channel/noise models and perform efficiently.

9.1 Learning Framework

Learning is innate to humans. Right from birth, we learn to crawl and then to walk accumulating experience from the previous endeavors and course correcting each time something does not work out as planned. Inspired by the self-taught learning approaches of humans, Arthur Samuel coined the term “machine learning” in 1959 at IBM [210]. He defined machine learning as “the field of study that gives computers the ability to learn without being explicitly programmed.” Stated another way, machine learning can be viewed as a problem of construction of computer programs that automatically improve with experience, E . Machine learning largely involves models that have trainable (tunable) parameters that need to be optimized, using data referred to as training data. Typically, a metric or cost function or a performance criterion is used to arrive at the optimum values of these parameters. A typical machine learning problem contains three main parts. First, objective of the machine learning program denoted by O ; second, the performance criterion given by C ; and, third, the training data or experience represented by D . For example, for a machine learning program employed to classify images of handwritten digits from 0 to 9, the O , C , and D are defined as follows [211]:

- **O :** Classify a given image into one of the ten classes labelled from 0 to 9.
- **C :** Accuracy of classification or percentage of the images correctly classified.
- **D :** Images of the handwritten digits with their corresponding labels.

A program that learns from E is called as a machine learning program or simply a learning program.

9.1.1 Classification of Machine Learning Programs

Machine learning programs can be classified based on (i) the structure of training dataset D and (ii) the nature of the objective O .

9.1.1.1 Classification Based on Training Dataset

Depending on how the training data D is generated, any machine learning program can be classified into the following four categories [212].

- **Supervised learning:** In this type of machine learning programs, corresponding to each input there is an output in the training dataset, D . The training is “supervised” using the labels available in D . Classification and regression tasks fall under the category of supervised learning. An example for supervised learning is as follows. Objective O is the classification of images of birds into three categories, say, parrot, peacock, and sparrow. Training data, D , consists of images of birds belonging to each of the three classes. Corresponding to each image, there is a label provided, which describes if the particular image is a parrot, peacock, or sparrow. The labels are used in the criterion C while training.
- **Unsupervised learning:** This type of learning programs is used to draw inferences from input data without labels in the training data. The goal in such learning programs is to cluster different data points into the same class. Consider the task (O) of clustering pictures of birds and dogs into two clusters. Training data (D) consists of images and dogs without having the labels for each image. The goal is to make clusters of bird images and dog images, i.e., recognize the images are from the same class.
- **Semi-supervised learning:** This subclass of learning programs falls in between the two extremes of supervised learning and unsupervised learning. Here, some of the data have labels, while the

majority do not. The training data can be thought of as “incomplete” in terms of labels. This is chosen when the dataset has large amount of data, but only few are labelled due to the cost and effort involved in labelling.

- **Reinforcement learning:** In this type of learning, there is no expected ground truth with labels. Rather, a training feedback is given if the output of the learning algorithm is favorable or not. That is, the learning is carried just by telling the learning algorithm if the output is right or wrong without explicitly telling the algorithm how to correct it or why is the output wrong.

9.1.1.2 Classification Based on Nature of Objective

Based on the nature of the objective for which the learning program is trained, a learning program can be classified into the following three categories.

- **Regression:** Regression learning programs are also supervised learning programs. That is, for such programs, all the samples in the training data have label. The output of the learning algorithm for such problems typically spans the subsets of real line. Consider a learning algorithm used for channel estimation in wireless communication systems. The input is the received pilot symbol, and the learning algorithm is trained to provide the channel coefficient value (which can span the real line).
- **Classification:** Classification programs also fall under the category of supervised learning programs. The goal is to classify the input data into different classes. As opposed to regression, the outputs are typically probability mass functions over the classes. During training, the learning program updates its weights such that the correct class is assigned high probability. The classification of received symbols in a communication system into one of the M classes for data symbols chosen from a modulation alphabet is an example for a classification program.
- **Clustering:** Clustering programs are unsupervised. Unlike classification, the classes to which the training data belongs is not known a priori. The goal here is to group the training data into meaningful clusters based on the features present in the data.

9.1.2 Traditional Program and Machine Learning Program

Figure 9.1 shows the block diagrams of the traditional program and machine learning program (supervised) at a high level. In a traditional program, data is fed as input along with a program or logic. The computing resource uses the program to act on the data to provide the required output. On the other hand, in machine learning programs, inputs in the form of data and the expected output (ground truth) are fed as input to the computing resource. The computing resource outputs the program (trained parameters of the network). Once the training concludes, the parameters are frozen and saved. Following this, during the testing phase, operation similar to the traditional program is followed, wherein the input program is the saved network parameters.

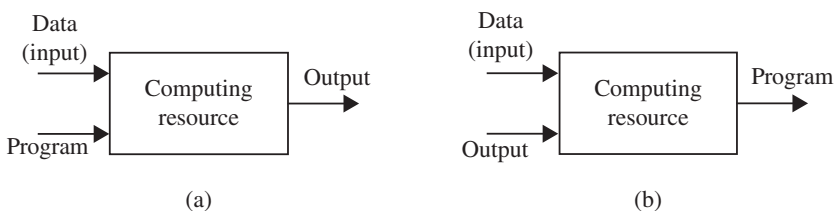


Figure 9.1 Block diagrams showing the difference between a traditional program (a) and a machine learning program (b).

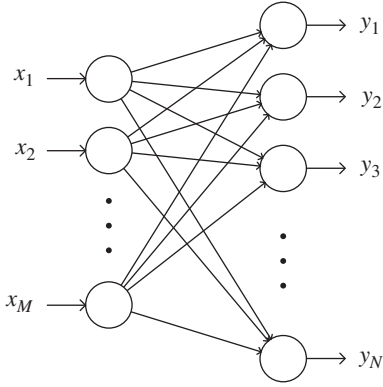


Figure 9.2 Architecture of one layer of FCNN.

9.1.3 Deep Learning Using Neural Networks

Classical machine learning algorithms require the manual extraction of the most important features from the training data. Later, these features are used to build a machine learning algorithm that makes use of these features for the classification or regression task. An example of this is a principal component analysis (PCA), where the most important features (principal components) in the training data can be extracted [213]. However, as the dimensionality of the training data grows, it might become infeasible to manually extract the most important features. NNs were introduced as a subset of machine learning algorithms to overcome this difficulty [214–217]. NNs can automatically retrieve the relevant characteristics/features from the training data through learning. Deep learning (DL) is the subset of machine learning algorithms, where NN having more than two layers (and therefore deep) is used. Three main architectures of NN exist in the literature. They include FCNN, RNN, and CNN, which are described below.

9.1.3.1 Fully Connected Neural Networks

FCNN architecture comprises neurons arranged in layers [218]. Each layer has an input dimension M and output dimension N , i.e., there are M neurons at the input of the layer and N neurons at the output. Each neuron at the input of a layer is connected to each neuron at the output of the layer. This is seen in Fig. 9.2, where the input is M -dimensional and the output is N -dimensional. The structure of a single neuron is shown in Fig. 9.3, where $u_1, u_2, \dots, u_n$ denote the inputs to the neuron which are multiplied by trainable weights $w_1, w_2, \dots, w_n$, respectively. The weighted inputs and a trainable bias b are added, after which it is passed through a nonlinear (or linear) function to obtain the output of the neuron. The output, $\hat{y}$, can be expressed as

$$\hat{y} = \psi \left(\sum_{j=1}^n w_j u_j + b \right). \quad (9.1)$$

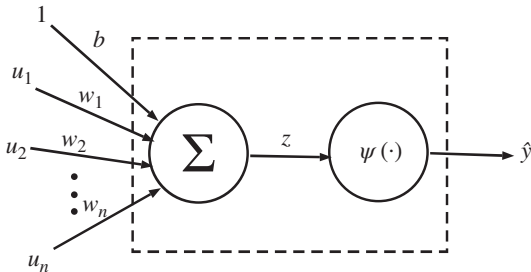


Figure 9.3 Structure of a single neuron.

Let $\mathbf{u} = [u_1, u_2, \dots, u_n]^T$ denote the vector containing the inputs of the neuron. The main goal of the neuron is to learn the weights $w_1, \dots, w_n$ and the bias b in order to mimic a real world system. The weights are learnt based on several training examples $(\mathbf{u}, y)$, where $\mathbf{u}$ is the training input and y is the expected output of the real world system when the input $\mathbf{u}$ is fed to it. The expected output is also called the ground truth in the machine learning literature. To find the optimal weights in order to mimic a real world system, a loss function is defined and is minimized with respect to the weights.

Loss Function and Weight Update A loss function is a differentiable function that captures the error between the actual output of the neuron $\hat{y}$ and the true expected output y . One of the common loss functions used in the machine learning literature is the mean square error (MSE), given by

$$L(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{m_T} \sum_{i=1}^{m_T} (y_i - \hat{y}_i)^2, \quad (9.2)$$

where y_i and $\hat{y}_i$ denote the true expected output and the actual output of the neuron for the i th training example, m_T denotes the number of training examples, $\mathbf{y} = [y_1, \dots, y_{m_T}]^T$, $\hat{\mathbf{y}} = [\hat{y}_1, \dots, \hat{y}_{m_T}]^T$, and $L(\mathbf{y}, \hat{\mathbf{y}})$ denotes the loss function. Another commonly used loss function is the binary cross entropy (BCE), given by

$$L(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{m_T} \sum_{i=1}^{m_T} (-y_i \log_2 \hat{y}_i - (1 - y_i) \log_2 (1 - \hat{y}_i)). \quad (9.3)$$

The goal of the training process is to adjust the weights of the neuron such that the actual output of the neuron $\hat{y}_i$ converges to the true output y_i for $i = 1, \dots, m_T$. This is achieved by minimizing the loss function $L(\mathbf{y}, \hat{\mathbf{y}})$ with respect to the weights $b, w_1, w_2, \dots, w_n$ for each training example. Note that, since each $\hat{y}_i$ is a function of all the weights, the loss function $L(\mathbf{y}, \hat{\mathbf{y}})$ is also a joint function of all the weights. Therefore, one way to optimize the function is to use the gradient descent algorithm, which first computes the partial derivative $\frac{\partial L(\mathbf{y}, \hat{\mathbf{y}})}{\partial w_i}$ for each weight w_i and then uses the following update rule in each iteration

$$w_i = w_i - \alpha \frac{\partial L(\mathbf{y}, \hat{\mathbf{y}})}{\partial w_i}, \quad (9.4)$$

where α is called the learning rate. The process of finding $\hat{y}$ given $\mathbf{u}$ using (9.1) is called forward propagation, and the process of finding the weights by minimizing $L(\mathbf{y}, \hat{\mathbf{y}})$ as above is called back propagation. The reason for these names lies in the use of computation graphs, which provide efficient means of computing the output $\hat{y}$ and the weights $b, w_1, \dots, w_n$ by moving in the forward and backward directions along the graphical representation of the NN.

Neural Networks A NN is a collection of several connected neurons, arranged in the form of layers. An example NN is shown in Fig. 9.4. A NN is composed of an input layer, an output layer, and several hidden layers. The NN in Fig. 9.4 has an input layer with two neurons, an output layer with two neurons, and two hidden layers with three neurons each. The main idea behind connecting several neurons to form a NN is that combining several non-linear units (neurons) will enable learning complex non-linear decision boundaries and hence help in learning the behavior of a complex non-linear real world system. Although there is no universally accepted definition, if the number of hidden layers in a NN is more than two, then it is called a deep NN (DNN). In the NN shown in Fig. 9.4, all the neurons in a given layer are connected to all the other neurons of the next layer. Such a NN is called a fully connected NN.

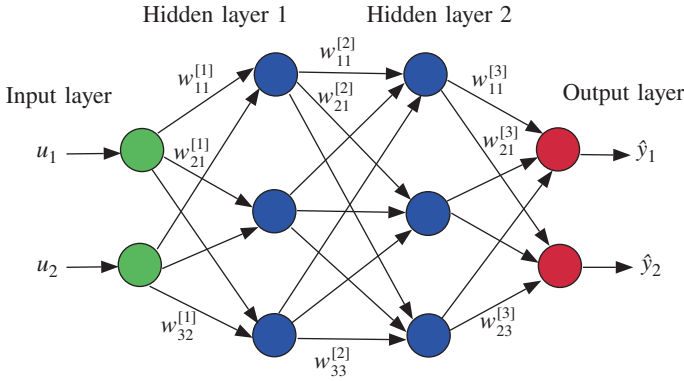


Figure 9.4 An example neural network.

Let n_i be the number of neurons in the i th layer. Also, let $\mathbf{W}^{[i]}$ denote the weight matrix which takes the output of the i th layer to the input of the $(i + 1)$ th layer, given by

$$\mathbf{W}^{[i]} = \begin{bmatrix} w_{11}^{[i]} & w_{12}^{[i]} & \cdots & w_{1n_i}^{[i]} \\ w_{21}^{[i]} & w_{22}^{[i]} & \cdots & w_{2n_i}^{[i]} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n_{i+1}1}^{[i]} & w_{n_{i+1}2}^{[i]} & \cdots & w_{n_{i+1}n_i}^{[i]} \end{bmatrix}, \quad (9.5)$$

where $w_{kl}^{[i]}$ denotes the weight for the connection from l th neuron of the i th layer to the k th neuron of the $(i + 1)$ th layer. For the example NN shown in Fig. 9.4, the output of the first hidden layer (second layer) is

$$\mathbf{y}^{[2]} = \psi^{[2]}(\mathbf{W}^{[1]} \mathbf{u} + \mathbf{b}^{[1]}), \quad (9.6)$$

where $\mathbf{u} = [u_1 \ u_2]^T$ is the input vector to the NN, $\psi^{[2]}(\cdot)$ is the activation function of the first hidden layer, and $\mathbf{W}^{[1]}$ and $\mathbf{b}^{[1]}$ are the weight matrix and bias vector, respectively, for the connections from the input layer (1st layer) to the first hidden layer (2nd layer). Similarly, the output of the second hidden layer (3rd layer) is given by

$$\mathbf{y}^{[3]} = \psi^{[3]}(\mathbf{W}^{[2]} \mathbf{y}^{[2]} + \mathbf{b}^{[2]}). \quad (9.7)$$

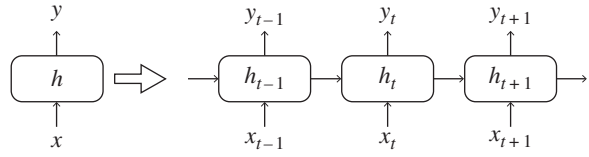
The final output at the output layer is given by

$$\mathbf{y}^{[3]} = \hat{\mathbf{y}} = \psi^{[4]}(\mathbf{W}^{[3]} \mathbf{y}^{[3]} + \mathbf{b}^{[3]}) = [\hat{y}_1 \ \hat{y}_2]^T. \quad (9.8)$$

The forward propagation involves the computation of $\hat{\mathbf{y}}$ from $\mathbf{u}$ using (9.6) to (9.8), and the backward propagation involves the computation of weight matrices and bias vectors $\mathbf{W}^{[1]}$, $\mathbf{b}^{[1]}$, $\mathbf{W}^{[2]}$, $\mathbf{b}^{[2]}$, $\mathbf{W}^{[3]}$, and $\mathbf{b}^{[3]}$, by minimizing the loss function, as discussed for a single neuron.

DL Programming Frameworks Although the implementation of forward propagation is generally straightforward, the implementation of back propagation is complex and is programming intensive. To reduce the programming complexity of NNs, several deep learning programming frameworks have been developed, such as TensorFlow (developed by Google Brain), Keras, and PyTorch (developed by Facebook). These frameworks internally take care of the back propagation during the training process once the NN architecture is defined. Furthermore, they also provide

Figure 9.5 A basic structure of RNN and time unfolding.



the implementations of advanced optimizers such as stochastic gradient descent (SGD) and Adam (derived from adaptive moment estimation), which can be easily used to achieve optimal learning of NN parameters.

9.1.3.2 Recurrent Neural Networks

RNNs are a special class of NNs that are used for learning involving time-series data [219]. The architecture of RNNs allows them to “unfold” over time, in which the output of the previous unfolding step becomes the input at the current unfolding step. Figure 9.5 shows the basic structure of the RNN on the left and the unfolding on the right. Following are the operations for the t th iteration:

$$h_t = \psi(w_{hh}h_{t-1} + w_{xh}x_t) \quad \text{and} \quad y_t = w_{yh}h_t, \quad (9.9)$$

where ψ is a non-linear (or linear) function, w_{hh} and w_{yh} are trainable parameters, h_t is called the hidden state, x_t is the input, and y_t is the output. Furthermore, across all unfolding indexed by t , although the weights remain constant, the hidden state and the output depend on all the unfolding till t , and, therefore, these networks are said to have “memory.” This property allows these networks to learn time-series data. This architecture of RNN, called the vanilla RNN, is affected by vanishing gradients that limits its capabilities. To overcome this and to learn long time-series data, two modified architectures are proposed, namely, long short-term memory (LSTM) [220] and gated recursive unit (GRU) [221].

9.1.3.3 Convolutional Neural Networks

CNNs were proposed to identify patterns from matrix-like datasets (image-like datasets) [222]. For a 2D CNN, the trainable parameters are the entries of a 2D kernel matrix that slides over the input multiplying the kernel entries with the corresponding input values and summing the values to get the output. In other words, the input values are filtered through the kernel. Figure 9.6 shows this operation. A 3×3 kernel and an input of size 5×5 are considered. y_1 is obtained as

$$y_1 = \psi(w_1x_1 + w_2x_2 + w_3x_3 + w_4x_6 + w_5x_7 + w_6x_8 + w_7x_{11} + w_8x_{12} + w_9x_{13}). \quad (9.10)$$

After this operation, the kernel is shifted one position to the right so that w_1 is aligned with x_2 to obtain y_2 . Once the row is done, the kernel is shifted to the next row (with w_1 aligning with x_6 to get y_4) and the process is continued until the entire input is filtered.

9.1.4 Common Terminologies in the DL Framework

Some common terminologies and their functions for training NNs are presented below.

- **Forward propagation:** A pass through the NN from the input to the output is called forward propagation.
- **Loss function:** A criterion evaluated at the output of the NN after the forward pass. This quantifies how far (or near) the output of the NN is to the expected output. Examples include MSE loss, BCE loss, L1 loss, among others.

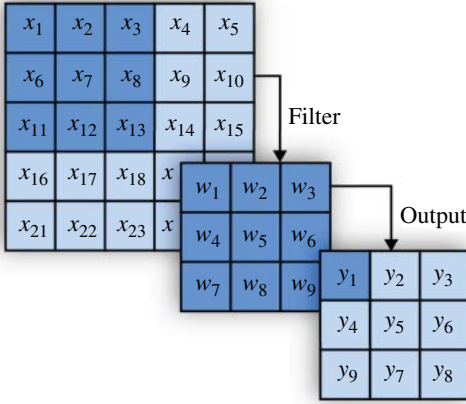


Figure 9.6 Pictorial representation of CNN.

- **Back propagation** [223]: Once the loss function value is evaluated, the gradient of this loss is computed with respect to each trainable parameter in the NN working from output to input. This process of evaluating the gradients from the output toward the input is called the back propagation.
- **Optimizer**: Optimizer is an algorithm that actually performs updates to all the trainable parameters based on the evaluated gradient. Common examples are Adam, SGD, and RMSProp.
- **Learning rate**: Learning rate defines how fast (or slow) the weight updates happen. It is desirable to have a large value of learning rate at first and decrease the value as the training progresses for fine tuning purposes.
- **Epoch**: The number of training iterations to perform. Each epoch contains multiple forward and backward propagations.
- **Batch size**: The number of training samples to be used in each epoch.
- **Activation function**: Typically a non-linear function is used at the output of the NN. In FCNN, RNN, and CNN descriptions, the function $\psi(\cdot)$ is the activation function. The non-linearity is what allows the NN to learn complex tasks. Common examples are rectified linear unit (ReLU), sigmoid, tanh, and softmax, where the ReLU activation function is given by

$$\text{ReLU}(x) = \max(0, x), \forall x \in (-\infty, \infty), \quad (9.11)$$

and sigmoid activation function is given by

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}, \forall x \in (-\infty, \infty). \quad (9.12)$$

- **Hyper-parameters**: All the parameters of training not pertaining to the NN are called hyper-parameters. For example, optimizer, loss function, and batch size are all hyper-parameters.
- **Parameters**: All the variables pertaining to the NN are referred to as parameters. The number of layers, activation function, and input and output size are all parameters.
- **Training dataset**: Dataset used for training.
- **Validation dataset**: Data not in the training dataset, used to validate the training. Used to handle issues like overfitting in the NN.
- **Test dataset**: Data used while testing after the training is complete. The data in this dataset is not present in both the training and validation datasets. The performance of the trained network is evaluated using this dataset.

- **Learning library:** This refers to libraries that contain all the basic codes required for training. That is, the codes for the NN, optimizer, and forward and back propagation are already written in these libraries. Popular examples are PyTorch [224] and TensorFlow [225].
- **Hardware for learning:** This refers to hardware that are designed to speed up training times. The hardware is generally referred to as graphical processor units (GPUs). Some of the most popular and powerful GPUs are Nvidia Titan, Nvidia RTX 3090, and Nvidia RTX 4090.

9.2 OTFS Signal Detection Using DNNs

Reliable detection of OTFS signals is a crucial task for the recovery of information symbols at the receiver. See Chapter 5 for a compilation of OTFS signal detection algorithms using traditional approaches. This section explores the use of DNNs for the detection of OTFS signals at the receiver. Toward this, identifying suitable network architectures and defining appropriate input/output layers and hyper-parameters for the chosen architectures are crucial. FCNN have been found to be suitable for signal detection purposes in the literature, and they are considered here as well. Start with the vectorized representation of the OTFS system model (see (4.10) in Chapter 4 which is based on multicarrier (MC)-OTFS modulation discussed in Chapter 3)

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}, \quad (9.13)$$

where $\mathbf{x}$ is the $MN \times 1$ transmitted vector, $\mathbf{H}$ is the $MN \times MN$ equivalent channel matrix, $\mathbf{v}$ is the $MN \times 1$ noise vector, and $\mathbf{y}$ is the $MN \times 1$ received vector. The goal is to detect $\mathbf{x}$, given $\mathbf{y}$ and the knowledge of $\mathbf{H}$. The received signal vector $\mathbf{y}$ forms the input to the network. The networks typically handle real variables, and hence, the complex vector $\mathbf{y}$ is converted in to a real vector by stacking the real and imaginary parts of the complex variables. Consequently, the number of nodes in the input layer is two times the OTFS frame size. Two DNN approaches for OTFS signal detection are presented in this section [226]. The first approach is a *full-DNN approach*, which employs a single large DNN for detection. The second approach is a *symbol-DNN approach*, which employs multiple smaller DNNs, one for each symbol in the frame. The symbol-DNN approach of using multiple smaller DNNs has the advantage of low complexity compared to the full-DNN approach of using a single large DNN, and hence, it can scale well for large/practical OTFS frame sizes. For small/medium frame sizes, both full-DNN and symbol-DNN approaches are found to perform close to the optimum maximum likelihood (ML) performance when noise is i.i.d. Gaussian. Additional benefits of using DNNs for OTFS detection show up when noise distribution deviates from the standard i.i.d. Gaussian model. In such cases, the ability of DNNs to learn the underlying noise distribution yields superior signal detection performance.

9.2.1 Full-DNN Detection Approach

In the full-DNN detection approach, the idea is to detect multi-dimensional signals using a single fully connected DNN, where the number of input neurons is decided by the size of the received/observed vector and the number of output neurons is decided by the size of the multi-dimensional signal set. Hence, each neuron in the output layer corresponds to one transmit signal vector from the OTFS signal set, enabling joint detection of the multi-dimensional symbol vector. The full-DNN architecture for OTFS signal detection is shown in Fig. 9.7. The DNN takes the real and imaginary parts of the received vector $\mathbf{y}$ as input and has $2MN$ input neurons,

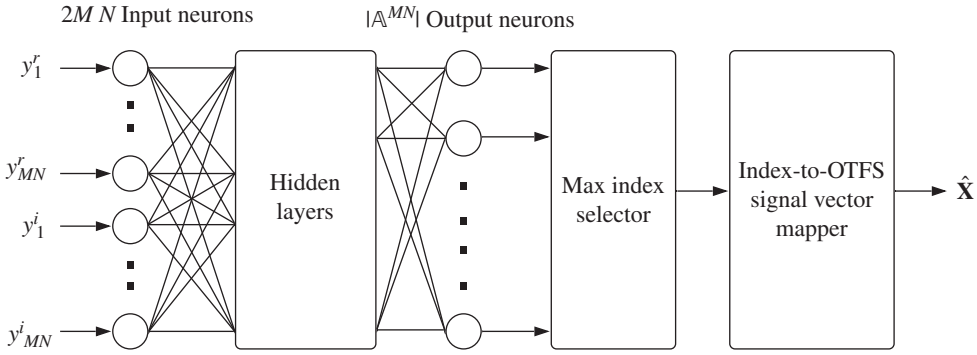


Figure 9.7 Full-DNN architecture for OTFS signal detection.

where M and N are the number of delay and Doppler bins, respectively, in the OTFS frame. The input layer is followed by several hidden layers. The number of neurons in the output layer is $|\mathbb{A}|^{MN}$, which is the number of signal vectors in the OTFS signal set. Here, $\mathbb{A}$ is the modulation alphabet (e.g., quadrature amplitude modulation [QAM]) from where the information symbols are drawn from. Each output neuron corresponds to one OTFS signal vector and gives the probability of that signal vector being transmitted. The “Max Index Selector” selects the output neuron which gives the highest probability value. The “Index-to-OTFS signal vector mapper” then takes the index of the selected output neuron and maps it to the corresponding OTFS signal vector. While this approach is architecturally simple and straightforward, it requires a large number of parameters (weights and biases) to be learned. This is because the number of output neurons grows exponentially in the size of transmit symbol vector, requiring a proportionately large number of neurons in the hidden layers. For example, an OTFS system with $M = N = 16$ will have 256 symbols per frame and $|\mathbb{A}|^{256}$ possible signal vectors. Full-DNN detection for this system will have $|\mathbb{A}|^{256}$ output neurons, and hence, the size of the DNN will be large with a large number of parameters to train during the training phase. The testing phase (signal detection) is also complex since the received vector has to pass through several layers with a large number of neurons.

9.2.2 Symbol-DNN Detection Approach

An alternate approach to the full-DNN approach is to devise multiple DNNs, one for each symbol in the transmit symbol vector. This approach has the benefit of low complexity because the number of DNNs grows only linearly with the size of the transmit symbol vector and the number of output neurons in each DNN also grows linearly in the size of the symbol alphabet. The symbol-DNN architecture for OTFS detection is shown in Fig. 9.8. It consists of MN DD symbol-DNNs, each corresponding to an individual information symbol in the DD grid. There are $2MN$ input neurons in all these symbol-DNNs through which the real and the imaginary parts of the received vector $\mathbf{y}$ are given as the input. The number of output neurons in each of the symbol-DNNs is equal to $|\mathbb{A}|$, the size of the modulation alphabet, such that each output neuron gives the corresponding probability of each symbol in the alphabet being transmitted. Softmax activation is used in the output layer such that all the probabilities of the symbols in the alphabet are dependent and sum to one. The “Max index selector” gives the index of the symbol corresponding to the highest probability, and the “Index to modulation symbol mapper” maps it to the corresponding transmitted symbol.

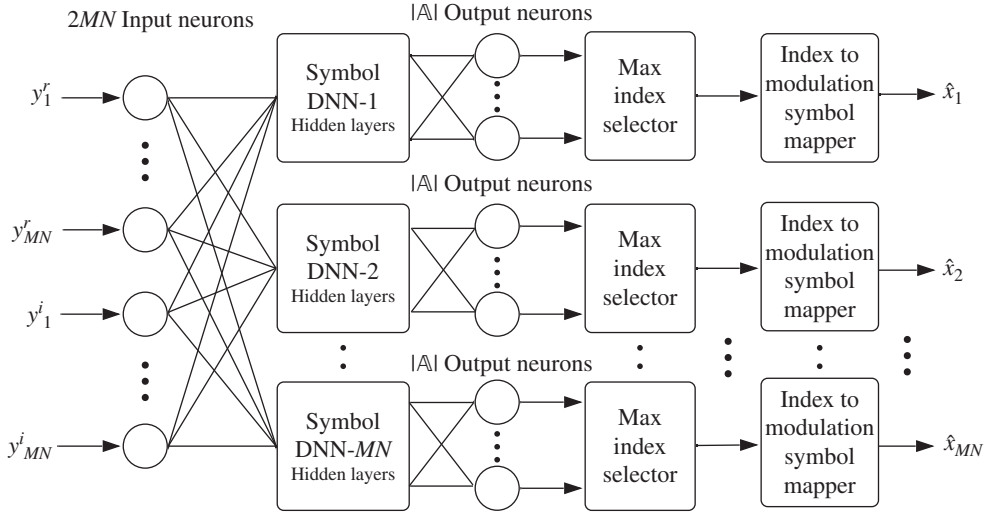


Figure 9.8 Symbol-DNN architecture for OTFS signal detection.

9.2.2.1 Training and Testing

In communication transceiver designs using DNNs, the networks are often trained using synthetically generated training examples following the underlying system model. In this case, the DNNs are trained using a set of randomly generated OTFS signal vectors ($\mathbf{x}_T$), channel matrices ($\mathbf{H}$), and corresponding received signal vectors ($\mathbf{y}$) as per the system model in (9.13) as training examples. The knowledge of $\mathbf{x}_T$ is made available to the receiver as ground truth during training. The stacked real and imaginary parts of $\mathbf{y}$ are given as input to the DNNs. The number of training examples to be used for training is chosen by trial. The training can start with a small number of examples which can be incremented until the performance of the trained DNN is good.

Once trained, the weights of the DNNs are frozen and the trained network is ready for signal detection (testing phase). During the testing phase, the receiver detects the data symbols by processing the received vector $\mathbf{y}$ using the already trained network. In the case of full-DNN, the network learns the mapping from the received vector $\mathbf{y}$ to the corresponding signal vector $\mathbf{x}$. In the case of symbol-DNNs, each symbol-DNN learns the mapping from the received vector to the corresponding symbol in the signal vector. For the performance results presented next, training and testing are carried out using the Tensorflow and Keras framework.

9.2.3 Performance Results

The bit error rate (BER) performance of the full-DNN- and symbol-DNN-based detection approaches are presented in this section. The corresponding performance of the optimum ML detection is also shown for comparison. When the noise distribution is i.i.d. Gaussian, the optimum ML detection rule is given by

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathcal{A}^{MN}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2. \quad (9.14)$$

The above decision rule becomes suboptimum when the noise distribution deviates from the standard i.i.d. Gaussian distribution. The effect of such deviations from the standard noise model on the achieved performance is also highlighted.

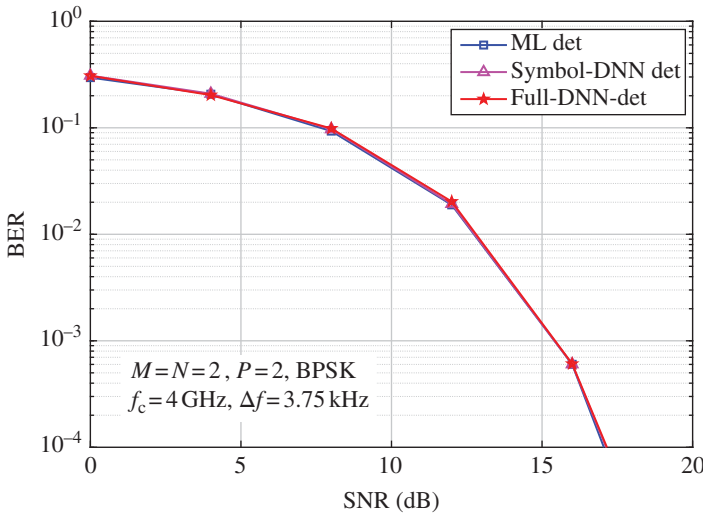


Figure 9.9 BER performance of OTFS with $M = N = 2$ using symbol-DNN, full-DNN, and ML detection in the static multipath channel with $P = 2$.

9.2.3.1 Static Channel with i.i.d. Gaussian Noise

Figure 9.9 shows the BER performance using symbol-DNN detection, full-DNN detection, and ML detection in an OTFS system with $M = N = 2$, binary phase shift keying (BPSK), and i.i.d. Gaussian noise. A carrier frequency (f_c) of 4 GHz and a subcarrier spacing (Δf) of 3.75 kHz are considered. A static (zero Doppler) multipath channel with $P = 2$ paths having uniform power profile is considered. For $M = N = 2$ and BPSK, there are $MN = 4$ BPSK information symbols in a frame. The symbol-DNN architecture has four DNNs, one for each BPSK symbol in the transmit symbol vector. The parameters used in the symbol-DNN as well as the full-DNN are listed in Table 9.1. The architectures used for symbol-DNN and full-DNN are as follows:

- (i) **Symbol-DNN**: Input $\rightarrow 8 \rightarrow \text{ReLU} \rightarrow 4 \rightarrow \text{ReLU} \rightarrow 2 \rightarrow \text{Softmax}$.
- (ii) **Full-DNN**: Input $\rightarrow 8 \rightarrow \text{ReLU} \rightarrow 12 \rightarrow \text{ReLU} \rightarrow 16 \rightarrow \text{Softmax}$.

The above notation of describing DNN architectures and parameters gives the sequence of layers and number of neurons in a given layer, followed by the activation function used in that layer. Figure 9.9 illustrates that the BER performance of symbol-DNN as well as full-DNN detection is almost the same as the performance of optimum ML detection. Although all the three detectors achieve very close performance, their complexities are different. Table 9.2 shows the complexities of the three detectors in the number of real operations. It is seen that ML detection has the highest complexity because of the exhaustive enumeration of all possible OTFS transmit vectors. Among the DNNs, symbol-DNN has a lower complexity compared to full-DNN. This is because the number of parameters to train in full-DNN is larger than in symbol-DNN, as observed in Table 9.2.

The performance of a larger OTFS system with $M = N = 16$, BPSK, $f_c = 4$ GHz, $\Delta f = 15$ kHz, $P = 8$, and i.i.d. Gaussian noise is considered in Fig. 9.10. Optimum ML detection for this system must do exhaustive search over 2^{256} signal vectors, which becomes computationally infeasible. Full-DNN detection for this system will also have $2^{MN} = 2^{256}$ output neurons, and hence, the size of

Table 9.1 Parameters of DNN detectors in Fig. 9.9.

Parameters	Symbol-DNN	Full-DNN
No. of input neurons	$2MN = 8$	$2MN = 8$
No. of output neurons	$ \mathbb{A} = 2$	$ \mathbb{A} ^{MN} = 16$
No. of hidden layers	1	1
Hidden layer activation	ReLU	ReLU
Output layer activation	Softmax	Softmax
Optimization	Adam	Adam
Loss function	Binary cross entropy	Categorical cross entropy
Training SNR	10 dB	10 dB
No. of training examples	30 000	30 000
No. of epochs	50	50

Table 9.2 Complexity (in no. of real operations) of detectors in Fig. 9.9.

Detector	ML det.	Symbol-DNN det.	Full-DNN det.
Complexity	1088	304	564
Trainable parameters	—	184	316

the full-DNN will be large with many trainable parameters. Therefore, symbol-DNN detection and minimum mean square error (MMSE) detection are considered here due to their low complexities. The symbol-DNN architecture used for detection in this system is as follows:

Symbol-DNN: Input $\rightarrow 512 \rightarrow \text{ReLU} \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 2 \rightarrow \text{Softmax}$.

There are $MN = 256$ such symbol-DNNs that constitute the detector. Each symbol-DNN is trained using the parameters in Table 9.3. In Fig. 9.10, it is observed that symbol-DNN detection achieves significantly better performance compared to MMSE detection. For example, symbol-DNN performs better by about 4 dB at 10^{-3} BER. Table 9.4 shows the complexities of the two detectors in a number of real operations. It is observed that the symbol-DNN detector performs better than the MMSE detector and is computationally efficient.

9.2.3.2 Static Channel with Non-Gaussian Noise

In order to illustrate the effect of non-Gaussian noise on detection performance, consider the case where the noise follows t -distribution with parameter μ . The t -distribution pdf resembles the Gaussian pdf with some deviation. The amount of deviation depends on the value of the parameter μ . Larger the value of μ , closer the t -distribution to Gaussian pdf. Figure 9.11 shows the performance of symbol-DNN detection and ML detection as per (9.14) for $M = N = 2$, $P = 2$, $f_c = 4$ GHz, $\Delta f = 3.75$ kHz, BPSK, and t -distributed noise with parameter $\mu = 5$. The performance of ML detection with i.i.d. Gaussian noise is also plotted for comparison. The DNN architecture used for Fig. 9.9 and the parameters in Table 9.2 are used. As expected, Fig. 9.11 shows the degraded

Table 9.3 Parameters of the symbol-DNN detector in Fig. 9.10.

Parameters	Symbol-DNN
No. of input neurons	$2MN = 512$
No. of output neurons	$ \mathbb{A} = 2$
No. of hidden layers	1
Hidden layer activation	ReLU
Output layer activation	Softmax
Optimization	Adam
Loss function	Binary cross entropy
Training SNR	8 dB
No. of training examples	80 000
No. of epochs	20

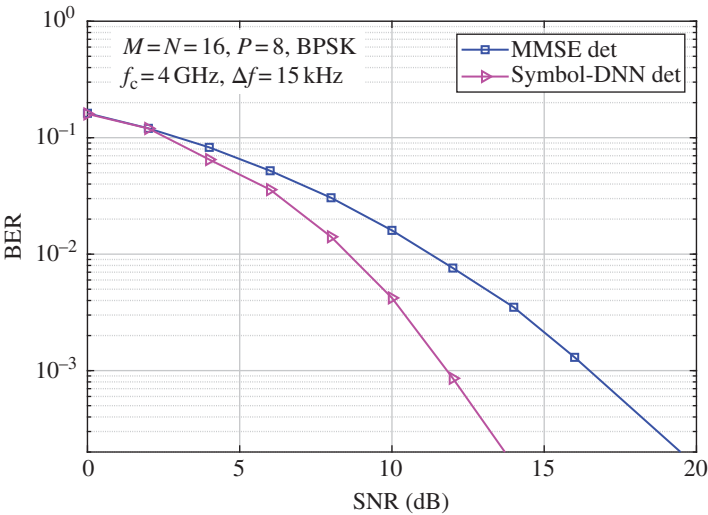


Figure 9.10 BER performance of OTFS with $M = N = 16$ using symbol-DNN and MMSE detection in the static multipath channel with $P = 8$.

performance of the ML detector when the noise model deviates from Gaussian distribution to t -distribution. This is because the ML decision rule in (9.14) is optimum only when noise is i.i.d. Gaussian and it is suboptimum when noise is non-Gaussian. On the other hand, the symbol-DNN performs significantly better in the non-Gaussian (t -distribution) noise case. This is because of the inherent ability of the DNN to learn the underlying noise distribution. A similar performance behavior is observed in a larger OTFS system with $M = N = 16$, $P = 8$, and BPSK in Fig. 9.12. The DNN architecture used for Fig. 9.10 and the parameters from Table 9.3 are used for Fig. 9.12. It can be seen that the symbol-DNN detector performs very well, while the performance of the MMSE detector gets degraded in the case of t -distributed noise.

Table 9.4 Complexity (in no. of real operations) of detectors in Fig. 9.10.

Detector	Complexity	Trainable parameters
MMSE det	83 951 616	—
Symbol DNN det	67 305 472	33 751 552

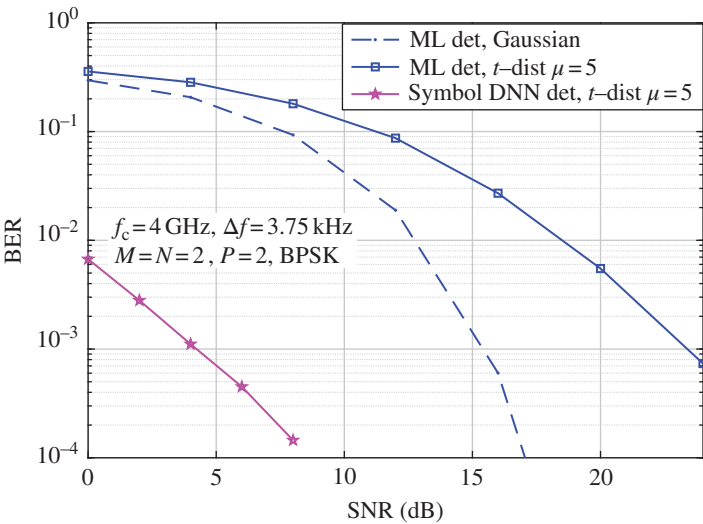


Figure 9.11 BER performance of OTFS with $M = N = 2$ in the static channel with $P = 2$ using symbol-DNN and ML detectors under t -distributed noise.

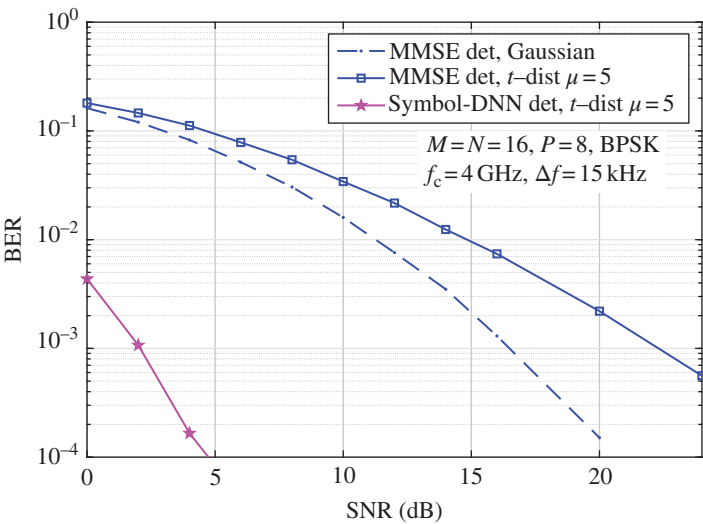


Figure 9.12 BER performance of OTFS with $M = N = 16$ in the static channel with $P = 8$ using symbol-DNN and MMSE detectors under t -distributed noise.

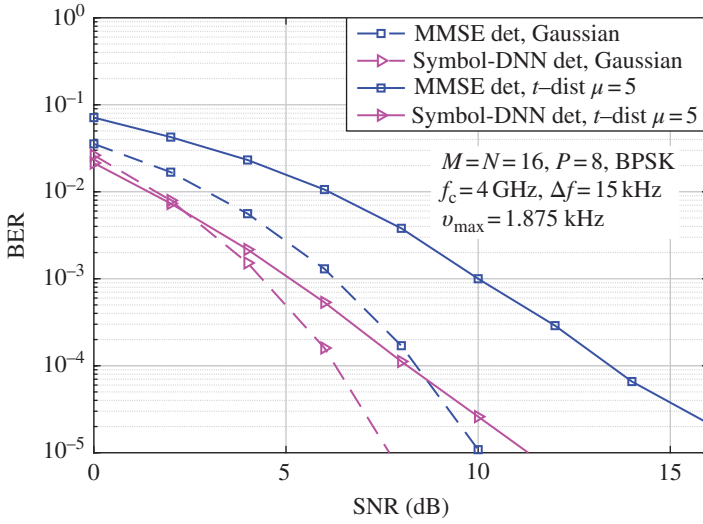


Figure 9.13 BER performance of OTFS with $M = N = 16$ in the non-zero Doppler channel with $P = 8$ under Gaussian and t -distributed noise.

Table 9.5 Delay-Doppler profile for Fig. 9.13.

Path, i	1	2	3	4	5	6	7	8
τ_i (μ s)	0	4.16	8.32	12.48	16.64	20.8	24.96	29.12
ν_i (Hz)	0	0	938.5	938.5	938.5	1875	1875	1875

9.2.3.3 Doppler Channel with Non-Gaussian Noise

Figure 9.13 shows the BER performance of symbol-DNN and MMSE detectors under non-zero Doppler conditions where the channel variations are slow in the DD domain. The system parameters considered are $M = N = 16$, $P = 8$, BPSK, $f_c = 4$ GHz, $\Delta f = 15$ kHz, and maximum Doppler $\nu_{\max} = 1.875$ kHz. The DD profile considered is given in Table 9.5. The DNN architecture of Fig. 9.10 and the parameters of Table 9.3 are considered for Fig. 9.13. The performance of symbol-DNN and MMSE detectors is compared for the cases of Gaussian noise and t -distributed noise. It is seen that the symbol-DNN detector outperforms the MMSE detector in both the cases, where the performance advantage of symbol-DNN is more in the case of t -distributed noise.

9.2.3.4 Correlated Gaussian Noise in MIMO-OTFS

In order to observe the effect of deviation from the i.i.d. assumption of noise, a multiple-input multiple-output (MIMO)-OTFS system model is considered where there is noise correlation across multiple receive antennas due to insufficient spacing between them at the receiver. Consider the correlation model [227], where the noise correlation matrix is given by

$$\mathbf{N}_c = \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n_r-1} \\ \rho & 1 & \rho & \dots & \rho^{n_r-2} \\ & & & \ddots & \\ \rho^{n_r-1} & \rho^{n_r-2} & \dots & & 1 \end{bmatrix} \quad (9.15)$$

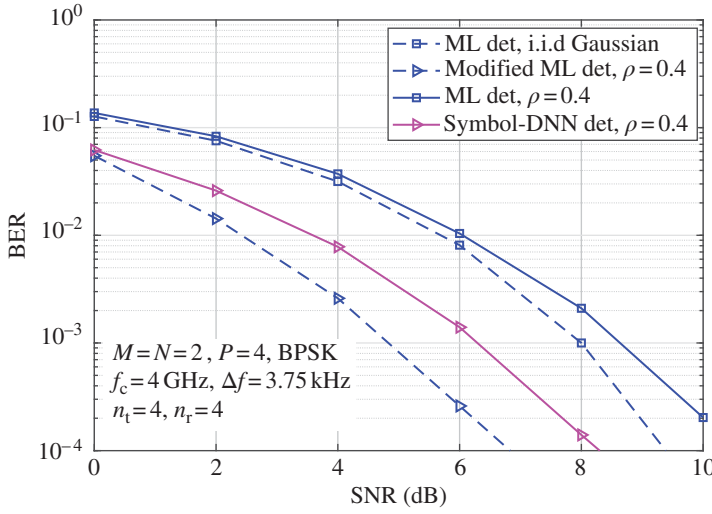


Figure 9.14 BER performance of the MIMO-OTFS system with $n_t = n_r = 4$, $M = N = 2$, $P = 4$, and BPSK in correlated noise using symbol-DNN and ML detectors.

and ρ is the correlation coefficient such that $0 \leq \rho \leq 1$. The correlated noise across the receive antennas is $\mathbf{n}_c = \mathbf{N}_c \mathbf{n}$, where $\mathbf{n}$ is the i.i.d. Gaussian noise.

Figure 9.14 shows the BER performance of symbol-DNN and ML detectors when the noise is correlated across receive antennas with correlation coefficient $\rho = 0.4$. The performance of the ML detector with i.i.d. Gaussian noise and the modified ML detector for the case of correlated noise is also plotted for comparison. The decision rule for the modified ML detector is given by

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathcal{A}^{MN}} (\mathbf{y} - \mathbf{H}\mathbf{x})^H \Sigma^{-1} (\mathbf{y} - \mathbf{H}\mathbf{x}), \quad (9.16)$$

where Σ is the estimated noise covariance matrix [228]. The system parameters considered are number of transmit antennas $n_t = 4$, number of receive antennas $n_r = 4$, $M = N = 2$, $P = 4$, BPSK, $f_c = 4$ GHz, $\Delta f = 3.75$ kHz, and $v_{\max} = 1.875$ kHz. The symbol-DNN architecture used for detection is as follows.

Symbol-DNN: Input $\rightarrow 32 \rightarrow \text{ReLU} \rightarrow 8 \rightarrow \text{ReLU} \rightarrow 16 \rightarrow \text{ReLU} \rightarrow 2 \rightarrow \text{Softmax}$.

The DNN is trained for 100 epochs with 80 000 training examples at a training SNR of 4 dB.

It is observed from Fig. 9.14 that the symbol-DNN detector performs better than the conventional ML detector in (9.13) because (i) the ML detection in (9.13) is suboptimum for correlated noise and (ii) the DNN learns the underlying noise model including the noise correlation associated with the receive antennas. The modified ML detector in (9.16) gives the best performance as it is optimum for the correlated noise case. However, it has high computational complexity. On the other hand, the symbol-DNN detector scales well for larger systems in the case of correlated noise.

In summary, symbol-level DNN architectures for OTFS signal detection are found to be attractive in terms of both error performance and complexity. Also, when the noise model deviates from the standard Gaussian i.i.d. model, the DNN-based detection approach is found to learn the underlying noise distribution and outperform conventional ML detection, which is optimum only for the standard i.i.d. Gaussian noise model.

9.3 IQI Estimation/Compensation Using DNNs

In wireless transceivers, radio frequency (RF) impairments such as IQ imbalance (IQI) at the transmitter and receiver can affect system performance. IQI occurs due to imperfect match/balance between I (in-phase) and Q (quadrature phase) arms in the RF chain because of which the orthogonality between I and Q arms is compromised. Unless compensated, this leads to performance degradation. The IQI impairment is modeled in terms of the gain and phase imbalances between the I and Q arms. The learning framework can be adopted for the estimation and compensation of these imbalances at the transmitter/receiver. This section presents the design of a multi-DNN architecture for IQI estimation and compensation in OTFS transceivers [170]. Refer Section 8.3 in Chapter 8 for the OTFS transceiver system model in the presence of transmit and receive IQI.

The transmit IQI compensation scheme considered here is realized using a single DNN at the transmitter which learns and provides a compensating modulation alphabet (to pre-rotate the modulation symbols before sending through the transmitter) without explicitly estimating the transmit gain and phase imbalances. The receive IQI compensation considered is realized using two DNNs at the receiver, one DNN for explicit estimation of receive gain and phase imbalances and another DNN for compensation.

9.3.1 DNN-Based OTFS Transceiver

An OTFS transceiver design that carries out IQI estimation/compensation and channel training and detection using DNNs is presented here. The transceiver performs the following tasks: (i) IQI compensation at the transmitter, (ii) IQI compensation at the receiver, and (iii) channel training and detection at the receiver.

9.3.1.1 Tx IQI Compensation

If $\mathbf{x}$ is the OTFS signal vector at the input of the transmit chain with IQI, then the corresponding vector at the output of the transmit chain is given by (see Fig. 8.9 and Eqn. (8.48))

$$\mathbf{x}_{IQ} = \alpha_T \mathbf{x} + \beta_T \mathbf{x}^*, \quad (9.17)$$

where $\mathbf{x}^*$ is the image signal vector generated due to Tx IQI, and the complex scalars α_T and β_T are dependent on the transmitter gain and phase imbalances ($\Delta G_T, \Delta \phi_T$) and are given by (8.46) and (8.47), respectively. While the intent is to transmit the vector $\mathbf{x}$, the actual transmitted vector becomes $\mathbf{x}_{IQ}$ because of Tx IQI. Assuming an ideal RF chain at the receiver (without receiver IQI), the received vector can be written as (see Eqn. (8.49))

$$\mathbf{y} = \mathbf{H}\mathbf{x}_{IQ} + \mathbf{v}. \quad (9.18)$$

Figure 9.15 shows a DNN-based Tx IQI compensation scheme. The DNN, referred to as DNN-1, is a fully connected DNN with $2|\mathbb{A}|$ input neurons and $2|\mathbb{A}|$ output neurons, where $|\mathbb{A}|$ is the size of the modulation alphabet $\mathbb{A}$. The alphabet $\mathbb{A}$ is given as input to the Tx IQI model in (8.48) which generates the alphabet affected by Tx IQI (denoted by $\mathbb{A}_{iq}$). The DNN takes the real and imaginary parts of $\mathbb{A}_{iq}$ and training data as input and generates a compensating alphabet (denoted by $\mathbb{A}_{comp}$), which is fed to the transmit chain. The transmit chain maps $\mathbb{A}$ to $\mathbb{A}_{comp}$ and sends the symbols from $\mathbb{A}_{comp}$ through the chain. This ensures that the alphabet symbols at the transmitter output are pre-compensated and the effect of IQI is nullified. The data used for training the DNN is multiple

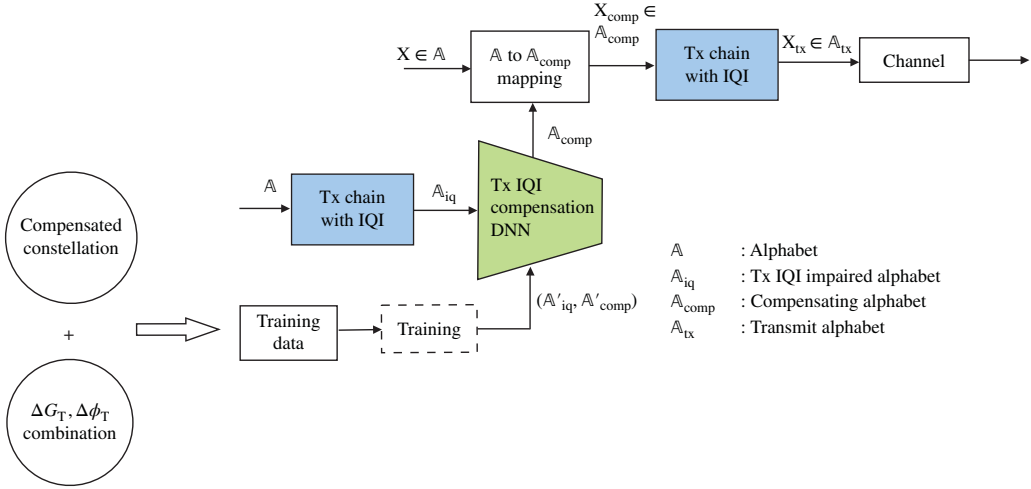


Figure 9.15 DNN-based architecture for Tx IQI compensation.

realizations of $(\mathbb{A}'_{iq}, \mathbb{A}'_{comp})$ pairs, where $\mathbb{A}'_{iq}$ are generated using the Tx IQI model for different $(\Delta G_T, \Delta \phi_T)$ values and $\mathbb{A}'_{comp}$ are obtained using the compensation model, given by [170]

$$\begin{bmatrix} \mathbb{A}'_{comp} \\ \mathbb{A}'^*_{comp} \end{bmatrix} = \begin{bmatrix} \alpha_T & \beta_T \\ \beta_T^* & \alpha_T^* \end{bmatrix}^{-1} \begin{bmatrix} \mathbb{A} \\ \mathbb{A}^* \end{bmatrix}. \quad (9.19)$$

9.3.1.2 Rx IQI Compensation

As in the case of transmitter with IQI, the receiver with IQI will also generate an image at the receiver output, which is modeled as (see Fig. 8.10 and Eqn. (8.50))

$$\mathbf{y}_{IQ} = \alpha_R \mathbf{y} + \beta_R \mathbf{y}^*, \quad (9.20)$$

where the complex scalars α_R and β_R are dependent on the receiver gain and phase imbalances $(\Delta G_R, \Delta \phi_R)$ and are given by (8.51) and (8.52), respectively.

At the receiver, Rx IQI compensation is done using two DNNs, namely, DNN-2 and DNN-3, as shown in Fig. 9.16. DNN-2 is a fully connected DNN with $2|\mathbb{A}|$ input neurons and 2 output neurons meant for estimating the gain and phase imbalances. DNN-3 is also a fully connected network with $2MN$ input neurons and $2MN$ output neurons meant for compensating the imbalances. In DNN-2, the Rx IQI impaired alphabet vector (denoted by $\mathbf{s}_{iq}$) is generated by using $\mathbb{A}$ (e.g., BPSK) in the IQI model in (8.50). It takes the real and imaginary parts of $\mathbf{s}_{iq}$ and training data as input and gives the estimated values of gain and phase imbalances (denoted by $\Delta \hat{G}_R$ and $\Delta \hat{\phi}_R$, respectively) as the output. The data used for training DNN-2 is multiple realizations of $(\mathbf{s}'_{iq}, \mathbf{e}'_{est})$ pairs, where $\mathbf{e}'_{est} = [\Delta G_R; \Delta \phi_R]$ and $\mathbf{s}'_{iq}$ are obtained using different $(\Delta G_R, \Delta \phi_R)$ values in (8.50). The estimated $\Delta \hat{G}_R$ and $\Delta \hat{\phi}_R$ obtained from DNN-2 are subsequently used for training DNN-3 meant for Rx IQI compensation. In DNN-3, the received signal vector $\mathbf{y}$ is given as input to the Rx IQI model in (8.50) which generates the received vector affected by IQI (denoted by $\mathbf{y}_{iq}$). It takes the real and imaginary parts of $\mathbf{y}_{iq}$ and training data as input and generates a compensated received vector $\mathbf{y}_{comp}$, which is expected to be the same as the vector $\mathbf{y}$ received in the absence of Rx IQI. The Rx IQI compensated vector $\mathbf{y}_{comp}$ is subsequently used for signal detection in DNN-4. The training data for DNN-3 consists of multiple realizations of $(\mathbf{y}'_{iq}, \mathbf{g}'_{est})$ pairs where $\mathbf{g}'_{est} = [\Delta G_R; \Delta \phi_R]$ and $\mathbf{y}'_{iq}$ are generated according to (8.50).

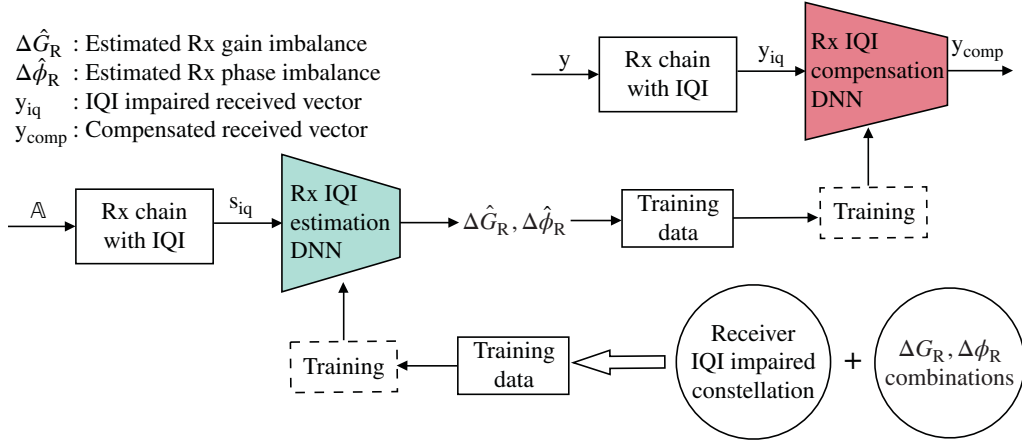


Figure 9.16 DNN-based architecture for Rx IQI compensation.

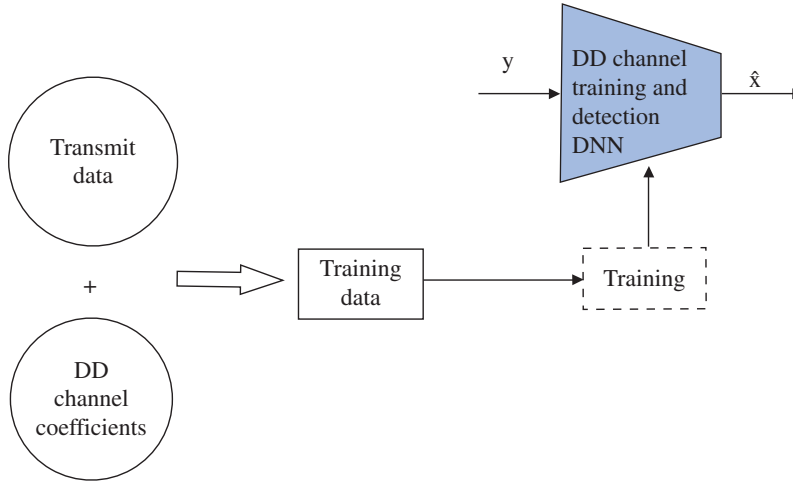


Figure 9.17 DNN-based architecture for channel training and detection.

9.3.1.3 Channel Training and Detection

The task of channel training and detection is carried out by DNN-4 in Fig. 9.17, which is a fully connected DNN with $4MN$ input neurons and MN output neurons. The input to the DNN consists of two OTFS frames. The first frame is a pilot frame consisting of pilot symbol(s), and the second frame is a data frame consisting of information symbols. The DD channel is considered to be constant over these two frames. The pilot frame is assumed to consist of one pilot symbol at a fixed pilot location in the frame. At the receiver, these two frames are vectorized to form the input to the DNN as $\mathbf{y} = [\mathbf{y}_p; \mathbf{y}_d]$, where $\mathbf{y}_p$ and $\mathbf{y}_d$ are the received vectors corresponding to the pilot frame and the data frame, respectively. The same pilot frame is used for all the data frames in a spatial coherence interval. The DNN takes the real and imaginary parts of the vector $\mathbf{y}$ as input and recovers the information bits in the data frame. The training data for the DNN consists of multiple realizations of $(\mathbf{y}', \mathbf{x}')$ pairs, where $\mathbf{x}'$'s are pseudo-randomly generated transmit vectors and $\mathbf{y}'$'s are obtained using (9.13) for different realizations of DD channel fade coefficients.

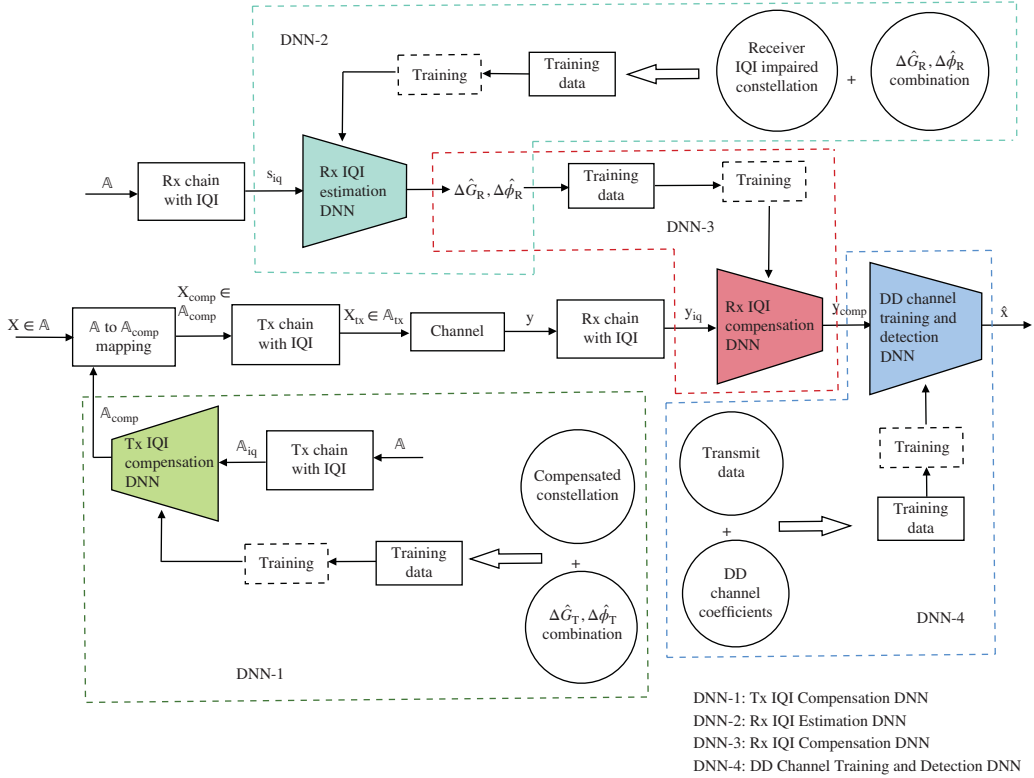


Figure 9.18 Combined DNN-based OTFS transceiver architecture.

Figure 9.18 shows the full OTFS transceiver architecture combining the four DNNs described above.

9.3.2 Performance Results

The BER performance of the OTFS transceiver without and with DNN-based IQI compensation are presented here. Table 9.6 shows the OTFS system parameters considered in the simulations.

Table 9.6 OTFS system parameters.

Parameter	Value
Frame size (M, N)	$(4, 4)$
Carrier frequency (GHz)	4
Subcarrier spacing (kHz)	3.75
No. of DD channel paths (P)	4
Delay-Doppler profile (τ_i, ν_i)	$(0, \frac{1}{NT}), (\frac{1}{M\Delta f}, \frac{1}{NT}),$ $(\frac{2}{M\Delta f}, \frac{1}{NT}), (\frac{3}{M\Delta f}, \frac{1}{NT})$
Delay power profile	Uniform

Table 9.7 Parameters of Tx IQI compensation DNN-1.

Parameters	BPSK-DNN	4QAM-DNN	16QAM-DNN
No. of input neurons	$2 \mathbb{A} = 4$	$2 \mathbb{A} = 8$	$2 \mathbb{A} = 32$
No. of output neurons	$2 \mathbb{A} = 4$	$2 \mathbb{A} = 8$	$2 \mathbb{A} = 32$
No. of hidden layers	4	3	3
Hidden layer activation	Tanh	Tanh	Tanh
Output layer activation	Linear	Linear	Linear
Optimization	Adam	Adam	Adam
Loss function	MSE	MSE	MSE
No. of training examples	1000	1000	1000
No. of epochs	5000	5000	5000
Batch size	5	5	5

9.3.2.1 Tx IQI Compensation

The performance of Tx IQI compensation DNN is considered here assuming the receiver RF chain to be ideal. Table 9.7 shows the parameters used in DNN-1 for BPSK, 4-QAM, and 16-QAM. DNN-1 has five layers with one input layer, three hidden layers, and one output layer. The activation function used in the hidden layers is Tanh activation function. The linear activation function is used for the output layer so that the real-valued output is not confined between any range. The mean-squared error loss function is chosen, which minimizes the average squared difference between the DNN output and the expected output. DNN-1 architecture for 4-QAM is as follows.

DNN-1 for 4-QAM: Input $\rightarrow 8 \rightarrow \text{Tanh} \rightarrow 64 \rightarrow \text{Tanh} \rightarrow 32 \rightarrow \text{Tanh} \rightarrow 16 \rightarrow \text{Tanh} \rightarrow 8 \rightarrow \text{Linear}$.

Figure 9.19 illustrates the effectiveness of the Tx IQI compensation achieved by DNN-1. It shows the ideal 4-QAM alphabet $\mathbb{A}$, the Tx IQI impaired alphabet $\mathbb{A}_{\text{iq}}$, the compensating alphabet $\mathbb{A}_{\text{comp}}$, and the transmit alphabet $\mathbb{A}_{\text{tx}}$ for $\Delta G_{\text{T}} = 0.4$ and $\Delta \phi_{\text{T}} = 40^\circ$. From Fig. 9.19, it is seen that the transmitted symbols after compensation are almost the same as the corresponding symbols from the ideal alphabet $\mathbb{A}$. This shows that DNN-1 is effective in compensating the Tx IQI. This effectiveness translates into very good BER performance, as shown in Fig. 9.20. Figure 9.20 shows three BER plots as a function of signal-to-noise ratio (SNR) for the following three cases: (i) ideal Tx with no IQI, (ii) Tx with IQI and with no IQI compensation, and (iii) Tx with IQI and with DNN-based IQI compensation. The symbols are detected using MMSE detection. It is seen that, though Tx IQI significantly degrades the BER performance, the performance achieved by the Tx IQI compensation using DNN-1 is almost the same as the ideal BER performance achieved in the absence of IQI.

A similar restoration of BER performance is achieved by DNN-1 in the case of BPSK and 16-QAM as well (see Fig. 9.21). ML and MMSE detection are used for BPSK and 16-QAM, respectively. The DNN architectures used for BPSK and 16-QAM are as follows.

DNN-1 for BPSK: Input $\rightarrow 4 \rightarrow \text{Tanh} \rightarrow 64 \rightarrow \text{Tanh} \rightarrow 32 \rightarrow \text{Tanh} \rightarrow 16 \rightarrow \text{Tanh} \rightarrow 8 \rightarrow \text{Tanh} \rightarrow 4 \rightarrow \text{Linear}$.

DNN-1 for 16-QAM: Input $\rightarrow 32 \rightarrow \text{Tanh} \rightarrow 256 \rightarrow \text{Tanh} \rightarrow 128 \rightarrow \text{Tanh} \rightarrow 64 \rightarrow \text{Tanh} \rightarrow 32 \rightarrow \text{Linear}$.

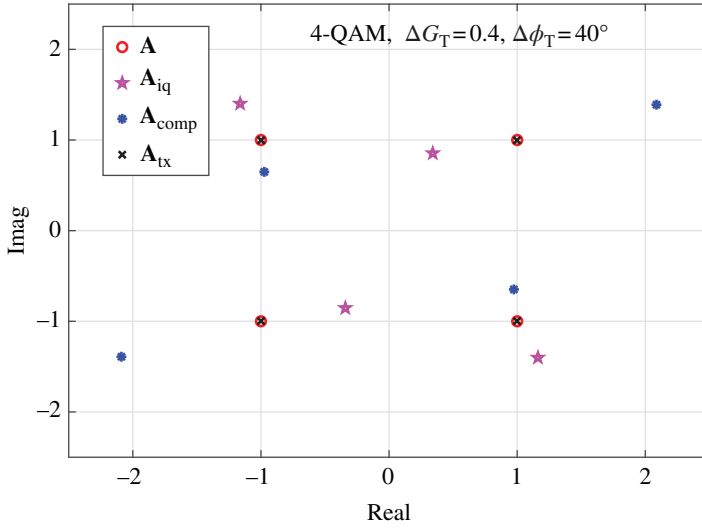


Figure 9.19 Illustration of the ideal 4-QAM alphabet ($\mathbb{A}$), Tx IQI impaired alphabet ($\mathbb{A}_{iq}$), compensating alphabet ($\mathbb{A}_{comp}$), and transmit alphabet ($\mathbb{A}_{tx}$) in DNN-1 for $\Delta G = 0.4$ and $\Delta\phi = 40^\circ$.

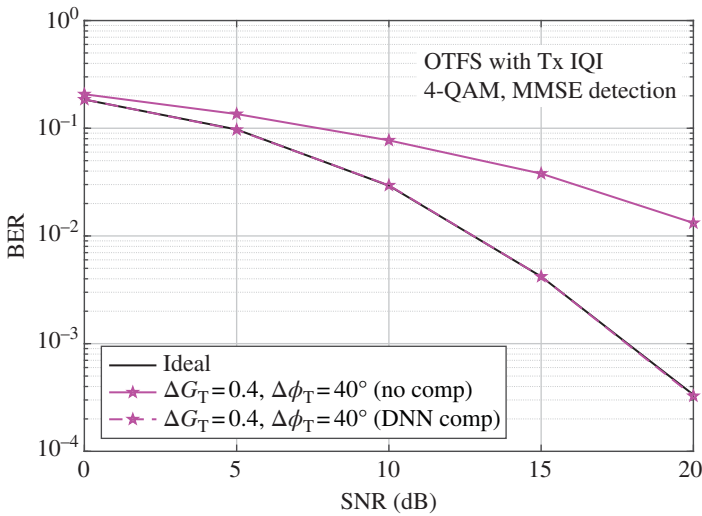


Figure 9.20 BER performance of OTFS with Tx IQI compensation using DNN-1 for 4-QAM.

From both Figs. 9.20 and 9.21, it is observed that DNN-1 has effectively nullified the effect of Tx IQI and achieved near-ideal performance.

9.3.2.2 Rx IQI Compensation

The performance of DNN-2 and DNN-3 for the estimation and compensation of Rx IQI in the absence of Tx IQI is considered here. The Rx IQI parameters, $\Delta\hat{G}_R$ and $\Delta\hat{\phi}_R$, are estimated first using DNN-2 using a test alphabet $\mathbb{A}$ as the input. DNN-3 is trained separately for BPSK, 4-QAM, and 16-QAM. The IQ-impaired received vector at the receiver is passed through DNN-3, and the output obtained is used for detection using conventional methods like ML and MMSE detection.

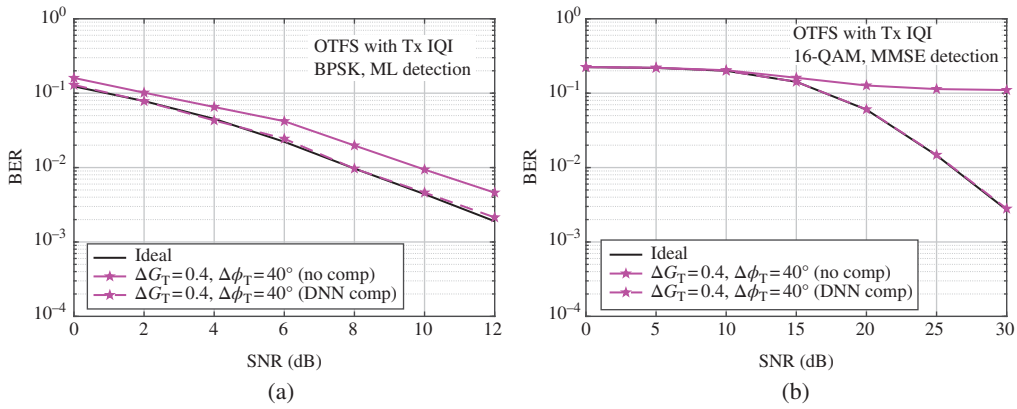


Figure 9.21 BER performance of OTFS with Tx IQI compensation using DNN-1 for BPSK (a) and 16-QAM (b).

Table 9.8 Parameters of Rx IQI DNN-2 and DNN-3.

Parameters	DNN-2	DNN-3
No. of input neurons	$2 \mathbb{A} = 4$	$2MN = 32$
No. of output neurons	2	$2MN = 32$
No. of hidden layers	1	1
Hidden layer activation	Tanh	Tanh
Output layer activation	Linear	Linear
Optimization	Adam	Adam
Loss function	MSE	MSE
No. of training examples	1000	50 000
No. of epochs	500	500
Batch size	5	50

Table 9.8 shows the parameters used in DNN-2 and DNN-3. DNN-2 and DNN-3 architectures used are as follows.

DNN-2: Input $\rightarrow 4 \rightarrow \text{Tanh} \rightarrow 8 \rightarrow \text{Tanh} \rightarrow 2 \rightarrow \text{Linear}$.

DNN-3: Input $\rightarrow 32 \rightarrow \text{Tanh} \rightarrow 64 \rightarrow \text{Tanh} \rightarrow 32 \rightarrow \text{Linear}$.

Figures 9.22(a), (b), and (c) show the combined BER performance of DNN-2 and DNN-3 for BPSK, 4-QAM, and 16-QAM, respectively. The values of gain and phase imbalances considered are $\Delta G_R = 0.4$ and $\Delta \phi_R = 40^\circ$. In all the three cases, it can be seen that the performance achieved by the proposed DNN-2 and DNN-3 in the presence of Rx IQI is almost the same as the BER performance of an ideal transceiver without IQI. This shows that the proposed Rx IQI DNNs are able to nullify the degrading effects caused by Rx IQI.

9.3.2.3 Channel Training and Detection

An OTFS system with $M = N = 4$ and BPSK is considered here to illustrate the channel training and detection performance of DNN-4. DNN-4 has four layers. The hidden layers use the ReLU activation

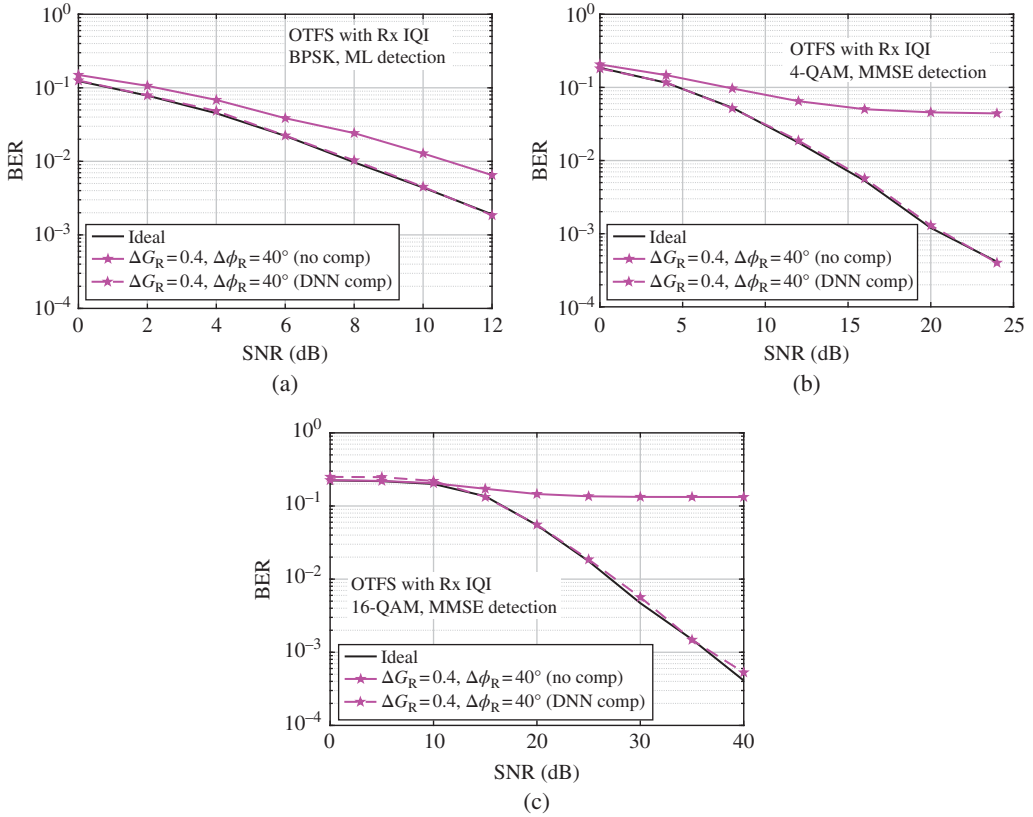


Figure 9.22 BER performance of OTFS with Rx IQI estimation and compensation using DNN-2 and DNN-3 for BPSK (a), 4-QAM (b), and 16-QAM (c).

function, and the output layer uses the sigmoid activation function. The output of the DNN has values in the interval $[0,1]$ due to the sigmoid activation at the output layer. The transmitted bits are recovered by thresholding the output at 0.5. Table 9.9 shows the parameters used in DNN-4. The architecture used in DNN-4 is as follows.

DNN-4: Input $\rightarrow 64 \rightarrow \text{ReLU} \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 64 \rightarrow \text{ReLU} \rightarrow 16 \rightarrow \text{sigmoid}$.

Figure 9.23 shows the BER performance of DNN-4 as a function of pilot SNR (SNR_p). The BER is computed at a data SNR (SNR_d) of 10 dB. DNN-4 is trained at $\text{SNR}_p = 10$ dB and $\text{SNR}_d = 10$ dB. DNN-4 once trained can be used over a spatial coherence interval. The performance of DNN-4 is compared with the BER performance of ML detection and MMSE detection which use the channel estimated using impulse-based channel estimation [39]. The BER with perfect channel knowledge is also shown for comparison. It can be seen that DNN-4 performs better than the other methods at pilot SNR lower than 20 dB. Beyond that, it performs worse than ML detection but performs slightly better than MMSE detection. Also, the DNN trained at a particular pilot SNR performs well when tested at a different pilot SNR.

9.3.2.4 Combined Performance of All DNNs

Figure 9.24 shows the BER performance of OTFS using the complete DNN-based transceiver architecture with Tx and Rx IQI compensation and DD channel training/detection. The Tx and Rx

Table 9.9 Parameters of channel training and detection DNN-4.

Parameters	DNN-4
No. of input neurons	$4MN = 64$
No. of output neurons	$MN = 16$
No. of hidden layers	2
Hidden layer activation	ReLU
Output layer activation	Sigmoid
Training data SNR	10 dB
Training pilot SNR	10 dB
Optimization	Adam
Loss function	MSE
No. of training examples	200 000
No. of epochs	500
Batch size	500

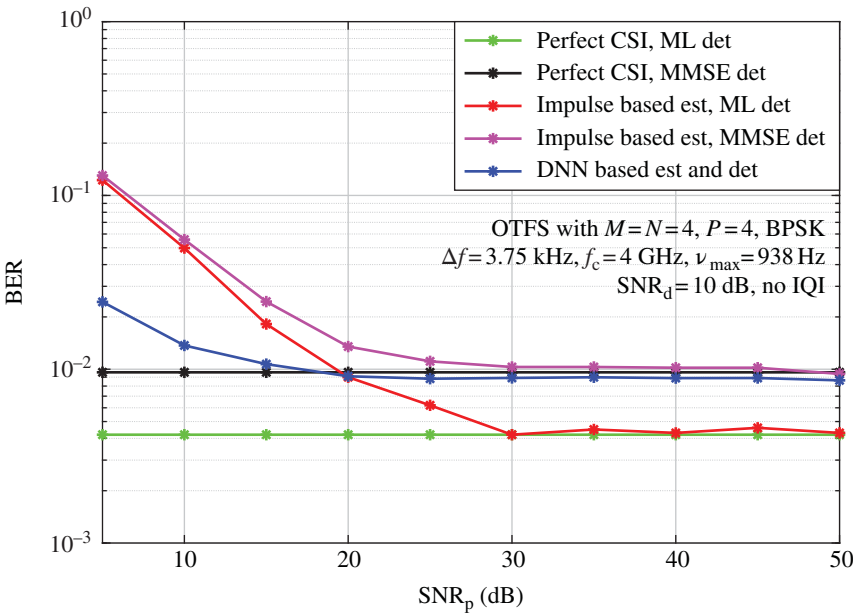


Figure 9.23 BER performance of DNN-4 as a function of pilot SNR.

IQI are compensated using DNN-1, DNN-2, and DNN-3. The BER performance for the following cases are obtained and compared: (i) impulse-based channel estimation and ML detection, (ii) impulse-based channel estimation and MMSE detection, and (iii) DNN-based channel training and detection using DNN-4. The performance of ML detection with perfect channel knowledge is also compared. The comparison is done at $\text{SNR}_p = 10$ dB, 20 dB, 30 dB. When the pilot SNR is small,

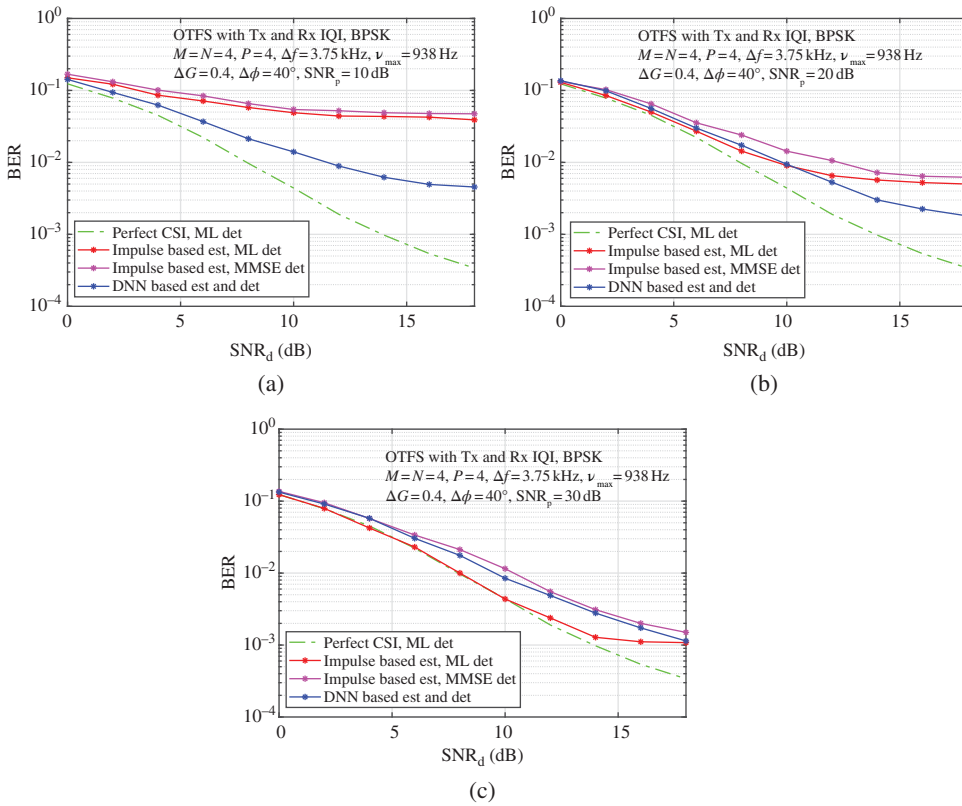


Figure 9.24 BER performance of OTFS using the complete DNN-based transceiver with Tx and Rx IQI compensation and DD channel training/detection at different pilot SNRs. (a) Pilot SNR = 10 dB, (b) pilot SNR = 20 dB, and (c) pilot SNR = 30 dB.

both ML detection and MMSE detection methods show degradation in the BER performance due to increased error in the impulse-based estimation of the channel fade coefficients. The DNN-based detection outperforms the other detection methods at $\text{SNR}_p = 10$ dB and $\text{SNR}_p = 20$ dB because of effective channel training. At $\text{SNR}_p = 30$ dB, ML detection with impulse-based estimation gives the best performance. Also, the performance of DNN-4 is comparable to that of MMSE detection with impulse-based channel estimation.

In summary, the presented multi-DNN-based transceiver architecture that carries out DD channel training and detection of OTFS signals along with IQI compensation could learn the IQIs present in the transmitter and receiver and effectively compensate them. Also, it could learn the DD channel effectively and detect the information symbols, suggesting that DNN-based techniques are attractive for the design of practical OTFS transceivers.

9.4 DD Channel Estimation Using DNNs

Accurate estimation of the DD domain channel is essential for reliable OTFS signal detection at the receiver. See Chapter 6 for a compilation of DD channel estimation techniques/algorithms using traditional approaches. This section deals with DD channel estimation using DNNs. The task of DD

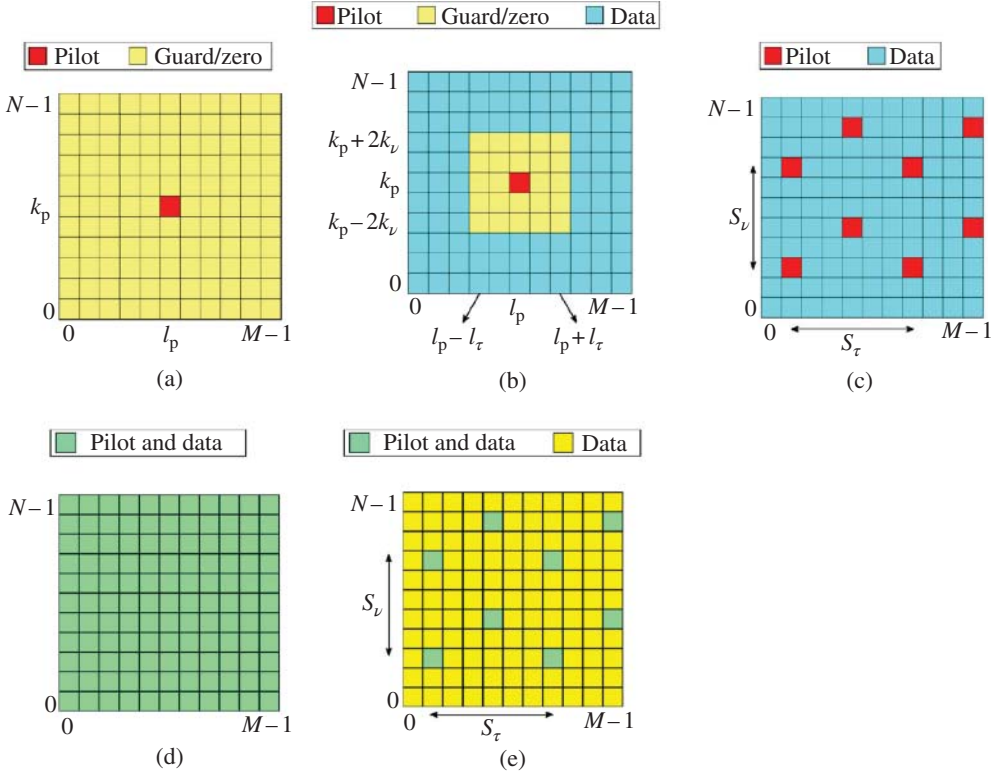


Figure 9.25 Pilot placement schemes: (a) exclusive pilot, (b) embedded pilot, (c) interleaved pilots, (d) fully superimposed pilots, and (e) sparse superimposed pilots.

channel estimation amounts to estimation of the delays (τ_i s) and Dopplers (ν_i s) of various paths and the corresponding channel fade coefficients (h_i s). Pilot symbols are sent in OTFS frames for this purpose. The number of pilot symbols in an OTFS frame and where they are placed in the frame influence the rate and performance. The placement of pilot and data symbols in an OTFS frame can be done in different ways. Widely considered pilot placement schemes in the OTFS literature include the exclusive pilot scheme [39], embedded pilot scheme [102], interleaved pilot scheme [229], fully superimposed pilot (FSP) scheme [112], and sparse superimposed pilot (SSP) schemes [114]. The above pilot placement schemes are illustrated in Fig. 9.25.

Exclusive pilot scheme: In the exclusive pilot scheme, a frame is used exclusively for pilot. A popularly considered pilot placement in this scheme is one where a single pilot symbol a_p is placed at the (k_p, l_p) th location in a frame and zeros elsewhere (see Fig. 9.25(a)), i.e., the (k, l) th entry in the frame $x[k, l]$ is given by

$$x[k, l] = \begin{cases} a_p, & \text{for } k = k_p, l = l_p, \\ 0, & \text{elsewhere,} \end{cases} \quad (9.21)$$

for $k = 0, \dots, N-1$ and $l = 0, \dots, M-1$. This scheme is simple, but it suffers rate loss because the full frame goes as overhead.

Embedded pilot scheme: In the embedded pilot scheme, each OTFS frame consists of a pilot symbol surrounded by guard bins, and the remaining bins carry data symbols (see Fig. 9.25(b)).

The entries of the frame are given by

$$x[k, l] = \begin{cases} a_p, & \text{for } k = k_p, l = l_p, \\ 0, & \text{for } k = k_g, l = l_g, \\ a_d \in \mathbb{A}, & \text{elsewhere,} \end{cases} \quad (9.22)$$

for $k = 0, \dots, N - 1$ and $l = 0, \dots, M - 1$, (k_p, l_p) is the location of the pilot symbol, (k_g, l_g) s are the locations of the guard symbols (zeros), and a_d s are data symbols from modulation alphabet $\mathbb{A}$. The guard bins are introduced to avoid pilot leakage into data bins, and vice versa. The number of guard bins is chosen to accommodate the maximum delay and Doppler spreads of the channel, and these guard bins contribute to some rate loss. Note that the exclusive pilot scheme becomes a special case of the embedded pilot scheme when all bins other than the pilot bin are guard bins.

Interleaved pilot scheme: In this scheme, pilot symbols are placed across each frame in a lattice-type arrangement (see Fig. 9.25(c)). The pilots are surrounded by data symbols without any guard bins in between. The pilots are separated in the delay domain by S_τ bins and in the Doppler domain by S_ν bins, which are chosen based on the number of pilots, N_p , in a frame, with the constraint $S_\tau > m_\tau$ and $S_\nu > n_\nu$, where m_τ and n_ν are the delay and Doppler tap number corresponding to the maximum delay spread $\tau_{\max}$ and maximum Doppler spread $\nu_{\max}$, respectively. While guard bins are avoided in this scheme, the receive signal processing must be capable of handling the effect of the leakage between pilot and data symbols. Also, the presence of pilots causes some rate loss.

Fully superimposed pilot scheme: In this scheme, all bins in each frame carry data as well as pilot symbols, i.e., each bin carries a data symbol and a pilot symbol superimposed on it (see Fig. 9.25(d)). Since pilots are superimposed on data symbols in all the bins, there is no rate loss in this scheme. However, inter-symbol leakage/interference needs to be handled and more sophisticated channel estimation and detection/equalization techniques are needed.

Sparse superimposed pilot scheme: In this scheme, as in the case of FSP scheme, all the bins in a frame carry data symbols. However, pilot symbols are superimposed only in some of these bins. This is in contrast with the FSP scheme where pilots are superimposed in all the bins. The rationale behind this scheme is the possibility of achieving a better inter-symbol leakage/interference profile while retaining full rate. The bins where pilot symbols are superimposed follow a interleaved lattice-type arrangement, as shown in Fig. 9.25(e). Since the pilot symbols per frame are fewer in this scheme compared to the FSP scheme, the energy per pilot symbol is high for the same total pilot energy per frame which can positively influence the estimation performance. Still, the leakage between the pilot and data symbols needs to be handled in this scheme as well.

In Sections 9.4.1–9.4.3, methods to estimate the channel parameters (τ_i, ν_i, h_i) using learning networks are presented. DNNs devised for this purpose for the cases of embedded pilot, interleaved pilot, and superimposed pilot schemes are presented. For illustration purposes, both integer DD estimation (using embedded and interleaved pilots schemes) and fractional DD estimation (using superimposed pilots) are considered.

9.4.1 Estimation Using Embedded Pilots

This section presents a multilayered RNN architecture for integer DD channel estimation. The embedded pilot scheme shown in Fig. 9.25(a) is considered. A key ingredient in this learning approach is a generalized training methodology that makes the network generalize for OTFS frames with different pilot arrangements. In other words, once trained, the network can seamlessly work for OTFS frames with different guard band sizes.

The symbols $x[k, l]$ s in the DD domain are converted to a time domain signal $x(t)$ at the OTFS transmitter and transmitted through a doubly spread channel. At the OTFS receiver, the received signal $y(t)$ is converted back to DD symbols $y[k, l]$ s, $k = 0, \dots, N - 1$, $l = 0, \dots, M - 1$. Let $y'[k', l']$ s, a subset of $y[k, l]$ s, denote the received DD symbols corresponding to the pilot and guard bins, where $k' = k_p - k_v, k_p - k_v + 1, \dots, k_p + k_v$ and $l' = l_p, l_p + 1, \dots, l_p + l_r$ for the embedded pilot scheme in (9.22) and $k' = 0, 1, \dots, N - 1$, $l' = 0, 1, \dots, M - 1$ for the exclusive pilot scheme in (9.21). The number of symbols in $y'[k', l']$ s is $(2k_v + 1)(l_r + 1)$ for the pilot scheme in (9.22) and MN for the pilot scheme in (9.21), respectively. These $y'[k', l']$ symbols are converted to a vector $\mathbf{y}'$ of length $(2k_v + 1)(l_r + 1)$ for the scheme in (9.22) and of length MN for the scheme in (9.21). The input to the network is the vector $\mathbf{y}'$. The output of the network is a vector $\mathbf{m}$, called mask, of length $(2k_v + 1)(l_r + 1)$ for the scheme in (9.22) and MN for the scheme in (9.21). Entries in $\mathbf{m}$ are values between 0 and 1. These entries are thresholded such that values above 0.5 are replaced by 1 and those below are replaced by 0. In the thresholded $\mathbf{m}$ vector, denoted by $\mathbf{m}'$, the indices corresponding to the location of 1s denote the presence of valid channel paths at those locations in the DD grid. These locations are used to obtain the estimates of the integers corresponding to delay taps (i.e., $\hat{\alpha}$ s) and Doppler frequencies ($\hat{\beta}$ s) in the DD grid. Finally, the vector $\mathbf{m}'$ is element-wise multiplied with the input vector $\mathbf{y}'$ and the non-zero values from the resulting vector are returned as the DD domain channel coefficient vector $\hat{\mathbf{h}}$. Using the estimates $\hat{\mathbf{h}}$, $\hat{\alpha}$, and $\hat{\beta}$, the estimated DD domain channel matrix $\hat{\mathbf{H}}$ is obtained. This matrix is used for the detection of data symbols.

9.4.1.1 Architecture

The architecture of the estimation network (called DDNet [102]) is shown in Fig. 9.26. The architecture consists of Q layers of LSTM [220], a variant of RNN. The output of the LSTM layers is passed through a ReLU activation function, given by (9.11). This is then passed on to a FCNN with one layer. The FCNN is employed to reduce the dimension of the output of the LSTM network to the required dimension. This is then followed by a sigmoid activation function, given by (9.12). The purpose of using a sigmoid function is to restrict the output between 0 and 1 and to determine if a particular DD bin contains a valid path. To achieve this, as mentioned above, the mask $\mathbf{m}$ at the output of sigmoid function is first thresholded to obtain the vector $\mathbf{m}'$, followed by element-wise multiplication with the input. The non-zero entries in the resulting vector constitute the estimated DD channel coefficients, denoted by $\hat{\mathbf{h}}$. The other parameters of the network architecture are given in Table 9.10. The variable c refers to the batch size and s is the sequence length, which is set to be MN for the exclusive pilot scheme in (9.21) and $(2k_v + 1)(l_r + 1)$ for the embedded pilot scheme in (9.22).

9.4.1.2 Training Methodology

Training data is obtained by generating multiple DD domain OTFS frames using pilot schemes in (9.21) and (9.22). Furthermore, various guard band realizations are also added to the training data for the scheme in (9.22). These frames are converted to the time domain and transmitted through

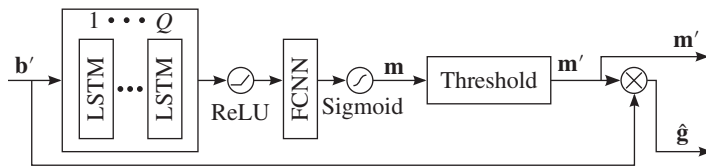


Figure 9.26 DDNet: RNN-based architecture for DD channel estimation.

Table 9.10 Parameters of the DDNet architecture.

Parameter	Value
Number of LSTM layers	Q
LSTM hidden size (h)	100
LSTM input dimensions	$(c, s, 2)$
LSTM output dimensions	$(c, s, 100)$
FCNN input neurons	100
FCNN output neurons	1

a doubly spread channel, and the received signal is converted back to the DD domain. From the received DD symbols, depending on the pilot scheme employed, s -length vector $\mathbf{y}'$ is obtained. The real and imaginary parts of this vector are concatenated before being fed to the network. For training the network, the ground truth is obtained by generating the s -length true mask, denoted by $\mathbf{z}$, whose i th entry is defined as

$$z_i = \begin{cases} 1, & \text{if DD bin corresponding to } i \text{ is a valid path,} \\ 0, & \text{else.} \end{cases} \quad (9.23)$$

During training, the weights of the network are updated such that the value of the BCE loss function between $\mathbf{z}$ and the output of the network, $\mathbf{m}$, is minimized. The BCE loss function for the i th index is given by

$$L(z_i, m_i) = -z_i \log(m_i) - (1 - z_i) \log(1 - m_i), \quad (9.24)$$

where $0 \leq m_i \leq 1$ is the output of the network and $z_i = \{0, 1\}$ is the ground truth. The other hyper-parameters used in the training of the network are presented in Table 9.11. This training needs to be carried out offline, only once. Subsequently, the network weights are frozen. New channel estimates are obtained from pilots in each OTFS frame using the same trained network.

9.4.1.3 Inference from the Network

Once the network is trained, the weights are frozen. During the inference (testing) phase, channel estimates, $\hat{\mathbf{h}}$, in the DD domain are obtained through element-wise multiplication of the input ($\mathbf{y}'$) with the thresholded mask ($\mathbf{m}'$), as shown in Fig. 9.31. To obtain the estimates of α and β , denoted by $\hat{\alpha}$ and $\hat{\beta}$, respectively, the following steps are followed for the case of the exclusive

Table 9.11 Hyper-parameters used for training the DDNet.

Parameter	Value
Epochs	20 000
Optimizer	Adam
Learning rate	0.001, divide by 2 every 4000 epochs
Batch size	1000
Mini-batch size	64

pilot scheme in (9.21). Let $\mathcal{J}$ denote the set of indices where the thresholded mask $\mathbf{m}'$ is 1, i.e., $\mathcal{J} = \{j : m'_j = 1, j = 0, 1, \dots, s-1\}$. Then, for the i th path index,

$$\hat{\alpha}_i = (\mathcal{J}_i)_M - l_p, \quad \hat{\beta}_i = \left\lfloor \frac{\mathcal{J}_i}{M} \right\rfloor - k_p. \quad (9.25)$$

For the embedded pilot scheme in (9.22) where $s = (2k_v + 1)(l_\tau + 1)$, the thresholded mask $\mathbf{m}'$ is reshaped into a matrix of shape $(2k_v + 1) \times (l_\tau + 1)$. Vectors $\mathbf{u}$ and $\mathbf{v}$ are defined as $\mathbf{u} = [k_p - k_v, k_p - k_v + 1, \dots, k_p + k_v]$ and $\mathbf{v} = [l_p, l_p + 1, \dots, l_p + l_\tau]$. Then, an index set, $\mathcal{I}$, is defined as $\mathcal{I} = \{u_i \times M + v_j : m'_{ij} = 1, i = 0, 1, \dots, 2k_v, j = 0, 1, \dots, l_\tau\}$. For the i th path index,

$$\hat{\alpha}_i = (\mathcal{I}_i)_M - l_p, \quad \hat{\beta}_i = \left\lfloor \frac{\mathcal{I}_i}{M} \right\rfloor - k_p. \quad (9.26)$$

9.4.1.4 Performance Results

To evaluate the performance of the network, the following system parameters are considered: carrier frequency of $f_c = 4$ GHz, subcarrier spacing of $\Delta f = 15$ kHz, Vehicular A (Veh-A) channel model defined by ITU-R [70] with $P = 6$ paths, and a maximum speed of 220 km/h. This speed at 4 GHz carrier frequency corresponds to a maximum Doppler, $\nu_{\max}$, of 815 Hz. Each path has a Doppler generated using Jakes model $\nu_i = \nu_{\max} \cos \theta_i$, where θ_i is assumed to be uniformly distributed in the range of $[-\pi, \pi]$. The number of Doppler bins (N) and delay bins (M) are 12 and 64, respectively. A symbol +1 is used as the pilot symbol, and data symbols are chosen from the 4-QAM alphabet. To train the network, the batch size (c) is chosen to be 1100 of which 1000 OTFS frames are used for training and 100 frames are used for validating the training. This training data is refreshed every 20 epochs, wherein the pilot schemes in (9.21) and (9.22) are chosen randomly with equal probability. The accuracy of the channel estimates provided by the network is evaluated by computing the normalized mean square error (NSME) for the DD domain channel matrix as follows. The estimates of the channel coefficients, delay taps, and Doppler taps are obtained from the network as described before. Using these values, an estimate for the matrix $\mathbf{H}$, denoted by $\hat{\mathbf{H}}$, is obtained. The NMSE is computed as $\text{NMSE} = \mathbb{E} \left[\frac{\|\mathbf{H} - \hat{\mathbf{H}}\|_F^2}{\|\mathbf{H}\|_F^2} \right]$. For evaluating the BER performance, the message passing (MP) detector in [38] is used.

Effect of Number of LSTM Layers Figure 9.27 shows the NMSE performance of the network as a function of pilot SNR for three different values of the number of LSTM layers, $Q = 1, 2, 3$. Other than the value of Q , the same parameters in Table 9.10 and the training hyper-parameters in Table 9.11 are used for all the values of Q . The number of parameters for Q layers can be computed as $N_Q = 4h^2(2Q - 1) + 4hi_d + 8Qh + 101$, where h is the hidden size (see Table 9.10), $i_d = 2$ is the input dimension, and 101 is the number of parameters in the FCNN layer. Therefore, the number of parameters to be learnt are 41 701, 122 501, and 203 301 for $Q = 1, 2$, and 3, respectively. The performance of embedded as well as exclusive pilot frames is shown. From Fig. 9.27, it can be seen that, while the NMSE performance is comparable for different values of Q , the performance for $Q = 3$ is slightly worse. This can be attributed to the steep increase in the number of parameters that need to be learnt for $Q = 3$, resulting in difficulty in training the network. As a good balance between training complexity and achieved performance, $Q = 2$ is considered in the subsequent performance evaluation experiments. During the inference (testing) stage, only 301 floating-point operations per second (FLOPs) are required to compute the mask from the network.

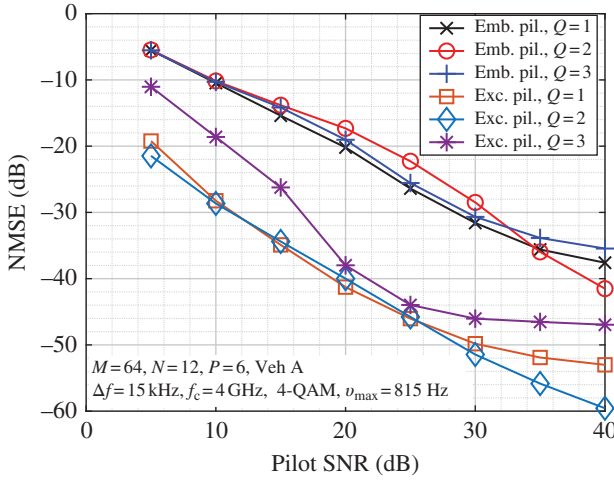


Figure 9.27 Effect of the number of LSTM layers, Q , on the NMSE performance of the DDNet.

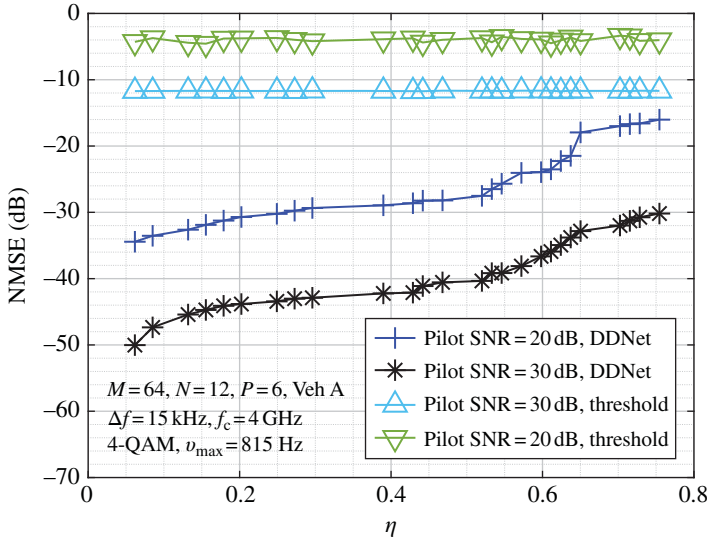


Figure 9.28 NMSE vs. spectral efficiency at different pilot SNRs.

NMSE vs. Spectral Efficiency Figure 9.28 shows the effect of the number of guard bins on the NMSE performance of the network. It shows NMSE as a function of spectral efficiency η , where η is defined as $\eta = 1 - \frac{N_g}{MN}$, and N_g is the number of guard bins in the frame. The pilot SNRs considered are 20 and 30 dB. The performance of the threshold-based channel estimation scheme in [102] is also plotted for comparison. It can be seen that the network achieves significantly better NMSE performance compared to the threshold-based scheme. Also, while the NMSE performance of the network improves as the number of guard bins N_g is increased (i.e., smaller values of η), the NMSE performance of the threshold-based scheme does not improve because of the inherent limitation in using a fixed threshold, for a given pilot SNR. However, the network is able to generalize for varying guard band sizes because of the training methodology employed.

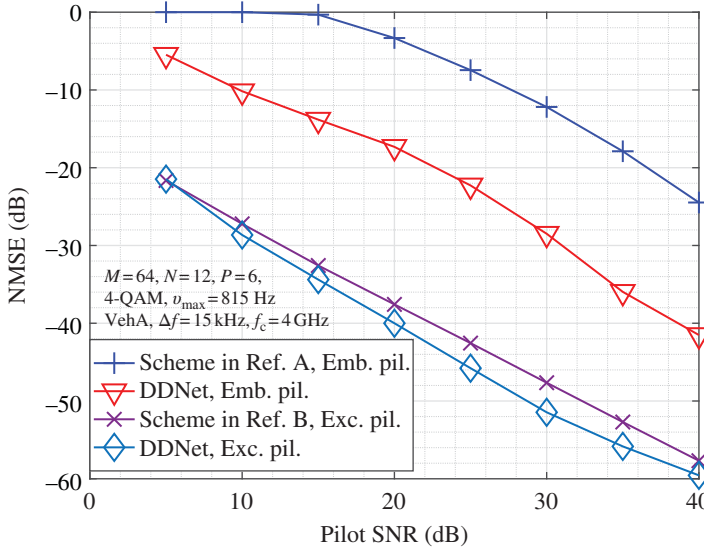


Figure 9.29 NMSE performance comparison between the DDNet and the estimation schemes in Ref. A [102] (embedded pilot) and Ref. B [39] (exclusive pilot).

NMSE Performance Comparison Figure 9.29 shows the NMSE comparison for the cases of the exclusive pilot frame and embedded pilot frame. For the exclusive pilot case, the comparison is made between the network and the scheme in [39]. For the embedded pilot case, the comparison is made between the network and the scheme in [102] for $\eta = 0.97$. The following observations can be made from Fig. 9.29. In the exclusive pilot case, the performance of the network is better compared to that of the scheme in [39]. For example, to achieve an NMSE of -50 dB, the network requires about 4 dB less pilot SNR. In the embedded pilot case, the performance of the network is far superior to that of the scheme in [102]. The scheme in [102] does not work well up to 15 dB pilot SNR because of the erroneous estimation of the delay and Doppler indices at these SNRs. For pilot SNRs greater than 15 dB, the NMSE is seen to reduce with pilot SNR. Even in this SNR region, there is a significant performance advantage for the network. For example, to achieve an NMSE of -20 dB, the network requires about 14 dB less pilot SNR.

BER Performance Comparison Figure 9.33 presents a comparison between the BER performance of the network and that of the scheme in [102] with embedded pilot frame with $\eta = 0.97$. The BER performance with perfect channel state information (CSI) is also presented for comparison. It is seen that the network outperforms the scheme in [102] by a large margin. For example, at 40 dB pilot SNR, there is about 4 dB advantage for the network at a BER of 10^{-2} . Also, the performance of the network at this pilot SNR is close to the performance with perfect CSI. When the pilot SNR is 20 dB, the scheme in [102] fails to perform, whereas the network performs much better. This corroborates with the NMSE performance advantage predicted in Fig. 9.29, where the scheme in [102] has a high NMSE value at 20 dB pilot SNR.

9.4.2 Estimation Using Interleaved Pilots

This section presents an RNN-based channel estimation scheme for the interleaved pilot scheme shown in Fig. 9.25(c) with integer DDs. The network used for this purpose (called IPNet [229]) is

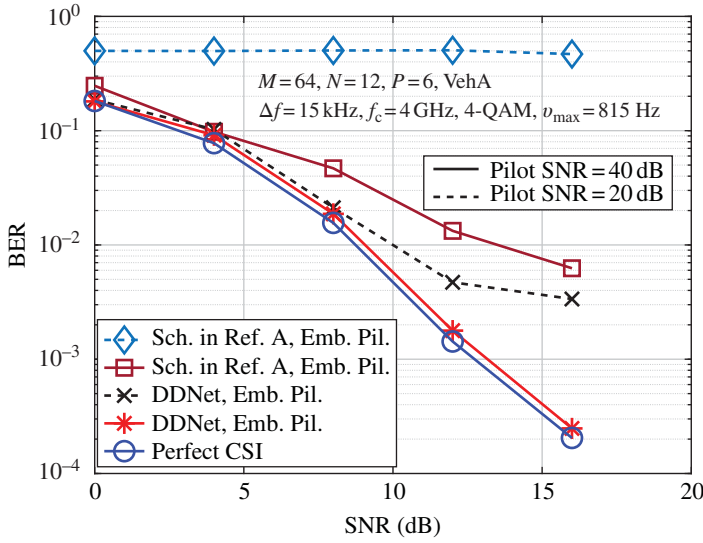


Figure 9.30 BER performance comparison between the DDNet and the thresholding scheme in Ref. A [102].

shown in Fig. 9.31. RNNs have been used in the deep learning literature for drawing inference from correlated sequences [230]. In the interleaved pilot scheme, the pilot symbols leak into the adjacent data bins, resulting in a sequence of correlated received symbols. Therefore, the use of RNNs is a natural approach for making inference (i.e., estimating the channel) from the correlated received symbols. The RNN is trained to overcome the effects of leakage from data symbols and provide channel estimates with good accuracy. The obtained channel estimates are then used to remove the effect of leakage from pilot symbols in the detection of data symbols. To further improve the accuracy of the channel estimates, the detected symbols are used to further reduce the leakage from data symbols and another set of channel estimates is obtained from the network. This process is carried out in an iterative fashion.

The following set of operations are carried out on the received DD frame to obtain the first set of channel estimates. Let n_{p_i} and m_{p_i} denote the Doppler and delay indices for the i th pilot symbol, respectively (see Fig. 9.25(c)), where $i = 1, \dots, N_p$. Due to the channel, the pilot symbol spreads into the nearby DD bins. For the i th pilot, the spread is contained within the indices $n_{p_i} - n_v$ to $n_{p_i} + n_v$ on the Doppler axis and m_{p_i} to $m_{p_i} + m_r$ on the delay axis. From the received OTFS frame, the symbols in these locations are extracted and vectorized to obtain the vector $\mathbf{y}'_i \in \mathbb{C}^{(2n_v+1)(m_r+1) \times 1}$ for the i th pilot. This is repeated for each pilot to obtain the vector $\mathbf{b}' \in \mathbb{C}^{N_p(2n_v+1)(m_r+1) \times 1}$, given by

$$\mathbf{y}' = [\mathbf{y}'_1, \mathbf{y}'_2, \dots, \mathbf{y}'_{N_p}]. \quad (9.27)$$

Note that the vector $\mathbf{y}'$ contains the effect of both pilot and data symbols. The input to the network is the vector $\mathbf{y}'$. The output of the network is a vector $\hat{\mathbf{h}} \in \mathbb{C}^{(2n_v+1)(m_r+1) \times 1}$ of channel estimates. Among the $(2n_v+1)(m_r+1)$ entries in this vector, only those channel estimates are picked as valid paths for which the absolute value is greater than 4% of the maximum absolute value in the vector, i.e., if $\hat{h}_{\max} = \max_i |\hat{h}_i|$, then

$$\hat{h}_i = \begin{cases} 0, & \text{if } |\hat{h}_i| \leq 0.04 \hat{h}_{\max}, \\ \hat{h}_i, & \text{otherwise.} \end{cases} \quad (9.28)$$

This operation is carried out because the absolute value of the output of the network for an invalid path is close to zero but not exactly zero. The locations of the valid paths are then used to obtain the estimates for integers corresponding to delay taps ($\hat{\alpha}_i s$) and Doppler taps ($\hat{\beta}_i s$) in the DD grid. Using the estimates $\hat{\mathbf{h}}$, $\hat{\alpha}$, and $\hat{\beta}$, the estimated DD domain channel matrix $\hat{\mathbf{H}}$ is obtained. This matrix is used for detection of data symbols. To further improve the accuracy of the channel estimates, the output of the detector, $\mathbf{x}'[n, m]$, is fed back to the network for cancelling the effect of data symbols. A new DD frame is constructed as

$$\mathbf{y}'' = \mathbf{y} - \hat{\mathbf{H}}\mathbf{x}', \quad (9.29)$$

where $\mathbf{x}' \in \mathbb{C}^{NM \times 1}$ and $\mathbf{y} \in \mathbb{C}^{NM \times 1}$ are the vectorized versions of $\mathbf{x}'[n, m]$ and $\mathbf{y}[n, m]$, respectively. Vector $\mathbf{y}'$ is computed again using (9.27) and $\mathbf{y}''$ as the received frame. This is provided as input to the network, and another set of refined channel estimates are obtained. This iterative procedure is repeated P times, and the output of the detector at the end of P th iteration, $\hat{\mathbf{x}}[n, m]$, is used to compute the bit error performance.

9.4.2.1 Architecture

The architecture of the network (called IPNet) is shown in Fig. 9.31. It consists of Q layers of LSTM, a variant of RNN. The output of the LSTM layers is passed through a ReLU activation function, given by $\text{ReLU}(x) = \max(0, x)$, $\forall x \in (-\infty, \infty)$. An FCNN with one layer is employed to reduce the dimension of the output of the LSTM network to the required dimension. Since the output of the FCNN layer needs to be the channel estimate, a linear activation function, with a range of $(-\infty, \infty)$, is used at the output of the FCNN. Valid paths are picked from the resulting vector at the output of the FCNN layer using (9.28). The resulting vector $\hat{\mathbf{h}}$ is then returned as the channel coefficient vector. The other parameters of the network are presented in Table 9.12. The variable c refers to the batch size, and $s = N_p(2n_v + 1)(m_\tau + 1)$ is the sequence length. The output of the FCNN is a vector of dimension $2(2n_v + 1)(m_\tau + 1)$, where the first $(2n_v + 1)(m_\tau + 1)$ dimensions are treated as real and the remaining as the imaginary part of the channel estimates.

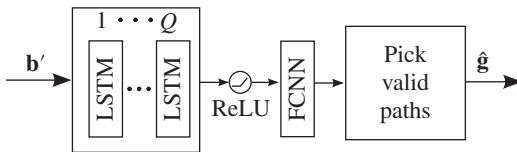


Figure 9.31 IPNet: RNN-based network architecture.

Table 9.12 Parameters of the IPNet architecture.

Parameter	Value
Number of LSTM layers (Q)	3
LSTM hidden size (h)	50
LSTM input dimensions	$(c, s, 2)$
LSTM output dimensions	$(c, s, 50)$
FCNN input neurons	50
FCNN output neurons	$2(2n_v + 1)(m_\tau + 1)$

9.4.2.2 Training Methodology

Training data is obtained by generating multiple OTFS frames with varying N_p (number of interleaved pilots). These frames are converted to the time domain and transmitted through the channel. The received signal is converted back to the DD domain. $N_p(2n_v + 1)(m_\tau + 1)$ symbols corresponding to the N_p transmitted pilots are extracted from the received frame as per (9.27) to obtain the vector $\mathbf{y}'$. The real and imaginary parts of $\mathbf{y}'$ are concatenated before being provided as input to the network. For training the network, the ground truth data is obtained by generating a $(2n_v + 1)(m_\tau + 1)$ -length true channel estimate vector, $\mathbf{h}$. This vector is constructed such that the entries are channel estimates only where there are valid paths and zeros elsewhere. During training, the weights of the network are updated such that the L1 loss between the output of network, $\hat{\mathbf{h}}$, and $\mathbf{h}$ is minimized. The L1 loss function is

$$L(\mathbf{h}, \hat{\mathbf{h}}) = \frac{1}{J} \sum_{i=1}^J |\mathbf{h} - \hat{\mathbf{h}}|, \quad (9.30)$$

where J is the number of samples in the training set. The other hyperparameters used while training are presented in Table 9.13. Once trained offline, the network weights are frozen and the same trained network is used to obtain channel estimates from pilots in each OTFS frame. Also, the trained network works well for various N_p values, owing to the construction of training data.

9.4.2.3 Estimation of Delay and Doppler Indices

During the inference (testing) phase, channel estimates, $\hat{\mathbf{h}}$, in the DD domain are obtained after picking the valid paths (based on (9.28)). To obtain the estimates of α and β , denoted by $\hat{\alpha}$ and $\hat{\beta}$, respectively, the following steps are followed. The channel estimate vector $\hat{\mathbf{g}} \in \mathbb{C}^{(2n_v+1)(m_\tau+1) \times 1}$ is reshaped into a matrix $\hat{\mathbf{H}} \in \mathbb{C}^{(2n_v+1) \times (m_\tau+1)}$. Index sets $\mathcal{I}$ and $\mathcal{J}$ are defined to store the row and column indices of the non-zero elements in $\hat{\mathbf{H}}$, respectively. That is, $\mathcal{I} = \{i : \hat{\mathbf{H}}[i, j] \neq 0, i = 0, \dots, 2n_v, j = 0, \dots, m_\tau\}$ and $\mathcal{J} = \{j : \hat{\mathbf{H}}[i, j] \neq 0, i = 0, \dots, 2n_v, j = 0, \dots, m_\tau\}$. Then, for the p th path index, the estimate of delay and Doppler indices are obtained as

$$\hat{\alpha}_p = \mathcal{J}_p, \quad \hat{\beta}_p = \mathcal{I}_p - n_v. \quad (9.31)$$

9.4.2.4 Complexity

The number of parameters to be learnt for Q layers can be computed as $N_Q = 4h^2(2Q - 1) + 4hi_d + 8Qh + 100 \times 2(2n_v + 1)(m_\tau + 1) + 2(2n_v + 1)(m_\tau + 1)$, where h is the hidden size (see Table 9.12), $i_d = 2$ is the input dimension, and $100 \times 2(2n_v + 1)(m_\tau + 1) + 2(2n_v + 1)(m_\tau + 1)$ is the number of

Table 9.13 Hyper-parameters used for training the IPNet.

Parameter	Value
Epochs	20 000
Optimizer	Adam
Learning rate	0.001, divide by 2 every 4000 epochs
Batch size	1000
Mini-batch size	64
Refresh training data	Every epoch

parameters in the FCNN layer. For $Q = 3$, $n_v = 1$, $m_\tau = 2$, $h = 50$, and $N_Q = 52\,518$. These parameters need to be learnt only once, offline. During the inference (testing) stage, only 1018 FLOPs are required to compute the channel estimate.

9.4.2.5 Performance Results

For assessing the NMSE and BER performance of the network, the system parameters and channel model used for the performance evaluation of embedded pilot scheme in Section 9.4.1 are used. Since the valid paths are chosen based on (9.28), which depends on the energy of channel estimates, the matrix $\hat{\mathbf{H}}$ may have more non-zero entries than the actual channel matrix $\mathbf{H}$. Therefore, at low pilot SNRs, where the noise energy is dominant, the NMSE can take values that are greater than 1.

NMSE Performance Figure 9.32 shows the NMSE performance of the network as a function of pilot SNR for $N_p = 8$ and 12. The pilot power is equally distributed among the N_p pilots. The NMSE performance of the embedded pilot scheme in [102] is also presented for comparison. The NMSE performance of the network without data cancellation (DC) for both the N_p values is observed to be close to that of the scheme in [102] with the performance being slightly inferior at SNRs above 35 dB. However, with one iteration of DC (see (9.29)), the NMSE performance improves beyond that of the scheme in [102] in the mid- and high-SNR regime. Note that, for the considered parameters, n_v is 1 and m_τ is 2. The scheme in [102], therefore, requires $(4n_v + 1)(2m_\tau + 1) = 25$ symbols for interference-free estimation of channel coefficients, while a better NMSE performance is achieved using the network scheme using fewer DD bins (12 and 8 bins for $N_p = 12$ and 8, respectively).

BER as a Function of Number of Pilots Figure 9.30 shows the BER performance of the network as a function of number of pilots, N_p , for a fixed pilot SNR of 30 dB and a fixed data SNR of 16 dB. Because of the training methodology adopted, the same trained network can work for different values of N_p . The performance of the scheme in [102], with 1 pilot and 24 guard symbols, is also presented for comparison. With the increase in the number of pilots, the BER performance of the network is observed to improve. This is because with the increase in N_p , the sequence length s

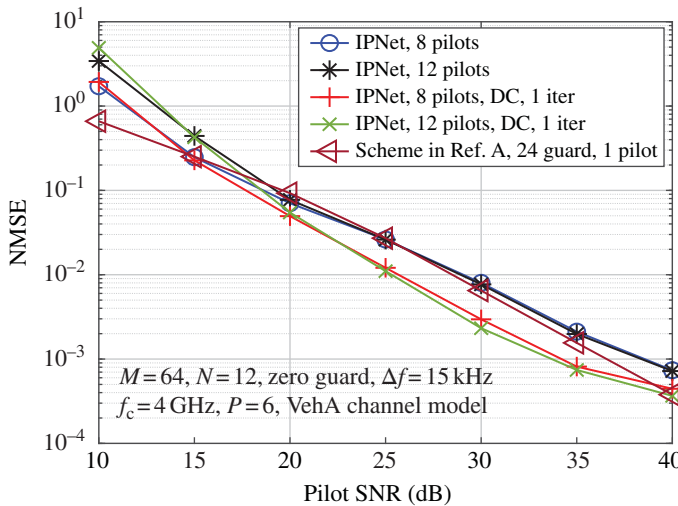


Figure 9.32 NMSE performance of the IPNet and the scheme in Ref. A [102] as a function of pilot SNR for a different number of pilots.

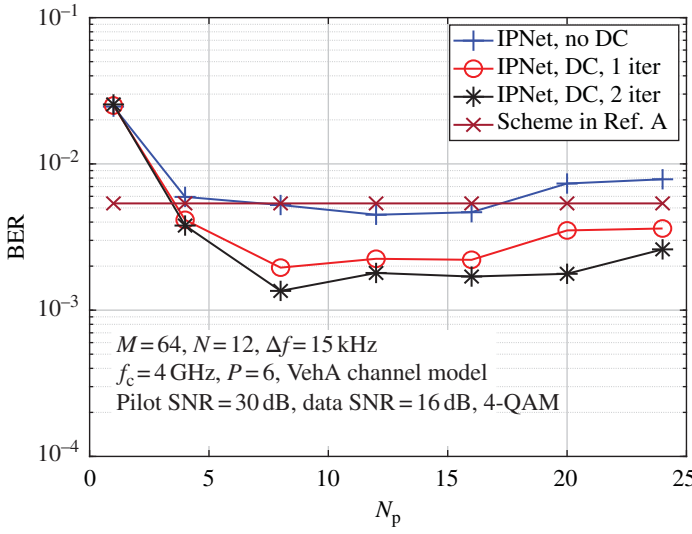


Figure 9.33 BER performance of the IPNet and the scheme in Ref. A [102] as a function of number of pilots, N_p .

at the input of the network also increases (see Table 9.12). This allows the network to provide estimates with better accuracy as more information is available at the input. However, with the further increase in the number of pilots ($N_p > 16$), the BER performance is observed to increase slightly, owing to the decrease in the energy per pilot symbol.

9.4.3 Estimation Using Superimposed Pilots

This section considers DD channel estimation with *fractional delay-Dopplers* using RNNs. The SSP scheme is considered. Both FSP and SSP schemes have the advantage of full rate (i.e., no rate loss due to guard/pilot symbols). Since pilot symbols are superimposed in all bins, the inter-symbol interference/leakage in FSP scheme can be severe. However, the SSP scheme can alleviate this issue since the number of superimposed pilots is fewer and the energy per pilot symbol is high for the same total pilot energy per frame.

Given a received frame, the task is to obtain estimates of the channel parameters (h_i, τ_i, ν_i) , $i = 0, \dots, P - 1$. Figure 9.34 shows the architecture of the network (called SSPNet [114]) for channel estimation. The vector of received symbols in the DD domain, $\mathbf{y}$ (generated as per the input-output relation for the case of fractional DD [114]) is used to generate the input vector to the network, $\mathbf{y}'$, as outlined below. Let n_{p_i} and m_{p_i} denote the Doppler and delay indices for the i th pilot

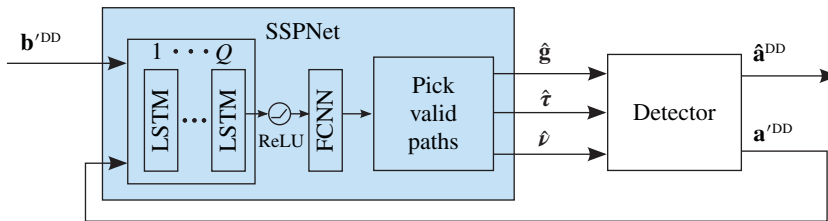


Figure 9.34 SSPNet: RNN-based channel estimation scheme.

symbol, respectively (see Fig. 9.25(e)), where $i = 1, \dots, N_p$, and N_p is the number of pilot symbols superimposed on the frame. The channel spreads the pilot symbols into their neighboring DD bins. For the i th pilot, the spread is contained within the indices $n_{p_i} - n_v$ to $n_{p_i} + n_v$ on the Doppler axis, and m_{p_i} to $m_{p_i} + m_\tau$ on the delay axis. Here, $m_\tau = \lceil \tau_{\max} M \Delta f \rceil$ and $n_v = \lceil \nu_{\max} N T \rceil$ are integers corresponding to the maximum delay and Doppler spread, respectively. The received symbols in the bins of the i th pilot's spread area in the frame are extracted and vectorized to obtain the vector $\mathbf{y}'_i \in \mathbb{C}^{(2n_v+1)(m_\tau+1) \times 1}$. This is done for each pilot. The concatenated vector $\mathbf{y}' \in \mathbb{C}^{N_p(2n_v+1)(m_\tau+1) \times 1}$, given by $\mathbf{y}' = [\mathbf{y}'_1 \mathbf{y}'_2 \dots \mathbf{y}'_{N_p}]^T$, is fed as the input to the network. The network obtains an estimate of the channel gain vector $\hat{\mathbf{h}} \in \mathbb{C}^{(2n_v+1)(m_\tau+1) \times 1}$. Among the $(2n_v + 1)(m_\tau + 1)$ entries in this vector, only those channel gain estimates whose absolute values are greater than a small threshold ϵ are picked as valid paths, i.e.,

$$\hat{h}_i = \begin{cases} 0, & \text{if } |\hat{h}_i| \leq \epsilon, \\ \hat{h}_i, & \text{otherwise.} \end{cases} \quad (9.32)$$

The locations of the valid paths are then used to obtain the delay and Doppler estimates, $(\hat{\tau}_i s)$ and $(\hat{\nu}_i s)$. Using the estimated vectors $\hat{\mathbf{h}}$, $\hat{\tau}$, and $\hat{\nu}$, the estimated DD domain channel matrix $\hat{\mathbf{H}}$ is obtained, which is used for the detection of data symbols.

9.4.3.1 Iterative Scheme

To further improve the accuracy of the channel estimates, the output of the detector, $\mathbf{x}'$, is fed back to the network for iteratively cancelling the effect of data symbols in channel estimation as follows. A new DD frame is constructed as

$$\mathbf{y}'' = \mathbf{y} - \hat{\mathbf{H}}\mathbf{x}'. \quad (9.33)$$

The vector $\mathbf{y}'$ is computed again using (9.27) with $\mathbf{y}''$ in (9.33) as the received frame. This updated $\mathbf{y}'$ vector is given as input to the network, and another set of refined channel estimates is obtained. This iterative procedure is repeated N_{iter} times or until a convergence criterion is met. The procedure is stopped at the i th iteration if $\|\hat{\mathbf{h}}^{(i)} - \hat{\mathbf{h}}^{(i-1)}\|^2 < \zeta$, i.e., the squared norm of the difference between the channel estimate vector at the i th iteration and the $(i - 1)$ th iteration is less than ζ . The output of the detector at the end of the iterations, denoted by $\hat{\mathbf{x}}$, is taken as the final detected output. The network architecture, hyper-parameters of the network, and the training methodology are the same as those used for the interleaved pilot scheme in Section 9.4.2.

9.4.3.2 Performance Results

The system parameters, channel model, and data detection scheme used for performance evaluation are same as those used in Sections 9.4.1 and 9.4.2 for the embedded and interleaved pilot schemes. Here, each data symbol and pilot symbol has energy denoted by σ_d^2 and σ_p^2 , respectively, with $\sigma_d^2 + \sigma_p^2 = 1$. This ensures that the average energy per frame is set to 1 for fair comparison between SSP and FSP schemes. The maximum number of iterations (N_{iter}) is 10, $S_\tau = 2m_\tau + 1 = 5$, $S_\nu = 4n_\nu + 1 = 5$, $\epsilon = 10^{-10}$, and $\zeta = 10^{-6}$.

Pilot Energy Spread in FSP and SSP Frames Figure 9.35 shows an example distribution of energy in various bins in FSP and SSP frames at the OTFS transmitter and receiver for integer DD. At the transmitter, there is no discernible difference between the pilot and data symbols in the FSP scheme (see Fig. 9.35(a)). However, in the SSP scheme, the pilot symbols have significantly higher energy ($MN\sigma_p^2/N_p$) than data symbols (see Fig. 9.35(b)). This is because of fewer pilots in the SSP

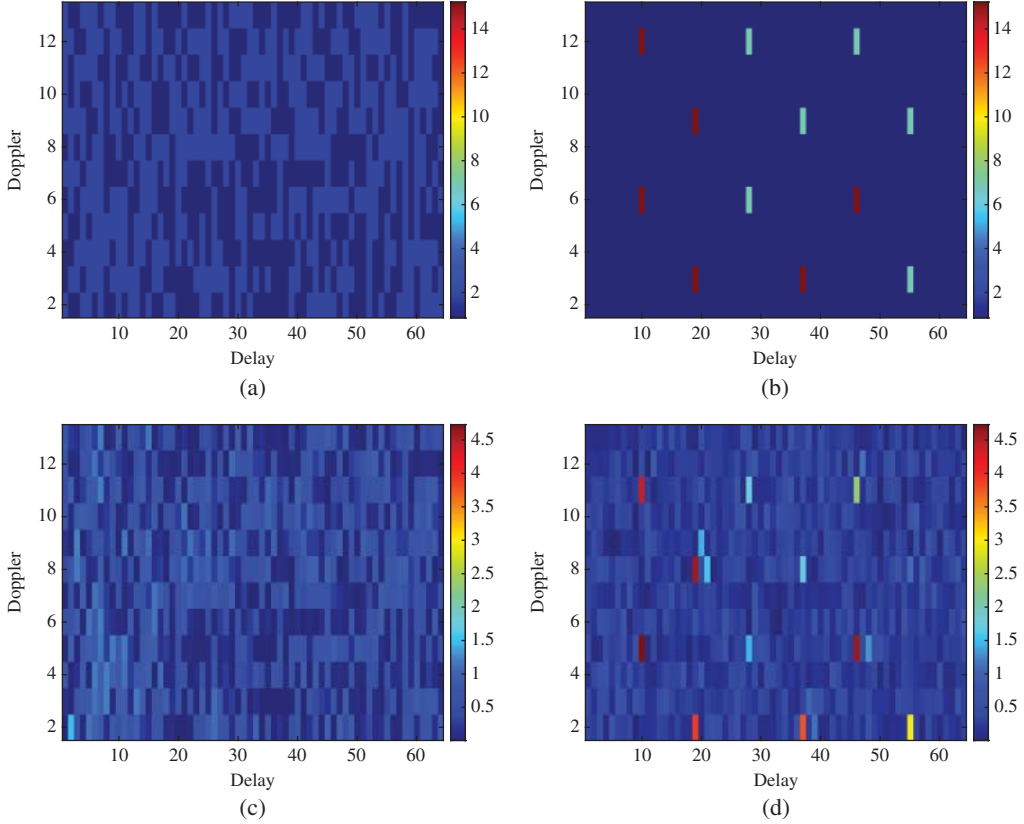


Figure 9.35 Energy distribution in FSP and SSP frames at the transmitter and receiver. (a) FSP Tx frame, $N_p = MN = 768$. (b) SSP Tx frame, $N_p = 12$. (c) FSP Rx frame, $N_p = MN = 768$. (d) SSP Rx frame, $N_p = 12$.

scheme for the same total pilot energy per frame. A similar trend is observed at the receiver as well (see Figs. 9.35(c) and (d)), where the pilots and data symbols in the FSP scheme have leaked into one another, while the interference among the pilots is alleviated in the SSP scheme. Due to higher energy per pilot symbol in the SSP scheme, the corresponding received frame also contains high energy DD bins that help to achieve improved channel estimation accuracy.

Choosing Optimal N_p and σ_d^2 Figure 9.36(a) shows the BER performance of using the SSPNet for channel estimation as a function of number of pilots, N_p , in a frame. When N_p is small, the network has very few observations (s) to work with and the MSE and consequently the BER performance is poor. The BER performance improves as N_p is increased. For $N_p > 20$, the energy per pilot symbol gets reduced and becomes comparable to the data symbol energy. This degrades the MSE of the channel estimates, and therefore, the BER increases. In this simulation setting, the BER attains its minimum value when $N_p = 12$. Figure 9.36(b) shows the BER performance as a function of the data energy, σ_d^2 . It is seen that the BER performance improves as the data energy σ_d^2 increases. Minimum BER is attained at $\sigma_d^2 = 0.84$, beyond which the BER increases. This is because $\sigma_p^2 < 0.16$ in this regime, and the energy per pilot symbol reduces, resulting in poor accuracy of channel estimates. The choice of $N_p = 12$ and $\sigma_d^2 = 0.84$ works well for both integer and fractional DDs. The optimal energy allocation for the FSP scheme in [112] is $\sigma_{p_{opt}}^2 = 0.3322$ and $\sigma_{d_{opt}}^2 = 0.6678$.

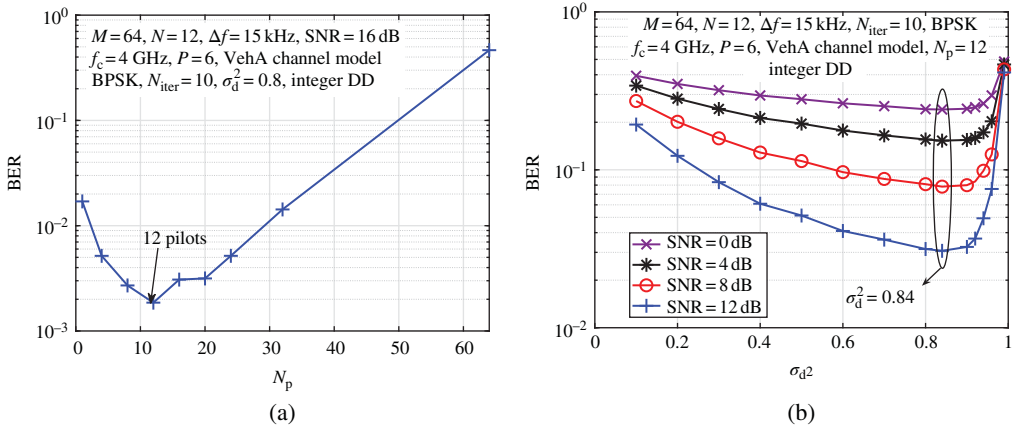


Figure 9.36 BER performance of SSPNet as a function of N_p and σ_d^2 for integer DD. (a) BER vs. N_p . (b) BER vs. σ_d^2 .

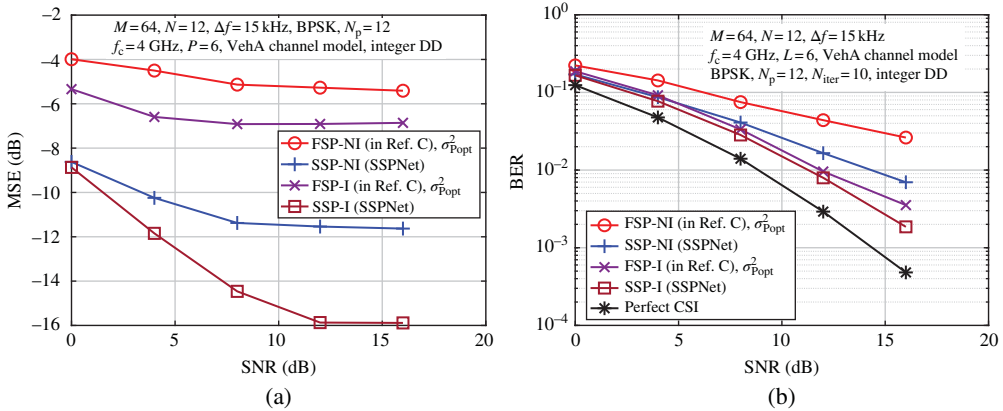


Figure 9.37 MSE and BER performance comparison between SSP scheme and FSP scheme in Ref. C [112] for integer DD. (a) MSE vs. SNR. (b) BER vs. SNR.

Performance Comparison Between SSP and FSP Schemes The MSE performance of the network without iterative cancellation (SSP-NI) and with iterative cancellation (SSP-I) with integer DD is presented in Fig. 9.37(a). The performance of the FSP-NI and FSP-I scheme in [112] is also plotted for comparison. It is seen that the MSE performance of the SSP-NI scheme is better than both the FSP-NI and FSP-I schemes. Also, the SSP-I scheme achieves even better MSE performance. Figure 9.37(b) shows the BER performance of the proposed SSPNet as a function of SNR for integer DD. The performance of FSP-NI, FSP-I, and perfect CSI schemes is also plotted in an integer DD channel for comparison. It can be observed that the performance of the SSP-NI scheme is better than the FSP-NI scheme, with a performance gain of about 4 dB. For example, a BER of 4×10^{-2} is attained at 8 dB with SSP-NI, while it is attained at 12 dB with the FSP-NI scheme. It can also be observed that the SSP-I scheme performs better than the FSP-I scheme.

The MSE and BER performance for the case of fractional DD are presented in Figs. 9.38(a) and (b). It is seen that the MSE performance of the SSP-NI scheme is superior compared to that of

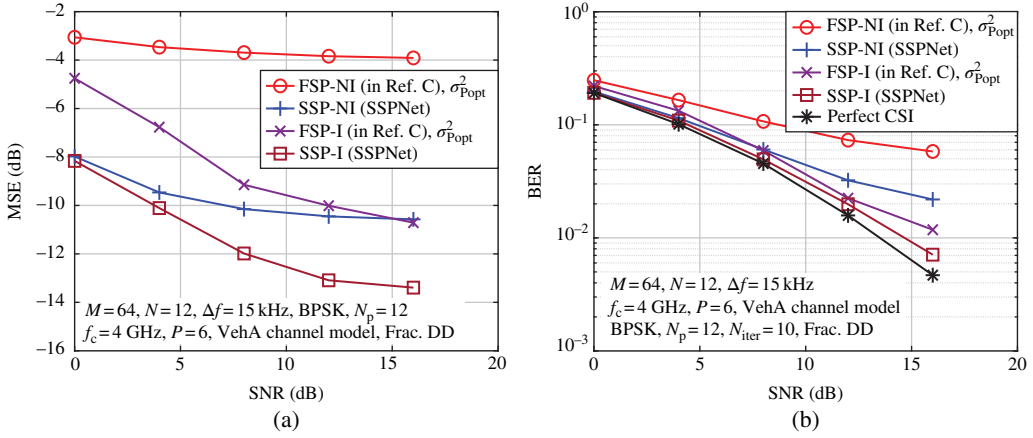


Figure 9.38 MSE and BER performance comparison between SSP scheme and FSP scheme in Ref. C [112] for fractional DD. (a) MSE vs. BER. (b) BER vs. SNR.

the FSP-NI scheme and FSP-I scheme in low- and mid-SNR regime, while the SSP-I scheme performs even better. The BER performance of the SSP-NI scheme is much better than the FSP-NI counterpart. For example, a BER of 6×10^{-2} is attained at 8 dB with the SSP-NI scheme, while it is attained at 16 dB with the FSP-NI scheme. It is also observed that the SSP-I scheme performs better than the FSP-I scheme. These performance improvements are significant given that the FSP schemes assume that the estimates of τ_i s and v_i s are perfectly known and only h_i s are estimated, whereas the network estimated all the three tuples (τ_i, v_i, h_i) . This shows that superimposing the pilots sparsely and using a learning-based network for channel estimation can achieve better MSE and BER performance.

10

OTFS for Radar Sensing

In Chapter 2, Zak-orthogonal time frequency space (OTFS) pulsones were introduced as information carriers naturally suited for predictable and non-fading communication. It was shown that choosing the Zak-OTFS modulation parameters τ_p and $\nu_p = 1/\tau_p$ to be larger than the channel delay spread and Doppler spread, respectively, guarantees a predictable and non-fading I/O relation. This condition on the Zak-OTFS parameters was referred to as the *crystallization condition*.

In communication systems designed for doubly spread channels, it is desired that the carrier waveform be localized along both time and frequency domains. While this is not possible due to the Heisenberg uncertainty principle, a quasi-periodic delay-Doppler (DD) pulse (DD representation of a Zak-OTFS pulsone) is quasi-localized along both delay and Doppler domains (i.e., it is localized within any interval of width τ_p and $\nu_p = 1/\tau_p$ along the delay and Doppler domains, respectively).

In radar sensing, the objective is to use the received echo of the transmitted radar waveform to estimate the delay and Doppler of multiple targets present in the radar scene [231, 232]. The radar scene is completely described by the DD spreading function of the echo channel between the radar transmitter and receiver. The DD location of a target is a point in the DD domain, whose delay and Doppler domain coordinates are simply the delay and Doppler shift induced in the received echo due to that target. The DD spreading function has peaks at the target locations in the DD domain. DD domain signal processing and waveform design/analysis is, therefore, central to radar sensing. Following this observation, in this chapter, a mathematical framework for the design of radar waveforms is presented based on the DD representation of signals.

Good radar waveforms must be localized in the DD domain. Time-domain (TD) and frequency-domain (FD) pulses are not good radar waveforms as they are not localized simultaneously along both delay and Doppler domains. TD pulses are localized along delay but not along Doppler, and, therefore, they are not suited for estimating Doppler shifts due to targets. Similarly, FD pulses are localized along Doppler but not along delay, and, therefore, they are not suited for estimating target delays. The ability of a radar waveform to resolve multiple targets unambiguously is determined by its ambiguity function in the DD domain [233]. The ambiguity function of a TD pulse is narrow along the delay domain but wide along the Doppler domain, whereas the ambiguity function of a FD pulse is narrow along the Doppler domain but wide along the delay domain.

A typical radar receiver detects targets from peaks of the magnitude of the cross-ambiguity function between the transmitted radar waveform and the received echo. The DD domain cross-ambiguity function is simply the twisted convolution between the DD spreading function

of the channel and the ambiguity function of the transmitted radar waveform. The ambiguity function for an ideal radar waveform should therefore be a DD domain Dirac-delta impulse at the origin since the magnitude of the resulting cross-ambiguity will then be the magnitude of the channel's DD spreading function itself, which has peaks at the exact target locations.

However, such an ideal waveform does not exist since the total volume of the squared magnitude of the ambiguity function of any radar waveform is finite (it is equal to the square of the total energy of the radar waveform), and, for practical waveforms, finite energy cannot be concentrated at a point in the DD domain. Also, such an ideal waveform would have an infinite time duration and bandwidth, which is not practically feasible. Since the total ambiguity volume/blur is finite, radar engineers are interested in manipulating/redistributing the ambiguity volume/blur so as to concentrate it in those regions of the DD domain that matter least for the operational task of the radar.

One such idea considered by P. M. Woodward in [52] is that of a train of narrow TD pulses (regularly spaced apart by τ_d seconds in time) modulated by a broad TD pulse. The width of the narrow pulse is inversely proportional to the bandwidth of the radar waveform, and the width of the broad pulse is proportional to its time duration. The ambiguity function of such a modulated pulse train is quasi-localized, i.e., the ambiguity volume within any DD domain interval of widths τ_d and $1/\tau_d$ along the delay and Doppler domains, respectively, is localized in a very small region of this interval. Such a construction redistributes all the ambiguity volume earlier concentrated around the origin (e.g., for TD/FD pulses) to points of a lattice in the DD domain, where the lattice points are separated by τ_d and $1/\tau_d$ along the delay and Doppler domains, respectively. Clearly, if τ_d is chosen appropriately so that τ_d and $1/\tau_d$ are greater than the spread in target delay and Doppler shift, respectively, then unambiguous target detection is possible. Otherwise, it is possible that non-existent targets may also get detected.

Even though modulated pulse trains give radar waveforms which are quasi-localized and can achieve unambiguous target detection, radar resolution (i.e., radar's ability to resolve targets whose DD locations are close to each other) depends on the shape of the ambiguity function of the radar waveform. The radar waveform design approach based on the modulated pulse train suffers from the fact that it does not appear to provide a simple analytical relation between the transmitted radar waveform and its ambiguity function. In the absence of a simple analytical relation, the design is mostly based on trial-and-error, i.e., by changing the shape of the narrow and broad TD pulses and observing the impact on the ambiguity function.

Zak-OTFS pulsones were studied extensively in Chapter 2 and are TD realizations of quasi-periodic DD domain pulses, which, by design, are quasi-localized in any DD domain interval of width, τ_p and $\nu_p = 1/\tau_p$ along the delay and Doppler domains, respectively. They are, therefore, good radar waveforms. In this chapter, the ambiguity function of Zak-OTFS pulsones is studied, and, indeed, just like the modulated pulse train radar waveform, Zak-OTFS pulsones can also achieve unambiguous target detection if τ_p and ν_p are greater than the spread in the target delay and Doppler shifts, respectively. This condition on τ_p is reminiscent of the crystallization condition in Chapter 2 in the context of Zak-OTFS modulation for the predictable communication of information.

Unlike the modulated pulse train-based approach, the rich theory of the Zak transform gives a very simple relation between the DD representation of the Zak-OTFS pulsone and its ambiguity function. The design of a Zak-OTFS pulsone involves the use of a DD domain pulse shaping filter/function $w_{\text{tx}}(\tau, \nu)$, which is used to limit the TD and FD realization of the pulsone in time and bandwidth, respectively. Given that the crystallization condition is satisfied, radar resolution depends primarily on the shape of the ambiguity function within the DD region $\{(\tau, \nu) \mid |\tau| < \tau_p/2, |\nu| < \nu_p/2\}$. The ambiguity function of the Zak-OTFS pulsone within this region is simply the twisted convolution of the pulse shaping filter $w_{\text{tx}}(\tau, \nu)$ with the corresponding

matched filter $w_{\text{tx}}^*(-\tau, -\nu) e^{j2\pi\nu\tau}$. Zak theory simplifies the radar waveform design to the design of the pulse shaping filter $w_{\text{tx}}(\tau, \nu)$. Therefore, Zak theory introduces a rigorous mathematical framework for the design of radar waveforms, which is a paradigm shift in how radar waveforms have been designed so far.

This chapter is organized as follows. The radar scene and preliminaries are discussed in Section 10.1. The radar ambiguity function is introduced in Section 10.2 as a measure of *goodness* of a radar waveform. The ambiguity functions for TD and FD pulses are presented in Section 10.2.2. The design of good radar waveforms based on the modulated pulse train is discussed in Section 10.3. A Zak theoretical framework for the design of good radar waveforms is discussed in Section 10.4.

10.1 Radar Scene

The radar scene is a collection of targets, whose distances and speeds relative to the radar location are to be estimated. Assume that the radar transmitter and receiver are at the same location. The radar transmitter emits a radio signal (i.e., radar waveform), whose echoes from the targets are processed at the radar receiver to estimate the time delay and Doppler shift encountered by the emitted radio signal from each of the targets. The time delay due to a particular target (τ) is equal to twice the distance between the radar and that target (d) divided by the speed of radio wave (i.e., speed of light, c). Hence, the estimated distance between the radar and that target ($\hat{d}$) is half of the product of the speed of light and the estimated target delay ($\hat{\tau}$), i.e.,

$$\hat{d} = \frac{c\hat{\tau}}{2}. \quad (10.1)$$

The Doppler shift due to a particular target (ν) equals twice the product of the frequency of the emitted radio signal (f_c) times the speed of the target relative to the radar (v) divided by the speed of light. Therefore, the estimated relative speed of a target ($\hat{v}$) is given by half of the speed of light times the estimated Doppler shift due to that target ($\hat{\nu}$) divided by the frequency of the emitted radio signal, i.e.,

$$\hat{v} = \frac{c\hat{\nu}}{2f_c}. \quad (10.2)$$

Let there be P targets with τ_i, ν_i denoting the delay and Doppler shift due to the i th target, $i = 1, 2, \dots, P$. The received TD echo signal is a superposition of echoes received from all targets, given by

$$r_{\text{id}}(t) = \sum_{i=1}^P h_i s_{\text{id}}(t - \tau_i) e^{j2\pi\nu_i(t - \tau_i)} + n_{\text{id}}(t), \quad (10.3)$$

where $s_{\text{id}}(t)$ is the transmitted radar waveform and h_i is the complex gain encountered by the radar waveform when reflected from the i th target and subsequently received at the radar. Also, $n_{\text{id}}(t)$ denotes the additive white Gaussian noise (AWGN) at the radar receiver. The *radar scene* is, therefore, completely described by (h_i, τ_i, ν_i) , $i = 1, 2, \dots, P$. In this radar scene, the i th target is said to be *located* at the DD location (τ_i, ν_i) .

In a general setting, the radar scene need not necessarily consist of targets located at discrete delay and Doppler shifts. Therefore, in general, a radar scene is described by the DD spreading function $h_{\text{phy}}(\tau, \nu)$ of the channel between the radar transmitter and receiver. For the general radar scene, the received echo signal at the radar receiver is given by

$$r_{\text{id}}(t) = \iint h_{\text{phy}}(\tau, \nu) s_{\text{id}}(t - \tau) e^{j2\pi\nu(t - \tau)} d\tau d\nu + n_{\text{id}}(t). \quad (10.4)$$

This is similar to the expression for the received TD signal at a communication receiver, where $s_{\text{td}}(t)$ is the transmitted communication signal carrying information (see (2.52)). For the special case where targets are located at discrete DD locations, the DD spreading function is given by

$$h_{\text{phy}}(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i). \quad (10.5)$$

This when used on the RHS of (10.4) gives the expression of $r_{\text{td}}(t)$ in (10.3).

10.2 The Radar Ambiguity Function

In this section, we quantify the *goodness* of a radar waveform through its *ambiguity* function. Consider a single target radar scene (i.e., $P = 1$). From (10.3), the received signal is given by

$$r_{\text{td}}(t) = h_1 s_{\text{td}}(t - \tau_1) e^{j2\pi\nu_1(t-\tau_1)} + n_{\text{td}}(t). \quad (10.6)$$

Since the noise is AWGN, the maximum likelihood (ML) estimate for (τ_1, ν_1) is given by

$$(\hat{\tau}_1, \hat{\nu}_1) = \arg \max_{\tau, \nu} |A_{r,s}(\tau, \nu)|, \quad (10.7)$$

where

$$A_{r,s}(\tau, \nu) \triangleq \int r_{\text{td}}(t) s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt \quad (10.8)$$

is the cross-ambiguity function between the transmitted radar waveform $s_{\text{td}}(t)$ and the received echo $r_{\text{td}}(t)$. A derivation of (10.7) is provided in Appendix 10.A.

Motivated by (10.8) for the single target radar scene, it follows that for a general radar scene with multiple targets, the DD locations for the targets are usually estimated from the peaks of the cross-ambiguity function $A_{r,s}(\tau, \nu)$ in the DD domain.

An important question here is under what conditions can multiple targets be resolved unambiguously. It is clear that the answer to this question depends on the transmitted radar waveform $s_{\text{td}}(t)$. This dependency is best understood in the DD domain. First, for any arbitrary $(\tau, \nu) = (\tau_0, \nu_0)$, the cross-ambiguity integral expression on the RHS of (10.8) can be expressed as a DD domain integral limited to the fundamental DD period $\mathcal{D}_0$ (see (2.18)), i.e.,

$$A_{r,s}(\tau_0, \nu_0) = \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau-\tau_0)} d\tau d\nu, \quad (10.9)$$

where $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(r_{\text{td}}(t))$ and $s_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(s_{\text{td}}(t))$ are the Zak transform/DD representations of the received echo $r_{\text{td}}(t)$ and the transmitted radar waveform $s_{\text{td}}(t)$, respectively. A derivation of this expression is provided in Appendix 10.B. The expression in (10.9) can be interpreted as follows. If there were a single target located at (τ_1, ν_1) , then the received AWGN-free echo signal from that target would be $s_{\text{td}}(t - \tau_1) e^{j2\pi\nu_1(t-\tau_1)}$, whose Zak transform/DD representation is $s_{\text{dd}}(\tau - \tau_1, \nu - \nu_1) e^{j2\pi\nu_1(\tau-\tau_1)}$ (see (2.27)). Therefore, the integral on the RHS of (10.9) would be

$$\int_0^{\tau_p} \int_0^{\nu_p} s_{\text{dd}}(\tau - \tau_1, \nu - \nu_1) e^{j2\pi\nu_1(\tau-\tau_1)} s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau-\tau_0)} d\tau d\nu, \quad (10.10)$$

whose peak is attained at $(\tau_0, \nu_0) = (\tau_1, \nu_1)$, i.e.,

$$(\tau_1, \nu_1) = \arg \max_{(\tau_0, \nu_0)} \left| \int_0^{\tau_p} \int_0^{\nu_p} s_{dd}(\tau - \tau_1, \nu - \nu_1) e^{j2\pi\nu_1(\tau - \tau_1)} s_{dd}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \right|. \quad (10.11)$$

Since a doubly spread channel acts on the DD domain representation of its input through twisted convolution, from (2.24) and (10.4), it follows that the DD representation of the received echo signal is given by

$$y_{dd}(\tau, \nu) = h_{\text{phy}}(\tau, \nu) *_{\sigma} s_{dd}(\tau, \nu) + n_{dd}(\tau, \nu), \quad (10.12)$$

where $n_{dd}(\tau, \nu) = \mathcal{Z}_t(n_{td}(t))$ is the Zak transform of the TD AWGN signal $n_{td}(t)$ and $*_{\sigma}$ denotes the twisted convolution operation (see (2.20)). Using this expression of $y_{dd}(\tau, \nu)$ on the RHS of (10.9) gives an expression for the cross-ambiguity function in terms of the auto-ambiguity function of the transmitted radar waveform and $h_{\text{phy}}(\tau, \nu)$. The following theorem states this result.

Theorem 10.1 The cross-ambiguity function is given by

$$A_{r,s}(\tau, \nu) = h_{\text{phy}}(\tau, \nu) *_{\sigma} A_{s,s}(\tau, \nu) + \int_0^{\tau_p} \int_0^{\nu_p} n_{dd}(\tau', \nu') s_{dd}^*(\tau' - \tau, \nu' - \nu) e^{-j2\pi\nu(\tau' - \tau)} d\tau' d\nu', \quad (10.13)$$

where $A_{s,s}(\tau, \nu)$ denotes the (auto)-ambiguity function of the transmitted radar waveform $s_{td}(t)$, which, at $(\tau, \nu) = (\tau_0, \nu_0)$, is given by

$$\begin{aligned} A_{s,s}(\tau_0, \nu_0) &\stackrel{\Delta}{=} \int_0^{\tau_p} \int_0^{\nu_p} s_{dd}(\tau, \nu) s_{dd}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\ &= \int s_{td}(t) s_{td}^*(t - \tau_0) e^{-j2\pi\nu_0(t - \tau_0)} dt \\ &= \int s_{fd}(f) s_{fd}^*(f - \nu_0) e^{j2\pi f \tau_0} df, \end{aligned} \quad (10.14)$$

where

$$s_{fd}(f) = \int s_{td}(t) e^{-j2\pi ft} dt \quad (10.15)$$

is the Fourier transform/FD representation of $s_{td}(t)$.

Proof: See Appendix 10.C.

Equation (10.13) is the continuous DD domain I/O relation for a radar sensing system and is analogous to the I/O relation of a communication system (see (2.24)) since, in both, the output DD domain signal is equal to the twisted convolution of the input with the DD spreading function of the channel. From (10.13), it is clear that the performance of a radar sensing system depends on the ambiguity function $A_{s,s}(\tau_0, \nu_0)$ of the radar waveform $s_{td}(t)$.

For delay-only or Doppler-only scenarios, the ambiguity function in (10.14) reduces to the well-known auto-correlation function in TD and FD. For example, for a delay-only scenario where

the targets are stationary, i.e., the target Doppler shifts are zero, substituting $\nu_0 = 0$ in the second line of (10.14) gives

$$A_{s,s}(\tau_0, 0) = \int s_{\text{td}}(t) s_{\text{td}}^*(t - \tau_0) dt, \quad (10.16)$$

which is the well-known TD auto-correlation function of the transmitted radar waveform $s_{\text{td}}(t)$. Similarly, for a Doppler-only scenario where the target delays are zero (or very close to zero), substituting $\tau_0 = 0$ in the third line of (10.14) gives

$$A_{s,s}(0, \nu_0) = \int s_{\text{fd}}(f) s_{\text{fd}}^*(f - \nu_0) df, \quad (10.17)$$

which is the FD auto-correlation function of the radar waveform.

10.2.1 Radar Resolution

The ambiguity function $A_{s,s}(\tau, \nu)$ places fundamental limits on the ability to accurately estimate target's delay/Doppler location. To understand this, considering a radar scene with two targets, i.e.,

$$h_{\text{phy}}(\tau, \nu) = \sum_{i=1}^2 h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (10.18)$$

where (τ_1, ν_1) and (τ_2, ν_2) are the two target locations in the DD domain. From Theorem 10.1, it follows that the noise-free cross-ambiguity is given by

$$\begin{aligned} h_{\text{phy}}(\tau, \nu) *_{\sigma} A_{s,s}(\tau, \nu) &= h_1 A_{s,s}(\tau - \tau_1, \nu - \nu_1) e^{j2\pi\nu_1(\tau - \tau_1)} \\ &\quad + h_2 A_{s,s}(\tau - \tau_2, \nu - \nu_2) e^{j2\pi\nu_2(\tau - \tau_2)}, \end{aligned} \quad (10.19)$$

and we *resolve* the two targets by limiting the overlap between the two terms appearing on the RHS. In other words, we require the delay domain spread of $A_{s,s}(\tau, \nu)$ to be less than $|\tau_1 - \tau_2|$ and the Doppler domain spread of $A_{s,s}(\tau, \nu)$ to be less than $|\nu_1 - \nu_2|$. Therefore, for resolvability, it appears that a good radar waveform should have an ambiguity function which is localized in a small region of the DD domain, i.e., ideally, a Dirac-delta function at $(0, 0)$ in the DD domain. Such an ambiguity function would result in a cross-ambiguity function $h_{\text{phy}}(\tau, \nu) *_{\sigma} \delta(\tau, \nu) = h_{\text{phy}}(\tau, \nu)$, which when plotted in the 2D DD domain would show two distinct peaks at the two target locations. However, for finite energy radar waveforms having finite duration and bandwidth, the ambiguity function cannot be a Dirac-delta at $(0, 0)$. Clearly, from the expression of the ambiguity function in (10.14), it is observed that

$$A_{s,s}(0, 0) = \int s_{\text{td}}(t) s_{\text{td}}^*(t) dt = \int |s_{\text{td}}(t)|^2 dt, \quad (10.20)$$

i.e., the value of the ambiguity function at $(0, 0)$ is equal to the energy of the radar waveform. Also, from Moyal's identity [234], it is known that the volume bounded by the surface of the squared ambiguity function ($|A_{s,s}(\tau, \nu)|^2$) is equal to the square of the radar waveform energy, i.e.,

$$\iint |A_{s,s}(\tau, \nu)|^2 d\tau d\nu = \left(\int |s_{\text{td}}(t)|^2 dt \right)^2. \quad (10.21)$$

See Appendix 10.D for a derivation of (10.21). Since the total volume of the squared ambiguity function is finite and the value of the ambiguity function at the origin is also finite, the ambiguity function cannot be a Dirac-delta at the origin.

Also, a radar waveform $s_{\text{id}}(t)$ that is limited in time and bandwidth to T seconds and B Hz, respectively, has a DD representation $s_{\text{dd}}(\tau, \nu)$ which, within the DD region $\{(\tau, \nu) \mid |\tau| < \tau_p/2, |\nu| < \nu_p/2\}$, has a spread of the order of $1/B$ and $1/T$ along the delay and Doppler domain, respectively (see Appendix 10.E for a discussion). Since $s_{\text{dd}}(\tau, \nu)$ has a finite delay and Doppler domain spread and is not ideally localized at the origin, from the integral expression in the first line of (10.14), it follows that the ambiguity function of $s_{\text{id}}(t)$ is also spread and is not ideally localized. This spreading of the ambiguity function *blurs* the location of the target and governs the radar resolution, as explained through an example below.

For the two target example, from (10.19), we know that the cross-ambiguity function $A_{r,s}(\tau, \nu)$ consists of two terms, each corresponding to one of the targets. The magnitude of the first term on the RHS, i.e., $|h_1||A_{s,s}(\tau - \tau_1, \nu - \nu_1)|$ is simply the magnitude of the ambiguity function shifted by the first target's location (τ_1, ν_1) . In Appendix 10.F, it is shown that, for any arbitrary radar waveform, the peak of its ambiguity function is always at the origin, and, therefore, the peak of $|h_1||A_{s,s}(\tau - \tau_1, \nu - \nu_1)|$ is at (τ_1, ν_1) and the distribution of its volume around (τ_1, ν_1) is the same as the distribution of the ambiguity volume of $A_{s,s}(\tau, \nu)$ around $(0, 0)$. This spread of $|h_1||A_{s,s}(\tau - \tau_1, \nu - \nu_1)|$ around (τ_1, ν_1) is what blurs the location of the first target, and the presence of noise shifts the peak location resulting in target estimation errors. Similarly, the magnitude of the second term on the RHS of (10.19) is the magnitude of the ambiguity function shifted by the second target's location (τ_2, ν_2) . If these two target locations are well separated along the delay and/or Doppler domain, then the two terms on the RHS do not interfere with each other (see Fig. 10.1). However, if the target locations are close to each other along both the delay

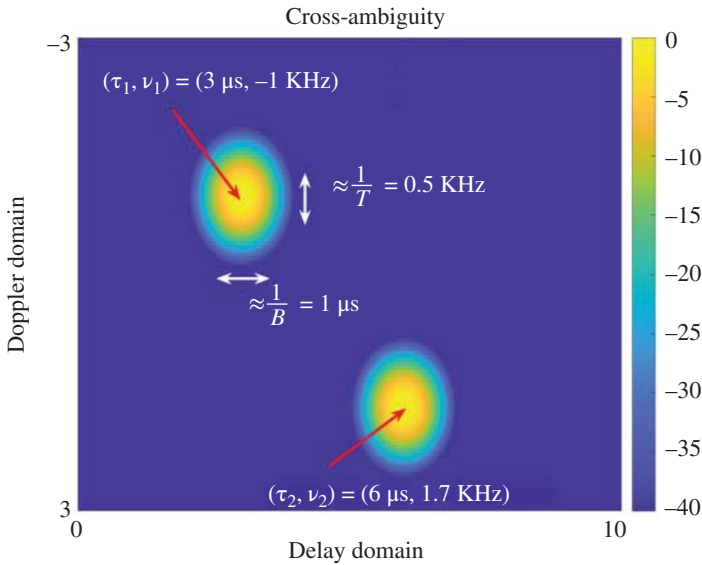


Figure 10.1 Magnitude of the cross-ambiguity function between the transmitted and the received echo waveform for a radar scene with two targets at $(\tau_1, \nu_1) = (3.0 \mu\text{s}, -1 \text{ kHz})$ and $(\tau_2, \nu_2) = (6.0 \mu\text{s}, 1.7 \text{ kHz})$. The radar waveform time duration and bandwidth are $T = 2 \text{ ms}$ and $B = 1 \text{ MHz}$, respectively. The cross-ambiguity volume/blur has peaks at the two target locations and is spread around them in a region of width equal to the delay and Doppler domain resolution (i.e., $\frac{1}{B} = 1 \mu\text{s}$ and $\frac{1}{T} = 0.5 \text{ kHz}$ along the delay and Doppler domains, respectively). The targets, therefore, appear *blurred*. Since the two target locations are separated by more than the resolution along the delay/Doppler domain, their blurs do not overlap, and, therefore, they are separately resolvable.

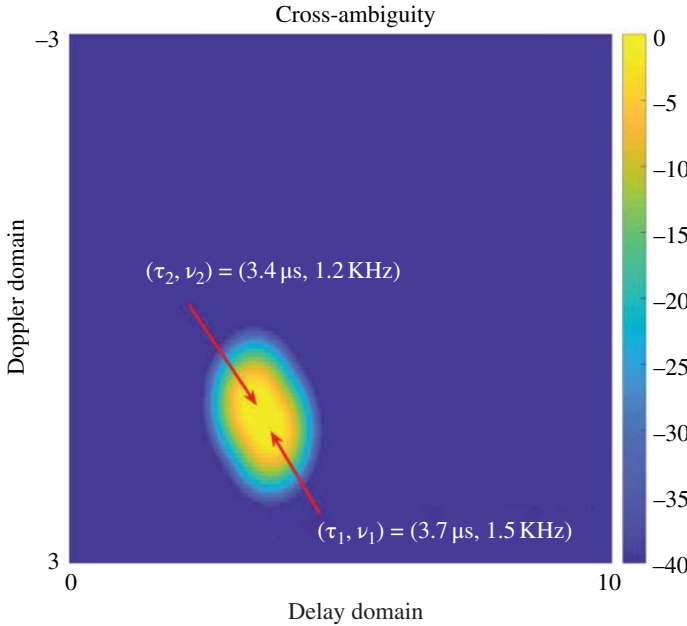


Figure 10.2 Magnitude of the cross-ambiguity function between the transmitted and the received echo waveform for a radar scene with two targets at $(\tau_1, \nu_1) = (3.7 \mu\text{s}, 1.5 \text{ kHz})$ and $(\tau_2, \nu_2) = (3.4 \mu\text{s}, 1.2 \text{ kHz})$. The radar waveform time duration and bandwidth are $T = 2 \text{ ms}$ and $B = 1 \text{ MHz}$, respectively. Since the two target locations are now separated by less than the resolution along both the delay and Doppler domain, their blurs overlap, and, therefore, they are not separately resolvable and they appear as a single target.

and Doppler domains (i.e., the spread of the ambiguity function along the delay and Doppler is larger than $|\tau_2 - \tau_1|$ and $|\nu_2 - \nu_1|$, respectively), then the two terms on the RHS of (10.19) would interfere and two distinct peaks might not be visible in the cross-ambiguity function (see Fig. 10.2).

The minimum delay domain separation of two targets having the same Doppler location, such that their peaks in the cross-ambiguity function are distinctly observable, is referred to as the “delay domain resolution” of the radar. Similarly, the minimum Doppler domain separation of two targets having the same delay location, such that their peaks in the cross-ambiguity function are distinctly observable, is referred to as the “Doppler domain resolution” of the radar. Clearly, with a radar waveform time and bandwidth constraint of T seconds and B Hz, the achievable radar resolution along the delay and Doppler domain are of the order of $1/B$ and $1/T$, respectively.

Since the distribution of the blur/ambiguity volume around each target depends on the distribution of volume/blur of the ambiguity function of the radar waveform, radar engineers are interested in manipulating/redistributing the conserved volume/blur of the ambiguity function (see ((10.21))) so as to concentrate it in those regions of the DD domain that matter least for the operational task of the radar.

In the following, we illustrate the limitations of using a TD pulse and a FD pulse as radar waveforms by studying their ambiguity functions in Section 10.2.2. Equation (10.14) gives a mathematical relation between the ambiguity function and the DD representation of the radar waveform. This forms the basis of the DD domain-based design of radar waveforms, which is discussed in Section 10.4. In Section 10.4, it is shown that Zak-OTFS pulsones which achieve predictable communication (see Chapter 2) are also good radar waveforms.

10.2.2 Ambiguity Functions for TD and FD Pulses

Consider a unit energy TD pulse $s(t) = s_{\text{td}}(t) = \sqrt{B} \text{sinc}(Bt)$ with bandwidth B , which is localized around $t = 0$. Its ambiguity function is given by the following theorem.

Theorem 10.2 The ambiguity function of the radar waveform $s_{\text{td}}(t) = \sqrt{B} \text{sinc}(Bt)$ (a TD pulse of bandwidth B) is given by

$$A_{s,s}^{\text{tdm}}(\tau, \nu) = \begin{cases} \left(1 - \frac{|\nu|}{B}\right) e^{j\pi\nu\tau} \text{sinc}((B - |\nu|)\tau), & |\nu| < B, \\ 0, & |\nu| \geq B. \end{cases} \quad (10.22)$$

Proof: See Appendix 10.G.

For a fixed ν , we consider the term $\text{sinc}((B - |\nu|)\tau)$ as a function of τ and conclude that the delay spread of the time division multiplexing (TDM) ambiguity function is about $1/B$. Hence, it is possible to separate two targets if their delays differ by more than $1/B$. On the other hand, the Doppler spread of the TD pulse ambiguity function is the bandwidth B . Hence, it is not possible to separate two targets with delays differing by less than $1/B$ unless the two Doppler shifts differ by more than B , which is unlikely in most scenarios. *The TD pulse waveform is localized in the TD, but not in the FD, and this is the reason it is unable to separate targets in the Doppler domain.* The ambiguity function for a TD pulse is illustrated in Fig. 10.3.

Next, consider a FD pulse of duration T , which is localized at $f = 0$ in FD, i.e., $s_{\text{fd}}(f) = \sqrt{T} \text{sinc}(fT)$. The corresponding TD radar waveform is the inverse Fourier transform of $\sqrt{T} \text{sinc}(fT)$, i.e.,

$$s_{\text{td}}(t) = \begin{cases} \frac{1}{\sqrt{T}}, & |t| < \frac{T}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (10.23)$$

The following theorem gives the ambiguity function of an FD pulse.

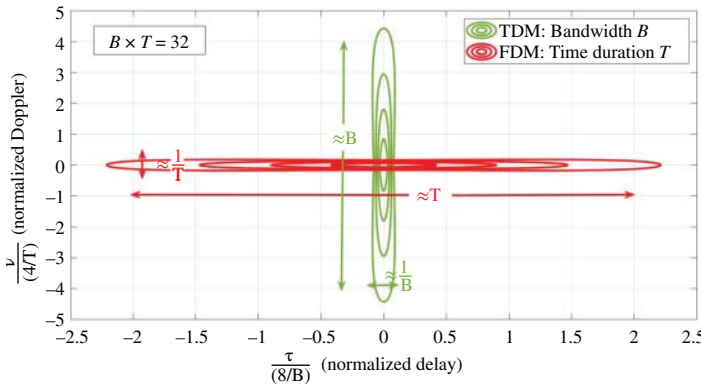


Figure 10.3 Squared magnitude $|A_{s,s}(\tau, \nu)|^2$ of the ambiguity functions for TD and FD pulses depicted in light and dark gray respectively. The TDM carrier waveform (i.e., TD pulse) is not able to separate targets in Doppler, and the FDM carrier waveform (i.e., FD pulse) is not able to separate targets in delay.

Theorem 10.3 The ambiguity function for an FD pulse $s_{fd}(f) = \sqrt{T} \text{sinc}(fT)$ is given by

$$A_{s,s}^{\text{fdm}}(\tau, \nu) = \begin{cases} \left(1 - \frac{|\tau|}{T}\right) e^{j\pi\nu\tau} \text{sinc}((T - |\tau|)\nu), & |\tau| < T, \\ 0, & |\tau| \geq T. \end{cases} \quad (10.24)$$

Proof: See Appendix 10.H.

Now, the Doppler spread is small (about $1/T$) and the delay spread is large (the duration T , see Fig. 10.3), and, hence, it is not possible to separate two targets with Dopplers differing by less than $1/T$ unless the two delay shifts differ by more than T . This is unlikely in most scenarios. The FD pulse waveform is localized in the FD, but not in the TD, and this is the reason it is unable to separate targets in the delay domain.

The volume under the (squared) magnitude ambiguity surface is fixed by Moyal's identity (10.21), but it can be redistributed to enable the resolution of radar targets. This possibility was known to P. M. Woodward more than 70 years ago. In his 1953 book [52], he described how the ambiguity function of a narrow TD Gaussian pulse (the light gray shaded ellipse in Fig. 10.4) can be redistributed into several constituent DD domain pulses (shown as black diamonds in Fig. 10.4). The idea was to modulate a train of narrow TD Gaussian pulses with a broad Gaussian envelope.

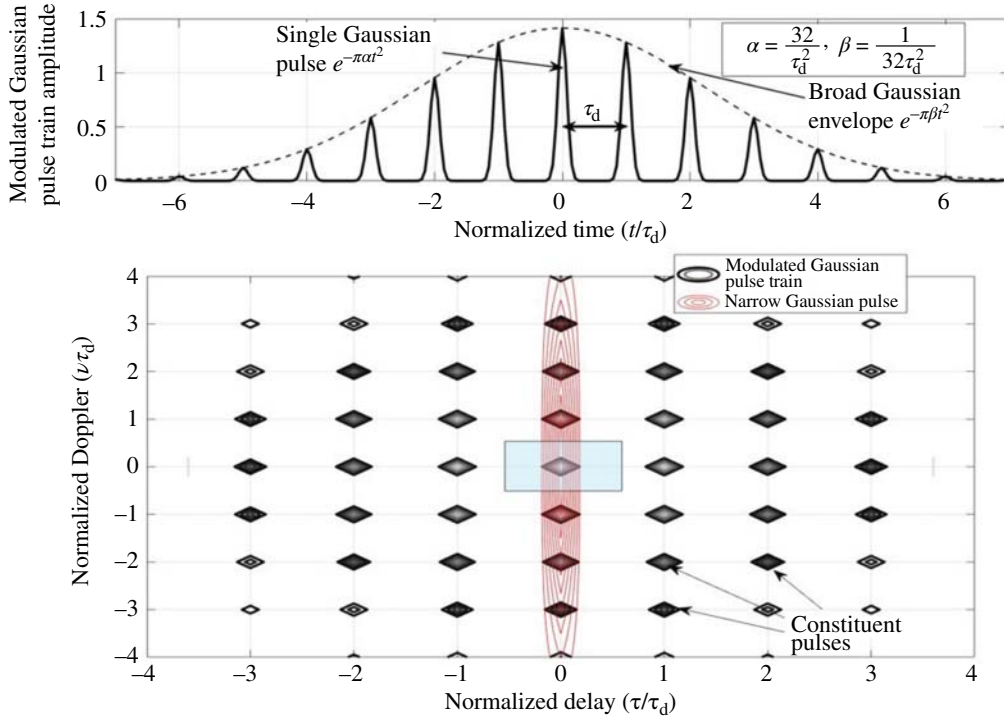


Figure 10.4 Redistributing the squared magnitude $|A_{s,s}(\tau, \nu)|^2$ of the ambiguity function of a narrow Gaussian pulse. Modulating a train of narrow TD Gaussian pulses with a broad Gaussian envelope produces an ambiguity function that is better able to separate targets in delay and Doppler.

10.3 A Good Radar Waveform: Quasi-localization in TD and FD Using Modulated Pulse Trains

As argued in the following, a modulated pulse train signal appears to be quasi-localized both in time and frequency. Let $f(t)$ be a narrow TD pulse having bandwidth B and approximate TD pulse width $1/B$. Clearly, as discussed earlier, such TD pulses have an ambiguity function whose delay spread is small ($1/B$) but whose Doppler spread is large, rendering them suitable only for delay estimation and not for Doppler estimation. This was because the pulse $f(t)$ is localized only in time and not in frequency. However, we extend this pulse to a pulse train with consecutive pulses separated by τ_d seconds, i.e.,

$$p(t) = \sum_{n=-\infty}^{\infty} f(t - n\tau_d). \quad (10.25)$$

Clearly, the pulse train $p(t)$ is a periodic TD signal with period τ_d , and, therefore, its Fourier series tells us that its frequency domain representation $P(f)$ consists of FD Dirac-delta impulses at $f = k/\tau_d$ (i.e., integer multiples of $1/\tau_d$), i.e.,

$$P(f) = \sum_{k \in \mathbb{Z}} p_k \delta\left(f - \frac{k}{\tau_d}\right),$$

$$p_k = \frac{1}{\tau_d} \int_0^{\tau_d} p(t) e^{-j2\pi k \frac{t}{\tau_d}} dt. \quad (10.26)$$

The Fourier series expansion for $p(t)$ is

$$p(t) = \sum_{k \in \mathbb{Z}} p_k e^{j2\pi k \frac{t}{\tau_d}}. \quad (10.27)$$

To limit the time duration and energy of $p(t)$, we modulate it with a broad pulse $g(t)$ having finite energy, resulting in the modulated pulse train (the radar waveform)

$$s_{td}(t) = p(t)g(t) = \sum_{k \in \mathbb{Z}} p_k g(t) e^{j2\pi k \frac{t}{\tau_d}}. \quad (10.28)$$

The time duration T of such a radar waveform is limited to the duration of $g(t)$. An example of such a modulated pulse train for a Gaussian narrow TD pulse $f(t) = e^{-\pi\alpha t^2}$ and a modulating broad Gaussian pulse $g(t) = e^{-\pi\beta t^2}$ ($\alpha \gg \beta$) is shown on the top of Fig. 10.4. As can be observed, by choosing τ_d to be sufficiently large, for any contiguous interval of τ_d seconds, the modulated pulse train is localized in a time interval $1/B$ (equal to the inverse bandwidth of $f(t)$) which is much smaller than τ_d , i.e., it is quasi-localized along TD. For this, τ_d must satisfy

$$\frac{1}{B} \ll \tau_d. \quad (10.29)$$

From (10.28), it follows that the Fourier transform of $s_{td}(t)$ is given by

$$s_{td}(f) = \sum_{k \in \mathbb{Z}} p_k G\left(f - \frac{k}{\tau_d}\right), \quad (10.30)$$

where $G(f)$ is the Fourier transform of $g(t)$. Since $g(t)$ is limited to a duration of T seconds, the FD spread of $G(f)$ is approximately $1/T$. From (10.30), it is clear that the FD representation of the modulated TD pulse train is itself a FD pulse train, where the k th FD pulse $p_k G(f - k/\tau_d)$ is essentially a scaled version of the FD pulse $G(f)$ shifted by an integer multiple of $1/\tau_d$. Consecutive

FD pulses in this pulse train are located $1/\tau_d$ Hz apart. With $1/\tau_d \gg 1/T$, it is clear that in any contiguous FD interval of width $1/\tau_d$, the FD pulse is localized to a small fraction of this FD interval, i.e., the modulated pulse train is also quasi-localized along FD. From the discussion above, this simultaneous quasi-localization of the modulated pulse train $s_{\text{id}}(t) = p(t)g(t)$ in both TD and FD is possible by choosing τ_d such that (see (10.29))

$$\frac{1}{B} \ll \tau_d \ll T. \quad (10.31)$$

Due to this simultaneous quasi-localization in both TD and FD, it follows that within any contiguous 2D DD domain region of widths τ_d and $1/\tau_d$ along the delay and Doppler domains, respectively, the ambiguity function of such a modulated pulse train is also quasi-localized. This is due to the fact that in (10.14), the ambiguity function value at (τ, ν) is essentially a TD or FD inner product between the radar waveform and its replica shifted by τ and ν along the delay and Doppler domains, respectively.

For example, in the second line of (10.14), the ambiguity function value at (τ, ν) is the TD inner product of the radar waveform $s_{\text{id}}(t)$ with $s_{\text{id}}(t - \tau)e^{j2\pi\nu(t-\tau)}$ (which is simply $s_{\text{id}}(t)$ Doppler shifted by ν to give $s_{\text{id}}(t)e^{j2\pi\nu t}$, which is then delayed by τ). The magnitude of this inner product is upper bounded by $\int |s_{\text{id}}(t)||s_{\text{id}}(t - \tau)| dt$, which is simply the auto-correlation of $|s_{\text{id}}(t)|$ for a time lag of τ . Next, for a given value ν , the variation of this correlation value with variation in the lag value τ is described in the following. In TD, the modulated pulse train radar waveform is quasi-localized to a small interval of width $1/B$ within any contiguous interval of width τ_d , and, therefore, for lag values $-\tau_d/2 < \tau < \tau_d/2$, the correlation value can take significant values only for $|\tau| < 1/B$ (i.e., quasi-localization along the delay domain since $1/B \ll \tau_d$). Since the pulses in the modulated TD pulse train repeat every τ_d seconds, the correlation exhibits peaks at lag values $\tau = n\tau_d, n \in \mathbb{Z}$. However, since the pulse train is time limited to T seconds, the correlation peaks at $\tau = n\tau_d$ exist only for $|n| \leq T/(2\tau_d)$.

Similarly, from the third line of (10.14), the magnitude of the ambiguity function value at (τ, ν) is upper bounded by the FD correlation value $\int |s_{\text{id}}(f)||s_{\text{id}}(f - \nu)| df$ at lag ν . In FD, the modulated pulse train radar waveform is quasi-localized to a small interval of width $1/T$ within any contiguous FD interval of width $1/\tau_d$, and, therefore, for lag values $-1/(2\tau_d) < \nu < 1/(2\tau_d)$, the correlation can take significant values only for $|\nu| < 1/T$ (i.e., quasi-localization along the Doppler domain since $1/T \ll 1/\tau_d$). Since the pulses in the FD pulse train repeat every $1/\tau_d$ Hz, the correlation exhibits peaks at lag values $\nu = m/\tau_d, m \in \mathbb{Z}$. However, since the FD pulse train is bandwidth limited to B Hz, the correlation peaks at $\nu = m/\tau_d$ exist only for $|m| \leq B\tau_d/2$.

This quasi-localization of the ambiguity function of a modulated pulse train with Gaussian pulses is illustrated in Fig. 10.4. The ambiguity function of a narrow Gaussian pulse $f(t) = e^{-\pi\alpha t^2}$, $\alpha \gg \frac{1}{\tau_d^2}$, has a pulse width ($\approx 1/\sqrt{\alpha}$) much smaller than τ_d and is, therefore, localized along the delay domain. It is, however, not localized along the Doppler domain (see the shaded ellipse in Fig. 10.4). By having an infinite train of such pulses separated by τ_d along TD (i.e., $p(t)$) and modulating it with a broad Gaussian pulse $g(t) = e^{-\pi\beta t^2}$, $\beta \ll 1/\tau_d^2$ ensures that the modulated pulse train $p(t)g(t)$ has a time duration $\approx \frac{1}{\sqrt{\beta}} \gg \tau_d$. As discussed above, from the figure, it is observed that the ambiguity function of this modulated pulse train is indeed quasi-localized in the DD domain. For example, within the 2D interval $|\tau| < \tau_d/2, |\nu| < 1/(2\tau_d)$ (shown as the rectangle in the middle in Fig. 10.4), ambiguity is present only within a small portion of this interval (highlighted as a diamond at the origin). Therefore, by modulating a pulse train in such a way that the total energy of TD pulse $f(t)$ and that of the modulated pulse train $p(t)g(t)$ is the same, the total conserved ambiguity volume/blur of the pulse $f(t)$ ($(\int |s_{\text{id}}(t)|^2 dt)^2$, i.e., square of total

energy, see (10.21)) is *redistributed* in such a way that the ambiguity function of the modulated pulse train becomes quasi-localized (the ambiguity volume is now localized at the locations $(\tau, \nu) = (n\tau_d, m/\tau_d)$, $n, m \in \mathbb{Z}$, see the diamonds at these locations in Fig. 10.4).

If the product of the delay and Doppler spreads of $h_{\text{phy}}(\tau, \nu)$ is less than one, it is possible to choose τ_d such that τ_d is larger than the delay spread and $1/\tau_d$ is larger than the Doppler spread. With such τ_d , all target locations are within the 2D rectangle shown in Fig. 10.4 and can be detected *unambiguously* as long as their locations are resolvable (which is possible by choosing a sufficiently large B and T so that the spread of the ambiguity volume/blur around each target, i.e., $1/B$ along delay and $1/T$ along Doppler, is small and does not overlap with the blur around other targets, see Fig. 10.1).

Since the cross-ambiguity function $A_{r,s}(\tau, \nu)$ is the twisted convolution of the ambiguity function $A_{s,s}(\tau, \nu)$ with the channel spreading function $h_{\text{phy}}(\tau, \nu)$, the ambiguity volume/blur at the locations $(\tau, \nu) = (n\tau_d, m/\tau_d)$, $n, m \in \mathbb{Z}$, interacts with the channel which spreads each one of them to a DD interval around the ambiguity location having width equal to the channel delay and Doppler spread. Let $I_{n,m}$ denote such a DD interval corresponding to the (n, m) th ambiguity location $(n\tau_d, m/\tau_d)$. Since the ambiguity locations are spaced apart by τ_d and $1/\tau_d$ along the delay and Doppler domains, respectively, and the spread of each DD interval $I_{n,m}$ around its respective ambiguity location $(n\tau_d, m/\tau_d)$ along both delay and Doppler domains is less than the corresponding spacing between adjacent ambiguity locations, these intervals do not overlap with each other. Since these intervals do not overlap, unambiguous target detection is possible from the DD locations of the peaks of the cross-ambiguity function within $I_{0,0}$. However, unambiguous target detection is possible only when τ_d satisfies the condition that it is greater than the channel delay spread and $1/\tau_d$ is greater than the channel Doppler spread. If this condition is not satisfied, the intervals overlap with each other due to which the cross-ambiguity function when observed within $I_{0,0}$ may exhibit peaks which are actually due to the channel spreading of the ambiguity blur at some ambiguity location in the vicinity of the ambiguity location $(0, 0)$, thereby resulting in the detection of non-existent targets. This condition on τ_d is similar to the crystallization condition on the Zak-OTFS parameters (τ_p, ν_p) for the predictable I/O relation of Zak-OTFS modulation in Chapter 2.

Although good radar waveforms (which are quasi-localized along both TD and FD) can be designed through this approach, for a given B and T , the accuracy of target delay and Doppler estimation with a modulated pulse train would depend on the ambiguity function $A_{s,s}(\tau, \nu)$ of the radar waveform which primarily depends on the narrow pulse $f(t)$. However, since the relation between $f(t)$ and the ambiguity function $A_{s,s}(\tau, \nu)$ of the corresponding modulated pulse train is not simple, it is challenging to start from a desired ambiguity function and then work backward to get the corresponding desired $f(t)$.

Interestingly, there is a striking resemblance between the modulated pulse train discussed in this section and the Zak-OTFS carrier waveform (pulsone) in Chapter 2, which is also a train of narrow pulses modulated by a sinusoid. In Section 10.4, the ambiguity function of a Zak-OTFS pulsone is derived. The Zak-based approach gives a rigorous mathematical framework, where good radar waveforms with desired ambiguity functions can be designed directly in the DD domain by choosing an appropriate DD domain pulse shaping filter $w_{\text{tx}}(\tau, \nu)$.

10.4 Zak Theoretical Framework for Design of Radar Waveforms

The Zak theoretical framework is centered around the Zak-OTFS modulation carrier waveform, also called the Zak-OTFS pulsone, which has been discussed in detail in Chapter 2 and is summarized here. The Zak-OTFS pulsone is the TD realization of a DD domain quasi-periodic pulse

filtered with a pulse shaping filter $w_{\text{tx}}(\tau, \nu)$. Pulse shaping is required to limit the time duration and bandwidth of the pulsone. From (2.39), it is known that a filtered quasi-periodic DD pulse located at (τ_0, ν_0) is given by

$$\begin{aligned} x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; \tau_0, \nu_0) \\ = \sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} w_{\text{tx}}(\tau - n\tau_p - \tau_0, \nu - m\nu_p - \nu_0) e^{j2\pi\nu(n\tau_p + \tau_0)} e^{-j2\pi\tau_0(\nu_0 + m\nu_p)}. \end{aligned} \quad (10.32)$$

Since a DD pulse at any location (τ_0, ν_0) can be generated from a DD pulse at origin through twisted convolution, subsequent discussion is restricted to the DD pulse at the origin. Substituting $(\tau_0, \nu_0) = (0, 0)$ in (10.32) gives

$$x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0) = \sum_{m \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi n\nu\tau_p} \quad (10.33)$$

since $\tau_p \nu_p = 1$. The Zak-OTFS TD pulsone corresponding to a quasi-periodic DD pulse at the origin is then the TD realization of $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0)$, i.e.,

$$s_{\text{td}}(t) = \mathcal{Z}_t^{-1} \left(x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0) \right), \quad (10.34)$$

where $\mathcal{Z}_t^{-1}$ is the inverse Zak transform. The following theorem gives the general expression for the ambiguity function of the Zak-OTFS pulsone.

Theorem 10.4 The ambiguity function of the Zak-OTFS pulsone corresponding to a quasi-periodic DD pulse at the origin is given by

$$A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) = w_{\text{tx}}(\tau, \nu) *_\sigma \left(\sum_{n,m \in \mathbb{Z}} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) \right) *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu), \quad (10.35)$$

where $*_\sigma$ denotes the twisted convolution, $w_{\text{tx}}(\tau, \nu)$ is the pulse shaping filter at the transmitter, and

$$w_{\text{tx}}^{\text{mf}}(\tau, \nu) = w_{\text{tx}}^*(-\tau, -\nu) e^{j2\pi\nu\tau} \quad (10.36)$$

is the corresponding matched filter (see (2.N.5)).

Proof: See Appendix 10.I.

Furthermore, (10.35) gives

$$\begin{aligned} A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) &= \left[w_{\text{tx}}(\tau, \nu) *_\sigma \left(\sum_{n,m \in \mathbb{Z}} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) \right) \right] *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu) \\ &= \left[\sum_{n,m \in \mathbb{Z}} w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi\nu n\tau_p} \right] *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu) \\ &= w_{\text{tx}}(\tau, \nu) *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu) \\ &\quad + \sum_{n,m \in \mathbb{Z}, (n,m) \neq (0,0)} \left[w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi\nu n\tau_p} \right] *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu). \end{aligned} \quad (10.37)$$

The bandwidth B and duration T of the Zak-OTFS TD pulsone are much larger than ν_p and τ_p , respectively. The bandwidth and duration of the Zak-OTFS TD pulsone is approximately the inverse of the delay and Doppler domain spread of the pulse shaping filter $w_{\text{tx}}(\tau, \nu)$ (see

(2.41), (2.44), Fig. 2.17, and the related discussion in Chapter 2). Hence, the delay and Doppler spread of $w_{\text{tx}}(\tau, \nu)$ is much smaller than τ_p and ν_p , respectively. Therefore, both $w_{\text{tx}}(\tau, \nu)$ and $w_{\text{tx}}^{\text{mf}}(\tau, \nu) = w_{\text{tx}}^*(-\tau, -\nu) e^{j2\pi\nu\tau}$ are approximately localized to the DD domain interval $(\tau, \nu) \in \{|\tau| < \frac{1}{2B}, |\nu| < \frac{1}{2T}\}$. Hence, for a given $(n, m) \in \mathbb{Z}$, $w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p)$ is localized at $(n\tau_p, m\nu_p)$ in a small interval of width $\approx 1/B$ and $1/T$ along the delay and Doppler domains, respectively. Therefore, for $(n, m) \neq (0, 0)$, there is an insignificant overlap between the support of $w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p)$ and $w_{\text{tx}}^{\text{mf}}(\tau, \nu)$.

Hence, for $(\tau, \nu) \in \{(\tau, \nu) \mid |\tau| < \tau_p/2, |\nu| < \nu_p/2\}$, in the expression for the ambiguity function of the Zak-OTFS pulsone on the RHS of (10.37), the first term $w_{\text{tx}}(\tau, \nu) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$ dominates over all other terms in the summation over $(n, m) \neq (0, 0)$. For (τ, ν) in the (n, m) th period, i.e., $\{(\tau, \nu) \mid |\tau - n\tau_p| < \tau_p/2, |\nu - m\nu_p| < \nu_p/2\}$, from (10.37), it follows that the ambiguity function of the Zak-OTFS pulsone is dominated by the term $[w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi\nu n\tau_p}] *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$, which is localized at $(n\tau_p, m\nu_p)$ since $w_{\text{tx}}(\tau - n\tau_p, \nu - m\nu_p)$ is localized at $(n\tau_p, m\nu_p)$. Therefore, it appears that the volume/blur of the ambiguity function of the Zak-OTFS pulsone is localized at the points $(n\tau_p, m\nu_p)$, $n, m \in \mathbb{Z}$, of the period lattice Λ_p (see (2.51)). This is similar to the ambiguity function of the modulated pulse train in Section 10.3, where τ_d played the role of τ_p .

Detected targets are given by the peak DD locations of the magnitude of the cross-ambiguity function $A_{r,s}(\tau, \nu)$ (see (10.8)) between the received echo waveform $r_{\text{id}}(t)$ and the transmitted radar waveform $s_{\text{id}}(t)$. From Theorem 10.1, it is known that the AWGN-free cross-ambiguity is simply the twisted convolution of the channel spreading function and the ambiguity function, i.e.,

$$\begin{aligned} A_{r,s}(\tau, \nu) &= h_{\text{phy}}(\tau, \nu) *_{\sigma} A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) \\ &= \sum_{n,m \in \mathbb{Z}} h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu) *_{\sigma} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu), \end{aligned} \quad (10.38)$$

where the second step follows from the expression of the ambiguity function of the Zak-OTFS pulsone in (10.37). From the discussion above, and from the properties of the support set of the twisted convolution between two DD domain functions (see Corollaries 2.3 and 2.4), it follows that $w_{\text{tx}}(\tau, \nu) *_{\sigma} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$ is localized around $(n\tau_p, m\nu_p)$ roughly in the interval $[n\tau_p - \frac{1}{B}, n\tau_p + \frac{1}{B}]$ along the delay domain and in the interval $[m\nu_p - \frac{1}{T}, m\nu_p + \frac{1}{T}]$ along the Doppler domain. Let $h_{\text{phy}}(\tau, \nu)$ have bounded support, i.e., it is localized to the interval $[\tau_{\min}, \tau_{\max}]$ along the delay domain and to the interval $[\nu_{\min}, \nu_{\max}]$ along the Doppler domain. Clearly, $(\tau_{\max} - \tau_{\min})$ and $(\nu_{\max} - \nu_{\min})$ are, respectively, the delay and Doppler spread of $h_{\text{phy}}(\tau, \nu)$. Therefore, the (n, m) th term $h_{\text{phy}}(\tau, \nu) *_{\sigma} w_{\text{tx}}(\tau, \nu) *_{\sigma} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$ in the expression for the cross-ambiguity in (10.38) is localized to the DD domain interval

$$\begin{aligned} \mathcal{I}_{n,m} &\triangleq \left\{ (\tau, \nu) \mid n\tau_p - \frac{1}{B} + \tau_{\min} < \tau < n\tau_p + \frac{1}{B} + \tau_{\max}, \right. \\ &\quad \left. m\nu_p - \frac{1}{T} + \nu_{\min} < \nu < m\nu_p + \frac{1}{T} + \nu_{\max} \right\}. \end{aligned} \quad (10.39)$$

Clearly, any two terms in (10.38) do not interfere with each other (i.e., $\mathcal{I}_{n_1, m_1} \cap \mathcal{I}_{n_2, m_2} = \emptyset$, $(n_1, m_1) \neq (n_2, m_2)$) if and only if

$$\tau_p > \frac{2}{B} + (\tau_{\max} - \tau_{\min}) \quad \text{and} \quad \nu_p > \frac{2}{T} + (\nu_{\max} - \nu_{\min}). \quad (10.40)$$

Clearly, when (τ_p, ν_p) satisfies this condition, unambiguous detection of targets can be carried out by detecting the peaks of the cross-ambiguity within the interval $\mathcal{I}_{0,0}$. If (τ_p, ν_p) do not satisfy (10.40), then the $(n, m) = (0, 0)$ th term in the summation in (10.38) is not separately observable since it

overlaps with the other terms which could result in the detection of targets which do not exist. This is explained in the following through an example.

Example 10.1 As an example, consider a channel with two targets at discrete DD locations $(0, 0)$ and (τ_1, ν_1) , $\tau_1 > 0$ and $\frac{1}{\tau_1} > \nu_1 > 0$ (i.e., $\tau_1 \nu_1 < 1$). The delay and Doppler spread of this channel is τ_1 and ν_1 , respectively. Also, let $\tau_p > \tau_1$ and $\nu_p = \frac{1}{\tau_p} > \nu_1$ be such that the crystallization condition in (10.40) is satisfied. The support sets for the $(n, m) = (0, 0)$ th and the $(n, m) = (-1, 0)$ th terms in the cross-ambiguity expression are given by

$$\begin{aligned} \mathcal{I}_{0,0} &\triangleq \left\{ (\tau, \nu) \mid -\frac{1}{B} < \tau < \frac{1}{B} + \tau_1, -\frac{1}{T} < \nu < \frac{1}{T} + \nu_1 \right\}, \\ \mathcal{I}_{-1,0} &\triangleq \left\{ (\tau, \nu) \mid -\tau_p - \frac{1}{B} < \tau < -\tau_p + \frac{1}{B} + \tau_1, -\frac{1}{T} < \nu < \frac{1}{T} + \nu_1 \right\}. \end{aligned} \quad (10.41)$$

The support set $\mathcal{I}_{n,m}$ for the general (n, m) th term in the cross-ambiguity expression is rectangular in the 2D DD domain. For this example, the support sets $\mathcal{I}_{-1,0}$, $\mathcal{I}_{0,0}$, and $\mathcal{I}_{1,0}$ are shown in Fig. 10.5 (T and B are assumed to be significantly larger than τ_p and ν_p , respectively). Clearly, as observed in the figure, for the crystalline regime, the support set for any two terms do not overlap. For target detection, the radar receiver looks out for peaks in the cross-ambiguity function within $\mathcal{I}_{0,0}$. Since the channel spreading function $h_{\text{phy}}(\tau, \nu)$ has two peaks at the target locations $(0, 0)$ and (τ_1, ν_1) , the cross-ambiguity function (which is the twisted convolution of $h_{\text{phy}}(\tau, \nu)$ with the ambiguity function of the Zak-OTFS pulsone) has two peaks due to each term on the RHS of (10.38). To be precise, the two peaks due to the (n, m) th term are located at $(n\tau_p, m\nu_p)$ and $(n\tau_p + \tau_1, m\nu_p + \nu_1)$, both of which lie within $\mathcal{I}_{n,m}$ irrespective of whether the crystallization condition is satisfied or not. The two peaks due to the $(0, 0)$ th term are located at $(0, 0)$ and (τ_1, ν_1) (depicted as stars at these locations), both of which lie within $\mathcal{I}_{0,0}$ (depicted by the small rectangle connecting these stars). The two peaks due to the $(-1, 0)$ th term lie at $(-\tau_p, 0)$ and $(\tau_1 - \tau_p, \nu_1)$ (depicted as stars at these locations). When the crystallization condition in (10.40) is satisfied, no two support sets

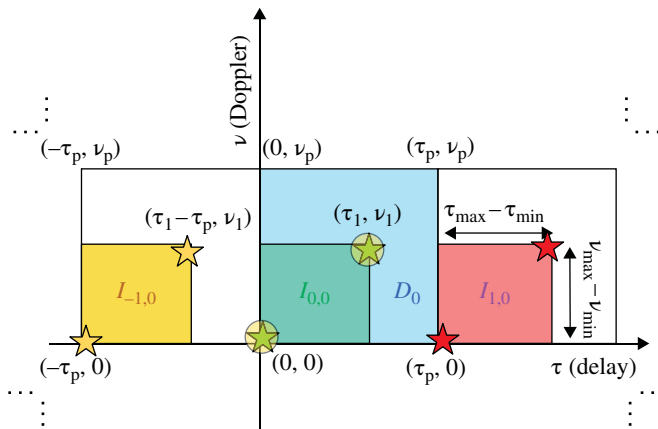
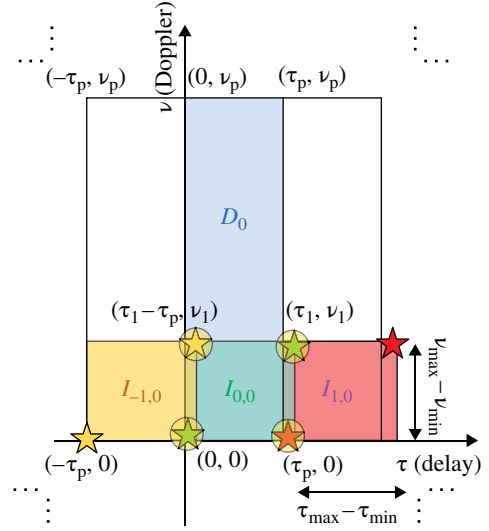


Figure 10.5 Unambiguous target detection in the crystalline regime, i.e., when the crystallization condition in (10.40) is satisfied. The stars with circle around them denote the location of the detected targets from the cross-ambiguity peaks in $\mathcal{I}_{0,0}$. When the crystallization condition in (10.40) is satisfied, the support sets $\mathcal{I}_{n,m}$, $n, m \in \mathbb{Z}$, do not overlap, and, therefore, target detection is unambiguous.

Figure 10.6 Detection of non-existent targets in the non-crystalline regime, i.e., when the crystallization condition in (10.40) is not satisfied.



I_{n_1, m_1} and I_{n_2, m_2} ($(n_1, m_1) \neq (n_2, m_2)$) overlap, and, hence, the peak of the cross-ambiguity function within $I_{0,0}$ is observed only at the two true targets (shown as stars with circles around them). No other false targets are detected.

Now, if τ_p is decreased (i.e., $\nu_p = 1/\tau_p$ is increased) such that $\tau_p < \tau_1$, then clearly the crystallization condition in (10.40) is not satisfied since τ_p is less than the channel delay spread τ_1 . This scenario is shown in Fig. 10.6. The dimension of the support sets $I_{n,m}$ is the same irrespective of the choice of (τ_p, ν_p) since it depends primarily on the delay and Doppler spread of the channel. Clearly, in the non-crystalline regime, as the delay domain spacing τ_p between the lattice points of the period lattice (where the ambiguity volume/blur is concentrated) becomes smaller, the support sets $I_{n,m}$ (whose dimension does not change) corresponding to the terms on the RHS of (10.38) (see also (10.39)) move closer to each other along the delay domain due to which adjacent support sets overlap with each other along the delay domain. This is clear from Fig. 10.6 where $I_{0,0} \cap I_{-1,0} \neq \emptyset$. The two peaks due to the $(-1, 0)$ th term lie at $(-\tau_p, 0)$ and $(\tau_1 - \tau_p, \nu_1)$, out of which the first peak $(-\tau_p, 0)$ does not lie within $I_{0,0}$. However, the second peak $(\tau_1 - \tau_p, \nu_1)$ lies within $I_{0,0}$ since $\tau_1 - \tau_p > 0 > -\frac{1}{B}$ and $\tau_1 - \tau_p < \tau_1 < \frac{1}{B} + \tau_1$ (see (10.41)). This results in the detection of a target at $(\tau_1 - \tau_p, \nu_1)$ which does not really exist (shown as a star at this location with circle around it in Fig. 10.6). Similarly, the $(1, 0)$ th term results in a peak at $(\tau_p, 0)$ which also lies within $I_{0,0}$ (since $\tau_p < \tau_1$), and, therefore, results in a non-existent target detected at $(\tau_p, 0)$ (shown by a star at this location with circle around it). Therefore, as shown in the figure, there are four peaks detected within $I_{0,0}$ (stars with a circle around them), out of which the peaks at $(0, 0)$ and (τ_1, ν_1) are true targets (shown by stars at these locations with circle around them), whereas the peaks at $(\tau_1 - \tau_p, \nu_1)$ and $(\tau_p, 0)$ correspond to the non-existent targets.

Remark The condition in (10.40) is exactly the same as the *crystallization condition* in (2.74) for ensuring a predictable and non-fading I/O relation for communication based on Zak-OTFS modulation. In the context of radar sensing, choosing (τ_p, ν_p) to satisfy $\tau_p \nu_p = 1$ and (10.40) ensures unambiguous detection of targets.

Next, the expression for the ambiguity function of the Zak-OTFS pulsone for a sinc pulse shaping filter is derived so as to illustrate the redistribution of the ambiguity volume/blur.

10.4.1 Ambiguity Function of the Zak-OTFS Pulsone for Sinc Pulse

Consider the pulse shaping filter

$$w_{\text{tx}}(\tau, \nu) = W_1(\tau) W_2(\nu),$$

$$W_1(\tau) \triangleq \sqrt{B} \text{sinc}(B\tau), \quad W_2(\nu) \triangleq \sqrt{T} \text{sinc}(T\nu), \quad (10.42)$$

where B and T are the bandwidth and time duration of the corresponding Zak-OTFS pulsone. Without loss of generality, let $M \triangleq B\tau_p$ and $N \triangleq T\nu_p$ be positive integers. Then, for the sinc pulse shaping filter, from (2.41), it follows that the Zak-OTFS pulsone located at $(\tau_0, \nu_0) = (0, 0)$ is given by (for even N)

$$s_{\text{td}}(t) = \sqrt{\frac{B}{N}} \sum_{n=-\frac{N}{2}}^{\frac{N}{2}-1} \text{sinc}(B(t - n\tau_p)). \quad (10.43)$$

The derivation of this expression starting from (2.41) is given in Appendix 10.J. Note that the summation index n in the above equation is an integer, and, therefore, for odd N , the summation starts with $n = -(N-1)/2$ and ends with $n = (N-1)/2$, i.e.,

$$s_{\text{td}}(t) = \sqrt{\frac{B}{N}} \sum_{n=-\left(\frac{N-1}{2}\right)}^{\frac{N-1}{2}} \text{sinc}(B(t - n\tau_p)). \quad (10.44)$$

The derivation for odd N is similar to that for even N .

The following theorem gives the expression for the ambiguity function of the Zak-OTFS pulsone corresponding to a quasi-periodic DD pulse located at the origin.

Theorem 10.5 The ambiguity function of a Zak-OTFS pulsone in (10.43) (i.e., for even N) is given by

$$A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) = \begin{cases} \frac{\left(1 - \frac{|\nu|}{B}\right)}{N} \sum_{n_1=-\frac{N}{2}}^{\frac{N}{2}-1} \sum_{n_2=-\frac{N}{2}}^{\frac{N}{2}-1} \left[e^{j\pi\nu(\tau - (n_1+n_2)\tau_p)} \text{sinc}((B - |\nu|)(\tau + (n_1 - n_2)\tau_p)) \right], & |\nu| < B, \\ 0, & |\nu| \geq B. \end{cases} \quad (10.45)$$

Proof: See Appendix 10.K.

The ambiguity function of a Zak-OTFS pulsone (see (10.45)) is plotted in Fig. 10.7 for the same time and bandwidth constraints as for the TDM and frequency division multiplexing (FDM) waveforms illustrated in Fig. 10.3.

From Fig. 10.7, it is clear that, as discussed earlier, the volume/blur of the ambiguity function of the Zak-OTFS pulsone is indeed localized at the points $(n\tau_p, m\nu_p)$, $n, m \in \mathbb{Z}$, of the period lattice Λ_p . Compared to the ambiguity function of TD/FD pulses where the blur is wide and contiguous along FD/TD, for Zak-OTFS pulsones, the blur is localized around the lattice points of the period lattice which are separated by τ_p and ν_p along the delay and Doppler domains, respectively. This redistribution of the blur allows for unambiguous target detection when (τ_p, ν_p) satisfies the crystallization condition in (10.40).

Therefore, just like the modulated pulse train in Section 10.3, the Zak-OTFS pulsone can also achieve unambiguous target detection by redistributing the ambiguity volume/blur on the period lattice. Given an arbitrary pulse shaping filter $w_{\text{tx}}(\tau, \nu)$, Zak theory gives a precise expression for

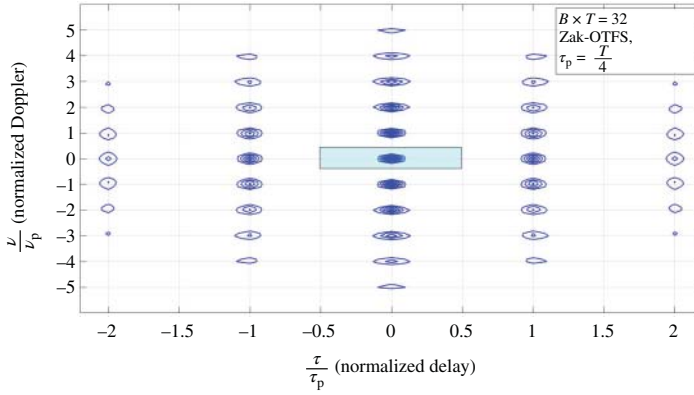


Figure 10.7 Plot of the ambiguity function (squared magnitude) for the Zak-OTFS carrier waveform. Simultaneous delay and Doppler resolvability can be achieved. Unambiguous target estimation is achievable in the crystalline regime.

the ambiguity function $A_{s,s}^{\text{Zak-OTFS}}$ of the Zak-OTFS pulsone in terms of the pulse shaping filter (see (10.35) in Theorem 10.4). Zak theory, therefore, reduces the radar waveform design to the design of the pulse shaping filter, as the shape of the ambiguity function within any period of the period lattice is related to the pulse shaping filter through simple analytical expressions. For example, in the crystalline regime (i.e., when the crystallization condition is satisfied), the ambiguity function within $\mathcal{I}_{0,0}$ is dominated by the $(n, m) = (0, 0)$ th term on the RHS of (10.38), which is simply $w_{\text{tx}}(\tau, \nu) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$. In the crystalline regime, the resolution of the radar therefore depends on the shape of this function $w_{\text{tx}}(\tau, \nu) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu)$ which can be precisely/analytically controlled by designing $w_{\text{tx}}(\tau, \nu)$ appropriately. Such precise control is not possible with the design of modulated pulse train-based radar waveform since the relation between the TD pulse $f(t)$, the broad pulse $g(t)$, and the ambiguity function of the modulated pulse train is not as simple as the relation between $w_{\text{tx}}(\tau, \nu)$ and the ambiguity function for the Zak-OTFS pulsone.

10.5 Summary

This chapter presented a Zak theory-based framework for the design of good waveforms for radar sensing. Since Zak-OTFS pulsones (studied in Chapter 2) are quasi-localized in the DD domain, in addition to being ideally suited for predictable communication in doubly spread channels, they also serve as good radar waveforms. Interestingly, the crystallization condition for achieving a predictable I/O relation in the context of communication systems is also the condition required to guarantee unambiguous target detection in radar sensing. At the same time, the shape of the ambiguity function of the radar waveform decides its resolution (i.e., the ability of the radar to resolve close targets). Zak theory reveals that the shape of the ambiguity function depends on the DD domain pulse shaping filter used to limit the Zak-OTFS pulsone in time and bandwidth. It turns out that within the DD region $\{(\tau, \nu) \mid |\tau| < \tau_p/2, |\nu| < \nu_p/2\}$, the ambiguity function of a radar waveform is simply the twisted convolution between the pulse shaping filter and its corresponding matched filter. Zak theory, therefore, simplifies the design of radar waveforms to that of designing pulse shaping filters. This is a paradigm shift in the manner in which radar waveforms have been designed so far.

The resolution of a Zak-OTFS pulsone-based radar is governed by the shape of the pulse shaping filter. Similarly, in the context of communication systems, the shape of the pulse shaping

Table 10.1 Role of the local and global behavior of quasi-periodic DD pulses in communication and radar sensing.

	Communication	Radar sensing
(Global behavior)		
Crystallization condition is satisfied	Predictable and non-fading I/O relation	Unambiguous target detection
(Local behavior)		
Smaller delay and Doppler spread of pulse shaping filter $w_{\text{tx}}(\tau, \nu)$	Implies lesser inter-symbol interference	Implies better radar resolution

filter decides the degree of interference between information symbols transmitted on Zak-OTFS pulsones corresponding to adjacent DD pulses on the DD domain information grid Λ_{dd} . Also, in both communication and radar sensing, the time duration and bandwidth of the transmitted waveform are inversely proportional to the Doppler and delay spread of the pulse shaping filter, respectively. This similarity between predictability in communication systems and unambiguous target detection in radar sensing, and the similarity between interference between information symbols in communication systems and resolution in radar sensing have been summarized in Table 10.1.

Since Zak-OTFS pulsones are good for both communication and radar sensing, they serve as potential candidate waveforms for joint communication and sensing.

Appendix 10.A Proof of (10.7)

The ML estimate of (τ_1, ν_1) is given by

$$\begin{aligned}
 (\hat{\tau}_1, \hat{\nu}_1) &= \arg \min_{h, \tau, \nu} \int \left| r_{\text{td}}(t) - h s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} \right|^2 dt \\
 &\stackrel{(a)}{=} \arg \min_{\tau, \nu} \left[\min_h \int \left| r_{\text{td}}(t) - h s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} \right|^2 dt \right].
 \end{aligned} \tag{10.A.1}$$

In step (a) above, the inner minimization with respect to h for a given (τ, ν) is given by

$$\begin{aligned}
 &\min_h \int \left| r_{\text{td}}(t) - h s_{\text{td}}(t - \tau) e^{j2\pi\nu(t-\tau)} \right|^2 dt \\
 &\stackrel{(a)}{=} \min_h \left[\underbrace{\int |r_{\text{td}}(t)|^2 dt}_{E_r} + |h|^2 \underbrace{\int |s_{\text{td}}(t - \tau)|^2 dt}_{E_t} + 2\Re(h^* A_{r,s}(\tau, \nu)) \right] \\
 &\stackrel{(b)}{=} \min_h \left[E_r + |h|^2 E_t + 2|h| |A_{r,s}(\tau, \nu)| \cos(\arg(A_{r,s}(\tau, \nu)) - \arg(h)) \right] \\
 &\stackrel{(c)}{=} \min_h \left[E_r + |h|^2 E_t - 2|h| |A_{r,s}(\tau, \nu)| \right] \\
 &\stackrel{(d)}{=} \left[E_r - \frac{|A_{r,s}(\tau, \nu)|^2}{E_t} \right],
 \end{aligned} \tag{10.A.2}$$

where in step (a), E_r and E_t denote the received and transmitted energy, respectively, and the cross-ambiguity $A_{r,s}(\tau, \nu)$ is given by (10.8). In step (b), $\arg(z)$, $z \in \mathbb{C}$ denotes the argument of complex number z . In step (c), for a given (τ, ν) , the argument of h is chosen such that $\arg(A_{r,s}(\tau, \nu)) - \arg(h)$ is an odd multiple of π . Finally, minimizing the RHS in step (c) with respect to $|h|$ gives the expression in step (d), where the optimal value of $|h|$ is $|A_{r,s}(\tau, \nu)|/E_t$.

Using (10.A.2) in (10.A.1), it follows that

$$\begin{aligned} (\hat{\tau}_1, \hat{\nu}_1) &= \arg \min_{\tau, \nu} \left[E_r - \frac{|A_{r,s}(\tau, \nu)|^2}{E_t} \right] \\ &= \arg \max_{\tau, \nu} |A_{r,s}(\tau, \nu)|^2 \\ &= \arg \max_{\tau, \nu} |A_{r,s}(\tau, \nu)| \end{aligned} \quad (10.A.3)$$

since E_t does not depend on (τ, ν) .

Appendix 10.B Proof of (10.9)

From Theorem 2.4, we know that the inner product between any two TD signals is conserved in the DD domain, i.e., for any two TD signals $x(t)$ and $y(t)$,

$$\int_{-\infty}^{\infty} x(t)y^*(t) dt = \int_0^{\tau_p} \int_0^{\nu_p} x_{\text{dd}}(\tau, \nu)y_{\text{dd}}^*(\tau, \nu) d\nu d\tau, \quad (10.B.1)$$

where $x_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(x(t))$ and $y_{\text{dd}}(\tau, \nu) = \mathcal{Z}_t(y(t))$ are the Zak transform/DD representation of $x(t)$ and $y(t)$, respectively. For any (τ_0, ν_0) , with $x(t) = r_{\text{id}}(t)$ and $y(t) = s_{\text{id}}(t - \tau_0)e^{j2\pi\nu_0(t - \tau_0)}$, from (10.8), it follows that

$$\begin{aligned} A_{r,s}(\tau_0, \nu_0) &= \int_{-\infty}^{\infty} r_{\text{id}}(t) \left[s_{\text{id}}(t - \tau_0)e^{j2\pi\nu_0(t - \tau_0)} \right]^* dt \\ &\stackrel{(a)}{=} \int_0^{\tau_p} \int_0^{\nu_p} \underbrace{\mathcal{Z}_t(r_{\text{id}}(t))}_{=y_{\text{dd}}(\tau, \nu)} \left[\mathcal{Z}_t(s_{\text{id}}(t - \tau_0)e^{j2\pi\nu_0(t - \tau_0)}) \right]^* d\nu d\tau, \end{aligned} \quad (10.B.2)$$

where step (a) follows from (10.B.1). Note that for any (τ_0, ν_0) , $s_{\text{id}}(t - \tau_0)e^{j2\pi\nu_0(t - \tau_0)}$ is the effect of a delay and Doppler shift of τ_0 and ν_0 , respectively, to the radar waveform $s_{\text{id}}(t)$. Therefore, from (2.27), it follows that

$$\begin{aligned} \mathcal{Z}_t(s_{\text{id}}(t - \tau_0)e^{j2\pi\nu_0(t - \tau_0)}) &= [\delta(\tau - \tau_0)\delta(\nu - \nu_0)] *_{\sigma} \underbrace{\mathcal{Z}_t(s_{\text{id}}(t))}_{s_{\text{dd}}(\tau, \nu)} \\ &= e^{j2\pi\nu_0(\tau - \tau_0)} s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0). \end{aligned} \quad (10.B.3)$$

Using (10.B.3) in (10.B.2) gives (10.9).

Appendix 10.C Proof of Theorem 10.1

Substituting the expression of $y_{\text{dd}}(\tau, \nu)$ from (10.12) into (10.9) gives

$$\begin{aligned}
 A_{r,s}(\tau_0, \nu_0) &= \int_0^{\tau_p} \int_0^{\nu_p} y_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} (h_{\text{phy}}(\tau, \nu) *_{\sigma} s_{\text{dd}}(\tau, \nu)) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\
 &\quad + \int_0^{\tau_p} \int_0^{\nu_p} n_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu.
 \end{aligned} \tag{10.C.1}$$

The first integral on the RHS of the above equation is

$$\begin{aligned}
 &\int_0^{\tau_p} \int_0^{\nu_p} (h_{\text{phy}}(\tau, \nu) *_{\sigma} s_{\text{dd}}(\tau, \nu)) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\
 &\stackrel{(a)}{=} \int_0^{\tau_p} \int_0^{\nu_p} \left[\left(\iint h_{\text{phy}}(\tau', \nu') s_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \right) \right. \\
 &\quad \left. s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} \right] d\tau d\nu \\
 &\stackrel{(b)}{=} \iint h_{\text{phy}}(\tau', \nu') \left[\int_0^{\tau_p} \int_0^{\nu_p} s_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} \right. \\
 &\quad \left. s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \right] e^{j2\pi\nu'(\tau_0 - \tau')} d\tau' d\nu',
 \end{aligned} \tag{10.C.2}$$

where step (a) follows from the twisted convolution integral expression for $h_{\text{phy}}(\tau, \nu) *_{\sigma} s_{\text{dd}}(\tau, \nu)$ and step (b) follows from interchanging the order of integration. For a given (τ', ν') , the inner integral in the last step above is given by

$$\begin{aligned}
 &\int_0^{\tau_p} \int_0^{\nu_p} s_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau_0)} s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\
 &= \int_{\tau'}^{\tau' + \tau_p} \int_{\nu'}^{\nu' + \nu_p} s_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau_0)} s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu,
 \end{aligned} \tag{10.C.3}$$

which follows from the fact that for any given (τ_0, ν_0) , the integrand in the above integral is a periodic DD function with τ_p and ν_p as the delay and Doppler domain period respectively, and therefore the integral value is the same over any delay-Doppler period (the DD period in the RHS is $[\tau', \tau' + \tau_p] \times [\nu', \nu' + \nu_p]$). To see that the integrand is a periodic function, note that, for any given (τ', ν') , $s_{\text{dd}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau_0)}$ is also a quasi-periodic function with respect to (τ, ν) . Since the integrand above is the product of a quasi-periodic function and the complex conjugate of another quasi-periodic function, it is a periodic function with respect to (τ_p, ν_p) . Therefore, the value of the integral is preserved as long as the 2D integral is over a period of the period lattice.

First, it is clear that $f_{\text{dd}}(\tau, \nu) \triangleq s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0) e^{j2\pi\nu_0(\tau - \tau_0)}$ is a quasi-periodic function of (τ, ν) with respect to the period lattice since for any $n, m \in \mathbb{Z}$,

$$\begin{aligned} f_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) &= s_{\text{dd}}(\tau - \tau_0 + n\tau_p, \nu - \nu_0 + m\nu_p) e^{j2\pi\nu_0(\tau + n\tau_p - \tau_0)} \\ &= s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0) e^{j2\pi n(\nu - \nu_0)\tau_p} e^{j2\pi\nu_0(\tau + n\tau_p - \tau_0)} \\ &= s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0) \underbrace{e^{j2\pi\nu_0(\tau - \tau_0)} e^{j2\pi n\nu\tau_p}}_{=f_{\text{dd}}(\tau, \nu)} \\ &= f_{\text{dd}}(\tau, \nu) e^{j2\pi n\nu\tau_p}, \end{aligned} \quad (10.C.4)$$

where the second step follows from the quasi-periodicity of $s_{\text{dd}}(\tau, \nu)$ as it is the Zak transform of the TD radar waveform $s_{\text{td}}(t)$. The last step clearly shows that $f_{\text{dd}}(\tau, \nu)$ satisfies the quasi-periodicity condition defined in (2.25). Also, the product of a quasi-periodic DD function $a_{\text{dd}}(\tau, \nu)$ with the conjugate of another quasi-periodic DD function $b_{\text{dd}}(\tau, \nu)$, i.e., $c_p(\tau, \nu) \triangleq a_{\text{dd}}(\tau, \nu) b_{\text{dd}}^*(\tau, \nu)$, is periodic since, for any $n, m \in \mathbb{Z}$,

$$\begin{aligned} c_p(\tau + n\tau_p, \nu + m\nu_p) &= a_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p) b_{\text{dd}}^*(\tau + n\tau_p, \nu + m\nu_p) \\ &= a_{\text{dd}}(\tau, \nu) e^{j2\pi n\nu\tau_p} \left(b_{\text{dd}}(\tau, \nu) e^{j2\pi n\nu\tau_p} \right)^* \\ &= a_{\text{dd}}(\tau, \nu) b_{\text{dd}}^*(\tau, \nu) = c_p(\tau, \nu), \end{aligned} \quad (10.C.5)$$

where the second step follows from the quasi-periodicity of $a_{\text{dd}}(\tau, \nu)$ and $b_{\text{dd}}(\tau, \nu)$.

In (10.C.3), substituting the integration variables (τ, ν) with (τ'', ν'') where $\tau'' \triangleq (\tau - \tau')$ and $\nu'' \triangleq (\nu - \nu')$, it follows that the integral is

$$\begin{aligned} &\int_0^{\tau_p} \int_0^{\nu_p} s_{\text{dd}}(\tau'', \nu'') s_{\text{dd}}^*(\tau'' - (\tau_0 - \tau'), \nu'' - (\nu_0 - \nu')) e^{-j2\pi(\nu_0 - \nu')(\tau'' - (\tau_0 - \tau'))} d\tau'' d\nu'' \\ &\stackrel{(a)}{=} A_{s,s}(\tau_0 - \tau', \nu_0 - \nu'), \end{aligned} \quad (10.C.6)$$

where step (a) follows from the definition of the auto-ambiguity function of $s_{\text{td}}(t)$ in (10.14). Hence, the inner integral in (10.C.2) is simply $A_{s,s}(\tau_0 - \tau', \nu_0 - \nu')$. Using this fact in (10.C.2) gives

$$\begin{aligned} &\int_0^{\tau_p} \int_0^{\nu_p} (h_{\text{phy}}(\tau, \nu) *_{\sigma} s_{\text{dd}}(\tau, \nu)) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\ &= \iint h_{\text{phy}}(\tau', \nu') A_{s,s}(\tau_0 - \tau', \nu_0 - \nu') e^{j2\pi\nu'(\tau_0 - \tau')} d\tau' d\nu' \\ &= (h_{\text{phy}} *_{\sigma} A_{s,s})(\tau_0, \nu_0), \end{aligned} \quad (10.C.7)$$

which is nothing but the twisted convolution between $h_{\text{phy}}(\tau, \nu)$ and $A_{s,s}(\tau, \nu)$ evaluated at $(\tau, \nu) = (\tau_0, \nu_0)$. Since this is the first integral on the RHS of (10.C.1), it follows that

$$\begin{aligned} A_{r,s}(\tau_0, \nu_0) &= (h_{\text{phy}} *_{\sigma} A_{s,s})(\tau_0, \nu_0) \\ &+ \int_0^{\tau_p} \int_0^{\nu_p} n_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu. \end{aligned} \quad (10.C.8)$$

Replacing the integration variables in the above expression with (τ', ν') and (τ_0, ν_0) with (τ, ν) gives the general expression on the RHS of (10.13).

The second equality in (10.14) follows from a derivation similar to the derivation in Appendix 10.B of the expression for $A_{r,s}(\tau_0, \nu_0)$ in (10.9). The expression in the third line of (10.14) follows from the expression in the second line since the inner product between two signals is conserved in the frequency domain (Parseval's relation). To be precise, for any two finite energy TD signals $x(t)$ and $y(t)$,

$$\int x(t) y^*(t) dt = \int X(f) Y^*(f) df, \quad (10.C.9)$$

where $X(f)$ and $Y(f)$ are the Fourier transform/FD representation of $x(t)$ and $y(t)$, respectively. With $x(t) = s_{\text{id}}(t)$ and $y(t) = s_{\text{id}}(t - \tau_0) e^{j2\pi\nu_0(t - \tau_0)}$, the inner product $\int x(t) y^*(t) dt$ is $A_{s,s}(\tau_0, \nu_0)$, which, therefore, is also equal to the inner product between the Fourier transforms $X(f)$ and $Y(f)$, i.e.,

$$A_{s,s}(\tau_0, \nu_0) = \int X(f) Y^*(f) df. \quad (10.C.10)$$

Since $x(t) = s_{\text{id}}(t)$,

$$X(f) = s_{\text{id}}(f), \quad (10.C.11)$$

where $s_{\text{id}}(f)$ is the Fourier transform of $s_{\text{id}}(t)$. Also, since $y(t) = s_{\text{id}}(t - \tau_0) e^{j2\pi\nu_0(t - \tau_0)}$,

$$Y(f) = s_{\text{id}}(f - \nu_0) e^{-j2\pi f \tau_0}. \quad (10.C.12)$$

Substituting the expressions for $X(f)$ and $Y(f)$ from (10.C.11) and (10.C.12) into (10.C.10) gives the expression for the ambiguity function in the third line of (10.14).

Appendix 10.D Proof of (10.21)

From (10.14), it follows that

$$\begin{aligned} \iint |A_{s,s}(\tau, \nu)|^2 d\tau d\nu &= \iint A_{s,s}(\tau, \nu) A_{s,s}^*(\tau, \nu) d\tau d\nu \\ &= \iiint \int s_{\text{id}}(t_1) s_{\text{id}}^*(t_2) s_{\text{id}}^*(t_1 - \tau) e^{-j2\pi\nu(t_1 - \tau)} s_{\text{id}}(t_2 - \tau) e^{j2\pi\nu(t_2 - \tau)} d\tau d\nu dt_1 dt_2 \\ &= \iiint \int s_{\text{id}}(t_1) s_{\text{id}}^*(t_2) s_{\text{id}}^*(t_1 - \tau) s_{\text{id}}(t_2 - \tau) \underbrace{\left[\int e^{j2\pi\nu(t_2 - t_1)} d\nu \right]}_{=\delta(t_2 - t_1)} d\tau dt_1 dt_2 \\ &= \iiint \int s_{\text{id}}(t_1) s_{\text{id}}^*(t_2) s_{\text{id}}^*(t_1 - \tau) s_{\text{id}}(t_2 - \tau) \delta(t_2 - t_1) d\tau dt_1 dt_2 \\ &= \iint |s_{\text{id}}(t_1)|^2 |s_{\text{id}}(t_1 - \tau)|^2 d\tau dt_1 = \int |s_{\text{id}}(t_1)|^2 \left[\int |s_{\text{id}}(t_1 - \tau)|^2 d\tau \right] dt_1 \\ &= \left[\int |s_{\text{id}}(t)|^2 dt \right]^2. \end{aligned} \quad (10.D.1)$$

Appendix 10.E Effect of Time and Bandwidth Limitation on the DD Representation of a Waveform

Limiting a waveform in time and bandwidth to T seconds and B Hz, respectively, is equivalent to multiplying it with a rectangular TD pulse limited to T seconds and then convolving the

time-limited waveform with a TD filter whose frequency response is a rectangular FD pulse limited to B Hz (see (12) in [60]).

The effect of time limitation on the DD domain representation of a waveform is the Doppler domain linear convolution of its DD representation $s_{\text{dd}}(\tau, \nu)$ with the DD representation of a rectangular TD pulse of duration T seconds (convolution is limited to one Doppler period, see Appendix F in [60]). Since the DD representation of a TD pulse of duration T is spread over the Doppler domain over an interval of width approximately $1/T$ Hz (see (17) in [60]), time limitation of a waveform to T seconds spreads its DD representation over an interval of width approximately $1/T$ along the Doppler domain.

The effect of bandwidth limitation on the DD representation of a waveform is the delay domain linear convolution of its DD representation with the DD representation of a FD pulse having bandwidth B Hz (convolution is limited to one delay period, see Appendix E in [60]). Since the DD representation of a FD pulse of width B is spread over the delay domain over an interval of width approximately $1/B$ (see (17) in [60]), bandwidth limitation of a waveform to B Hz spreads its DD representation over an interval of width approximately $1/B$ along the delay domain.

Therefore, simultaneously limiting a waveform in time and bandwidth to T seconds and B Hz spreads its DD representation by $1/T$ and $1/B$ along the Doppler and delay domains, respectively.

Appendix 10.F Peaks of the Ambiguity Function

From the expression of the ambiguity function in the first line of (10.14), it follows that, for any (τ_0, ν_0) ,

$$\begin{aligned}
 |A_{s,s}(\tau_0, \nu_0)| &= \left| \int_0^{\tau_p} \int_0^{\nu_p} s_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \right| \\
 &\leq \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu) s_{\text{dd}}^*(\tau - \tau_0, \nu - \nu_0) e^{-j2\pi\nu_0(\tau - \tau_0)}| d\tau d\nu \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)| |s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0)| d\tau d\nu \\
 &\leq \sqrt{\left[\int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu \right] \left[\int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0)|^2 d\tau d\nu \right]} \\
 &= \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu, \tag{10.F.1}
 \end{aligned}$$

where the inequality in the second last step follows from the Cauchy–Schwartz inequality (i.e., for any two non-negative valued functions $a(\tau, \nu)$ and $b(\tau, \nu)$, $|\int \int a(\tau, \nu) b(\tau, \nu) d\tau d\nu|^2 \leq \int \int a^2(\tau, \nu) d\tau d\nu \int \int b^2(\tau, \nu) d\tau d\nu$). The equality in the last step follows from the fact that $|s_{\text{dd}}(\tau, \nu)|^2$ is a periodic function with respect to the period lattice Λ_p , i.e., for any $n, m \in \mathbb{Z}$,

$$\begin{aligned}
 |s_{\text{dd}}(\tau + n\tau_p, \nu + m\nu_p)|^2 &= |s_{\text{dd}}(\tau, \nu) e^{j2\pi n\nu\tau_p}|^2 \\
 &= |s_{\text{dd}}(\tau, \nu)|^2, \tag{10.F.2}
 \end{aligned}$$

where the first step follows from the quasi-periodicity of $s_{\text{dd}}(\tau, \nu)$ (see (2.25)). Since $|s_{\text{dd}}(\tau, \nu)|^2$ is a periodic function with respect to the period lattice Λ_p , the integral of $|s_{\text{dd}}(\tau, \nu)|^2$ over any period is the same, i.e., for any (τ_0, ν_0) ,

$$\begin{aligned} \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu &= \int_{-\tau_0}^{\tau_p - \tau_0} \int_{-\nu_0}^{\nu_p - \nu_0} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu \\ &= \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau - \tau_0, \nu - \nu_0)|^2 d\tau d\nu, \end{aligned} \quad (10.F.3)$$

where the second step is obtained by substituting the integration variables (τ, ν) on the RHS of the first step by $(\tau + \tau_0, \nu + \nu_0)$. Therefore, from (10.F.1), it follows that

$$|A_{s,s}(\tau_0, \nu_0)| \leq \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu. \quad (10.F.4)$$

Furthermore, the ambiguity function when evaluated at $(\tau_0, \nu_0) = (0, 0)$ gives

$$A_{s,s}(0, 0) = \int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu. \quad (10.F.5)$$

Comparing (10.F.4) and (10.F.5), it follows that, for any (τ_0, ν_0) ,

$$|A_{s,s}(\tau_0, \nu_0)| \leq A_{s,s}(0, 0), \quad (10.F.6)$$

i.e., the origin is always a peak for the ambiguity function of any arbitrary radar waveform. Furthermore, the value of this peak is

$$\int_0^{\tau_p} \int_0^{\nu_p} |s_{\text{dd}}(\tau, \nu)|^2 d\tau d\nu = \int |s_{\text{td}}(t)|^2 dt, \quad (10.F.7)$$

which is the energy of the radar waveform (the RHS follows by substituting $\tau_0 = \nu_0 = 0$ in the second line of (10.14)).

Appendix 10.G Proof of Theorem 10.2

From (10.14), it follows that the ambiguity function for the sinc pulse is given by

$$\begin{aligned} A_{s,s}^{\text{tdm}}(\tau, \nu) &= \int s_{\text{td}}(t) s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt \\ &= B \int \text{sinc}(Bt) [\text{sinc}(B(t - \tau)) e^{j2\pi\nu(t-\tau)}]^* dt. \end{aligned} \quad (10.G.1)$$

From Parseval's relation, we know that for any two finite energy TD signals $x(t)$ and $y(t)$, the inner product is conserved in the frequency domain, i.e.,

$$\int x(t) y^*(t) dt = \int X(f) Y^*(f) df, \quad (10.G.2)$$

where $X(f)$ and $Y(f)$ are the Fourier transforms of $x(t)$ and $y(t)$, respectively. Let $x(t) = \sqrt{B} \text{sinc}(Bt)$ and $y(t) = \sqrt{B} \text{sinc}(B(t - \tau)) e^{j2\pi\nu(t-\tau)}$. Clearly,

$$X(f) = \begin{cases} \frac{1}{\sqrt{B}}, & |f| < \frac{B}{2}, \\ 0, & \text{otherwise,} \end{cases} \quad (10.G.3)$$

and

$$Y(f) = \begin{cases} \frac{e^{-j2\pi f\tau}}{\sqrt{B}}, & -\frac{B}{2} + \nu < f < \frac{B}{2} + \nu, \\ 0, & \text{otherwise.} \end{cases} \quad (10.G.4)$$

Therefore, for $0 \leq \nu < B$, using Parseval's relation in (10.G.2), the integral in (10.G.1) evaluates to

$$\begin{aligned} A_{s,s}^{\text{tdm}}(\tau, \nu) &= \int X(f) Y^*(f) df = \frac{1}{B} \int_{-\frac{B}{2} + \nu}^{\frac{B}{2}} e^{j2\pi f\tau} df \\ &= e^{j\pi\tau\nu} \frac{\sin(\pi\tau(B - \nu))}{\pi B\tau} = \frac{B - \nu}{B} e^{j\pi\tau\nu} \text{sinc}((B - \nu)\tau) \\ &= \left(1 - \frac{|\nu|}{B}\right) e^{j\pi\tau\nu} \text{sinc}((B - |\nu|)\tau), \end{aligned} \quad (10.G.5)$$

where the last step follows from the fact that for $\nu \geq 0$, $\nu = |\nu|$. For $-B \leq \nu < 0$, using Parseval's relation, the integral in (10.G.1) evaluates to

$$\begin{aligned} A_{s,s}^{\text{tdm}}(\tau, \nu) &= \int X(f) Y^*(f) df = \frac{1}{B} \int_{-\frac{B}{2}}^{\frac{B}{2} + \nu} e^{j2\pi f\tau} df = \frac{B + \nu}{B} e^{j\pi\tau\nu} \text{sinc}((B + \nu)\tau) \\ &= \left(1 - \frac{|\nu|}{B}\right) e^{j\pi\tau\nu} \text{sinc}((B - |\nu|)\tau), \end{aligned} \quad (10.G.6)$$

where the last step follows from the fact that for $\nu < 0$, $|\nu| = -\nu$. For $|\nu| > B$, it is clear that $A_{s,s}^{\text{tdm}}(\tau, \nu) = \int X(f) Y^*(f) df$ is zero since the support set for $X(f)$ and $Y(f)$ do not overlap (see (10.G.3) and (10.G.4)). Using this fact along with (10.G.5) and (10.G.6) completes the proof.

Appendix 10.H Proof of Theorem 10.3

The FD pulse $s_{\text{fd}}(f) = \sqrt{T} \text{sinc}(fT)$ has a TD realization which is a rectangular TD pulse limited to the time interval $(-T/2, T/2)$. From the second line of (10.14), it follows that the ambiguity function for an FD pulse is given by

$$\begin{aligned} A_{s,s}^{\text{fdm}}(\tau, \nu) &= \int s_{\text{td}}(t) s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt \\ &\stackrel{(a)}{=} \frac{1}{\sqrt{T}} \int_{-\frac{T}{2}}^{\frac{T}{2}} s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt, \end{aligned} \quad (10.H.1)$$

where step (a) follows from the expression of $s_{\text{td}}(t)$ in (10.23). For $|\tau| > T$, the support set for $s_{\text{td}}^*(t - \tau)$ does not overlap with the interval $(-T/2, T/2)$, and, therefore, the integral on the RHS

above evaluates to zero. For $0 \leq \tau < T$, the integral on the RHS above evaluates to

$$\begin{aligned} \frac{1}{\sqrt{T}} \int_{-\frac{T}{2}}^{\frac{T}{2}} s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt &= \frac{1}{T} \int_{\tau - \frac{T}{2}}^{\frac{T}{2}} e^{-j2\pi\nu(t-\tau)} dt \\ &= \left(1 - \frac{|\tau|}{T}\right) e^{j\pi\nu\tau} \text{sinc}((T - |\tau|)\nu). \end{aligned} \quad (10.H.2)$$

Similarly, for $-T < \tau < 0$,

$$\begin{aligned} \frac{1}{\sqrt{T}} \int_{-\frac{T}{2}}^{\frac{T}{2}} s_{\text{td}}^*(t - \tau) e^{-j2\pi\nu(t-\tau)} dt &= \frac{1}{T} \int_{-\frac{T}{2}}^{\tau + \frac{T}{2}} e^{-j2\pi\nu(t-\tau)} dt \\ &= \left(1 - \frac{|\tau|}{T}\right) e^{j\pi\nu\tau} \text{sinc}((T - |\tau|)\nu). \end{aligned} \quad (10.H.3)$$

The fact that the ambiguity function is zero for $|\tau| > T$, along with (10.H.2) and (10.H.3), completes the proof.

Appendix 10.I Proof of Theorem 10.4

The DD representation of the Zak-OTFS pulsone corresponding to a quasi-periodic DD pulse at the origin is given by $x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0)$ in (10.33). From (10.14), it follows that the ambiguity function of this Zak-OTFS pulsone evaluated at (τ_0, ν_0) is given by

$$\begin{aligned} A_{s,s}^{\text{Zak-OTFS}}(\tau_0, \nu_0) &= \int_0^{\tau_p} \int_0^{\nu_p} x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0) \left(x_{\text{dd}}^{w_{\text{tx}}}(\tau - \tau_0, \nu - \nu_0; 0, 0) \right)^* e^{-j2\pi\nu_0(\tau - \tau_0)} d\tau d\nu \\ &\stackrel{(a)}{=} \sum_{n_1, n_2, m_1, m_2 \in \mathbb{Z}} \int_0^{\tau_p} \int_0^{\nu_p} \left[w_{\text{tx}}(\tau - n_1\tau_p, \nu - m_1\nu_p) e^{j2\pi\nu n_1\tau_p} w_{\text{tx}}^*(\tau - n_2\tau_p - \tau_0, \nu - m_2\nu_p - \nu_0) \right. \\ &\quad \left. e^{-j2\pi(\nu - \nu_0)n_2\tau_p} e^{-j2\pi\nu_0(\tau - \tau_0)} \right] d\tau d\nu \\ &\stackrel{(b)}{=} \sum_{n_1, n_2, m_1, m_2 \in \mathbb{Z}} \int_{-n_1\tau_p}^{-(n_1-1)\tau_p} \int_{-m_1\nu_p}^{-(m_1-1)\nu_p} \left[w_{\text{tx}}(\tau', \nu') e^{j2\pi\nu' n_1\tau_p} e^{-j2\pi(\nu' - \nu_0)n_2\tau_p} e^{-j2\pi\nu_0(\tau' + n_1\tau_p - \tau_0)} \right. \\ &\quad \left. w_{\text{tx}}^*(\tau' + (n_1 - n_2)\tau_p - \tau_0, \nu' + (m_1 - m_2)\nu_p - \nu_0) \right] d\tau' d\nu' \\ &\stackrel{(c)}{=} \sum_{n_1, n_2, m_1, m_2 \in \mathbb{Z}} \int_{-n_1\tau_p}^{-(n_1-1)\tau_p} \int_{-m_1\nu_p}^{-(m_1-1)\nu_p} \left[w_{\text{tx}}(\tau', \nu') e^{j2\pi\nu' n_1\tau_p} e^{-j2\pi(\nu' - \nu_0)n_2\tau_p} e^{-j2\pi\nu_0(\tau' + n_1\tau_p - \tau_0)} \right. \\ &\quad \left. w_{\text{tx}}^{\text{mf}}(\tau_0 + (n_2 - n_1)\tau_p - \tau', \nu_0 + (m_2 - m_1)\nu_p - \nu') \right. \\ &\quad \left. e^{-j2\pi(\tau_0 + (n_2 - n_1)\tau_p - \tau')(\nu_0 + (m_2 - m_1)\nu_p - \nu')} \right] d\tau' d\nu' \\ &\stackrel{(d)}{=} \sum_{n_1, m_1 \in \mathbb{Z}} \int_{-n_1\tau_p}^{-(n_1-1)\tau_p} \int_{-m_1\nu_p}^{-(m_1-1)\nu_p} \left(w_{\text{tx}}(\tau', \nu') \left[\sum_{n_2, m_2 \in \mathbb{Z}} e^{-j2\pi(\tau_0 - \tau') (m_2 - m_1)\nu_p} \right. \right. \end{aligned}$$

$$\left. w_{\text{tx}}^{\text{mf}}(\tau_0 + (n_2 - n_1)\tau_p - \tau', \nu_0 + (m_2 - m_1)\nu_p - \nu') \right] e^{j2\pi\nu'(\tau_0 - \tau')} d\tau' d\nu', \quad (10.I.1)$$

where step (a) follows from the expression of $X_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0)$ in (10.33). In step (b), for each (n_1, m_1) , the integration variables (τ, ν) are substituted with (τ', ν') , $\tau' = (\tau - n_1\tau_p)$ and $\nu' = (\nu - m_1\nu_p)$. In step (c), the fact $\tau_p\nu_p = 1$ and also the expression for $w_{\text{tx}}^{\text{mf}}$ in (10.36) are used. Step (d) follows by moving the summation over (n_2, m_2) inside the integral and by simplifying the exponential terms. In the inner summation in the last step above, the terms inside the summation depend on (n_2, m_2) only through the difference $(n_2 - n_1)$ and $(m_2 - m_1)$. Therefore, for a given (n_1, m_1) in the outer summation, substituting inner summation variables n_2 and m_2 with new variables $n = (n_1 - n_2)$ and $m = (m_1 - m_2)$, respectively, gives the following expression for the inner summation:

$$\sum_{n, m \in \mathbb{Z}} w_{\text{tx}}^{\text{mf}}(\tau_0 - n\tau_p - \tau', \nu_0 - m\nu_p - \nu') e^{j2\pi(\tau_0 - \tau')m\nu_p}. \quad (10.I.2)$$

The expression above is a function of $(\tau_0 - \tau')$ and $(\nu_0 - \nu')$. Let

$$w_{\text{tx,dd}}^{\text{mf}}(\tau, \nu) \triangleq \sum_{n, m \in \mathbb{Z}} w_{\text{tx}}^{\text{mf}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi\tau m\nu_p}. \quad (10.I.3)$$

Then, the expression in (10.I.2) is simply $w_{\text{tx,dd}}^{\text{mf}}(\tau_0 - \tau', \nu_0 - \nu')$. Furthermore, note that $w_{\text{tx,dd}}^{\text{mf}}(\tau, \nu)$ is nothing but the twisted convolution of a quasi-periodic Dirac-delta impulse at the origin with the matched filter $w_{\text{tx}}^{\text{mf}}(\tau, \nu)$ since

$$\begin{aligned} & \left[\sum_{(n, m) \in \mathbb{Z}} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) \right] *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu) \\ &= \sum_{(n, m) \in \mathbb{Z}} [\delta(\tau - n\tau_p) \delta(\nu - m\nu_p)] *_\sigma w_{\text{tx}}^{\text{mf}}(\tau, \nu) \\ &\stackrel{(a)}{=} \sum_{(n, m) \in \mathbb{Z}} \iint \delta(\tau' - n\tau_p) \delta(\nu' - m\nu_p) w_{\text{tx}}^{\text{mf}}(\tau - \tau', \nu - \nu') e^{j2\pi\nu'(\tau - \tau')} d\tau' d\nu' \\ &\stackrel{(b)}{=} \sum_{(n, m) \in \mathbb{Z}} w_{\text{tx}}^{\text{mf}}(\tau - n\tau_p, \nu - m\nu_p) e^{j2\pi\tau m\nu_p} \\ &\stackrel{(c)}{=} w_{\text{tx,dd}}^{\text{mf}}(\tau, \nu), \end{aligned} \quad (10.I.4)$$

where step (a) follows from the definition of twisted convolution operation and step (b) uses the fact that $\tau_p\nu_p = 1$. Step (c) follows from (10.I.3).

Substituting the inner summation in (10.I.1) with $w_{\text{tx,dd}}^{\text{mf}}(\tau_0 - \tau', \nu_0 - \nu')$, it follows that

$$\begin{aligned} A_{s,s}^{\text{Zak-OTFS}}(\tau_0, \nu_0) &= \sum_{n_1, m_1 \in \mathbb{Z}} \int_{-n_1\tau_p}^{-(n_1-1)\tau_p} \int_{-m_1\nu_p}^{-(m_1-1)\nu_p} w_{\text{tx}}(\tau', \nu') w_{\text{tx,dd}}^{\text{mf}}(\tau_0 - \tau', \nu_0 - \nu') e^{j2\pi\nu'(\tau_0 - \tau')} d\tau' d\nu' \\ &= \iint w_{\text{tx}}(\tau', \nu') w_{\text{tx,dd}}^{\text{mf}}(\tau_0 - \tau', \nu_0 - \nu') e^{j2\pi\nu'(\tau_0 - \tau')} d\tau' d\nu' \end{aligned} \quad (10.I.5)$$

since the integrand does not depend on (n_1, m_1) . Note that the integral in the last step is nothing but twisted convolution between $w_{\text{tx}}(\tau, \nu)$ and $w_{\text{tx,dd}}^{\text{mf}}(\tau, \nu)$ evaluated at (τ_0, ν_0) . Hence,

$$\begin{aligned} A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) &= w_{\text{tx}}(\tau, \nu) *_{\sigma} w_{\text{tx,dd}}^{\text{mf}}(\tau, \nu) \\ &= w_{\text{tx}}(\tau, \nu) *_{\sigma} \left(\sum_{n,m \in \mathbb{Z}} \delta(\tau - n\tau_p) \delta(\nu - m\nu_p) \right) *_{\sigma} w_{\text{tx}}^{\text{mf}}(\tau, \nu), \end{aligned} \quad (10.I.6)$$

where the last step follows from (10.I.4).

Appendix 10.J Proof of (10.43) and (10.44)

Starting from (2.41), with $\tau_0 = \nu_0 = 0$ and $w_{\text{tx}}(\tau, \nu) = W_1(\tau) W_2(\nu)$ given by (10.42), the corresponding Zak-OTFS pulsone is

$$\begin{aligned} s_{\text{id}}(t) &= \mathcal{Z}_t^{-1} \left(x_{\text{dd}}^{w_{\text{tx}}}(\tau, \nu; 0, 0) \right) = \sqrt{\tau_p} \sum_{n=-\infty}^{\infty} g(n\tau_p) W_1(t - n\tau_p), \\ g(t) &\triangleq \mathcal{F}^{-1}(W_2(f)) = \int W_2(f) e^{j2\pi ft} df, \end{aligned} \quad (10.J.1)$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$, respectively, denote the Fourier transform and its inverse. The inverse Fourier transform of $W_2(\cdot)$ is

$$\begin{aligned} g(t) &= \int W_2(f) e^{j2\pi ft} df = \int \sqrt{T} \text{sinc}(fT) e^{j2\pi ft} df \\ &= \begin{cases} \frac{1}{\sqrt{T}}, & -\frac{T}{2} \leq t < \frac{T}{2}, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (10.J.2)$$

Therefore, for even N and $n \in \mathbb{Z}$,

$$\sqrt{\tau_p} g(n\tau_p) = \begin{cases} \frac{1}{\sqrt{N}}, & -\frac{N}{2} \leq n \leq \left(\frac{N}{2} - 1\right), \\ 0, & \text{otherwise.} \end{cases} \quad (10.J.3)$$

Similarly, for odd N ,

$$\sqrt{\tau_p} g(n\tau_p) = \begin{cases} \frac{1}{\sqrt{N}}, & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (10.J.4)$$

For even N , substituting (10.J.3) in (10.J.1) gives (10.43). Similarly, for odd N , substituting (10.J.4) in (10.J.1) gives (10.44).

Appendix 10.K Proof of Theorem 10.5

From (10.14) and the expression for the Zak-OTFS pulsone in (10.43), it follows that, for even N ,

$$\begin{aligned} A_{s,s}^{\text{Zak-OTFS}}(\tau, \nu) &= \int s_{\text{id}}(t) s_{\text{id}}^*(t - \tau) e^{-j2\pi \nu(t - \tau)} dt \\ &= \frac{1}{N} \sum_{n_1 = -\frac{N}{2}}^{\frac{N}{2}-1} \sum_{n_2 = -\frac{N}{2}}^{\frac{N}{2}-1} \int B \text{sinc}(B(t - n_1\tau_p)) \text{sinc}(B(t - n_2\tau_p - \tau)) e^{-j2\pi \nu(t - \tau)} dt. \end{aligned} \quad (10.K.1)$$

Let

$$x(t) \triangleq \sqrt{B} \operatorname{sinc}(B(t - n_1 \tau_p)) , \quad y(t) \triangleq \sqrt{B} \operatorname{sinc}(B(t - n_2 \tau_p - \tau) e^{j2\pi\nu(t - n_2 \tau_p - \tau)}) . \quad (10.K.2)$$

Then, the integral on the RHS of (10.K.1) is simply $e^{-j2\pi\nu n_2 \tau_p}$ times the inner product $\int x(t)y^*(t) dt$ between $x(t)$ and $y(t)$. From Parseval's equivalence between the inner product of two signals in TD and FD, it follows that

$$\int x(t)y^*(t) dt = \int X(f) Y^*(f) df, \quad (10.K.3)$$

where the Fourier transforms of $x(t)$ and $y(t)$ are

$$X(f) = \begin{cases} \frac{1}{\sqrt{B}} e^{-j2\pi f n_1 \tau_p}, & -\frac{B}{2} \leq f < \frac{B}{2}, \\ 0, & \text{otherwise,} \end{cases} \quad (10.K.4)$$

and

$$Y(f) = \begin{cases} \frac{1}{\sqrt{B}} e^{-j2\pi f(n_2 \tau_p + \tau)}, & -\frac{B}{2} + \nu \leq f < \frac{B}{2} + \nu, \\ 0, & \text{otherwise,} \end{cases} \quad (10.K.5)$$

respectively. Clearly, for $|\nu| > B$, the support sets of $X(f)$ and $Y(f)$ do not overlap, and, therefore, the inner product is clearly zero. For $0 \leq \nu < B$, the inner product when evaluated in FD gives

$$\begin{aligned} & \frac{1}{B} \int_{-\frac{B}{2} + \nu}^{\frac{B}{2}} e^{-j2\pi f n_1 \tau_p} e^{j2\pi f(n_2 \tau_p + \tau)} df \\ &= \left(1 - \frac{|\nu|}{B}\right) \left[e^{j\pi\nu(\tau - (n_1 - n_2)\tau_p)} \operatorname{sinc}((B - |\nu|)(\tau + (n_1 - n_2)\tau_p)) \right]. \end{aligned} \quad (10.K.6)$$

For $-B < \nu < 0$, the inner product when evaluated in FD gives the same expression, i.e.,

$$\begin{aligned} & \frac{1}{B} \int_{-\frac{B}{2}}^{\frac{B}{2} + \nu} e^{-j2\pi f n_1 \tau_p} e^{j2\pi f(n_2 \tau_p + \tau)} df \\ &= \left(1 - \frac{|\nu|}{B}\right) \left[e^{j\pi\nu(\tau - (n_1 - n_2)\tau_p)} \operatorname{sinc}((B - |\nu|)(\tau + (n_1 - n_2)\tau_p)) \right]. \end{aligned} \quad (10.K.7)$$

Using the inner product expressions in (10.K.6) and (10.K.7) along with the fact that the integral on the RHS of (10.K.1) is $e^{-j2\pi\nu n_2 \tau_p}$ times this inner product gives the expression for the ambiguity function in (10.45).

References

- 1 W. R. Young, "Advanced mobile phone system: introduction, background, objectives," *Bell Syst. Tech. J.*, vol. 58, no. 1, pp. 1–14, Jan. 1979.
- 2 H. V. Macdonald, "The cellular concept," *Bell Syst. Tech. J.*, vol. 58, no. 1, pp. 15–41, Jan. 1979.
- 3 T. S. Rappaport, *Wireless Communications - Principles and Practice*, Prentice Hall, 1996.
- 4 M. Mouly and M.-B. Pautet, *The GSM System for Mobile Communications*, Telecom Pub., June 1992.
- 5 A. J. Viterbi, *CDMA - Principles of Spread Spectrum Communication*, Addison-Wesley, 1995.
- 6 D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*, Cambridge University Press, 2005.
- 7 A. Goldsmith, *Wireless Communications*, Cambridge University Press, 2005.
- 8 U. Madhow, *Fundamentals of Digital Communication*, Cambridge University Press, 2008.
- 9 G. J. Foschini and M. J. Gans, "On limits of wireless communications in a fading environment when using multiple antennas," *Wireless Pers. Commun.*, vol. 6, no. 3, pp. 311–335, 1998.
- 10 E. Telatar, "Capacity of multi-antenna Gaussian channels," *Eur. Trans. Telecommun.*, vol. 10, no. 6, pp. 585–595, Nov. 1999.
- 11 V. Tarokh, N. Seshadri, and A. Calderbank, "Space-time codes for high data rate wireless communication: performance criterion and code construction," *IEEE Trans. Inf. Theory*, vol. 44, no. 2, pp. 744–765, Feb. 1998.
- 12 A. Paulraj, R. Nabar, and D. Gore, *Introduction to Space-Time Wireless Communications*, Cambridge University Press, May 2003.
- 13 L. Cimini, "Analysis and simulation of a digital mobile channel using orthogonal frequency division multiplexing," *IEEE Trans. Commun.*, vol. 33, no. 7, pp. 665–675, Jul. 1985.
- 14 J. Li, X. Wu, and R. Laroia, *OFDMA Mobile Broadband Communications: A Systems Approach*, Cambridge University Press, 2013.
- 15 C. Berrou, A. Glavieux, and P. Thitimajshima, "Near Shannon limit error correcting coding and decoding: turbo-codes," *Proc. IEEE ICC'1993*, vol. 2, pp. 1064–1070, May 1993.
- 16 A. Chockalingam and B. S. Rajan, *Large MIMO Systems*, Cambridge University Press, 2014.
- 17 T. L. Marzetta, E. G. Larsson, H. Yang, and H. Q. Ngo, *Fundamentals of Massive MIMO*, Cambridge University Press, 2016.
- 18 R. Gallager, "Low-density parity-check codes," *IRE Trans. Inf. Theory*, vol. 8, no. 1, pp. 21–28, 1962.
- 19 E. Arıkan, "Channel polarization: a method for constructing capacity achieving codes for symmetric binary-input memoryless channels," *IEEE Trans. Inf. Theory*, vol. 55, no. 7, pp. 3051–3073, Jul. 2009.

- 20 W. C. Jakes, Ed. *Microwave Mobile Communications*, Reprint, IEEE Press, 1994.
- 21 H. Sari, G. Karam, and I. Jeanclaude, "Transmission techniques for digital terrestrial TV broadcasting," *IEEE Commun. Mag.*, vol. 33, no. 2, pp. 100–109, Feb. 1995.
- 22 J. A. C. Bingham, "Multicarrier modulation for data transmission: an idea whose time has come," *IEEE Commun. Mag.*, vol. 28, no. 5, pp. 5–14, May 1990.
- 23 Z. Wang and G. B. Giannakis, "Wireless multicarrier communications," *IEEE Signal Process. Mag.*, vol. 17, no. 3, pp. 29–48, May 2000.
- 24 N. Benvenuto, R. Dinis, D. Falconer, and S. Tomasin, "Single carrier modulation with nonlinear frequency domain equalization: an idea whose time has come - again," *Proc. of the IEEE*, vol. 98, no. 1, pp. 69–96, Jan. 2010.
- 25 P. Robertson and S. Kaiser, "Analysis of the loss of orthogonality through Doppler spread in OFDM systems," *Proc. IEEE GLOBECOM'99*, vol. 1b, pp. 701–706, Dec. 1999.
- 26 B. Stantchev and G. Fettweis, "Time-variant distortions in OFDM," *IEEE Commun. Lett.*, vol. 4, no. 10, pp. 312–314, Oct. 2000.
- 27 F. Hlawatsch and G. Matz, *Wireless Communications Over Rapidly Time-varying Channels*, Academic Press, 2011.
- 28 R. Hadani, S. Rakib, M. Tsatsanis, A. Monk, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," *Proc. IEEE WCNC'2017*, pp. 1–6, Mar. 2017.
- 29 R. Hadani and A. Monk, "OTFS: a new generation of modulation addressing the challenges of 5G," arXiv:1802.02623v1 [cs.IT], 7 Feb. 2018.
- 30 R. Hadani, S. Rakib, S. Kons, M. Tsatsanis, A. Monk, C. Ibars, J. Delfeld, Y. Hebron, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," arXiv:1808.00519v1 [cs.IT], 1 Aug. 2018.
- 31 R. Hadani, S. Rakib, A. F. Molisch, C. Ibars, A. Monk, M. Tsatsanis, J. Delfeld, A. Goldsmith, and R. Calderbank, "Orthogonal time frequency space (OTFS) modulation for millimeter-wave communications systems," *Proc. IEEE MTT-S Intl. Microwave Symp.*, pp. 681–683, Jun. 2017.
- 32 A. Monk, R. Hadani, M. Tsatsanis, and S. Rakib, "OTFS - orthogonal time frequency space: a novel modulation meeting 5G high mobility and massive MIMO challenges," arXiv:1608.02993 [cs.IT], 9 Aug. 2016.
- 33 S. K. Mohammed, R. Hadani, A. Chockalingam, and R. Calderbank, "OTFS—a mathematical foundation for communication and radar sensing in the delay-Doppler domain," *IEEE BITS Inf. Theory Mag.*, vol. 2, no. 2, pp. 36–55, 1 Nov. 2022.
- 34 S. K. Mohammed, R. Hadani, A. Chockalingam, and R. Calderbank, "OTFS—predictability in the delay-Doppler domain and its value to communication and radar sensing," *IEEE BITS Inf. Theory Mag.*, vol. 3, no. 2, pp. 7–31, Jun. 2023.
- 35 IEEE ComSoc®IEEE Communications Society, "Best readings in orthogonal time frequency space (OTFS) and delay Doppler signal processing," Jun. 2022. Available: <https://www.comsoc.org/publications/best-readings/orthogonal-timefrequency-space-otfs-and-delay-doppler-signal-processing>.
- 36 K. R. Murali and A. Chockalingam, "On OTFS modulation for high-Doppler fading channels," *Proc. ITA'2018*, pp. 1–10, Feb. 2018.
- 37 F. Wiffen, L. Sayer, M. Z. Bocus, A. Doufexi, and A. Nix, "Comparison of OTFS and OFDM in ray launched sub-6 GHz and mmWave line-of-sight mobility channels," *Proc. IEEE PIMRC'2018*, pp. 73–79, Sep. 2018.
- 38 P. Raviteja, K. T. Phan, Y. Hong, and E. Viterbo, "Interference cancellation and iterative detection for orthogonal time frequency space modulation," *IEEE Trans. Wireless Commun.*, vol. 17, no. 10, pp. 6501–6515, Oct. 2018.

- 39 M. K. Ramachandran and A. Chockalingam, "MIMO-OTFS in high-Doppler fading channels: signal detection and channel estimation," *Proc. IEEE GLOBECOM'2018*, pp. 206–212, Dec. 2018.
- 40 L. Li, Y. Liang, P. Fan, and Y. Guan, "Low complexity detection algorithms for OTFS under rapidly time-varying channel," *Proc. IEEE VTC'2019-Spring*, pp. 1–5, Apr.-May 2019.
- 41 G. D. Surabhi, R. M. Augustine, and A. Chockalingam, "On the diversity of uncoded OTFS modulation in doubly-dispersive channels," *IEEE Trans. Wireless Commun.*, vol. 18, no. 6, pp. 3049–3063, Jun. 2019.
- 42 T. Thaj and E. Viterbo, "Low complexity iterative rake detector for orthogonal time frequency space modulation," *Proc. IEEE WCNC'2020*, pp. 1–6, May 2020.
- 43 W. Yuan, Z. Wei, J. Yuan, and D. W. K. Ng, "A simple variational Bayes detector for orthogonal time frequency space (OTFS) modulation," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 7976–7980, Jul. 2020.
- 44 M. K. Ramachandran, G. D. Surabhi, and A. Chockalingam, "OTFS: a new modulation scheme for high-mobility use cases," *J. Indian Inst. Sci.*, vol. 100, pp. 315–336, 2020.
- 45 A. Gunturu, A. R. Godala, A. K. Sahoo, and A. K. R. Chavva, "Performance analysis of OTFS waveform for 5G NR mmWave communication system," *Proc. IEEE WCNC'2021*, pp. 1–6, Mar.-Apr. 2021.
- 46 H. Zhang and T. Zhang, "A low-complexity message passing detector for OTFS modulation with probability clipping," *IEEE Wireless Commun. Lett.*, vol. 10, no. 6, pp. 1271–1275, Jun. 2021.
- 47 L. Xiang, Y. Liu, L.-L. Yang, and L. Hanzo, "Gaussian approximate message passing detection of orthogonal time frequency space modulation," *IEEE Trans. Veh. Technol.*, vol. 70, no. 10, pp. 10999–11004, Oct. 2021.
- 48 D. Ma, N. Shlezinger, T. Huang, Y. Liu, and Y. C. Eldar, "Joint radar-communications strategies for autonomous vehicles: combining two key automotive technologies," *IEEE Signal Process. Mag.*, vol. 37, no. 4, pp. 85–97, Jul. 2020.
- 49 Y. Zeng, Y. Ma, and S. Sun, "Joint radar-communication with cyclic prefixed single carrier waveforms," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4069–4079, Apr. 2020.
- 50 J. Li and P. Stoica, *MIMO Radar Signal Processing*, IEEE-Wiley Press, 2009.
- 51 Y. Yu, A. P. Petropulu, and H. V. Poor, "MIMO radar using compressive sampling," *IEEE J. Sel. Top. Signal Process.*, vol. 4, no. 1, pp. 146–163, Feb. 2010.
- 52 P. M. Woodward, *Probability and Information Theory, with Applications to Radar*, Pergamon Press, 1964.
- 53 L. Zheng, M. Lops, Y. C. Eldar, and X. Wang, "Radar and communication coexistence: an overview: a review of recent methods," *IEEE Signal Process. Mag.*, vol. 36, no. 5, pp. 85–99, Sep. 2019.
- 54 F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint radar and communication design: applications, state-of-the-art, and the road ahead," *IEEE Trans. Commun.*, vol. 68, no. 6, pp. 3834–3862, Jun. 2020.
- 55 P. A. Bello, "Characterization of randomly time-variant linear channels," *IEEE Trans. Commun. Syst.*, vol. 11, pp. 360–393, 1963.
- 56 A. J. E. M. Janssen, "The Zak transform: a signal transform for sampled time-continuous signals," *Philips J. Res.*, vol. 43, pp. 23–69, 1988.
- 57 J. Zak, "Finite translations in solid state physics," *Phys. Rev. Lett.*, 19, pp. 1385–1387, 1967.
- 58 D. Slepian and H. O. Pollak, "Prolate spheroidal wave functions, Fourier analysis and uncertainty - I," *The Bell Syst. Tech. J.*, vol. 40, no. 1, pp. 43–63, Jan. 1961.

- 59 H. J. Landau and H. O. Pollak, "Prolate spheroidal wave functions, Fourier analysis and uncertainty - III: the dimension of the space of essentially time- and band-limited signals," *The Bell Syst. Tech. J.*, vol. 41, no. 4, pp. 1295–1336, Jul. 1962.
- 60 S. K. Mohammed, "Derivation of OTFS modulation from first principles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 7619–7636, Mar. 2021.
- 61 S. K. Mohammed, "Time-domain to delay-Doppler domain conversion of OTFS signals in very high mobility scenarios," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 617–6183, Jun. 2021.
- 62 V. S. Bhat, G. Harshavardhan, and A. Chockalingam, "Input-output relation and performance of RIS-aided OTFS with fractional delay-Doppler," *IEEE Commun. Lett.*, vol. 27, no. 1, pp. 337–341, Jan. 2023.
- 63 A. S. Bondre and C. D. Richmond, "On the Zak transform-based interpretation of OTFS modulation," *Proc. Asilomar Conf. on Signals, Systems, and Computers*, Oct.-Nov. 2022.
- 64 F. Lampel, A. Avarado and F. M. J. Willems, "On OTFS using the discrete Zak transform," *Proc. IEEE ICC'2022 Workshops*, pp. 729–734, May 2022.
- 65 V. Yogesh, V. S. Bhat, S. R. Mattu, and A. Chockalingam, "On the bit error performance of OTFS modulation using discrete Zak transform," *Proc. IEEE ICC'2023*, May-Jun. 2023.
- 66 P. Bello, "Measurement of random time-variant linear channels," *IEEE Trans. Inf. Theory*, vol. 15, no. 4, pp. 469–475, Jul. 1969.
- 67 G. E. Pfander and D. F. Walnut, "Measurement of time-variant linear channels," *IEEE Trans. Inf. Theory*, vol. 52, no. 11, pp. 4808–4820, Nov. 2006.
- 68 R. Heckel and H. Bolcskei, "Identification of sparse linear operators," *IEEE Trans. Inf. Theory*, vol. 59, no. 12, pp. 7985–8000, Dec. 2013.
- 69 S. Haykin, *Digital Communication Systems*, John Wiley & Sons, 2014.
- 70 ITU-R M.1225, "Guidelines for the Evaluation of Radio Transmission Technologies for IMT-2000," International Telecommunication Union Radio communication, 1997.
- 71 P. J. Davis, *Circulant Matrices*, Providence, RI, USA: American Mathematical Society, 2012.
- 72 M. O. Damen, A. Tewfik, and J.-C. Belfiore, "A construction of a space-time code based on number theory," *IEEE Trans. Inf. Theory*, vol. 48, no. 3, pp. 753–760, Mar. 2002.
- 73 H. Jafarkhani, *Space-Time Coding Theory and Practice*, Cambridge University Press, 2005.
- 74 S. M. Alamouti, "A simple transmit diversity technique for wireless communications," *IEEE J. Sel. Areas Commun.*, vol. 16, no. 8, pp. 1451–1458, Aug. 1998.
- 75 R. M. Augustine, G. D. Surabhi, and A. Chockalingam, "Space-time coded OTFS modulation in high-Doppler channels," *Proc. IEEE VTC'2019-Spring*, pp. 1–6, Apr.-May 2019.
- 76 V. S. Bhat, G. D. Surabhi, and A. Chockalingam, "Performance analysis of OTFS modulation with receive antenna selection," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3382–3395, Apr. 2021.
- 77 I. Bahceci, T. M. Duman, and Y. Altunbasak, "Antenna selection for multiple-antenna transmission systems: performance analysis and code construction," *IEEE Trans. Inf. Theory*, vol. 49, no. 10, pp. 2669–2681, Oct. 2003.
- 78 I. Bahceci, T. M. Duman, and Y. Altunbasak, "Correction to "Antenna selection for multiple-antenna transmission systems: performance analysis and code construction"," *IEEE Trans. Inf. Theory*, vol. 54, no. 12, pp. 5788–5789, Dec. 2008.
- 79 G. D. Surabhi and A. Chockalingam, "Low-complexity linear equalization for OTFS modulation," *IEEE Commun. Lett.*, vol. 24, no. 2, pp. 330–334, Feb. 2020.
- 80 G. D. Surabhi and A. Chockalingam, "Low-complexity linear equalization for 2×2 MIMO-OTFS signals," *Proc. IEEE SPAWC'2020*, pp. 1–5, May 2020.

- 81 P. Singh, A. Gupta, H. B. Mishra, and R. Budhiraja, "Low-complexity ZF/MMSE MIMO-OTFS receivers for high-speed vehicular communication," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 209–227, 2022.
- 82 S. Tiwari, S. S. Das, and V. Rangamgari, "Low complexity LMMSE receiver for OTFS," *IEEE Commun. Lett.*, vol. 23, no. 12, pp. 2205–2209, Dec. 2019.
- 83 P. Raviteja, Y. Hong, E. Viterbo, and E. Biglieri, "Practical pulse-shaping waveforms for reduced-cyclic-prefix OTFS," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 957–961, Jan. 2019.
- 84 P. Som, T. Datta, N. Srinidhi, A. Chockalingam, and B. S. Rajan, "Low-complexity detection in large-dimension MIMO-ISI channels using graphical models," *IEEE J. Sel. Top. Signal Process.*, vol. 5, no. 8, pp. 1497–1511, Dec. 2011.
- 85 M. Pretti, "A message-passing algorithm with damping," *J. Stat. Mech.: Theory Exp.*, vol. 2005, p. P11008, Nov. 2005.
- 86 S. Kullback and R. A. Leibler, "On information and sufficiency," *Ann. Math. Stat.*, vol. 22, no. 1, pp. 79–86, 1951.
- 87 R. Giordano, T. Broderick, and M. I. Jordan, "Covariances, robustness and variational Bayes," *J. Mach. Learn. Res.*, vol. 19, no. 1, pp. 1981–2029, 2018.
- 88 R. J. Giordano, T. Broderick, and M. I. Jordan, "Linear response methods for accurate covariance estimates from mean field variational Bayes," *Proc. Advances in Neural Information Processing Systems 28 (NIPS 2015)*, pp. 1441–1449, 2015.
- 89 F. R. Kschischang, B. J. Frey, and H. -A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 498–519, Feb. 2001.
- 90 S. Li, W. Yuan, Z. Wei, J. Yuan, B. Bai, D. W. Kwan, and Y. Xie, "Hybrid MAP and PIC detection for OTFS modulation," *IEEE Trans. Veh. Technol.*, vol. 70, no. 7, pp. 7193–7198, Jul. 2021.
- 91 B. Farhang-Boroujeny, H. Zhu, and Z. Shi, "Markov chain Monte Carlo algorithms for CDMA and MIMO communication systems," *IEEE Trans. Signal Process.*, vol. 54, no. 5, pp. 1896–1909, May 2006.
- 92 R. Peng, R.-R. Chen, and B. Farhang-Boroujeny, "Markov chain Monte Carlo detectors for channels with intersymbol interference," *IEEE Trans. Signal Process.*, vol. 58, no. 4, pp. 2206–2217, Apr. 2010.
- 93 M. Hansen, B. Hassibi, A. G. Dimakis, and W. Xu, "Near-optimal detection in MIMO systems using Gibbs sampling," *Proc. IEEE GLOBECOM2009*, pp. 1–6, Dec. 2009.
- 94 T. Datta, N. Ashok Kumar, A. Chockalingam, and B. Sundar Rajan, "A novel Monte-Carlo-sampling-based receiver for large-scale uplink multiuser MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 62, no. 7, pp. 3019–3038, Sep. 2013.
- 95 S. Li, W. Yuan, Z. Wei, and J. Yuan, "Cross domain iterative detection for orthogonal time frequency space modulation," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2227–2242, Apr. 2022.
- 96 Q. Guo and D. D. Huang, "A concise representation for the soft-in soft-out LMMSE detector," *IEEE Commun. Lett.*, vol. 15, no. 5, pp. 566–568, May 2011.
- 97 J. Ma, X. Yuan, and L. Ping, "Turbo compressed sensing with partial DFT sensing matrix," *IEEE Signal Process. Lett.*, vol. 22, no. 2, pp. 158–161, Feb. 2015.
- 98 A. Fish, S. Gurevich, R. Hadani, A. M. Sayeed, and O. Schwartz, "Delay-Doppler channel estimation in almost linear complexity," *IEEE Trans. Inf. Theory*, vol. 59, no. 11, pp. 7632–7644, Nov. 2013.

- 99 I. A. Khan and S. K. Mohammed, "A low complexity OTFS channel estimation method for fractional delay-Doppler scenarios," *IEEE Wireless Commun. Lett.*, vol. 12, no. 9, pp. 1484–1488, Sep. 2023.
- 100 I. A. Khan and S. K. Mohammed, "Low complexity channel estimation for OTFS modulation with fractional delay and Doppler," arXiv:2111.06009 [cs.IT], Nov. 2021.
- 101 S. P. Muppaneni, S. R. Mattu, and A. Chockalingam, "Channel and radar parameter estimation with fractional delay-Doppler using OTFS," *IEEE Commun. Lett.*, vol. 27, no. 5, pp. 1392–1396, May 2023.
- 102 P. Raviteja, K. T. Phan, Y. Hong, and E. Viterbo, "Embedded pilot-aided channel estimation for OTFS in delay-Doppler channels," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4906–4917, May 2019.
- 103 L. Zhao, W.-J. Gao, and W. Guo, "Sparse Bayesian learning of delay-Doppler channel for OTFS system," *IEEE Commun. Lett.*, vol. 24, no. 12, pp. 2766–2769, Dec. 2020.
- 104 P. Raviteja, K. T. Phan, Q. Jin, Y. Hong, and E. Viterbo, "Low-complexity iterative detection for orthogonal time frequency space modulation," *Proc. IEEE WCNC'2018*, pp. 1–6, Apr. 2018.
- 105 S. D. Babacan, R. Molina, and A. K. Katsaggelos, "Bayesian compressive sensing using Laplace priors," *IEEE Trans. Image Process.*, vol. 19, no. 1, pp. 53–63, Jan. 2010.
- 106 M. E. Tipping, "Sparse Bayesian learning and the relevance vector machine," *J. Mach. Learn. Res.*, vol. 1, pp. 211–244, Sep. 2001.
- 107 Z. Yang, L. Xie, and C. Zhang, "Off-grid direction of arrival estimation using sparse Bayesian inference," *IEEE Trans. Signal Process.*, vol. 61, no. 1, pp. 38–43, Jan. 2013.
- 108 K. You, W. Guo, Y. Liu, W. Wang, and Z. Sun, "Grid evolution: joint dictionary learning and sparse Bayesian recovery for multiple off-grid targets localization," *IEEE Commun. Lett.*, vol. 22, no. 10, pp. 2068–2071, Oct. 2018.
- 109 J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.
- 110 LTE, "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) Radio Transmission and Reception," May 2019.
- 111 L. Zhao, J. Yang, Y. Liu, and W. Guo, "Block sparse Bayesian learning-based channel estimation for MIMO-OTFS systems," *IEEE Commun. Lett.*, vol. 26, no. 4, pp. 892–896, Apr. 2022.
- 112 H. B. Mishra, P. Singh, A. K. Prasad, and R. Budhiraja, "OTFS channel estimation and data detection designs with superimposed pilots," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2258–2274, Apr. 2022.
- 113 W. Yuan, S. Li, Z. Wei, J. Yuan, and D. W. K. Ng, "Data-aided channel estimation for OTFS systems with a superimposed pilot and data transmission scheme," *IEEE Wireless Commun. Lett.*, vol. 10, no. 9, pp. 1954–1958, Sep. 2021.
- 114 S. R. Mattu and A. Chockalingam, "Fractional delay-Doppler channel estimation in OTFS with sparse superimposed pilots using RNNs," *Proc. IEEE VTC'2023-Spring*, pp. 1–6, Jun. 2023.
- 115 F. Jesbin, S. Rao Mattu, and A. Chockalingam, "Sparse superimposed pilot based channel estimation in OTFS systems," *Proc. IEEE WCNC'2023*, pp. 1–6, Mar. 2023.
- 116 J. K. Tugnait and W. Luo, "On channel estimation using superimposed training and first-order statistics," *IEEE Commun. Lett.*, vol. 7, no. 9, pp. 413–415, Sep. 2003.
- 117 S. He, J. K. Tugnait, and X. Meng, "On superimposed training for MIMO channel estimation and symbol detection," *IEEE Trans. Signal Process.*, vol. 55, no. 6, pp. 3007–3021, Jun. 2007.

- 118 S. He and J. K. Tugnait, "On doubly selective channel estimation using superimposed training and discrete prolate spheroidal sequences," *IEEE Trans. Signal Process.*, vol. 56, no. 7, pp. 3214–3228, Jul. 2008.
- 119 S. M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*, Upper Saddle River, NJ, USA: Prentice-Hall, 1993.
- 120 S. Rakib and R. Hadani, "Multiple access in wireless telecommunications system for high-mobility applications," U.S. Patent 9 722 741 B1, Aug. 2017.
- 121 V. Khammammetti and S. K. Mohammed, "OTFS-based multiple-access in high Doppler and delay spread wireless channels," *IEEE Wireless Commun. Lett.*, vol. 8, no. 2, pp. 528–531, Apr. 2019.
- 122 V. Khammammetti and S. K. Mohammed, "Spectral efficiency of OTFS based orthogonal multiple access with rectangular pulses," *IEEE Trans. Veh. Technol.*, vol. 71, no. 12, pp. 12989–13006, Dec. 2022.
- 123 T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd Edition, John Wiley and Sons, 2006.
- 124 R. M. Augustine and A. Chockalingam, "Interleaved time-frequency multiple access using OTFS modulation," *Proc. IEEE VTC'2019-Fall*, pp. 1–5, Sep. 2019.
- 125 LTE, "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) Radio Transmission and Reception (3GPP TS 36.104 v8.6.0)," ETSI TS, vol. 136, no. 104, Jul. 2009.
- 126 O. E. Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath, "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, pp. 1499–1513, Mar. 2014.
- 127 F. Long, K. Niu, C. Dong, and J. Lin, "Low complexity iterative LMMSE-PIC equalizer for OTFS," *Proc. IEEE ICC'2019*, pp. 1–6, Jun. 2019.
- 128 J. Feng, H. Q. Ngo, M. F. Flanagan, and M. Matthaiou, "Performance analysis of OTFS-based uplink massive MIMO with ZF receivers," *Proc. IEEE ICC'2021 Workshops*, pp. 1–7, Jun. 2021.
- 129 A. M. Sayeed, "Deconstructing multiantenna fading channels," *IEEE Trans. Signal Process.*, vol. 50, no. 10, pp. 2563–2579, Oct. 2002.
- 130 M. Li, S. Zhang, P. Fan, and O. A. Dobre, "Multiple access for massive MIMO-OTFS networks over angle-delay-Doppler Domain," *Proc. IEEE GLOBECOM'2020*, pp. 1–6, Dec. 2020.
- 131 M. Li, S. Zhang, F. Gao, P. Fan and, O. A. Dobre, "A new path division multiple access for the massive MIMO-OTFS networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 4, pp. 903–918, Apr. 2021.
- 132 W. Shao, S. Zhang, C. Zhong, X. Lei, and P. Fan, "Angle-delay-Doppler domain NOMA over massive MIMO-OTFS networks," *Proc. IEEE/CIC ICC'2020*, pp. 74–79, Aug. 2020.
- 133 C. B. Peel, B. M. Hochwald, and A. L. Swindlehurst, "A vector-perturbation technique for near-capacity multiantenna multiuser communication-part I: channel inversion and regularization," *IEEE Trans. Commun.*, vol. 53, no. 1, pp. 195–202, Jan. 2005.
- 134 H. Sampath, P. Stoica, and A. Paulraj, "Generalized linear precoder and decoder design for MIMO channels using the weighted MMSE criterion," *IEEE Trans. Commun.*, vol. 49, no. 12, pp. 2198–2206, Dec. 2001.
- 135 F. Rusek, D. Persson, B. K. Lau, E. G. Larsson, T. L. Marzetta, O. Edfors, and F. Tufvesson, "Scaling up MIMO: opportunities and challenges with very large arrays," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 40–60, Jan. 2013.
- 136 H. Yang and T. L. Marzetta, "Performance of conjugate and zero-forcing beamforming in large-scale antenna systems," *IEEE J. Sel. Areas Commun.*, vol. 31, no. 2, pp. 172–179, Feb. 2013.

- 137 B. M. Hochwald, C. B. Peel, and A. L. Swindlehurst, "A vector-perturbation technique for near-capacity multiantenna multiuser communication-part II: perturbation," *IEEE Trans. Commun.*, vol. 53, no. 3, pp. 537–544, Mar. 2005.
- 138 M. Tomlinson, "New automatic equaliser employing modulo arithmetic," *Electron. Lett.*, vol. 7, pp. 138–139, Mar. 1971.
- 139 H. Harashima and H. Miyakawa, "Matched-transmission technique for channels with inter-symbol interference," *IEEE Trans. Commun.*, vol. 20, no. 4, pp. 774–780, Aug. 1972.
- 140 E. Agrell, T. Eriksson, A. Vardy, and K. Zeger, "Closest point search in lattices," *IEEE Trans. Inf. Theory*, vol. 48, no. 8, pp. 2201–2214, Aug. 2002.
- 141 O. Damen, A. Chkeif, and J.-C. Belfiore, "Lattice code decoder for space-time codes," *IEEE Commun. Lett.*, vol. 4, no. 5, pp. 161–163, May 2000.
- 142 C. Windpassinger, R. F. H. Fischer, and J. B. Huber, "Lattice-reduction-aided broadcast precoding," *IEEE Trans. Commun.*, vol. 52, no. 12, pp. 2057–2060, Dec. 2004.
- 143 B. C. Pandey, S. K. Mohammed, P. Raviteja, Y. Hong, and E. Viterbo, "Low complexity precoding and detection in multi-user massive MIMO OTFS downlink," *IEEE Trans. Veh. Technol.*, vol. 70, no. 5, pp. 4389–4405, May 2021.
- 144 W. Shen, L. Dai, J. An, P. Fan, and R. W. Heath, "Channel estimation for orthogonal time frequency space (OTFS) massive MIMO," *IEEE Trans. Signal Process.*, vol. 67, no. 16, pp. 4204–4217, Aug. 2019.
- 145 O. K. Rasheed, G. D. Surabhi, and A. Chockalingam, "Sparse delay-Doppler channel estimation in rapidly time-varying channels for multiuser OTFS on the uplink," *Proc. IEEE VTC'2020-Spring*, pp. 1–5, May 2020.
- 146 D. Shi, W. Wang, L. You, X. Song, Y. Hong, X. Gao, and G. Fettweis, "Deterministic pilot design and channel estimation for downlink massive MIMO-OTFS systems in presence of the fractional Doppler," *IEEE Trans. Wireless Commun.*, vol. 20, no. 11, pp. 7151–7165, Nov. 2021.
- 147 F. Liu, Z. Yuan, Q. Guo, Z. Wang, and P. Sun, "Message passing-based structured sparse signal recovery for estimation of OTFS channels with fractional Doppler shifts," *IEEE Trans. Wireless Commun.*, vol. 20, no. 12, pp. 7773–7785, Dec. 2021.
- 148 Z. Wei, W. Yuan, S. Li, J. Yuan, and D. W. Kwan Ng, "A new off-grid channel estimation method with sparse Bayesian learning for OTFS systems," *Proc. IEEE GLOBECOM'2021*, pp. 1–7, Dec. 2021.
- 149 Z. Zhou, L. Liu, J. Xu, and R. Calderbank, "Learning to equalize OTFS," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7723–7736, Sep. 2022.
- 150 E. Dahlman, S. Parkvall, and J. Skold, *4G, LTE-Advanced Pro and the Road to 5G*, Academic Press, Aug. 2016.
- 151 E. Dahlman, S. Parkvall, and J. Skold, *5G NR: The Next Generation Wireless Access Technology*, Academic Press, Aug. 2018.
- 152 J. G. Proakis and M. Salehi, *Digital Communication*, McGraw-Hill, 2008.
- 153 T. M. Schmidl and D. C. Cox, "Robust frequency and timing synchronization for OFDM," *IEEE Trans. Commun.*, vol. 45, no. 12, pp. 1613–1621, Dec. 1997.
- 154 M. Wahlqvist, R. Larsson, and C. Ostberg, "Time synchronization in the uplink of an OFDM system," *Proc. IEEE VTC'1996*, pp. 1569–1573, Apr.-May 1996.
- 155 D. Chu, "Polyphase codes with good periodic correlation properties," *IEEE Trans. Inf. Theory*, vol. 18, no. 4, pp. 531–532, Jul. 1972.
- 156 M. Hua, M. Wang, K. W. Yang, and K. J. Zou, "Analysis of the frequency-offset effect on Zadoff–Chu sequence timing performance," *IEEE Trans. Commun.*, vol. 62, no. 11, pp. 4024–4039, Nov. 2014.

- 157 A. K. Sinha, S. K. Mohammed, P. Raviteja, Y. Hong, and E. Viterbo, "OTFS based random access preamble transmission for high mobility scenarios," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 15078–15094, Dec. 2020.
- 158 LTE, "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) Radio Transmission and Reception," ETSI TS, Tech. Rep. 3GPP TS 36.104 v8.6.0, vol. 136, no. 104, Jul. 2009.
- 159 LTE, "Technical Specification Group Radio Access Network; NR; Phys. Channels Modulation (Release 15)," Standard 38.211, Rev. 15.2.0, Jun. 2018.
- 160 M. S. Khan, Y. J. Kim, Q. Sultan, J. Joung, and Y. S. Cho, "Downlink synchronization for OTFS-based cellular systems in high Doppler environments," *IEEE Access*, vol. 9, pp. 73575–73589, 2021.
- 161 G. D. Surabhi, R. M. Augustine, and A. Chockalingam, "Peak-to-average power ratio of OTFS modulation," *IEEE Commun. Lett.*, vol. 23, no. 6, pp. 999–1002, Jun. 2019.
- 162 P. Wei, Y. Xiao, W. Feng, N. Ge, and M. Xiao, "Charactering the peak-to-average power ratio of OTFS signals: a large system analysis," *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 3705–3720, Jun. 2022.
- 163 M. N. Hossain, Y. Sugiura, T. Shimamura, and H.-G. Ryu, "DFT-spread OTFS communication system with the reductions of PAPR and nonlinear degradation," *Wireless Pers. Commun.*, vol. 115, no. 3, pp. 2211–2228, Aug. 2020.
- 164 J. Su, S. Liu, Y. Huang, and J. Yuan, "Peak-to-average power ratio reduction via symbol pre-coding in OTFS modulation," *Proc. IEEE VTC'2022-Spring*, pp. 1–5, Jun. 2022.
- 165 Y. I. Tek and E. Basar, "PAPR reduction precoding for orthogonal time frequency space modulation," *Int. Conf. Telecommun. Signal Process.*, pp. 172–176, Jul. 2023.
- 166 S. Gao and J. Zheng, "Peak-to-average power ratio reduction in pilot-embedded OTFS modulation through iterative clipping and filtering," *IEEE Commun. Lett.*, vol. 24, no. 9, pp. 2055–2059, Sep. 2020.
- 167 M. Liu, M.-M. Zhao, M. Lei, and M.-J. Zhao, "Autoencoder based PAPR reduction for OTFS modulation," *Proc. IEEE VTC'2021-Fall*, pp. 1–5, Sep.-Oct. 2021.
- 168 A. S. Sumer, T. Yilmaz, E. Memisoglu, and H. Arslan, "Exploiting OTFS frame structure for PAPR reduction," *Proc. IEEE VTC'2022-Fall*, pp. 1–5, Sep. 2022.
- 169 G. D. Surabhi, M. K. Ramachandran, and A. Chockalingam, "OTFS modulation with phase noise in mmWave communications," *Proc. IEEE VTC'2019-Spring*, pp. 1–5, Apr.-May 2019.
- 170 A. Naikoti and A. Chockalingam, "A DNN-based OTFS transceiver with delay-Doppler channel training and IQI compensation," *Proc. IEEE PIMRC'2021*, pp. 628–634, Sep. 2021.
- 171 A. Farhang, A. RezazadehReyhani, L. E. Doyle, and B. Farhang-Boroujeny, "Low complexity modem structure for OFDM-based orthogonal time frequency space modulation," *IEEE Wireless Commun. Lett.*, vol. 7, no. 3, pp. 344–347, Jun. 2018.
- 172 A. RezazadehReyhani, A. Farhang, M. Ji, R. R. Chen, and B. Farhang-Boroujeny, "Analysis of discrete-time MIMO OFDM-based orthogonal time frequency space modulation," *Proc. IEEE ICC'2018*, pp. 1–6, May 2018.
- 173 T. J. Roupheal, *Wireless Receiver Architectures and Design: Antenna, RF, Synthesizers, Mixed Signal and Digital Signal Processing*, Academic Press, 2014.
- 174 T. C. W. Schenk, *RF Impairments in High-Rate Wireless Systems*, Springer, 2008.
- 175 L. Smaini, *RF Analog Impairments Modeling for Communication Systems Simulation: Application to OFDM-based Transceivers*, John-Wiley & Sons, 2012.
- 176 A. Nimr, M. Chafii, M. Matthe, and G. Fettweis, "Extended GFDM framework: OTFS and GFDM comparison," *Proc. IEEE GLOBECOM'2018*, pp. 1–6, Dec. 2018.

- 177 R. V. Nee and R. Prasad, *OFDM for Wireless Multimedia Communications*, Norwood, MA, USA: Artech House, 2000.
- 178 N. Michailow, M. Matthe, I. S. Gaspar, A. N. Caldevilla, L. L. Mendes, A. Festag, and G. Fettweis, "Generalized frequency division multiplexing for 5th generation cellular networks," *IEEE Trans. Commun.*, vol. 62, no. 9, pp. 3045–3061, Sep. 2014.
- 179 S. S. Das and S. Tiwari, "Discrete Fourier transform spreading-based generalised frequency division multiplexing," *Electron. Lett.*, vol. 51, no. 10, pp. 789–791, May 2015.
- 180 S. B. Slimane, "Peak-to-average power ratio reduction of OFDM signals using pulse shaping," *Proc. IEEE GLOBECOM'2000*, pp. 1412–1416, Dec. 2000.
- 181 T. S. Rappaport, R. W. Heath, Jr., J. N. Murdock, and R. C. Daniels, *Millimeter Wave Wireless Communications*, Prentice Hall, 2014.
- 182 Y. Niu, Y. Li, D. Jin, L. Su, and A. V. Vasilakos, "A survey of millimeter wave (mmWave) communications for 5G: opportunities and challenges," *Wireless Netw.*, vol. 21, no. 8, pp. 2657–2676, Aug. 2015.
- 183 3GPP, "TR 138.900 V14.2.0: Study on channel model for frequency spectrum above 6 GHz," Release 14 (2017-06).
- 184 3GPP, "R1-163984: Discussion on phase noise modeling," 3GPP TSG RAN WG1 #85, May 2016.
- 185 R. Marsalek, J. Blumenstein, D. Schtzenhfer, and M. Pospisil, "OTFS modulation and influence of wideband RF impairments measured on a 60 GHz testbed," *Proc. IEEE SPAWC'2020*, pp. 1–5, May 2020.
- 186 S. G. Neelam and P. R. Sahu, "Error performance of OTFS in the presence of IQI and PA non-linearity," *Proc. NCC'2020*, pp. 1–6, Feb. 2020.
- 187 J. Tubbax, B. Come, L. Van der Perre, S. Donnay, M. Moonen, and H. De Man, "Compensation of transmitter IQ imbalance for OFDM systems," *Proc. IEEE ICASSP'2004*, pp. 325–328, May 2004.
- 188 H. Bolcskei and F. Hlawatsch, "Discrete Zak transforms, polyphase transforms, and applications," *IEEE Trans. Signal Process.*, vol. 45, no. 4, pp. 851–866, Apr. 1997.
- 189 R. Horn and C. Johnson, *Matrix Analysis*, Cambridge University Press, 2013.
- 190 V. Yogesh, S. R. Mattu, and A. Chockalingam, "Low-complexity delay-Doppler channel estimation in discrete Zak transform based OTFS," *IEEE Commun. Lett.*, vol. 28, no. 3, pp. 672–676, Jan. 2024.
- 191 C. Jiang, H. Zhang, Y. Ren, Z. Han, K. C. Chen, and L. Hanzo, "Machine learning paradigms for next-generation wireless networks," *IEEE Wireless Commun.*, vol. 24, no. 2, pp. 98–105, Apr. 2017.
- 192 S. Dorner, S. Cammerer, J. Hoydis, and S. ten Brink, "Deep learning-based communication over the air," *IEEE J. Sel. Top. Signal Process.*, vol. 12, no. 1, pp. 132–143, Feb. 2018.
- 193 T. Oshea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cognit. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.
- 194 F. A. Aoudia and J. Hoydis, "End-to-end learning of communications systems without a channel model," *Proc. Asilomar Conf. on Signals, Systems, and Computers*, pp. 298–303, Oct. 2018.
- 195 H. Kim, Y. Jiang, R. Rana, S. Kannan, and P. Viswanath, "Communication algorithms via deep learning," *Proc. ICLR'2018*, pp. 1–17, 2018.
- 196 Y. Jiang, H. Kim, H. Asnani, S. Kannan, S. Oh, and P. Viswanath, "LEARN codes: inventing low-latency codes via recurrent neural networks," *Proc. IEEE ICC'2019*, pp. 1–7, Jul. 2019.

- 197 E. Nachmani, E. Marciano, L. Lugosch, W. J. Gross, D. Burshtein, and Y. Beery, "Deep learning methods for improved decoding of linear codes," *IEEE J. Sel. Top. Signal Process.*, vol. 12, no. 1, pp. 119–131, Feb. 2018.
- 198 F. Liang, C. Shen, and F. Wu, "An iterative BP-CNN architecture for channel decoding," *IEEE J. Sel. Top. Signal Process.*, vol. 12, no. 1, pp. 144–159, Feb. 2018.
- 199 N. Farsad and A. Goldsmith, "Neural network detection of data sequences in communication systems," *IEEE Trans. Signal Process.*, vol. 66, no. 21, pp. 5663–5678, Sep. 2018.
- 200 N. T. Nguyen and K. Lee, "Deep learning-aided tabu search detection for large MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 19, no. 6, pp. 4262–4275, Jun. 2020.
- 201 M. Khani, M. Alizadeh, J. Hoydis, and P. Fleming, "Adaptive neural signal detection for massive MIMO," *IEEE Trans. Wireless Commun.*, vol. 19, no. 8, pp. 5635–5648, Aug. 2020.
- 202 B. Shamasundar and A. Chockalingam, "A DNN architecture for the detection of generalized spatial modulation signals," *IEEE Commun. Lett.*, vol. 24, no. 12, pp. 2770–2774, Dec. 2020.
- 203 H. Ye, G. Y. Li, and B. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 114–117, Feb. 2018.
- 204 M. Soltani, V. Pourahmadi, A. Mirzaei, and H. Sheikhzadeh, "Deep learning-based channel estimation," *IEEE Commun. Lett.*, vol. 23, no. 4, pp. 652–655, Apr. 2019.
- 205 X. Ma and Z. Gao, "Data-driven deep learning to design pilot and channel estimator for massive MIMO," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5677–5682, May 2020.
- 206 Q. Hu, F. Gao, H. Zhang, S. Jin, and G. Y. Li, "Deep learning for channel estimation: interpretation, performance, and comparison," *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, pp. 2398–2412, Apr. 2021.
- 207 W. Xia, G. Zheng, Y. Zhu, J. Zhang, J. Wang, and A. P. Petropulu, "A deep learning framework for optimization of MISO downlink beamforming," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1866–1880, Mar. 2020.
- 208 W. Ma, C. Qi, Z. Zhang, and J. Cheng, "Sparse channel estimation and hybrid precoding using deep learning for millimeter wave massive MIMO," *IEEE Trans. Commun.*, vol. 65, no. 8, pp. 2838–2849, May 2020.
- 209 S. R. Mattu, L. N. Theagarajan, and A. Chockalingam, "Deep channel prediction: a DNN framework for receiver design in time-varying fading channels," *IEEE Trans. Veh. Technol.*, vol. 71, no. 6, pp. 6439–6453, Jun. 2022.
- 210 A. L. Samuel, "Some studies in machine learning using the game of checkers," *IBM J. Res. Dev.*, vol. 3, no. 3, pp. 210–229, Jul. 1959.
- 211 C. M. Bishop, *Pattern Recognition and Machine Learning*, vol. 4, no. 4. Springer, 2006.
- 212 R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*, Wiley, 2001.
- 213 K. Pearson, "On lines and planes of closest fit to systems of points in space," *Philos. Mag.*, vol. 2, no. 11, p. 559–572, 1901.
- 214 W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity" *Bull. Math. Biophys.*, vol. 5, 115–133, Dec. 1943.
- 215 B. Widrow and M. E. Hoff, "Adaptive switching circuits," *IRE WESCON Convention Record*, vol. 4, no. 1, 1960.
- 216 F. Rosenblatt, *Principles of Neurodynamics: Perceptrons and the Theory of Brain Mechanisms*, Spartan Books, 1962.
- 217 D. E. Rumelhart and J. L. McClelland, *Parallel Distributed Processing: Explorations in the Microstructure of Cognition: Foundations*, MIT Press, 1987.
- 218 A. G. Ivakhnenko and V. G. Lapa, *Cybernetic Predicting Devices*, Joint Publications Research Service, 1965.

- 219 S.-I. Amari, "Learning patterns and pattern sequences by self-organizing nets of threshold elements," *IEEE Trans. Computers*, vol. C-21, no. 11, pp. 1197–1206, Nov. 1972.
- 220 S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- 221 J. Chung, G. Caglar, K. Cho, and Y. Bengio. "Empirical evaluation of gated recurrent neural networks on sequence modeling," online: arXiv:1412.3555 [cs.LG], 11 Dec 2014.
- 222 K. Fukushima, "Neocognitron: a self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position," *Biol. Cybern.*, vol. 36, p. 193–202, Apr. 1980.
- 223 S. Linnainmaa, "Taylor expansion of the accumulated rounding error," *BIT Numer. Math.*, vol. 16, p. 146–160, Jun. 1976.
- 224 A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, and B. Ste, "Pytorch: an imperative style, high-performance deep learning library," *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pp. 1–12, Dec. 2019.
- 225 M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G. S. Corrado, A. Davis, J. Dean, M. Devin, S. Ghemawat, I. Goodfellow, A. Harp, G. Irving, M. Isard, Y. Jia, R. Jozefowicz, L. Kaiser, M. Kudlur, J. Levenberg, D. Mane, R. Monga, S. Moore, D. Murray, C. Olah, M. Schuster, J. Shlens, B. Steiner, I. Sutskever, K. Talwar, P. Tucker, V. Vanhoucke, V. Vasudevan, F. Viegas, O. Vinyals, P. Warden, M. Wattenberg, M. Wicke, Y. Yu, and X. Zheng, "TensorFlow: large-scale machine learning on heterogeneous distributed systems," software available from [tensorflow.org](https://www.tensorflow.org), 2015.
- 226 A. Naikoti and A. Chockalingam, "Low-complexity delay-Doppler symbol DNN for OTFS signal detection," *Proc. IEEE VTC'2021-Spring*, pp. 1–6, Apr.-May 2021.
- 227 S. Krusevac, P. Rapajic, and R. A. Kennedy, "Channel capacity estimation for MIMO systems with correlated noise," *Proc. IEEE GLOBECOM'2005*, pp. 2812–2816, Dec. 2005.
- 228 M. L. Ammari and P. Fortier, "Analysis of MIMO receiver using generalized least squares method in colored environments," *J. Comput. Networks Commun.*, vol. 2014, pp. 1–6, Aug. 2014.
- 229 S. R. Mattu and A. Chockalingam, "Learning based delay-Doppler channel estimation with interleaved pilots in OTFS," *Proc. IEEE VTC'2022-Fall*, pp. 1–6, Sep. 2022.
- 230 C. Sun, J. Fan, C. Chen, W. Li, and W. Chen, "A two-stage neural network for sleep stage classification based on feature learning, sequence learning, and data augmentation," *IEEE Access*, vol. 7, pp. 109386–109397, 2019.
- 231 N. Levanon and E. Mozeson, *Radar Signals*, Hoboken, NJ, USA: Wiley, 2004.
- 232 M. J. Skolnik, *An Introduction and Overview of Radar*, 3rd edition, New York: McGraw-Hill, 2008.
- 233 L. Auslander and R. Tolmieri, "Radar ambiguity functions and group theory," *SIAM J. Math. Anal.*, vol. 16, no. 3, pp. 577–601, 1985.
- 234 W. Moran, The mathematics of radar, in *Twentieth Century Harmonic Analysis: A Celebration (NATO Science Series, NATO)*, J. S. Byrnes Ed., Norwell, MA, USA: Kluwer, pp. 295–328, 2001.

Index

a

ACF. *See* Auto-correlation function (ACF)
 Activation function 334
 Additive white Gaussian noise (AWGN) 373
 Advanced mobile phone system (AMPS) 2–5
 AI. *See* Artificial intelligence (AI)
 Alamouti space–time code matrix 165
 Aliases 38
 Ambiguity function 18
 peaks of 395–296
 radar sensing 374–380
 radar waveform 379
 for TD and FD pulses 379–380
 Zak-OTFS pulsone 385
 for sinc pulse 388–389, 396–397
 AMPS. *See* Advanced mobile phone system (AMPS)
 Angle-delay-Doppler domain 264, 265
 Angles of arrival (AoA) 256
 Angles of departure (AoD) 256
 AoA. *See* Angles of arrival (AoA)
 AoD. *See* Angles of departure (AoD)
 AR. *See* Augmented reality (AR)
 Artificial intelligence (AI) 1
 Augmented reality (AR) 4
 Auto-ambiguity function 375, 393
 Auto-correlation function (ACF) 8, 9, 210
 and Doppler spectrum 9
 AWGN. *See* Additive white Gaussian noise (AWGN)

b

Bandwidth-limited DD domain carrier waveforms 43–51
 Base station (BS) 28, 233, 256
 antenna array 256
 Basic circulant permutation matrix (BPCM) 305
 Bayesian hierarchical modeling 225

Bayesian inference 224
 BCE. *See* Binary cross entropy (BCE)
 BER performance. *See* Bit error rate (BER) performance
 Binary cross entropy (BCE) 331
 Binary phase shift keying (BPSK) 3, 10, 214, 309, 338, 350
 Bi-orthogonal MC generators 127–128, 145
 Bit error rate (BER) performance 26, 71, 73, 106, 131
 comparison between OTFS and OFDM 298
 of DZT-OTFS system 307–311
 with estimated channel parameters 220
 full-DNN-and symbol-DNN-based detection 337–343
 of linear minimum mean squared error 261
 of MIMO-OTFS with RAS 180
 OTFS transceiver architecture 347–353
 performance comparison 360
 with PN-based channel estimation 212
 of SIMO-OTFS with RAS 179
 of SISO-OTFS 155
 of STC-OTFS 169
 for Zak-OTFS and MC-OTFS 131
 of ZF and LMMSE multiuser precoders 270
 Block circulant matrix 153
 BPCM. *See* Basic circulant permutation matrix (BPCM)
 BPSK. *See* Binary phase shift keying (BPSK)
 BS. *See* Base station (BS)

c

Cauchy–Schwartz inequality 288, 395
 CCDF. *See* Complementary cumulative distribution function (CCDF)
 CDMA. *See* Code division multiple access (CDMA)
 Cellular mobile communication technology 1

- Cellular mobile evolution 2–5
 - Moore's law 4–5
 - Central limit theorem (CLT) 7
 - Channel coefficients 12
 - Channel estimation 21
 - embedded pilot approach 21, 221–227
 - exclusive pilot approach 21, 208, 212–221
 - impulse-based 212–214, 277
 - for interleaved pilot scheme 360–365
 - multiuser 276–277
 - PN pilot approach 208–212
 - received symbols for 222
 - superimposed pilot approach 21, 208, 227–232, 365–369
 - threshold-based 221–223
 - using DNNs 353–369
 - Channel estimation error 229
 - Channel path delay shifts 39
 - Channel path Doppler shift 23
 - Channel state information (CSI) 154, 207, 226
 - Channel statistics 154
 - Channel training and detection 346–347
 - Classical machine learning algorithms 330
 - Classification learning programs 329
 - Clustering learning programs 329
 - CMOS technology. *See* Complementary metal-oxide semiconductor (CMOS) technology
 - CNN. *See* Convolutional neural networks (CNN)
 - Coarse estimation 217
 - Code division multiple access (CDMA) 2–4
 - Coherence bandwidth 6
 - Coherence time 8
 - Communication reliability 256
 - Communication theory 16
 - Communication waveforms 18
 - for doubly selective channels 14–17
 - Complementary cumulative distribution function (CCDF) 289–290
 - Complementary metal-oxide semiconductor (CMOS) technology 286
 - Complex Gaussian distributed 8
 - computational complexity for cost function 219
 - Conventional MMSE detection 188
 - Convolutional neural networks (CNN) 333
 - Cost function, computational complexity for 219
 - Covariance matrix 204
 - of error vector 229
 - mean vector and 229
 - CP. *See* Cyclic prefix (CP)
 - CPSC schemes. *See* Cyclic prefixed single carrier (CPSC) schemes
 - Cross-ambiguity function 375–378, 383
 - Cross-domain detection 184, 198–205
 - Crystalline decomposition 62–63
 - Crystalline regime
 - impact of fading in 70–73
 - model-free operation in 73–77
 - prediction in 60, 102
 - unambiguous target detection in 386
 - Zak-OTFS I/O relation in 58–66
 - Crystallization condition 25, 42, 60, 64, 371, 387
 - Cyclic prefix (CP) 293, 303
 - Cyclic prefixed single carrier (CPSC) schemes 10, 18
- d**
- Data-aided channel estimator 230
 - Data detection 222
 - DD channel spreading function 23
 - DD domain. *See* Delay-Doppler (DD) domain
 - DD domain aliasing 25, 42, 43, 77
 - DD domain IPI cancellation (DDIPIC) algorithm
 - estimation 215–221
 - DD domain pulse 38, 41
 - channel interaction of 39–43
 - non-fading 43
 - non-stationary 43
 - predictable 40–43
 - discrete time realization 320
 - and doubly spread channel 42
 - DD domain resource element (DDRE) 235, 256, 264, 282
 - DDIPIC algorithm estimation. *See* DD domain IPI cancellation (DDIPIC) algorithm estimation
 - DDNet architecture 356–357
 - DDRE. *See* DD domain resource element (DDRE)
 - DD spreading function 28, 36, 79, 80
 - Deep learning (DL) 330. *See also* Deep neural networks (DNNs)
 - convolutional neural networks 333
 - fully connected neural networks 330–333
 - programming frameworks 332–333
 - recurrent neural networks 333
 - Deep neural networks (DNNs) 327
 - communication transceiver designs using 337
 - DD channel estimation using 353–369
 - IQI estimation/compensation using 344–353
 - BER performance 347–353
 - channel training and detection 346–347
 - Rx IQI compensation 345–346
 - Tx IQI compensation 344–345

- for OTFS transceiver design
 - learning framework 328–335
 - signal detection using DNNs 335–343
 - Delay domain resolution 378
 - Delay-Doppler diversity
 - in MIMO-OTFS 161–165
 - opportunities 149
 - in OTFS with antenna selection 170–181
 - in SISO OTFS 150–161
 - space-time coding OTFS 165–170
 - Delay-Doppler (DD) domain 28–29, 327
 - channel action on 36, 37
 - channel estimation. *See* Channel estimation
 - channel sparsity 198
 - characterization 12–14
 - detection 204
 - for doubly spread channels 23
 - information signal 24
 - modulation 34–38, 44
 - sparsity and slow variability of 14
 - time and bandwidth limitation on 43–51, 394–395
 - and time domain realization 15
 - time-varying channel response in 13
 - transmit symbols 257
 - transmit vector 227
 - using transform 35
 - vectorized 304–305
 - Delay-Doppler signaling
 - channel action on 36, 37, 83–84
 - DD domain modulation 34–38
 - delay-Doppler domain 28–29
 - in doubly spread channels 23–24
 - for integer and fractional DDs 216
 - single path on 38
 - Delay-only (Doppler-only) channels 23
 - Delay-only Veh-A channel 72
 - Delay spread 6, 25
 - DFRC. *See* Dual-function radar communication (DFRC)
 - DFT. *See* Discrete Fourier transform (DFT)
 - Diagonal matrix 185, 269
 - Digital implementation, Zak-OTFS modulation 311–316
 - Dirac-delta functions 28, 134, 376
 - Dirac-delta impulses 38, 96, 372, 399
 - Dirac-delta pulse train 320
 - Direct conversion architecture 296
 - Discrete DD domain signals 94–96, 130
 - channel output 319
 - impulse signal 58–59, 95–96
 - information signal 56
 - quasi-periodic 316
 - Discrete DD realization 316
 - Discrete delay shifts 318–319
 - Discrete Fourier transform (DFT) 10, 199
 - discrete time pulsone 324
 - Discrete pulsones 319–326
 - Discrete-time baseband model 294, 295
 - Discrete time-domain signal 304, 313
 - Discrete time pulsones 324
 - Discrete time realization 320
 - Discrete twisted convolution 317–318
 - Discrete Zak transform (DZT) 22, 285, 313, 316–326
 - Diversity analysis, of DZT-OTFS 305–307
 - Diversity opportunities 149
 - DNNs. *See* Deep neural networks (DNNs)
 - Doppler domain resolution 378
 - Doppler-only Veh-A channel 72
 - Doppler shifts 7–8, 10, 20, 28, 29, 33, 151, 262, 276, 318–319, 373
 - path delays and 33
 - Doppler spectrum 9
 - Doppler spread 11, 25, 381
 - Doubly selective channels 8
 - communication waveforms for 14–17
 - Doubly spread channels 23, 25, 29, 35, 36, 99
 - delay-Doppler signaling in 23–24
 - Downlink multiuser communication 265–276
 - Downlink orthogonal communication 234
 - Downlink precoding 234, 266–271
 - Downlink timing synchronization 283–284
 - Dual-function radar communication (DFRC) 19
 - DZT. *See* Discrete Zak transform (DZT)
 - DZT-OTFS 302–311
 - bit error rate performance of 307–311
 - diversity analysis of 305–307
 - and MC-OTFS, complexity comparison between 310–311
 - system model 302–304
 - transceiver 302
 - vectorized DD domain I/O relation 304–305
- e**
- Effective DD domain I/O relation 101
 - Eigenvalues decomposition 186
 - ELBO. *See* Evidence lower bound (ELBO)
 - Embedded frame structure 224
 - Embedded pilot approach/frames 21, 208, 221–227
 - channel estimation using DNNs 354–360
 - with guard symbols 221–223
 - without guard symbols 223–227
 - Emerging intelligent transportation systems 17
 - Encode wireless channel dynamics 50–51

End-to-end system model 294
 Epoch 334
 Estimation network 356
 Evidence lower bound (ELBO) 192, 193
 Exclusive pilot approach/frames 21, 208, 212–221
 channel estimation using DNNs 354
 DDIPIC-based estimation 215–221
 impulse-based channel estimation 212–215
 Expectation maximization (EM) algorithm 225

f

Factorizable DD domain function 45, 46
 Fading 23
 Fast Fourier transform (FFT) 184
 FCNNs. *See* Fully connected neural networks (FCNNs)
 FD. *See* Frequency domain (FD)
 FDM. *See* Frequency domain modulation (FDM)
 FDMA technology. *See* Frequency division multiple access (FDMA) technology
 FFT. *See* Fast Fourier transform (FFT)
 Fifth generation (5G) standard 1
 Fine estimation 217–218
 First generation (1G) technology 2
 Fourier series expansion 381
 Fourier transform 36, 51, 381
 Fractional delay-Dopplers 214–216, 223, 226, 365
 Frame synchronization 234
 Frequency division multiple access (FDMA) technology 2
 Frequency domain (FD) 10, 15
 characterization 6–7
 pulsone 25, 46, 48, 90, 325
 quasi-localization in 381–383
 Frequency domain modulation (FDM) 14–15, 29–34, 69–70
 attributes of 78
 doubly spread channel with 30
 FD domain information signal 35
 I/O relation 92–94
 as limiting case 50
 in modulation domain and time domain 29
 signal processing steps in 54
 Frequency domain (FD) pulse 32–34
 Frequency-flat fading model 7, 8
 Frequency selectivity 6, 7
 Frequency synchronization 234, 277
 Frobenius norm 211
 Full-DNN detection approach 335–336
 Fully connected neural networks (FCNNs) 327, 330–333, 356

channel estimation using DNNs 355
 DL programming frameworks 332–333
 loss function and weight update 331
 neural networks 331–332

g

Gamma distribution 225
 Gaussian distribution 289, 337, 340
 Gaussian minimum shift keying (GMSK) modulation 2
 Gaussian noise 338–339
 in MIMO-OTFS 342–343
 Gaussian noise model 327, 335, 343
 GB-based OMA method 233
 Generalized frequency division multiplexing (GFDM) 22, 285
 and orthogonal frequency division multiplexing 292
 transmit waveform of 292
 Generalized transceiver signal processing 52–54
 GFDM. *See* Generalized frequency division multiplexing (GFDM)
 Gibbs sampling 197
 Global System for Mobile communication (GSM) 2
 GMSK modulation. *See* Gaussian minimum shift keying (GMSK) modulation
 Good radar waveforms 381–383
 GSM. *See* Global System for Mobile communication (GSM)
 Guard band (GB)-based OMA 237–240
 Guard symbols 221–227

h

Heisenberg's uncertainty principle 34
 Heisenberg transform 101–103, 108, 133, 140
 pulsones generated using 110–113
 time and bandwidth TD pulsones using 104, 106
 Hermitian matrix 155
 Hermitian operation 184
 Heterodyne architecture 298
 High Doppler 9–12
 Higher-order modulation 3
 Hybrid MAP detection 194–197
 Hyper-parameters 357

i

ICI. *See* Inter-carrier interference (ICI)
 IDD-OMA. *See* Interleaved delay-Doppler OMA (IDD-OMA)
 Ideal MC generators 128–129

- IDFT. *See* Inverse discrete Fourier transform (IDFT)
- IDZT. *See* Inverse discrete Zak transform (IDZT)
- IFFT. *See* Inverse fast Fourier transform (IFFT)
- Impulse-based channel estimation 212–214, 229
- “Index-to-OTFS signal vector mapper” 336
- Information grid 51
- Information lattice 51
- Information symbols 10, 56
- Integer DD 214–216
- Integrated sensing and communications (ISAC) 18–19
- Inter-carrier interference (ICI) 11
- Interference plus noise 190, 229
- Interleaved delay-Doppler OMA (IDD-OMA) 240–245, 265
- Interleaved pilot scheme 360–365
- Interleaved time-frequency OMA (ITF-OMA) 245–249
- Internet of Things (IoT) devices 4
- Inter-path interference (IPI) 207, 215
- Inter-symbol interference (ISI) 6, 9
- In-vehicle mobile broadband service 5
- Inverse discrete Fourier transform (IDFT) 10
- Inverse discrete Zak transform (IDZT) 302, 315, 317, 326
- Inverse fast Fourier transform (IFFT) 184–185
- Inverse frequency-Zak transforms 25, 46, 51, 88
- Inverse symplectic finite Fourier transform (ISFFT) 16, 151, 200, 257, 293
- Inverse symplectic Fourier transform (ISFT) 135–138
- Inverse time-Zak transform 87
- Inverse Zak transform (IZT) 17, 43, 44, 87–88, 134
- I/O relation
- frequency domain modulation 92–94
 - MC-OTFS modulation 126–130
 - time domain modulation 90–92
 - for two-step MC-OTFS transceiver 146
 - vectorized DD domain 304–305
 - Zak-OTFS. *See* Zak-OTFS I/O relation
- IoT devices. *See* Internet of Things (IoT) devices
- IPI. *See* Inter-path interference (IPI)
- IPNet 363
- IQI. *See* IQ imbalance (IQI)
- IQ imbalance (IQI) 298–302
- estimation/compensation using DNNs 344–353
 - OTFS system model with 299–300
 - sensitivity of OTFS and OFDM to 300–302
- ISAC. *See* Integrated sensing and communications (ISAC)
- ISFFT. *See* Inverse symplectic finite Fourier transform (ISFFT)
- ISFT. *See* Inverse symplectic Fourier transform (ISFT)
- ITF-OMA. *See* Interleaved time-frequency OMA (ITF-OMA)
- IZT. *See* Inverse Zak transform (IZT)
- j**
- Jakes’ formula 226
- Jakes’ spectrum 8
- k**
- Kronecker product operation 293
- l**
- Large-scale fading 5
- LDPC codes. *See* Low density parity check (LDPC) codes
- Learning library 335
- Learning rate 331, 334
- Likelihood function 225
- Linear convolution operation 313
- Linear detection methods 184–189
- Linear minimum mean squared error (LMMSE) 71, 202–203, 261, 262
- BER performance of 262
 - equalization 188
 - estimation 199, 203
 - mean and covariance matrix 203
 - and ZF precoders 270
- Linear MMSE precoder 269
- Linear precoding 267–270
- Linear time varying (LTV) channel 12, 58, 63–64, 293
- Linear transformation matrix 184
- LMMSE. *See* Linear minimum mean squared error (LMMSE)
- Loss function 331, 333
- Low-complexity MMSE detection algorithm 187
- Low density parity check (LDPC) codes 4
- Low-power pilot symbols 227
- LTV channel. *See* Linear time varying (LTV) channel
- m**
- Machine learning program 328
- classification of 328–329
 - deep learning using neural networks 330–333
 - traditional program and 329–330
- MAP-based detection 189–198

- Markov chain Monte Carlo (MCMC) 21
 - sampling techniques 184, 197–198
 - Massive MIMO systems 4, 234, 263–265, 271–276
 - Matrix operation 293
 - Matrix-vector, of I/O relation 67–70
 - Maximum-likelihood (ML) detection 154
 - decision rule 183
 - Maximum-likelihood sequence estimation (MLSE) detection 203
 - Maximum ratio combining (MRC) 234
 - Max Index Selector 336
 - MCMC. *See* Markov chain Monte Carlo (MCMC)
 - MC-OTFS. *See* Multicarrier OTFS (MC-OTFS)
 - MC pulsones 104
 - MC transmission approach. *See* Multicarrier transmission approach
 - Mean and covariance matrix 205
 - Mean square error (MSE) 331
 - Mean vector and covariance matrix 229
 - Message passing (MP) detector 189–192, 295, 358
 - MIMO-OTFS
 - BER performance of 215
 - diversity in 161–165
 - Gaussian noise in 342–343
 - impulse-based pilot scheme in 214
 - non-orthogonal multiple access system 261
 - MIMO techniques. *See* Multiple-input multiple-output (MIMO) techniques
 - Minimum mean squared error (MMSE) 21, 183, 309
 - complexity reduction 185–189
 - time domain equalizer 11
 - transformation matrix for 184
 - Mixed-Gibbs sampling detection 199
 - ML detection rule 203
 - MLSE detection. *See* Maximum-likelihood sequence estimation (MLSE) detection
 - MMSE. *See* Minimum mean squared error (MMSE)
 - mmWave frequencies 296
 - Mobility requirements 5
 - Model-dependent operation 26, 73
 - Model-free operation 26, 74–77
 - in crystalline regime 73–77
 - uncoded BER performance of 132
 - Modern communication systems 311
 - Modified ML detector 343
 - Moore's law 4–5
 - MSE. *See* Mean square error (MSE)
 - Multi-antenna OTFS systems 170, 173–178
 - Multicarrier OTFS (MC-OTFS) modulation 16, 17, 19–21, 101, 118–123, 235
 - complexity comparison between DZT-OTFS and 310–311
 - I/O relation 106, 126–130
 - two-step transceiver 126–129
 - for Zak transform-based transceiver 129–130
 - low-complexity MMSE detection algorithm for 187
 - PAPR of 286–292
 - predictability attribute of 106
 - receiver 123–126
 - transmit signal 287
 - transmitter multiplexes 150
 - uncoded BER performance for 131
 - to Zak-OTFS comparison 130–132
 - Multicarrier (MC) transmission approach 10
 - Multilayered RNN architecture 355
 - Multipath fading channels 5–14
 - delay-Doppler domain characterization 12–14
 - frequency domain characterization 6–7
 - high Doppler 9–12
 - time domain characterization 7–9
 - Multiple access techniques 276
 - Multiple antennas 21
 - Multiple-input multiple-output (MIMO) techniques 3, 149
 - wireless systems 170
 - Multisuser OTFS
 - channel estimation 276–277
 - downlink 265–276
 - resource elements 234–236
 - synchronization 277–284
 - uplink 236–265
- n**
- Near-far problem 3
 - Neural networks (NNs) 327
 - deep learning using 330–333
 - NMSE. *See* Normalized mean square error (NMSE)
 - NMSE vs. spectral efficiency 359–360
 - Noise correlation matrix 342
 - Noise-free cross-ambiguity 376
 - NOMA methods. *See* Non-orthogonal multiple access (NOMA) methods
 - Non-crystalline regime 387
 - Non-diagonal matrix 202
 - Non-Gaussian noise 339–342
 - Non-GB-based OMA methods 233
 - Nonlinear detection methods 187, 189
 - Nonlinear precoding techniques 270–271
 - Non-orthogonal multiple access (NOMA) methods 2, 21, 233–234, 256–265

Non-overlapping received impulses 86
 Non-predictable taps 63
 Non-zero delays 15
 Non-zero Doppler spreads 11
 Normalized error 225
 Normalized mean square error (NMSE) 219, 358
 performance comparison 360
 vs. spectral efficiency 359–360
 Nyquist sampling 288, 290

O

ODC. *See* Orthogonal downlink communication (ODC)
 OFDM. *See* Orthogonal frequency division multiplexing (OFDM)
 OFDMA. *See* Orthogonal frequency division multiple access (OFDMA)
 OMA. *See* Orthogonal multiple access (OMA)
 Optimizer 334
 Order of complexity 187
 Orthogonal downlink communication (ODC) 265–266
 Orthogonal frequency division multiple access (OFDMA) 3–4
 Orthogonal frequency division multiplexing (OFDM) 3, 10, 18, 22, 292
 Orthogonality 11
 of pulsons 49–50
 Orthogonal multiple access (OMA) 21, 233, 237–255
 guard band (GB)-based 237–240
 with ideal pulses 249–251
 interleaved delay-Doppler OMA 240–245
 interleaved time-frequency OMA 245–249
 with practical rectangular pulses 251–255
 Orthogonal time frequency space (OTFS) modulation 15
 approximations of 101–147
 Heisenberg transform 110–113
 MC-OTFS modulation 118–123
 MC-OTFS receiver 123–126
 pulsons. *See* Pulsons
 time- and bandwidth-limited pulsons 113–118
 crystalline regime. *See* Crystalline regime
 DD domain modulation 34–38
 delay-Doppler diversity in. *See* Delay-Doppler diversity
 discrete Zak approach to 302–311
 for doubly selective channels 16
 frame synchronization 234
 frequency domain modulation 29–34
 impact of phase noise on 292–298

for integrated sensing and communication 18–19
 IQ imbalance on 298–302
 MC-OTFS. *See* Multicarrier OTFS (MC-OTFS) and OFDM, BER performance comparison 298
 for radar sensing. *See* Radar sensing
 realization 16–17
 research on 18
 signal detection 21
 cross-domain detection 198–205
 linear detection methods 184–189
 MAP-based detection 189–198
 using DNNs 335–343
 system model with IQI 299–300
 time domain modulation 29–34
 Oscillator phase noise 292–298
 performance evaluation with 295
 OTFS modulation. *See* Orthogonal time frequency space (OTFS) modulation

P

Pairwise error probability (PEP) 150, 154, 306
 PAPR. *See* Peak-to-average power ratio (PAPR)
 Parallel interference cancellation (PIC) detection 194–197
 Parseval's theorem 288, 394, 401
 Path delay 28, 33
 Path division multiple access (PDMA)
 multiuser detection 265
 principle 264
 Path Doppler shift 28
 PDMA. *See* Path division multiple access (PDMA)
 PDP. *See* Power-delay profile (PDP)
 Peak-to-average power ratio (PAPR) 10, 281, 285
 complementary cumulative distribution function 289–290
 effect of pulse shaping on 291–292
 of MC-OTFS signals 286–292
 reduction techniques 285
 upper bound on 288
 PEP. *See* Pairwise error probability (PEP)
 Periodic DD domain function 105, 135, 138
 impact of windowing on 140–141
 Permutation matrix 200
 Phase locked loop (PLL) 296
 Phase noise 292–298
 OTFS and OFDM systems with 297
 presence of 295–298
 system model with 294–295
 system model without 293–294
 variance of 297

Phase rotation 157–161
 Physical layer waveform 1
 Physical resource 234
 PIC detection. *See* Parallel interference cancellation (PIC) detection
 Pilot SNR 219–220
 Pilot symbol(s) 11, 21, 208, 222, 227, 354
 PLL. *See* Phase locked loop (PLL)
 PN sequences. *See* Pseudo-noise (PN) sequences
 Polynomial complexity 189
 Power-delay profile (PDP) 7, 70
 Power spectral density (PSD) 8, 237
 of oscillator phase noise 296
 Practical rectangular pulses 251–255
 Predictable taps 63
 Principal component analysis (PCA) 330
 Probabilistic graphical model 195
 Probability factorization 195, 204
 PSD. *See* Power spectral density (PSD)
 Pseudo-noise (PN) sequences 207
 auto-correlation function of 210
 pilot-based estimation 208–212
 Pulse-based waveforms 18
 Pulse rate 15
 Pulsones 15, 44, 109
 discrete 319–326
 frequency domain 25, 46, 48, 325
 generated using Heisenberg transform 110–113
 orthogonality of 49–50
 time-and bandwidth-limited 113–118
 time domain 25, 45–48, 50–51, 109–110
 Zak-OTFS 372
 PyTorch 332

q

QPSK modulation. *See* Quadrature phase shift keying (QPSK) modulation
 Quadrature phase shift keying (QPSK) modulation 3
 Quasi-periodic delay-Doppler (DD) pulse 371
 Quasi-periodicity 37–38
 communication and radar sensing 390
 DD domain pulse 20, 25, 38, 39, 63, 101, 109, 319
 DD functions 25, 51, 144, 392
 Dirac-delta impulse 399
 discrete DD domain signal 326
 impulse signal 95
 Zak-OTFS information signal 321

r

Radar sensing 373–374
 ambiguity function 374–380
 good waveform 381–383
 Zak theoretical framework 383–389
 Radar sensing, waveforms for 17–19
 Radio frequency (RF)
 band 2
 impairments 344
 Rank profile, of difference matrices 307
 RAS. *See* Receive antenna selection (RAS)
 Rayleigh distribution 296
 Rayleigh fading 282
 Receive antenna selection (RAS) 21
 MIMO-OTFS with 170–172
 multiantenna OTFS with 173–178
 STC-OTFS with 172–173
 Receive IQI compensation 345–346
 Receiver oscillator phase noise 295
 Recurrent neural networks (RNN) 333
 Regression learning programs 329
 Reinforcement learning 329
 ReLU activation function 334, 350–351, 356
 REs. *See* Resource elements (REs)
 Resource elements (REs) 233–236
 Revolutionary telecommunication technology 2
 Rician fading distribution 296
 RNN. *See* Recurrent neural networks (RNN)
 Root raised cosine (RRC) 64–65
 RRC. *See* Root raised cosine (RRC)

s

Sampling distributions 198
 SBL. *See* Sparse Bayesian learning (SBL)
 SBL-based channel estimation 224–227
 SC approach. *See* Single carrier (SC) approach
 Second generation (2G) systems 2
 Semi-supervised learning 328–329
 SFT. *See* Symplectic Fourier transform (SFT)
 Sigmoid activation function 334
 Signal detection algorithms 11, 295
 Signal-to-noise ratios (SNRs) 11, 16, 71, 149, 219
 Sinc pulse, ambiguity function for 388–389, 396–397
 Single carrier (SC) approach 10
 Single-input single-output (SISO) 149
 Single neuron 330
 SISO-OTFS 150–161
 BER performance of 155
 diversity analysis 154–157
 full diversity with phase rotation 157–161
 system model 150–154

- Small-scale fading 5
- Space-time coded OTFS (STC-OTFS) 165–170
- Sparse Bayesian learning (SBL) 277
- SP-I scheme 230–231
- SP-NI scheme 227, 229
- SSPNet 363
- Stalling 198
- State-of-the-art methods 233
- STC-OTFS. *See* Space-time coded OTFS (STC-OTFS)
- Superimposed pilot approach/frames 21, 208, 227–232, 355, 365–369
- Supervised learning 328
- Symbol-by-symbol decision 190
- Symbol-by-symbol detection 205
- Symbol-DNN detection approach 335–337, 340, 343
- Symbol-wise MAP decision rule 195–196
- Symplectic finite Fourier transform (SFFT) 193, 286, 294
- Symplectic Fourier transform (SFT) 135–138, 142
- Synchronization 277–284
 - frequency 277
 - timing 277–284
- t**
- Tapped-delay line (TDL) channel model 6–8
- Tapped delay line-D (TDL-D) model 296
- TD. *See* Time domain (TD)
- TDL-D model. *See* Tapped delay line-D (TDL-D) model
- TDM. *See* Time domain modulation (TDM)
- TDMA. *See* Time division multiple access (TDMA)
- TensorFlow 332
- TEP. *See* Timing error probability (TEP)
- Terahertz (THz) bands 5
- Testing phase (signal detection) 336
- TFRE. *See* Time-frequency resource element (TFRE)
- THP. *See* Tomlinson–Harashima precoding (THP)
- Third generation (3G) systems 3
- Threshold-based channel estimation 221–223
- Time division multiple access (TDMA) 2
- Time division multiplexing (TDM) 130
- Time domain (TD) 14–15
 - carrier waveform 43
 - characterization 7–9
 - Fourier transform of 33
 - pulsone 25, 45–48, 50–51, 89, 101, 109
 - quasi-localization in 381–383
 - transmit signal 235
 - using IDZT 303
- Time domain modulation (TDM) 14–15, 29–34, 68–69
 - attributes of 78
 - doubly spread channel with 30
 - I/O relation 90–92
 - as limiting case 50
 - in modulation domain and time domain 29
 - signal processing steps in 54
 - waveform 15
- Time domain (TD) pulse 18
 - channel interaction of 30–32
 - doubly spread channel with 34
- Time-frequency representation 13–14
- Time-frequency resource element (TFRE) 236, 256
- Time-limited DD domain carrier waveforms 43–51
- Time-selective fading 8
- Time-varying channel response
 - in DD domain characterization 13
 - in time-frequency representation 13
- Timing error probability (TEP) 282
 - vs. transmit SNR 283
- Timing synchronization 277–284
 - in OFDM-based systems 278
 - in OTFS-based systems 278–283
- Tomlinson–Harashima precoding (THP) 270
- Tone-based waveforms 18
- Traditional block SBL method 227
- Traditional frequency domain modulation (FDM) 23
- Traditional program 329–330
- Traditional time domain modulation (TDM) 23
- Transceiver signal processing 53
- Transmit DD pulse 40
- Transmit IQI compensation 344–345
- Transmit signal processing 257
- Transmit SNR 282
- Transmitted TD signal 2, 32, 150
- Transmitter oscillator 294
- Twisted convolution 23, 35, 78–83
 - in discrete DD domain 318
 - preserves quasi-periodicity 88–89
- Two-dimensional DD domain interval 235
- Two doubly spread channels 78–83
- Two-step MC-OTFS transceiver 126–129, 146
- u**
- UE. *See* User equipment (UE)
- Uniform rectangular array (URA) 261

Unitary transformation 199
 Unsupervised learning 328
 Uplink MA methods 234
 Uplink multiuser communication 236–265
 non-orthogonal multiple access 256–265
 orthogonal multiple access (OMA) 237–255
 Uplink timing synchronization 278–283
 URA. *See* Uniform rectangular array (URA)
 Urban multi-lanes 14
 User equipment (UE) 28
 User terminals (UTs) 233, 236, 256, 262, 271

V

Validation dataset 334
 Variational Bayes (VB) detection 192–194
 VB detection. *See* Variational Bayes (VB) detection
 Vectorized DD domain I/O relation 304–305
 Vector-matrix form 153
 Veh-A channel model 70, 72, 76, 131
 Virtual reality (VR) 4
 VR. *See* Virtual reality (VR)

W

Waveforms for radar sensing 17–19
 Wigner transform 150, 258
 Wireless channel response 149
 Wireless channels, characteristic of 5
 Woodbury matrix identity 225

Z

Zak-orthogonal time frequency space (Zak-OTFS)
 17–21, 26, 67–68
 comparing MC-OTFS to 130–132

 digital implementation of 311–316
 modulation 51–58
 generalized transceiver signal processing 52–54
 signal processing steps in 55
 modulator architecture 96–97
 pulsone carrier waveform 22
 radar sensing 22
 receiver 97–99
 digital implementation of 311–314
 signal processing 56–57
 TD pulsone 102, 383–384, 388–389, 398
 transmitter implementation 314–316
 uncoded BER performance for 131
 Zak-OTFS. *See* Zak-orthogonal time frequency space (Zak-OTFS)
 Zak-OTFS I/O relation 26, 57–58, 102
 attributes of 78
 in crystalline regime 58–66
 crystalline decomposition 62–63
 error in prediction of 64–66
 linear time-varying channels 63–64
 non-predictability of 61–62
 matrix-vector description of 67–70
 predictable and non-fading attribute of 78
 Zak theory 101, 373
 Zak transform approach/framework 17, 25, 36, 44, 84–85, 134, 150, 235, 286, 302
 definition of 312
 radar sensing 383–389
 Zak transform-based transceiver 129–130, 146
 Zero-forcing (ZF) detection 21, 183
 Zero-forcing precoder 268–269
 Zero-IF architecture 298
 Zeroth-order Bessel function 8

WILEY END USER LICENSE AGREEMENT

Go to www.wiley.com/go/eula to access Wiley's ebook EULA.